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# Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

Version x.y.z

where:

- x the first digit:
  - 1 presented to TSG for information;
  - 2 presented to TSG for approval;
  - 3 or greater indicates TSG approved document under change control.
- y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.
- z the third digit is incremented when editorial only changes have been incorporated in the document.

In the present document, modal verbs have the following meanings:

**shall** indicates a mandatory requirement to do something

**shall not** indicates an interdiction (prohibition) to do something

The constructions "shall" and "shall not" are confined to the context of normative provisions, and do not appear in Technical Reports.

The constructions "must" and "must not" are not used as substitutes for "shall" and "shall not". Their use is avoided insofar as possible, and they are not used in a normative context except in a direct citation from an external, referenced, non-3GPP document, or so as to maintain continuity of style when extending or modifying the provisions of such a referenced document.

**should** indicates a recommendation to do something

**should not** indicates a recommendation not to do something

**may** indicates permission to do something

**need not** indicates permission not to do something

The construction "may not" is ambiguous and is not used in normative elements. The unambiguous constructions "might not" or "shall not" are used instead, depending upon the meaning intended.

**can** indicates that something is possible

**cannot** indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

**will** indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

**will not** indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

**might** indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

**might not** indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

**is** (or any other verb in the indicative mood) indicates a statement of fact

**is not** (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

# 1 Scope

The present document establishes the minimum RF characteristics and minimum performance requirements of NR and NB-IoT operation in NR in-band Base Station (BS).

# 2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document.*

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] ITU-R Recommendation SM.329: "Unwanted emissions in the spurious domain".

[3] Recommendation ITU-R SM.328: "Spectra and bandwidth of emissions".

[4] 3GPP TR 25.942: "RF system scenarios".

[5] 3GPP TS 38.141-1: "NR; Base Station (BS) conformance testing; Part 1: Conducted conformance testing".

[6] 3GPP TS 38.141-2: "NR; Base Station (BS) conformance testing; Part 2: Radiated conformance testing".

[7] Recommendation ITU-R M.1545: "Measurement uncertainty as it applies to test limits for the terrestrial component of International Mobile Telecommunications-2000".

[8] "Title 47 of the Code of Federal Regulations (CFR)", Federal Communications Commission.

[9] 3GPP TS 38.211: "NR; Physical channels and modulation".

[10] 3GPP TS 38.213: "NR; Physical layer procedures for control".

[11] 3GPP TS 38.331: "NR; Radio Resource Control (RRC); Protocol specification".

[12] ECC/DEC/(17)06: "The harmonised use of the frequency bands 1427-1452 MHz and 1492-1518 MHz for Mobile/Fixed Communications Networks Supplemental Downlink (MFCN SDL)"

[13] 3GPP TS 36.104: "Evolved Universal Terrestrial Radio Access (E-UTRA); Base Station (BS) radio transmission and reception".

[14] 3GPP TS 37.105: "Active Antenna System (AAS) Base Station (BS) transmission and reception".

[15] 3GPP TS 38.212: "NR; Multiplexing and channel coding".

[16] 3GPP TR 38.901: "Study on channel model for frequencies from 0.5 to 100 GHz"

[17] 3GPP TS 38.101-1: "NR; User Equipment (UE) radio transmission and reception; Part 1: Range 1 Standalone".

[18] 3GPP TS 38.101-2: "NR; User Equipment (UE) radio transmission and reception; Part 2: Range 2 Standalone"

[19] ERC Recommendation 74-01, "Unwanted emissions in the spurious domain".

[20] 3GPP TS 37.213: "Physical layer procedures for shared spectrum channel access".

[21] ECC Decision(20)02: “Harmonised use of the paired frequency bands 874.4-880.0 MHz and 919.4-925.0 MHz and of the unpaired frequency band 1900-1910 MHz for Railway Mobile Radio (RMR)”

[22] 3GPP TR 38.852: Introduction of 1900MHz NR band for Europe for Rail Mobile Radio (RMR)

[23] 3GPP TR 38.853: Introduction of 900MHz NR band for Europe for Rail Mobile Radio (RMR)

[24] FCC Report And Order And Further Notice Of Proposed Rulemaking FCC 20-51, April 2020.

[25] FCC Order DA 20-48, April 2020

[26] Commission Implementing Decision (EU) 2020/590 of 24 April 2020 amending Decision (EU) 2019/784 as regards an update of relevant technical conditions applicable to the 24,25-27,5 GHz frequency band

# 3 Definitions, symbols and abbreviations

## 3.1 Definitions

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

**Aggregated BS Channel Bandwidth:**The RF bandwidth in which a Base Station transmits and receives multiple contiguously aggregated carriers. The *aggregated BS channel bandwidth* is measured in MHz.

**antenna connector:** connector at the conducted interface of the *BS type 1-C*

**active transmitter unit:** transmitter unit which is ON, and has the ability to send modulated data streams that are parallel and distinct to those sent from other transmitter units to a *BS type 1-C antenna connector*, or to one or more *BS type 1-H TAB connectors* at the *transceiver array boundary*

**Base Station RF Bandwidth**: RF bandwidth in which a base station transmits and/or receives single or multiple carrier(s) within a supported *operating band*

NOTE: In single carrier operation, the *Base Station RF Bandwidth* is equal to the *BS channel bandwidth*.

**Base Station RF Bandwidth edge:** frequency of one of the edges of the *Base Station RF Bandwidth*.

**basic limit:** emissions limit relating to the power supplied by a single transmitter to a single antenna transmission line in ITU-R SM.329 [2] used for the formulation of unwanted emission requirements for FR1

**beam:** beam (of the antenna) is the main lobe of the radiation pattern of an *antenna array*

NOTE: For certain BS *antenna array*, there may be more than one beam.

**beam centre direction:** direction equal to the geometric centre of the half-power contour of the beam

**beam direction pair:** data set consisting of the *beam centre direction* and the related *beam peak direction*

**beam peak direction:** direction where the maximum EIRP is found

**beamwidth:** beam which has a half-power contour that is essentially elliptical, the half-power beamwidths in the two pattern cuts that respectively contain the major and minor axis of the ellipse

**BS channel bandwidth**: RF bandwidth supporting a single NR RF carrier with the *transmission bandwidth* configured in the uplink or downlink

NOTE 1: The *BS channel bandwidth* is measured in MHz and is used as a reference for transmitter and receiver RF requirements.

NOTE 2: It is possible for the BS to transmit to and/or receive from one or more UE bandwidth parts that are smaller than or equal to the *BS transmission bandwidth configuration*, in any part of the *BS transmission bandwidth configuration*.

**BS transmission bandwidth configuration**: set of resource blocks located within the *BS channel bandwidth* which may be used for transmitting or receiving by the BS

**BS type 1-C:** NR base station operating at FR1 with requirements set consisting only of conducted requirements defined at individual *antenna connectors*

**BS type 1-H:** NR base station operating at FR1 with a *requirement set* consisting of conducted requirements defined at individual *TAB connectors* and OTA requirements defined at RIB

**BS type 1-O:** NR base station operating at FR1 with a *requirement set* consisting only of OTA requirements defined at the RIB

**BS type 2-O:** NR base station operating at FR2 with a *requirement set* consisting only of OTA requirements defined at the RIB

**Channel edge:** lowest or highest frequency of the NR carrier, separated by the *BS channel bandwidth*.

**Carrier aggregation:** aggregation of two or more component carriers in order to support wider *transmission bandwidths*

**Carrier aggregation configuration:** a set of one or more *operating bands* across which the BS aggregates carriers with a specific set of technical requirements

**co-location reference antenna**: a passive antenna used as reference for base station to base station co-location requirements

**Contiguous carriers:** set of two or more carriers configured in a spectrum block where there are no RF requirements based on co-existence for un-coordinated operation within the spectrum block.

**Contiguous spectrum:** spectrum consisting of a contiguous block of spectrum with no *sub-block gap(s).*

**directional requirement:** requirement which is applied in a specific direction within the *OTA coverage range* for the Tx and when the AoA of the incident wave of a received signal is within the *OTA REFSENS RoAoA* or the *minSENS RoAoA* as appropriate for the receiver

**equivalent isotropic radiated power:** equivalent power radiated from an isotropic directivity device producing the same field intensity at a point of observation as the field intensity radiated in the direction of the same point of observation by the discussed device

NOTE: Isotropic directivity is equal in all directions (i.e. 0 dBi).

**equivalent isotropic sensitivity:** sensitivity for an isotropic directivity device equivalent to the sensitivity of the discussed device exposed to an incoming wave from a defined AoA

NOTE 1: The sensitivity is the minimum received power level at which specific requirement is met.

NOTE 2: Isotropic directivity is equal in all directions (i.e. 0 dBi).

**fractional bandwidth:** *fractional bandwidth* FBW is defined as $FBW = 200 \cdot \frac{F_{FBWhigh} - F_{FBWlow}}{F_{FBWhigh} + F_{FBWlow}}\%$

**Highest Carrier:** The carrier with the highest carrier frequency transmitted/received in a specified frequency band.

**Inter-band carrier aggregation:** *carrier aggregation* of component carriers in different *operating bands***.**

NOTE: Carriers aggregated in each band can be contiguous or non-contiguous.

**Inter-band gap**: The frequency gap between two supported consecutive *operating bands*.

**Intra-band contiguous carrier aggregation:** *contiguous carriers* aggregated in the same *operating band*.

**Intra-band non-contiguous carrier aggregation:** non-contiguous carriers aggregated in the same *operating band*.

**Inter RF Bandwidth gap:** frequency gap between two consecutive *Base Station RF Bandwidths* that are placed within two supported *operating bands*

**Lowest Carrier:** The carrier with the lowest carrier frequency transmitted/received in a specified frequency band.

**Lower sub-block edge:** frequency at the lower edge of one *sub-block.*

NOTE: It is used as a frequency reference point for both transmitter and receiver requirements.

**maximum carrier output power:** mean power level measured per carrier at the indicated interface, during the *transmitter ON period* in a specified reference condition

**maximum carrier TRP output power:** mean power level measured per RIB during the *transmitter ON period* for a specific carrier in a specified reference condition and corresponding to the declared *rated carrier TRP output* power ($P_{rated,c,TRP}$)

**maximum total output power:** mean power level measured within the *operating band* at the indicated interface, during the *transmitter ON period* in a specified reference condition

**maximum total TRP output power:** mean power level measured per RIB during the *transmitter ON period* in a specified reference condition and corresponding to the declared *rated total TRP output* power ($P_{rated,t,TRP}$)

**measurement bandwidth**: RF bandwidth in which an emission level is specified

**minSENS:** the lowest declared EIS value for the OSDD's declared for OTA sensitivity requirement.

**minSENS RoAoA:** The *reference RoAoA* associated with the OSDD with the lowest declared EIS

**multi-band connector**: *Antenna Connector* of *BS type 1-C* or *TAB connector* of *BS type 1-H* associated with a transmitter or receiver that is characterized by the ability to process two or more carriers in common active RF components simultaneously, where at least one carrier is configured at a different *operating band* than the other carrier(s) and where this different *operating band* is not a *sub-band* or *superseding-band* of another supported *operating band*

**multi-band RIB:** *operating band* specific RIB associated with a transmitter or receiver that is characterized by the ability to process two or more carriers in common active RF components simultaneously, where at least one carrier is configured at a different *operating band* than the other carrier(s) and where this different *operating band* is not a *sub-band* or *superseding-band* of another supported *operating band*

**Multi-carrier transmission configuration:** set of one or more contiguous or non-contiguous carriers that a NR BS is able to transmit simultaneously according to the manufacturer's specification.

**NB-IoT operation in NR in-band:** NB-IoT is operating in-band when it is located within a NR transmission bandwidth configuration plus 15 kHz at each edge but not within the NR minimum guard band $GB_{Channel}$.

**NB-IoT operation in NR guard band:** NB-IoT is operating in guard band when it is located within a NR BS channel bandwidth but is not NB-IoT operation in NR in-band.

**Non-contiguous spectrum:** spectrum consisting of two or more *sub-blocks* separated by *sub-block gap(s)*.

**operating band:** frequency range in which NR operates (paired or unpaired), that is defined with a specific set of technical requirements

NOTE: The *operating band*(s) for a BS is declared by the manufacturer according to the designations in tables 5.2-1 and 5.2-2.

**OTA coverage range**: a common range of directions within which TX OTA requirements that are neither specified in the *OTA peak directions sets* nor as *TRP requirement* are intended to be met

**OTA peak directions set:** set(s) of *beam peak directions* within which certain TX OTA requirements are intended to be met, where all *OTA peak directions set(s)* are subsets of the *OTA coverage range*

NOTE: The *beam peak directions* are related to a corresponding contiguous range or discrete list of *beam centre directions* by the *beam direction pairs* included in the set.

**OTA REFSENS RoAoA:** the RoAoA determined by the contour defined by the points at which the achieved EIS is 3dB higher than the achieved EIS in the reference direction assuming that for any AoA, the receiver gain is optimized for that AoA

NOTE: This contour will be related to the average element/sub-array radiation pattern 3dB beamwidth.

**OTA sensitivity directions declaration:** set of manufacturer declarations comprising at least one set of declared minimum EIS values (with *BS channel bandwidth*), and related directions over which the EIS applies

NOTE: All the directions apply to all the EIS values in an OSDD.

**polarization match:** condition that exists when a plane wave, incident upon an antenna from a given direction, has a polarization that is the same as the receiving polarization of the antenna in that direction

**radiated interface boundary**: *operating band* specific radiated requirements reference where the radiated requirements apply

NOTE: For requirements based on EIRP/EIS, the *radiated interface boundary* is associated to the far-field region

**Radio Bandwidth:** frequency difference between the upper edge of the highest used carrier and the lower edge of the lowest used carrier

**Railway Mobile Radio:** railway operations encompassing GSM-R and its successor(s), including the Future Railway Mobile Communication System (FRMCS); in the context of this specification the Railway Mobile Radio is limited to NR operation in band n100, or n101

**rated beam EIRP:** For a declared beam and *beam direction pair*, the *rated beam EIRP* level is the maximum power that the base station is declared to radiate at the associated *beam peak direction* during the *transmitter ON period*

**rated carrier output power:** mean power level associated with a particular carrier the manufacturer has declared to be available at the indicated interface, during the *transmitter ON period* in a specified reference condition

**rated carrier TRP output power:** mean power level declared by the manufacturer per carrier, for BS operating in single carrier, multi-carrier, or carrier aggregation configurations that the manufacturer has declared to be available at the RIB during the *transmitter ON period*

**rated total output power:** mean power level associated with a particular *operating band* the manufacturer has declared to be available at the indicated interface, during the *transmitter ON period* in a specified reference condition

**rated total TRP output power:** mean power level declared by the manufacturer, that the manufacturer has declared to be available at the RIB during the *transmitter ON period*

**reference beam direction pair:** declared *beam direction pair*, including reference *beam centre direction* and reference *beam peak direction* where the reference *beam peak direction* is the direction for the intended maximum EIRP within the *OTA peak directions set*

**receiver target:** AoA in which reception is performed by *BS types 1-H* or *BS type 1-O*

**receiver target redirection range:** union of all the *sensitivity RoAoA* achievable through redirecting the *receiver target* related to particular OSDD

**receiver target reference direction:** direction inside the *OTA sensitivity directions declaration* declared by the manufacturer for conformance testing. For an OSDD without *receiver target redirection range*, this is a direction inside the *sensitivity RoAoA*

**reference RoAoA**: the *sensitivity RoAoA* associated with the *receiver target reference direction* for each OSDD.

**requirement set:**one of the NR base station requirement's set as defined for *BS type 1-C*, *BS type 1-H*, *BS type 1-O*, and *BS type 2-O*

**sensitivity RoAoA:** RoAoA within the *OTA sensitivity directions declaration*, within which the declared EIS(s) of an OSDD is intended to be achieved at any instance of time for a specific BS direction setting

**single-band connector:** *BS type 1-C antenna connector* or *BS type 1-H TAB connector* supporting operation either in a single *operating band* only, or in multiple *operating bands* but does not meet the conditions for a *multi-band connector*.

**single-band RIB:** *operating band* specific RIB supporting operation either in a single *operating band* only, or in multiple *operating bands* but does not meet the conditions for a *multi-band RIB*.

**sub-band**: A *sub-band* of an operating band contains a part of the uplink and downlink frequency range of the operating band.

**sub-block bandwidth:** bandwidth of one *sub-block*.

**sub-block:** one contiguous allocated block of spectrum for transmission and reception by the same base station

NOTE: There may be multiple instances of *sub-blocks* within a *Base Station RF Bandwidth*.

**sub-block gap:** frequency gap between two consecutive sub-blocks within a *Base Station RF Bandwidth*, where the RF requirements in the gap are based on co-existence for un-coordinated operation

**superseding-band**: A *superseding-band* of an operating band includes the whole of the uplink and downlink frequency range of the operating band.

**TAB connector:** *transceiver array boundary* connector

**TAB connector RX min cell group:** *operating band* specific declared group of *TAB connectors* to which *BS type 1-H* conducted RX requirements are applied

NOTE: Within this definition, the group corresponds to the group of *TAB connectors* which are responsible for receiving a cell when the *BS type 1-H* setting corresponding to the declared minimum number of cells with reception on all *TAB connectors* supporting an *operating band*, but its existence is not limited to that condition

**TAB connector TX min cell group:** *operating band* specific declared group of *TAB connectors* to which *BS type 1-H* conducted TX requirements are applied.

NOTE: Within this definition, the group corresponds to the group of *TAB connectors* which are responsible for transmitting a cell when the *BS type 1-H* setting corresponding to the declared minimum number of cells with transmission on all *TAB connectors* supporting an *operating band*, but its existence is not limited to that condition

**total radiated power:** is the total power radiated by the antenna

NOTE: The *total radiated power* is the power radiating in all direction for two orthogonal polarizations. *Total radiated power* is defined in both the near-field region and the far-field region

**transceiver array boundary:** conducted interface between the transceiver unit array and the composite antenna

**transmission bandwidth:** RF Bandwidth of an instantaneous transmission from a UE or BS, measured in resource block units

**transmitter OFF period:** time period during which the BS transmitter is not allowed to transmit

**transmitter ON period:** time period during which the BS transmitter is transmitting data and/or reference symbols

**transmitter transient period:** time period during which the transmitter is changing from the OFF period to the ON period or vice versa

**UE transmission bandwidth configuration:** set of resource blocks located within the UE channel bandwidth which may be used for transmitting or receiving by the UE

**upper sub-block edge:** frequency at the upper edge of one *sub-block*.

NOTE: It is used as a frequency reference point for both transmitter and receiver requirements.

# 3.2 Symbols

For the purposes of the present document, the following symbols apply:

| | |
|---|---|
| $\beta$ | Percentage of the mean transmitted power emitted outside the occupied bandwidth on the assigned channel |
| $BeW_{\theta,REFSENS}$ | Beamwidth equivalent to the *OTA REFSENS RoAoA* in the θ-axis in degrees. Applicable for FR1 only. |
| $BeW_{\varphi,REFSENS}$ | Beamwidth equivalent to the *OTA REFSENS RoAoA* in the φ-axis in degrees. Applicable for FR1 only. |
| $BW_{Channel}$ | *BS channel bandwidth* |
| $BW_{Channel_CA}$ | *Aggregated BS Channel Bandwidth*, expressed in MHz. $BW_{Channel_CA} = F_{edge,high} - F_{edge,low}$. |
| $BW_{Channel,block}$ | *Sub-block bandwidth*, expressed in MHz. $BW_{Channel,block} = F_{edge,block,high} - F_{edge,block,low}$. |
| $BW_{Config}$ | *Transmission bandwidth configuration*, where $BW_{Config} = N_{RB}$ x SCS x 12 |
| $BW_{Contiguous}$ | Contiguous *transmission bandwidth*, i.e. *BS channel bandwidth* for single carrier or *Aggregated BS channel bandwidth* for contiguously aggregated carriers. For non-contiguous operation within a band the term is applied per *sub-block*. |
| $BW_{GB,low}$ | The minimum guard band defined in clause 5.3.3 for lowest assigned component carrier |
| $BW_{GB,high}$ | The minimum guard band defined in clause 5.3.3 for highest assigned component carrier |
| $\Delta f$ | Separation between the *channel edge* frequency and the nominal -3 dB point of the measuring filter closest to the carrier frequency |
| $\Delta f_{BE_offset}$ | Separation between the edge of the last transmitted channel of the channels assigned for NR-U channel bandwidth and the nominal -3 dB point of the measuring filter closest to the carrier frequency |
| $\Delta F_{Global}$ | Global frequency raster granularity |
| $\Delta f_{max}$ | $f_offset_{max}$ minus half of the bandwidth of the measuring filter |
| $\Delta f_{OBUE}$ | Maximum offset of the *operating band* unwanted emissions mask from the downlink *operating band* edge |
| $\Delta f_{OOB}$ | Maximum offset of the out-of-band boundary from the uplink *operating band* edge |
| $\Delta_{FR2_REFSENS}$ | Offset applied to the FR2 OTA REFSENS depending on the AoA |
| $\Delta_{minSENS}$ | Difference between conducted reference sensitivity and minSENS |
| $\Delta_{OTAREFSENS}$ | Difference between conducted reference sensitivity and OTA REFSENS |
| $\Delta F_{Raster}$ | Channel raster granularity |
| $\Delta_{shift}$ | Channel raster offset for SUL |
| $EIS_{minSENS}$ | The EIS declared for the *minSENS RoAoA* |
| $EIS_{REFSENS}$ | OTA REFSENS EIS value |
| $EIS_{REFSENS_50M}$ | Declared OTA reference sensitivity basis level for FR2 based on a reference measurement channel with 50MHz *BS channel bandwidth* |
| $F_{FBWhigh}$ | Highest supported frequency within supported *operating band*, for which *fractional bandwidth* support was declared |
| $F_{FBWlow}$ | Lowest supported frequency within supported *operating band*, for which *fractional bandwidth* support was declared |
| $F_C$ | *RF reference frequency* on the channel raster, given in table 5.4.2.2-1 |
| $F_{C,block,high}$ | Fc of the highest transmitted/received carrier in a *sub-block*. |
| $F_{C,block,low}$ | Fc of the lowest transmitted/received carrier in a *sub-block*. |
| $F_{C,low}$ | The Fc of the *lowest carrier*, expressed in MHz. |
| $F_{C,high}$ | The Fc of the *highest carrier*, expressed in MHz. |
| $F_{DL,low}$ | The lowest frequency of the downlink *operating band* |
| $F_{DL,high}$ | The highest frequency of the downlink *operating band* |
| $F_{edge,low}$ | The lower edge of *Aggregated BS Channel Bandwidth*, expressed in MHz. $F_{edge,low} = F_{C,low} - F_{offset,low}$. |
| $F_{edge,high}$ | The upper edge of *Aggregated BS Channel Bandwidth*, expressed in MHz. $F_{edge,high} = F_{C,high} + F_{offset,high}$. |
| $F_{edge,block,low}$ | The *lower sub-block edge*, where $F_{edge,block,low} = F_{C,block,low} - F_{offset,low}$. |
| $F_{edge,block,high}$ | The *upper sub-block edge*, where $F_{edge,block,high} = F_{C,block,high} + F_{offset,high}$. |
| $F_{filter}$ | Filter centre frequency |
| $F_{offset,high}$ | Frequency offset from $F_{C,high}$ to the upper *Base Station RF Bandwidth edge*, or from $F_{C,block,high}$ to the *upper sub-block edge* |
| $F_{offset,low}$ | Frequency offset from $F_{C,low}$ to the lower *Base Station RF Bandwidth edge*, or from $F_{C,block,low}$ to the *lower sub-block edge*. |

| | |
|---|---|
| f_BE_offset | Separation between the edge of the last transmitted channel of the channels assigned for NR-U channel bandwidth and the centre of the measuring |
| f_offset | Separation between the *channel edge* frequency and the centre of the measuring |
| $f_offset_{max}$ | The offset to the frequency $\Delta f_{OBUE}$ outside the downlink *operating band* |
| $F_{REF}$ | RF reference frequency |
| $F_{REF-Offs}$ | Offset used for calculating $F_{REF}$ |
| $F_{REF,shift}$ | RF reference frequency for Supplementary Uplink (SUL) bands |
| $F_{step,X}$ | Frequency steps for the OTA transmitter spurious emissions (Category B) |
| $F_{UL,low}$ | The lowest frequency of the uplink *operating band* |
| $F_{UL,high}$ | The highest frequency of the uplink *operating band* |
| $GB_{Channel}$ | Minimum guard band defined in clause 5.3.3 |
| $N_{cells}$ | The declared number corresponding to the minimum number of cells that can be transmitted by an *BS type 1-H* in a particular *operating band* |
| $n_{PRB}$ | Physical resource block number |
| $N_{RB}$ | *Transmission bandwidth configuration*, expressed in resource blocks |
| $N_{RB,high}$ | *Transmission bandwidth configuration* for the highest assigned component carrier within a *sub-block* in CA |
| $N_{RB,low}$ | *Transmission bandwidth configuration* for the lowest assigned component carrier within a *sub-block* in CA |
| $N_{REF}$ | NR Absolute Radio Frequency Channel Number (NR-ARFCN) |
| $N_{REF-Offs}$ | Offset used for calculating $N_{REF}$ |
| $N_{RXU,active}$ | The number of active receiver units. The same as the number of *demodulation branches* to which compliance is declared for chapter 8 performance requirements |
| $N_{RXU,counted}$ | The number of active receiver units that are taken into account for conducted Rx spurious emission scaling, as calculated in clause 7.6.1 |
| $N_{RXU,countedpercell}$ | The number of active receiver units that are taken into account for conducted RX spurious emissions scaling per cell, as calculated in clause 7.6.1 |
| $N_{TXU,counted}$ | The number of *active transmitter units* as calculated in clause 6.1, that are taken into account for conducted TX output power limit in clause 6.2.1, and for unwanted TX emissions scaling |
| $N_{TXU,countedpercell}$ | The number of *active transmitter units* that are taken into account for conducted TX emissions scaling per cell, as calculated in clause 6.1 |
| $P_{EM,n50/n75,ind}$ | Declared emission level for Band n50/n75; ind = a, b |
| $P_{EIRP,N}$ | EIRP level for channel N |
| $P_{max,c,AC}$ | *Maximum carrier output power* measured per *antenna connector* |
| $P_{max,c,cell}$ | The *maximum carrier output power* per *TAB connector TX min cell group* |
| $P_{max,c,TABC}$ | The *maximum carrier output power per TAB connector* |
| $P_{max,c,TRP}$ | *Maximum carrier TRP output power* measured at the RIB(s), and corresponding to the declared *rated carrier TRP output power* ($P_{rated,c,TRP}$) |
| $P_{max,c,EIRP}$ | The maximum carrier EIRP when the NR BS is configured at the maximum rated carrier output TRP ($P_{rated,c,TRP}$) |
| $P_{rated,c,AC}$ | The *rated carrier output power per antenna connector* |
| $P_{rated,c,cell}$ | The *rated carrier output power* per *TAB connector TX min cell group* |
| $P_{rated,c,FBWhigh}$ | The rated carrier EIRP for the higher supported frequency range within supported *operating band*, for which *fractional bandwidth* support was declared |
| $P_{rated,c,FBWlow}$ | The rated carrier EIRP for the lower supported frequency range within supported *operating band*, for which *fractional bandwidth* support was declared |
| $P_{rated,c,sys}$ | The sum of $P_{rated,c,TABC}$ for all *TAB connectors* for a single carrier |
| $P_{rated,c,TABC}$ | The *rated carrier output power per TAB connector* |
| $P_{rated,c,TRP}$ | *Rated carrier TRP output power* declared per RIB |
| $P_{rated,t,AC}$ | The *rated total output power* declared at the *antenna connector* |
| $P_{rated,t,TABC}$ | The *rated total output power* declared at *TAB connector* |
| $P_{rated,t,TRP}$ | *Rated total TRP output power* declared per RIB |
| $P_{REFSENS}$ | Conducted Reference Sensitivity power level |
| $SCS_{low}$ | Sub-Carrier Spacing for the lowest assigned component carrier within a *sub-block* in CA |
| $SCS_{high}$ | Sub-Carrier Spacing for the highest assigned component carrier within a *sub-block* in CA |
| $SS_{REF}$ | SS block reference frequency position |
| $W_{gap}$ | *Sub-block gap* or *Inter RF Bandwidth gap* size |

## 3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

| | |
|---|---|
| AA | Antenna Array |
| AAS | Active Antenna System |
| ACLR | Adjacent Channel Leakage Ratio |
| ACS | Adjacent Channel Selectivity |
| AoA | Angle of Arrival |
| AWGN | Additive White Gaussian Noise |
| BS | Base Station |
| BW | Bandwidth |
| CA | Carrier Aggregation |
| CACLR | Cumulative ACLR |
| CPE | Common Phase Error |
| CP-OFDM | Cyclic Prefix-OFDM |
| CW | Continuous Wave |
| DFT-s-OFDM | Discrete Fourier Transform-spread-OFDM |
| DM-RS | Demodulation Reference Signal |
| EIS | Equivalent Isotropic Sensitivity |
| EIRP | Effective Isotropic Radiated Power |
| E-UTRA | Evolved UTRA |
| EVM | Error Vector Magnitude |
| FBW | Fractional Bandwidth |
| FR | Frequency Range |
| FRC | Fixed Reference Channel |
| FRMCS | Future Railway Mobile Communication System |
| GSCN | Global Synchronization Channel Number |
| GSM | Global System for Mobile communications |
| HAPS | High Altitude Platform Station |
| ITU-R | Radiocommunication Sector of the International Telecommunication Union |
| ICS | In-Channel Selectivity |
| LA | Local Area |
| LNA | Low Noise Amplifier |
| MCS | Modulation and Coding Scheme |
| MR | Medium Range |
| NB-IoT | Narrowband – Internet of Things |
| NR | New Radio |
| NR-ARFCN | NR Absolute Radio Frequency Channel Number |
| OBUE | Operating Band Unwanted Emissions |
| OCC | Orthogonal Covering Code |
| OOB | Out-of-band |
| OSDD | OTA Sensitivity Directions Declaration |
| OTA | Over-The-Air |
| PRB | Physical Resource Block |
| PT-RS | Phase Tracking Reference Signal |
| QAM | Quadrature Amplitude Modulation |
| RB | Resource Block |
| RDN | Radio Distribution Network |
| RE | Resource Element |
| REFSENS | Reference Sensitivity |
| RF | Radio Frequency |
| RIB | Radiated Interface Boundary |
| RMR | Railway Mobile Radio |
| RMS | Root Mean Square (value) |
| RoAoA | Range of Angles of Arrival |
| QAM | Quadrature Amplitude Modulation |
| RB | Resource Block |
| RX | Receiver |

| | |
|---|---|
| SCS | Sub-Carrier Spacing |
| SDL | Supplementary Downlink |
| SS | Synchronization Symbol |
| SSB | Synchronization Signal Block |
| SUL | Supplementary Uplink |
| TAB | Transceiver Array Boundary |
| TAE | Time Alignment Error |
| TDL | Tapped Delay Line |
| TX | Transmitter |
| TRP | Total Radiated Power |
| UCI | Uplink Control Information |
| UEM | Unwanted Emissions Mask |
| UTRA | Universal Terrestrial Radio Access |
| WA | Wide Area |
| ZF | Zero Forcing |

# 4 General

## 4.1 Relationship with other core specifications

The present document is a single-RAT specification for a BS, covering RF characteristics and minimum performance requirements. Conducted and radiated core requirements are defined for the BS architectures and BS types defined in clause 4.3.

The applicability of each requirement is described in clause 5.

## 4.2 Relationship between minimum requirements and test requirements

Conformance to the present specification is demonstrated by fulfilling the test requirements specified in the conformance specification TS 38.141-1 [5] and TS 38.141-2 [6].

The minimum requirements given in this specification make no allowance for measurement uncertainty. The test specifications TS 38.141-1 [5] and TS 38.141-2 [6] define test tolerances. These test tolerances are individually calculated for each test. The test tolerances are used to relax the minimum requirements in this specification to create test requirements. For some requirements, including regulatory requirements, the test tolerance is set to zero.

The measurement results returned by the test system are compared - without any modification - against the test requirements as defined by the shared risk principle.

The shared risk principle is defined in recommendation ITU-R M.1545 [7].

## 4.3 Conducted and radiated requirement reference points

### 4.3.1 *BS type 1-C*

For *BS type 1-C*, the requirements are applied at the BS *antenna connector* (port A) for a single transmitter or receiver with a full complement of transceivers for the configuration in normal operating conditions. If any external apparatus such as an amplifier, a filter or the combination of such devices is used, requirements apply at the far end *antenna connector* (port B).

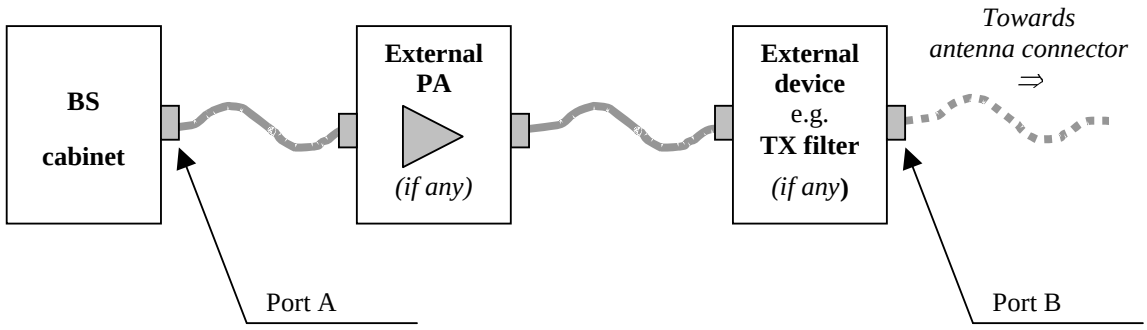



**Figure 4.3.1-1: *BS type 1-C* transmitter interface**

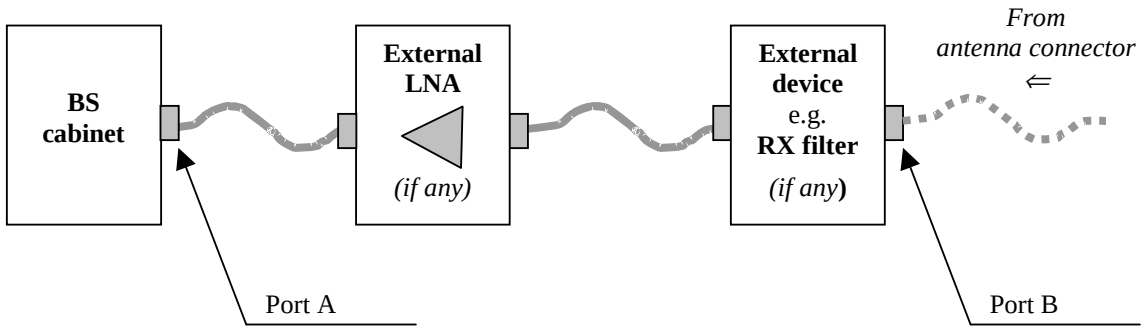



**Figure 4.3.1-2: *BS type 1-C* receiver interface**

## 4.3.2 *BS type 1-H*

For *BS type 1-H*, the requirements are defined for two points of reference, signified by radiated requirements and conducted requirements.

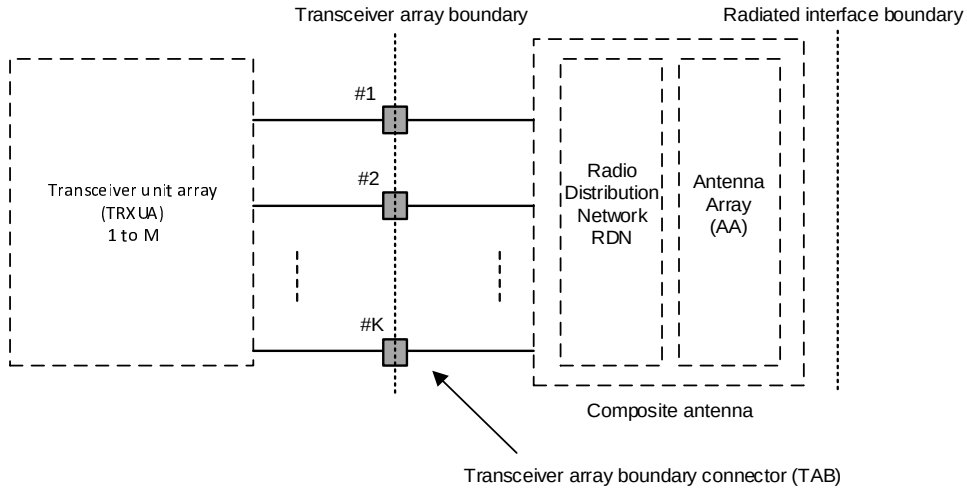



**Figure 4.3.2-1: Radiated and conducted reference points for *BS type 1-H***

Radiated characteristics are defined over the air (OTA), where the *operating band* specific radiated interface is referred to as the *Radiated Interface Boundary* (RIB). Radiated requirements are also referred to as OTA requirements. The (spatial) characteristics in which the OTA requirements apply are detailed for each requirement.

Conducted characteristics are defined at individual or groups of *TAB connectors* at the *transceiver array boundary*, which is the conducted interface between the transceiver unit array and the composite antenna.

The transceiver unit array is part of the composite transceiver functionality generating modulated transmit signal structures and performing receiver combining and demodulation.

The transceiver unit array contains an implementation specific number of transmitter units and an implementation specific number of receiver units. Transmitter units and receiver units may be combined into transceiver units. The transmitter/receiver units have the ability to transmit/receive parallel independent modulated symbol streams.

The composite antenna contains a radio distribution network (RDN) and an antenna array. The RDN is a linear passive network which distributes the RF power generated by the transceiver unit array to the antenna array, and/or distributes the radio signals collected by the antenna array to the transceiver unit array, in an implementation specific way.

How a conducted requirement is applied to the *transceiver array boundary* is detailed in the respective requirement clause.

### 4.3.3 *BS type 1-O* and *BS type 2-O*

For *BS type 1-O* and *BS type 2-O*, the radiated characteristics are defined over the air (OTA), where the *operating band* specific radiated interface is referred to as the *Radiated Interface Boundary* (RIB). Radiated requirements are also referred to as OTA requirements. The (spatial) characteristics in which the OTA requirements apply are detailed for each requirement.

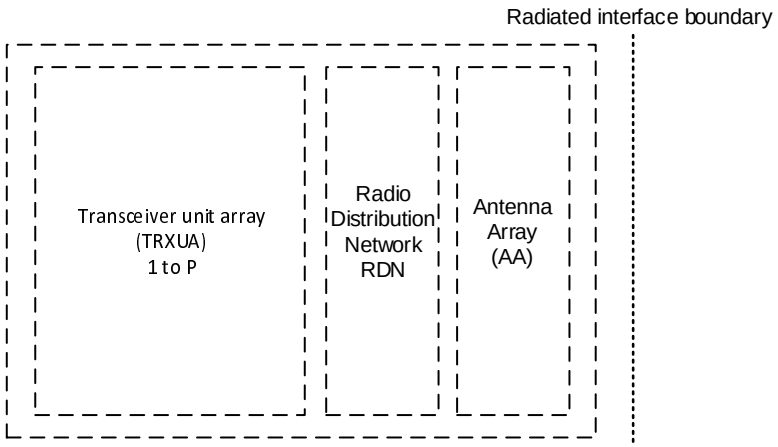



**Figure 4.3.3-1: Radiated reference points for *BS type 1-O* and *BS type 2-O***

Co-location requirements are specified at the conducted interface of the *co-location reference antenna*, the c*o-location reference antenna* does not form part of the BS under test but is a means to provide OTA power levels which are representative of a co-located system, further defined in clause 4.9.

For a *BS type 1-O* the transceiver unit array must contain at least 8 transmitter units and at least 8 receiver units. Transmitter units and receiver units may be combined into transceiver units. The transmitter/receiver units have the ability to transmit/receive parallel independent modulated symbol streams.

## 4.4 Base station classes

The requirements in this specification apply to Wide Area Base Stations, Medium Range Base Stations and Local Area Base Stations unless otherwise stated. The associated deployment scenarios for each class are exactly the same for BS with and without connectors.

For BS *type 1-O* and 2-O, BS classes are defined as indicated below:

- Wide Area Base Stations are characterised by requirements derived from Macro Cell scenarios with a BS to UE minimum distance along the ground equal to 35 m.
- Medium Range Base Stations are characterised by requirements derived from Micro Cell scenarios with a BS to UE minimum distance along the ground equal to 5 m.
- Local Area Base Stations are characterised by requirements derived from Pico Cell scenarios with a BS to UE minimum distance along the ground equal to 2 m.

For *BS type 1-C* and 1-H, BS classes are defined as indicated below:

- Wide Area Base Stations are characterised by requirements derived from Macro Cell scenarios with a BS to UE minimum coupling loss equal to 70 dB.
- Medium Range Base Stations are characterised by requirements derived from Micro Cell scenarios with a BS to UE minimum coupling loss equals to 53 dB.
- Local Area Base Stations are characterised by requirements derived from Pico Cell scenarios with a BS to UE minimum coupling loss equal to 45 dB.

NOTE: Local Are Base stations are applicable also in Femto Cell scenarios.

For BS *type 1-C, 1-H and 1-O*, HAPS BS class is defined as indicated below:

- HAPS Base Stations are characterised by requirements derived from High Altitude Platform scenarios with a BS to ground UE minimum distance of typically around 20km.
- Unless otherwise stated, HAPS BS class would refer to Wide Area BS class, which is specified in clause 4.4.

## 4.5 Regional requirements

Some requirements in the present document may only apply in certain regions either as optional requirements, or as mandatory requirements set by local and regional regulation. It is normally not stated in the 3GPP specifications under what exact circumstances the regional requirements apply, since this is defined by local or regional regulation.

Table 4.5-1 lists all requirements in the present specification that may be applied differently in different regions.

**Table 4.5-1: List of regional requirements**

| Clause number | Requirement | Comments |
|---|---|---|
| 5.2 | *Operating bands* | Some NR *operating bands* may be applied regionally. |
| 6.2.1, 9.3.1 | Base station output power, OTA base station output power: | For Band n41 and n90 operation in Japan, additional output power limits shall be applied. |
| 6.2.4, 9.3.4 | Base station output power, OTA base station output power: Additional requirements | These requirements may be applied regionally as additional base station output power requirements. For operation with shared spectrum channel access, the BS may have to comply with the applicable BS power limits established regionally, when deployed in regions where those limits apply and under the conditions declared by the manufacturer. |
| 6.6.2, 9.7.2 | Occupied bandwidth, OTA occupied bandwidth | The requirement may be applied regionally. There may also be regional requirements to declare the occupied bandwidth according to the definition in present specification. |
| 6.6.3.3 | Adjacent Channel Leakage Power Ratio | For Band n41 and n90 operation in Japan, absolute ACLR limits shall be applied to the sum of the absolute ACLR power over all *antenna connectors* for *BS type 1-C*. |
| 6.6.4.2, 9.7.4.2 | Operating band unwanted emission, OTA operating band unwanted emissions | Category A or Category B operating band unwanted emissions limits may be applied regionally. In addition, for operation with shared spectrum channel access, the BS may have to comply with the applicable operating band unwanted emission limits established regionally, when deployed in regions where those limits apply and under the conditions declared by the manufacturer. |
| 6.6.4.2.5.1, 9.7.4.2.1.2 | Operating band unwanted emission, OTA operating band unwanted emissions: Limits in FCC Title 47 | The BS may have to comply with the additional requirements, when deployed in regions where those limits are applied, and under the conditions declared by the manufacturer. |
| 6.6.4.2.5.2, 9.7.4.2.1.1 | Operating band unwanted emission, OTA operating band unwanted emissions Protection of DTT | The BS operating in Band n20 may have to comply with the additional requirements for protection of DTT, when deployed in certain regions. |
| 6.6.4.3 | Operating band unwanted emissions | For Band n41 and n90 operation in Japan, the operating band unwanted emissions limits shall be applied to the sum of the emission power over all *antenna connectors* for *BS type 1-C.* |
| 6.6.5.2.1, 9.7.5.2 | Tx spurious emissions, OTA Tx spurious emissions | Category A or Category B spurious emission limits, as defined in ITU-R Recommendation SM.329 [2], may apply regionally. The emission limits for *BS type 1-H* and *BS type 1-O* specified as the *basic limit* + X (dB) are applicable, unless stated differently in regional regulation. In addition, for operation with shared spectrum channel access, the BS may have to comply with the applicable spurious emission limits established regionally, when deployed in regions where those limits apply and under the conditions declared by the manufacturer. |
| 6.6.5.2.3, 9.7.5.3.3 | Tx spurious emissions: additional requirements, OTA Tx spurious emissions: additional requirements | These requirements may be applied for the protection of system operating in frequency ranges other than the BS *operating band*. |
| 6.6.5.3 | Transmitter spurious emissions | For Band n41 and n90 operation in Japan, the sum of the spurious emissions over all *antenna connectors* for *BS type 1-C* shall not exceed the *basic limits*. |
| 6.7.2.1.1, 6.7.3.1.1 9.8.2 | Transmitter intermodulation, OTA transmitter intermodulation | Interfering signal positions that are partially or completely outside of any downlink *operating band* of the base station are not excluded from the requirement in Japan in Band n77, n78, n79. |
| 6.7.2.2, 6.7.3.3 | Transmitter intermodulation | For Band n41 and n90 operation in Japan, the BS may have to comply with the additional requirements, when deployed in certain regions. |
| 6.7.2.2, 6.7.3.3, 9.8.3 | Transmitter intermodulation, OTA transmitter intermodulation | For Band n79 operation in Japan, the BS shall comply with the additional requirements. |
| 7.6.3 | Rx spurious emissions, | For Band n41 and n90 operation in Japan, the emission limits for *BS type 1-C* may apply to the sum of the emission power over all *antenna connectors*. |

| Clause number | Requirement | Comments |
|---|---|---|
| 7.6.4,<br>10.7.2<br>10.7.3 | Rx spurious emissions,<br>OTA Rx spurious emissions | The emission limits for BS type 1-H and BS type 1-O specified as the basic limit + X (dB) are applicable, unless stated differently in regional regulation.<br>Additional limits for BS type 2-O may apply regionally. |

# 4.6 Applicability of requirements

In table 4.6-1, the requirement applicability for each *requirement set* is defined. For each requirement, the applicable requirement clause in the specification is identified. Requirements not included in a *requirement set* is marked not applicable (NA).

**Table 4.6-1: *Requirement set* applicability**

| Requirement | Requirement set | | | |
|---|---|---|---|---|
| | ***BS type 1-C*** | ***BS type 1-H*** | ***BS type 1-O*** | ***BS type 2-O*** |
| BS output power | 6.2 | 6.2 | NA | NA |
| Output power dynamics | 6.3 | 6.3 | | |
| Transmit ON/OFF power | 6.4 | 6.4 | | |
| Transmitted signal quality | 6.5 | 6.5 | | |
| Occupied bandwidth | 6.6.2 | 6.6.2 | | |
| ACLR | 6.6.3 | 6.6.3 | | |
| Operating band unwanted emissions | 6.6.4 | 6.6.4 | | |
| Transmitter spurious emissions | 6.6.5 | 6.6.5 | | |
| Transmitter intermodulation | 6.7 | 6.7 | | |
| Reference sensitivity level | 7.2 | 7.2 | | |
| Dynamic range | 7.3 | 7.3 | | |
| In-band selectivity and blocking | 7.4 | 7.4 | | |
| Out-of-band blocking | 7.5 | 7.5 | | |
| Receiver spurious emissions | 7.6 | 7.6 | | |
| Receiver intermodulation | 7.7 | 7.7 | | |
| In-channel selectivity | 7.8 | 7.8 | | |
| Performance requirements | 8 | 8 | | |
| Radiated transmit power | NA | 9.2 | 9.2 | 9.2 |
| OTA base station output power | | NA | 9.3 | 9.3 |
| OTA output power dynamics | | | 9.4 | 9.4 |
| OTA transmit ON/OFF power | | | 9.5 | 9.5 |
| OTA transmitted signal quality | | | 9.6 | 9.6 |
| OTA occupied bandwidth | | | 9.7.2 | 9.7.2 |
| OTA ACLR | | | 9.7.3 | 9.7.3 |
| OTA out-of-band emission | | | 9.7.4 | 9.7.4 |
| OTA transmitter spurious emission | | | 9.7.5 | 9.7.5 |
| OTA transmitter intermodulation | | | 9.8 | NA |
| OTA sensitivity | | 10.2 | 10.2 | NA |
| OTA reference sensitivity level | | NA | 10.3 | 10.3 |
| OTA dynamic range | | | 10.4 | NA |
| OTA in-band selectivity and blocking | | | 10.5 | 10.5 |
| OTA out-of-band blocking | | | 10.6 | 10.6 |
| OTA receiver spurious emission | | | 10.7 | 10.7 |
| OTA receiver intermodulation | | | 10.8 | 10.8 |
| OTA in-channel selectivity | | | 10.9 | 10.9 |
| Radiated performance requirements | | | 11 | 11 |

# 4.7 Requirements for contiguous and *non-contiguous spectrum*

A spectrum allocation where a BS operates can either be contiguous or non-contiguous. Unless otherwise stated, the requirements in the present specification apply for BS configured for both *contiguous spectrum* operation and *non-contiguous spectrum* operation.

For BS operation in *non-contiguous spectrum*, some requirements apply both at the *Base Station RF Bandwidth edges* and inside the *sub-block gaps*. For each such requirement, it is stated how the limits apply relative to the *Base Station RF Bandwidth edges* and the *sub-block* edges respectively.

## 4.8 Requirements for BS capable of multi-band operation

For *multi-band connector* or *multi-band RIB*, the RF requirements in clause 6, 7, 9 and 10 apply separately to each supported *operating band* unless otherwise stated. For some requirements, it is explicitly stated that specific additions or exclusions to the requirement apply at *multi-band connector(s)*, and *multi-band RIB(s)* as detailed in the requirement clause. For *BS* capable of multi-band operation, various structures in terms of combinations of different transmitter and receiver implementations (multi-band or single band) with mapping of transceivers to one or more *antenna connectors* for *BS type 1-C* or *TAB connectors* for *BS type 1-H* in different ways are possible. For *multi-band connector(s)* the exclusions or provisions for multi-band apply. For *single-band connector(s)*, the following applies:

- Single-band transmitter spurious emissions, *operating band* unwanted emissions, ACLR, transmitter intermodulation and receiver spurious emissions requirements apply to this *connector* that is mapped to single-band.

- If the BS is configured for single-band operation, single-band requirements shall apply to this *connector* configured for single-band operation and no exclusions or provisions for multi-band capable BS are applicable. Single-band requirements are tested separately at the *connector* configured for single-band operation, with all other *antenna connectors* terminated.

A *BS type 1-H* may be capable of supporting operation in multiple *operating bands* with one of the following implementations of *TAB connectors* in the *transceiver array boundary*:

- All *TAB connectors* are *single-band connectors*.

  - Different sets of *single-band connectors* support different *operating bands*, but each *TAB connector* supports only operation in one single *operating band*.

  - Sets of *single-band connectors* support operation in multiple *operating bands* with some *single-band connectors* supporting more than one *operating band*.

- All *TAB connectors* are multi-band *connectors*.

- A combination of single-band sets and multi-band sets of *TAB connectors* provides support of the type *BS type 1-H* capability of operation in multiple *operating bands*.

Unless otherwise stated all requirements specified for an *operating band* apply only to the set of *TAB connectors* supporting that *operating band*.

In the case of an *operating band* being supported only by *single-band connectors* in a *TAB connector TX min cell group* or a *TAB connector RX min cell group*, *single-band requirements* apply to that set of *TAB connectors*.

In the case of an *operating band* being supported only by *multi-band connector*s supporting the same *operating band* combination in a *TAB connector TX min cell group* or a *TAB connector RX min cell group*, *multi-band requirements* apply to that set of *TAB connectors*.

The case of an *operating band* being supported by both *multi-band connectors* and *single-band connectors* in a *TAB connector TX min cell group* or a *TAB connector RX min cell group* is not covered by the present release of this specification.

The case of an *operating band* being supported by *multi-band connectors* which are not all supporting the same *operating band* combination in a *TAB connector TX min cell group* or a *TAB connector RX min cell group* is not covered by the present release of this specification.

*BS type 1-O* may be capable of supporting operation in multiple *operating bands* with one of the following implementations at the *radiated interface boundary*:

- All RIBs are *single-band RIBs*.

- All RIBs are *multi-band RIBs*.

- A combination of single-band *RIBs* and *multi-band RIBs* provides support of the *BS type 1-O* capability of operation in multiple *operating bands*.

For *multi-band connectors* and *multi-band RIBs* supporting the bands for TDD, the RF requirements in the present specification assume no simultaneous uplink and downlink occur between the bands.

The RF requirements for *multi-band connectors* and *multi-band RIBs* supporting bands for both FDD and TDD are not covered by the present release of this specification.

## 4.9 OTA co-location with other base stations

Co-location requirements are requirements which are based on assuming the *BS type 1-O* is co-located with another BS of the same base station class, they ensure that both co-located systems can operate with minimal degradation to each other.

Unwanted emission and out of band blocking co-location requirements are optional requirements based on declaration. TX OFF and TX IMD are mandatory requirements and have the form of a co-location requirement as it represents the worst-case scenario of all the interference cases.

NOTE: Due to the low level of the unwanted emissions for the spurious emissions and TX OFF level co-location is the most suitable method to show conformance.

The *co-location reference antenna* shall be a single column passive antenna which has the same vertical radiating dimension (h), frequency range, polarization, as the composite antenna of the *BS type 1-O* and nominal 65° horizontal half-power beamwidth (suitable for 3-sector deployment) and is placed at a distance *d* from the edge of the *BS type 1-O*, as shown in figure 4.9-1.

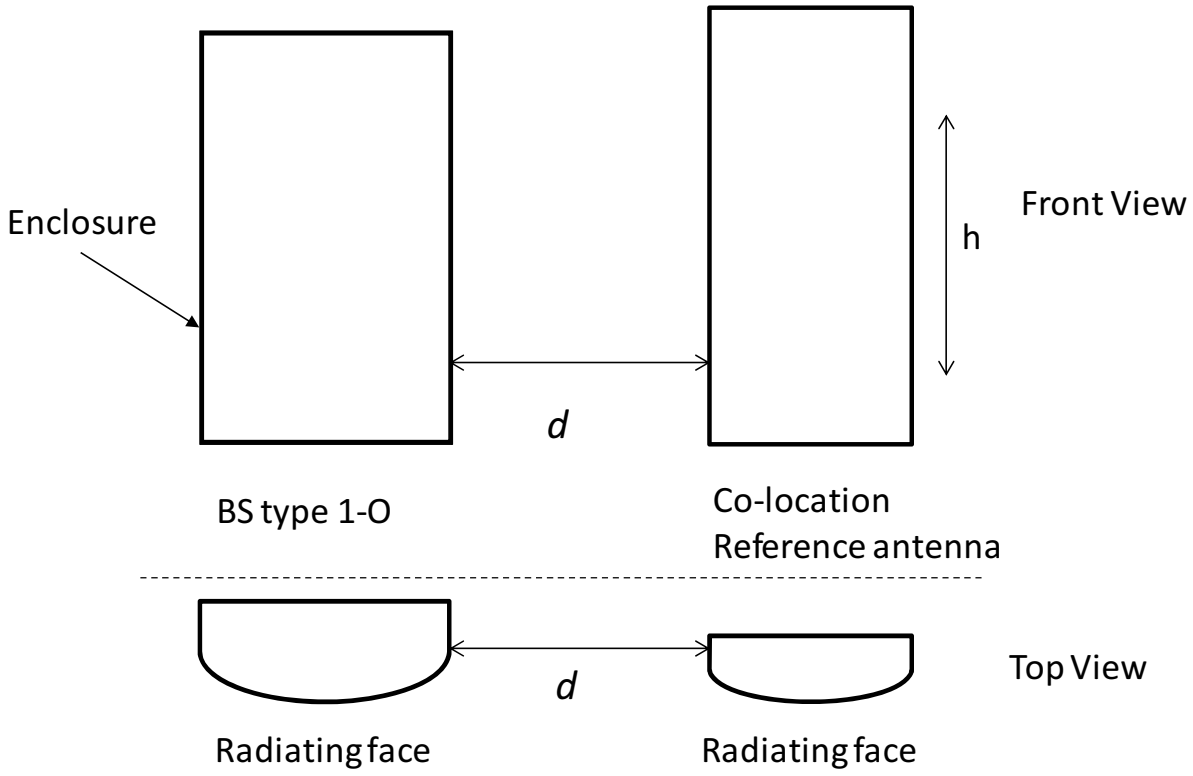



**Figure 4.9-1: Illustration of *BS type 1-O* enclosure and co-location reference antenna**

Edge-to-edge separation *d* between the *BS type 1-O* and the *co-location reference antenna* shall be set to 0.1 m.

The *BS type 1-O* and the *co-location reference antenna* shall be aligned in a common plane perpendicular to the mechanical bore-sight direction, as shown in figure 4.9-1.

The *co-location reference antenna* and the *BS type 1-O* can have different width.

The vertical radiating regions of the *co-location reference antenna* and the *BS type 1-O* composite antenna shall be aligned.

For co-location requirements where the frequency range of the signal at the *co-location reference antenna* is different from the *BS type 1-O*, a *co-location reference antenna* suitable for the frequency stated in the requirement is assumed.

OTA co-location requirements are based on the power at the conducted interface of a *co-location reference antenna*, depending on the requirement this interface is either an input or an output. For *BS type 1-O* with dual polarization *the co-location reference antenna* has two conducted interfaces each representing one polarization.

# 5 *Operating bands* and channel arrangement

## 5.1 General

The channel arrangements presented in this clause are based on the *operating bands* and *BS channel bandwidths* defined in the present release of specifications.

NOTE: Other *operating bands* and *BS channel bandwidth*s may be considered in future releases.

Requirements throughout the RF specifications are in many cases defined separately for different frequency ranges (FR). The frequency ranges in which NR can operate according to the present version of the specification are identified as described in table 5.1-1. Whenever FR2 is referred, both FR2-1 and FR2-2 frequency sub-ranges shall be applied, unless otherwise stated.

**Table 5.1-1: Definition of frequency ranges**

| Frequency range designation | | Corresponding frequency range |
|---|---|---|
| FR1 | | 410 MHz – 7125 MHz |
| FR2 | FR2-1 | 24250 MHz – 52600 MHz |
| | FR2-2 | 52600 MHz – 71000 MHz |

## 5.2 *Operating bands*

NR is designed to operate in the *operating bands* defined in table 5.2-1 and 5.2-2.

NR operating band n1, which is defined in Table 5.2-1, can be applied for HAPS operation.

NB-IoT is designed to operate in the NR operating bands n1, n2, n3, n5, n7, n8, n12, n13, n14, n18, n20, n25, n26, n28, n41, n65, n66, n70, n71, n74, n85, n90 which are defined in Table 5.2-1.

**Table 5.2-1: NR *operating bands* in FR1**

| NR *operating band* | Uplink (UL) *operating band* BS receive / UE transmit $F_{UL,low}$ – $F_{UL,high}$ (MHz) | Downlink (DL) *operating band* BS transmit / UE receive $F_{DL,low}$ – $F_{DL,high}$ (MHz) | Duplex mode | Notes |
|---|---|---|---|---|
| n1 | 1920 – 1980 | 2110 – 2170 | FDD | |
| n2 | 1850 – 1910 | 1930 – 1990 | FDD | |
| n3 | 1710 – 1785 | 1805 – 1880 | FDD | |
| n5 | 824 – 849 | 869 – 894 | FDD | |
| n7 | 2500 – 2570 | 2620 – 2690 | FDD | |
| n8 | 880 – 915 | 925 – 960 | FDD | |
| n12 | 699 – 716 | 729 – 746 | FDD | |
| n13 | 777 – 787 | 746 – 756 | FDD | |
| n14 | 788 – 798 | 758 – 768 | FDD | |
| n18 | 815 – 830 | 860 – 875 | FDD | |
| n20 | 832 – 862 | 791 – 821 | FDD | |
| n24 | 1626.5 – 1660.5 | 1525 – 1559 | FDD | Note 7 |
| n25 | 1850 – 1915 | 1930 – 1995 | FDD | |
| n26 | 814 – 849 | 859 – 894 | FDD | |
| n28 | 703 – 748 | 758 – 803 | FDD | |
| n29 | N/A | 717 – 728 | SDL | |
| n30 | 2305 – 2315 | 2350 – 2360 | FDD | |
| n34 | 2010 – 2025 | 2010 – 2025 | TDD | |
| n38 | 2570 – 2620 | 2570 – 2620 | TDD | |
| n39 | 1880 – 1920 | 1880 – 1920 | TDD | |
| n40 | 2300 – 2400 | 2300 – 2400 | TDD | |
| n41 | 2496 – 2690 | 2496 – 2690 | TDD | |
| n46 | 5150 – 5925 | 5150 – 5925 | TDD | Note 3 |
| n48 | 3550 – 3700 | 3550 – 3700 | TDD | |
| n50 | 1432 – 1517 | 1432 – 1517 | TDD | |
| n51 | 1427 – 1432 | 1427 – 1432 | TDD | |
| n53 | 2483.5 – 2495 | 2483.5 – 2495 | TDD | |
| n65 | 1920 – 2010 | 2110 – 2200 | FDD | |
| n66 | 1710 – 1780 | 2110 – 2200 | FDD | |
| n67 | N/A | 738 – 758 | SDL | |
| n70 | 1695 – 1710 | 1995 – 2020 | FDD | |
| n71 | 663 – 698 | 617 – 652 | FDD | |
| n74 | 1427 – 1470 | 1475 – 1518 | FDD | |
| n75 | N/A | 1432 – 1517 | SDL | |
| n76 | N/A | 1427 – 1432 | SDL | |
| n77 | 3300 – 4200 | 3300 – 4200 | TDD | Note 10 |
| n78 | 3300 – 3800 | 3300 – 3800 | TDD | |
| n79 | 4400 – 5000 | 4400 – 5000 | TDD | |
| n80 | 1710 – 1785 | N/A | SUL | |
| n81 | 880 – 915 | N/A | SUL | |
| n82 | 832 – 862 | N/A | SUL | |
| n83 | 703 – 748 | N/A | SUL | |
| n84 | 1920 – 1980 | N/A | SUL | |
| n85 | 698 – 716 | 728 – 746 | FDD | |
| n86 | 1710 – 1780 | N/A | SUL | |
| n89 | 824 – 849 | N/A | SUL | |
| n90 | 2496 – 2690 | 2496 – 2690 | TDD | |
| n91 | 832 – 862 | 1427 – 1432 | FDD | Note 2 |
| n92 | 832 – 862 | 1432 – 1517 | FDD | Note 2 |
| n93 | 880 – 915 | 1427 – 1432 | FDD | Note 2 |
| n94 | 880 – 915 | 1432 – 1517 | FDD | Note 2 |
| n95 | 2010 – 2025 | N/A | SUL | Note 1 |
| n96 | 5925 – 7125 | 5925 – 7125 | TDD | Note 3, Note 4 |
| n97 | 2300 – 2400 | N/A | SUL | Note 5 |
| n98 | 1880 – 1920 | N/A | SUL | Note 5 |
| n99 | 1626.5 – 1660.5 | N/A | SUL | Note 6 |
| n100 | 874.4 – 880 | 919.4 – 925 | FDD | Note 9 |
| n101 | 1900 – 1910 | 1900 – 1910 | TDD | Note 9 |
| n102 | 5925 – 6425 | 5925 – 6425 | TDD | Note 3, Note 4 |
| n104 | 6425 – 7125 | 6425 – 7125 | TDD | |

NOTE 1: This band is applicable in China only.
NOTE 2: Variable duplex operation does not enable dynamic variable duplex configuration by the network, and is used such that DL and UL frequency ranges are supported independently in any valid frequency range for the band.
NOTE 3: This band is restricted to operation with shared spectrum channel access as defined in TS 37.213 [20].
NOTE 4: This band is applicable only in countries/regions designating this band for shared-spectrum access use subject to country-specific conditions.
NOTE 5: The requirements for this band are applicable only where no other NR or E-UTRA TDD operating band(s) are used within the frequency range of this band in the same geographical area. For scenarios where other NR or E-UTRA TDD operating band(s) are used within the frequency range of this band in the same geographical area, special co-existence requirements may apply that are not covered by the 3GPP specifications.
NOTE 6: UL operation is restricted to 1627.5 – 1637.5 MHz and 1646.5 – 1656.5 MHz per FCC Order DA 20-48 [25].
NOTE 7: DL operation is restricted to 1526-1536 MHz frequency range. UL operation is restricted to 1627.5 – 1637.5 MHz and 1646.5 – 1656.5 MHz per FCC Order 20-51 [24].
NOTE 8: Void
NOTE 9: This band is applicable to CEPT countries subject to ECC Decision (20)02 [21], for the RMR application.
NOTE 10: In the USA this band is restricted to 3450 – 3550 MHz and 3700 – 3980 MHz. In Canada this band is restricted to 3450 – 3650 MHz and 3650 – 3980 MHz.

**Table 5.2-2: NR *operating bands* in FR2**

| NR *operating band* | Uplink (UL) and Downlink (DL) *operating band* BS transmit/receive UE transmit/receive $F_{UL,low}$ – $F_{UL,high}$ $F_{DL,low}$ – $F_{DL,high}$ (MHz) | Duplex mode |
|---|---|---|
| n257 | 26500 – 29500 | TDD |
| n258 | 24250 – 27500 | TDD |
| n259 | 39500 – 43500 | TDD |
| n260 | 37000 – 40000 | TDD |
| n261 | 27500 – 28350 | TDD |
| n262 | 47200 – 48200 | TDD |
| n263 | 57000 – 71000 | TDD |

# 5.3 *BS channel bandwidth*

## 5.3.1 General

The *BS channel bandwidth* supports a single NR RF carrier in the uplink or downlink at the Base Station. Different UE channel bandwidths may be supported within the same spectrum for transmitting to and receiving from UEs connected to the BS. The placement of the UE channel bandwidth is flexible but can only be completely within the *BS channel bandwidth*. The BS shall be able to transmit to and/or receive from one or more UE bandwidth parts that are smaller than or equal to the number of carrier resource blocks on the RF carrier, in any part of the carrier resource blocks.

The relationship between the channel bandwidth, the guardband and the *transmission bandwidth configuration* is shown in figure 5.3.1-1.

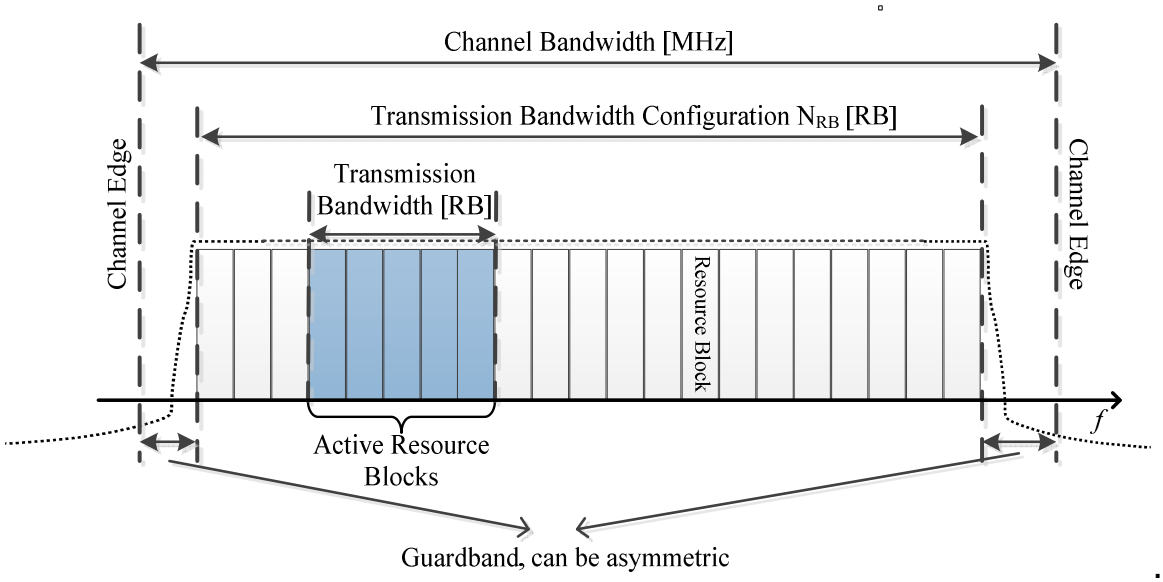



**Figure 5.3.1-1: Definition of channel bandwidth and *transmission bandwidth configuration* for one NR channel**

## 5.3.2 *Transmission bandwidth configuration*

The *transmission bandwidth configuration* $N_{RB}$ for each *BS channel bandwidth* and subcarrier spacing is specified in table 5.3.2.-1 for FR1 and table 5.3.2-2 for FR2.

**Table 5.3.2-1: *Transmission bandwidth configuration* $N_{RB}$ for FR1**

| SCS (kHz) | 5 MHz | 10 MHz | 15 MHz | 20 MHz | 25 MHz | 30 MHz | 35 MHz | 40 MHz | 45 MHz | 50 MHz | 60 MHz | 70 MHz | 80 MHz | 90 MHz | 100 MHz |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $N_{RB}$ | $N_{RB}$ | $N_{RB}$ | $N_{RB}$ | $N_{RB}$ | $N_{RB}$ | $N_{RB}$ | $N_{RB}$ | $N_{RB}$ | $N_{RB}$ | $N_{RB}$ | $N_{RB}$ | $N_{RB}$ | $N_{RB}$ | $N_{RB}$ |
| 15 | 25 | 52 | 79 | 106 | 133 | 160 | 188 | 216 | 242 | 270 | N/A | N/A | N/A | N/A | N/A |
| 30 | 11 | 24 | 38 | 51 | 65 | 78 | 92 | 106 | 119 | 133 | 162 | 189 | 217 | 245 | 273 |
| 60 | N/A | 11 | 18 | 24 | 31 | 38 | 44 | 51 | 58 | 65 | 79 | 93 | 107 | 121 | 135 |

**Table 5.3.2-2: *Transmission bandwidth configuration* $N_{RB}$ for FR2-1**

| SCS (kHz) | 50 MHz | 100 MHz | 200 MHz | 400 MHz |
|---|---|---|---|---|
| | $N_{RB}$ | $N_{RB}$ | $N_{RB}$ | $N_{RB}$ |
| 60 | 66 | 132 | 264 | N/A |
| 120 | 32 | 66 | 132 | 264 |

**Table 5.3.2-3: Transmission bandwidth configuration $N_{RB}$ for FR2-2**

| SCS (kHz) | 100 MHz | 400 MHz | 800 MHz | 1600 MHz | 2000 MHz |
|---|---|---|---|---|---|
| | $N_{RB}$ | $N_{RB}$ | $N_{RB}$ | $N_{RB}$ | $N_{RB}$ |
| 120 | 66 | 264 | N/A | N/A | N/A |
| 480 | N/A | 66 | 124 | 248 | N/A |
| 960 | N/A | 33 | 62 | 124 | 148 |

NOTE: All Tx and Rx requirements are defined based on *transmission bandwidth configuration* specified in table 5.3.2-1 for FR1 and table 5.3.2-2 and table 5.3.2-3 for FR2.

The transmission bandwidth configuration for NB-IoT is specified in TS 36.104 [13] clause 5.6.

## 5.3.3 Minimum guardband and *transmission bandwidth configuration*

The minimum guardband for each *BS channel bandwidth* and SCS is specified in table 5.3.3-1 for FR1 and in table 5.3.3-2 and table 5.3.3-2a for FR2.

**Table 5.3.3-1: Minimum guardband (kHz) (FR1)**

| SCS (kHz) | 5 MHz | 10 MHz | 15 MHz | 20 MHz | 25 MHz | 30 MHz | 35 MHz | 40 MHz | 45 MHz | 50 MHz | 60 MHz | 70 MHz | 80 MHz | 90 MHz | 100 MHz |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 15 | 242.5 | 312.5 | 382.5 | 452.5 | 522.5 | 592.5 | 572.5 | 552.5 | 712.5 | 692.5 | N/A | N/A | N/A | N/A | N/A |
| 30 | 505 | 665 | 645 | 805 | 785 | 945 | 925 | 905 | 1065 | 1045 | 825 | 965 | 925 | 885 | 845 |
| 60 | N/A | 1010 | 990 | 1330 | 1310 | 1290 | 1630 | 1610 | 1590 | 1570 | 1530 | 1490 | 1450 | 1410 | 1370 |

**Table 5.3.3-2: Minimum guardband (kHz) (FR2-1)**

| SCS (kHz) | 50 MHz | 100 MHz | 200 MHz | 400 MHz |
|---|---|---|---|---|
| 60 | 1210 | 2450 | 4930 | N/A |
| 120 | 1900 | 2420 | 4900 | 9860 |

**Table 5.3.3-2a: Minimum guardband (kHz) (FR2-2)**

| SCS (kHz) | 100 MHz | 400 MHz | 800 MHz | 1600 MHz | 2000 MHz |
|---|---|---|---|---|---|
| 120 | 2420 | 9860 | N/A | N/A | N/A |
| 480 | N/A | 9680 | 42640 | 85520 | N/A |
| 960 | N/A | 9440 | [42400] | 85280 | 147040 |

The minimum guardband of SCS 240 kHz SS/PBCH block for each *BS channel bandwidth* is specified in table 5.3.3-3 for FR2.

**Table 5.3.3-3: Minimum guardband (kHz) of SCS 240 kHz SS/PBCH block (FR2)**

| SCS (kHz) | 100 MHz | 200 MHz | 400 MHz |
|---|---|---|---|
| 240 | 3800 | 7720 | 15560 |

NOTE: The minimum guardband in Table 5.3.3-3 is applicable only when the SCS 240 kHz SS/PBCH block is placed adjacent to the edge of the *BS channel bandwidth* within which the SS/PBCH block is located.

The number of RBs configured in any *BS channel bandwidth* shall ensure that the minimum guardband specified in this clause is met.

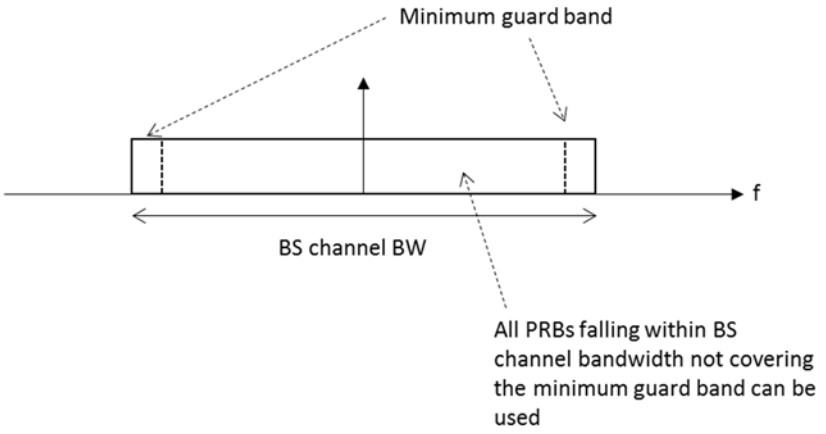



**Figure 5.3.3-1: BS PRB utilization**

In the case that multiple numerologies are multiplexed in the same symbol, the minimum guardband on each side of the carrier is the guardband applied at the configured *BS channel bandwidth* for the numerology that is transmitted/received immediately adjacent to the guard band.

For FR1, if multiple numerologies are multiplexed in the same symbol and the *BS channel bandwidth* is >50 MHz, the guardband applied adjacent to 15 kHz SCS shall be the same as the guardband defined for 30 kHz SCS for the same *BS channel bandwidth*.

For FR2, if multiple numerologies are multiplexed in the same symbol and the *BS channel bandwidth* is >200 MHz, the guardband applied adjacent to 60 kHz SCS shall be the same as the guardband defined for 120 kHz SCS for the same *BS channel bandwidth*.

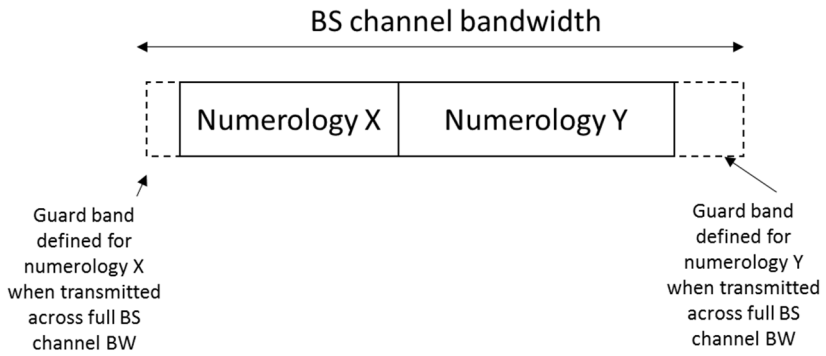



**Figure 5.3.3-2: Guard band definition when transmitting multiple numerologies**

NOTE: Figure 5.3.3-2 is not intended to imply the size of any guard between the two numerologies. Inter-numerology guard band within the carrier is implementation dependent.

**Figure 5.3.3-3: Void**

**Figure 5.3.3-4: Void**

**Figure 5.3.3-5: Void**

## 5.3.4 RB alignment

For each *BS channel bandwidth* and each numerology, *BS transmission bandwidth configuration* must fulfil the minimum guardband requirement specified in clause 5.3.3.

For each numerology, its common resource blocks are specified in clause 4.4.4.3 in [9], and the starting point of its *transmission bandwidth configuration* on the common resource block grid for a given channel bandwidth is indicated by an offset to “Reference point A” in the unit of the numerology.

For each numerology, all *UE transmission bandwidth configurations* indicated to UEs served by the BS by higher layer parameter *carrierBandwidth* defined in TS 38.331 [11] shall fall within the *BS transmission bandwidth configuration*.

## 5.3.5 *BS channel bandwidth* per *operating band*

The requirements in this specification apply to the combination of *BS channel bandwidths*, SCS and *operating bands* shown in table 5.3.5-1 for FR1 and in table 5.3.5-2 and table 5.3.5-3 for FR2. The *transmission bandwidth configuration* in table 5.3.2-1, table 5.3.2-2 and table 5.3.2-3 shall be supported for each of the *BS channel bandwidths* within the BS capability. The *BS channel bandwidths* are specified for both the Tx and Rx path.

**Table 5.3.5-1: *BS channel bandwidths* and SCS per *operating band* in FR1**

| NR Band | SCS (kHz) | *BS channel bandwidth* (MHz) | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | **5** | **10** | **15** | **20** | **25** | **30** | **35** | **40** | **45** | **50** | **60** | **70** | **80** | **90** | **100** |
| n1 | 15 | 5 | 10 | 15 | 20 | 25 | 30 | | 40 | 45 | 50 | | | | | |
| | 30 | | 10 | 15 | 20 | 25 | 30 | | 40 | 45 | 50 | | | | | |
| | 60 | | 10 | 15 | 20 | 25 | 30 | | 40 | 45 | 50 | | | | | |
| n2 | 15 | 5 | 10 | 15 | 20 | 25 | 30 | 35 | 40 | | | | | | | |
| | 30 | | 10 | 15 | 20 | 25 | 30 | 35 | 40 | | | | | | | |
| | 60 | | 10 | 15 | 20 | 25 | 30 | 35 | 40 | | | | | | | |
| n3 | 15 | 5 | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 | 50 | | | | | |
| | 30 | | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 | 50 | | | | | |
| | 60 | | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 | 50 | | | | | |
| n5 | 15 | 5 | 10 | 15 | 20 | 25[7] | | | | | | | | | | |
| | 30 | | 10 | 15 | 20 | 25[7] | | | | | | | | | | |
| | 60 | | | | | | | | | | | | | | | |
| n7 | 15 | 5 | 10 | 15 | 20 | 25 | 30 | 35 | 40 | | 50 | | | | | |
| | 30 | | 10 | 15 | 20 | 25 | 30 | 35 | 40 | | 50 | | | | | |
| | 60 | | 10 | 15 | 20 | 25 | 30 | 35 | 40 | | 50 | | | | | |
| n8 | 15 | 5 | 10 | 15 | 20 | | | 35 | | | | | | | | |
| | 30 | | 10 | 15 | 20 | | | 35 | | | | | | | | |
| | 60 | | | | | | | | | | | | | | | |
| n12 | 15 | 5 | 10 | 15 | | | | | | | | | | | | |
| | 30 | | 10 | 15 | | | | | | | | | | | | |
| | 60 | | | | | | | | | | | | | | | |
| n13 | 15 | 5 | 10 | | | | | | | | | | | | | |
| | 30 | | 10 | | | | | | | | | | | | | |
| | 60 | | | | | | | | | | | | | | | |
| n14 | 15 | 5 | 10 | | | | | | | | | | | | | |
| | 30 | | 10 | | | | | | | | | | | | | |
| | 60 | | | | | | | | | | | | | | | |
| n18 | 15 | 5 | 10 | 15 | | | | | | | | | | | | |
| | 30 | | 10 | 15 | | | | | | | | | | | | |
| | 60 | | | | | | | | | | | | | | | |
| n20 | 15 | 5 | 10 | 15 | 20 | | | | | | | | | | | |
| | 30 | | 10 | 15 | 20 | | | | | | | | | | | |
| | 60 | | | | | | | | | | | | | | | |
| n24 | 15 | 5 | 10 | | | | | | | | | | | | | |
| | 30 | | 10 | | | | | | | | | | | | | |
| | 60 | | 10 | | | | | | | | | | | | | |
| n25 | 15 | 5 | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 | | | | | | |
| | 30 | | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 | | | | | | |
| | 60 | | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 | | | | | | |
| n26 | 15 | 5 | 10 | 15 | 20 | 25[7] | 30[7] | | | | | | | | | |
| | 30 | | 10 | 15 | 20 | 25[7] | 30[7] | | | | | | | | | |
| n28 | 15 | 5 | 10 | 15 | 20 | 25 | 30 | | 40 | | | | | | | |
| | 30 | | 10 | 15 | 20 | 25 | 30 | | 40 | | | | | | | |
| | 60 | | | | | | | | | | | | | | | |
| n29 | 15 | 5 | 10 | | | | | | | | | | | | | |
| | 30 | | 10 | | | | | | | | | | | | | |
| | 60 | | | | | | | | | | | | | | | |
| n30 | 15 | 5 | 10 | | | | | | | | | | | | | |
| | 30 | | 10 | | | | | | | | | | | | | |
| | 60 | | | | | | | | | | | | | | | |
| n34 | 15 | 5 | 10 | 15 | | | | | | | | | | | | |
| | 30 | | 10 | 15 | | | | | | | | | | | | |
| | 60 | | 10 | 15 | | | | | | | | | | | | |
| n38 | 15 | 5 | 10 | 15 | 20 | 25 | 30 | | 40 | | | | | | | |
| | 30 | | 10 | 15 | 20 | 25 | 30 | | 40 | | | | | | | |
| | 60 | | 10 | 15 | 20 | 25 | 30 | | 40 | | | | | | | |
| n39 | 15 | 5 | 10 | 15 | 20 | 25 | 30 | | 40 | | | | | | | |
| | 30 | | 10 | 15 | 20 | 25 | 30 | | 40 | | | | | | | |
| | 60 | | 10 | 15 | 20 | 25 | 30 | | 40 | | | | | | | |
| n40 | 15 | 5[2] | 10 | 15 | 20 | 25 | 30 | | 40 | | 50 | | | | | |
| | 30 | | 10 | 15 | 20 | 25 | 30 | | 40 | | 50 | 60 | 70 | 80 | 90 | 100 |

| NR Band | SCS (kHz) | *BS channel bandwidth* (MHz) | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | **5** | **10** | **15** | **20** | **25** | **30** | **35** | **40** | **45** | **50** | **60** | **70** | **80** | **90** | **100** |
| n41 | 60 | | 10 | 15 | 20 | 25 | 30 | | 40 | | 50 | 60 | 70 | 80 | 90 | 100 |
| | 15 | 5[8] | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 | 50 | | | | | |
| | 30 | | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 | 50 | 60 | 70 | 80 | 90 | 100 |
| | 60 | | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 | 50 | 60 | 70 | 80 | 90 | 100 |
| n46 | 15 | | 10[6] | | 20 | | | | 40 | | | | | | | |
| | 30 | | 10[6] | | 20 | | | | 40 | | | 60 | | 80 | | 100 |
| | 60 | | 10[6] | | 20 | | | | 40 | | | 60 | | 80 | | 100 |
| n48 | 15 | 5[2] | 10 | 15 | 20 | | 30 | | 40 | | 50[1] | | | | | |
| | 30 | | 10 | 15 | 20 | | 30 | | 40 | | 50[1] | 60[1] | 70[1] | 80[1] | 90[1] | 100[1] |
| | 60 | | 10 | 15 | 20 | | 30 | | 40 | | 50[1] | 60[1] | 70[1] | 80[1] | 90[1] | 100[1] |
| n50 | 15 | 5[2] | 10 | 15 | 20 | | 30 | | 40 | | 50 | | | | | |
| | 30 | | 10 | 15 | 20 | | 30 | | 40 | | 50 | 60 | | 80 | | |
| | 60 | | 10 | 15 | 20 | | 30 | | 40 | | 50 | 60 | | 80 | | |
| n51 | 15 | 5 | | | | | | | | | | | | | | |
| | 30 | | | | | | | | | | | | | | | |
| | 60 | | | | | | | | | | | | | | | |
| n53 | 15 | 5 | 10 | | | | | | | | | | | | | |
| | 30 | | 10 | | | | | | | | | | | | | |
| | 60 | | 10 | | | | | | | | | | | | | |
| n65 | 15 | 5 | 10 | 15 | 20 | | | | | | 50 | | | | | |
| | 30 | | 10 | 15 | 20 | | | | | | 50 | | | | | |
| | 60 | | 10 | 15 | 20 | | | | | | 50 | | | | | |
| n66 | 15 | 5 | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 | | | | | | |
| | 30 | | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 | | | | | | |
| | 60 | | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 | | | | | | |
| n67 | 15 | 5 | 10 | 15 | 20 | | | | | | | | | | | |
| | 30 | | 10 | 15 | 20 | | | | | | | | | | | |
| | 60 | | | | | | | | | | | | | | | |
| n70 | 15 | 5 | 10 | 15 | 20 | 25 | | | | | | | | | | |
| | 30 | | 10 | 15 | 20 | 25 | | | | | | | | | | |
| | 60 | | 10 | 15 | 20 | 25 | | | | | | | | | | |
| n71 | 15 | 5 | 10 | 15 | 20 | 25 | 30 | 35 | | | | | | | | |
| | 30 | | 10 | 15 | 20 | 25 | 30 | 35 | | | | | | | | |
| | 60 | | | | | | | | | | | | | | | |
| n74 | 15 | 5 | 10 | 15 | 20 | | | | | | | | | | | |
| | 30 | | 10 | 15 | 20 | | | | | | | | | | | |
| | 60 | | 10 | 15 | 20 | | | | | | | | | | | |
| n75 | 15 | 5 | 10 | 15 | 20 | 25 | 30 | | 40 | | 50 | | | | | |
| | 30 | | 10 | 15 | 20 | 25 | 30 | | 40 | | 50 | | | | | |
| | 60 | | 10 | 15 | 20 | 25 | 30 | | 40 | | 50 | | | | | |
| n76 | 15 | 5 | | | | | | | | | | | | | | |
| | 30 | | | | | | | | | | | | | | | |
| | 60 | | | | | | | | | | | | | | | |
| n77 | 15 | | 10 | 15 | 20 | 25 | 30 | | 40 | | 50 | | | | | |
| | 30 | | 10 | 15 | 20 | 25 | 30 | | 40 | | 50 | 60 | 70 | 80 | 90 | 100 |
| | 60 | | 10 | 15 | 20 | 25 | 30 | | 40 | | 50 | 60 | 70 | 80 | 90 | 100 |
| n78 | 15 | | 10 | 15 | 20 | 25 | 30 | | 40 | | 50 | | | | | |
| | 30 | | 10 | 15 | 20 | 25 | 30 | | 40 | | 50 | 60 | 70 | 80 | 90 | 100 |
| | 60 | | 10 | 15 | 20 | 25 | 30 | | 40 | | 50 | 60 | 70 | 80 | 90 | 100 |
| n79 | 15 | | 10 | | 20 | | 30 | | 40 | | 50 | | | | | |
| | 30 | | 10 | | 20 | | 30 | | 40 | | 50 | 60 | 70 | 80 | 90 | 100 |
| | 60 | | 10 | | 20 | | 30 | | 40 | | 50 | 60 | 70 | 80 | 90 | 100 |
| n80 | 15 | 5 | 10 | 15 | 20 | 25 | 30 | | 40 | | | | | | | |
| | 30 | | 10 | 15 | 20 | 25 | 30 | | 40 | | | | | | | |
| | 60 | | 10 | 15 | 20 | 25 | 30 | | 40 | | | | | | | |
| n81 | 15 | 5 | 10 | 15 | 20 | | | | | | | | | | | |
| | 30 | | 10 | 15 | 20 | | | | | | | | | | | |
| | 60 | | | | | | | | | | | | | | | |
| n82 | 15 | 5 | 10 | 15 | 20 | | | | | | | | | | | |
| | 30 | | 10 | 15 | 20 | | | | | | | | | | | |
| | 60 | | | | | | | | | | | | | | | |
| n83 | 15 | 5 | 10 | 15 | 20 | 25 | 30 | | 40 | | | | | | | |
| | 30 | | 10 | 15 | 20 | 25 | 30 | | 40 | | | | | | | |

| NR Band | SCS (kHz) | *BS channel bandwidth* (MHz) | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | **5** | **10** | **15** | **20** | **25** | **30** | **35** | **40** | **45** | **50** | **60** | **70** | **80** | **90** | **100** |
| | 60 | | | | | | | | | | | | | | | |
| n84 | 15 | 5 | 10 | 15 | 20 | 25 | 30 | | 40 | | 50 | | | | | |
| | 30 | | 10 | 15 | 20 | 25 | 30 | | 40 | | 50 | | | | | |
| | 60 | | 10 | 15 | 20 | 25 | 30 | | 40 | | 50 | | | | | |
| n85 | 15 | 5 | 10 | 15 | | | | | | | | | | | | |
| | 30 | | 10 | 15 | | | | | | | | | | | | |
| | 60 | | | | | | | | | | | | | | | |
| n86 | 15 | 5 | 10 | 15 | 20 | | | | 40 | | | | | | | |
| | 30 | | 10 | 15 | 20 | | | | 40 | | | | | | | |
| | 60 | | 10 | 15 | 20 | | | | 40 | | | | | | | |
| n89 | 15 | 5 | 10 | 15 | 20 | | | | | | | | | | | |
| | 30 | | 10 | 15 | 20 | | | | | | | | | | | |
| | 60 | | | | | | | | | | | | | | | |
| n90 | 15 | 5 | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 | 50 | | | | | |
| | 30 | | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 | 50 | 60 | 70 | 80 | 90 | 100 |
| | 60 | | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 | 50 | 60 | 70 | 80 | 90 | 100 |
| n91 | 15 | 5 | 10[3] | | | | | | | | | | | | | |
| | 30 | | | | | | | | | | | | | | | |
| | 60 | | | | | | | | | | | | | | | |
| n92 | 15 | 5 | 10 | 15 | 20 | | | | | | | | | | | |
| | 30 | | 10 | 15 | 20 | | | | | | | | | | | |
| | 60 | | | | | | | | | | | | | | | |
| n93 | 15 | 5 | 10[3] | | | | | | | | | | | | | |
| | 30 | | | | | | | | | | | | | | | |
| | 60 | | | | | | | | | | | | | | | |
| n94 | 15 | 5 | 10 | 15 | 20 | | | | | | | | | | | |
| | 30 | | 10 | 15 | 20 | | | | | | | | | | | |
| | 60 | | | | | | | | | | | | | | | |
| n95 | 15 | 5 | 10 | 15 | | | | | | | | | | | | |
| | 30 | | 10 | 15 | | | | | | | | | | | | |

| NR Band | SCS (kHz) | *BS channel bandwidth* (MHz) | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 5 | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 | 50 | 60 | 70 | 80 | 90 | 100 |
| | 60 | | 10 | 15 | | | | | | | | | | | | |
| n96 | 15 | | | | 20 | | | | 40 | | | | | | | |
| | 30 | | | | 20 | | | | 40 | | | 60 | | 80 | | 100 |
| | 60 | | | | 20 | | | | 40 | | | 60 | | 80 | | 100 |
| n97 | 15 | 5 | 10 | 15 | 20 | 25 | 30 | | 40 | | 50 | | | | | |
| | 30 | | 10 | 15 | 20 | 25 | 30 | | 40 | | 50 | 60 | 70 | 80 | 90 | 100 |
| | 60 | | 10 | 15 | 20 | 25 | 30 | | 40 | | 50 | 60 | 70 | 80 | 90 | 100 |
| n98 | 15 | 5 | 10 | 15 | 20 | 25 | 30 | | 40 | | | | | | | |
| | 30 | | 10 | 15 | 20 | 25 | 30 | | 40 | | | | | | | |
| | 60 | | 10 | 15 | 20 | 25 | 30 | | 40 | | | | | | | |
| n99 | 15 | 5 | 10 | | | | | | | | | | | | | |
| | 30 | | 10 | | | | | | | | | | | | | |
| | 60 | | 10 | | | | | | | | | | | | | |
| n100 | 15 | 5 | | | | | | | | | | | | | | |
| | 30 | | | | | | | | | | | | | | | |
| | 60 | | | | | | | | | | | | | | | |
| n101 | 15 | 5 | 10 | | | | | | | | | | | | | |
| | 30 | | 10 | | | | | | | | | | | | | |
| | 60 | | | | | | | | | | | | | | | |
| n102 | 15 | | | | 20 | | | | 40 | | | | | | | |
| | 30 | | | | 20 | | | | 40 | | | 60 | | 80 | | 100 |
| | 60 | | | | 20 | | | | 40 | | | 60 | | 80 | | 100 |
| n104 | 15 | | | | 20 | | 30 | | 40 | | 50 | | | | | |
| | 30 | | | | 20 | | 30 | | 40 | | 50 | 60 | 70 | 80 | 90 | 100 |
| | 60 | | | | 20 | | 30 | | 40 | | 50 | 60 | 70 | 80 | 90 | 100 |

NOTE 1: For this bandwidth, the minimum requirements are restricted to operation when carrier is configured as an downlink SCell part of CA configuration.
NOTE 2: For this bandwidth, the minimum requirements are restricted to operation when carrier is configured as an SCell part of DC or CA configuration.
NOTE 3: For this bandwidth, it only applies for UL transmission.
NOTE 4: Void.
NOTE 5: Void.
NOTE 6: This bandwidth can only be applied in certain regions where the absence of non 3GPP technologies can be guaranteed on a long term basis in this version of specification.
NOTE 7: For this bandwidth, it only applies for DL transmission.
NOTE 8: Not all frequency positions of 5 MHz carriers are possible due limitations of the SSB position relative to the 5 MHz channels. 5 MHz channels with $F_c$ such that $2499+N*1.2 \le F_c < 2499.3+N*1.2$MHz for $0 \le N < 157$ are not compatible with SSB positions and cannot be used for 5 MHz n41.

**Table 5.3.5-2: *BS channel bandwidths* and SCS per *operating band* in FR2-1**

| NR Band | SCS (kHz) | *BS channel bandwidth* (MHz) | | | |
|---|---|---|---|---|---|
| | | 50 | 100 | 200 | 400 |
| n257 | 60 | 50 | 100 | 200 | |
| | 120 | 50 | 100 | 200 | 400 |
| n258 | 60 | 50 | 100 | 200 | |
| | 120 | 50 | 100 | 200 | 400 |
| n259 | 60 | 50 | 100 | 200 | |
| | 120 | 50 | 100 | 200 | 400 |
| n260 | 60 | 50 | 100 | 200 | |
| | 120 | 50 | 100 | 200 | 400 |
| n261 | 60 | 50 | 100 | 200 | |
| | 120 | 50 | 100 | 200 | 400 |
| n262 | 60 | 50 | 100 | 200 | |
| | 120 | 50 | 100 | 200 | 400 |

**Table 5.3.5-3: *BS channel bandwidths* and SCS per *operating band* in FR2-2**

| NR Band | SCS | *BS channel bandwidth* (MHz) |
|---|---|---|

| | (kHz) | 100 | 400 | 800 | 1600 | 2000 |
|---|---|---|---|---|---|---|
| n263 | 120 | 100 | 400 | | | |
| | 480 | | 400 | 800 | 1600 | |
| | 960 | | 400 | 800 | 1600 | 2000 |

# 5.3A *BS channel bandwidth* for CA

## 5.3A.1 *Transmission bandwidth configuration* for CA

For *carrier aggregation*, the *transmission bandwidth configuration* is defined per component carrier and the requirement is specified in clause 5.3.2.

## 5.3A.2 Minimum guardband and *transmission bandwidth configuration* for CA

For intra-band contiguous *carrier aggregation*, *Aggregated BS Channel Bandwidth* and *Guard Bands* are defined as follows, see Figure 5.3A.2-1.

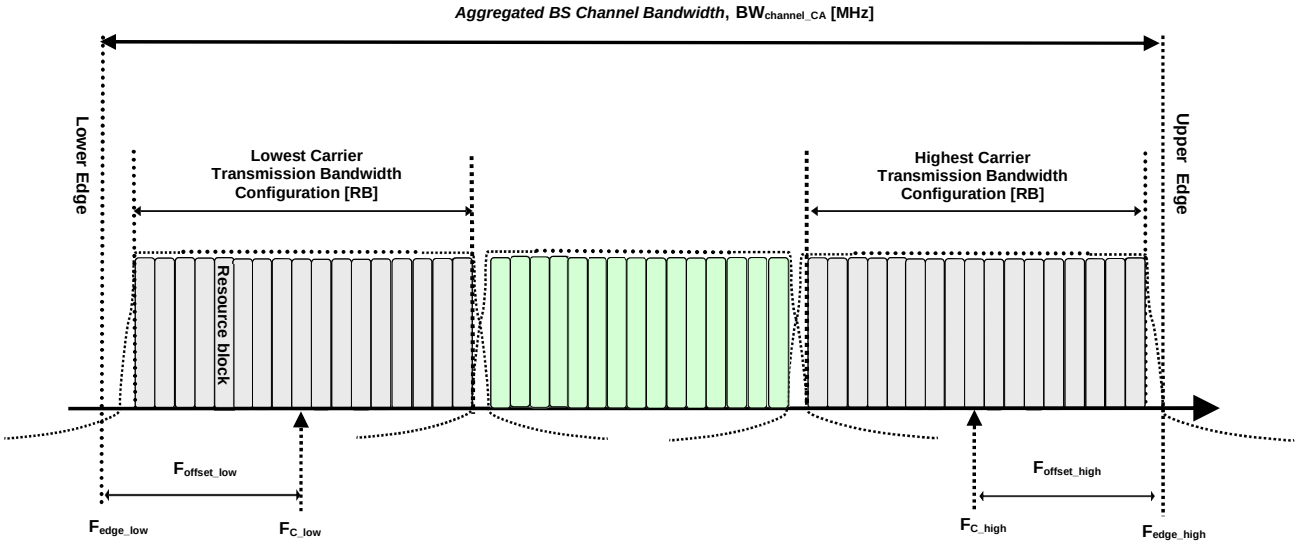



**Figure 5.3A.2-1: Definition of *Aggregated BS Channel Bandwidth* for intra-band *carrier aggregation***

The *aggregated BS Channel Bandwidth*, $BW_{Channel_CA}$, is defined as

$$BW_{Channel_CA} = F_{edge,high} - F_{edge,low} \text{ (MHz)}$$

The lower bandwidth edge $F_{edge,\ low}$ and the upper bandwidth edge $F_{edge,high}$ of the *aggregated BS channel bandwidth* are used as frequency reference points for transmitter and receiver requirements and are defined by

$$F_{edge,low} = F_{C,low} - F_{offset,low}$$

$$F_{edge,high} = F_{C,high} + F_{offset,high}$$

The lower and upper frequency offsets depend on the *transmission bandwidth configurations* of the lowest and highest assigned edge component carrier and are defined as

$$F_{offset,low} = (N_{RB,low}*12 + 1)*SCS_{low}/2 + BW_{GB,low} \text{ (MHz)}$$

$$F_{offset,high} = (N_{RB,high}*12 - 1)*SCS_{high}/2 + BW_{GB,high} \text{ (MHz)}$$

$N_{RB,low}$ and $N_{RB,high}$ are the *transmission bandwidth configurations* according to Table 5.3.2-1 or Table 5.3.2-2 or Table 5.3.2-3 for the lowest and highest assigned component carrier, $SCS_{low}$ and $SCS_{high}$ are the sub-carrier spacing for the lowest and highest assigned component carrier respectively. $SCS_{low}$, $SCS_{high}$, $N_{RB,low}$, $N_{RB,high}$, $BW_{GB,low}$ and $BW_{GB,high}$ use the largest μ value among the subcarrier spacing configurations supported in the operating band for both of the channel bandwidths according to Table 5.3.5-1 and $BW_{GB,low}$ and $BW_{GB,high}$ are the minimum guard band for lowest and highest assigned component carrier according to Table 5.3.3-1 for the said $\mu$ value. In case there is no common μ value for both of the channel bandwidths, $\mu$=1 is used for $SCS_{low}$, $SCS_{high}$, $N_{RB,low}$, $N_{RB,high}$, $BW_{GB,low}$ and $BW_{GB,high}$.

For *intra-band non-contiguous carrier aggregation sub-block bandwidth* and *sub-block edges* are defined as follows, see figure 5.3A.2-2.

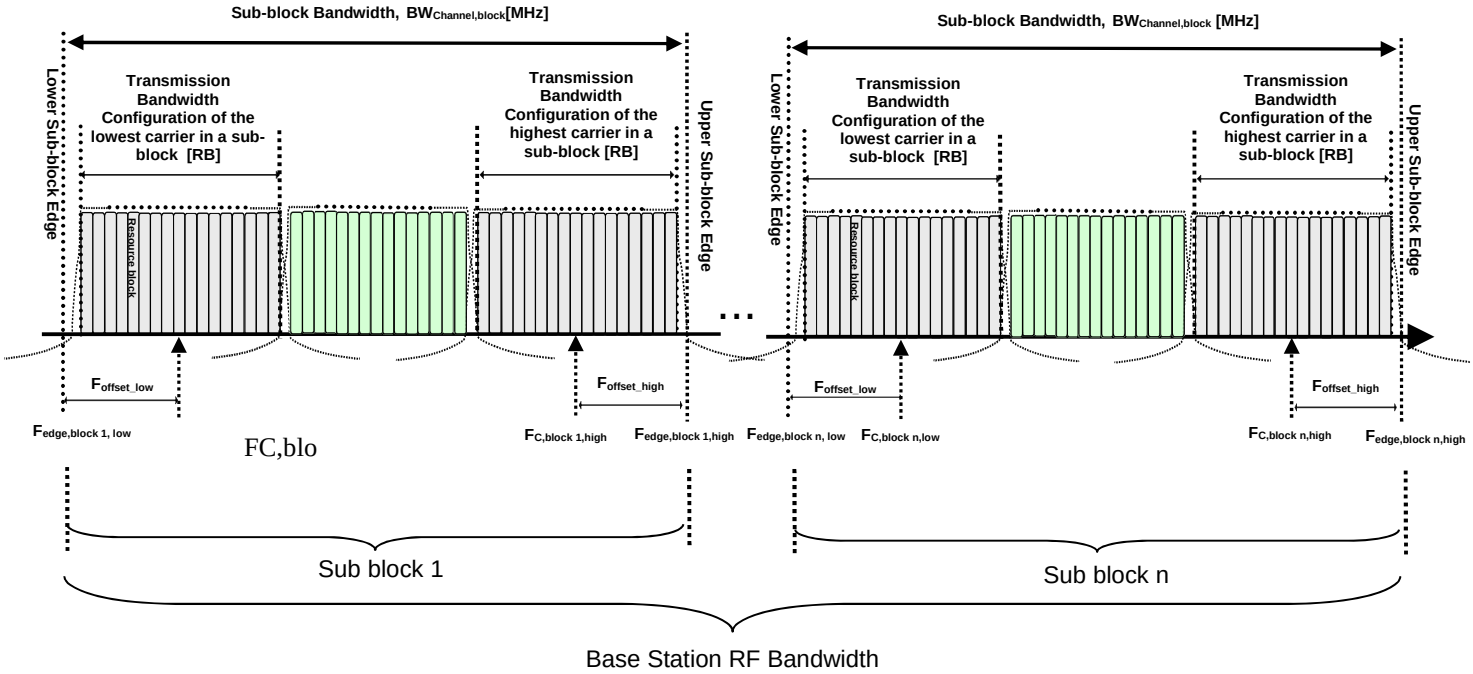



**Figure 5.3A.2-2: Definition of *sub-block bandwidth* for intra-band *non-contiguous spectrum***

The *lower sub-block edge* of the *sub-block bandwidth* ($BW_{Channel,block}$) is defined as follows:

$$F_{edge,block,\ low} = F_{C,block,low} - F_{offset,low}$$

The upper *sub-block* edge of the *sub-block bandwidth* is defined as follows:

$$F_{edge,block,high} = F_{C,block,high} + F_{offset,high}$$

The *sub-block bandwidth*, $BW_{Channel,block}$, is defined as follows:

$$BW_{Channel,block} = F_{edge,block,high} - F_{edge,block,low}\ (MHz)$$

The lower and upper frequency offsets $F_{offset,block,low}$ and $F_{offset,block,high}$ depend on the *transmission bandwidth configurations* of the lowest and highest assigned edge component carriers within a *sub-block* and are defined as

$$F_{offset,block,low} = (N_{RB,low}*12 + 1)*SCS_{low}/2 + BW_{GB,low}\ (MHz)$$

$$F_{offset,block,high} = (N_{RB,high}*12 - 1)*SCS_{high}/2 + BW_{GB,high}\ (MHz)$$

where $N_{RB,low}$ and $N_{RB,high}$ are the *transmission bandwidth configurations* according to Table 5.3.2-1 or Table 5.3.2-2 for the lowest and highest assigned component carrier within a *sub-block*, respectively. $SCS_{low}$ and $SCS_{high}$ are the sub-carrier spacing for the lowest and highest assigned component carrier within a *sub-block*, respectively. $SCS_{low}$, $SCS_{high}$, $N_{RB,low}$, $N_{RB,high}$, $BW_{GB,low}$ and $BW_{GB,high}$ use the largest μ value among the subcarrier spacing configurations supported in the operating band for both of the channel bandwidths according to Table 5.3.5-1 and $BW_{GB,low}$ and $BW_{GB,high}$ are the minimum guard band for lowest and highest assigned component carrier according to Table 5.3.3-1 for the said $\mu$ value.

In case there is no common μ value for both of the channel bandwidths, $\mu$=1 is used for $SCS_{low}$, $SCS_{high}$, $N_{RB,low}$, $N_{RB,high}$, $BW_{GB,low}$ and $BW_{GB,high}$.

The *sub-block gap size* between two consecutive *sub-blocks* $W_{gap}$ is defined as follows:

$$W_{gap} = F_{edge,block\ n+1,low} - F_{edge,block\ n,high}\ (MHz)$$

# 5.4 Channel arrangement

## 5.4.1 Channel spacing

### 5.4.1.1 Channel spacing for adjacent NR carriers

The spacing between carriers will depend on the deployment scenario, the size of the frequency block available and the *BS channel bandwidths*. The nominal channel spacing between two adjacent NR carriers is defined as following:

- For NR FR1 *operating bands* with 100 kHz channel raster,
  - Nominal Channel spacing = $(BW_{Channel(1)} + BW_{Channel(2)})/2$
- For NR FR1 *operating bands* with 15 kHz channel raster,
  - Nominal Channel spacing = $(BW_{Channel(1)} + BW_{Channel(2)})/2$ + {-5 kHz, 0 kHz, 5 kHz} for $\Delta F_{Raster}$ equals to 15 kHz
  - Nominal Channel spacing = $(BW_{Channel(1)} + BW_{Channel(2)})/2$ + {-10 kHz, 0 kHz, 10 kHz} for $\Delta F_{Raster}$ equals to 30 kHz
- For NR FR2 *operating bands* with 60 kHz channel raster,
  - Nominal Channel spacing = $(BW_{Channel(1)} + BW_{Channel(2)})/2$ + {-20 kHz, 0 kHz, 20 kHz} for $\Delta F_{Raster}$ equals to 60 kHz
  - Nominal Channel spacing = $(BW_{Channel(1)} + BW_{Channel(2)})/2$ + {-40 kHz, 0 kHz, 40 kHz} for $\Delta F_{Raster}$ equals to 120 kHz
- For operating band n263
  - Nominal Channel spacing = ceil($(BW_{Channel(1)} + BW_{Channel(2)})/100.8$)*50.4 MHz,

where $BW_{Channel(1)}$ and $BW_{Channel(2)}$ are the *BS channel bandwidths* of the two respective NR carriers. The channel spacing can be adjusted depending on the channel raster to optimize performance in a particular deployment scenario.

For NR bands restricted to operation with shared-spectrum channel access, the maximum deviation from the nominal channel spacing is 40 kHz.

### 5.4.1.2 Channel spacing for CA

For intra-band contiguously aggregated carriers, the channel spacing between adjacent component carriers shall be multiple of least common multiple of channel raster and sub-carrier spacing.

The nominal channel spacing between two adjacent aggregated NR carriers is defined as follows:

For NR *operating bands* with 100 kHz channel raster:

$$\text{Nominal channel spacing} = \left\lfloor \frac{BW_{Channel(1)} + BW_{Channel(2)} - 2\left|GB_{Channel(1)} - GB_{Channel(2)}\right|}{0.6} \right\rfloor 0.3\ (\text{MHz})$$

For NR *operating bands* with 15 kHz channel raster:

$$\text{Nominal channel spacing} = \left\lfloor \frac{BW_{Channel(1)} + BW_{Channel(2)} - 2\left|GB_{Channel(1)} - GB_{Channel(2)}\right|}{0.015 * 2^{n+1}} \right\rfloor 0.015 * 2^{n}\ (\text{MHz})$$

with

$$n = \mu_0$$

For NR *operating bands* with 60kHz channel raster:

$$\text{Nominal channel spacing} = \left\lfloor \frac{BW_{Channel(1)} + BW_{Channel(2)} - 2\left|GB_{Channel(1)} - GB_{Channel(2)}\right|}{0.06 * 2^{n+1}} \right\rfloor 0.06 * 2^n \ (\text{MHz})$$

with

$$n = \mu_0 - 2$$

For operating band n263

Nominal Channel spacing = ceil((BW$_{Channel(1)}$ + BW$_{Channel(2)}$)/100.8)*50.4 MHz,

where BW$_{Channel(1)}$ and BW$_{Channel(2)}$ are the *BS channel bandwidths* of the two respective NR component carriers according to Table 5.3.2-1, 5.3.2-2 and 5.3.2-3 with values in MHz, $\mu_0$ the largest $\mu$ value among the subcarrier spacing configurations supported in the operating band for both of the channel bandwidths according to Table 5.3.5-1 and Table 5.3.5-2 and $GB_{Channel(i)}$ the minimum guard band for channel bandwidth *i* according to Table 5.3.3-1, Table 5.3.3-2 and Table 5.3.3-2a for the said $\mu$ value, with $\mu$ as defined in TS 38.211 [9]. In case there is no common μ value for both of the channel bandwidths, $\mu_0$=1 is selected for NR *operating bands* with 15 kHz channel raster and $GB_{Channel(i)}$ is the minimum guard band for channel bandwidth i according to Table 5.3.3-1 for $\mu$=1 with $\mu$ as defined in TS 38.211[9].

The channel spacing for *intra-band contiguous carrier aggregation* can be adjusted to any multiple of least common multiple of channel raster and sub-carrier spacing less than the nominal channel spacing to optimize performance in a particular deployment scenario.

For intra-band contiguous carrier aggregation in NR bands restricted to operation with shared-spectrum channel access, the maximum deviation from the nominal channel spacing is 300 kHz.

For *intra-band non-contiguous carrier aggregation*, the channel spacing between two NR component carriers in different *sub-blocks* shall be larger than the nominal channel spacing defined in this clause.

## 5.4.2 Channel raster

### 5.4.2.1 NR-ARFCN and channel raster

The global frequency raster defines a set of *RF reference frequencies* $F_{REF}$. The *RF reference frequency* is used in signalling to identify the position of RF channels, SS blocks and other elements. The global frequency raster is defined for all frequencies from 0 to 100 GHz. The granularity of the global frequency raster is $\Delta F_{Global}$.

*RF reference frequencies* are designated by an NR Absolute Radio Frequency Channel Number (NR-ARFCN) in the range [0…3279165] on the global frequency raster. The relation between the NR-ARFCN and the *RF reference frequency* $F_{REF}$ in MHz is given by the following equation, where $F_{REF-Offs}$ and $N_{Ref-Offs}$ are given in table 5.4.2.1-1 and $N_{REF}$ is the NR-ARFCN.

$$F_{REF} = F_{REF-Offs} + \Delta F_{Global} (N_{REF} - N_{REF-Offs})$$

**Table 5.4.2.1-1: NR-ARFCN parameters for the global frequency raster**

| Range of frequencies (MHz) | $\Delta F_{Global}$ (kHz) | $F_{REF-Offs}$ (MHz) | $N_{REF-Offs}$ | Range of $N_{REF}$ |
|---|---|---|---|---|
| 0 – 3000 | 5 | 0 | 0 | 0 – 599999 |
| 3000 – 24250 | 15 | 3000 | 600000 | 600000 – 2016666 |
| 24250 – 100000 | 60 | 24250.08 | 2016667 | 2016667 – 3279165 |

The *channel raster* defines a subset of *RF reference frequencies* that can be used to identify the RF channel position in the uplink and downlink. The *RF reference frequency* for an RF channel maps to a resource element on the carrier. For each *operating band*, a subset of frequencies from the global frequency raster are applicable for that band and forms a channel raster with a granularity $\Delta F_{Raster}$, which may be equal to or larger than $\Delta F_{Global}$.

For SUL bands except n95, n97, n98 and for the uplink of all FDD bands defined in table 5.2-1, for TDD bands n34, n38, n39, n48, n90, and n40,

$F_{REF,shift} = F_{REF} + \Delta_{shift}$, where $\Delta_{shift}$ = 0 kHz or 7.5 kHz

where $\Delta_{shift}$ is signalled by the network in higher layer parameter *frequencyShift7p5khz* as defined in TS 38.331 [11].

For bands n34, n38, n39, n48 and n40, $F_{REF, shift}$ is only applicable to uplink transmissions using a 15 kHz SCS.

The mapping between the *channel raster* and corresponding resource element is given in clause 5.4.2.2. The applicable entries for each *operating band* are defined in clause 5.4.2.3.

### 5.4.2.1A NB-IoT carrier frequency numbering

The NB-IoT carrier frequency numbering (EARFCN) is defined in clause 5.7 of TS 36.104 [4].

### 5.4.2.2 Channel raster to resource element mapping

The mapping between the *RF reference frequency* on the channel raster and the corresponding resource element is given in table 5.4.2.2-1 and can be used to identify the RF channel position. The mapping depends on the total number of RBs that are allocated in the channel and applies to both UL and DL. The mapping must apply to at least one numerology supported by the BS.

**Table 5.4.2.2-1: Channel Raster to Resource Element Mapping**

| | $N_{RB} \, mod \, 2 = 0$ | $N_{RB} \, mod \, 2 = 1$ |
|---|---|---|
| Resource element index $k$ | 0 | 6 |
| Physical resource block index $n_{PRB}$ | $n_{PRB} = \left\lfloor \frac{N_{RB}}{2} \right\rfloor$ | $n_{PRB} = \left\lfloor \frac{N_{RB}}{2} \right\rfloor$ |

$N_{RB}$ is the transmission bandwidth configuration specified in sub-clause 5.3.2, $n_{PRB}$ is the PRB index within the $N_{RB}$, and $k$ is the resource element index within this PRB

### 5.4.2.3 Channel raster entries for each *operating band*

The RF channel positions on the channel raster in each NR *operating band* are given through the applicable NR-ARFCN in table 5.4.2.3-1 for FR1 and table 5.4.2.3-2 for FR2, using the channel raster to resource element mapping in clause 5.4.2.2.

- For NR *operating bands* with 100 kHz channel raster, $\Delta F_{Raster} = 20 \times \Delta F_{Global}$. In this case, every 20$^{th}$ NR-ARFCN within the *operating band* are applicable for the channel raster within the *operating band* and the step size for the channel raster in table 5.4.2.3-1 is given as <20>.

- For NR *operating bands* with 15 kHz channel raster below 3 GHz, $\Delta F_{Raster} = I \times \Delta F_{Global}$, where $I \in \{3,6\}$. In this case, every $I^{th}$ NR-ARFCN within the *operating band* are applicable for the channel raster within the *operating band* and the step size for the channel raster in table 5.4.2.3-1 is given as <*I*>.

- For NR *operating bands* with 15 kHz and 60 kHz channel raster above 3 GHz, $\Delta F_{Raster} = I \times \Delta F_{Global}$, where $I \in \{1, 2\}$. In this case, every $I^{th}$ NR-ARFCN within the *operating band* are applicable for the channel raster within the *operating band* and the step size for the channel raster in table 5.4.2.3-1 and table 5.4.2.3-2 is given as <*I*>.

- For frequency bands with two $\Delta F_{Raster}$ in FR1, the higher $\Delta F_{Raster}$ applies to channels using only the SCS that is equal to or larger than the higher $\Delta F_{Raster}$ and SSB SCS is equal to the higher $\Delta F_{Raster}$.

- For frequency bands with two $\Delta F_{Raster}$ in FR2, the higher $\Delta F_{Raster}$ applies to channels using only the SCS that is equal to the higher $\Delta F_{Raster}$ and the SSB SCS that is equal to or larger than the higher $\Delta F_{Raster}$.

**Table 5.4.2.3-1: Applicable NR-ARFCN per *operating band* in FR1**

| NR *operating band* | $\Delta F_{Raster}$ (kHz) | Uplink range of $N_{REF}$ (First – <Step size> – Last) | Downlink range of $N_{REF}$ (First – <Step size> – Last) |
|---|---|---|---|
| n1 | 100 | 384000 – <20> – 396000 | 422000 – <20> – 434000 |
| n2 | 100 | 370000 – <20> – 382000 | 386000 – <20> – 398000 |
| n3 | 100 | 342000 – <20> – 357000 | 361000 – <20> – 376000 |
| n5 | 100 | 164800 – <20> – 169800 | 173800 – <20> – 178800 |
| n7 | 100 | 500000 – <20> – 514000 | 524000 – <20> – 538000 |
| n8 | 100 | 176000 – <20> – 183000 | 185000 – <20> – 192000 |
| n12 | 100 | 139800 – <20> – 143200 | 145800 – <20> – 149200 |
| n13 | 100 | 155400 – <20> – 157400 | 149200 – <20> – 151200 |
| n14 | 100 | 157600 – <20> –159600 | 151600 – <20> – 153600 |
| n18 | 100 | 163000 – <20> – 166000 | 172000 – <20> – 175000 |
| n20 | 100 | 166400 – <20> – 172400 | 158200 – <20> – 164200 |
| n25 | 100 | 370000 – <20> – 383000 | 386000 – <20> – 399000 |
| n24 | 100 | 325300 – <20> – 332100 | 305000 – <20> – 311800 |
| n26 | 100 | 162800 – <20> – 169800 | 171800 – <20> – 178800 |
| n28 | 100 | 140600 – <20> – 149600 | 151600 – <20> – 160600 |
| | | 144608[4] | 155608[4] |
| n29 | 100 | N/A | 143400 – <20> – 145600 |
| n30 | 100 | 461000 – <20> – 463000 | 470000 – <20> – 472000 |
| n34 | 100 | 402000 – <20> – 405000 | 402000 – <20> – 405000 |
| n38 | 100 | 514000 – <20> – 524000 | 514000 – <20> – 524000 |
| n39 | 100 | 376000 – <20> – 384000 | 376000 – <20> – 384000 |
| n40 | 100 | 460000 – <20> – 480000 | 460000 – <20> – 480000 |
| n41 | 15 | 499200 – <3> – 537999 | 499200 – <3> – 537999 |
| | 30 | 499200 – <6> – 537996 | 499200 – <6> – 537996 |
| n46[1] | 15 | 743334 – <1> – 795000 | 743334 – <1> – 795000 |
| n48 | 15 | 636667 – <1> – 646666 | 636667 – <1> – 646666 |
| | 30 | 636668 – <2> – 646666 | 636668 – <2> – 646666 |
| n50 | 100 | 286400 – <20> – 303400 | 286400 – <20> – 303400 |
| n51 | 100 | 285400 – <20> – 286400 | 285400 – <20> – 286400 |
| n53 | 100 | 496700 – <20> – 499000 | 496700 – <20> – 499000 |
| n65 | 100 | 384000 – <20> – 402000 | 422000 – <20> – 440000 |
| n66 | 100 | 342000 – <20> – 356000 | 422000 – <20> – 440000 |
| n67 | 100 | N/A | 147600 – <20> – 151600 |
| n70 | 100 | 339000 – <20> – 342000 | 399000 – <20> – 404000 |
| n71 | 100 | 132600 – <20> – 139600 | 123400 – <20> – 130400 |
| n74 | 100 | 285400 – <20> – 294000 | 295000 – <20> – 303600 |
| n75 | 100 | N/A | 286400 – <20> – 303400 |
| n76 | 100 | N/A | 285400 – <20> – 286400 |
| n77 | 15 | 620000 – <1> – 680000 | 620000 – <1> – 680000 |
| | 30 | 620000 – <2> – 680000 | 620000 – <2> – 680000 |
| n78 | 15 | 620000 – <1> – 653333 | 620000 – <1> – 653333 |
| | 30 | 620000 – <2> – 653332 | 620000 – <2> – 653332 |
| n79 | 15 | 693334 – <1> – 733333 | 693334 – <1> – 733333 |
| | 30 | 693334 – <2> – 733332 | 693334 – <2> – 733332 |
| n80 | 100 | 342000 – <20> – 357000 | N/A |
| n81 | 100 | 176000 – <20> – 183000 | N/A |
| n82 | 100 | 166400 – <20> – 172400 | N/A |
| n83 | 100 | 140600 – <20> –149600 | N/A |
| n84 | 100 | 384000 – <20> – 396000 | N/A |
| n85 | 100 | 139600 – <20> – 143200 | 145600 – <20> – 149200 |
| n86 | 100 | 342000 – <20> – 356000 | N/A |
| n89 | 100 | 164800 – <20> – 169800 | N/A |
| n90 | 15 | 499200 – <3> – 537999 | 499200 – <3> – 537999 |
| | 30 | 499200 – <6> – 537996 | 499200 – <6> – 537996 |
| | 100 | 499200 – <20> – 538000 | 499200 – <20> – 538000 |
| n91 | 100 | 166400 – <20> – 172400 | 285400 – <20> – 286400 |
| n92 | 100 | 166400 – <20> – 172400 | 286400 – <20> – 303400 |
| n93 | 100 | 176000 – <20> – 183000 | 285400 – <20> – 286400 |
| n94 | 100 | 176000 – <20> – 183000 | 286400 – <20> – 303400 |
| n95 | 100 | 402000 – <20> – 405000 | N/A |
| n96[2] | 15 | 795000 – <1> – 875000 | 795000 – <1> – 875000 |
| n97 | 100 | 460000 – <20> – 480000 | N/A |

| | | | |
|---|---|---|---|
| n98 | 100 | 376000 – <20> – 384000 | N/A |
| n99 | 100 | 325300 -- <20> – 332100 | N/A |
| n100 | 100 | 174880 – <20> – 176000 | 183880 – <20> – 185000 |
| n101 | 100 | 380000 – <20> – 382000 | 380000 – <20> – 382000 |
| n102[3] | 15 | 795000 – <1> – 828333 | 795000– <1> – 828333 |
| n104 | 15 | 828334 – <1> – 875000 | 828334 – <1> – 875000 |
| | 30 | 828334 – <2> – 875000 | 828334 – <2> – 875000 |

NOTE 1: Applicable NR-ARFCN for band n46
for 10 MHz channel bandwidth, $N_{REF}$ = {782000, 788668}
for 20 MHz channel bandwidth, $N_{REF}$ = {744000, 745332, 746668, 748000, 749332, 750668, 752000, 753332, 754668, 756000, 765332, 766668, 768000, 769332, 770668, 772000, 773332, 774668, 776000, 777332, 778668, 780000, 781332, 783000, 784332, 785668, 787000, 788332, 789668, 791000, 792332, 793668};
for 40 MHz channel bandwidth, $N_{REF}$ = {744668, 746000, 748668, 751332, 754000, 755332, 766000, 767332, 770000, 772668, 775332, 778000, 780668, 783668, 786332, 787668, 790332, 793000};
for 60 MHz channel bandwidth, $N_{REF}$ = {745332, 746668, 748000, 752000, 753332, 754668, 766668, 768000, 769332, 773332, 774668, 778668, 780000, 784332, 785668, 791000, 792332};
for 80 MHz channel bandwidth, $N_{REF}$ = {746000, 747332, 752668, 754000, 767332, 768668, 774000, 779332, 785000, 791668}
for 100 MHz channel bandwidth, $N_{REF}$ = {746668, 753332, 768000, 791000}

NOTE 2: Applicable NR-ARFCN for band n96
for 20 MHz channel bandwidth, $N_{REF}$ = {797000, 798332, 799668, 801000, 802332, 803668, 805000, 806332, 807668, 809000, 810332, 811668, 813000, 814332, 815668, 817000, 818332, 819668, 821000, 822332, 823668, 825000, 826332, 827668, 829000, 830332, 831668, 833000, 834332, 835668, 837000, 838332, 839668, 841000, 842332, 843668, 845000, 846332, 847668, 849000, 850332, 851668, 853000, 854332, 855668, 857000, 858332, 859668, 861000, 862332, 863668, 865000, 866332, 867668, 869000, 870332, 871668, 873000, 874332}
for 40 MHz channel bandwidth, $N_{REF}$ = {797668, 800332, 803000, 805668, 808332, 811000, 813668, 816332, 819000, 821668, 824332, 827000, 829668, 832332, 835000, 837668, 840332, 843000, 845668, 848332, 851000, 853668, 856332, 859000, 861668, 864332, 867000, 869668, 872332}
for 60 MHz channel bandwidth, $N_{REF}$ = {798332, 799668, 803668, 805000, 809000, 810332, 814332, 815668, 819668, 821000, 825000, 826332, 830332, 831668, 835668, 837000, 841000, 842332, 846332, 847668, 851668, 853000, 857000, 858332, 862332, 863668, 867668, 869000, 873000}
for 80 MHz channel bandwidth, $N_{REF}$ = {799000, 804332, 809668, 815000, 820332, 825668, 831000, 836332, 841668, 847000, 852332, 857668, 863000, 868332}
for 100 MHz channel bandwidth, $N_{REF}$ = {799668, 803668, 810332, 814332, 821000, 825000, 831668, 835668, 842332, 846332, 853000, 857000, 863668, 867668, 869000, 870332, 871668}

NOTE 3: Applicable NR-ARFCN for band n102
for 20 MHz channel bandwidth, $N_{REF}$ = {797000, 798332, 799668, 801000, 802332, 803668, 805000, 806332, 807668, 809000, 810332, 811668, 813000, 814332, 815668, 817000, 818332, 819668, 821000, 822332, 823668, 825000, 826332, 827668}
for 40 MHz channel bandwidth, $N_{REF}$ = {797668, 800332, 803000, 805668, 808332, 811000, 813668, 816332, 819000, 821668, 824332, 827000}
for 60 MHz channel bandwidth, $N_{REF}$ = {798332, 799668, 803668, 805000, 809000, 810332, 814332, 815668, 819668, 821000, 825000, 826332}
for 80 MHz channel bandwidth, $N_{REF}$ = {799000, 804332, 809668, 815000, 820332, 825668}
for 100 MHz channel bandwidth, $N_{REF}$ = {799668, 803668, 810332, 814332, 821000, 825000}

NOTE 4: This exceptional raster point is applicable only to n28 and is only applicable for 40MHz BS channel bandwidth to ensure the guardband with 30MHz UE channel bandwidth.

**Table 5.4.2.3-2: Applicable NR-ARFCN per *operating band* in FR2**

| NR *operating band* | $\Delta F_{Raster}$ (kHz) | Uplink and Downlink range of $N_{REF}$ (First – <Step size> – Last) |
|---|---|---|
| n257 | 60 | 2054166 – <1> – 2104165 |
| | 120 | 2054167 – <2> – 2104165 |
| n258 | 60 | 2016667 – <1> – 2070832 |
| | 120 | 2016667 – <2> – 2070831 |
| n259 | 60 | 2270833 – <1> – 2337499 |
| | 120 | 2270833 – <2> – 2337499 |
| n260 | 60 | 2229166 – <1> – 2279165 |
| | 120 | 2229167 – <2> – 2279165 |
| n261 | 60 | 2070833 – <1> – 2084999 |
| | 120 | 2070833 – <2> – 2084999 |
| n262 | 60 | 2399166 – <1> – 2415832 |
| | 120 | 2399167 – <2> – 2415831 |
| n263 | 120 | See Table 5.4.2.3-3 |
| | 480 | |
| | 960 | |

**Table 5.4.2.3-3: Applicable NR-ARFCN for operation in band n263**

| Channel Bandwidth | Applicable NR-ARFCN |
|---|---|
| 100 MHz | 2564083 + 1680 * N, N = 0:137 |
| 400 MHz | 2566603 + 6720 * N, N = 0:33 |
| 800 MHz | 2569963 + 6720 * N, N = 0:32 |
| 1600 MHz | 2576683 + 6720 * N, N =0:30 |
| 2000 MHz | 2580043 + 6720 * N, N=0:29,<br>2585083, 2655643, 2692603, 2764843 |

## 5.4.3 Synchronization raster

### 5.4.3.1 Synchronization raster and numbering

The synchronization raster indicates the frequency positions of the synchronization block that can be used by the UE for system acquisition when explicit signalling of the synchronization block position is not present.

A global synchronization raster is defined for all frequencies. The frequency position of the SS block is defined as $SS_{REF}$ with corresponding number GSCN. The parameters defining the $SS_{REF}$ and GSCN for all the frequency ranges are in table 5.4.3.1-1.

The resource element corresponding to the SS block reference frequency $SS_{REF}$ is given in clause 5.4.3.2. The synchronization raster and the subcarrier spacing of the synchronization block is defined separately for each band.

**Table 5.4.3.1-1: GSCN parameters for the global frequency raster**

| Range of frequencies (MHz) | SS block frequency position $SS_{REF}$ | GSCN | Range of GSCN |
|---|---|---|---|
| 0 – 3000 | N * 1200 kHz + M * 50 kHz,<br>N = 1:2499, M ϵ {1,3,5} (Note) | 3N + (M-3)/2 | 2 – 7498 |
| 3000 – 24250 | 3000 MHz + N * 1.44 MHz,<br>N = 0:14756 | 7499 + N | 7499 – 22255 |
| 24250 – 100000 | 24250.08 MHz + N * 17.28 MHz,<br>N = 0:4383 | 22256 + N | 22256 – 26639 |
| NOTE: The default value for *operating bands* which only support SCS spaced channel raster(s) is M=3. | | | |

### 5.4.3.2 Synchronization raster to synchronization block resource element mapping

The mapping between the synchronization raster and the corresponding resource element of the SS block is given in table 5.4.3.2-1.

**Table 5.4.3.2-1: Synchronization Raster to SS block Resource Element Mapping**

| Resource element index k | 120 |
|---|---|
| | |

*k* is the subcarrier number of SS/PBCH block defined in TS 38.211 clause 7.4.3.1 [9].

### 5.4.3.3 Synchronization raster entries for each operating band

The synchronization raster for each band is give in table 5.4.3.3-1. The distance between applicable GSCN entries is given by the <Step size> indicated in table 5.4.3.3-1 for FR1 and table 5.4.3.3-2 for FR2.

**Table 5.4.3.3-1: Applicable SS raster entries per *operating band* (FR1)**

| NR *operating band* | SS Block SCS | SS Block pattern (NOTE 1) | Range of GSCN (First – <Step size> – Last) |
|---|---|---|---|
| n1 | 15 kHz | Case A | 5279 – <1> – 5419 |
| n2 | 15 kHz | Case A | 4829 – <1> – 4969 |
| n3 | 15 kHz | Case A | 4517 – <1> – 4693 |
| n5 | 15 kHz | Case A | 2177 – <1> – 2230 |
| | 30 kHz | Case B | 2183 – <1> – 2224 |
| n7 | 15 kHz | Case A | 6554 – <1> – 6718 |
| n8 | 15 kHz | Case A | 2318 – <1> – 2395 |
| n12 | 15 kHz | Case A | 1828 – <1> – 1858 |
| n13 | 15 kHz | Case A | 1871 – <1> – 1885 |
| n14 | 15 kHz | Case A | 1901 – <1> – 1915 |
| n18 | 15 kHz | Case A | 2156 – <1> – 2182 |
| n20 | 15 kHz | Case A | 1982 – <1> – 2047 |
| n24 | 15 kHz | Case A | 3818 – <1> – 3892 |
| | 30 kHz | Case B | 3824 – <1> – 3886 |
| n25 | 15 kHz | Case A | 4829 – <1> – 4981 |
| n26 | 15 kHz | Case A | 2153 – <1> – 2230 |
| n28 | 15 kHz | Case A | 1901 – <1> – 2002 |
| n29 | 15 kHz | Case A | 1798 – <1> – 1813 |
| n30 | 15 kHz | Case A | 5879 – <1> – 5893 |
| n34 | 15 kHz | Case A | NOTE 3 |
| | 30 kHz | Case C | 5036 – <1> – 5050 |
| n38 | 15 kHz | Case A | NOTE 2 |
| | 30 kHz | Case C | 6437 – <1> – 6538 |
| n39 | 15 kHz | Case A | NOTE 4 |
| | 30 kHz | Case C | 4712 – <1> – 4789 |
| n40 | 30 kHz | Case C | 5762 – <1> – 5989 |
| n41 | 15 kHz | Case A | 6246 – <3> – 6717 |
| | 30 kHz | Case C | 6252 – <3> – 6714 |
| n46[5] | 30 kHz | Case C | 8993 – <1> – 9530 |
| n48 | 30 kHz | Case C | 7884 – <1> – 7982 |
| n50 | 30 kHz | Case C | 3590 – <1> – 3781 |
| n51 | 15 kHz | Case A | 3572 – <1> – 3574 |
| n53 | 15 kHz | Case A | 6215 – <1> – 6232 |
| | 30 kHz | Case C | 6221 – <1> – 6226 |
| n65 | 15 kHz | Case A | 5279 – <1> – 5494 |
| n66 | 15 kHz | Case A | 5279 – <1> – 5494 |
| | 30 kHz | Case B | 5285 – <1> – 5488 |
| n67 | 15 kHz | Case A | 1850 – <1> – 1888 |
| n70 | 15 kHz | Case A | 4993 – <1> – 5044 |
| n71 | 15 kHz | Case A | 1547 – <1> – 1624 |
| n74 | 15 kHz | Case A | 3692 – <1> – 3790 |
| n75 | 15 kHz | Case A | 3584 – <1> – 3787 |
| n76 | 15 kHz | Case A | 3572 – <1> – 3574 |
| n77 | 30 kHz | Case C | 7711 – <1> – 8329 |
| n78 | 30 kHz | Case C | 7711 – <1> – 8051 |
| n79 | 30 kHz | Case C | 8480 – <16> – 8880[7] |
| | | | 8475 – <1> – 8884[8] |
| n85 | 15 kHz | Case A | 1826 – <1> – 1858 |
| n90 | 15 kHz | Case A | 6246 – <1> – 6717[10] |
| | 15 kHz | Case A | 6245 – <1> – 6718[11] |
| | 30 kHz | Case C | 6252 – <1> – 6714 |
| n91 | 15 kHz | Case A | 3572 – <1> – 3574 |
| n92 | 15 kHz | Case A | 3584 – <1> – 3787 |
| n93 | 15 kHz | Case A | 3572 – <1> – 3574 |
| n94 | 15 kHz | Case A | 3584 – <1> – 3787 |
| n96[6] | 30 kHz | Case C | 9531 – <1> – 10363 |
| n100 | 15 kHz | Case A | 2303 – <1> – 2307 |
| n101 | 15 kHz | Case A | 4754 – <1> – 4768 |
| | 30 kHz | Case C | 4760 – <1> – 4764 |
| n102[9] | 30 kHz | Case C | 9531 – <1> – 9877 |
| n104 | 30 kHz | Case C | 9882 – <7> – 10358 |

NOTE 1: SS Block pattern is defined in clause 4.1 in TS 38.213 [10].
NOTE 2: The applicable SS raster entries are GSCN = {6432, 6443, 6457, 6468, 6479, 6493, 6507, 6518, 6532, 6543}
NOTE 3: The applicable SS raster entries are GSCN = {5032, 5043, 5054}
NOTE 4: The applicable SS raster entries are GSCN = {4707, 4715, 4718, 4729, 4732, 4743, 4747, 4754, 4761, 4768, 4772, 4782, 4786, 4793}
NOTE 5: The following GSCN are allowed for operation in band n46:
GSCN = {8996, 9010, 9024, 9038, 9051, 9065, 9079, 9093, 9107, 9121, 9218, 9232, 9246, 9260, 9274, 9288, 9301, 9315, 9329, 9343, 9357, 9371, 9385, 9402, 9416, 9430, 9444, 9458, 9472, 9485, 9499, 9513}.
NOTE 6: The following GSCN are allowed for operation in band n96:
GSCN = { 9548, 9562, 9576, 9590, 9603, 9617, 9631, 9645, 9659, 9673, 9687, 9701, 9714, 9728, 9742, 9756, 9770, 9784, 9798, 9812, 9826, 9840, 9853, 9867, 9881, 9895, 9909, 9923, 9937, 9951, 9964, 9978, 9992, 10006, 10020, 10034, 10048, 10062, 10076, 10090, 10103, 10117, 10131, 10145, 10159, 10173, 10187, 10201, 10214, 10228, 10242, 10256, 10270, 10284, 10298, 10312, 10325, 10339, 10353}.
NOTE 7: The SS raster entries apply for channel bandwidths larger than or equal to 40 MHz.
NOTE 8: The SS raster entries apply for channel bandwidths smaller than 40 MHz.
NOTE 9: The following GSCN are allowed for operation in band n102:
GSCN = {9535, 9548, 9562, 9576, 9590, 9603, 9617, 9631, 9645, 9659, 9673, 9687, 9701, 9714, 9728, 9742, 9756, 9770, 9784, 9798, 9812, 9826, 9840, 9853, 9867}
NOTE 10: The SS raster entries apply for channel bandwidths larger than or equal to 10 MHz.
NOTE 11: The SS raster entries apply for channel bandwidth equal to 5 MHz.

**Table 5.4.3.3-2: Applicable SS raster entries per *operating band* (FR2)**

| NR *operating band* | SS Block SCS | SS Block pattern (note 1) | Range of GSCN (First – <Step size> – Last) |
|---|---|---|---|
| n257 | 120 kHz | Case D | 22388 – <1> – 22558 |
| | 240 kHz | Case E | 22390 – <2> – 22556 |
| n258 | 120 kHz | Case D | 22257 – <1> – 22443 |
| | 240 kHz | Case E | 22258 – <2> – 22442 |
| n259 | 120 kHz | Case D | 23140 – <1> – 23369 |
| | 240 kHz | Case E | 23142 – <2> – 23368 |
| n260 | 120 kHz | Case D | 22995 – <1> – 23166 |
| | 240 kHz | Case E | 22996 – <2> – 23164 |
| n261 | 120 kHz | Case D | 22446 – <1> – 22492 |
| | 240 kHz | Case E | 22446 – <2> – 22490 |
| n262 | 120 kHz | Case D | 23586 – <1> – 23641 |
| | 240 kHz | Case E | 23588 – <2> – 23640 |
| n263 | 120 kHz | Case D | Table 5.4.3.3-3 |
| | 480 kHz | Case F | |
| | 960 kHz[2] | Case G | 24162 – <6> – 24954 |

NOTE 1: SS Block pattern is defined in section 4.1 in TS 38.213 [10].
NOTE 2: SS Block SCS of 960 kHz is not used for initial access.

**Table 5.4.3.3-3: Allowed GSCN for operation in band n263 for 120 kHz and 480 kHz**

| SS Block SCS | Range of GSCN |
|---|---|
| 120 kHz | 24156 + 6 * N – 3 * floor((N+5)/18), N=0:137 |
| 480 kHz | 24162 + 24 * N – 12 * floor((N+4)/18), N=0:33 |

# 6 Conducted transmitter characteristics

## 6.1 General

Unless otherwise stated, the conducted transmitter characteristics are specified at the *antenna connector* for *BS type 1-C* and at the *TAB connector* for *BS type 1-H*, with a full complement of transceiver units for the configuration in normal operating conditions.

For *BS type 1-H* the manufacturer shall declare the minimum number of supported geographical cells (i.e. geographical areas covered by beams). The minimum number of supported geographical cells ($N_{cells}$) relates to the BS setting with the minimum amount of cell splitting supported with transmission on all *TAB connectors* supporting the *operating band*, or with minimum amount of transmitted beams.

For *BS type 1-H* manufacturer shall also declare *TAB connector TX min cell groups*. Every *TAB connector* of the *BS type 1-H* supporting transmission in an *operating band* shall map to one *TAB connector TX min cell group* supporting the same *operating band*, where mapping of *TAB connector*s to cells/beams is implementation dependent.

The number of *active transmitter units* that are considered when calculating the conducted TX emissions limits ($N_{TXU,counted}$) for *BS type 1-H* is calculated as follows:

$$N_{TXU,counted} = min(N_{TXU,active}, 8 \times N_{cells})$$

$N_{TXU,countedpercell}$ is used for scaling of *basic limits* and is derived as $N_{TXU,countedpercell} = N_{TXU,counted} / N_{cells}$

NOTE: $N_{TXU,active}$ depends on the actual number of *active transmitter unit*s and is independent to the declaration of $N_{cells}$.

For BS *type 1-H* there is no requirement specified for band n46, n100 and n101 and n102.

## 6.2 Base station output power

### 6.2.1 General

The BS conducted output power requirement is at *antenna connector* for *BS type 1-C*, or at *TAB connector* for *BS type 1-H*.

The *rated carrier output power* of the *BS type 1-C* shall be as specified in table 6.2.1-1.

**Table 6.2.1-1: *BS type 1-C* rated output power limits for BS classes**

| BS class | $P_{rated,c,AC}$ |
|---|---|
| Wide Area BS | (Note) |
| Medium Range BS | ≤ 38 dBm |
| Local Area BS | ≤ 24 dBm |
| NOTE: There is no upper limit for the $P_{rated,c,AC}$ rated output power of the Wide Area Base Station. | |

For operation in bands n100 and n101 in CEPT countries subject to the ECC Decision (20)02 [21], the WA BS requirement in table 6.2.1-1 apply in case of coordinated RMR BS deployments, while the requirements in clause 6.2.4 apply in case of uncoordinated RMR BS deployments.

The *rated carrier output power* of the *BS type 1-H* shall be as specified in table 6.2.1-2.

**Table 6.2.1-2: *BS type 1-H* rated output power limits for BS classes**

| BS class | $P_{rated,c,sys}$ | $P_{rated,c,TABC}$ |
|---|---|---|
| Wide Area BS | (Note) | (Note) |
| Medium Range BS | ≤ 38 dBm +10log($N_{TXU,counted}$) | ≤ 38 dBm |
| Local Area BS | ≤ 24 dBm +10log($N_{TXU,counted}$) | ≤ 24 dBm |
| NOTE: There is no upper limit for the $P_{rated,c,sys}$ or $P_{rated,c,TABC}$ of the Wide Area Base Station. | | |

In addition, for operation with shared spectrum channel access operation, the BS may have to comply with the applicable BS power limits established regionally, when deployed in regions where those limits apply and under the conditions declared by the manufacturer.

## 6.2.2 Minimum requirement for *BS type 1-C*

In normal conditions, $P_{max,c,AC}$ shall remain within +2 dB and -2 dB of the *rated carrier output power* $P_{rated,c,AC}$, declared by the manufacturer.

In extreme conditions, $P_{max,c,AC}$ shall remain within +2.5 dB and -2.5 dB of the *rated carrier output power* $P_{rated,c,AC}$, declared by the manufacturer.

In certain regions, the minimum requirement for normal conditions may apply also for some conditions outside the range of conditions defined as normal.

NOTE: For NB-IoT operation in NR in-band, the NR carrier and NB-IoT carrier shall be seen as a single carrier occupied NR channel bandwidth, the output power over this carrier is shared between NR and NB-IoT. This note shall apply for $P_{max,c,AC}$ and $P_{rated,c,AC}$.

## 6.2.3 Minimum requirement for *BS type 1-H*

In normal conditions, $P_{max,c,TABC}$ shall remain within +2 dB and -2 dB of the *rated carrier output power* $P_{rated,c,TABC}$ for each *TAB connector* as declared by the manufacturer.

In extreme conditions, $P_{max,c,TABC}$ shall remain within +2.5 dB and -2.5 dB of the *rated carrier output power* $P_{rated,c,TABC}$ for each *TAB connector* as declared by the manufacturer.

In certain regions, the minimum requirement for normal conditions may apply also for some conditions outside the range of conditions defined as normal.

## 6.2.4 Additional requirements (regional)

In certain regions, additional regional requirements may apply.

For Band n41 and n90 operation in Japan, the rated output power, $P_{rated,c.sys}$ for BS type 1-H or sum of $P_{rated,c,AC}$ over all *antenna connectors* for BS type 1-C declared by the manufacturer shall be equal to or less than 20 W per 10 MHz bandwidth.

For band n100 in CEPT countries subject to the ECC Decision (20)02 [21], $P_{rated,c,AC}$ of the WA BS shall not exceed 51.5 dBm/5MHz + ($f_{DL}$-922.1) x 40/3 dB, with $f_{DL}$ being the centre frequency in MHz. This limit is derived from ECC Decision (20)02 [21] assuming a 17 dBi maximum antenna gain and 4 dB losses, and assuming one antenna connector. The above rated output power limit for band n100 applies to uncoordinated deployments and in case of coordinated deployments, higher output power values may be allowed.

For CEPT countries subject to the ECC Decision (20)02 [21], administrations wishing to allow multiple carriers, i.e., more than one wideband carrier (LTE, NR, NB-IoT standalone, or NR/LTE with NB-IoT in-band) in band n100, should consider the implementation of a coordination procedure or other mitigation measures.

NOTE 1: For more details on the maximum level derivation, refer to TR 38.853 [23].

NOTE 2: Standalone NB-IoT operation is specified in TS 36.104 [13].

For band n101 in CEPT countries subject to the ECC Decision (20)02 [21], $P_{rated,c,AC}$ shall not exceed 51 dBm/10MHz or 48 dBm/5MHz. This limit is derived from ECC Decision (20)02 [21] assuming a 18 dBi maximum antenna gain and

4 dB losses, and assuming one antenna connector. The above rated output power limit for band n101 applies to uncoordinated deployments and in case of coordinated deployments, higher output power values may be allowed.

For CEPT countries subject to the ECC Decision (20)02 [21], administrations wishing to allow multiple carriers, i.e., more than one wideband carrier (LTE, NR, NB-IoT standalone, or NR/LTE with NB-IoT in-band) in band n101, should consider the implementation of a coordination procedure or other mitigation measures.

NOTE 3: For more details on the maximum level derivation, refer to TR 38.852 [22].

NOTE 4: Standalone NB-IoT operation is specified in TS 36.104 [13].

# 6.3 Output power dynamics

## 6.3.1 General

The requirements in clause 6.3 apply during the *transmitter ON period*. Transmitted signal quality (as specified in clause 6.5) shall be maintained for the output power dynamics requirements of this clause.

Power control is used to limit the interference level.

## 6.3.2 RE power control dynamic range

### 6.3.2.1 General

The RE power control dynamic range is the difference between the power of an RE and the average RE power for a BS at maximum output power ($P_{max,c,AC}$ or $P_{max,c,TABC}$) for a specified reference condition.

For *BS type 1-C* this requirement shall apply at the *antenna connector* supporting transmission in the *operating band*.

For *BS type 1-H* this requirement shall apply at each *TAB connector* supporting transmission in the *operating band*.

### 6.3.2.2 Minimum requirement for *BS type 1-C* and *BS type 1-H*

RE power control dynamic range:

**Table 6.3.2.2-1: RE power control dynamic range**

| Modulation scheme used on the RE | RE power control dynamic range (dB) | |
|---|---|---|
| | (down) | (up) |
| QPSK (PDCCH) | -6 | +4 |
| QPSK (PDSCH) | -6 | +3 |
| 16QAM (PDSCH) | -3 | +3 |
| 64QAM (PDSCH) | 0 | 0 |
| 256QAM (PDSCH) | 0 | 0 |
| 1024QAM (PDSCH) | 0 | 0 |
| NOTE: The output power per carrier shall always be less or equal to the maximum output power of the base station. | | |

## 6.3.3 Total power dynamic range

### 6.3.3.1 General

The BS total power dynamic range is the difference between the maximum and the minimum transmit power of an OFDM symbol for a specified reference condition.

For *BS type 1-C* this requirement shall apply at the *antenna connector* supporting transmission in the *operating band*.

For *BS type 1-H* this requirement shall apply at each *TAB connector* supporting transmission in the *operating band*.

NOTE 1: The upper limit of the dynamic range is the OFDM symbol power for a BS when transmitting on all RBs at maximum output power. The lower limit of the total power dynamic range is the average power for single RB transmission. The OFDM symbol shall carry PDSCH and not contain RS or SSB.

NOTE 2: The requirement does not apply to operation with shared spectrum channel access.

#### 6.3.3.2 Minimum requirement for *BS type 1-C* and *BS type 1-H*

The downlink (DL) total power dynamic range for each NR carrier shall be larger than or equal to the level in table 6.3.3.2-1.

**Table 6.3.3.2-1: Total power dynamic range**

| ***BS channel bandwidth*** **(MHz)** | **Total power dynamic range (dB)** | | |
|---|---|---|---|
| | **15 kHz SCS** | **30 kHz SCS** | **60 kHz SCS** |
| 5 | 13.9 | 10.4 | N/A |
| 10 | 17.1 | 13.8 | 10.4 |
| 15 | 18.9 | 15.7 | 12.5 |
| 20 | 20.2 | 17 | 13.8 |
| 25 | 21.2 | 18.1 | 14.9 |
| 30 | 22 | 18.9 | 15.7 |
| 35 | 22.7 | 19.6 | 16.4 |
| 40 | 23.3 | 20.2 | 17 |
| 45 | 23.8 | 20.7 | 17.6 |
| 50 | 24.3 | 21.2 | 18.1 |
| 60 | N/A | 22 | 18.9 |
| 70 | N/A | 22.7 | 19.6 |
| 80 | N/A | 23.3 | 20.2 |
| 90 | N/A | 23.8 | 20.8 |
| 100 | N/A | 24.3 | 21.3 |

### 6.3.4 NB-IoT RB power dynamic range for NB-IoT operation in NR in-band

#### 6.3.4.1 General

The NB-IoT RB power dynamic range (or NB-IoT power boosting) is the difference between the average power of NB-IoT REs (which occupy certain REs within a NR transmission bandwidth configuration plus 15 kHz at each edge but not within the NR minimum guard band $GB_{Channel}$) and the average power over all REs (from both NB-IoT and the NR carrier containing the NB-IoT REs).

#### 6.3.4.2 Minimum Requirement

NB-IoT RB power dynamic range for NB-IoT operation in NR in-band shall be larger than or equal to the level specified in Table 6.3.4.2-1. This power dynamic range level is only required for one NB-IoT RB.

**Table 6.3.4.2-1: NB-IoT RB power dynamic range for NB-IoT operation in NR in-band**

| **BS channel bandwidth (MHz)** | **NB-IoT RB frequency position** | **NB-IoT RB power dynamic range (dB)** |
|---|---|---|
| 5, 10 | Any | +6 |
| 15 | Within center 77*180kHz+15kHz at each edge | +6 |
| | Other | +3 |
| 20 | Within center 102*180kHz+15kHz at each edge | +6 |
| | Other | +3 |
| 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100 | Within center 90% of BS channel bandwidth | +6 |
| | Other | +3 |

# 6.4 Transmit ON/OFF power

## 6.4.1 Transmitter OFF power

### 6.4.1.1 General

Transmit OFF power requirements apply only to TDD operation of the BS.

Transmitter OFF power is defined as the mean power measured over 70/N us filtered with a square filter of bandwidth equal to the *transmission bandwidth configuration* of the BS ($BW_{Config}$) centred on the assigned channel frequency during the *transmitter OFF period*. N = SCS/15, where SCS is Sub Carrier Spacing in kHz.

For *multi-band connectors* and for *single band connectors* supporting transmission in multiple *operating bands*, the requirement is only applicable during the *transmitter OFF period* in all supported *operating bands*.

For BS supporting intra-band contiguous CA, the transmitter OFF power is defined as the mean power measured over 70/N us filtered with a square filter of bandwidth equal to the *Aggregated BS Channel Bandwidth* $BW_{Channel_CA}$ centred on $(F_{edge,high}+F_{edge,low})/2$ during the *transmitter OFF period*. N = SCS/15, where SCS is the smallest supported Sub Carrier Spacing in kHz in the *Aggregated BS Channel Bandwidth*.

### 6.4.1.2 Minimum requirement for *BS type 1-C*

For *BS type 1-C*, the requirements for transmitter OFF power spectral density shall be less than -85 dBm/MHz per *antenna connector*.

### 6.4.1.3 Minimum requirement for *BS type 1-H*

For *BS type 1-H*, the requirements for transmitter OFF power spectral density shall be less than -85 dBm/MHz per *TAB connector*.

## 6.4.2 *Transmitter transient period*

### 6.4.2.1 General

*Transmitter transient period* requirements apply only to TDD operation of the BS.

The *transmitter transient period* is the time period during which the transmitter is changing from the *transmitter OFF period* to the *transmitter ON period* or vice versa. The *transmitter transient period* is illustrated in figure 6.4.2.1-1.

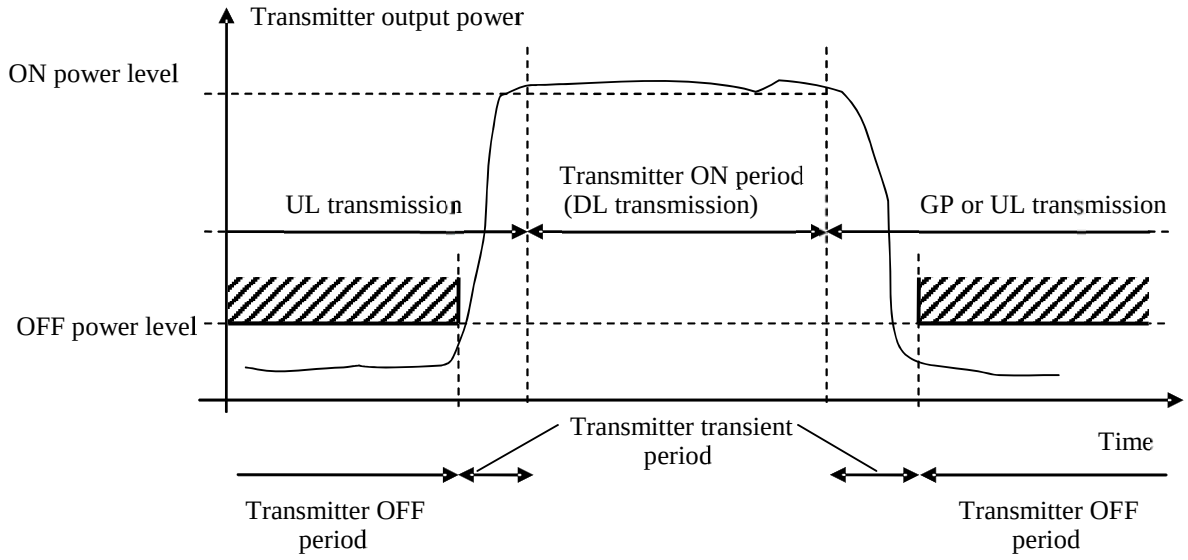



**Figure 6.4.2.1-1: Example of relations between transmitter ON period, transmitter OFF period and *transmitter transient period***

For *BS type 1-C* this requirement shall be applied at the *antenna connector* supporting transmission in the *operating band*.

For *BS type 1-H* this requirement shall be applied at each *TAB connector* supporting transmission in the *operating band.*

### 6.4.2.2 Minimum requirement for *BS type 1-C* and *BS type 1-H* 1-H

For *BS type 1-C* and *BS type 1-H*, the *transmitter transient period* shall be shorter than the values listed in the minimum requirement table 6.4.2.2-1.

**Table 6.4.2.2-1: Minimum requirement for the *transmitter transient period* for *BS type 1-C* and *BS type 1-H***

| Transition | Transient period length (μs) |
|---|---|
| OFF to ON | 10 |
| ON to OFF | 10 |

### 6.4.2.3 Void

## 6.5 Transmitted signal quality

### 6.5.1 Frequency error

#### 6.5.1.1 General

The requirements in clause 6.5.1 apply to the *transmitter ON period*.

Frequency error is the measure of the difference between the actual BS transmit frequency and the assigned frequency. The same source shall be used for RF frequency and data clock generation.

For *BS type 1-C* this requirement shall be applied at the *antenna connector* supporting transmission in the *operating band*.

For *BS type 1-H* this requirement shall be applied at each *TAB connector* supporting transmission in the *operating band.*

### 6.5.1.2 Minimum requirement for *BS type 1-C* and *BS type 1-H*

For *BS type 1-C* and *BS type* 1-H, the modulated carrier frequency of each NR carrier configured by the BS shall be accurate to within the accuracy range given in table 6.5.1.2-1 observed over 1 ms.

The frequency error requirement for NB-IoT are specified in TS 36.104 [13] clause 6.5.1.

**Table 6.5.1.2-1: Frequency error minimum requirement**

| BS class | Accuracy |
|---|---|
| Wide Area BS | ±0.05 ppm |
| Medium Range BS | ±0.1 ppm |
| Local Area BS | ±0.1 ppm |

## 6.5.2 Modulation quality

### 6.5.2.1 General

Modulation quality is defined by the difference between the measured carrier signal and an ideal signal. Modulation quality can e.g. be expressed as Error Vector Magnitude (EVM). The Error Vector Magnitude is a measure of the difference between the ideal symbols and the measured symbols after the equalization. This difference is called the error vector. Details about how the EVM is determined are specified in Annex B.

For *BS type 1-C* this requirement shall be applied at the *antenna connector* supporting transmission in the *operating band*.

For *BS type 1-H* this requirement shall be applied at each *TAB connector* supporting transmission in the *operating band.*

### 6.5.2.2 Minimum Requirement for *BS type 1-C* and *BS type 1-H*

For *BS type 1-C* and *1-H*, the EVM levels of each NR carrier for different modulation schemes on PDSCH outlined in table 6.5.2.2-1 shall be met using the frame structure described in clause 6.5.2.3.

**Table 6.5.2.2-1: EVM requirements for *BS type 1-C* and *BS type 1-H* carrier**

| Modulation scheme for PDSCH | Required EVM |
|---|---|
| QPSK | 17.5 % |
| 16QAM | 12.5 % |
| 64QAM | 8 % |
| 256QAM | 3.5 % |
| 1024QAM | 2.5 %[1]<br>2.8 %[2] |
| Note1: This requirement is applicable for frequencies equal to or below 4.2 GHz.<br>Note 2: This requirement is applicable for frequencies above 4.2 GHz. | |

The modulation quality requirements for NB-IoT are specified in TS 36.104 [13] clause 6.5.2.

### 6.5.2.3 EVM frame structure for measurement

EVM shall be evaluated for each NR carrier over all allocated resource blocks and downlink subframes. Different modulation schemes listed in table 6.5.2.2-1 shall be considered for rank 1.

For NR, for all bandwidths, the EVM measurement shall be performed for each NR carrier over all allocated resource blocks and downlink subframes within 10 ms measurement periods. The boundaries of the EVM measurement periods need not be aligned with radio frame boundaries.

### 6.5.3 Time alignment error

#### 6.5.3.1 General

This requirement shall apply to frame timing in MIMO transmission, *carrier aggregation* and their combinations.

Frames of the NR signals present at the BS transmitter *antenna connectors* or *TAB connectors* are not perfectly aligned in time. The RF signals present at the BS transmitter *antenna connectors* or *transceiver array boundary* may experience certain timing differences in relation to each other.

The TAE is specified for a specific set of signals/transmitter configuration/transmission mode.

For *BS type 1-C*, the TAE is defined as the largest timing difference between any two signals belonging to different *antenna connectors* for a specific set of signals/transmitter configuration/transmission mode.

For *BS type 1-H*, the TAE is defined as the largest timing difference between any two signals belonging to *TAB connectors* belonging to different transmitter groups at the *transceiver array boundary*, where transmitter groups are associated with the *TAB connectors* in the transceiver unit array corresponding to MIMO transmission, *carrier aggregation* for a specific set of signals/transmitter configuration/transmission mode.

#### 6.5.3.2 Minimum requirement for *BS type 1-C* and *BS type* 1-H

For MIMO transmission, at each carrier frequency, TAE shall not exceed 65 ns.

For *intra-band contiguous carrier aggregation*, with or without MIMO, TAE shall not exceed 260ns.

For *intra-band non-contiguous carrier aggregation*, with or without MIMO, TAE shall not exceed 3μs.

For inter-band *carrier aggregation*, with or without MIMO, TAE shall not exceed 3μs.

The time alignment error requirements for NB-IoT are specified in TS 36.104 [13] clause 6.5.3.

**Table 6.5.3.2-1: Void**

**Table 6.5.3.2-2: Void**

**Table 6.5.3.2-3: Void**

## 6.6 Unwanted emissions

### 6.6.1 General

Unwanted emissions consist of out-of-band emissions and spurious emissions according to ITU definitions [2]. In ITU terminology, out of band emissions are unwanted emissions immediately outside the *BS channel bandwidth* resulting from the modulation process and non-linearity in the transmitter but excluding spurious emissions. Spurious emissions are emissions which are caused by unwanted transmitter effects such as harmonics emission, parasitic emission, intermodulation products and frequency conversion products, but exclude out of band emissions.

The out-of-band emissions requirement for the BS transmitter is specified both in terms of Adjacent Channel Leakage power Ratio (ACLR) and *operating band* unwanted emissions (OBUE).

The maximum offset of the *operating band* unwanted emissions mask from the *operating band* edge is $\Delta f_{OBUE}$. The Operating band unwanted emissions define all unwanted emissions in each supported downlink *operating band* plus the frequency ranges $\Delta f_{OBUE}$ above and $\Delta f_{OBUE}$ below each band. Unwanted emissions outside of this frequency range are limited by a spurious emissions requirement.

The values of $\Delta f_{OBUE}$ are defined in table 6.6.1-1 for the NR *operating bands*.

**Table 6.6.1-1: Maximum offset of OBUE outside the downlink *operating band***

| BS type | *Operating band* characteristics | $\Delta f_{OBUE}$ (MHz) |
|---|---|---|
| *BS type 1-H* | $F_{DL,high} - F_{DL,low} < 100$ MHz | 10 |
| | 100 MHz $\leq F_{DL,high} - F_{DL,low} \leq 900$ MHz | 40 |
| | | |
| *BS type 1-C* | $F_{DL,high} - F_{DL,low} \leq 200$ MHz | 10 |
| | 200 MHz $< F_{DL,high} - F_{DL,low} \leq 900$ MHz | 40 |
| | | |

For band n46, n96 and n102, the values of $\Delta f_{OBUE}$ are defined in table 6.6.1-1a.

**Table 6.6.1-1a: Maximum offset of OBUE outside the downlink *operating band* for band n46, n96 and n102**

| *Operating band* | $\Delta f_{OBUE}$ (MHz) |
|---|---|
| n46, n102 | 40 |
| n96 | 50 |

For band n104, the values of $\Delta f_{OBUE}$ are defined in table 6.6.1-1b.

**Table 6.6.1-1b: Maximum offset of OBUE outside the downlink *operating band* for band n104**

| BS type | *Operating band* | $\Delta f_{OBUE}$ (MHz) |
|---|---|---|
| *BS type 1-H* | n104 | 100 |
| *BS type 1-C* | n104 | 40 |

For *BS type 1-H* the unwanted emission requirements are applied per the *TAB connector TX min cell groups* for all the configurations supported by the BS. The *basic limits* and corresponding emissions scaling are defined in each relevant clause.

There is in addition a requirement for occupied bandwidth.

## 6.6.2 Occupied bandwidth

### 6.6.2.1 General

The occupied bandwidth is the width of a frequency band such that, below the lower and above the upper frequency limits, the mean powers emitted are each equal to a specified percentage $\beta/2$ of the total mean transmitted power. See also Recommendation ITU-R SM.328 [3].

The value of $\beta/2$ shall be taken as 0.5%.

The occupied bandwidth requirement shall apply during the *transmitter ON period* for a single transmitted carrier. The minimum requirement below may be applied regionally. There may also be regional requirements to declare the occupied bandwidth according to the definition in the present clause.

For *BS type 1-C* this requirement shall be applied at the *antenna connector* supporting transmission in the *operating band*.

For *BS type 1-H* this requirement shall be appliedat each *TAB connector* supporting transmission in the *operating band*.

### 6.6.2.2 Minimum requirement for *BS type 1-C* and *BS type 1-H*

The occupied bandwidth for each NR carrier shall be less than the *BS channel bandwidth*. For intra-band contiguous CA, the occupied bandwidth shall be less than or equal the *Aggregated BS Channel Bandwidth*.

For NB.IoT operation in NR in-band, the occupied bandwidth for each NR carrier with NB-IoT shall be less than than the *BS channel bandwidth*.

## 6.6.3 Adjacent Channel Leakage Power Ratio

### 6.6.3.1 General

Adjacent Channel Leakage power Ratio (ACLR) is the ratio of the filtered mean power centred on the assigned channel frequency to the filtered mean power centred on an adjacent channel frequency.

The requirements shall apply outside the *Base Station RF Bandwidth* or *Radio Bandwidth* whatever the type of transmitter considered (single carrier or multi-carrier) and for all transmission modes foreseen by the manufacturer's specification.

The requirements shall also apply if the BS supports NB-IoT operation in NR in-band.

For a BS operating in *non-contiguous spectrum*, the ACLR or the CACLR requirement in Tables 6.6.3.2-2a to 6.6.3.2-3aa shall apply in *sub-block gaps*, depending on the *operating band* and the *sub-block gap size* ($W_{gap}$) where the limit applies.

For a *multi-band connector*, the ACLR or the CACLR requirement in Tables 6.6.3.2-2a to 6.6.3.2-3aa shall apply in *Inter RF Bandwidth gaps*, depending on the *operating band* and the *Inter RF Bandwidth gap size* ($W_{gap}$) where the limit applies.

The requirement shall apply during the *transmitter ON period*.

### 6.6.3.2 Limits and *Basic limits*

The ACLR is defined with a square filter of bandwidth equal to the transmission bandwidth configuration of the transmitted signal ($BW_{Config}$) centred on the assigned channel frequency and a filter centred on the adjacent channel frequency according to the tables below.

For operation in paired and unpaired spectrum, the ACLR shall be higher than the value specified in table 6.6.3.2-1 in any operating band except for band n46, n96 and n102.

**Table 6.6.3.2-1: Base station ACLR limit**

| ***BS channel bandwidth* of *lowest/highest carrier* transmitted $BW_{Channel}$ (MHz)** | **BS adjacent channel centre frequency offset below the lowest or above the highest carrier centre frequency transmitted** | **Assumed adjacent channel carrier (informative)** | **Filter on the adjacent channel frequency and corresponding filter bandwidth** | **ACLR limit** |
|---|---|---|---|---|
| 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90,100 | $BW_{Channel}$ | NR of same BW (Note 2) | Square ($BW_{Config}$) | 45 dB, 38 dB (Note 4) |
| | 2 x $BW_{Channel}$ | NR of same BW (Note 2) | Square ($BW_{Config}$) | 45 dB, 38 dB (Note 4) |
| | $BW_{Channel}$ /2 + 2.5 MHz | 5 MHz E-UTRA | Square (4.5 MHz) | 45 dB (Note 3) |
| | $BW_{Channel}$ /2 + 7.5 MHz | 5 MHz E-UTRA | Square (4.5 MHz) | 45 dB (Note 3) |
| NOTE 1: $BW_{Channel}$ and $BW_{Config}$ are the *BS channel bandwidth* and *transmission bandwidth configuration* of the *lowest/highest carrier* transmitted on the assigned channel frequency.<br>NOTE 2: With SCS that provides largest transmission bandwidth configuration ($BW_{Config}$).<br>NOTE 3: The requirements are applicable when the band is also defined for E-UTRA or UTRA.<br>NOTE 4: For BS operating in band n104, ACLR requirement 38 dB applies. For BS operating in other bands, ACLR requirement 45 dB applies. | | | | |

For band n46, n96 and n102, the ACLR shall be higher than the value specified in Table 6.6.3.2-1a.

**Table 6.6.3.2-1a: Base station ACLR limit for band n46, n96 and n102**

| ***BS channel bandwidth*** **of lowest/highest NR carrier transmitted $BW_{Channel}$ (MHz)** | **BS adjacent channel centre frequency offset below the lowest or above the highest carrier centre frequency transmitted** | **Assumed adjacent channel carrier (informative)** | **Filter on the adjacent channel frequency and corresponding filter bandwidth** | **ACLR limit** |
|---|---|---|---|---|
| 10, 20, 40, 60, 80 | $BW_{Channel}$ | NR of same BW (Note 2) | Square ($BW_{Config}$) | 35 dB |
| | 2 x $BW_{Channel}$ | NR of same BW (Note 2) | Square ($BW_{Config}$) | 40 dB |
| NOTE 1: $BW_{Channel}$ and $BW_{Config}$ are the *BS channel bandwidth* and transmission bandwidth configuration of the lowest/highest NR carrier transmitted on the assigned channel frequency.<br>NOTE 2: With SCS that provides largest transmission bandwidth configuration ($BW_{Config}$). | | | | |

The ACLR absolute *basic limit* is specified in table 6.6.3.2-2.

**Table 6.6.3.2-2: Base station ACLR absolute *basic limit***

| **BS category / BS class** | **ACLR absolute *basic limit*** |
|---|---|
| Category A Wide Area BS | -13 dBm/MHz |
| Category B Wide Area BS | -15 dBm/MHz |
| Medium Range BS | -25 dBm/MHz |
| Local Area BS | -32 dBm/MHz |

For operation in non-contiguous spectrum or multiple bands, the ACLR shall be higher than the value specified in Table 6.6.3.2-2a in any operating band except for band n46, n96 and n102.

**Table 6.6.3.2-2a: Base Station ACLR limit in non-contiguous spectrum or multiple bands**

| ***BS channel bandwidth*** **of *carrier*** **transmitted adjacent to *sub-block gap* or *inter RF Bandwidth gap*** **$BW_{Channel}$ (MHz)** | **Sub-block or Inter RF Bandwidth gap size ($W_{gap}$) where the limit applies (MHz)** | **BS adjacent channel centre frequency offset below or above the sub-block or Base Station RF Bandwidth edge (inside the gap)** | **Assumed adjacent channel carrier** | **Filter on the adjacent channel frequency and corresponding filter bandwidth** | **ACLR limit** |
|---|---|---|---|---|---|
| 5, 10, 15, 20 | $W_{gap}$ ≥ 15 (Note 3)<br>$W_{gap}$ ≥ 45 (Note 4) | 2.5 MHz | 5 MHz NR (Note 2) | Square ($BW_{Config}$) | 45 dB, 38 dB (Note 5) |
| | $W_{gap}$ ≥ 20 (Note 3)<br>$W_{gap}$ ≥ 50 (Note 4) | 7.5 MHz | 5 MHz NR (Note 2) | Square ($BW_{Config}$) | 45 dB, 38 dB (Note 5) |
| 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100 | $W_{gap}$ ≥ 60 (Note 4)<br>$W_{gap}$ ≥ 30 (Note 3) | 10 MHz | 20 MHz NR (Note 2) | Square ($BW_{Config}$) | 45 dB, 38 dB (Note 5) |
| | $W_{gap}$ ≥ 80 (Note 4)<br>$W_{gap}$ ≥ 50 (Note 3) | 30 MHz | 20 MHz NR (Note 2) | Square ($BW_{Config}$) | 45 dB, 38 dB (Note 5) |
| NOTE 1: $BW_{Config}$ is the transmission bandwidth configuration of the assumed adjacent channel carrier.<br>NOTE 2: With SCS that provides largest transmission bandwidth configuration ($BW_{Config}$).<br>NOTE 3: Applicable in case the *BS channel bandwidth* of the NR carrier transmitted at the other edge of the gap is 5, 10, 15, 20 MHz.<br>NOTE 4: Applicable in case the *BS channel bandwidth* of the NR carrier transmitted at the other edge of the gap is 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100 MHz.<br>NOTE 5: For BS operating in band n104, ACLR requirement 38 dB applies. For BS operating in other bands, ACLR requirement 45 dB applies. | | | | | |

For operation in non-contiguous spectrum for band n46, n96 and n102, the ACLR shall be higher than the value specified in Table 6.6.3.2-2b.

**Table 6.6.3.2-2b: Base Station ACLR limit in non-contiguous spectrum for band n46, n96 and n102**

| ***BS channel bandwidth*** **of NR carrier transmitted $BW_{Channel}$ adjacent to s*ub-block gap* or *inter RF Bandwidth gap* (MHz)** | **Sub-block or Inter RF Bandwidth gap size ($W_{gap}$) where the limit applies (MHz)** | **BS adjacent channel centre frequency offset below or above the sub-block or Base Station RF Bandwidth edge (inside the gap)** | **Assumed adjacent channel carrier** | **Filter on the adjacent channel frequency and corresponding filter bandwidth** | **ACLR limit** |
|---|---|---|---|---|---|
| 10, 20, 40, 60, 80 | $W_{gap} \geq 60$ | 10 MHz | 20 MHz NR (Note 2) | Square ($BW_{Config}$) | 35 dB |
| | $W_{gap} \geq 80$ | 30 MHz | 20 MHz NR (Note 2) | Square ($BW_{Config}$) | 40 dB |
| NOTE 1: $BW_{Config}$ is the transmission bandwidth configuration of the assumed adjacent channel carrier.<br>NOTE 2: With SCS that provides largest transmission bandwidth configuration ($BW_{Config}$). | | | | | |

The Cumulative Adjacent Channel Leakage power Ratio (CACLR) in a *sub-block gap* or the *Inter RF Bandwidth gap* is the ratio of:

a) the sum of the filtered mean power centred on the assigned channel frequencies for the two carriers adjacent to each side of the *sub-block gap* or the *Inter RF Bandwidth gap*, and

b) the filtered mean power centred on a frequency channel adjacent to one of the respective *sub-block* edges or *Base Station RF Bandwidth edges*.

The assumed filter for the adjacent channel frequency is defined in table 6.6.3.2-3 and the filters on the assigned channels are defined in table 6.6.3.2-4.

For operation in *non-contiguous spectrum* or multiple bands, the CACLR for NR carriers located on either side of the *sub-block gap* or the *Inter RF Bandwidth gap* shall be higher than the value specified in table 6.6.3.2-3.

The CACLR requirements in Table 6.6.3.2-3 apply to BS that supports NR, in any operating band except for band n46, n96 and n102. The CACLR requirements for band n46, n96 and n102 are in Table 6.6.3.2-3aa.

**Table 6.6.3.2-3: Base Station CACLR limit**

| ***BS channel bandwidth*** **of *carrier* transmitted adjacent to s*ub-block gap* or *inter RF Bandwidth gap* $BW_{Channel}$ (MHz)** | **Sub-block or Inter RF Bandwidth gap size ($W_{gap}$) where the limit applies (MHz)** | **BS adjacent channel centre frequency offset below or above the sub-block or Base Station RF Bandwidth edge (inside the gap)** | **Assumed adjacent channel carrier** | **Filter on the adjacent channel frequency and corresponding filter bandwidth** | **CACLR limit** |
|---|---|---|---|---|---|
| 5, 10, 15, 20 | $5 \leq W_{gap} < 15$ (Note 3)<br>$5 \leq W_{gap} < 45$ (Note 4) | 2.5 MHz | 5 MHz NR (Note 2) | Square ($BW_{Config}$) | 45 dB, 38 dB (Note 5) |
| | $10 < W_{gap} < 20$ (Note 3)<br>$10 \leq W_{gap} < 50$ (Note 4) | 7.5 MHz | 5 MHz NR (Note 2) | Square ($BW_{Config}$) | 45 dB, 38 dB (Note 5) |
| 25, 30, 35, 40, 45, 50, 60, 70, 80,90, 100 | $20 \leq W_{gap} < 60$ (Note 4)<br>$20 \leq W_{gap} < 30$ (Note 3) | 10 MHz | 20 MHz NR (Note 2) | Square ($BW_{Config}$) | 45 dB, 38 dB (Note 5) |
| | $40 < W_{gap} < 80$ (Note 4)<br>$40 \leq W_{gap} < 50$ (Note 3) | 30 MHz | 20 MHz NR (Note 2) | Square ($BW_{Config}$) | 45 dB, 38 dB (Note 5) |
| NOTE 1: $BW_{Config}$ is the transmission bandwidth configuration of the assumed adjacent channel carrier.<br>NOTE 2: With SCS that provides largest transmission bandwidth configuration ($BW_{Config}$).<br>NOTE 3: Applicable in case the *BS channel bandwidth* of the NR carrier transmitted at the other edge of the gap is 5, 10, 15, 20 MHz.<br>NOTE 4: Applicable in case the *BS channel bandwidth* of the NR carrier transmitted at the other edge of the gap is 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100 MHz.<br>NOTE 5: For BS operating in band n104, ACLR requirement 38 dB applies. For BS operating in other bands, ACLR requirement 45 dB applies. | | | | | |

For operation in non-contiguous spectrum for band n46, n96 and n102, the CACLR for NR carriers located on either side of the *sub-block gap* shall be higher than the value specified in Table 6.6.3.2-3aa.

**Table 6.6.3.2-3aa: Base Station CACLR limit for band n46, n96 and n102**

| ***BS channel bandwidth*** **of NR carrier transmitted $BW_{Channel}$ adjacent to s*ub-block gap* or *inter RF Bandwidth gap* (MHz)** | **Sub-block or Inter RF Bandwidth gap size ($W_{gap}$) where the limit applies (MHz)** | **BS adjacent channel centre frequency offset below or above the sub-block or Base Station RF Bandwidth edge (inside the gap)** | **Assumed adjacent channel carrier** | **Filter on the adjacent channel frequency and corresponding filter bandwidth** | **CACLR limit** |
|---|---|---|---|---|---|
| 10, 20, 40, 60, 80 | $20 \le W_{gap} < 60$ | 10 MHz | 20 MHz NR (Note 2) | Square ($BW_{Config}$) | 35 dB |
| | $40 < W_{gap} < 80$ | 30 MHz | 20 MHz NR (Note 2) | Square ($BW_{Config}$) | 40 dB |
| NOTE 1: $BW_{Config}$ is the transmission bandwidth configuration of the assumed adjacent channel carrier.<br>NOTE 2: With SCS that provides largest transmission bandwidth configuration ($BW_{Config}$). | | | | | |

The CACLR absolute *basic limit* is specified in table 6.6.3.2-3a.

**Table 6.6.3.2-3a: Base station CACLR absolute *basic limit***

| **BS category / BS class** | **CACLR absolute *basic limit*** |
|---|---|
| Category A Wide Area BS | -13 dBm/MHz |
| Category B Wide Area BS | -15 dBm/MHz |
| Medium Range BS | -25 dBm/MHz |
| Local Area BS | -32 dBm/MHz |

**Table 6.6.3.2-4: Filter parameters for the assigned channel**

| **RAT of the carrier adjacent to the *sub-block* or *Inter RF Bandwidth gap*** | **Filter on the assigned channel frequency and corresponding filter bandwidth** |
|---|---|
| NR | NR of same BW with SCS that provides largest *transmission bandwidth configuration* |

### 6.6.3.3 Minimum requirement for *BS type 1-C*

The ACLR (CACLR) absolute *basic limits* in table 6.6.3.2-2, 6.6.3.2-3a or the ACLR (CACLR) *limits* in table 6.6.3.2-1, 6.6.3.2-2a or 6.6.3.2-3, whichever is less stringent, shall apply for each *antenna connector*.

For Band n41 and n90 operation in Japan, absolute ACLR limits shall be applied to the sum of the absolute ACLR power over all *antenna connectors* for *BS type 1-C*.

### 6.6.3.4 Minimum requirement for *BS type 1-H*

The ACLR (CACLR) absolute *basic limits* in table 6.6.3.2-2 + X, 6.6.3.2-3a + X (where $X = 10\log_{10}(N_{TXU,countedpercell})$) or the ACLR (CACLR) *limits* in table 6.6.3.2-1, 6.6.3.2-2a or 6.6.3.2-3, whichever is less stringent, shall apply for each *TAB connector TX min cell group*.

NOTE: Conformance to the *BS type 1-H* ACLR requirement can be demonstrated by meeting at least one of the following criteria as determined by the manufacturer:

1) The ratio of the sum of the filtered mean power measured on each *TAB connector* in the *TAB connector TX min cell group* at the assigned channel frequency to the sum of the filtered mean power measured on each *TAB connector* in the *TAB connector TX min cell group* at the adjacent channel frequency shall be greater than or equal to the ACLR *basic limit* of the BS. This shall apply for each *TAB connector TX min cell group*.

Or

2) The ratio of the filtered mean power at the *TAB connector* centred on the assigned channel frequency to the filtered mean power at this *TAB connector* centred on the adjacent channel frequency shall be greater than or equal to the ACLR *basic limit* of the BS for every *TAB connector* in the *TAB connector TX min cell group*, for each *TAB connector TX min cell group*.

In case the ACLR (CACLR) absolute *basic limit* of *BS type 1-H* are applied, the conformance can be demonstrated by meeting at least one of the following criteria as determined by the manufacturer:

1) The sum of the filtered mean power measured on each *TAB connector* in the *TAB connector TX min cell group* at the adjacent channel frequency shall be less than or equal to the ACLR (CACLR) absolute ba*sic limit* + X of the BS. This shall apply to each *TAB* connector *TX min cell group.*

Or

2) The filtered mean power at each *TAB connector* centred on the adjacent channel frequency shall be less than or equal to the ACLR (CACLR) absolute *basic limit* of the BS scaled by X $-10\log_{10}(n)$ for every *TAB connector* in the *TAB connector TX min cell group*, for each *TAB connector TX min cell group*, where *n* is the number of *TAB connectors* in the *TAB connector TX min cell group.*

## 6.6.4 Operating band unwanted emissions

### 6.6.4.1 General

Unless otherwise stated, the operating band unwanted emission (OBUE) limits in FR1 are defined from $\Delta f_{OBUE}$ below the lowest frequency of each supported downlink *operating band* up to $\Delta f_{OBUE}$ above the highest frequency of each supported downlink *operating band*. The values of $\Delta f_{OBUE}$ are defined in table 6.6.1-1 for the NR *operating bands*.

The requirements shall apply whatever the type of transmitter considered and for all transmission modes foreseen by the manufacturer's specification. In addition, for a BS operating in *non-contiguous spectrum*, the requirements apply inside any *sub-block gap*. In addition, for a BS operating in multiple bands, the requirements apply inside any *Inter RF Bandwidth gap*.

*Basic limits* are specified in the tables below, where:

- $\Delta f$ is the separation between the *channel edge* frequency and the nominal -3dB point of the measuring filter closest to the carrier frequency.
- f_offset is the separation between the *channel edge* frequency and the centre of the measuring filter.
- $f_offset_{max}$ is the offset to the frequency $\Delta f_{OBUE}$ outside the downlink *operating band*, where $\Delta f_{OBUE}$ is defined in table 6.6.1-1.
- $\Delta f_{max}$ is equal to $f_offset_{max}$ minus half of the bandwidth of the measuring filter.

For a *multi-band connector* inside any *Inter RF Bandwidth gaps* with $W_{gap} < 2*\Delta f_{OBUE}$, a combined *basic* limit shall be applied which is the cumulative sum of the *basic limit*s specified at the *Base Station RF Bandwidth edges* on each side of the *Inter RF Bandwidth gap*. The *basic limit* for *Base Station RF Bandwidth edge* is specified in clauses 6.6.4.2.1 to 6.6.4.2.4 below, where in this case:

- $\Delta f$ is the separation between the *Base Station RF Bandwidth edge* frequency and the nominal -3 dB point of the measuring filter closest to the *Base Station RF Bandwidth edge*.
- f_offset is the separation between the *Base Station RF Bandwidth edge* frequency and the centre of the measuring filter.
- $f_offset_{max}$ is equal to the *Inter RF Bandwidth gap* minus half of the bandwidth of the measuring filter.
- $\Delta f_{max}$ is equal to $f_offset_{max}$ minus half of the bandwidth of the measuring filter.

For a *multi-band connector*, the operating band unwanted emission limits apply also in a supported *operating band* without any carrier transmitted, in the case where there are carrier(s) transmitted in another supported *operating band*. In this case, no cumulative *basic limit* is applied in the *inter-band gap* between a supported downlink *operating band* with carrier(s) transmitted and a supported downlink *operating band* without any carrier transmitted and

- In case the *inter-band gap* between a supported downlink *operating band* with carrier(s) transmitted and a supported downlink *operating band* without any carrier transmitted is less than $2*\Delta f_{OBUE}$, $f_offset_{max}$ shall be the offset to the frequency $\Delta f_{OBUE}$ MHz outside the outermost edges of the two supported downlink *operating bands* and the operating band unwanted emission *basic limits* of the band where there are carriers transmitted, as defined in the tables of the present clause, shall apply across both downlink bands.

- In other cases, the operating band unwanted emission *basic limits* of the band where there are carriers transmitted, as defined in the tables of the present clause for the largest frequency offset ($\Delta f_{max}$), shall apply from $\Delta f_{OBUE}$ MHz below the lowest frequency, up to $\Delta f_{OBUE}$ MHz above the highest frequency of the supported downlink *operating band* without any carrier transmitted.

For a multi-carrier *single-band connector* or a *single-band connector* configured for intra-band contiguous or non-contiguous *carrier aggregation* the definitions above apply to the lower edge of the carrier transmitted at the *lowest carrier* frequency and the upper edge of the carrier transmitted at the *highest carrier* frequency within a specified frequency band.

In addition inside any *sub-block gap* for a *single-band connector* operating in *non-contiguous spectrum*, a combined *basic* limit shall be applied which is the cumulative sum of the *basic limit*s specified for the adjacent *sub-block*s on each side of the *sub-block gap*. The *basic limit* for each *sub-block* is specified in clauses 6.6.4.2.1 to 6.6.4.2.4 below, where in this case:

- $\Delta f$ is the separation between the *sub-block* edge frequency and the nominal -3 dB point of the measuring filter closest to the *sub-block* edge.

- f_offset is the separation between the *sub-block* edge frequency and the centre of the measuring filter.

- $f_offset_{max}$ is equal to the *sub-block gap* bandwidth minus half of the bandwidth of the measuring filter.

- $\Delta f_{max}$ is equal to $f_offset_{max}$ minus half of the bandwidth of the measuring filter.

For Wide Area BS, the requirements of either clause 6.6.4.2.1 (Category A limits) or clause 6.6.4.2.2 (Category B limits) shall apply.

For Medium Range BS, the requirements in clause 6.6.4.2.3 shall apply (Category A and B).

For Local Area BS, the requirements of clause 6.6.4.2.4 shall apply (Category A and B).

The requirements shall also apply if the BS supports NB-IoT operation in NR in-band.

The application of either Category A or Category B *basic limits* shall be the same as for Transmitter spurious emissions in clause 6.6.5.

### 6.6.4.2 *Basic limits*

#### 6.6.4.2.1 *Basic limits* for Wide Area BS (Category A)

For BS operating in Bands n5, n8, n12, n13, n14, n18, n26, n28, n29, n71, n85, *basic limits* are specified in table 6.6.4.2.1-1.

**Table 6.6.4.2.1-1: Wide Area BS operating band unwanted emission limits (NR bands below 1 GHz) for Category A**

| Frequency offset of measurement filter -3dB point, Δf | Frequency offset of measurement filter centre frequency, f_offset | *Basic limits* (Note 1, 2) | *Measurement bandwidth* |
|---|---|---|---|
| 0 MHz ≤ Δf < 5 MHz | 0.05 MHz ≤ f_offset < 5.05 MHz | $-7dBm - \frac{7}{5} \cdot \left( \frac{f_offset}{MHz} - 0.05 \right) dB$ | 100 kHz |
| 5 MHz ≤ Δf < min(10 MHz, $\Delta f_{max}$) | 5.05 MHz ≤ f_offset < min(10.05 MHz, $f_offset_{max}$) | -14 dBm | 100 kHz |
| 10 MHz ≤ Δf ≤ $\Delta f_{max}$ | 10.05 MHz ≤ f_offset < $f_offset_{max}$ | -13 dBm (Note 3) | 100 kHz |

NOTE 1: For a BS supporting *non-contiguous spectrum* operation within any *operating band*, the emission limits within *sub-block gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* on each side of the *sub-block gap*. Exception is Δf ≥ 10MHz from both adjacent *sub-blocks* on each side of the *sub-block gap*, where the emission limits within *sub-block gaps* shall be -13 dBm/100 kHz.
NOTE 2: For a *multi-band connector* with *Inter RF Bandwidth gap* < 2*$\Delta f_{OBUE}$ the emission limits within the *Inter RF Bandwidth gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* or RF Bandwidth on each side of the *Inter RF Bandwidth gap*.
NOTE 3: The requirement is not applicable when $\Delta f_{max}$ < 10 MHz.

For BS operating in Bands n1, n2, n3, n7, n24, n25, n30, n34, n38, n39, n40, n41, n48, n50, n65, n66, n70, n74, n75, n77, n78, n79, n90, n92, n94, *basic limits* are specified in table 6.6.4.2.1-2:

**Table 6.6.4.2.1-2: Wide Area BS *operating band* unwanted emission limits (NR bands above 1 GHz) for Category A**

| Frequency offset of measurement filter -3dB point, Δf | Frequency offset of measurement filter centre frequency, f_offset | *Basic limits* (Note 1, 2) | *Measurement bandwidth* |
|---|---|---|---|
| 0 MHz ≤ Δf < 5 MHz | 0.05 MHz ≤ f_offset < 5.05 MHz | $-7dBm - \frac{7}{5} \cdot \left( \frac{f_offset}{MHz} - 0.05 \right) dB$ | 100 kHz |
| 5 MHz ≤ Δf < min(10 MHz, $\Delta f_{max}$) | 5.05 MHz ≤ f_offset < min(10.05 MHz, $f_offset_{max}$) | -14 dBm | 100 kHz |
| 10 MHz ≤ Δf ≤ $\Delta f_{max}$ | 10.5 MHz ≤ f_offset < $f_offset_{max}$ | -13 dBm (Note 3) | 1MHz |

NOTE 1: For a BS supporting *non-contiguous spectrum* operation within any *operating band*, the emission limits within *sub-block gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* on each side of the *sub-block gap*, where the contribution from the far-end *sub-block* shall be scaled according to the *measurement bandwidth* of the near-end *sub-block*. Exception is Δf ≥ 10MHz from both adjacent *sub-blocks* on each side of the *sub-block gap*, where the emission limits within *sub-block gaps* shall be -13 dBm/1 MHz.
NOTE 2: For a *multi-band connector* with *Inter RF Bandwidth gap* < 2*$\Delta f_{OBUE}$ the emission limits within the *Inter RF Bandwidth gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* or RF Bandwidth on each side of the *Inter RF Bandwidth gap*, where the contribution from the far-end *sub-block* or RF Bandwidth shall be scaled according to the *measurement bandwidth* of the near-end *sub-block* or RF Bandwidth.
NOTE 3: The requirement is not applicable when $\Delta f_{max}$ < 10 MHz.

### 6.6.4.2.2 Basic limits for Wide Area BS (Category B)

For Category B Operating band unwanted emissions, there are two options for the *basic limits* that may be applied regionally. Either the *basic limits* in clause 6.6.4.2.2.1 or clause 6.6.4.2.2.2 shall be applied.

#### 6.6.4.2.2.1 Category B requirements (Option 1)

For BS operating in Bands n5, n8, n12, n20, n26, n28, n29, n67, n71, n85, the *basic limits* are specified in table 6.6.4.2.2.1-1:

**Table 6.6.4.2.2.1-1: Wide Area BS operating band unwanted emission limits (NR bands below 1 GHz) for Category B**

| Frequency offset of measurement filter -3dB point, Δf | Frequency offset of measurement filter centre frequency, f_offset | *Basic limits* (Note 1, 2) | *Measurement bandwidth* |
|---|---|---|---|
| 0 MHz ≤ Δf < 5 MHz | 0.05 MHz ≤ f_offset < 5.05 MHz | $-7dBm - \frac{7}{5} \cdot \left( \frac{f_offset}{MHz} - 0.05 \right) dB$ | 100 kHz |
| 5 MHz ≤ Δf < min(10 MHz, $\Delta f_{max}$) | 5.05 MHz ≤ f_offset < min(10.05 MHz, $f_offset_{max}$) | -14 dBm | 100 kHz |
| 10 MHz ≤ Δf ≤ $\Delta f_{max}$ | 10.05 MHz ≤ f_offset < $f_offset_{max}$ | -16 dBm (Note 3) | 100 kHz |

NOTE 1: For a BS supporting *non-contiguous spectrum* operation within any *operating band*, the emission limits within *sub-block gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* on each side of the *sub-block gap*. Exception is Δf ≥ 10MHz from both adjacent *sub-blocks* on each side of the *sub-block gap*, where the emission limits within *sub-block gaps* shall be -16 dBm/100 kHz.

NOTE 2: For a *multi-band connector* with *Inter RF Bandwidth gap* < 2*$\Delta f_{OBUE}$ the emission limits within the *Inter RF Bandwidth gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* or RF Bandwidth on each side of the *Inter RF Bandwidth gap*, where the contribution from the far-end *sub-block* or RF Bandwidth shall be scaled according to the *measurement bandwidth* of the near-end *sub-block* or RF Bandwidth.

NOTE 3: The requirement is not applicable when $\Delta f_{max}$ < 10 MHz.

For BS operating in Bands n1, n2, n3, n7, n25, n34, n38, n39, n40, n41, n48, n50, n65, n66, n70, n75, n77, n78, n79, n90, n92, n94, *basic limits* are specified in tables 6.6.4.2.2.1-2:

**Table 6.6.4.2.2.1-2: Wide Area BS operating band unwanted emission limits (NR bands above 1 GHz) for Category B**

| Frequency offset of measurement filter -3dB point, Δf | Frequency offset of measurement filter centre frequency, f_offset | *Basic limits* (Note 1, 2) | *Measurement bandwidth* |
|---|---|---|---|
| 0 MHz ≤ Δf < 5 MHz | 0.05 MHz ≤ f_offset < 5.05 MHz | $-7dBm - \frac{7}{5} \cdot \left( \frac{f_offset}{MHz} - 0.05 \right) dB$ | 100 kHz |
| 5 MHz ≤ Δf < min(10 MHz, $\Delta f_{max}$) | 5.05 MHz ≤ f_offset < min(10.05 MHz, $f_offset_{max}$) | -14 dBm | 100 kHz |
| 10 MHz ≤ Δf ≤ $\Delta f_{max}$ | 10.5 MHz ≤ f_offset < $f_offset_{max}$ | -15 dBm (Note 3) | 1MHz |

NOTE 1: For a BS supporting *non-contiguous spectrum* operation within any *operating band*, the emission limits within *sub-block gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* on each side of the *sub-block gap*, where the contribution from the far-end *sub-block* shall be scaled according to the *measurement bandwidth* of the near-end *sub-block*. Exception is Δf ≥ 10MHz from both adjacent *sub-blocks* on each side of the *sub-block gap*, where the emission limits within *sub-block gaps* shall be -15 dBm/1 MHz.

NOTE 2: For a *multi-band connector* with *Inter RF Bandwidth gap* < 2*$\Delta f_{OBUE}$ the emission limits within the *Inter RF Bandwidth gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* or RF Bandwidth on each side of the *Inter RF Bandwidth gap*, where the contribution from the far-end *sub-block* or RF Bandwidth shall be scaled according to the *measurement bandwidth* of the near-end *sub-block* or RF Bandwidth.

NOTE 3: The requirement is not applicable when $\Delta f_{max}$ < 10 MHz.

For *BS type 1-C* operating in Band n104, the limits are specified in tables 6.6.4.2.2.1-2a:

**Table 6.6.4.2.2.1-2a: Wide Area *BS type 1-C* operating band unwanted emission limits for band n104 for Category B**

| Frequency offset of measurement filter -3dB point, Δf | Frequency offset of measurement filter centre frequency, f_offset | Basic limits | Measurement bandwidth |
|---|---|---|---|
| 0 MHz ≤ Δf < 20 MHz | 0.05 MHz ≤ f_offset < 20.05 MHz | $-7\text{dBm} - \frac{7}{20}\left(\frac{f_{offset}}{MHz} - 0.05\right)$ | 100 kHz |
| 20 MHz ≤ Δf < min(40 MHz, $\Delta f_{max}$) | 20.05 MHz ≤ f_offset < min(40.05 MHz, $f_offset_{max}$) | -14 dBm | 100 kHz |
| 40 MHz ≤ Δf ≤ $\Delta f_{max}$ | 40.5 MHz ≤ f_offset < $f_offset_{max}$ | -15 dBm (Note 3) | 1MHz |
| NOTE 1: For a BS supporting *non-contiguous spectrum* operation within any *operating band* the emission limits within *sub-block gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* on each side of the *sub-block gap*. Exception is Δf ≥ 40MHz from both adjacent *sub-blocks* on each side of the *sub-block gap*, where the emission limits within *sub-block gaps* shall be -15 dBm/1 MHz.<br>NOTE 2: For a *multi-band connector* with *Inter RF Bandwidth gap* < 2*$\Delta f_{OBUE}$ the emission limits within the *Inter RF Bandwidth gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* or RF Bandwidth on each side of the *Inter RF Bandwidth gap*.<br>NOTE 3: The requirement is not applicable when $\Delta f_{max}$ <40 MHz. | | | |

For *BS type 1-H* operating in Band n104, *basic limits* are specified in tables 6.6.4.2.2.1-2b:

**Table 6.6.4.2.2.1-2b: Wide Area *BS type 1-H* operating band unwanted emission limits for band n104 for Category B**

| Frequency offset of measurement filter -3dB point, Δf | Frequency offset of measurement filter centre frequency, f_offset | Basic limits | Measurement bandwidth |
|---|---|---|---|
| 0 MHz ≤ Δf < 50 MHz | 0.05 MHz ≤ f_offset < 50.05 MHz | $-7\text{dBm} - \frac{7}{50}\left(\frac{f_{offset}}{MHz} - 0.05\right)$ | 100 kHz |
| 50 MHz ≤ Δf < min(100 MHz, $\Delta f_{max}$) | 50.05 MHz ≤ f_offset < min(100.05 MHz, $f_offset_{max}$) | -14 dBm | 100 kHz |
| 100 MHz ≤ Δf ≤ $\Delta f_{max}$ | 100.5 MHz ≤ f_offset < $f_offset_{max}$ | -15 dBm (Note 3) | 1MHz |
| NOTE 1: For a BS supporting *non-contiguous spectrum* operation within any *operating band* the emission limits within *sub-block gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* on each side of the *sub-block gap*. Exception is Δf ≥ 100MHz from both adjacent *sub-blocks* on each side of the *sub-block gap*, where the emission limits within *sub-block gaps* shall be -15 dBm/1 MHz.<br>NOTE 2: For a *multi-band connector* with *Inter RF Bandwidth gap* < 2*$\Delta f_{OBUE}$ the emission limits within the *Inter RF Bandwidth gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* or RF Bandwidth on each side of the *Inter RF Bandwidth gap*.<br>NOTE 3: The requirement is not applicable when $\Delta f_{max}$ < 100 MHz. | | | |

#### 6.6.4.2.2.2 Category B requirements (Option 2)

The limits in this clause are intended for Europe and may be applied regionally for BS operating in bands n1, n3, n7, n8, n38, n65, n100, n101.

For a BS operating in bands n1, n3, n8, n65 or *BS type 1-C* operating in bands n7, n38, n100 or n101, *basic limits* are specified in Table 6.6.4.2.2.2-1:

**Table 6.6.4.2.2.2-1: Regional Wide Area BS operating band unwanted emission limits for Category B**

| **Frequency offset of measurement filter -3dB point, Δf** | **Frequency offset of measurement filter centre frequency, f_offset** | ***Basic limits* (Note 1, 2)** | ***Measurement bandwidth*** |
|---|---|---|---|
| 0 MHz ≤ Δf < 0.2 MHz | 0.015 MHz ≤ f_offset < 0.215 MHz | -14 dBm | 30 kHz |
| 0.2 MHz ≤ Δf < 1 MHz | 0.215 MHz ≤ f_offset < 1.015 MHz | $-14dBm-15\cdot\left(\frac{f_offset}{MHz}-0.215\right)dB$ (Note 5) | 30 kHz |
| (Note 4) | 1.015 MHz ≤ f_offset < 1.5 MHz | -26 dBm (Note 5) | 30 kHz |
| 1 MHz ≤ Δf ≤ min( 10 MHz, $\Delta f_{max}$) | 1.5 MHz ≤ f_offset < min(10.5 MHz, $f_offset_{max}$) | -13 dBm (Note 5) | 1 MHz |
| 10 MHz ≤ Δf ≤ $\Delta f_{max}$ | 10.5 MHz ≤ f_offset < $f_offset_{max}$ | -15 dBm (Note 3) (Note 5) | 1 MHz |

NOTE 1: For a BS supporting *non-contiguous spectrum* operation within any *operating band*, the minimum requirement within *sub-block gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* on each side of the *sub-block gap*, where the contribution from the far-end *sub-block* shall be scaled according to the *measurement bandwidth* of the near-end *sub-block*. Exception is Δf ≥ 10MHz from both adjacent *sub-blocks* on each side of the *sub-block gap*, where the minimum requirement within *sub-block gaps* shall be -15dBm/1MHz. For BS supporting multi-band operation, either this limit or -16dBm/100kHz (f_offset adjusted according to the measurement bandwidth), whichever is less stringent, shall apply at Δf ≥ 10MHz for operating bands <1GHz.

NOTE 2: For a *multi-band connector* with *Inter RF Bandwidth gap* < 2*$\Delta f_{OBUE}$ the minimum requirement within the *Inter RF Bandwidth gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* or RF Bandwidth on each side of the *Inter RF Bandwidth gap*, where the contribution from the far-end *sub-block* or RF Bandwidth shall be scaled according to the *measurement bandwidth* of the near-end *sub-block* or RF Bandwidth.

NOTE 3: The requirement is not applicable when $\Delta f_{max}$ < 10 MHz.

NOTE 4: This frequency range ensures that the range of values of f_offset is continuous.

NOTE 5: For BS supporting multi-band operation, either this limit or -16dBm/100kHz (f_offset adjusted according to the measurement bandwidth), whichever is less stringent, shall apply for operating bands <1GHz.

### 6.6.4.2.3 *Basic limits* for Medium Range BS (Category A and B)

For Medium Range BS, *basic limits* are specified in table 6.6.4.2.3-1 and table 6.6.4.2.3-2 except for Band n104.

For the tables in this clause for *BS type 1-C* $P_{rated,x} = P_{rated,c,AC}$, and for *BS type 1-H* $P_{rated,x} = P_{rated,c,cell} - 10*\log_{10}(N_{TXU,countedpercell})$, and for *BS type 1-O* $P_{rated,x} = P_{rated,c,TRP} - 9$ dB.

**Table 6.6.4.2.3-1: Medium Range BS *operating band* unwanted emission limits, 31< $P_{rated,x}$ ≤ 38 dBm**

| Frequency offset of measurement filter -3dB point, Δf | Frequency offset of measurement filter centre frequency, f_offset | *Basic limits* (Note 1, 2) | *Measurement bandwidth* |
|---|---|---|---|
| 0 MHz ≤ Δf < 5 MHz | 0.05 MHz ≤ f_offset < 5.05 MHz | $P_{rated,x} - 53dB - \frac{7}{5}\left(\frac{f_offset}{MHz} - 0.05\right) dB$ | 100 kHz |
| 5 MHz ≤ Δf < min(10 MHz, $\Delta f_{max}$) | 5.05 MHz ≤ f_offset < min(10.05 MHz, $f_offset_{max}$) | $P_{rated,x}$ - 60dB | 100 kHz |
| 10 MHz ≤ Δf ≤ $\Delta f_{max}$ | 10.05 MHz ≤ f_offset < $f_offset_{max}$ | Min($P_{rated,x}$ - 60dB, -25dBm) (Note 3) | 100 kHz |

NOTE 1: For a BS supporting *non-contiguous spectrum* operation within any *operating band* the emission limits within *sub-block gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* on each side of the *sub-block gap*. Exception is Δf ≥ 10MHz from both adjacent *sub-blocks* on each side of the *sub-block gap*, where the emission limits within *sub-block gaps* shall be Min($P_{rated,x}$ -60dB, -25dBm)/100kHz.

NOTE 2: For a *multi-band connector* with *Inter RF Bandwidth gap* < 2*$\Delta f_{OBUE}$ the emission limits within the *Inter RF Bandwidth gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* or RF Bandwidth on each side of the *Inter RF Bandwidth gap*.

NOTE 3: The requirement is not applicable when $\Delta f_{max}$ < 10 MHz.

For *BS type 1-C* operating in Band n104, the limits are specified in Table 6.6.4.2.3-1a and Table 6.6.4.2.3-2a.

**Table 6.6.4.2.3-1a. Medium Range *BS type 1-C operating band* unwanted emission limits for band n104, 31< $P_{rated,x}$ ≤ 38 dBm**

| Frequency offset of measurement filter -3dB point, Δf | Frequency offset of measurement filter centre frequency, f_offset | *Basic limits* | *Measurement bandwidth* |
|---|---|---|---|
| 0 MHz ≤ Δf < 20 MHz | 0.05 MHz ≤ f_offset < 20.05 MHz | $\mathrm{Prated,x} - 53\mathrm{dB} - \frac{7}{20}\left(\frac{f_offset}{MHz} - 0.05\right)$ | 100 kHz |
| 20 MHz ≤ Δf < min(40 MHz, $\Delta f_{max}$) | 20.05 MHz ≤ f_offset < min(40.05 MHz, $f_offset_{max}$) | $P_{rated,x}$ - 60dB | 100 kHz |
| 40 MHz ≤ Δf ≤ $\Delta f_{max}$ | 40.05 MHz ≤ f_offset < $f_offset_{max}$ | Min($P_{rated,x}$ - 60dB, -25dBm) (Note 3) | 100 kHz |

NOTE 1: For a BS supporting *non-contiguous spectrum* operation within any *operating band* the emission limits within *sub-block gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* on each side of the *sub-block gap*. Exception is Δf ≥ 40MHz from both adjacent *sub-blocks* on each side of the *sub-block gap*, where the emission limits within *sub-block gaps* shall be Min($P_{rated,x}$ -60dB, -25dBm)/100kHz.

NOTE 2: For a *multi-band connector* with *Inter RF Bandwidth gap* < 2*$\Delta f_{OBUE}$ the emission limits within the *Inter RF Bandwidth gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* or RF Bandwidth on each side of the *Inter RF Bandwidth gap*.

NOTE 3: The requirement is not applicable when $\Delta f_{max}$ < 40 MHz.

For *BS type 1-H* operating in Band n104, *basic limits* are specified in Table 6.6.4.2.3-1b and Table 6.6.4.2.3-2b:

**Table 6.6.4.2.3-1b. Medium Range *BS type 1-H operating band* unwanted emission limits for band n104, 31< $P_{rated,x}$ ≤ 38 dBm**

| Frequency offset of measurement filter -3dB point, Δf | Frequency offset of measurement filter centre frequency, f_offset | *Basic limits* | *Measurement bandwidth* |
|---|---|---|---|
| 0 MHz ≤ Δf < 50 MHz | 0.05 MHz ≤ f_offset < 50.05 MHz | $\mathrm{Prated,x} - 53\mathrm{dB} - \frac{7}{50}\left(\frac{f_offset}{MHz} - 0.05\right)$ | 100 kHz |
| 50 MHz ≤ Δf < min(100 MHz, $\Delta f_{max}$) | 50.05 MHz ≤ f_offset < min(100.05 MHz, $f_offset_{max}$) | $P_{rated,x}$ - 60dB | 100 kHz |
| 100 MHz ≤ Δf ≤ $\Delta f_{max}$ | 100.05 MHz ≤ f_offset < $f_offset_{max}$ | Min($P_{rated,x}$ - 60dB, -25dBm) (Note 3) | 100 kHz |

NOTE 1: For a BS supporting *non-contiguous spectrum* operation within any *operating band* the emission limits within *sub-block gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* on each side of the *sub-block gap*. Exception is Δf ≥ 100MHz from both adjacent *sub-blocks* on each side of the *sub-block gap*, where the emission limits within *sub-block gaps* shall be Min($P_{rated,x}$ -60dB, -25dBm)/100kHz.
NOTE 2: For a *multi-band connector* with *Inter RF Bandwidth gap* < 2*$\Delta f_{OBUE}$ the emission limits within the *Inter RF Bandwidth gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* or RF Bandwidth on each side of the *Inter RF Bandwidth gap*.
NOTE 3: The requirement is not applicable when $\Delta f_{max}$ < 100 MHz.

**Table 6.6.4.2.3-2: Medium Range BS operating band unwanted emission limits, $P_{rated,x}$ ≤ 31 dBm**

| Frequency offset of measurement filter -3dB point, Δf | Frequency offset of measurement filter centre frequency, f_offset | *Basic limits* (Note 1, 2) | *Measurement bandwidth* |
|---|---|---|---|
| 0 MHz ≤ Δf < 5 MHz | 0.05 MHz ≤ f_offset < 5.05 MHz | $-22\mathrm{dBm} - \frac{7}{5}\left(\frac{f_offset}{MHz} - 0.05\right) dB$ | 100 kHz |
| 5 MHz ≤ Δf < min(10 MHz, $\Delta f_{max}$) | 5.05 MHz ≤ f_offset < min(10.05 MHz, $f_offset_{max}$) | -29 dBm | 100 kHz |
| 10 MHz ≤ Δf ≤ $\Delta f_{max}$ | 10.05 MHz ≤ f_offset < $f_offset_{max}$ | -29 dBm (Note 3) | 100 kHz |

NOTE 1: For a BS supporting *non-contiguous spectrum* operation within any *operating band* the emission limits within *sub-block gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* on each side of the *sub-block gap*. Exception is Δf ≥ 10MHz from both adjacent *sub-blocks* on each side of the *sub-block gap*, where the emission limits within *sub-block gaps* shall be -29dBm/100kHz.
NOTE 2: For a *multi-band connector* with *Inter RF Bandwidth gap* < 2*$\Delta f_{OBUE}$ the emission limits within the *Inter RF Bandwidth gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* or RF Bandwidth on each side of the *Inter RF Bandwidth gap*.
NOTE 3: The requirement is not applicable when $\Delta f_{max}$ < 10 MHz.

**Table 6.6.4.2.3-2a. Medium Range *BS type 1-C* operating band unwanted emission limits for band 104, $P_{rated,x}$ ≤ 31 dBm**

| Frequency offset of measurement filter -3dB point, Δf | Frequency offset of measurement filter centre frequency, f_offset | *Basic limits* | *Measurement bandwidth* |
|---|---|---|---|
| 0 MHz ≤ Δf < 20 MHz | 0.05 MHz ≤ f_offset < 20.05 MHz | $-22\mathrm{dBm} - \frac{7}{20}\left(\frac{f_offset}{MHz} - 0.05\right)$ | 100 kHz |
| 20 MHz ≤ Δf < min(40 MHz, $\Delta f_{max}$) | 20.05 MHz ≤ f_offset < min(40.05 MHz, $f_offset_{max}$) | -29 dBm | 100 kHz |
| 40 MHz ≤ Δf ≤ $\Delta f_{max}$ | 40.05 MHz ≤ f_offset < $f_offset_{max}$ | -29 dBm | 100 kHz |

NOTE 1: For a BS supporting *non-contiguous spectrum* operation within any *operating band* the emission limits within *sub-block gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* on each side of the *sub-block gap*. Exception is Δf ≥ 40MHz from both adjacent *sub-blocks* on each side of the *sub-block gap*, where the emission limits within *sub-block gaps* shall be -29 dBm/100kHz.
NOTE 2: For a *multi-band connector* with *Inter RF Bandwidth gap* < 2*$\Delta f_{OBUE}$ the emission limits within the *Inter RF Bandwidth gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* or RF Bandwidth on each side of the *Inter RF Bandwidth gap*.
NOTE 3: The requirement is not applicable when $\Delta f_{max}$ < 40 MHz.

**Table 6.6.4.2.3-2b. Medium Range *BS type 1-H* operating band unwanted emission limits for band 104, $P_{rated,x} \leq 31$ dBm**

| Frequency offset of measurement filter -3dB point, Δf | Frequency offset of measurement filter centre frequency, f_offset | *Basic limits* | *Measurement bandwidth* |
|---|---|---|---|
| 0 MHz ≤ Δf < 50 MHz | 0.05 MHz ≤ f_offset < 50.05 MHz | $-22\text{dBm} - \frac{7}{50}\left(\frac{f_offset}{MHz} - 0.05\right)$ | 100 kHz |
| 50 MHz ≤ Δf < min(100 MHz, $\Delta f_{max}$) | 50.05 MHz ≤ f_offset < min(100.05 MHz, $f_offset_{max}$) | -29 dBm | 100 kHz |
| 100 MHz ≤ Δf ≤ $\Delta f_{max}$ | 100.05 MHz ≤ f_offset < $f_offset_{max}$ | -29 dBm | 100 kHz |
| NOTE 1: For a BS supporting *non-contiguous spectrum* operation within any *operating band* the emission limits within *sub-block gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* on each side of the *sub-block gap*. Exception is Δf ≥ 100MHz from both adjacent *sub-blocks* on each side of the *sub-block gap*, where the emission limits within *sub-block gaps* shall be -29 dBm/100kHz.<br>NOTE 2: For a *multi-band connector* with *Inter RF Bandwidth gap* < 2*$\Delta f_{OBUE}$ the emission limits within the *Inter RF Bandwidth gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* or RF Bandwidth on each side of the *Inter RF Bandwidth gap*.<br>NOTE 3: The requirement is not applicable when $\Delta f_{max}$ < 100 MHz. | | | |

### 6.6.4.2.4 *Basic limits* for Local Area BS (Category A and B)

For Local Area BS, *basic limits* are specified in table 6.6.4.2.4-1 except for n46, n96, n102 and n104.

**Table 6.6.4.2.4-1: Local Area BS operating band unwanted emission limits**

| Frequency offset of measurement filter -3dB point, Δf | Frequency offset of measurement filter centre frequency, f_offset | *Basic limits* (Note 1, 2) | *Measurement bandwidth* |
|---|---|---|---|
| 0 MHz ≤ Δf < 5 MHz | 0.05 MHz ≤ f_offset < 5.05 MHz | $-30dBm - \frac{7}{5}\left(\frac{f_offset}{MHz} - 0.05\right)dB$ | 100 kHz |
| 5 MHz ≤ Δf < min(10 MHz, $\Delta f_{max}$) | 5.05 MHz ≤ f_offset < min(10.05 MHz, $f_offset_{max}$) | -37 dBm | 100 kHz |
| 10 MHz ≤ Δf ≤ $\Delta f_{max}$ | 10.05 MHz ≤ f_offset < $f_offset_{max}$ | -37 dBm (Note 10) | 100 kHz |
| NOTE 1: For a BS supporting *non-contiguous spectrum* operation within any *operating band* the emission limits within *sub-block gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* on each side of the *sub-block gap*. Exception is Δf ≥ 10MHz from both adjacent *sub-blocks* on each side of the *sub-block gap*, where the emission limits within *sub-block gaps* shall be -37dBm/100kHz.<br>NOTE 2: For a *multi-band connector* with *Inter RF Bandwidth gap* < 2*$\Delta f_{OBUE}$ the emission limits within the *Inter RF Bandwidth gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* or RF Bandwidth on each side of the *Inter RF Bandwidth gap*<br>NOTE 3: The requirement is not applicable when $\Delta f_{max}$ < 10 MHz. | | | |

For *BS type 1-C* operating in Band n104, the limits are specified in Table 6.6.4.2.4-1a.

**Table 6.6.4.2.4-1a. Local Area *BS type 1-C* operating band unwanted emission limits for band n104**

| Frequency offset of measurement filter -3dB point, Δf | Frequency offset of measurement filter centre frequency, f_offset | *Basic limits* (Note 1, 2) | *Measurement bandwidth* |
|---|---|---|---|
| 0 MHz ≤ Δf < 20 MHz | 0.05 MHz ≤ f_offset < 20.05 MHz | $-30\text{dBm} - \frac{7}{20}\left(\frac{f_offset}{MHz} - 0.05\right)$ | 100 kHz |
| 20 MHz ≤ Δf < min(40 MHz, $\Delta f_{max}$) | 20.05 MHz ≤ f_offset < min(40.05 MHz, f_offset$_{max}$) | -37 dBm | 100 kHz |
| 40 MHz ≤ Δf ≤ $\Delta f_{max}$ | 40.05 MHz ≤ f_offset < f_offset$_{max}$ | -37 dBm | 100 kHz |

NOTE 1: For a BS supporting *non-contiguous spectrum* operation within any *operating band* the emission limits within *sub-block gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* on each side of the *sub-block gap*. Exception is Δf ≥ 40MHz from both adjacent *sub-blocks* on each side of the *sub-block gap*, where the emission limits within *sub-block gaps* shall be -37dBm/100kHz.

NOTE 2: For a *multi-band connector* with *Inter RF Bandwidth gap* < 2*$\Delta f_{OBUE}$ the emission limits within the *Inter RF Bandwidth gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* or RF Bandwidth on each side of the *Inter RF Bandwidth gap*

NOTE 3: The requirement is not applicable when $\Delta f_{max}$ < 40 MHz.

For *BS type 1-H* operating in Band n104, *basic limits* are specified in Table 6.6.4.2.4-1b:

**Table 6.6.4.2.4-1b. Local Area *BS type 1-H* operating band unwanted emission limits for band n104**

| Frequency offset of measurement filter -3dB point, Δf | Frequency offset of measurement filter centre frequency, f_offset | *Basic limits* (Note 1, 2) | *Measurement bandwidth* |
|---|---|---|---|
| 0 MHz ≤ Δf < 50 MHz | 0.05 MHz ≤ f_offset < 50.05 MHz | $-30\text{dBm} - \frac{7}{50}\left(\frac{f_offset}{MHz} - 0.05\right)$ | 100 kHz |
| 50 MHz ≤ Δf < min(100 MHz, $\Delta f_{max}$) | 50.05 MHz ≤ f_offset < min(100.05 MHz, f_offset$_{max}$) | -37 dBm | 100 kHz |
| 100 MHz ≤ Δf ≤ $\Delta f_{max}$ | 100.05 MHz ≤ f_offset < f_offset$_{max}$ | -37 dBm | 100 kHz |

NOTE 1: For a BS supporting *non-contiguous spectrum* operation within any *operating band* the emission limits within *sub-block gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* on each side of the *sub-block gap*. Exception is Δf ≥ 100MHz from both adjacent *sub-blocks* on each side of the *sub-block gap*, where the emission limits within *sub-block gaps* shall be -37dBm/100kHz.

NOTE 2: For a *multi-band connector* with *Inter RF Bandwidth gap* < 2*$\Delta f_{OBUE}$ the emission limits within the *Inter RF Bandwidth gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* or RF Bandwidth on each side of the *Inter RF Bandwidth gap*

NOTE 3: The requirement is not applicable when $\Delta f_{max}$ < 100 MHz.

### 6.6.4.2.4A Basic limits for Local Area and Medium Range BS for bands n46, n96 and n102 (Category A and B)

For Local Area and Medium Range BS operating in band n46, basic limits for 10 MHz channel bandwidth are specified in table 6.6.4.2.4A-1. For Local Area and Medium Range BS operating in bands n46, n96 and n102, basic limits for 20 MHz, 40 MHz, 60 MHz, 80 MHz and 100 MHz channel bandwidth are specified in table 6.6.4.2.4A-2. The nominal bandwidth N = $BW_{Channel}$ of the transmitted carrier. For one non-transmitted channel basic limits are specified in table 6.6.4.2.4A-3, and for two non-transmitted channels basic limits are specified in table 6.6.4.2.4A-4, and for three non-transmitted channels basic limits are specified in table 6.6.4.2.4A-5.

**Table 6.6.4.2.4A-1: Medium Range BS and Local Area BS operating band unwanted emission limits for 10 MHz channel bandwidth for band n46**

| Frequency offset of measurement filter -3dB point, Δf | Frequency offset of measurement filter centre frequency, f_offset | Basic limits (Note 1) | Measurement bandwidth |
|---|---|---|---|
| 0 MHz ≤ Δf < 0.5 MHz | 0.05 MHz ≤ f_offset < 0.55 MHz | $\mathrm{Max}(P_{\mathrm{rated,x}}\text{-}19.5\mathrm{dB} - 20\left(\frac{f_offset}{MHz} - 0.05\right)dB$, -40dBm) | 100 kHz |
| 0.5 MHz ≤ Δf < 5 MHz | 0.55 MHz ≤ f_offset < min(5.05 MHz, f_offset$_{max}$) | $\mathrm{Max}(P_{\mathrm{rated,x}}\text{-}29.5\mathrm{dB} - \frac{16}{9}\left(\frac{f_offset}{MHz} - 0.55\right)dB$, -40dBm) | 100 kHz |
| 5 MHz ≤ Δf < min(10 MHz, Δf$_{max}$) | 5.05 MHz ≤ f_offset < min(10.05 MHz, f_offset$_{max}$) | $\mathrm{Max}(P_{\mathrm{rated,x}}\text{-}37.5\mathrm{dB} - \frac{12}{5}\left(\frac{f_offset}{MHz} - 5.05\right)dB$, -40dBm) | 100 kHz |
| 10 MHz ≤ Δf < min(85 MHz, Δf$_{max}$) | 10.05 MHz ≤ f_offset < min(85.05 MHz, f_offset$_{max}$) | Max(P$_{rated,x}$ – 59.5dB, -40dBm) | 100 kHz |
| 85 MHz ≤ Δf < min(103 MHz, Δf$_{max}$) | 85.05 MHz ≤ f_offset < min(103.05 MHz, f_offset$_{max}$) | Max(P$_{rated,x}$ – 61.5dB, -40dBm) | 100 kHz |
| 103 MHz ≤ Δf ≤ Δf$_{max}$ | 103.05 MHz ≤ f_offset < f_offset$_{max}$ | Max(P$_{rated,x}$ – 66.5dB, -40dBm) | 100 kHz |
| NOTE 1: For a BS supporting non-contiguous spectrum operation within any operating band, the minimum requirement within *sub-block gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* on each side of the *sub-block gap*. Exception is Δf ≥ 10 MHz from both adjacent *sub-blocks* on each side of the *sub-block gap*, where the minimum requirement within *sub-block gaps* shall be Max (P$_{rated,x}$ – 59.5dB, -40 dBm)/100kHz. | | | |

**Table 6.6.4.2.4A-2: Medium Range BS and Local Area BS operating band unwanted emission limits for 20 MHz, 40 MHz, 60 MHz, 80 MHz and 100 MHz channel bandwidth for bands n46, n96 and n102**

| Frequency offset of measurement filter -3dB point, Δf | Frequency offset of measurement filter centre frequency, f_offset | Basic limits (Note 1) | Measurement bandwidth |
|---|---|---|---|
| 0 MHz ≤ Δf < 1 MHz | 0.05 MHz ≤ f_offset < 1.05 MHz | $\mathrm{Max}(P_{\mathrm{rated,x}}\text{-}10\log10\left(\frac{\mathrm{BW_{channel}}}{100kHz}\right) - 20\left(\frac{f_offset}{MHz} - 0.05\right)dB$, -40dBm) | 100 kHz |
| 1 MHz ≤ Δf < min(0.5N MHz, Δf$_{max}$) | 1.05 MHz ≤ f_offset < min((0.5N+0.05) MHz, f_offset$_{max}$) | $\mathrm{Max}(P_{\mathrm{rated,x}}\text{-}10\log10\left(\frac{\mathrm{BW_{Channel}}}{100kHz}\right) - 20 - \frac{8}{0.5N-1}\left(\frac{f_offset}{MHz} - 1.05\right)dB$, -40dBm) | 100 kHz |
| 0.5N MHz ≤ Δf < min(N MHz, Δf$_{max}$) | (0.5N+0.05) MHz ≤ f_offset < min((N+0.05) MHz, f_offset$_{max}$) | Max($P_{\mathrm{rated,x}}\text{-}10\log10\left(\frac{\mathrm{BW_{Channel}}}{100kHz}\right) - 28 - \frac{12}{0.5N}\left(\frac{f_offset}{MHz} - 0.5N - 0.05\right)dB$, -40dBm) | 100 kHz |
| N MHz ≤ Δf < min(8.5N MHz, Δf$_{max}$) | (N+0.05) MHz ≤ f_offset < min((8.5N+0.05) MHz, f_offset$_{max}$) | $\mathrm{Max}\left(P_{\mathrm{rated,x}}\text{-}10\log10\left(\frac{\mathrm{BW_{Channel}}}{100kHz}\right) - 40dB, -40dBm\right)$ | 100 kHz |
| 8.5N MHz ≤ Δf < min(10.3N MHz, Δf$_{max}$) | (8.5N+0.05) MHz ≤ f_offset < min((10.3N+0.05) MHz, f_offset$_{max}$) | $\mathrm{Max}\left(P_{\mathrm{rated,x}}\text{-}10\log10\left(\frac{\mathrm{BW_{Channel}}}{100kHz}\right) - 42dB, -40dBm\right)$ | 100 kHz |
| 10.3N MHz ≤ Δf ≤ Δf$_{max}$ | (10.3N+0.05) MHz ≤ f_offset < f_offset$_{max}$ | $\mathrm{Max}\left(P_{\mathrm{rated,x}}\text{-}10\log10\left(\frac{\mathrm{BW_{Channel}}}{100kHz}\right) - 47dB, -40dBm\right)$ | 100 kHz |
| NOTE 1: For a BS supporting non-contiguous spectrum operation within any operating band, the minimum requirement within *sub-block gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* on each side of the *sub-block gap*. Exception is Δf ≥ N MHz from both adjacent *sub-blocks* on each side of the *sub-block gap*, where the minimum requirement within *sub-block gaps* shall be $\mathrm{Max}\left(P_{\mathrm{rated,x}}\text{-}10\log10\left(\frac{\mathrm{BW_{Channel}}}{100kHz}\right) - 40dB, -40dBm\right)$ | | | |

In the case of one, two or three non-transmitted 20 MHz channels between transmitted channels, when a NR-U channel bandwidth of 60 MHz, 80 MHz or 100 MHz have been assigned, the spectrum emission mask for non-transmitted

channels specified in Table 6.6.4.2.4A-3,Table 6.6.4.2.4A-4 and Table 6.6.4.2.4A-5 applies for one, two, and three non-transmitted channels respectively. The spectrum emission mask for non-transmitted channels apply to frequencies ($\Delta f_{BE_offset}$) starting from the edge of the last transmitted channel of the channels assigned for NR-U channel bandwidth. The relative power of any BS emission shall not exceed the most stringent levels given by Table 6.6.4.2.4A-3, Table 6.6.4.2.4A-4 and Table 6.6.4.2.4A-5 in the case of non-transmitted channels between transmitted channels.

**Table 6.6.4.2.4A-3: Medium Range BS and Local Area BS operating band unwanted emission limits for one non-transmitted channel for 60 MHz, 80MHz and 100 MHz channel bandwidth for bands n46, n96 and n102**

| Frequency offset of measurement filter -3dB point, $\Delta f_{BE_offset}$ | Frequency offset of measurement filter centre frequency, f_BE_offset | *Basic limits* | Measurement bandwidth |
|---|---|---|---|
| 0 MHz ≤ $\Delta f_{BE_offset}$ < 1 MHz | 0.05 MHz ≤ f_BE_offset < 1.05 MHz | $P_{\text{rated,x}}\text{-}10\log10\left(\frac{\text{BW}_{\text{Channel}}}{100kHz}\right) - 20\left(\frac{f_BE_offset}{MHz} - 0.05\right)dB$ | 100 kHz |
| 1 MHz ≤ $\Delta f_{BE_offset}$ < 10 MHz | 1.05 MHz ≤ f_BE_offset < 10.05 MHz | $P_{\text{rated,x}}\text{-}10\log10\left(\frac{\text{BW}_{\text{Channel}}}{100kHz}\right) - 20 - \frac{1}{3}\left(\frac{f_BE_offset}{MHz} - 1.05\right)dB$ | 100 kHz |
| 10 MHz ≤ $\Delta f_{BE_offset}$ < 19 MHz | 10.05 MHz ≤ f_BE offset < 19.05 MHz | $P_{\text{rated,x}}\text{-}10\log10\left(\frac{\text{BW}_{\text{Channel}}}{100kHz}\right) - 23 + \frac{1}{3}\left(\frac{f_BE_offset}{MHz} - 10.05\right)dB$ | 100 kHz |
| 19 MHz ≤ $\Delta f_{BE_offset}$ < 19.9 MHz | 19.05 MHz ≤ f_BE_offset < 19.95 MHz | $P_{\text{rated,x}}\text{-}10\log10\left(\frac{\text{BW}_{\text{Channel}}}{100kHz}\right) - 20 + 20\left(\frac{f_BE_offset}{MHz} - 19.05\right)dB$ | 100 kHz |

**Table 6.6.4.2.4A-4: Medium Range BS and Local Area BS operating band unwanted emission limits for two non-transmitted channels for 80 MHz and 100 MHz channel bandwidth for bands n46, n96 and n102**

| Frequency offset of measurement filter -3dB point, $\Delta f_{BE_offset}$ | Frequency offset of measurement filter centre frequency, f_BE_offset | *Basic limits* | Measurement bandwidth |
|---|---|---|---|
| 0 MHz ≤ $\Delta f_{BE_offset}$ < 1 MHz | 0.05 MHz ≤ f_BE_offset < 1.05 MHz | $P_{\text{rated,x}}\text{-}10\log10\left(\frac{\text{BW}_{\text{Channel}}}{100kHz}\right) - 20\left(\frac{f_BE_offset}{MHz} - 0.05\right)dB$ | 100 kHz |
| 1 MHz ≤ $\Delta f_{BE_offset}$ <10 MHz | 1.05 MHz ≤ f_BE_offset < 10.05 MHz | $P_{\text{rated,x}}\text{-}10\log10\left(\frac{\text{BW}_{\text{Channel}}}{100kHz}\right) - 20 - \frac{5}{9}\left(\frac{f_BE_offset}{MHz} - 1.05\right)dB$ | 100 kHz |
| 10 MHz ≤ $\Delta f_{BE_offset}$ <30 MHz | 10.05 MHz ≤ f_BE offset < 30.05 MHz | $P_{\text{rated,x}}\text{-}10\log10\left(\frac{\text{BW}_{\text{Channel}}}{100kHz}\right) - 25dB$ | 100 kHz |
| 30 MHz ≤ $\Delta f_{BE_offset}$ < 39 MHz | 30.05 MHz ≤ f_BE_offset < 39.05 MHz | $P_{\text{rated,x}}\text{-}10\log10\left(\frac{\text{BW}_{\text{Channel}}}{100kHz}\right) - 25 + \frac{5}{9}\left(\frac{f_BE_offset}{MHz} - 30.05\right)dB$ | 100 kHz |
| 39 MHz ≤ $\Delta f_{BE_offset}$ < 39.9 MHz | 39.05 MHz ≤ f_BE_offset < 39.95 MHz | $P_{\text{rated,x}}\text{-}10\log10\left(\frac{\text{BW}_{\text{Channel}}}{100kHz}\right) - 20 + 20\left(\frac{f_BE_offset}{MHz} - 39.05\right)dB$ | 100 kHz |

**Table 6.6.4.2.4A-5: Medium Range BS and Local Area BS operating band unwanted emission limits for three non-transmitted channels for 100 MHz channel bandwidth for bands n46, n96 and n102**

| Frequency offset of measurement filter -3dB point, $\Delta f_{BE_offset}$ | Frequency offset of measurement filter centre frequency, f_BE_offset | *Basic limits* | Measurement bandwidth |
|---|---|---|---|
| 0 MHz ≤ $\Delta f_{BE_offset}$ < 1 MHz | 0.05 MHz ≤ f_BE_offset < 1.05 MHz | $P_{\text{rated,x}}$-10log10$\left(\frac{\text{BW}_{\text{Channel}}}{100kHz}\right) - 20\left(\frac{f_BE_offset}{MHz} - 0.05\right)dB$ | 100 kHz |
| 1 MHz ≤ $\Delta f_{BE_offset}$ <10 MHz | 1.05 MHz ≤ f_BE_offset < 10.05 MHz | $P_{\text{rated,x}}$-10log10$\left(\frac{\text{BW}_{\text{Channel}}}{100kHz}\right) - 20 - \frac{5}{9}\left(\frac{f_BE_offset}{MHz} - 1.05\right)dB$ | 100 kHz |
| 10 MHz ≤ $\Delta f_{BE_offset}$ <50 MHz | 10.05 MHz ≤ f_BE offset < 50.05 MHz | $P_{\text{rated,x}}$-10log10$\left(\frac{\text{BW}_{\text{Channel}}}{100kHz}\right) - 25dB$ | 100 kHz |
| 50 MHz ≤ $\Delta f_{BE_offset}$ < 59 MHz | 50.05 MHz ≤ f_BE_offset < 59.05 MHz | $P_{\text{rated,x}}$-10log10$\left(\frac{\text{BW}_{\text{Channel}}}{100kHz}\right) - 25 + \frac{5}{9}\left(\frac{f_BE_offset}{MHz} - 50.05\right)dB$ | 100 kHz |
| 59 MHz ≤ $\Delta f_{BE_offset}$ < 59.9 MHz | 59.05 MHz ≤ f_BE_offset < 59.95 MHz | $P_{\text{rated,x}}$-10log10$\left(\frac{\text{BW}_{\text{Channel}}}{100kHz}\right) - 20 + 20\left(\frac{f_BE_offset}{MHz} - 59.05\right)dB$ | 100 kHz |

In the case of non-transmitted 20 MHz channel(s) on the edges of an assigned NR-U channel bandwidth the general spectrum emission mask specified in Table 6.6.4.2.4A-2 is applied to the remaining transmitted channels to form an additional spectrum emission mask. The additional spectrum emission mask is applied to the total bandwidth of the remaining transmitted channels.

The additional spectrum emission mask is floored at: $P_{\text{rated,x}}$-10log10$\left(\frac{\text{BW}_{\text{Channel}}}{100kHz}\right) - 28dB$.

The relative power of any BS emission shall not exceed the most stringent levels given by the initial general spectrum emission mask with full channel bandwidth and the additional spectrum emission mask with the channel bandwidth of the transmitted channels in the case of non-transmitted channels at the edge of an assigned NR-U channel bandwidth.

### 6.6.4.2.5 *Basic limits* for additional requirements

#### 6.6.4.2.5.1 Limits in FCC Title 47

In addition to the requirements in clauses 6.6.4.2.1, 6.6.4.2.2, 6.6.4.2.3 and 6.6.4.2.4, the BS may have to comply with the applicable emission limits established by FCC Title 47 [8], when deployed in regions where those limits are applied, and under the conditions declared by the manufacturer.

#### 6.6.4.2.5.2 Protection of DTT

In certain regions the following requirement may apply for protection of DTT. For *BS type 1-C* or *BS type 1-H* operating in Band n20, the level of emissions in the band 470-790 MHz, measured in an 8 MHz filter bandwidth on centre frequencies $F_{filter}$ according to table 6.6.4.2.5.2-1, a *basic limits* $P_{EM,N}$ is declared by the manufacturer. This requirement applies in the frequency range 470-790 MHz even though part of the range falls in the spurious domain.

**Table 6.6.4.2.5.2-1: Declared emissions *basic limit* for protection of DTT**

| Filter centre frequency, $F_{filter}$ | *Measurement bandwidth* | Declared emission *basic limit* (dBm) |
|---|---|---|
| $F_{filter}$ = 8*N + 306 (MHz); 21 ≤ N ≤ 60 | 8 MHz | $P_{EM,N}$ |

Note: The regional requirement is defined in terms of EIRP (effective isotropic radiated power), which is dependent on both the BS emissions at the *antenna connector* and the deployment (including antenna gain and feeder loss). The requirement defined above provides the characteristics of the BS needed to verify compliance with the regional requirement. Compliance with the regional requirement can be determined using the method outlined in TS 36.104 [13], annex F.

#### 6.6.4.2.5.3 Additional operating band unwanted emissions limits for Band n48

The following requirement may apply to BS operating in Band n48 in certain regions. Emissions shall not exceed the maximum levels specified in table 6.6.4.2.5.3-1.

**Table 6.6.4.2.5.3-1: Additional operating band unwanted emission limits for Band n48**

| Channel bandwidth | Frequency offset of measurement filter -3dB point, Δf | Frequency offset of measurement filter centre frequency, f_offset | Minimum requirement | *Measurement bandwidth* (Note) |
|---|---|---|---|---|
| All | 0 MHz ≤ Δf < 10 MHz | 0.5 MHz ≤ f_offset < 9.5 MHz | -13 dBm | 1 MHz |

NOTE: The resolution bandwidth of the measuring equipment should be equal to the *measurement bandwidth*. However, to improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth may be smaller than the *measurement bandwidth*. When the resolution bandwidth is smaller than the *measurement bandwidth*, the result should be integrated over the *measurement bandwidth* in order to obtain the equivalent noise bandwidth of the *measurement bandwidth*.

#### 6.6.4.2.5.4 Additional operating band unwanted emissions limits for Band n53

The following requirement may apply to BS operating in Band n53 in certain regions. Emissions shall not exceed the maximum levels specified in table 6.6.4.2.5.4-1.

**Table 6.6.4.2.5.4-1: Additional operating band unwanted emission limits for Band n53**

| Channel bandwidth [MHz] | Frequency range [MHz] | Frequency offset of measurement filter -3dB point, Δf | Frequency offset of measurement filter centre frequency, f_offset | Minimum requirement | Measurement bandwidth (Note) |
|---|---|---|---|---|---|
| 5 | 2400 - 2477.5 | 6 MHz ≤ Δf < 83.5 MHz | 6.5 MHz ≤ f_offset < 83 MHz | -25 dBm | 1 MHz |
| 10 | 2400 - 2473.5 | 10 MHz ≤ Δf < 83.5 MHz | 10.5 MHz ≤ f_offset < 83 MHz | -25 dBm | 1 MHz |
| 5 | 2477.5 - 2478.5 | 5 MHz ≤ Δf < 6 MHz | 5.5 MHz | -13 dBm | 1 MHz |
| 10 | 2473.5 - 2478.5 | 5 MHz ≤ Δf < 10 MHz | 5.5 MHz ≤ f_offset < 9.5 MHz | -13 dBm | 1 MHz |
| All | 2478.5 - 2483.5 | 0 MHz ≤ Δf < 5 MHz | 0.5 MHz ≤ f_offset < 4.5 MHz | -10 dBm | 1 MHz |
| 5 | 2495 - 2501 | 0 MHz ≤ Δf < 6 MHz | 0.5 MHz ≤ f_offset < 5.5 MHz | -13 dBm | 1 MHz |
| 10 | 2495 - 2505 | 0 MHz ≤ Δf < 10 MHz | 0.5 MHz ≤ f_offset < 9.5 MHz | -13 dBm | 1 MHz |
| 5 | 2501 - 2690 | 6 MHz ≤ Δf < 195 MHz | 6.5 MHz ≤ f_offset < 194.5 MHz | -25 dBm | 1 MHz |
| 10 | 2505 - 2690 | 10 MHz ≤ Δf < 195 MHz | 10.5 MHz ≤ f_offset < 194.5 MHz | -25 dBm | 1 MHz |

NOTE: The resolution bandwidth of the measuring equipment should be equal to the *measurement bandwidth*. However, to improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth may be smaller than the *measurement bandwidth*. When the resolution bandwidth is smaller than the *measurement bandwidth*, the result should be integrated over the *measurement bandwidth* in order to obtain the equivalent noise bandwidth of the *measurement bandwidth*.

#### 6.6.4.2.5.5 Additional operating band unwanted emissions limits for operation with shared spectrum channel access

In addition, for operation with shared spectrum channel access, the BS may have to comply with the applicable operating band unwanted emission limits established regionally, when deployed in regions where those limits apply and under the conditions declared by the manufacturer. The regional requirements may be in the form of conducted power, power spectral density, EIRP and other types of limits. In case of regulatory limits based on EIRP, assessment of the EIRP level is described in Annex F.2.

#### 6.6.4.2.5.6 Additional operating band unwanted emissions limits for Band n24

In regions where FCC regulation applies, requirements for protection of GPS according to FCC Order DA 20-48 [25] applies for operation in Band n24. The following normative requirement covers the base station, to be used together with other information about the site installation to verify compliance with the requirement in FCC Order DA 20-48 [25]. The requirement applies to BS operating in Band n24 to ensure that appropriate interference protection is provided to the 1541 – 1650 MHz band. This requirement applies to the frequency range 1541-1650 MHz, even though part of this range falls within the spurious domain.

The level of emissions in the 1541 – 1650 MHz band, measured in measurement bandwidth according to Table 6.6.4.2.5.6-1 shall not exceed the maximum emission levels $P_{EM,B24,a}$, $P_{EM,B24,b}$, $P_{EM,B24,c}$, $P_{EM,B24,d}$, $P_{EM,B24,e}$ and $P_{EM,B24,f}$ declared by the manufacturer.

**Table 6.6.4.2.5.6-1: Declared emissions levels for protection of the 1541-1650 MHz band**

| Operating Band | Frequency range | Declared emission level (dBW) (Measurement bandwidth = 1 MHz) | Declared emission level (dBW) of discrete emissions of less than 700 Hz bandwidth (Measurement bandwidth = 1 kHz) | Declared emission level (dBW) of discrete emissions of less than 2 kHz bandwidth (Measurement bandwidth = 1 kHz) |
|---|---|---|---|---|
| n24 | 1541 - 1559 MHz | $P_{EM,B24,a}$ | | $P_{EM,B24,f}$ |
| | 1559 - 1610 MHz | $P_{EM,B24,b}$ | $P_{EM,B24,d}$ | |
| | 1610 - 1650 MHz | $P_{EM,B24,c}$ | $P_{EM,B24,e}$ | |

Note: The regional requirements in FCC Order DA 20-48 [25] are defined in terms of EIRP (effective isotropic radiated power), which is dependent on both the BS emissions at the antenna connector and the deployment (including antenna gain and feeder loss). The EIRP level is calculated using: $P_{EIRP} = P_E + G_{ant}$ where $P_E$ denotes the BS unwanted emission level at the antenna connector, $G_{ant}$ equals the BS antenna gain minus feeder loss. The requirement defined above provides the characteristics of the base station needed to verify compliance with the regional requirement.

#### 6.6.4.2.5.7 Void

### 6.6.4.3 Minimum requirements for *BS type 1-C*

The operating band unwanted emissions for *BS type 1-C* for each *antenna connector* shall be below the applicable *basic limits* defined in clause 6.6.4.2.

For Band n41 and n90 operation in Japan, the operating band unwanted emissions limits shall be applied to the sum of the emission power over all *antenna connectors* for *BS type 1-C.*

### 6.6.4.4 Minimum requirements for *BS type 1-H*

The operating band unwanted emissions requirements for *BS type 1-H* are that for each *TAB connector TX min cell group* and each applicable *basic limit* in clause 6.6.4.2, the power summation emissions at the *TAB connectors* of the *TAB connector TX min cell group* shall not exceed a BS limit specified as the *basic limit* + X, where X = $10\log_{10}(N_{TXU,countedpercell})$.

NOTE: Conformance to the *BS type 1-H* spurious emission requirement can be demonstrated by meeting at least one of the following criteria as determined by the manufacturer:

1) The sum of the emissions power measured on each *TAB connector* in the *TAB connector TX min cell group* shall be less than or equal to the limit as defined in this clause for the respective frequency span.

Or

2) The unwanted emissions power at each *TAB connector* shall be less than or equal to the *BS type 1-H* limit as defined in this clause for the respective frequency span, scaled by $-10\log_{10}(n)$, where n is the number of *TAB connectors* in the *TAB connector TX min cell group*.

## 6.6.5 Transmitter spurious emissions

### 6.6.5.1 General

The transmitter spurious emission limits shall apply from 9 kHz to 12.75 GHz, excluding the frequency range from $\Delta f_{OBUE}$ below the lowest frequency of each supported downlink *operating band*, up to $\Delta f_{OBUE}$ above the highest frequency of each supported downlink *operating band*, where the $\Delta f_{OBUE}$ is defined in table 6.6.1-1. For some *operating bands*, the upper limit is higher than 12.75 GHz in order to comply with the 5th harmonic limit of the downlink *operating band*, as specified in ITU-R recommendation SM.329 [2].

For a *multi-band connector*, for each supported *operating band* together with $\Delta f_{OBUE}$ around the band is excluded from the transmitter spurious emissions requirement.

The requirements shall apply whatever the type of transmitter considered (single carrier or multi-carrier). It applies for all transmission modes foreseen by the manufacturer's specification.

The requirements shall also apply if the BS supports NB-IoT operation in NR in-band.

Unless otherwise stated, all requirements are measured as mean power (RMS).

### 6.6.5.2 *Basic limits*

#### 6.6.5.2.1 General transmitter spurious emissions requirements

The *basic limits* of either table 6.6.5.2.1-1 (Category A limits) or table 6.6.5. 2.1-2 (Category B limits) shall apply. The application of either Category A or Category B limits shall be the same as for operating band unwanted emissions in clause 6.6.4.

**Table 6.6.5.2.1-1: General BS transmitter spurious emission limits in FR1, Category A**

| Spurious frequency range | ***Basic limit*** | ***Measurement bandwidth*** | **Notes** |
|---|---|---|---|
| 9 kHz – 150 kHz | -13 dBm | 1 kHz | Note 1, Note 4 |
| 150 kHz – 30 MHz | | 10 kHz | Note 1, Note 4 |
| 30 MHz – 1 GHz | | 100 kHz | Note 1 |
| 1 GHz 12.75 GHz | | 1 MHz | Note 1, Note 2 |
| 12.75 GHz – 5th harmonic of the upper frequency edge of the DL *operating band* in GHz | | 1 MHz | Note 1, Note 2, Note 3 |
| 12.75 GHz - 26 GHz | -13 dBm | 1 MHz | Note 1, Note 2, Note 5 |

NOTE 1: *Measurement bandwidth*s as in ITU-R SM.329 [2], s4.1.
NOTE 2: Upper frequency as in ITU-R SM.329 [2], s2.5 table 1.
NOTE 3: Applies for Band for which the upper frequency edge of the DL *operating band* is greater than 2.55 GHz and less than or equal to 5.2 GHz..
NOTE 4: This spurious frequency range applies only to *BS type 1-C* and *BS type 1-H*.
NOTE 5: Applies for Band for which the upper frequency edge of the DL *operating band* is greater than 5.2 GHz.

**Table 6.6.5.2.1-2: General BS transmitter spurious emission limits in FR1, Category B**

| Spurious frequency range | *Basic limit* | *Measurement bandwidth* | Notes |
|---|---|---|---|
| 9 kHz – 150 kHz | -36 dBm | 1 kHz | Note 1, Note 4 |
| 150 kHz – 30 MHz | | 10 kHz | Note 1, Note 4 |
| 30 MHz – 1 GHz | | 100 kHz | Note 1 |
| 1 GHz – 12.75 GHz | -30 dBm | 1 MHz | Note 1, Note 2 |
| 12.75 GHz – 5th harmonic of the upper frequency edge of the DL *operating band* in GHz | | 1 MHz | Note 1, Note 2, Note 3 |
| 12.75 GHz - 26 GHz | - 30 dBm | 1 MHz | Note 1, Note 2, Note 5 |

NOTE 1: *Measurement bandwidth*s as in ITU-R SM.329 [2], s4.1.
NOTE 2: Upper frequency as in ITU-R SM.329 [2], s2.5 table 1.
NOTE 3: Applies for Band for which the upper frequency edge of the DL *operating band* is greater than 2.55 GHz and less than or equal to 5.2 GHz.
NOTE 4: This spurious frequency range applies only to *BS type 1-C* and *BS type 1-H*.
NOTE 5: Applies for Band for which the upper frequency edge of the DL *operating band* is greater than 5.2 GHz.

### 6.6.5.2.2 Protection of the BS receiver of own or different BS

This requirement shall be applied for NR FDD operation in order to prevent the receivers of the BSs being desensitised by emissions from a BS transmitter. It is measured at the transmit *antenna connector* for *BS type 1-C* or at the *TAB connector* for *BS type 1-H* for any type of BS which has common or separate Tx/Rx *antenna connectors* / *TAB connectors*.

The spurious emission *basic limits* are provided in table 6.6.5.2.2-1.

**Table 6.6.5.2.2-1: BS spurious emissions *basic limits* for protection of the BS receiver**

| BS class | Frequency range | *Basic limits* | *Measurement bandwidth* | Note |
|---|---|---|---|---|
| Wide Area BS | $F_{UL,low}$ – $F_{UL,high}$ | -96 dBm | 100 kHz | |
| Medium Range BS | $F_{UL,low}$ – $F_{UL,high}$ | -91 dBm | 100 kHz | |
| Local Area BS | $F_{UL,low}$ – $F_{UL,high}$ | -88 dBm | 100 kHz | |

NOTE 1: For BS operating in band n104, the basic limit is increased by 1dB.
NOTE 2: For BS operating in regions where a band is only partially allocated for NR operations (e.g. band n28), this requirement only applies in the UL frequency range of the partial allocation.
NOTE 3: For BS capable of multi-band operation, Table 6.6.5.2.2-1 assumes that the supported *operating band*s, where the corresponding BS transmit and receive frequency ranges in table 5.2-1 would be overlapping, are not deployed in the same geographical area. For such a case of operation with overlapping frequency arrangements in the same geographical area, special protection requirements may apply that are not covered by the 3GPP specifications.

### 6.6.5.2.3 Additional spurious emissions requirements

These requirements may be applied for the protection of system operating in frequency ranges other than the BS downlink *operating band*. The limits may apply as an optional protection of such systems that are deployed in the same geographical area as the BS, or they may be set by local or regional regulation as a mandatory requirement for an NR *operating band*. It is in some cases not stated in the present document whether a requirement is mandatory or under what exact circumstances that a limit applies, since this is set by local or regional regulation. An overview of regional requirements in the present document is given in clause 4.5.

Some requirements may apply for the protection of specific equipment (UE, MS and/or BS) or equipment operating in specific systems (GSM, CDMA, UTRA, E-UTRA, NR, etc.) as listed below.

The spurious emission *basic limits* are provided in table 6.6.5.2.3 -1 for a BS where requirements for co-existence with the system listed in the first column apply. For a *multi-band connector*, the exclusions and conditions in the Note column of table 6.6.5.2.3 -1 apply for each supported *operating band*.

**Table 6.6.5.2.3-1: BS spurious emissions *basic limits* for BS for co-existence with systems operating in other frequency bands**

| System type for NR to co-exist with | Frequency range for co-existence requirement | *Basic limits* | *Measurement bandwidth* | Note |
|---|---|---|---|---|
| GSM900 | 921 – 960 MHz | -57 dBm | 100 kHz | This requirement does not apply to BS operating in band n8 |
| | 876 – 915 MHz | -61 dBm | 100 kHz | For the frequency range 880-915 MHz, this requirement does not apply to BS operating in band n8, since it is already covered by the requirement in clause 6.6.5.2.2. |
| DCS1800 | 1805 – 1880 MHz | -47 dBm | 100 kHz | This requirement does not apply to BS operating in band n3. |
| | 1710 – 1785 MHz | -61 dBm | 100 kHz | This requirement does not apply to BS operating in band n3, since it is already covered by the requirement in clause 6.6.5.2.2. |
| PCS1900 | 1930 – 1990 MHz | -47 dBm | 100 kHz | This requirement does not apply to BS operating in band n2, n25 or band n70. |
| | 1850 – 1910 MHz | -61 dBm | 100 kHz | This requirement does not apply to BS operating in band n2 or n25 since it is already covered by the requirement in clause 6.6.5.2.2. |
| GSM850 or CDMA850 | 869 – 894 MHz | -57 dBm | 100 kHz | This requirement does not apply to BS operating in band n5 or n26. |
| | 824 – 849 MHz | -61 dBm | 100 kHz | This requirement does not apply to BS operating in band n5 or n26, since it is already covered by the requirement in clause 6.6.5.2.2. |
| UTRA FDD Band I or E-UTRA Band 1 or NR Band n1 | 2110 – 2170 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n1 or n65 |
| | 1920 – 1980 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n1 or n65, since it is already covered by the requirement in clause 6.6.5.2.2. |
| UTRA FDD Band II or E-UTRA Band 2 or NR Band n2 | 1930 – 1990 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n2 or n70. |
| | 1850 – 1910 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n2, since it is already covered by the requirement in clause 6.6.5.2.2. |
| UTRA FDD Band III or E-UTRA Band 3 or NR Band n3 | 1805 – 1880 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n3. |
| | 1710 – 1785 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n3, since it is already covered by the requirement in clause 6.6.5.2.2. |
| UTRA FDD Band IV or E-UTRA Band 4 | 2110 – 2155 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n66 |
| | 1710 – 1755 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n66, since it is already covered by the requirement in clause 6.6.5.2.2. |
| UTRA FDD Band V or E-UTRA Band 5 or NR Band n5 | 869 – 894 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n5 or n26. |
| | 824 – 849 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n5 or n26, since it is already covered by the requirement in clause 6.6.5.2.2. |
| UTRA FDD Band VI, XIX or E-UTRA Band 6, 18, 19 or NR Band n18 | 860 – 890 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n18. |
| | 815 – 830 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n18, since it is already covered by the requirement in clause 6.6.5.2.2. |
| | 830 – 845 MHz | -49 dBm | 1 MHz | |
| UTRA FDD Band VII or E-UTRA Band 7 or NR Band n7 | 2620 – 2690 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n7. |
| | 2500 – 2570 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n7, since it is already covered by the requirement in clause 6.6.5.2.2. |
| UTRA FDD Band VIII or | 925 – 960 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n8 or n100. |

| System type for NR to co-exist with | Frequency range for co-existence requirement | *Basic limits* | *Measurement bandwidth* | Note |
|---|---|---|---|---|
| E-UTRA Band 8 or NR Band n8 | 880 – 915 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n8, since it is already covered by the requirement in clause 6.6.5.2.2. |
| UTRA FDD Band IX or E-UTRA Band 9 | 1844.9 – 1879.9 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n3. |
| | 1749.9 – 1784.9 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n3, since it is already covered by the requirement in clause 6.6.5.2.2. |
| UTRA FDD Band X or E-UTRA Band 10 | 2110 – 2170 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n66 |
| | 1710 – 1770 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n66, since it is already covered by the requirement in clause 6.6.5.2.2. |
| UTRA FDD Band XI or XXI or E-UTRA Band 11 or 21 | 1475.9 – 1510.9 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n50, n74, n75, n92 or n94. |
| | 1427.9 – 1447.9 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n50, n51, n74, n75, n76, n91, n92, n93 or n94. |
| | 1447.9 – 1462.9 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n50, n74, n75, n92 or n94. |
| UTRA FDD Band XII or E-UTRA Band 12 or NR Band n12 | 729 – 746 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n12 or n85. |
| | 699 – 716 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n12 or n85, since it is already covered by the requirement in clause 6.6.5.2.2. For NR BS operating in n29, it applies 1 MHz below the Band n29 downlink operating band (Note 5). |
| UTRA FDD Band XIII or E-UTRA Band 13 or NR Band n13 | 746 – 756 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n13. |
| | 777 – 787 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n13, since it is already covered by the requirement in clause 6.6.5.2.2. |
| UTRA FDD Band XIV or E-UTRA Band 14 or NR band n14 | 758 – 768 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n14. |
| | 788 – 798 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n14, since it is already covered by the requirement in clause 6.6.5.2.2. |
| E-UTRA Band 17 | 734 – 746 MHz | -52 dBm | 1 MHz | |
| | 704 – 716 MHz | -49 dBm | 1 MHz | For NR BS operating in n29, it applies 1 MHz below the Band n29 downlink operating band (Note 5). |
| UTRA FDD Band XX or E-UTRA Band 20 or NR Band n20 | 791 – 821 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n20 or n28. |
| | 832 – 862 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n20, since it is already covered by the requirement in clause 6.6.5.2.2. |
| UTRA FDD Band XXII or E-UTRA Band 22 | 3510 – 3590 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n48, n77 or n78. |
| | 3410 – 3490 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n77 or n78. |
| E-UTRA Band 24 or NR Band n24 | 1525 – 1559 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n24. |
| | 1626.5 – 1660.5 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n24, since it is already covered by the requirement in clause 6.6.5.2.2. |
| UTRA FDD Band XXV or E-UTRA Band 25 or NR band n25 | 1930 – 1995 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n2, n25 or n70. |
| | 1850 – 1915 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n25 since it is already covered by the requirement in clause 6.6.5.2.2. For BS operating in Band n2, it applies for 1910 MHz to 1915 MHz, while the rest is covered in clause 6.6.5.2.2. |

| System type for NR to co-exist with | Frequency range for co-existence requirement | *Basic limits* | *Measurement bandwidth* | Note |
|---|---|---|---|---|
| UTRA FDD Band XXVI or E-UTRA Band 26 or NR Band n26 | 859 – 894 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n5 or n26. |
| | 814 – 849 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n26 since it is already covered by the requirement in clause 6.6.5.2.2. For BS operating in Band n5, it applies for 814 MHz to 824 MHz, while the rest is covered in clause 6.6.5.2.2. |
| E-UTRA Band 27 | 852 – 869 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in Band n5. |
| | 807 – 824 MHz | -49 dBm | 1 MHz | This requirement also applies to BS operating in Band n28, starting 4 MHz above the Band n28 downlink *operating band* (Note 5). |
| E-UTRA Band 28 or NR Band n28 | 758 – 803 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n20, n67 or n28. |
| | 703 – 748 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n28, since it is already covered by the requirement in clause 6.6.5.2.2.<br>For BS operating in band n67, it applies for 703 MHz to 736 MHz. |
| E-UTRA Band 29 or NR Band n29 | 717 – 728 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in Band n29 or n85 |
| E-UTRA Band 30 or NR Band n30 | 2350 – 2360 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n30 |
| | 2305 – 2315 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n30, since it is already covered by the requirement in clause 6.6.5.2.2. |
| E-UTRA Band 31 | 462.5 – 467.5 MHz | -52 dBm | 1 MHz | |
| | 452.5 – 457.5 MHz | -49 dBm | 1 MHz | |
| UTRA FDD band XXXII or E-UTRA band 32 | 1452 – 1496 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n50, n74, n75, n92 or n94. |
| UTRA TDD Band a) or E-UTRA Band 33 | 1900 – 1920 MHz | -52 dBm | 1 MHz | |
| UTRA TDD Band a) or E-UTRA Band 34 or NR band n34 | 2010 – 2025 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in Band n34. |
| UTRA TDD Band b) or E-UTRA Band 35 | 1850 – 1910 MHz | -52 dBm | 1 MHz | |
| UTRA TDD Band b) or E-UTRA Band 36 | 1930 – 1990 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in Band n2 or n25. |
| UTRA TDD Band c) or E-UTRA Band 37 | 1910 – 1930 MHz | -52 dBm | 1 MHz | |
| UTRA TDD Band d) or E-UTRA Band 38 or NR Band n38 | 2570 – 2620 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in Band n38. |

| System type for NR to co-exist with | Frequency range for co-existence requirement | *Basic limits* | *Measurement bandwidth* | Note |
|---|---|---|---|---|
| UTRA TDD Band f) or E-UTRA Band 39 or NR band n39 | 1880 – 1920MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in Band n39. |
| UTRA TDD Band e) or E-UTRA Band 40 or NR Band n40 | 2300 – 2400MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in Band n30 or n40. |
| E-UTRA Band 41 or NR Band n41, n90 | 2496 – 2690 MHz | -52 dBm | 1 MHz | This is not applicable to BS operating in Band n41, n53 or [n90]. |
| E-UTRA Band 42 | 3400 – 3600 MHz | -52 dBm | 1 MHz | This is not applicable to BS operating in Band n48, n77 or n78. |
| E-UTRA Band 43 | 3600 – 3800 MHz | -52 dBm | 1 MHz | This is not applicable to BS operating in Band n48, n77 or n78. |
| E-UTRA Band 44 | 703 – 803 MHz | -52 dBm | 1 MHz | This is not applicable to BS operating in Band n28. |
| E-UTRA Band 45 | 1447 – 1467 MHz | -52 dBm | 1 MHz | |
| E-UTRA Band 46 or NR Band n46 | 5150 – 5925 MHz | -52 dBm | 1 MHz | This is not applicable to BS operating in Band n46, n96 or n102. |
| E-UTRA Band 47 | 5855 – 5925 MHz | -52 dBm | 1 MHz | |
| E-UTRA Band 48 or NR Band n48 | 3550 – 3700 MHz | -52 dBm | 1 MHz | This is not applicable to BS operating in Band n48, n77 or n78. |
| E-UTRA Band 50 or NR band n50 | 1432 – 1517 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in Band n50, n51, n74, n75, n76, n91, n92, n93 or n94. |
| E-UTRA Band 51 or NR Band n51 | 1427 – 1432 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in Band n50, n51, n75, n76, n91, n92, n93 or n94. |
| E-UTRA Band 53 or NR Band n53 | 2483.5 - 2495 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in Band n41, n53 or n90. |
| E-UTRA Band 65 or NR Band n65 | 2110 – 2200 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n1 or n65. |
| | 1920 – 2010 MHz | -49 dBm | 1 MHz | For BS operating in Band n1, it applies for 1980 MHz to 2010 MHz, while the rest is covered in clause 6.6.5.2.2.<br>This requirement does not apply to BS operating in band n65, since it is already covered by the requirement in clause 6.6.5.2.2. |
| E-UTRA Band 66 or NR Band n66 | 2110 – 2200 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n66. |
| | 1710 – 1780 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n66, since it is already covered by the requirement in clause 6.6.5.2.2. |
| E-UTRA Band 67 or NR Band n67 | 738 – 758 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in Band n28 or n67. |
| E-UTRA Band 68 | 753 -783 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n28. |
| | 698-728 MHz | -49 dBm | 1 MHz | For BS operating in Band n28, this requirement applies between 698 MHz and 703 MHz, while the rest is covered in clause 6.6.5.2.2. |
| E-UTRA Band 69 | 2570 – 2620 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in Band n38. |
| E-UTRA Band 70 or | 1995 – 2020 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n2, n25 or n70 |

| System type for NR to co-exist with | Frequency range for co-existence requirement | *Basic limits* | *Measurement bandwidth* | Note |
|---|---|---|---|---|
| NR Band n70 | 1695 – 1710 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n70, since it is already covered by the requirement in clause 6.6.5.2.2. |
| E-UTRA Band 71 or NR Band n71 | 617 – 652 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n71 |
| | 663 – 698 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n71, since it is already covered by the requirement in clause 6.6.5.2.2. |
| E-UTRA Band 72 | 461 – 466 MHz | -52 dBm | 1 MHz | |
| | 451 – 456 MHz | -49 dBm | 1 MHz | |
| E-UTRA Band 74 or NR Band n74 | 1475 – 1518 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n50, n74, n75, n92 or n94. |
| | 1427 – 1470 MHz | -49 dBm | 1MHz | This requirement does not apply to BS operating in band n50, n51, n74, n75, n76, n91, n92, n93 or n94. |
| E-UTRA Band 75 or NR Band n75 | 1432 – 1517 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in Band n50, n51, n74, n75, n76, n91, n92, n93 or n94. |
| E-UTRA Band 76 or NR Band n76 | 1427 – 1432 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in Band n50, n51, n75, n76, n91, n92, n93 or n94. |
| NR Band n77 | 3.3 – 4.2 GHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in Band n48, n77 or n78 |
| NR Band n78 | 3.3 – 3.8 GHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in Band n48, n77 or n78 |
| NR Band n79 | 4.4 – 5.0 GHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in Band n79 |
| NR Band n80 | 1710 – 1785 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n3, since it is already covered by the requirement in clause 6.6.5.2.2. |
| NR Band n81 | 880 – 915 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n8, since it is already covered by the requirement in clause 6.6.5.2.2. |
| NR Band n82 | 832 – 862 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n20, since it is already covered by the requirement in clause 6.6.5.2.2. |
| NR Band n83 | 703 – 748 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n28, since it is already covered by the requirement in clause 6.6.5.2.2.<br>For BS operating in Band n67, it applies for 703 MHz to 736 MHz. |
| NR Band n84 | 1920 – 1980 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n1, since it is already covered by the requirement in clause 6.6.5.2.2. |
| E-UTRA Band 85 or NR Band n85 | 728 – 746 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in band n12 or n85.<br>For NR BS operating in n29, it applies 1 MHz below the Band n29 downlink operating band (Note 5). |
| | 698 – 716 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n12 or n85, since it is already covered by the requirement in clause 6.6.5.2.2. |
| NR Band n86 | 1710 – 1780 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n66, since it is already covered by the requirement in clause 6.6.5.2.2. |
| E-UTRA Band 87 | 420 – 425 MHz | -52 dBm | 1 MHz | |
| | 410 – 415 MHz | -49 dBm | 1 MHz | |
| E-UTRA Band 88 | 422 – 427 MHz | -52 dBm | 1 MHz | |
| | 412 – 417 MHz | -49 dBm | 1 MHz | |

| System type for NR to co-exist with | Frequency range for co-existence requirement | *Basic limits* | *Measurement bandwidth* | Note |
|---|---|---|---|---|
| NR Band n89 | 824 – 849 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n5, since it is already covered by the requirement in clause 6.6.5.2.2. |
| NR Band n91 | 1427 – 1432 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in Band n50, n51, n75 or n76. |
| | 832 – 862 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n20, since it is already covered by the requirement in clause 6.6.5.2.2. |
| NR Band n92 | 1432 – 1517 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in Band n50, n51, n74, n75 or n76. |
| | 832 – 862 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n20, since it is already covered by the requirement in clause 6.6.5.2.2. |
| NR Band n93 | 1427 – 1432 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in Band n50, n51, n75 or n76. |
| | 880 – 915 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n8, since it is already covered by the requirement in clause 6.6.5.2.2. |
| NR Band n94 | 1432 – 1517 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in Band n50, n51, n74, n75 or n76. |
| | 880 – 915 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n8, since it is already covered by the requirement in clause 6.6.5.2.2. |
| NR Band n95 | 2010 – 2025 MHz | -52 dBm | 1 MHz | |
| NR Band n96 | 5925 – 7125 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in Band n46, n96, n102 or n104. |
| NR Band n97 | 2300 – 2400MHz | -52 dBm | 1 MHz | |
| NR Band n98 | 1880 – 1920MHz | -52 dBm | 1 MHz | |
| NR Band n99 | 1626.5 – 1660.5 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n24, since it is already covered by the requirement in clause 6.6.5.2.2. |
| NR band n100 | 919.4 – 925 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in Band n8 or n100. |
| | 874.4 – 880 MHz | -49 dBm | 1 MHz | This requirement does not apply to BS operating in band n100, since it is already covered by the requirement in clause 6.6.5.2.2. |
| NR band n101 | 1900 – 1910 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in Band n101. |
| NR Band n102 | 5925 – 6425 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in Band n46, n96, n102 or n104. |
| E-UTRA Band 103 | 757 – 758 MHz | -52 dBm | 1 MHz | |
| | 787 – 788 MHz | -49 dBm | 1 MHz | |
| NR Band n104 | 6425 – 7125 MHz | -52 dBm | 1 MHz | This requirement does not apply to BS operating in Band n96, n102 or n104 |

NOTE 1: As defined in the scope for spurious emissions in this clause, except for the cases where the noted requirements apply to a BS operating in Band n28, the co-existence requirements in table 6.6.5.2.3 -1 do not apply for the $\Delta f_{OBUE}$ frequency range immediately outside the downlink *operating band* (see table 5.2-1). Emission limits for this excluded frequency range may be covered by local or regional requirements.

NOTE 2: Table 6.6.5.2.3 -1 assumes that two *operating bands*, where the frequency ranges in table 5.2-1 would be overlapping, are not deployed in the same geographical area. For such a case of operation with overlapping frequency arrangements in the same geographical area, special co-existence requirements may apply that are not covered by the 3GPP specifications.

NOTE 3: TDD base stations deployed in the same geographical area, that are synchronized and use the same or adjacent *operating bands* can transmit without additional co-existence requirements. For unsynchronized base stations, special co-existence requirements may apply that are not covered by the 3GPP specifications.

NOTE 4: For NR Band n28 BS, specific solutions may be required to fulfil the spurious emissions limits for BS for co-existence with E-UTRA Band 27 UL *operating band*.

NOTE 5: For NR Band n29 BS, specific solutions may be required to fulfil the spurious emissions limits for NR BS for co-existence with UTRA Band XII, E-UTRA Band 12 or NR Band n12 UL operating band, E-UTRA Band 17 UL operating band or E-UTRA Band 85 UL or NR Band n85 UL operating band.

The following requirement may be applied for the protection of PHS. This requirement is also applicable at specified frequencies falling between $\Delta f_{OBUE}$ below the lowest BS transmitter frequency of the downlink *operating band* and $\Delta f_{OBUE}$ above the highest BS transmitter frequency of the downlink *operating band*. $\Delta f_{OBUE}$ is defined in clause 6.6.1.

The spurious emission *basic limit* for this requirement is:

**Table 6.6.5.2.3-2: BS spurious emissions *basic limits* for BS for co-existence with PHS**

| Frequency range | *Basic limit* | *Measurement Bandwidth* | Note |
|---|---|---|---|
| 1884.5 – 1915.7 MHz | -41 dBm | 300 kHz | Applicable when co-existence with PHS system operating in 1884.5 – 1915.7 MHz |

**Table 6.6.5.2.3-3: Void**

In certain regions, the following requirement may apply to NR BS operating in Band n50 and n75 within the 1432 – 1452 MHz, and in Band n51 and Band n76. The *basic limit is* specified in Table 6.6.5.2.3-4. This requirement is also applicable at the frequency range from $\Delta f_{OBUE}$ below the lowest frequency of the BS downlink *operating band* up to $\Delta f_{OBUE}$ above the highest frequency of the BS downlink *operating band*.

**Table 6.6.5.2.3-4: Additional operating band unwanted emission *basic limit* for NR BS operating in Band n50 and n75 within 1432 – 1452 MHz, and in Band n51 and n76**

| Filter centre frequency, $F_{filter}$ | *Basic limit* | *Measurement Bandwidth* |
|---|---|---|
| $F_{filter}$ = 1413.5 MHz | -42 dBm | 27 MHz |

In certain regions, the following requirement may apply to BS operating in NR Band n50 and n75 within 1492-1517 MHz and in Band n74 within 1492-1518 MHz. The maximum level of emissions, measured on centre frequencies $F_{filter}$ with filter bandwidth according to Table 6.6.5.2.3-5, shall be defined according to the *basic limits* $P_{EM,n50/n75,a}$ nor $P_{EM,n50/n75,b}$ declared by the manufacturer.

**Table 6.6.5.2.3-5: *Operating band* n50, n74 and n75 declared emission above 1518 MHz**

| Filter centre frequency, $F_{filter}$ | Declared *basic limits* (dBm) | *Measurement bandwidth* |
|---|---|---|
| 1518.5 MHz ≤ $F_{filter}$ ≤ 1519.5 MHz | $P_{EM,\ n50/n75,a}$ | 1 MHz |
| 1520.5 MHz ≤ $F_{filter}$ ≤ 1558.5 MHz | $P_{EM,n50/n75,b}$ | 1 MHz |

In certain regions, the following requirement shall be applied to BS operating in Band n13 and n14 to ensure that appropriate interference protection is provided to 700 MHz public safety operations. This requirement is also applicable at the frequency range from 10 MHz below the lowest frequency of the BS downlink operating band up to 10 MHz above the highest frequency of the BS downlink operating band.

The power of any spurious emission shall not exceed:

**Table 6.6.5.2.3-6: BS Spurious emissions limits for protection of 700 MHz public safety operations**

| Operating Band | Frequency range | *Basic limit* | *Measurement Bandwidth* |
|---|---|---|---|
| n13 | 763 - 775 MHz | -46 dBm | 6.25 kHz |
| n13 | 793 - 805 MHz | -46 dBm | 6.25 kHz |
| n14 | 769 - 775 MHz | -46 dBm | 6.25 kHz |
| n14 | 799 - 805 MHz | -46 dBm | 6.25 kHz |

In certain regions, the following requirement may apply to NR BS operating in Band n30. This requirement is also applicable at the frequency range from 10 MHz below the lowest frequency of the BS downlink operating band up to 10 MHz above the highest frequency of the BS downlink operating band.

The power of any spurious emission shall not exceed:

**Table 6.6.5.2.3-7: Additional NR BS Spurious emissions limits for Band n30**

| Frequency range | *Basic limit* | *Measurement Bandwidth* | Note |
|---|---|---|---|
| 2200 – 2345 MHz | -45 dBm | 1 MHz | |
| 2362.5 – 2365 MHz | -25 dBm | 1 MHz | |
| 2365 – 2367.5 MHz | -40 dBm | 1 MHz | |
| 2367.5 – 2370 MHz | -42 dBm | 1 MHz | |
| 2370 – 2395 MHz | -45 dBm | 1 MHz | |

The following requirement may apply to BS operating in Band n48 in certain regions. The power of any spurious emission shall not exceed:

**Table 6.6.5.2.3-8: Additional BS Spurious emissions limits for Band n48**

| Frequency range | *Basic limit* | *Measurement Bandwidth* (NOTE 1) | Note |
|---|---|---|---|
| 3530 MHz – 3720 MHz | -25 dBm | 1 MHz | Applicable 10 MHz from the assigned *channel edge* |
| 3100 MHz – 3530 MHz<br>3720 MHz – 4200 MHz | -40 dBm | 1 MHz | |

NOTE 1: The resolution bandwidth of the measuring equipment should be equal to the measurement bandwidth. However, to improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth may be smaller than the measurement bandwidth. When the resolution bandwidth is smaller than the measurement bandwidth, the result should be integrated over the measurement bandwidth in order to obtain the equivalent noise bandwidth of the measurement bandwidth.

NOTE 2: The regional requirement, included in [12], is defined in terms of EIRP, which is dependent on both the BS emissions at the *antenna connector* and the deployment (including antenna gain and feeder loss). The requirement defined above provides the characteristics of the base station needed to verify compliance with the regional requirement. The assessment of the EIRP level is described in Annex F.

The following requirement shall be applied to BS operating in Band n26 to ensure that appropriate interference protection is provided to 800 MHz public safety operations. This requirement is also applicable at the frequency range from 10 MHz below the lowest frequency of the BS downlink operating band up to 10 MHz above the highest frequency of the BS downlink operating band.

The power of any spurious emission shall not exceed:

**Table 6.6.5.2.3-9: BS Spurious emissions limits for protection of 800 MHz public safety operations**

| Operating Band | Frequency range | *Basic limit* | Measurement Bandwidth | Note |
|---|---|---|---|---|
| n26 | 851 - 859 MHz | -13 dBm | 100 kHz | Applicable for offsets > 37.5kHz from the channel edge |

The following requirement may apply to BS for Band n41 and n90 operation in Japan. This requirement is also applicable at the frequency range from $\Delta f_{OBUE}$ below the lowest frequency of the BS downlink operating band up to $\Delta f_{OBUE}$ above the highest frequency of the BS downlink operating band.

The power of any spurious emission shall not exceed:

**Table 6.6.5.2.3-10: Additional BS Spurious emissions limits for Band n41 and n90**

| Frequency range | *Basic limit* | *Measurement Bandwidth* |
|---|---|---|
| 2505 MHz – 2535 MHz | -42 dBm | 1 MHz |
| NOTE: This requirement applies for carriers allocated within 2545-2645 MHz. | | |

The following requirement may apply to BS operating in 3.45-3.55 GHz in Band n77 in certain regions. The power of any spurious emission shall not exceed:

**Table 6.6.5.2.3-11: Additional BS spurious emissions limits for Band n77**

| Channel bandwidth [MHz] | Frequency range [MHz] | Filter centre frequency, $F_{filter}$ [MHz] | *Basic limit* [dBm] | *Measurement bandwidth* [MHz] |
|---|---|---|---|---|
| All | 3430 – 3440<br>3560 – 3570 | $3430.5 \le F_{filter} < 3439.5$<br>$3560.5 \le F_{filter} < 3569.5$ | -25 | 1 |
| All | ≤ 3430<br>> 3570 | $F_{filter} < 3429.5$<br>$3570.5 \le F_{filter}$ | -40 | 1 |

NOTE: The resolution bandwidth of the measuring equipment should be equal to the measurement bandwidth. However, to improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth may be smaller than the measurement bandwidth. When the resolution bandwidth is smaller than the measurement bandwidth, the result should be integrated over the measurement bandwidth in order to obtain the equivalent noise bandwidth of the measurement bandwidth.

The following requirement shall apply to BS operating in Band n101 in CEPT countries. The power of any spurious emission shall not exceed:

**Table 6.6.5.2.3-12: Additional BS Spurious emissions limits for Band n101**

| Frequency range | Maximum Level | *Measurement Bandwidth* | Note |
|---|---|---|---|
| 1920 MHz – 1980 MHz | -57 dBm | 5 MHz | This limit is derived from ECC Decision(20)02 [21] assuming a 18 dBi maximum antenna gain and 4 dB losses, and assuming one antenna connector. For more details on the maximum level derivation, refer to TR 38.852 [23] |

The following requirement shall apply to BS operating in Band n100 in CEPT countries. The power of any spurious emission shall not exceed:

**Table 6.6.5.2.3-13: Additional BS Spurious emissions limits for Band n100**

| Frequency range | Maximum Level | *Measurement Bandwidth* | Note |
|---|---|---|---|
| 880 MHz – 915 MHz | -62 dBm | 5 MHz | This limit is derived from ECC Decision(20)02 [21] assuming a 17 dBi maximum antenna gain and 4 dB losses, and assuming one antenna connector. For more details on the maximum level derivation, refer to TR 38.853 [23]. |

### 6.6.5.2.4 Co-location with other base stations

These requirements may be applied for the protection of other BS receivers when GSM900, DCS1800, PCS1900, GSM850, CDMA850, UTRA FDD, UTRA TDD, E-UTRA and/or NR BS are co-located with a BS.

The requirements assume a 30 dB coupling loss between transmitter and receiver and are based on co-location with base stations of the same class.

The *basic limits* are in table 6.6.5.2.4-1 for a BS where requirements for co-location with a BS type listed in the first column apply, depending on the declared Base Station class. For a *multi-band connector*, the exclusions and conditions in the Note column of table 6.6.5.2.4-1 shall apply for each supported *operating band*.

**Table 6.6.5.2.4-1: BS spurious emissions *basic* limits for BS co-located with another BS**

| Type of co-located BS | Frequency range for co-location requirement | *Basic limits* | | | Measurement bandwidth | Note |
|---|---|---|---|---|---|---|
| | | WA BS | MR BS | LA BS | | |
| GSM900 | 876 – 915 MHz | -98 dBm | -91 dBm | -70 dBm | 100 kHz | |
| DCS1800 | 1710 – 1785 MHz | -98 dBm | -91 dBm | -80 dBm | 100 kHz | |
| PCS1900 | 1850 – 1910 MHz | -98 dBm | -91 dBm | -80 dBm | 100 kHz | |
| GSM850 or CDMA850 | 824 – 849 MHz | -98 dBm | -91 dBm | -70 dBm | 100 kHz | |
| UTRA FDD Band I or E-UTRA Band 1 or NR Band n1 | 1920 – 1980 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| UTRA FDD Band II or E-UTRA Band 2 or NR Band n2 | 1850 – 1910 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| UTRA FDD Band III or E-UTRA Band 3 or NR Band n3 | 1710 – 1785 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| UTRA FDD Band IV or E-UTRA Band 4 | 1710 – 1755 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| UTRA FDD Band V or E-UTRA Band 5 or NR Band n5 | 824 – 849 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| UTRA FDD Band VI, XIX or E-UTRA Band 6, 19 | 830 – 845 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| UTRA FDD Band VII or E-UTRA Band 7 or NR Band n7 | 2500 – 2570 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| UTRA FDD Band VIII or E-UTRA Band 8 or NR Band n8 | 880 – 915 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| UTRA FDD Band IX or E-UTRA Band 9 | 1749.9 – 1784.9 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| UTRA FDD Band X or E-UTRA Band 10 | 1710 – 1770 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| UTRA FDD Band XI or E-UTRA Band 11 | 1427.9 –1447.9 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | This is not applicable to BS operating in Band n50, n75, n91, n92, n93 or n94 |
| UTRA FDD Band XII or E-UTRA Band 12 or NR Band n12 | 699 – 716 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| UTRA FDD Band XIII or E-UTRA Band 13 or NR Band n13 | 777 – 787 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| UTRA FDD Band XIV or E-UTRA Band 14 or NR Band n14 | 788 – 798 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| E-UTRA Band 17 | 704 – 716 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| E-UTRA Band 18 or NR Band n18 | 815 – 830 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| UTRA FDD Band XX or E-UTRA Band 20 or NR Band n20 | 832 – 862 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| UTRA FDD Band XXI or E-UTRA Band 21 | 1447.9 – 1462.9 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | This is not applicable to BS operating in Band n50, n75, n92 or n94 |

| | | | | | | |
|---|---|---|---|---|---|---|
| UTRA FDD Band XXII or E-UTRA Band 22 | 3410 – 3490 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | This is not applicable to BS operating in Band n48, n77 or n78 |
| E-UTRA Band 24 or NR Band n24 | 1626.5 – 1660.5 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| UTRA FDD Band XXV or E-UTRA Band 25 or NR Band n25 | 1850 – 1915 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| UTRA FDD Band XXVI or E-UTRA Band 26 or NR Band n26 | 814 – 849 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| E-UTRA Band 27 | 807 – 824 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| E-UTRA Band 28 or NR Band n28 | 703 – 748 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| E-UTRA Band 30 or NR Band n30 | 2305 – 2315 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| E-UTRA Band 31 | 452.5 – 457.5 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| UTRA TDD Band a) or E-UTRA Band 33 | 1900 – 1920 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| UTRA TDD Band a) or E-UTRA Band 34 or NR band n34 | 2010 – 2025 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | This is not applicable to BS operating in Band n34 |
| UTRA TDD Band b) or E-UTRA Band 35 | 1850 – 1910 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| UTRA TDD Band b) or E-UTRA Band 36 | 1930 – 1990 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | This is not applicable to BS operating in Band n2 or band n25 |
| UTRA TDD Band c) or E-UTRA Band 37 | 1910 – 1930 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| UTRA TDD Band d) or E-UTRA Band 38 or NR Band n38 | 2570 – 2620 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | This is not applicable to BS operating in Band n38. |
| UTRA TDD Band f) or E-UTRA Band 39 or NR band n39 | 1880 – 1920MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | This is not applicable to BS operating in Band n39 |
| UTRA TDD Band e) or E-UTRA Band 40 or NR Band n40 | 2300 – 2400MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | This is not applicable to BS operating in Band n30 or n40. |
| E-UTRA Band 41 or NR Band n41, n90 | 2496 – 2690 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | This is not applicable to BS operating in Band n41, n53 or [n90] |
| E-UTRA Band 42 | 3400 – 3600 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | This is not applicable to BS operating in Band n48, n77 or n78 |
| E-UTRA Band 43 | 3600 – 3800 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | This is not applicable to BS operating in Band n48, n77 or n78 |
| E-UTRA Band 44 | 703 – 803 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | This is not applicable to BS operating in Band n28 |

| | | | | | | |
|---|---|---|---|---|---|---|
| E-UTRA Band 45 | 1447 – 1467 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| E-UTRA Band 46 or NR Band n46 | 5150 – 5925 MHz | N/A | -91 dBm | -88 dBm | 100 kHz | This is not applicable to BS operating in Band n46, n96 or n102 |
| E-UTRA Band 48 or NR Band n48 | 3550 – 3700 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | This is not applicable to BS operating in Band n48, n77 or n78 |
| E-UTRA Band 50 or NR Band n50 | 1432 – 1517 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | This is not applicable to BS operating in Band n51, n74, n75, n91, n92, n93 or n94 |
| E-UTRA Band 51 or NR Band n51 | 1427 – 1432 MHz | N/A | N/A | -88 dBm | 100 kHz | This is not applicable to BS operating in Band n50, n74, n75, n76, n91, n92, n93 or n94 |
| E-UTRA Band 53 or NR Band n53 | 2483.5 – 2495 MHz | N/A | -91 dBm | -88 dBm | 100 kHz | This is not applicable to BS operating in Band n41, n53 or n90 |
| E-UTRA Band 65 or NR Band n65 | 1920 – 2010 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| E-UTRA Band 66 or NR Band n66 | 1710 – 1780 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| E-UTRA Band 68 | 698 – 728 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| E-UTRA Band 70 or NR Band n70 | 1695 – 1710 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| E-UTRA Band 71 or NR Band n71 | 663 – 698 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| E-UTRA Band 72 | 451 – 456 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| E-UTRA Band 74 or NR Band n74 | 1427 – 1470 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | This is not applicable to BS operating in Band n50, n51, n91, n92, n93 or n94 |
| NR Band n77 | 3.3 – 4.2 GHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | This is not applicable to BS operating in Band n48, n77 or n78 |
| NR Band n78 | 3.3 – 3.8 GHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | This is not applicable to BS operating in Band n48, n77 or n78 |
| NR Band n79 | 4.4 – 5.0 GHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| NR Band n80 | 1710 – 1785 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| NR Band n81 | 880 – 915 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| NR Band n82 | 832 – 862 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| NR Band n83 | 703 – 748 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |

| | | | | | | |
|---|---|---|---|---|---|---|
| NR Band n84 | 1920 – 1980 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| E-UTRA Band 85 or NR Band 85 | 698 – 716 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| NR Band n86 | 1710 – 1780 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| E-UTRA Band 87 | 410 – 415 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| E-UTRA Band 88 | 412 – 417 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| NR Band n89 | 824 – 849 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| NR Band n91 | 832 – 862 MHz | N/A | N/A | -88 dBm | 100 kHz | |
| NR Band n92 | 832 – 862 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| NR Band n93 | 880 – 915 MHz | N/A | N/A | -88 dBm | 100 kHz | |
| NR Band n94 | 880 – 915 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| NR Band n95 | 2010 – 2025 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| NR Band n96 | 5925 – 7125 MHz | N/A | -90 dBm | -87 dBm | 100 kHz | This is not applicable to BS operating in Band n46, n96, n102 or n104 |
| NR Band n97 | 2300 – 2400MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| NR Band n98 | 1880 – 1920MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| NR Band n99 | 1626.5 – 1660.5 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| NR Band n100 | 874.4 – 880 MHz | -96 dBm | NA | NA | 100 kHz | |
| NR Band n101 | 1900 – 1910 MHz | -96 dBm | NA | NA | 100 kHz | |
| NR Band n102 | 5925 – 6425 MHz | N/A | -90 dBm | -87 dBm | 100 kHz | This is not applicable to BS operating in Band n46, n96, n102 or n104 |
| E-UTRA Band 103 | 787 – 788 MHz | -96 dBm | -91 dBm | -88 dBm | 100 kHz | |
| NR Band n104 | 6425 – 7125 MHz | -95 dBm | -90 dBm | -87 dBm | 100 kHz | This requirement does not apply to BS operating in Band n96, n102 or n104. |

NOTE 1: As defined in the scope for spurious emissions in this clause, the co-location requirements in table 6.6.5.2.4-1 do not apply for the frequency range extending $\Delta f_{OBUE}$ immediately outside the BS transmit frequency range of a downlink *operating band* (see table 5.2-1). The current state-of-the-art technology does not allow a single generic solution for co-location with other system on adjacent frequencies for 30dB BS-BS minimum coupling loss. However, there are certain site-engineering solutions that can be used. These techniques are addressed in TR 25.942 [4].

NOTE 2: Table 6.6.5.2.4-1 assumes that two *operating bands*, where the corresponding BS transmit and receive frequency ranges in table 5.2-1 would be overlapping, are not deployed in the same geographical area. For such a case of operation with overlapping frequency arrangements in the same geographical area, special co-location requirements may apply that are not covered by the 3GPP specifications.

NOTE 3: Co-located TDD base stations that are synchronized and using the same or adjacent *operating band* can transmit without special co-locations requirements. For unsynchronized base stations, special co-location requirements may apply that are not covered by the 3GPP specifications.

#### 6.6.5.3 Minimum requirements for *BS type 1-C*

The Tx spurious emissions for *BS type 1-C* for each *antenna connector* shall not exceed the *basic limits* specified in clause 6.6.5.2.

For Band n41 and n90 operation in Japan, the sum of the spurious emissions over all *antenna connectors* for *BS type 1-C* shall not exceed the *basic limits* defined in clause 6.6.5.2.

#### 6.6.5.4 Minimum requirements for *BS type 1-H*

The Tx spurious emissions requirements for *BS type 1-H* are that for each *TAB connector TX min cell group* and each applicable *basic limit* in clause 6.6.5.2, the power summation emissions at the *TAB connectors* of the *TAB connectors* of the *TAB connector TX min cell group* shall not exceed a limit specified as the *basic limit* + X, where X = $10\log_{10}(N_{TXU,countedpercell})$, unless stated differently in regional regulation.

NOTE: Conformance to the *BS type 1-H* spurious emission requirement can be demonstrated by meeting at least one of the following criteria as determined by the manufacturer:

- 1) The sum of the emissions power measured on each *TAB connector* in the *TAB connector TX min cell group* shall be less than or equal to the limit as defined in this clause for the respective frequency span.

- Or

- 2) The unwanted emissions power at each *TAB connector* shall be less than or equal to the *BS type 1-H* limit as defined in this clause for the respective frequency span, scaled by $-10\log_{10}(n)$, where n is the number of *TAB connectors* in the *TAB connector TX min cell group*.

## 6.7 Transmitter intermodulation

### 6.7.1 General

The transmitter intermodulation requirement is a measure of the capability of the transmitter unit to inhibit the generation of signals in its non-linear elements caused by presence of the wanted signal and an interfering signal reaching the transmitter unit via the antenna, RDN and antenna array. The requirement shall apply during the *transmitter ON period* and the *transmitter transient period*.

For *BS type 1-C*, the transmitter intermodulation level is the power of the intermodulation products when an interfering signal is injected into the *antenna connector*.

For *BS type 1-H*, the transmitter intermodulation level is the power of the intermodulation products when an interfering signal is injected into the *TAB connector*.

For *BS type 1-H*, there are two types of transmitter intermodulation cases captured by the transmitter intermodulation requirement:

1) Co-location transmitter intermodulation in which the interfering signal is from a co-located base station.

2) Intra-system transmitter intermodulation in which the interfering signal is from other transmitter units within the *BS type 1-H*.

For *BS type 1-H*, the co-location transmitter intermodulation requirement is considered sufficient if the interference signal for the co-location requirement is higher than the declared interference signal for intra-system transmitter intermodulation requirement.

## 6.7.2 Minimum requirements for *BS type 1-C*

### 6.7.2.1 Co-location minimum requirements

For *BS type 1-C*, the wanted signal and interfering signal centre frequency is specified in table 6.7.2.1-1, where interfering signal level is *Rated total output power* ($P_{rated,t,AC}$) at *antenna connector* in the *operating band* – 30 dB.

The requirement is applicable outside the *Base Station RF Bandwidth* or *Radio Bandwidth*. The interfering signal offset is defined relative to the *Base Station RF Bandwidth edges* or *Radio Bandwidth* edges.

For a BS operating in *non-contiguous spectrum*, the requirement is also applicable inside a *sub-block gap* for interfering signal offsets where the interfering signal falls completely within the *sub-block gap*. The interfering signal offset is defined relative to the *sub-block* edges.

For a *multi-band connector*, the requirement shall apply relative to the *Base Station RF Bandwidth edges* of each supported *operating band*. In case the *Inter RF Bandwidth gap* is less than $3*BW_{Channel}$ (where $BW_{Channel}$ is the minimal *BS channel bandwidth* of the band), the requirement in the gap shall apply only for interfering signal offsets where the interfering signal falls completely within the *Inter RF Bandwidth gap*.

The transmitter intermodulation level shall not exceed the unwanted emission limits in clauses 6.6.3, 6.6.4 and 6.6.5 in the presence of an NR interfering signal according to table 6.7.2.1-1.

**Table 6.7.2.1-1: Interfering and wanted signals for the co-location transmitter intermodulation requirement**

| Parameter | Value |
|---|---|
| Wanted signal type | NR single carrier, or multi-carrier, or multiple intra-band contiguously or non-contiguously aggregated carriers, with NB-IoT operation in NR in-band if supported. |
| Interfering signal type | NR signal, the minimum *BS channel bandwidth* ($BW_{Channel}$) with 15 kHz SCS of the band defined in clause 5.3.5. |
| Interfering signal level | *Rated total output power* ($P_{rated,t,AC}$) in the *operating band* – 30 dB |
| Interfering signal centre frequency offset from the lower/upper edge of the wanted signal or edge of *sub-block* inside a *sub-block gap* | $f_{offset} = \pm BW_{Channel}\left(n-\frac{1}{2}\right)$, for n=1, 2 and 3 |
| NOTE 1: Interfering signal positions that are partially or completely outside of any downlink *operating band* of the base station are excluded from the requirement, unless the interfering signal positions fall within the frequency range of adjacent downlink *operating bands* in the same geographical area. In case that none of the interfering signal positions fall completely within the frequency range of the downlink *operating band*, TS 38.141-1 [5] provides further guidance regarding appropriate test requirements.<br>NOTE 2: In Japan, NOTE 1 is not applied in Band n77, n78, n79. | |

### 6.7.2.2 Additional requirements

For Band n41 and n90 operation in Japan, the sum of transmitter intermodulation level over all *antenna connectors* shall not exceed the unwanted emission limits in clauses 6.6.3, 6.6.4 and 6.6.5 in the presence of an NR interfering signal according to table 6.7.2.2-1.

**Table 6.7.2.2-1 Interfering and wanted signals for the additional transmitter intermodulation requirement for Band n41 and n90**

| Parameter | Value |
|---|---|
| Wanted signal | NR single carrier (NOTE) |
| Interfering signal type | NR signal of 10 MHz *channel bandwidth* |
| Interfering signal level | Rated total output power in the operating band – 30 dB |
| Interfering signal centre frequency offset from the lower (upper) edge of the wanted signal | ± 5 MHz<br>± 15 MHz<br>± 25 MHz |
| NOTE: This requirement applies for NR carriers allocated within 2545-2645 MHz. | |

**Table 6.7.2.2-2: Void**

For Band n79 operation in Japan, the transmitter intermodulation level shall not exceed the unwanted emission limits in clauses 6.6.3, 6.6.4 and 6.6.5 in the presence of an NR interfering signal according to table 6.7.2.2-3.

**Table 6.7.2.2-3: Interfering and wanted signals for the additional transmitter intermodulation requirement for Band n79**

| Parameter | Value |
|---|---|
| Wanted signal | NR single carrier |
| Interfering signal type | NR signal of 40 MHz *channel bandwidth* |
| Interfering signal level | Rated total output power in the operating band – 30 dB |
| Interfering signal centre frequency offset from the lower/upper edge of the wanted signal | ± 20 MHz<br>± 60 MHz<br>± 100 MHz |

## 6.7.3 Minimum requirements for *BS type 1-H*

### 6.7.3.1 Co-location minimum requirements

The transmitter intermodulation level shall not exceed the unwanted emission limits in clauses 6.6.3, 6.6.4 and 6.6.5 in the presence of an NR interfering signal according to table 6.7.3.1-1

The requirement is applicable outside the *Base Station RF Bandwidth edges*. The interfering signal offset is defined relative to the *Base Station RF Bandwidth edges* or *Radio Bandwidth* edges.

For *TAB connectors* supporting operation in *non-contiguous spectrum*, the requirement is also applicable inside a *sub-block gap* for interfering signal offsets where the interfering signal falls completely within the *sub-block gap*. The interfering signal offset is defined relative to the *sub-block* edges.

For *multi-band connector*, the requirement shall apply relative to the *Base Station RF Bandwidth edges* of each *operating band*. In case the *inter RF Bandwidth gap* is less than $3*BW_{Channel}$ (where $BW_{Channel}$ is the minimal *BS channel bandwidth* of the band), the requirement in the gap shall apply only for interfering signal offsets where the interfering signal falls completely within the *inter RF Bandwidth gap*.

**Table 6.7.3.1-1: Interfering and wanted signals for the co-location transmitter intermodulation requirement**

| Parameter | Value |
|---|---|
| Wanted signal type | NR single carrier, or multi-carrier, or multiple intra-band contiguously or non-contiguously aggregated carriers |
| Interfering signal type | NR signal, the minimum *BS channel bandwidth* ($BW_{Channel}$) with 15 kHz SCS of the band defined in clause 5.3.5. |
| Interfering signal level | *Rated total output power* per *TAB connector* ($P_{rated,t,TABC}$) in the *operating band* – 30 dB |
| Interfering signal centre frequency offset from the lower/upper edge of the wanted signal or edge of *sub-block* inside a gap | $f_{offset} = \pm BW_{Channel}\left(n - \frac{1}{2}\right)$, for n=1, 2 and 3 |
| NOTE 1: Interfering signal positions that are partially or completely outside of any downlink *operating band* of the *TAB connector* are excluded from the requirement, unless the interfering signal positions fall within the frequency range of adjacent downlink *operating bands* in the same geographical area. In case that none of the interfering signal positions fall completely within the frequency range of the downlink *operating band*, TS 38.141-1 [5] provides further guidance regarding appropriate test requirements.<br>NOTE 2: In Japan, NOTE 1 is not applied in Band n77, n78, n79. | |

## 6.7.3.2 Intra-system minimum requirements

The transmitter intermodulation level shall not exceed the unwanted emission limits in clauses 6.6.3 and 6.6.4 in the presence of an NR interfering signal according to table 6.7.3.2-1.

**Table 6.7.3.2-1: Interfering and wanted signals for intra-system transmitter intermodulation requirement**

| Parameter | Value |
|---|---|
| Wanted signal type | NR signal |
| Interfering signal type | NR signal of the same *BS channel bandwidth* and SCS as the wanted signal (Note 1). |
| Interfering signal level | Power level declared by the base station manufacturer (Note 2). |
| Frequency offset between interfering signal and wanted signal | 0 MHz |
| NOTE 1: The interfering signal shall be incoherent with the wanted signal.<br>NOTE 2: The declared interfering signal power level at each *TAB connector* is the sum of the co-channel leakage power coupled via the combined RDN and Antenna Array from all the other *TAB connectors*, but does not comprise power radiated from the Antenna Array and reflected back from the environment. The power at each of the interfering *TAB connectors* is $P_{rated,c,TABC}$. | |

## 6.7.3.3 Additional requirements

For Band n41 and n90 operation in Japan, the transmitter intermodulation level shall not exceed the unwanted emission limits in clauses 6.6.3, 6.6.4 and 6.6.5 in the presence of an NR interfering signal according to table 6.7.3.3-1.

**Table 6.7.3.3-1 Interfering and wanted signals for the additional transmitter intermodulation requirement for Band n41 and n90**

| Parameter | Value |
|---|---|
| Wanted signal | NR single carrier (NOTE) |
| Interfering signal type | NR signal of 10 MHz *channel bandwidth* |
| Interfering signal level | Rated total output power in the operating band – 30 dB |
| Interfering signal centre frequency offset from the lower (upper) edge of the wanted signal | ± 5 MHz<br>± 15 MHz<br>± 25 MHz |
| NOTE: This requirement applies for NR carriers allocated within 2545-2645 MHz. | |

**Table 6.7.3.3-2: Void**

For Band n79 operation in Japan, the transmitter intermodulation level shall not exceed the unwanted emission limits in clauses 6.6.3, 6.6.4 and 6.6.5 in the presence of an NR interfering signal according to table 6.7.3.3-3.

**Table 6.7.3.3-3: Interfering and wanted signals for the additional transmitter intermodulation requirement for Band n79**

| Parameter | Value |
|---|---|
| Wanted signal | NR single carrier |
| Interfering signal type | NR signal of 40 MHz *channel bandwidth* |
| Interfering signal level | *Rated total output power* per *TAB connector* ($P_{rated,t,TABC}$) in the *operating band* – 30 dB |
| Interfering signal centre frequency offset from the lower/upper edge of the wanted signal | ± 20 MHz<br>± 60 MHz<br>± 100 MHz |

# 7 Conducted receiver characteristics

## 7.1 General

Conducted receiver characteristics are specified at the *antenna connector* for *BS type 1-C* and at the *TAB connector* for *BS type 1-H*, with full complement of transceivers for the configuration in normal operating condition.

Unless otherwise stated, the following arrangements apply for conducted receiver characteristics requirements in clause 7:

- Requirements apply during the BS receive period.
- Requirements shall be met for any transmitter setting.
- For FDD operation the requirements shall be met with the transmitter unit(s) ON.
- Throughput requirements defined for the radiated receiver characteristics do not assume HARQ retransmissions.
- When BS is configured to receive multiple carriers, all the throughput requirements are applicable for each received carrier.
- For ACS, blocking and intermodulation characteristics, the negative offsets of the interfering signal apply relative to the lower *Base Station RF Bandwidth* edge or *sub-block* edge inside a *sub-block gap*, and the positive offsets of the interfering signal apply relative to the upper *Base Station RF Bandwidth* edge or *sub-block* edge inside a *sub-block gap*.
- Requirements shall also apply for BS supporting NB-IoT operation in NR in-band. The corresponding NB-IoT requirements are specified in TS 36.104 [13] clause 7.

NOTE 1: In normal operating condition the BS in FDD operation is configured to transmit and receive at the same time.

NOTE 2: In normal operating condition the BS in TDD operation is configured to TX OFF power during *receive period*.

For BS *type 1-H* there is no requirement specified for band n46 and n102.

## 7.2 Reference sensitivity level

### 7.2.1 General

The reference sensitivity power level $P_{REFSENS}$ is the minimum mean power received at the *antenna connector* for *BS type 1-C* or *TAB connector* for *BS type 1-H* at which a throughput requirement shall be met for a specified reference measurement channel.

### 7.2.2 Minimum requirements for *BS type 1-C* and *BS type 1-H*

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel as specified in annex A.1 with parameters specified in table 7.2.2-1 for Wide Area BS, in table 7.2.2-2 for Medium Range BS and in table 7.2.2-3 for Local Area BS in any operating band except for band n46, n96, n102 and n104.

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel as specified in annex A.1 with parameters specified in table 7.2.2-2a for Medium Range BS and in table 7.2.2-3a for Local Area BS, for band n46 and n102.

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel as specified in annex A.1 with parameters specified in table 7.2.2-2b for Medium Range BS and in table 7.2.2-3b for Local Area BS, for band n96.

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel as specified in annex A.1 with parameters specified in table 7.2.2-1a for Wide Area BS, in table 7.2.2-2c for Medium Range BS and in table 7.2.2-3c for Local Area BS for band n104.

**Table 7.2.2-1: NR Wide Area BS reference sensitivity levels**

| ***BS channel bandwidth*** **(MHz)** | **Sub-carrier spacing (kHz)** | **Reference measurement channel** | **Reference sensitivity power level, $P_{REFSENS}$ (dBm)** |
|---|---|---|---|
| 5, 10, 15 | 15 | G-FR1-A1-1 (Note 1) | -101.7 |
| | | G-FR1-A1-10 (Note 3) | -101.7 (Note 2) |
| 10, 15 | 30 | G-FR1-A1-2 (Note 1) | -101.8 |
| 10, 15 | 60 | G-FR1-A1-3 (Note 1) | -98.9 |
| 20, 25, 30, 35, 40, 45, 50 | 15 | G-FR1-A1-4 (Note 1) | -95.3 |
| | | G-FR1-A1-11 (Note 4) | -95.3 (Note 2) |
| 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100 | 30 | G-FR1-A1-5 (Note 1) | -95.6 |
| 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100 | 60 | G-FR1-A1-6 (Note 1) | -95.7 |

NOTE 1: $P_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.
NOTE 2: The requirements apply to BS that supports NB-IoT operation in NR in-band.
NOTE 3: $P_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for a single instance of G-FR1-A1-10 mapped to the 24 NR resource blocks adjacent to the NB-IoT PRB, and for each consecutive application of a single instance of G-FR1-A1-1 mapped to disjoint frequency ranges with a width of 25 resource blocks each.
NOTE 4: $P_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for a single instance of G-FR1-A1-11 mapped to the 105 NR resource blocks adjacent to the NB-IoT PRB, and for each consecutive application of a single instance of G-FR1-A1-4 mapped to disjoint frequency ranges with a width of 106 resource blocks each.
NOTE 5: Void.

**Table 7.2.2-1a: NR Wide Area BS reference sensitivity levels for band n104**

| **BS channel bandwidth (MHz)** | **Sub-carrier spacing (kHz)** | **Reference measurement channel** | **Reference sensitivity power level, $P_{REFSENS}$ (dBm) (Note 6)** |
|---|---|---|---|
| 20, 30, 40, 50 | 15 | G-FR1-A1-4 (Note 1) | -94.3 |
| 20, 30, 40, 50, 60, 70, 80, 90, 100 | 30 | G-FR1-A1-5 (Note 1) | -94.6 |
| 20, 30, 40, 50, 60, 70, 80, 90, 100 | 60 | G-FR1-A1-6 (Note 1) | -94.7 |

Note 1: $P_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

**Table 7.2.2-2: NR Medium Range BS reference sensitivity levels**

| ***BS channel bandwidth*** **(MHz)** | **Sub-carrier spacing (kHz)** | **Reference measurement channel (Note 5)** | **Reference sensitivity power level, $P_{REFSENS}$ (dBm)** |
|---|---|---|---|
| 5, 10, 15 | 15 | G-FR1-A1-1 (Note 1) | -96.7 |
| | | G-FR1-A1-10 (Note 3) | -96.7 (Note 2) |
| 10, 15 | 30 | G-FR1-A1-2 (Note 1) | -96.8 |
| 10, 15 | 60 | G-FR1-A1-3 (Note 1) | -93.9 |
| 20, 25, 30, 35, 40, 45, 50 | 15 | G-FR1-A1-4 (Note 1) | -90.3 |
| | | G-FR1-A1-11 (Note 4) | -90.3 (Note 2) |
| 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100 | 30 | G-FR1-A1-5 (Note 1) | -90.6 |
| 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100 | 60 | G-FR1-A1-6 (Note 1) | -90.7 |

Note 1: $P_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

Note 2: The requirements apply to BS that supports NB-IoT operation in NR in-band.

Note 3: $P_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for a single instance of G-FR1-A1-10 mapped to the 24 NR resource blocks adjacent to the NB-IoT PRB, and for each consecutive application of a single instance of G-FR1-A1-1 mapped to disjoint frequency ranges with a width of 25 resource blocks each.

Note 4: $P_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for a single instance of G-FR1-A1-11 mapped to the 105 NR resource blocks adjacent to the NB-IoT PRB, and for each consecutive application of a single instance of G-FR1-A1-4 mapped to disjoint frequency ranges with a width of 106 resource blocks each.

Note 5: These reference measurement channels are not applied for band n46, n96 and n102.

**Table 7.2.2-2a: NR Medium Range BS reference sensitivity levels for band n46**

| *BS channel bandwidth* (MHz) | Sub-carrier spacing (kHz) | Reference measurement channel | **Reference sensitivity power level, $P_{REFSENS}$** (dBm) |
|---|---|---|---|
| 10 | 15 | G-FR1-A1-12 (Note 2) | -103.0 |
| | 30 | G-FR1-A1-13 (Note 2) | -100.7 |
| | 60 | G-FR1-A1-3 (Note 1) | -93.9 |
| 20 | 15 | G-FR1-A1-14 (Note 2) | -100.1 |
| | 30 | G-FR1-A1-15 (Note 2) | -97.1 |
| | 60 | G-FR1-A1-6 (Note 1) | -90.7 |
| 40 | 15 | G-FR1-A1-16 (Note 2) | -97.0 |
| | 30 | G-FR1-A1-17 (Note 2) | -94.0 |
| | 60 | G-FR1-A1-6 (Note 1) | -90.7 |
| 60 | 30 | G-FR1-A1-18 (Note 2) | -92.4 |
| | 60 | G-FR1-A1-6 (Note 1) | -90.7 |
| 80 | 30 | G-FR1-A1-19 (Note 2) | -91.1 |
| | 60 | G-FR1-A1-6 (Note 1) | -90.7 |

Note 1: $P_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

Note 2: $P_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for each interleaved application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

Note 3: For 60kHz SCS reference measurement channel is reused from Table 7.2.2-2.$P_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for each single interlace of FRC G-FR1-A1-12 and G-FR1-A1-19, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

**Table 7.2.2-2b: NR Medium Range BS reference sensitivity levels for band n96 and n102**

| *BS channel bandwidth* (MHz) | Sub-carrier spacing (kHz) | Reference measurement channel | **Reference sensitivity power level, $P_{REFSENS}$** (dBm) |
|---|---|---|---|
| 20 | 15 | G-FR1-A1-14 (Note 2) | -99.1 |
| | 30 | G-FR1-A1-15 (Note 2) | -96.1 |
| | 60 | G-FR1-A1-6 (Note 1) | -89.7 |
| 40 | 15 | G-FR1-A1-16 (Note 2) | -96.0 |
| | 30 | G-FR1-A1-17 (Note 2) | -93.0 |
| | 60 | G-FR1-A1-6 (Note 1) | -89.7 |
| 60 | 30 | G-FR1-A1-18 (Note 2) | -91.4 |
| | 60 | G-FR1-A1-6 (Note 1) | -89.7 |
| 80 | 30 | G-FR1-A1-19 (Note 2) | -90.1 |
| | 60 | G-FR1-A1-6 (Note 1) | -89.7 |

Note 1: $P_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

Note 2: $P_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for each interleaved application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

Note 3: For 60kHz SCS reference measurement channel is reused from Table 7.2.2-2.

**Table 7.2.2-2c: NR Medium Range BS reference sensitivity levels for band n104**

| **BS channel bandwidth (MHz)** | **Sub-carrier spacing (kHz)** | **Reference measurement channel (Note 5)** | **Reference sensitivity power level, $P_{REFSENS}$ (dBm)** |
|---|---|---|---|
| 20, 30, 40, 50 | 15 | G-FR1-A1-4 (Note 1) | -89.3 |
| 20, 30, 40, 50, 60, 70, 80, 90, 100 | 30 | G-FR1-A1-5 (Note 1) | -89.6 |
| 20, 30, 40, 50, 60, 70, 80, 90, 100 | 60 | G-FR1-A1-6 (Note 1) | -89.7 |

Note 1: $P_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

**Table 7.2.2-3: NR Local Area BS reference sensitivity levels**

| ***BS channel bandwidth*** **(MHz)** | **Sub-carrier spacing (kHz)** | **Reference measurement channel (Note 5)** | **Reference sensitivity power level, $P_{REFSENS}$ (dBm)** |
|---|---|---|---|
| 5, 10, 15 | 15 | G-FR1-A1-1 (Note 1) | -93.7 |
| | | G-FR1-A1-10 (Note 3) | -93.7 (Note 2) |
| 10, 15 | 30 | G-FR1-A1-2 (Note 1) | -93.8 |
| 10, 15 | 60 | G-FR1-A1-3 (Note 1) | -90.9 |
| 20, 25, 30, 35, 40, 45, 50 | 15 | G-FR1-A1-4 (Note 1) | -87.3 |
| | | G-FR1-A1-11 (Note 4) | -87.3 (Note 2) |
| 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100 | 30 | G-FR1-A1-5 (Note 1) | -87.6 |
| 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100 | 60 | G-FR1-A1-6 (Note 1) | -87.7 |

Note 1: $P_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

Note 2: The requirements apply to BS that supports NB-IoT operation in NR in-band.

Note 3: $P_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for a single instance of G-FR1-A1-10 mapped to the 24 NR resource blocks adjacent to the NB-IoT PRB, and for each consecutive application of a single instance of G-FR1-A1-1 mapped to disjoint frequency ranges with a width of 25 resource blocks each.

Note 4: $P_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for a single instance of G-FR1-A1-11 mapped to the 105 NR resource blocks adjacent to the NB-IoT PRB, and for each consecutive application of a single instance of G-FR1-A1-4 mapped to disjoint frequency ranges with a width of 106 resource blocks each.

Note 5: These reference measurement channels are not applied for band n46, n96 and n102.

**Table 7.2.2-3a: NR Local Area BS reference sensitivity levels for band n46**

| *BS channel bandwidth* (MHz) | Sub-carrier spacing (kHz) | Reference measurement channel | **Reference sensitivity power level, $P_{REFSENS}$** (dBm) |
|---|---|---|---|
| 10 | 15 | G-FR1-A1-12 (Note 2) | -100.0 |
| | 30 | G-FR1-A1-13 (Note 2) | -97.7 |
| | 60 | G-FR1-A1-3 (Note 1) | -90.9 |
| 20 | 15 | G-FR1-A1-14 (Note 2) | -97.1 |
| | 30 | G-FR1-A1-15 (Note 2) | -94.1 |
| | 60 | G-FR1-A1-6 (Note 1) | -87.7 |
| 40 | 15 | G-FR1-A1-16 (Note 2) | -94.0 |
| | 30 | G-FR1-A1-17 (Note 2) | -91.0 |
| | 60 | G-FR1-A1-6 (Note 1) | -87.7 |
| 60 | 30 | G-FR1-A1-18 (Note 2) | -89.4 |
| | 60 | G-FR1-A1-6 (Note 1) | -87.7 |
| 80 | 30 | G-FR1-A1-19 (Note 2) | -88.1 |
| | 60 | G-FR1-A1-6 (Note 1) | -87.7 |

Note 1: $P_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full BS channel bandwidth.

Note 2: $P_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for each interleaved application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

Note 3: For 60kHz SCS reference measurement channel is reused from Table 7.2.2-3.

**Table 7.2.2-3b: NR Local Area BS reference sensitivity levels for band n96 and n102**

| *BS channel bandwidth* (MHz) | Sub-carrier spacing (kHz) | Reference measurement channel | **Reference sensitivity power level, $P_{REFSENS}$** (dBm) |
|---|---|---|---|
| 20 | 15 | G-FR1-A1-14 (Note 2) | -96.1 |
| | 30 | G-FR1-A1-15 (Note 2) | -93.1 |
| | 60 | G-FR1-A1-6 (Note 1) | -86.7 |
| 40 | 15 | G-FR1-A1-16 (Note 2) | -93.0 |
| | 30 | G-FR1-A1-17 (Note 2) | -90.0 |
| | 60 | G-FR1-A1-6 (Note 1) | -86.7 |
| 60 | 30 | G-FR1-A1-18 (Note 2) | -88.4 |
| | 60 | G-FR1-A1-6 (Note 1) | -86.7 |
| 80 | 30 | G-FR1-A1-19 (Note 2) | -87.1 |
| | 60 | G-FR1-A1-6 (Note 1) | -86.7 |

Note 1: $P_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

Note 2: $P_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for each interleaved application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

Note 3: For 60kHz SCS reference measurement channel is reused from Table 7.2.2-3.

**Table 7.2.2-3c: NR Local Area BS reference sensitivity levels for band n104**

| **BS channel bandwidth (MHz)** | **Sub-carrier spacing (kHz)** | **Reference measurement channel (Note 5)** | **Reference sensitivity power level, $P_{REFSENS}$ (dBm)** |
|---|---|---|---|
| 20, 30, 40, 50 | 15 | G-FR1-A1-4 (Note 1) | -86.3 |
| 20, 30, 40, 50, 60, 70, 80, 90, 100 | 30 | G-FR1-A1-5 (Note 1) | -86.6 |
| 20, 30, 40, 50, 60, 70, 80, 90, 100 | 60 | G-FR1-A1-6 (Note 1) | -86.7 |

Note 1: $P_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

# 7.3 Dynamic range

## 7.3.1 General

The dynamic range is specified as a measure of the capability of the receiver to receive a wanted signal in the presence of an interfering signal at the *antenna connector* for *BS type 1-C* or *TAB connector* for *BS type 1-H* inside the received *BS channel bandwidth*. In this condition, a throughput requirement shall be met for a specified reference measurement channel. The interfering signal for the dynamic range requirement is an AWGN signal.

## 7.3.2 Minimum requirement for *BS type 1-C* and *BS type 1-H*

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel as specified in annex A.2 with parameters specified in table 7.3.2-1 for Wide Area BS, in table 7.3.2-2 for Medium Range BS and in table 7.3.2-3 for Local Area BS in any operating band except for band n46, n96, n102 and n104.

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel as specified in annex A.2 with parameters specified in table 7.3.2-2b for Medium Range BS and in table 7.3.2-3b for Local Area BS, for band n46.

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel as specified in annex A.2 with parameters specified in table 7.3.2-2c for Medium Range BS and in table 7.3.2-3c for Local Area BS, for band n96 and n102.

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel as specified in annex A.2 with parameters specified in table 7.3.2-1b for Wide Area BS, in table 7.3.2-2d for Medium Range BS and in table 7.3.2-3d for Local Area BS in band n104.

For NB-IoT operation in NR in-band, the throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel as specified in Annex A of TS 36.104 [13] with parameters specified in table 7.3.2-1a for Wide Area BS, in table 7.3.2-2a for Medium Range BS and in table 7.3.2-3a for Local Area BS.

**Table 7.3.2-1: Wide Area BS dynamic range**

| ***BS channel bandwidth*** **(MHz)** | **Subcarrier spacing (kHz)** | **Reference measurement channel** | **Wanted signal mean power (dBm)** | **Interfering signal mean power (dBm) / $BW_{Config}$** | **Type of interfering signal** |
|---|---|---|---|---|---|
| 5 | 15 | G-FR1-A2-1 | -70.7 | -82.5 | AWGN |
| | 30 | G-FR1-A2-2 | -71.4 | | |
| 10 | 15 | G-FR1-A2-1 | -70.7 | -79.3 | AWGN |
| | 30 | G-FR1-A2-2 | -71.4 | | |
| | 60 | G-FR1-A2-3 | -68.4 | | |
| 15 | 15 | G-FR1-A2-1 | -70.7 | -77.5 | AWGN |
| | 30 | G-FR1-A2-2 | -71.4 | | |
| | 60 | G-FR1-A2-3 | -68.4 | | |
| 20 | 15 | G-FR1-A2-4 | -64.5 | -76.2 | AWGN |
| | 30 | G-FR1-A2-5 | -64.5 | | |
| | 60 | G-FR1-A2-6 | -64.8 | | |
| 25 | 15 | G-FR1-A2-4 | -64.5 | -75.2 | AWGN |
| | 30 | G-FR1-A2-5 | -64.5 | | |
| | 60 | G-FR1-A2-6 | -64.8 | | |
| 30 | 15 | G-FR1-A2-4 | -64.5 | -74.4 | AWGN |
| | 30 | G-FR1-A2-5 | -64.5 | | |
| | 60 | G-FR1-A2-6 | -64.8 | | |
| 35 | 15 | G-FR1-A2-4 | -64.5 | -73.7 | AWGN |
| | 30 | G-FR1-A2-5 | -64.5 | | |
| | 60 | G-FR1-A2-6 | -64.8 | | |
| 40 | 15 | G-FR1-A2-4 | -64.5 | -73.1 | AWGN |
| | 30 | G-FR1-A2-5 | -64.5 | | |
| | 60 | G-FR1-A2-6 | -64.8 | | |
| 45 | 15 | G-FR1-A2-4 | -64.5 | -72.6 | AWGN |
| | 30 | G-FR1-A2-5 | -64.5 | | |
| | 60 | G-FR1-A2-6 | -64.8 | | |
| 50 | 15 | G-FR1-A2-4 | -64.5 | -72.1 | AWGN |
| | 30 | G-FR1-A2-5 | -64.5 | | |
| | 60 | G-FR1-A2-6 | -64.8 | | |
| 60 | 30 | G-FR1-A2-5 | -64.5 | -71.3 | AWGN |
| | 60 | G-FR1-A2-6 | -64.8 | | |
| 70 | 30 | G-FR1-A2-5 | -64.5 | -70.7 | AWGN |
| | 60 | G-FR1-A2-6 | -64.8 | | |
| 80 | 30 | G-FR1-A2-5 | -64.5 | -70.1 | AWGN |
| | 60 | G-FR1-A2-6 | -64.8 | | |
| 90 | 30 | G-FR1-A2-5 | -64.5 | -69.5 | AWGN |
| | 60 | G-FR1-A2-6 | -64.8 | | |
| 100 | 30 | G-FR1-A2-5 | -64.5 | -69.1 | AWGN |
| | 60 | G-FR1-A2-6 | -64.8 | | |

NOTE: The wanted signal mean power is the power level of a single instance of the corresponding reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

**Table 7.3.2-1a: Wide Area BS dynamic range for NB-IoT operation in NR in-band**

| *BS channel bandwidth* (MHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) / $BW_{Config}$ | Type of interfering signal |
|---|---|---|---|---|
| 5 | FRC A15-1 in Annex A.15 in TS 36.104 [13] | -99.7 | -82.5 | AWGN |
| 10 | | | -79.3 | |
| 15 | | | -77.5 | |
| 20 | | | -76.2 | |
| 25 | | | -75.2 | |
| 30 | | | -74.4 | |
| 35 | | | -73.7 | |
| 40 | | | -73.1 | |
| 45 | | | -72.6 | |
| 50 | | | -72.1 | |
| 5 | FRC A15-2 in Annex A.15 in TS 36.104 [13] | -105.6 | -82.5 | AWGN |
| 10 | | | -79.3 | |
| 15 | | | -77.5 | |
| 20 | | | -76.2 | |
| 25 | | | -75.2 | |
| 30 | | | -74.4 | |
| 35 | | | -73.7 | |
| 40 | | | -73.1 | |
| 45 | | | -72.6 | |
| 50 | | | -72.1 | |

**Table 7.3.2-1b: Wide Area BS dynamic range for n104**

| BS channel bandwidth (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) / $BW_{Config}$ | Type of interfering signal |
|---|---|---|---|---|---|
| 20 | 15 | G-FR1-A2-4 | -63.5 | -75.2 | AWGN |
| | 30 | G-FR1-A2-5 | -63.5 | | |
| | 60 | G-FR1-A2-6 | -63.8 | | |
| 30 | 15 | G-FR1-A2-4 | -63.5 | -73.4 | AWGN |
| | 30 | G-FR1-A2-5 | -63.5 | | |
| | 60 | G-FR1-A2-6 | -63.8 | | |
| 40 | 15 | G-FR1-A2-4 | -63.5 | -72.1 | AWGN |
| | 30 | G-FR1-A2-5 | -63.5 | | |
| | 60 | G-FR1-A2-6 | -63.8 | | |
| 50 | 15 | G-FR1-A2-4 | -63.5 | -71.1 | AWGN |
| | 30 | G-FR1-A2-5 | -63.5 | | |
| | 60 | G-FR1-A2-6 | -63.8 | | |
| 60 | 30 | G-FR1-A2-5 | -63.5 | -70.3 | AWGN |
| | 60 | G-FR1-A2-6 | -63.8 | | |
| 70 | 30 | G-FR1-A2-5 | -63.5 | -69.7 | AWGN |
| | 60 | G-FR1-A2-6 | -63.8 | | |
| 80 | 30 | G-FR1-A2-5 | -63.5 | -69.1 | AWGN |
| | 60 | G-FR1-A2-6 | -63.8 | | |
| 90 | 30 | G-FR1-A2-5 | -63.5 | -68.5 | AWGN |
| | 60 | G-FR1-A2-6 | -63.8 | | |
| 100 | 30 | G-FR1-A2-5 | -63.5 | -68.1 | AWGN |
| | 60 | G-FR1-A2-6 | -63.8 | | |

NOTE 1: The wanted signal mean power is the power level of a single instance of the corresponding reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

**Table 7.3.2-2: Medium Range BS dynamic range**

| *BS channel bandwidth* (MHz) | Subcarrier spacing (kHz) | Reference measurement channel (Note 2) | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) / $BW_{Config}$ | Type of interfering signal |
|---|---|---|---|---|---|
| 5 | 15 | G-FR1-A2-1 | -65.7 | -77.5 | AWGN |
| | 30 | G-FR1-A2-2 | -66.4 | | |
| 10 | 15 | G-FR1-A2-1 | -65.7 | -74.3 | AWGN |
| | 30 | G-FR1-A2-2 | -66.4 | | |
| | 60 | G-FR1-A2-3 | -63.4 | | |
| 15 | 15 | G-FR1-A2-1 | -65.7 | -72.5 | AWGN |
| | 30 | G-FR1-A2-2 | -66.4 | | |
| | 60 | G-FR1-A2-3 | -63.4 | | |
| 20 | 15 | G-FR1-A2-4 | -59.5 | -71.2 | AWGN |
| | 30 | G-FR1-A2-5 | -59.5 | | |
| | 60 | G-FR1-A2-6 | -59.8 | | |
| 25 | 15 | G-FR1-A2-4 | -59.5 | -70.2 | AWGN |
| | 30 | G-FR1-A2-5 | -59.5 | | |
| | 60 | G-FR1-A2-6 | -59.8 | | |
| 30 | 15 | G-FR1-A2-4 | -59.5 | -69.4 | AWGN |
| | 30 | G-FR1-A2-5 | -59.5 | | |
| | 60 | G-FR1-A2-6 | -59.8 | | |
| 35 | 15 | G-FR1-A2-4 | -59.5 | -68.7 | AWGN |
| | 30 | G-FR1-A2-5 | -59.5 | | |
| | 60 | G-FR1-A2-6 | -59.8 | | |
| 40 | 15 | G-FR1-A2-4 | -59.5 | -68.1 | AWGN |
| | 30 | G-FR1-A2-5 | -59.5 | | |
| | 60 | G-FR1-A2-6 | -59.8 | | |
| 45 | 15 | G-FR1-A2-4 | -59.5 | -67.6 | AWGN |
| | 30 | G-FR1-A2-5 | -59.5 | | |
| | 60 | G-FR1-A2-6 | -59.8 | | |
| 50 | 15 | G-FR1-A2-4 | -59.5 | -67.1 | AWGN |
| | 30 | G-FR1-A2-5 | -59.5 | | |
| | 60 | G-FR1-A2-6 | -59.8 | | |
| 60 | 30 | G-FR1-A2-5 | -59.5 | -66.3 | AWGN |
| | 60 | G-FR1-A2-6 | -59.8 | | |
| 70 | 30 | G-FR1-A2-5 | -59.5 | -65.7 | AWGN |
| | 60 | G-FR1-A2-6 | -59.8 | | |
| 80 | 30 | G-FR1-A2-5 | -59.5 | -65.1 | AWGN |
| | 60 | G-FR1-A2-6 | -59.8 | | |
| 90 | 30 | G-FR1-A2-5 | -59.5 | -64.5 | AWGN |
| | 60 | G-FR1-A2-6 | -59.8 | | |
| 100 | 30 | G-FR1-A2-5 | -59.5 | -64.1 | AWGN |
| | 60 | G-FR1-A2-6 | -59.8 | | |

NOTE 1: The wanted signal mean power is the power level of a single instance of the corresponding reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

NOTE 2: These reference measurement channels are not applied for band n46, n96 and n102.

**Table 7.3.2-2a: Medium Range BS dynamic range for NB-IoT operation in NR in-band**

| *BS channel bandwidth* (MHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) / $BW_{Config}$ | Type of interfering signal |
|---|---|---|---|---|
| 5 | FRC A15-1 in Annex A.15 in TS 36.104 [13] | -94.7 | -77.5 | AWGN |
| 10 | | | -74.3 | |
| 15 | | | -72.5 | |
| 20 | | | -71.2 | |
| 25 | | | -70.2 | |
| 30 | | | -69.4 | |
| 35 | | | -68.7 | |
| 40 | | | -68.1 | |
| 45 | | | -67.6 | |
| 50 | | | -67.1 | |
| 5 | FRC A15-2 in Annex A.15 in TS 36.104 [13] | -100.6 | -77.5 | AWGN |
| 10 | | | -74.3 | |
| 15 | | | -72.5 | |
| 20 | | | -71.2 | |
| 25 | | | -70.2 | |
| 30 | | | -69.4 | |
| 35 | | | -68.7 | |
| 40 | | | -68.1 | |
| 45 | | | -67.6 | |
| 50 | | | -67.1 | |

**Table 7.3.2-2b: Medium Range BS dynamic range for band n46**

| *BS channel bandwidth* (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) / $BW_{Config}$ | Type of interfering signal |
|---|---|---|---|---|---|
| 10 | 15 | G-FR1-A2-7 (Note 2) | -72.8 | -74.3 | AWGN |
| | 30 | G-FR1-A2-8 (Note 2) | -70.6 | | |
| | 60 | G-FR1-A2-3 (Note 1, 3) | -63.4 | | |
| 20 | 15 | G-FR1-A2-9 (Note 2) | -69.8 | -71.2 | AWGN |
| | 30 | G-FR1-A2-10 (Note 2) | -66.8 | | |
| | 60 | G-FR1-A2-6 (Note 1, 3) | -59.8 | | |
| 40 | 15 | G-FR1-A2-11 (Note 2) | -66.7 | -68.1 | AWGN |
| | 30 | G-FR1-A2-12 (Note 2) | -63.7 | | |
| | 60 | G-FR1-A2-6 (Note 1, 3) | -59.8 | | |
| 60 | 30 | G-FR1-A2-13 (Note 2) | -61.9 | -66.3 | AWGN |
| | 60 | G-FR1-A2-6 (Note 1, 3) | -59.8 | | |
| 80 | 30 | G-FR1-A2-14 (Note 2) | -60.7 | -65.1 | AWGN |
| | 60 | G-FR1-A2-6 (Note 1, 3) | -59.8 | | |

Note 1: The wanted signal mean power is the power level of a single instance of the corresponding reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

Note 2: The wanted signal mean power is the power level of a single instance of the corresponding reference measurement channel. This requirement shall be met for each interleaved application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

Note 3: For 60kHz SCS reference measurement channel is reused from Table 7.3.2-2.

**Table 7.3.2-2c: Medium Range BS dynamic range for band n96 and n102**

| *BS channel bandwidth* (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) / $BW_{Config}$ | Type of interfering signal |
|---|---|---|---|---|---|
| 20 | 15 | G-FR1-A2-9 (Note 2) | -68.8 | -70.2 | AWGN |
| | 30 | G-FR1-A2-10 (Note 2) | -65.8 | | |
| | 60 | G-FR1-A2-6 (Note 1, 3) | -58.8 | | |
| 40 | 15 | G-FR1-A2-11 (Note 2) | -65.7 | -67.1 | AWGN |
| | 30 | G-FR1-A2-12 (Note 2) | -62.7 | | |
| | 60 | G-FR1-A2-6 (Note 1, 3) | -58.8 | | |
| 60 | 30 | G-FR1-A2-13 (Note 2) | -60.9 | -65.3 | AWGN |
| | 60 | G-FR1-A2-6 (Note 1, 3) | -58.8 | | |
| 80 | 30 | G-FR1-A2-14 (Note 2) | -59.7 | -64.1 | AWGN |
| | 60 | G-FR1-A2-6 (Note 1, 3) | -58.8 | | |

Note 1: The wanted signal mean power is the power level of a single instance of the corresponding reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

Note 2: The wanted signal mean power is the power level of a single instance of the corresponding reference measurement channel. This requirement shall be met for each interleaved application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

Note 3: For 60kHz SCS reference measurement channel is reused from Table 7.3.2-2.

**Table 7.3.2-2d: Medium Range BS dynamic range for n104**

| BS channel bandwidth (MHz) | Subcarrier spacing (kHz) | Reference measurement channel (Note 2) | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) / $BW_{Config}$ | Type of interfering signal |
|---|---|---|---|---|---|
| 20 | 15 | G-FR1-A2-4 | -58.5 | -70.2 | AWGN |
| | 30 | G-FR1-A2-5 | -58.5 | | |
| | 60 | G-FR1-A2-6 | -58.8 | | |
| 30 | 15 | G-FR1-A2-4 | -58.5 | -68.4 | AWGN |
| | 30 | G-FR1-A2-5 | -58.5 | | |
| | 60 | G-FR1-A2-6 | -58.8 | | |
| 40 | 15 | G-FR1-A2-4 | -58.5 | -67.1 | AWGN |
| | 30 | G-FR1-A2-5 | -58.5 | | |
| | 60 | G-FR1-A2-6 | -58.8 | | |
| 50 | 15 | G-FR1-A2-4 | -58.5 | -66.1 | AWGN |
| | 30 | G-FR1-A2-5 | -58.5 | | |
| | 60 | G-FR1-A2-6 | -58.8 | | |
| 60 | 30 | G-FR1-A2-5 | -58.5 | -65.3 | AWGN |
| | 60 | G-FR1-A2-6 | -58.8 | | |
| 70 | 30 | G-FR1-A2-5 | -58.5 | -64.7 | AWGN |
| | 60 | G-FR1-A2-6 | -58.8 | | |
| 80 | 30 | G-FR1-A2-5 | -58.5 | -64.1 | AWGN |
| | 60 | G-FR1-A2-6 | -58.8 | | |
| 90 | 30 | G-FR1-A2-5 | -58.5 | -63.5 | AWGN |
| | 60 | G-FR1-A2-6 | -58.8 | | |
| 100 | 30 | G-FR1-A2-5 | -58.5 | -63.1 | AWGN |
| | 60 | G-FR1-A2-6 | -58.8 | | |

NOTE 1: The wanted signal mean power is the power level of a single instance of the corresponding reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

**Table 7.3.2-3: Local Area BS dynamic range**

| *BS channel bandwidth* (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) / $BW_{Config}$ | Type of interfering signal |
|---|---|---|---|---|---|
| 5 | 15 | G-FR1-A2-1 | -62.7 | -74.5 | AWGN |
| | 30 | G-FR1-A2-2 | -63.4 | | |
| 10 | 15 | G-FR1-A2-1 | -62.7 | -71.3 | AWGN |
| | 30 | G-FR1-A2-2 | -63.4 | | |
| | 60 | G-FR1-A2-3 | -60.4 | | |
| 15 | 15 | G-FR1-A2-1 | -62.7 | -69.5 | AWGN |
| | 30 | G-FR1-A2-2 | -63.4 | | |
| | 60 | G-FR1-A2-3 | -60.4 | | |
| 20 | 15 | G-FR1-A2-4 | -56.5 | -68.2 | AWGN |
| | 30 | G-FR1-A2-5 | -56.5 | | |
| | 60 | G-FR1-A2-6 | -56.8 | | |
| 25 | 15 | G-FR1-A2-4 | -56.5 | -67.2 | AWGN |
| | 30 | G-FR1-A2-5 | -56.5 | | |
| | 60 | G-FR1-A2-6 | -56.8 | | |
| 30 | 15 | G-FR1-A2-4 | -56.5 | -66.4 | AWGN |
| | 30 | G-FR1-A2-5 | -56.5 | | |
| | 60 | G-FR1-A2-6 | -56.8 | | |
| 35 | 15 | G-FR1-A2-4 | -56.5 | -65.7 | AWGN |
| | 30 | G-FR1-A2-5 | -56.5 | | |
| | 60 | G-FR1-A2-6 | -56.8 | | |
| 40 | 15 | G-FR1-A2-4 | -56.5 | -65.1 | AWGN |
| | 30 | G-FR1-A2-5 | -56.5 | | |
| | 60 | G-FR1-A2-6 | -56.8 | | |
| 45 | 15 | G-FR1-A2-4 | -56.5 | -64.6 | AWGN |
| | 30 | G-FR1-A2-5 | -56.5 | | |
| | 60 | G-FR1-A2-6 | -56.8 | | |
| 50 | 15 | G-FR1-A2-4 | -56.5 | -64.1 | AWGN |
| | 30 | G-FR1-A2-5 | -56.5 | | |
| | 60 | G-FR1-A2-6 | -56.8 | | |
| 60 | 30 | G-FR1-A2-5 | -56.5 | -63.3 | AWGN |
| | 60 | G-FR1-A2-6 | -56.8 | | |
| 70 | 30 | G-FR1-A2-5 | -56.5 | -62.7 | AWGN |
| | 60 | G-FR1-A2-6 | -56.8 | | |
| 80 | 30 | G-FR1-A2-5 | -56.5 | -62.1 | AWGN |
| | 60 | G-FR1-A2-6 | -56.8 | | |
| 90 | 30 | G-FR1-A2-5 | -56.5 | -61.5 | AWGN |
| | 60 | G-FR1-A2-6 | -56.8 | | |
| 100 | 30 | G-FR1-A2-5 | -56.5 | -61.1 | AWGN |
| | 60 | G-FR1-A2-6 | -56.8 | | |

NOTE 1: The wanted signal mean power is the power level of a single instance of the corresponding reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

NOTE 2: These reference measurement channels are not applied for band n46, n96 and n102.

**Table 7.3.2-3a: Local Area BS dynamic range for NB-IoT operation in NR in-band**

| *BS channel bandwidth* (MHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) / $BW_{Config}$ | Type of interfering signal |
|---|---|---|---|---|
| 5 | FRC A15-1 in Annex A.15 in TS 36.104 [13] | -91.7 | -74.5 | AWGN |
| 10 | | | -71.3 | |
| 15 | | | -69.5 | |
| 20 | | | -68.2 | |
| 25 | | | -67.2 | |
| 30 | | | -66.4 | |
| 35 | | | -65.7 | |
| 40 | | | -65.1 | |
| 45 | | | -64.6 | |
| 50 | | | -64.1 | |
| 5 | FRC A15-2 in Annex A.15 in TS 36.104 [13] | -97.6 | -74.5 | AWGN |
| 10 | | | -71.3 | |
| 15 | | | -69.5 | |
| 20 | | | -68.2 | |
| 25 | | | -67.2 | |
| 30 | | | -66.4 | |
| 35 | | | -65.7 | |
| 40 | | | -65.1 | |
| 45 | | | -64.6 | |
| 50 | | | -64.1 | |

**Table 7.3.2-3b: Local Area BS dynamic range for band n46**

| *BS channel bandwidth* (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) / $BW_{Config}$ | Type of interfering signal |
|---|---|---|---|---|---|
| 10 | 15 | G-FR1-A2-7 (Note 2) | -69.8 | -71.3 | AWGN |
| | 30 | G-FR1-A2-8 (Note 2) | -67.6 | | |
| | 60 | G-FR1-A2-3 (Note 1, 3) | -60.4 | | |
| 20 | 15 | G-FR1-A2-9 (Note 2) | -66.8 | -68.2 | AWGN |
| | 30 | G-FR1-A2-10 (Note 2) | -63.8 | | |
| | 60 | G-FR1-A2-6 (Note 1, 3) | -56.8 | | |
| 40 | 15 | G-FR1-A2-11 (Note 2) | -63.7 | -65.1 | AWGN |
| | 30 | G-FR1-A2-12 | -60.7 | | |
| | 60 | G-FR1-A2-6 (Note 1, 3) | -56.8 | | |
| 60 | 30 | G-FR1-A2-13 (Note 2) | -58.9 | -63.3 | AWGN |
| | 60 | G-FR1-A2-6 (Note 1, 3) | -56.8 | | |
| 80 | 30 | G-FR1-A2-14 (Note 2) | -57.7 | -62.1 | AWGN |
| | 60 | G-FR1-A2-6 (Note 1, 3) | -56.8 | | |

Note 1: The wanted signal mean power is the power level of a single instance of the corresponding reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

Note 2: The wanted signal mean power is the power level of a single instance of the corresponding reference measurement channel. This requirement shall be met for each interleaved application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

Note 3: For 60kHz SCS reference measurement channel is reused from Table 7.3.2-3.

**Table 7.3.2-3c: Local area BS dynamic range for band n96 and n102**

| *BS channel bandwidth* (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) / $BW_{Config}$ | Type of interfering signal |
|---|---|---|---|---|---|
| 20 | 15 | G-FR1-A2-9 (Note 2) | -65.8 | -67.2 | AWGN |
| | 30 | G-FR1-A2-10 (Note 2) | -62.8 | | |
| | 60 | G-FR1-A2-6 (Note 1, 3) | -55.8 | | |
| 40 | 15 | G-FR1-A2-11 (Note 2) | -62.7 | -64.1 | AWGN |
| | 30 | G-FR1-A2-12 (Note 2) | -59.7 | | |
| | 60 | G-FR1-A2-6 (Note 1, 3) | -55.8 | | |
| 60 | 30 | G-FR1-A2-13 (Note 2) | -57.9 | -62.3 | AWGN |
| | 60 | G-FR1-A2-6 (Note 1, 3) | -55.8 | | |
| 80 | 30 | G-FR1-A2-14 (Note 2) | -56.7 | -61.1 | AWGN |
| | 60 | G-FR1-A2-6 (Note 1, 3) | -55.8 | | |

Note 1: The wanted signal mean power is the power level of a single instance of the corresponding reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

Note 2: The wanted signal mean power is the power level of a single instance of the corresponding reference measurement channel. This requirement shall be met for each interleaved application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

Note 3: For 60kHz SCS reference measurement channel is reused from Table 7.3.2-3.

**Table 7.3.2-3d: Local Area BS dynamic range for n104**

| BS channel bandwidth (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) / $BW_{Config}$ | Type of interfering signal |
|---|---|---|---|---|---|
| 20 | 15 | G-FR1-A2-4 | -55.5 | -67.2 | AWGN |
| | 30 | G-FR1-A2-5 | -55.5 | | |
| | 60 | G-FR1-A2-6 | -55.8 | | |
| 30 | 15 | G-FR1-A2-4 | -55.5 | -65.4 | AWGN |
| | 30 | G-FR1-A2-5 | -55.5 | | |
| | 60 | G-FR1-A2-6 | -55.8 | | |
| 40 | 15 | G-FR1-A2-4 | -55.5 | -64.1 | AWGN |
| | 30 | G-FR1-A2-5 | -55.5 | | |
| | 60 | G-FR1-A2-6 | -55.8 | | |
| 50 | 15 | G-FR1-A2-4 | -55.5 | -63.1 | AWGN |
| | 30 | G-FR1-A2-5 | -55.5 | | |
| | 60 | G-FR1-A2-6 | -55.8 | | |
| 60 | 30 | G-FR1-A2-5 | -55.5 | -62.3 | AWGN |
| | 60 | G-FR1-A2-6 | -55.8 | | |
| 70 | 30 | G-FR1-A2-5 | -55.5 | -61.7 | AWGN |
| | 60 | G-FR1-A2-6 | -55.8 | | |
| 80 | 30 | G-FR1-A2-5 | -55.5 | -61.1 | AWGN |
| | 60 | G-FR1-A2-6 | -55.8 | | |
| 90 | 30 | G-FR1-A2-5 | -55.5 | -60.5 | AWGN |
| | 60 | G-FR1-A2-6 | -55.8 | | |
| 100 | 30 | G-FR1-A2-5 | -55.5 | -60.1 | AWGN |
| | 60 | G-FR1-A2-6 | -55.8 | | |

NOTE 1: The wanted signal mean power is the power level of a single instance of the corresponding reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

# 7.4 In-band selectivity and blocking

## 7.4.1 Adjacent Channel Selectivity (ACS)

### 7.4.1.1 General

Adjacent channel selectivity (ACS) is a measure of the receiver's ability to receive a wanted signal at its assigned channel frequency at the *antenna connector* for *BS type 1-C* or *TAB connector* for *BS type 1-H* in the presence of an adjacent channel signal with a specified centre frequency offset of the interfering signal to the band edge of a victim system.

### 7.4.1.2 Minimum requirement for *BS type 1-C* and *BS type 1-H*

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel.

For BS operating except for band n46, n96, n102 and n104, the wanted and the interfering signal coupled to the *BS type 1-C antenna connector* or *BS type 1-H TAB connector* are specified in table 7.4.1.2-1 and the frequency offset between the wanted and interfering signal in table 7.4.1.2-2 for ACS. The reference measurement channel for the wanted signal is identified in table 7.2.2-1, 7.2.2-2 and 7.2.2-3 for each *BS channel bandwidth* in any operating band except for band n46, n96, n102 and n104 and further specified in annex A.1. The characteristics of the interfering signal is further specified in annex D.

For BS operating in band n46, n96 and n102, the wanted and the interfering signal coupled to the BS type 1-C antenna connector or BS type 1-H TAB connector are specified in table 7.4.1.2-1a and the frequency offset between the wanted and interfering signal in table 7.4.1.2-2a for ACS. The reference measurement channel for the wanted signal is

identified in table 7.2.2-2a and 7.2.2-3a for each *BS channel bandwidth* and further specified in annex A.1a. The characteristics of the interfering signal is further specified in annex D.

For BS operating in band n104, the wanted and the interfering signal coupled to the BS type 1-C antenna connector or BS type 1-H TAB connector are specified in table 7.4.1.2-1b and the frequency offset between the wanted and interfering signal in table 7.4.1.2-2 for ACS. The reference measurement channel for the wanted signal is identified in 7.2.2-1a, 7.2.2-2c, 7.2.2-3c for each *BS channel bandwidth* and further specified in annex A.1. The characteristics of the interfering signal is further specified in annex D.

For BS supporting NB-IoT operation in NR in-band, the wanted and the interfering signal coupled to the *BS type 1-C antenna connector* are specified in table 7.4.1.2-1 and the frequency offset between the wanted and interfering signal in table 7.4.1.2-2 for ACS. The reference measurement channel for the NB-IoT wanted signal is identified in clause 7.2.1 of TS 36.104 [13]. The characteristics of the interfering signal is further specified in annex D.

The ACS requirement is applicable outside the *Base Station RF Bandwidth* or *Radio Bandwidth*. The interfering signal offset is defined relative to the *Base station RF Bandwidth* edges or *Radio Bandwidth* edges.

For a BS operating in *non-contiguous spectrum* within any *operating band*, the ACS requirement shall apply in addition inside any *sub-block gap*, in case the *sub-block gap size* is at least as wide as the NR interfering signal in table 7.4.1.2-2. The interfering signal offset is defined relative to the *sub-block* edges inside the *sub-block gap*.

For a *multi-band connector*, the ACS requirement shall apply in addition inside any *Inter RF Bandwidth gap*, in case the *Inter RF Bandwidth gap* size is at least as wide as the NR interfering signal in table 7.4.1.2-2. The interfering signal offset is defined relative to the *Base Station RF Bandwidth edges* inside the *Inter RF Bandwidth gap*.

Minimum conducted requirement is defined at the *antenna connector* for *BS type 1-C* and at the *TAB connector* for *BS type 1-H*.

**Table 7.4.1.2-1: Base station ACS requirement**

| ***BS channel bandwidth* of the lowest/*highest carrier* received (MHz)** | **Wanted signal mean power (dBm)** | **Interfering signal mean power (dBm)** |
|---|---|---|
| 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100 (Note 1) | $P_{REFSENS}$ + 6 dB | Wide Area BS: -52<br>Medium Range BS: -47<br>Local Area BS: -44 |
| NOTE 1: The SCS for the lowest/highest carrier received is the lowest SCS supported by the BS for that bandwidth.<br>NOTE 2: $P_{REFSENS}$ depends on the RAT. For NR, $P_{REFSENS}$ depends also on the *BS channel bandwidth* as specified in tables 7.2.2-1, 7.2.2-2, 7.2.2-3. For NB-IoT, $P_{REFSENS}$ depends also on the *sub-carrier spacing* as specified in tables 7.2.1-5, 7.2.1-5a and 7.2.1-5c of TS 36.104 [13]. | | |

**Table 7.4.1.2-1a: Base station ACS requirement for band n46, n96 and n102**

| ***BS channel bandwidth* of the lowest/*highest carrier* received (MHz)** | **Wanted signal mean power (dBm)** | **Interfering signal mean power (dBm)** |
|---|---|---|
| 10, 20, 40, 60, 80 (Note 1) | $P_{REFSENS}$ + 6 dB | Medium Range BS: -47<br>Local Area BS: -44 |
| NOTE 1: The SCS for the lowest/highest carrier received is the lowest SCS supported by the BS for that bandwidth.<br>NOTE 2: $P_{REFSENS}$ depends on the RAT. For NR, $P_{REFSENS}$ depends also on the *BS channel bandwidth* as specified in tables 7.2.2-2a, 7.2.2-2b, 7.2.2-3a, 7.2.2-3b.<br>NOTE 3: Void. | | |

**Table 7.4.1.2-1b: Base station ACS requirement for band n104**

| BS channel bandwidth of the lowest/highest carrier received (MHz) | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) |
|---|---|---|
| 20, 30, 40, 50, 60, 70, 80, 90, 100 (Note 1) | $P_{REFSENS}$ + 6 dB | Wide Area BS: -55<br>Medium Range BS: -50<br>Local Area BS: -47 |
| NOTE 1: The SCS for the lowest/highest carrier received is the lowest SCS supported by the BS for that bandwidth.<br>NOTE 2: $P_{REFSENS}$ depends on the RAT. For NR, $P_{REFSENS}$ depends also on the *BS channel bandwidth* as specified in tables 7.2.2-1a, 7.2.2-2c, 7.2.2-3c. | | |

**Table 7.4.1.2-2: Base Station ACS interferer frequency offset values**

| ***BS channel bandwidth* of the *lowest/highest carrier* received (MHz)** | **Interfering signal centre frequency offset from the lower/upper *Base Station RF Bandwidth edge* or *sub-block* edge inside a *sub-block gap* (MHz)** | **Type of interfering signal** |
|---|---|---|
| 5 | ±2.5025 | 5 MHz DFT-s-OFDM NR signal 15 kHz SCS, 25 RBs |
| 10 | ±2.5075 | |
| 15 | ±2.5125 | |
| 20 | ±2.5025 | |
| 25 | ±9.4675 | 20 MHz DFT-s-OFDM NR signal 15 kHz SCS, 100 RBs |
| 30 | ±9.4725 | |
| 35 | ±9.4625 | |
| 40 | ±9.4675 | |
| 45 | ±9.4725 | |
| 50 | ±9.4625 | |
| 60 | ±9.4725 | |
| 70 | ±9.4675 | |
| 80 | ±9.4625 | |
| 90 | ±9.4725 | |
| 100 | ±9.4675 | |

**Table 7.4.1.2-2: Base Station ACS interferer frequency offset values for band n46, n96 and n102**

| ***BS channel bandwidth* of the *lowest/highest carrier* received (MHz)** | **Interfering signal centre frequency offset from the lower/upper *Base Station RF Bandwidth edge* or *sub-block* edge inside a *sub-block gap* (MHz)** | **Type of interfering signal** |
|---|---|---|
| 10 | ±9.4675 | 20 MHz DFT-s-OFDM NR signal 15 kHz SCS, 100 RBs |
| 20 | ±9.4625 | |
| 40 | ±9.4675 | |
| 60 | ±9.4725 | |
| 80 | ±9.4625 | |

#### 7.4.1.3 Void

#### 7.4.1.4 Void

### 7.4.2 In-band blocking

#### 7.4.2.1 General

The in-band blocking characteristics is a measure of the receiver's ability to receive a wanted signal at its assigned channel at the *antenna connector* for *BS type 1-C* or *TAB connector* for *BS type 1-H* in the presence of an unwanted interferer, which is an NR signal for general blocking or an NR signal with one resource block for narrowband blocking.

#### 7.4.2.2 Minimum requirement for *BS type 1-C* and *BS type 1-H*

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel, with a wanted and an interfering signal coupled to *BS type 1-C antenna connector* or *BS type 1-H TAB connector* using the parameters in tables 7.4.2.2-1, 7.4.2.2-2 and 7.4.2.2-3 for general blocking and narrowband blocking requirements. Narrowband blocking requirements are not applied for band n46, n96, n102 and n104. The reference measurement channel for the wanted signal is identified in clause 7.2.2 for each *BS channel bandwidth* and further specified in annex A.1. The characteristics of the interfering signal is further specified in annex D.

For NB-IoT operation in NR in-band, the throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel, with a wanted and an interfering signal coupled to *BS type 1-C antenna connector* using the parameters in tables 7.4.2.2-1, 7.4.2.2-2a and 7.4.2.2-3 for general blocking and narrowband blocking requirements. The reference measurement channel for the NB-IoT wanted signal is identified in clause 7.2.1 of TS 36.104 [13]. The characteristics of the interfering signal is further specified in annex D.

The in-band blocking requirements apply outside the *Base Station RF Bandwidth* or *Radio Bandwidth*. The interfering signal offset is defined relative to the *Base Station RF Bandwidth edges* or *Radio Bandwidth* edges.

The in-band blocking requirement shall apply from $F_{UL,low}$ - $\Delta f_{OOB}$ to $F_{UL,high}$ + $\Delta f_{OOB}$, excluding the downlink frequency range of the FDD *operating band*. The $\Delta f_{OOB}$ for *BS type 1-C* and *BS type 1-H* is defined in table 7.4.2.2-0.

Minimum conducted requirement is defined at the *antenna connector* for *BS type 1-C* and at the *TAB connector* for *BS type 1-H.*

**Table 7.4.2.2-0: $\Delta f_{OOB}$ offset for NR *operating bands***

| BS type | *Operating band* characteristics | $\Delta f_{OOB}$ (MHz) |
|---|---|---|
| *BS type 1-C* | $F_{UL,high} - F_{UL,low} \leq$ 200 MHz | 20 |
| | 200 MHz < $F_{UL,high} - F_{UL,low} \leq$ 900 MHz | 60 |
| | | |
| *BS type 1-H* | $F_{UL,high} - F_{UL,low} <$ 100 MHz | 20 |
| | 100 MHz ≤ $F_{UL,high} - F_{UL,low} \leq$ 900 MHz | 60 |
| | | |

For band n46, n96 and n102, $\Delta f_{OOB}$ is defined in table 7.4.2.2-0a.

**Table 7.4.2.2-0a: $\Delta f_{OOB}$ offset for NR *operating bands* for band n46, n96 and n102**

| Operating band | $\Delta f_{OOB}$ (MHz) |
|---|---|
| n46, n102 | 60 |
| n96 | 70 |

For band n104, $\Delta f_{OOB}$ for *BS type 1-C* and *BS type 1-H* is defined in table 7.4.2.2-0b.

**Table 7.4.2.2-0b: $\Delta f_{OOB}$ offset for NR *operating bands* for band n104**

| BS type | *Operating band* | $\Delta f_{OOB}$ (MHz) |
|---|---|---|
| *BS type 1-H* | n104 | 100 |
| *BS type 1-C* | n104 | 60 |

For a BS operating in *non-contiguous spectrum* within any *operating band*, the in-band blocking requirements apply in addition inside any *sub-block gap*, in case the *sub-block gap* size is at least as wide as twice the interfering signal minimum offset in tables 7.4.2.2-1. The interfering signal offset is defined relative to the *sub-block* edges inside the *sub-block gap*.

For a *multi-band connector*, the blocking requirements apply in the in-band blocking frequency ranges for each supported *operating band*. The requirement shall apply in addition inside any *Inter RF Bandwidth gap*, in case the *Inter RF Bandwidth gap* size is at least as wide as twice the interfering signal minimum offset in tables 7.4.2.2-1.

For a BS operating in *non-contiguous spectrum* within any *operating band*, the narrowband blocking requirement shall apply in addition inside any *sub-block gap*, in case the *sub-block gap size* is at least as wide as the *channel bandwidth* of the NR interfering signal in Table 7.4.2.2-3. The interfering signal offset is defined relative to the *sub-block* edges inside the *sub-block gap*.

For a *multi-band connector*, the narrowband blocking requirement shall apply in addition inside any *Inter RF Bandwidth gap*, in case the *Inter RF Bandwidth gap* size is at least as wide as the NR interfering signal in Table 7.4.2.2-3. The interfering signal offset is defined relative to the *Base Station RF Bandwidth* edges inside the *Inter RF Bandwidth gap*.

**Table 7.4.2.2-1: Base station general blocking requirement**

| ***BS channel bandwidth* of the *lowest/highest carrier* received (MHz)** | **Wanted signal mean power (dBm) (Note 2)** | **Interfering signal mean power (dBm)** | **Interfering signal centre frequency minimum offset from the lower/upper *Base Station RF Bandwidth edge* or *sub-block* edge inside a *sub-block gap* (MHz)** | **Type of interfering signal** |
|---|---|---|---|---|
| 5, 10, 15, 20 | $P_{REFSENS}$ + x dB | Wide Area BS: -43<br>Medium Range BS: -38<br>Local Area BS: -35 | ±7.5 | 5 MHz DFT-s-OFDM NR signal<br>15 kHz SCS, 25 RBs |
| 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100 | $P_{REFSENS}$ + x dB | Wide Area BS: -43<br>Medium Range BS: -38<br>Local Area BS: -35 | ±30 | 20 MHz DFT-s-OFDM NR signal<br>15 kHz SCS, 100 RBs |

NOTE 1: $P_{REFSENS}$ depends on the RAT. For NR, $P_{REFSENS}$ depends also on the *BS channel bandwidth* as specified in tables 7.2.2-1, 7.2.2-2 and 7.2.2-3. For band n104, $P_{REFSENS}$ depends on the *BS channel bandwidth* as specified in tables 7.2.2-1a, 7.2.2-2c, and 7.2.2-3c. For NB-IoT, $P_{REFSENS}$ depends also on the *sub-carrier spacing* as specified in tables 7.2.1-5, 7.2.1-5a and 7.2.1-5c of TS 36.104 [13].

NOTE 2: For a BS capable of single band operation only, "x" is equal to 6 dB. For a BS capable of multi-band operation, "x" is equal to 6 dB in case of interfering signals that are in the in-band blocking frequency range of the operating band where the wanted signal is present or in the in-band blocking frequency range of an adjacent or overlapping operating band. For other in-band blocking frequency ranges of the interfering signal for the supported operating bands, "x" is equal to 1.4 dB.

**Table 7.4.2.2-1a: Base station general blocking requirement for n46**

| BS channel bandwidth of the lowest/highest carrier received (MHz) | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) | Interfering signal centre frequency minimum offset from the lower/upper Base Station RF Bandwidth edge or sub-block edge inside a sub-block gap (MHz) | Type of interfering signal |
|---|---|---|---|---|
| 10, 20, 40, 60, 80 | $P_{REFSENS}$ + 6 dB | Medium Range BS: -38<br>Local Area BS: -35 | ±30 | 20 MHz DFT-s-OFDM NR signal<br>15 kHz SCS, 100 RBs |
| NOTE: $P_{REFSENS}$ depends on the RAT. For NR, $P_{REFSENS}$ depends also on the *BS channel bandwidth* as specified in tables 7.2.2-2a and 7.2.2-3a. | | | | |

**Table 7.4.2.2-1b: Base station general blocking requirement for n96 and n102**

| BS channel bandwidth of the lowest/highest carrier received (MHz) | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) | Interfering signal centre frequency minimum offset from the lower/upper Base Station RF Bandwidth edge or sub-block edge inside a sub-block gap (MHz) | Type of interfering signal |
|---|---|---|---|---|
| 20, 40, 60, 80 | $P_{REFSENS}$ + 6 dB | Medium Range BS: -38<br>Local Area BS: -35 | ±30 | 20 MHz DFT-s-OFDM NR signal<br>15 kHz SCS, 100 RBs |
| NOTE 1: $P_{REFSENS}$ depends on the RAT. For NR, $P_{REFSENS}$ depends also on the *BS channel bandwidth* as specified in tables 7.2.2-2b and 7.2.2-3b. | | | | |

**Table 7.4.2.2-2: Base Station narrowband blocking requirement**

| ***BS channel bandwidth* of the *lowest/highest carrier* received (MHz)** | **Wanted signal mean power (dBm)** | **Interfering signal mean power (dBm)** |
|---|---|---|
| 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80,90, 100 (Note 1) | $P_{REFSENS}$ + 6 dB | Wide Area BS: -49<br>Medium Range BS: -44<br>Local Area BS: -41 |
| NOTE 1: The SCS for the *lowest/highest carrier* received is the lowest SCS supported by the BS for that *BS channel bandwidth*<br>NOTE 2: $P_{REFSENS}$ depends on the *BS channel bandwidth* as specified in tables 7.2.2-1, 7.2.2-2 and 7.2.2-3.<br>NOTE 3: 7.5 kHz shift is not applied to the wanted signal. | | |

**Table 7.4.2.2-2a: Base Station narrowband blocking requirement for NB-IoT operation in NR in-band**

| ***BS channel bandwidth*** **(MHz)** | **Wanted signal mean power (dBm)** | **Interfering signal mean power (dBm)** |
|---|---|---|
| 5, 10, 15, 20, 25, 30, 35, 40, 45, 50 | $P_{REFSENS}$ + x dB (Note 2) | Wide Area: -49<br>Medium Range: -44<br>Local Area: -41 |
| NOTE 1: $P_{REFSENS}$ depends on the *sub-carrier spacing* as specified in tables 7.2.1-5, 7.2.1-5a and 7.2.1-5c of TS 36.104 [13].<br>NOTE 2: "x" is equal to 8 in case of 5 MHz channel bandwidth and equal to 6 otherwise. | | |

**Table 7.4.2.2-3: Base Station narrowband blocking interferer frequency offsets**

| ***BS channel bandwidth*** **of the *lowest/highest carrier* received (MHz)** | **Interfering RB centre frequency offset to the lower/upper *Base Station RF Bandwidth edge* or *sub-block* edge inside a *sub-block gap* (kHz) (Note 2)** | **Type of interfering signal** |
|---|---|---|
| 5 | ±(350+m*180),<br>m=0, 1, 2, 3, 4, 9, 14, 19, 24 | 5 MHz DFT-s-OFDM NR signal, 15 kHz SCS, 1 RB |
| 10 | ±(355+m*180),<br>m=0, 1, 2, 3, 4, 9, 14, 19, 24 | |
| 15 | ±(360+m*180),<br>m=0, 1, 2, 3, 4, 9, 14, 19, 24 | |
| 20 | ±(350+m*180),<br>m=0, 1, 2, 3, 4, 9, 14, 19, 24 | |
| 25 | ±(565+m*180),<br>m=0, 1, 2, 3, 4, 29, 54, 79, 99 | 20 MHz DFT-s-OFDM NR signal, 15 kHz SCS, 1 RB |
| 30 | ±(570+m*180),<br>m=0, 1, 2, 3, 4, 29, 54, 79, 99 | |
| 35 | ±(560+m*180),<br>m=0, 1, 2, 3, 4, 29, 54, 79, 99 | |
| 40 | ±(565+m*180),<br>m=0, 1, 2, 3, 4, 29, 54, 79, 99 | |
| 45 | ±(570+m*180),<br>m=0, 1, 2, 3, 4, 29, 54, 79, 99 | |
| 50 | ±(560+m*180),<br>m=0, 1, 2, 3, 4, 29, 54, 79, 99 | |
| 60 | ±(570+m*180),<br>m=0, 1, 2, 3, 4, 29, 54, 79, 99 | |
| 70 | ±(565+m*180),<br>m=0, 1, 2, 3, 4, 29, 54, 79, 99 | |
| 80 | ±(560+m*180),<br>m=0, 1, 2, 3, 4, 29, 54, 79, 99 | |
| 90 | ±(570+m*180),<br>m=0, 1, 2, 3, 4, 29, 54, 79, 99 | |
| 100 | ±(565+m*180),<br>m=0, 1, 2, 3, 4, 29, 54, 79, 99 | |
| NOTE 1: Interfering signal consisting of one resource block positioned at the stated offset, the *channel bandwidth* of the interfering signal is located adjacently to the lower/upper *Base Station RF Bandwidth edge* or *sub-block* edge inside a *sub-block gap*.<br>NOTE 2: The centre of the interfering RB refers to the frequency location between the two central subcarriers. | | |

### 7.4.2.3 Void

### 7.4.2.4 Void

### 7.4.2.5 Additional narrowband blocking requirement for Band n100

The following requirement shall apply to BS operating in Band n100 in CEPT countries. For the wanted and interfering signal coupled to the antenna connector, using the parameters in table 7.4.2.5-1 and 7.4.2.5-2, the throughput shall be ≥ 95 % of the maximum throughput of the reference measurement channel.

**Table 7.4.2.5-1: Additional narrowband blocking requirement for RMR BS operating in n100**

| BS channel bandwidth of the lowest carrier received (MHz) | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) |
|---|---|---|
| 5 | $P_{REFSENS}$ + 6 dB | Wide Area BS: -39<br>(Note 2) |
| NOTE 1: $P_{REFSENS}$ depends on the *BS channel bandwidth* as specified in clause 7.2.2.<br>NOTE 2: Interfering signal mean power level was derived considering an interfering signal with 10% duty cycle. For more details, refer to TR 38.853 [23]. | | |

**Table 7.4.2.5-2: Interferer frequency for additional narrowband blocking requirement for RMR BS operating in n100**

| BS channel bandwidth of the lowest carrier received (MHz) | Interfering RB centre frequency (Note 2) | Type of interfering signal |
|---|---|---|
| 5 | 874.4 MHz - (350 kHz +m*180 kHz),<br>m=0, 1, 2, 3, 4, 9, 14, 19 | 5 MHz DFT-s-OFDM NR signal,<br>15 kHz SCS, 1 RB |
| NOTE 1: Interfering signal consisting of one resource block positioned at the stated frequency, the *channel bandwidth* of the interfering signal is located adjacently to the lower UL *operating band* edge.<br>NOTE 2: The centre of the interfering RB refers to the frequency location between the two central subcarriers. | | |

## 7.5 Out-of-band blocking

### 7.5.1 General

The out-of-band blocking characteristics is a measure of the receiver ability to receive a wanted signal at its assigned channel at the *antenna connector* for *BS type 1-C* or *TAB connector* for *BS type 1-H* in the presence of an unwanted interferer out of the *operating band*, which is a CW signal for out-of-band blocking.

## 7.5.2 Minimum requirement for *BS type 1-C* and *BS type 1-H*

Except for band n46, n96, n102 and n104, the throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel, with a wanted and an interfering signal coupled to *BS type 1-C antenna connector* or *BS type 1-H TAB connector* using the parameters in table 7.5.2-1.

For band n46, n96 and n102, the throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel, with a wanted and an interfering signal coupled to BS type 1-C antenna connector or BS type 1-H TAB connector using the parameters in table 7.5.2-1a.

For *BS type 1-C* operating in band n104, the throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel, with a wanted and an interfering signal coupled to BS type 1-C antenna connector using the parameters in table 7.5.2-1a.

For *BS type 1-H* operating in band n104, the throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel, with a wanted and an interfering signal coupled to BS type 1-H TAB connector using the parameters in table 7.5.2-1.

The reference measurement channel for the wanted signal is identified in clause 7.2.2 for each *BS channel bandwidth* and further specified in annex A.1. The characteristics of the interfering signal is further specified in annex D.

For NB-IoT operation in NR in-band, the throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel, with a wanted and an interfering signal coupled to *BS type 1-C antenna connector* using the parameters in table 7.5.2-1. The reference measurement channel for the NB-IoT wanted signal is identified in clause 7.2.1 of TS 36.104 [13].

The out-of-band blocking requirement apply from 1 MHz to $F_{UL,low}$ - $\Delta f_{OOB}$ and from $F_{UL,high}$ + $\Delta f_{OOB}$ up to 12750 MHz, including the downlink frequency range of the FDD *operating band* for BS supporting FDD. The $\Delta f_{OOB}$ for *BS type 1-C* and *BS type 1-H* is defined in table 7.4.2.2-0.

Minimum conducted requirement is defined at the *antenna connector* for *BS type 1-C* and at the *TAB connector* for *BS type 1-H.*

For a *multi-band connector*, the requirement in the out-of-band blocking frequency ranges apply for each *operating band*, with the exception that the in-band blocking frequency ranges of all supported *operating bands* according to clause 7.4.2.2 shall be excluded from the out-of-band blocking requirement.

**Table 7.5.2-1: Out-of-band blocking performance requirement for NR**

| Wanted Signal mean power (dBm) | Interfering Signal mean power (dBm) | Type of Interfering Signal |
|---|---|---|
| $P_{REFSENS}$ +6 dB (Note) | -15 | CW carrier |
| NOTE 1: $P_{REFSENS}$ depends on the RAT. For NR, $P_{REFSENS}$ depends also on the *BS channel bandwidth* as specified in Table 7.2.2-1, 7.2.2-2, and 7.2.2-3. For band n104, $P_{REFSENS}$ depends on the *BS channel bandwidth* as specified in tables 7.2.2-1a, 7.2.2-2c, and 7.2.2-3c. For NB-IoT, $P_{REFSENS}$ depends also on the *sub-carrier spacing* as specified in tables 7.2.1-5, 7.2.1-5a and 7.2.1-5c of TS 36.104 [13]. | | |
| NOTE 2: For NB-IoT, up to 24 exceptions are allowed for spurious response frequencies in each wanted signal frequency when measured using a 1MHz step size. For these exceptions the above throughput requirement shall be met when the blocking signal is set to a level of -40 dBm for 15 kHz subcarrier spacing and -46 dBm for 3.75 kHz subcarrier spacing. In addition, each group of exceptions shall not exceed three contiguous measurements using a 1MHz step size. | | |
| NOTE 3: Void | | |

**Table 7.5.2-1a: Out-of-band blocking performance requirement for NR band n46, n96, n102 and n104 for BS type 1-C**

| Operating Band | Centre Frequency of Interfering Signal [MHz] | | | Wanted Signal mean power (dBm) | Interfering Signal mean power (dBm) | Type of Interfering Signal |
|---|---|---|---|---|---|---|
| n46, n96, n102 | ($F_{UL_low}$ -500) to ($F_{UL_low}$ -$\Delta f_{OOB}$) | | | $P_{REFSENS}$ +6dB | -35 | CW carrier |
| | ($F_{UL_high}$ +$\Delta f_{OOB}$) to ($F_{UL_high}$ +500) | | | | | |
| | 1 to ($F_{UL_low}$ -500) | | | $P_{REFSENS}$ +6dB | -15 | CW carrier |
| | ($F_{UL_high}$ +500) to 12750 | | | | | |
| n104 | ($F_{UL_low}$ -100) to ($F_{UL_low}$ -$\Delta f_{OOB}$) | | | $P_{REFSENS}$ +6dB | -35 | CW carrier |
| | ($F_{UL_high}$ +$\Delta f_{OOB}$) to ($F_{UL_high}$ +100) | | | | | |
| | 1 to ($F_{UL_low}$ -100) | | | $P_{REFSENS}$ +6dB | -15 | CW carrier |
| | ($F_{UL_high}$ +100) to 12750 | | | | | |
| NOTE 1: For band n46, n96, n102, $P_{REFSENS}$ depends on the *BS channel bandwidth* as specified in tables 7.2.2-2a, 7.2.2-2b, 7.2.2-3a, 7.2.2-3b. | | | | | | |
| NOTE 2: For band n104, $P_{REFSENS}$ depends on the *BS channel bandwidth* as specified in tables 7.2.2-1a, 7.2.2-2c, 7.2.2-3c. | | | | | | |

## 7.5.3 Co-location minimum requirements for *BS type 1-C* and *BS type 1-H*

This additional blocking requirement may be applied for the protection of NR BS receivers when GSM, CDMA, UTRA, E-UTRA or NR BS operating in a different frequency band are co-located with a NR BS. The requirement is applicable to all *BS channel bandwidths* supported by the NR BS.

The requirements in this clause assume a 30 dB coupling loss between interfering transmitter and NR BS receiver and are based on co-location with base stations of the same class.

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel, with a wanted and an interfering signal coupled to *BS type 1-C antenna connector* or *BS type 1-H TAB connector* input using the parameters in table 7.5.3-1 for all the BS classes. The reference measurement channel for the wanted signal is identified in tables 7.2.2-1, 7.2.2-2 and 7.2.2-3 for each *BS channel bandwidth* and further specified in annex A.1.

The blocking requirement for co-location with BS in other bands is applied for all *operating bands* for which co-location protection is provided.

Minimum conducted requirement is defined at the *antenna connector* for *BS type 1-C* and at the *TAB connector* for *BS type 1-H.*

**Table 7.5.3-1: Blocking performance requirement for NR BS when co-located with BS in other frequency bands.**

| Frequency range of interfering signal | Wanted signal mean power (dBm) | Interfering signal mean power for WA BS (dBm) | Interfering signal mean power for MR BS (dBm) | Interfering signal mean power for LA BS (dBm) | Type of interfering signal |
|---|---|---|---|---|---|
| Frequency range of co-located downlink *operating band* | $P_{REFSENS}$ +6dB (Note 1) | +16 | +8 | x (Note 2) | CW carrier |
| NOTE 1: $P_{REFSENS}$ depends on the *BS channel bandwidth* as specified in Table 7.2.2-1, 7.2.2-2, and 7.2.2-3. | | | | | |
| NOTE 2: x = -7 dBm for NR BS co-located with Pico GSM850 or Pico CDMA850<br>x = -4 dBm for NR BS co-located with Pico DCS1800 or Pico PCS1900<br>x = -6 dBm for NR BS co-located with UTRA bands or E-UTRA bands or NR bands | | | | | |
| NOTE 3: The requirement does not apply when the interfering signal falls within any of the supported uplink *operating band(s)* or in $\Delta f_{OOB}$ immediately outside any of the supported uplink *operating band(s)*. | | | | | |
| NOTE 4: For unsynchronized base stations (except in band n46, n96 and n102), special co-location requirements may apply that are not covered by the 3GPP specifications. | | | | | |

### 7.5.4 Void

### 7.5.5 Additional out-of-band blocking requirements for the use of RMR bands

For the additional out-of-band blocking requirements, the interfering signal differs from the one used for the general out-of-band blocking requirement.

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel, with a wanted and an interfering signal coupled to *BS type 1-C antenna connector* using the parameters in table 7.5.5-1. The reference measurement channel for the wanted signal is identified in clause 7.2.2 for each *BS channel bandwidth* and further specified in annex A.1.

The following requirement may apply to BS operating in band n101 in certain regions.

**Table 7.5.5-1: Additional out-of-band blocking requirement for RMR BS operating in n101**

| BS channel bandwidth of the lowest/highest carrier received (MHz) | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) | Centre frequency of interfering signal (MHz) | Type of interfering signal |
|---|---|---|---|---|
| 5, 10 | $P_{REFSENS}$ + 3 dB | Wide Area BS: -20 | 1807.5 - 1877.5 | 5 MHz LTE signal |
| NOTE: $P_{REFSENS}$ depends on the *BS channel bandwidth* as specified in table 7.2.2-1. | | | | |

## 7.6 Receiver spurious emissions

### 7.6.1 General

The receiver spurious emissions power is the power of emissions generated or amplified in a receiver unit that appear at the *antenna connector* (for *BS type 1-C*) or at the *TAB connector* (for *BS type 1-H*). The requirements apply to all BS with separate RX and TX *antenna connectors* / *TAB connectors*.

NOTE: In this case for FDD operation the test is performed when both TX and RX are ON, with the TX *antenna connectors* / *TAB connectors* terminated.

For a*ntenna connectors* / *TAB connectors* supporting both RX and TX in TDD, the requirements apply during the *transmitter OFF period*. For *antenna connectors* / *TAB connectors* supporting both RX and TX in FDD, the RX spurious emissions requirements are superseded by the TX spurious emissions requirements, as specified in clause 6.6.5.

For RX-only *multi-band connectors*, the spurious emissions requirements are subject to exclusion zones in each supported *operating band*. For *multi-band connectors* that both transmit and receive in *operating band* supporting TDD, RX spurious emissions requirements are applicable during the *TX OFF period*, and are subject to exclusion zones in each supported *operating band*.

For *BS type 1-H* manufacturer shall declare *TAB connector RX min cell groups*. Every *TAB connector* of *BS type 1-H* supporting reception in an *operating band* shall map to one *TAB connector RX min cell group*, where mapping of *TAB connectors* to cells/beams is implementation dependent.

The number of active receiver units that are considered when calculating the conducted RX spurious emission limits ($N_{RXU,counted}$) for *BS type 1-H* is calculated as follows:

$$N_{RXU,counted} = min(N_{RXU,active}, 8 \times N_{cells})$$

$N_{RXU,countedpercell}$ is used for scaling of *basic limits* and is derived as $N_{RXU,countedpercell} = N_{RXU,counted} / N_{cells}$, where $N_{cells}$ is defined in clause 6.1.

NOTE: $N_{RXU,active}$ is the number of actually active receiver units and is independent to the declaration of $N_{cells}$.

### 7.6.2 *Basic limits*

The receiver spurious emissions *basic limits* are provided in table 7.6.2-1.

**Table 7.6.2-1: General BS receiver spurious emissions limits**

| Spurious frequency range | *Basic limits* | *Measurement bandwidth* | Note |
|---|---|---|---|
| 30 MHz – 1 GHz | -57 dBm | 100 kHz | Note 1 |
| 1 GHz – 12.75 GHz | -47 dBm | 1 MHz | Note 1, Note 2 |
| 12.75 GHz – 5th harmonic of the upper frequency edge of the UL *operating band* in GHz | -47 dBm | 1 MHz | Note 1, Note 2, Note 3 |
| 12.75 GHz - 26 GHz | -47 dBm | 1 MHz | Note 1, Note 2, Note 5 |
| NOTE 1: *Measurement bandwidth*s as in ITU-R SM.329 [2], s4.1.<br>NOTE 2: Upper frequency as in ITU-R SM.329 [2], s2.5 table 1.<br>NOTE 3: Applies for Band for which the upper frequency edge of the UL *operating band* is greater than 2.55 GHz and less than or equal to 5.2 GHz.<br>NOTE 4: The frequency range from $\Delta f_{OBUE}$ below the lowest frequency of the BS transmitter *operating band* to $\Delta f_{OBUE}$ above the highest frequency of the BS transmitter *operating band* may be excluded from the requirement. $\Delta f_{OBUE}$ is defined in clause 6.6.1. For *multi-band connectors*, the exclusion applies for all supported *operating bands*.<br>NOTE 5: Applies for Band for which the upper frequency edge of the UL *operating band* is greater than 5.2 GHz. | | | |

## 7.6.3 Minimum requirement for *BS type 1-C*

The RX spurious emissions requirements for *BS type 1-C* are that for each *antenna connector,* the power of emissions shall not exceed *basic limits* specified in table 7.6.2-1.

For Band n41 and n90 operation in Japan, the sum of RX spurious emissions requirements over all *antenna connectors* for *BS type 1-C* shall not exceed *basic limits* specified in table 7.6.2-1.

The Rx spurious emissions requirements shall apply to BS that support NR or NR with NB-IoT operation in NR in-band.

## 7.6.4 Minimum requirement for *BS type 1-H*

The RX spurious emissions requirements for *BS type 1-H* are that for each applicable *basic limit* specified in table 7.6.2-1 for each *TAB connector RX min cell group*, the power sum of emissions at respective *TAB connectors* shall not exceed the BS limits specified as the *basic limit*s + X, where $X = 10\log_{10}(N_{RXU,countedpercell})$, unless stated differently in regional regulation.

The RX spurious emission requirements are applied per the *TAB connector RX min cell group* for all the configurations supported by the BS.

NOTE: Conformance to the BS receiver spurious emissions requirement can be demonstrated by meeting at least one of the following criteria as determined by the manufacturer:

1) The sum of the spurious emissions power measured on each *TAB connector* in the *TAB connector RX min cell group* shall be less than or equal to the BS limit above for the respective frequency span.

Or

2) The spurious emissions power at each *TAB connector* shall be less than or equal to the BS limit as defined above for the respective frequency span, scaled by $-10\log_{10}(n)$, where *n* is the number of *TAB connectors* in the *TAB connector RX min cell group*.

# 7.7 Receiver intermodulation

## 7.7.1 General

Third and higher order mixing of the two interfering RF signals can produce an interfering signal in the band of the desired channel. Intermodulation response rejection is a measure of the capability of the receiver to receive a wanted signal on its assigned channel frequency at the *antenna connector* for *BS type 1-C* or *TAB connector* for *BS type 1-H* in the presence of two interfering signals which have a specific frequency relationship to the wanted signal.

## 7.7.2 Minimum requirement for *BS type 1-C* and *BS type 1-H*

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel, with a wanted signal at the assigned channel frequency and two interfering signals coupled to the *BS type 1-C antenna connector* or *BS type 1-H TAB connector*, with the conditions specified in tables 7.7.2-1 and 7.7.2-2 for intermodulation performance in any operating band except for band n46, n96 and n102, and 7.7.2-2a for band n46, n96 and n102 and in tables 7.7.2-3, and 7.7.2-4 for narrowband intermodulation performance. Narrowband intermodulation requirements are not applied for band n46, n96, n102 and n104. The reference measurement channel for the wanted signal is identified in tables 7.2.2-1, 7.2.2-2 and 7.2.2-3 for each *BS channel bandwidth* and further specified in annex A.1. The characteristics of the interfering signal is further specified in annex D.

For NB-IoT operation in NR in-band, the throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel, with a wanted signal at the assigned channel frequency and two interfering signals coupled to the *BS type 1-C antenna connector*, with the conditions specified in tables 7.7.2-1 and 7.7.2-2 for intermodulation performance and in tables 7.7.2-3, and 7.7.2-4 for narrowband intermodulation performance. The reference measurement channel for the NB-IoT wanted signal is identified in clause 7.2.1 of TS 36.104 [13]. The characteristics of the interfering signal is further specified in annex D.

The subcarrier spacing for the modulated interfering signal shall in general be the same as the subcarrier spacing for the wanted signal, except for the case of wanted signal subcarrier spacing 60 kHz and *BS channel bandwidth* <=20MHz, for which the subcarrier spacing of the interfering signal shall be 30 kHz.

The receiver intermodulation requirement is applicable outside the *Base Station RF Bandwidth* or *Radio Bandwidth edges*. The interfering signal offset is defined relative to the *Base Station RF Bandwidth edges* or *Radio Bandwidth* edges.

For a BS operating in *non-contiguous spectrum* within any *operating band*, the narrowband intermodulation requirement shall apply in addition inside any *sub-block gap* in case the *sub-block gap* is at least as wide as the *channel bandwidth* of the NR interfering signal in table 7.7.2-2 or 7.7.2-4. The interfering signal offset is defined relative to the *sub-block* edges inside the *sub-block gap*.

For a *multi-band connector*, the intermodulation requirement shall apply in addition inside any *Inter RF Bandwidth gap*, in case the gap size is at least twice as wide as the NR interfering signal centre frequency offset from the *Base Station RF Bandwidth edge*.

For a *multi-band connector*, the narrowband intermodulation requirement shall apply in addition inside any *Inter RF Bandwidth gap* in case the gap size is at least as wide as the NR interfering signal in tables 7.7.2-2 and 7.7.2-4. The interfering signal offset is defined relative to the Base Station RF Bandwidth edges inside the *Inter RF Bandwidth gap*.

**Table 7.7.2-1: General intermodulation requirement**

| Base Station Type | Wanted Signal mean power (dBm) | Mean power of interfering signals (dBm) | Type of interfering signals |
|---|---|---|---|
| Wide Area BS | $P_{REFSENS}$ +6 dB | -52 | See Table 7.7.2-2 |
| Medium Range BS | $P_{REFSENS}$ +6 dB | -47 | |
| Local Area BS | $P_{REFSENS}$ +6 dB | -44 | |
| NOTE 1: $P_{REFSENS}$ depends on the RAT and the BS class. For NR, $P_{REFSENS}$ depends also on the *BS channel bandwidth*, see clause 7.2. For NB-IoT, $P_{REFSENS}$ depends also on the *sub-carrier spacing* as specified in tables 7.2.1-5, 7.2.1-5a and 7.2.1-5c of TS 36.104 [13]. | | | |

**Table 7.7.2-1a: General intermodulation requirement for band n46, n96 and n102**

| Base Station Type | Wanted Signal mean power (dBm) | Mean power of interfering signals (dBm) | Type of interfering signals |
|---|---|---|---|
| Medium Range BS | $P_{REFSENS}$ +6 dB | -47 | See Table 7.7.2-2a |
| Local Area BS | $P_{REFSENS}$ +6 dB | -44 | |
| NOTE 1: $P_{REFSENS}$ depends on the RAT and the BS class. For NR, $P_{REFSENS}$ depends also on the *BS channel bandwidth*, see clause 7.2.5. | | | |

**Table 7.7.2-2: Interfering signals for intermodulation requirement**

| ***BS channel bandwidth*** **of the** ***lowest/highest carrier*** **received (MHz)** | **Interfering signal centre frequency offset from the lower/upper** ***Base Station RF Bandwidth*** **edge (MHz)** | **Type of interfering signal (Note 3)** |
|---|---|---|
| 5 | ±7.5 | CW |
| | ±17.5 | 5 MHz DFT-s-OFDM NR signal (Note 1) |
| 10 | ±7.465 | CW |
| | ±17.5 | 5 MHz DFT-s-OFDM NR signal (Note 1) |
| 15 | ±7.43 | CW |
| | ±17.5 | 5 MHz DFT-s-OFDM NR signal (Note 1) |
| 20 | ±7.395 | CW |
| | ±17.5 | 5 MHz DFT-s-OFDM NR signal (Note 1) |
| 25 | ±7.465 | CW |
| | ±25 | 20MHz DFT-s-OFDM NR signal (Note 2) |
| 30 | ±7.43 | CW |
| | ±25 | 20 MHz DFT-s-OFDM NR signal (Note 2) |
| 35 | ±7.44 | CW |
| | ±25 | 20 MHz DFT-s-OFDM NR signal (Note 2) |
| 40 | ±7.45 | CW |
| | ±25 | 20 MHz DFT-s-OFDM NR signal (Note 2) |
| 45 | ±7.37 | CW |
| | ±25 | 20 MHz DFT-s-OFDM NR signal (Note 2) |
| 50 | ±7.38 | CW |
| | ±25 | 20 MHz DFT-s-OFDM NR signal (Note 2) |
| 60 | ±7.49 | CW |
| | ±25 | 20 MHz DFT-s-OFDM NR signal (Note 2) |
| 70 | ±7.42 | CW |
| | ±25 | 20 MHz DFT-s-OFDM NR signal (Note 2) |
| 80 | ±7.44 | CW |
| | ±25 | 20 MHz DFT-s-OFDM NR signal (Note 2) |
| 90 | ±7.46 | CW |
| | ±25 | 20 MHz DFT-s-OFDM NR signal (Note 2) |
| 100 | ±7.48 | CW |
| | ±25 | 20 MHz DFT-s-OFDM NR signal (Note 2) |

NOTE 1: Number of RBs is 25 for 15 kHz subcarrier spacing and 10 for 30 kHz subcarrier spacing.
NOTE 2: Number of RBs is 100 for 15 kHz subcarrier spacing, 50 for 30 kHz subcarrier spacing and 24 for 60 kHz subcarrier spacing.
NOTE 3: The RBs shall be placed adjacent to the transmission bandwidth configuration edge which is closer to the *Base Station RF Bandwidth* edge.

**Table 7.7.2-2a: Interfering signals for intermodulation requirement for band n46, n96 and n102**

| ***BS channel bandwidth of the lowest/highest carrier*** **received (MHz)** | **Interfering signal centre frequency offset from the lower/upper *Base Station RF Bandwidth* edge (MHz)** | **Type of interfering signal (Note 2)** |
|---|---|---|
| 10 | ±7.57 | CW (Note 3) |
| | ±25 | 20 MHz DFT-s-OFDM NR signal (Note 1, 3) |
| 20 | ±7.50 | CW |
| | ±25 | 20 MHz DFT-s-OFDM NR signal (Note 1) |
| 40 | ±7.45 | CW |
| | ±25 | 20 MHz DFT-s-OFDM NR signal (Note 1) |
| 60 | ±7.49 | CW |
| | ±25 | 20 MHz DFT-s-OFDM NR signal (Note 1) |
| 80 | ±7.44 | CW |
| | ±25 | 20 MHz DFT-s-OFDM NR signal (Note 1) |

NOTE 1: Number of RBs is 100 for 15 kHz subcarrier spacing and 50 for 30 kHz subcarrier spacing.
NOTE 2: The RBs shall be placed adjacent to the transmission bandwidth configuration edge which is closer to the *Base Station RF Bandwidth* edge.
NOTE 3: This type of interfering signal is only applied for band n46.

**Table 7.7.2-3: Narrowband intermodulation performance requirement in FR1**

| **BS type** | **Wanted signal mean power (dBm)** | **Interfering signal mean power (dBm)** | **Type of interfering signals** |
|---|---|---|---|
| Wide Area BS | $P_{REFSENS}$ + 6dB (Note 1) | -52 | See Table 7.7.2-4 |
| Medium Range BS | $P_{REFSENS}$ + 6dB (Note 2) | -47 | |
| Local Area BS | $P_{REFSENS}$ + 6dB (Note 3) | -44 | |

NOTE 1: $P_{REFSENS}$ depends on the RAT.
For NR, $P_{REFSENS}$ depends also on the *BS channel bandwidth* as specified in table 7.2.2-1.
For NB-IoT, $P_{REFSENS}$ depends also on the *sub-carrier spacing* as specified in tables 7.2.1-5 of TS 36.104 [13].
NOTE 2: $P_{REFSENS}$ depends on the RAT.
For NR, $P_{REFSENS}$ depends also on the *BS channel bandwidth* as specified in table 7.2.2-2.
For NB-IoT, $P_{REFSENS}$ depends also on the *sub-carrier spacing* as specified in tables 7.2.1-5c of TS 36.104 [13].
NOTE 3: $P_{REFSENS}$ depends on the RAT.
For NR, $P_{REFSENS}$ depends also on the *BS channel bandwidth* as specified in table 7.2.2-3.
For NB-IoT, $P_{REFSENS}$ depends also on the *sub-carrier spacing* as specified in tables 7.2.1-5a of TS 36.104 [13].
NOTE 4: For NB-IoT, the requirement shall apply only for a FRC A1-3 of TS 36.104 [13] mapped to the frequency range at the channel edge adjacent to the interfering signals.
NOTE 5: For NB-IoT, the frequency offset shall be adjusted to accommodate the IMD product to fall in the NB-IoT RB for NB-IoT operation in NR in-band.
NOTE 6: For NB-IoT, if a BS RF receiver fails the test of the requirement, the test shall be performed with the CW interfering signal frequency shifted away from the wanted signal by 180 kHz and the NR interfering signal frequency shifted away from the wanted signal by 360 kHz. If the BS RF receiver still fails the test after the frequency shift, then the BS RF receiver shall be deemed to fail the requirement.

**Table 7.7.2-4: Interfering signals for narrowband intermodulation requirement in FR1**

| ***BS channel bandwidth*** **of the** ***lowest/highest carrier*** **received (MHz)** | **Interfering RB centre frequency offset from the lower/upper** ***Base Station RF Bandwidth edge*** **or** ***sub-block*** **edge inside a** ***sub-block gap*** **(kHz) (Note 3)** | **Type of interfering signal** |
|---|---|---|
| 5 | ±360 | CW |
| | ±1420 | 5 MHz DFT-s-OFDM NR signal, 1 RB (Note 1) |
| 10 | ±370 | CW |
| | ±1960 | 5 MHz DFT-s-OFDM NR signal, 1 RB (Note 1) |
| 15 (Note 2) | ±380 | CW |
| | ±1960 | 5 MHz DFT-s-OFDM NR signal, 1 RB (Note 1) |
| 20 (Note 2) | ±390 | CW |
| | ±2320 | 5 MHz DFT-s-OFDM NR signal, 1 RB (Note 1) |
| 25 (Note 2) | ±325 | CW |
| | ±2350 | 20 MHz DFT-s-OFDM NR signal, 1 RB (Note 1) |
| 30 (Note 2) | ±335 | CW |
| | ±2350 | 20 MHz DFT-s-OFDM NR signal, 1 RB (Note 1) |
| 35 (Note 2) | ±345 | CW |
| | ±2710 | 20 MHz DFT-s-OFDM NR signal, 1 RB (Note 1) |
| 40 (Note 2) | ±355 | CW |
| | ±2710 | 20 MHz DFT-s-OFDM NR signal, 1 RB (Note 1) |
| 45 (Note 2) | ±365 | CW |
| | ±2710 | 20 MHz DFT-s-OFDM NR signal, 1 RB (Note 1) |
| 50 (Note 2) | ±375 | CW |
| | ±2710 | 20 MHz DFT-s-OFDM NR signal, 1 RB (Note 1) |
| 60 (Note 2) | ±395 | CW |
| | ±2710 | 20 MHz DFT-s-OFDM NR signal, 1 RB (Note 1) |
| 70 (Note 2) | ±415 | CW |
| | ±2710 | 20 MHz DFT-s-OFDM NR signal, 1 RB (Note 1) |
| 80 (Note 2) | ±435 | CW |
| | ±2710 | 20 MHz DFT-s-OFDM NR signal, 1 RB (Note 1) |
| 90 (Note 2) | ±365 | CW |
| | ±2530 | 20 MHz DFT-s-OFDM NR signal, 1 RB (Note 1) |
| 100 (Note 2) | ±385 | CW |
| | ±2530 | 20 MHz DFT-s-OFDM NR signal, 1 RB (Note 1) |

NOTE 1: Interfering signal consisting of one resource block positioned at the stated offset, the *BS channel bandwidth* of the interfering signal is located adjacently to the lower/upper *Base Station RF Bandwidth edge* or *sub-block* edge inside a *sub-block gap*.

NOTE 2: This requirement shall apply only for a G-FRC mapped to the frequency range at the *channel edge* adjacent to the interfering signals.

NOTE 3: The centre of the interfering RB refers to the frequency location between the two central subcarriers.

### 7.7.3 Additional narrowband intermodulation requirement for Band n100

The following requirement shall apply to BS operating in Band n100 in CEPT countries. The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel, with a wanted signal at the assigned channel frequency and two interfering signals coupled to the *BS type 1-C antenna connector* operating in Band n100, with the conditions specified in tables 7.7.3-1 and 7.7.3-2 for narrowband intermodulation performance. The reference measurement channel for the wanted signal is identified in table 7.2.2-1 and further specified in annex A.1. The characteristics of the interfering signal is further specified in annex D.

The receiver intermodulation requirement is applicable outside the *Base Station RF Bandwidth*. The interfering signal offset is defined relative to the *Base Station RF Bandwidth edges*.

**Table 7.7.3-1: Additional narrowband intermodulation requirement for RMR BS operating in n100**

| Wanted signal mean power (dBm) | Interfering signal mean power (dBm) | Type of interfering signals |
|---|---|---|
| $P_{REFSENS}$ + 6dB | Wide Area BS: -39<br>(Note 2) | See Table 7.7.3-2 |
| NOTE 1: $P_{REFSENS}$ depends also on the *BS channel bandwidth* as specified in clause 7.2.2.<br>NOTE 2: Interfering signal mean power level was derived considering an interfering signal with 10% duty cycle. For more details, refer to TR 38.853 [23]. | | |

**Table 7.7.3-2: Interfering signals for the additional narrowband intermodulation requirement for RMR BS operating in n100**

| BS channel bandwidth of the lowest carrier received (MHz) | Interfering RB centre frequency offset from the lower Base Station RF Bandwidth edge (kHz)<br>(Note 3) | Type of interfering signal |
|---|---|---|
| 5 | -360 | CW |
| | -1420 | 5 MHz DFT-s-OFDM NR signal, 15 kHz SCS, 1 RB<br>(Note 1) |
| NOTE 1: Interfering signal consisting of one resource block positioned at the stated offset, the *BS channel bandwidth* of the interfering signal is located adjacently to the lower *Base Station RF Bandwidth edge*.<br>NOTE 2: This requirement shall apply only for a G-FRC mapped to the frequency range at the *channel edge* adjacent to the interfering signals.<br>NOTE 3: The centre of the interfering RB refers to the frequency location between the two central subcarriers. | | |

## 7.8 In-channel selectivity

### 7.8.1 General

In-channel selectivity (ICS) is a measure of the receiver ability to receive a wanted signal at its assigned resource block locations at the *antenna connector* for *BS type 1-C* or *TAB connector* for *BS type 1-H* in the presence of an interfering signal received at a larger power spectral density. In this condition a throughput requirement shall be met for a specified reference measurement channel. The interfering signal shall be an NR signal which is time aligned with the wanted signal.

### 7.8.2 Minimum requirement for *BS type 1-C* and *BS type 1-H*

For *BS type 1-C* and *BS type 1-H*, the throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel as specified in annex A.1 with parameters specified in table 7.8.2-1 for Wide Area BS except for band n104, in table 7.8.2-1b for Wide Area BS for band n104, in table 7.8.2-2 for Medium Range BS except for band n46, n96, n102 and n104, in table 7.8.2-2b for Medium Range BS for band n46, in table 7.8.2-2c for Medium Range BS for band n96 and n102, in table 7.8.2-2d for Medium Range BS for band n104, in table 7.8.2-3 for Local Area BS except for band n46, n96, n102 and n104, in table 7.8.2-3b for Local Area BS for band n46, in table 7.8.2-3c for Local Area BS for band n96 and n102 and in table 7.8.2-3d for Local Area BS. The characteristics of the interfering signal is further specified in annex D.

For NB-IoT operation in NR in-band, the throughput shall be ≥ 95% of the maximum throughput of the NB-IoT reference measurement channel as specified in Annex A of TS 36.104 [13] with parameters specified in table 7.8.2-1a for Wide Area BS, in table 7.8.2-2a for Medium Range BS and in table 7.8.2-3a for Local Area BS.

**Table 7.8.2-1: Wide Area BS in-channel selectivity**

| ***BS channel bandwidth*** **(MHz)** | **Subcarrier spacing (kHz)** | **Reference measurement channel** | **Wanted signal mean power (dBm)** | **Interfering signal mean power (dBm)** | **Type of interfering signal** |
|---|---|---|---|---|---|
| 5 | 15 | G-FR1-A1-7 | -100.6 | -81.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 10 RBs |
| 10, 15, 20, 25, 30, 35 | 15 | G-FR1-A1-1 | -98.7 | -77.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 25 RBs |
| 40, 45, 50 | 15 | G-FR1-A1-4 | -92.3 | -71.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 100 RBs |
| 5 | 30 | G-FR1-A1-8 | -101.3 | -81.4 | DFT-s-OFDM NR signal, 30 kHz SCS, 5 RBs |
| 10, 15, 20, 25, 30, 35 | 30 | G-FR1-A1-2 | -98.8 | -78.4 | DFT-s-OFDM NR signal, 30 kHz SCS, 10 RBs |
| 40, 45, 50, 60, 70, 80, 90, 100 | 30 | G-FR1-A1-5 | -92.6 | -71.4 | DFT-s-OFDM NR signal, 30 kHz SCS, 50 RBs |
| 10, 15, 20, 25, 30, 35 | 60 | G-FR1-A1-9 | -98.2 | -78.4 | DFT-s-OFDM NR signal, 60 kHz SCS, 5 RBs |
| 40, 45, 50, 60, 70, 80, 90, 100 | 60 | G-FR1-A1-6 | -92.7 | -71.6 | DFT-s-OFDM NR signal, 60 kHz SCS, 24 RBs |

NOTE: Wanted and interfering signal are placed adjacently around $F_c$, where the $F_c$ is defined for *BS channel bandwidth* of the wanted signal according to the table 5.4.2.2-1. The aggregated wanted and interferer signal shall be centred in the *BS channel bandwidth* of the wanted signal.

**Table 7.8.2-1a: Wide Area BS in-channel selectivity for NB-IoT operation in NR in-band**

| *BS channel bandwidth* (MHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) / $BW_{Config}$ | Type of interfering signal |
|---|---|---|---|---|
| 5 | FRC A14-1 in Annex A.14 in TS 36.104 [13] | -124.3 | -81.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 10 RBs |
| 10, 15, 20, 25, 30, 35 | | | -77.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 25 RBs |
| 40, 45, 50 | | | -71.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 100 RBs |
| 5 | FRC A14-2 in Annex A.14 in TS 36.104 [13] | -130.2 | -81.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 10 RBs |
| 10, 15, 20, 25, 30, 35 | | | -77.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 25 RBs |
| 40, 45, 50 | | | -71.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 100 RBs |
| NOTE: Interfering signal is placed in one side of the $F_c$, while the NB-IoT PRB is placed on the other side. Both interfering signal and NB-IoT PRB are placed at the middle of the available PRB locations. The wanted NB-IoT tone is placed at the centre of this NB-IoT PRB. | | | | |

**Table 7.8.2-1b: Wide Area BS in-channel selectivity for band n104**

| BS channel bandwidth (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) | Type of interfering signal |
|---|---|---|---|---|---|
| 20, 30 | 15 | G-FR1-A1-1 | -97.7 | -76.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 25 RBs |
| 40, 50 | 15 | G-FR1-A1-4 | -91.3 | -70.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 100 RBs |
| 20, 30 | 30 | G-FR1-A1-2 | -97.8 | -77.4 | DFT-s-OFDM NR signal, 30 kHz SCS, 10 RBs |
| 40, 50, 60, 70, 80, 90, 100 | 30 | G-FR1-A1-5 | -91.6 | -70.4 | DFT-s-OFDM NR signal, 30 kHz SCS, 50 RBs |
| 20, 30 | 60 | G-FR1-A1-9 | -97.2 | -77.4 | DFT-s-OFDM NR signal, 60 kHz SCS, 5 RBs |
| 40, 50, 60, 70, 80, 90, 100 | 60 | G-FR1-A1-6 | -91.7 | -70.6 | DFT-s-OFDM NR signal, 60 kHz SCS, 24 RBs |
| NOTE 1: Wanted and interfering signal are placed adjacently around $F_c$, where the $F_c$ is defined for *BS channel bandwidth* of the wanted signal according to the table 5.4.2.2-1. The aggregated wanted and interferer signal shall be centred in the *BS channel bandwidth* of the wanted signal. | | | | | |

**Table 7.8.2-2: Medium Range BS in-channel selectivity**

| *BS channel bandwidth* (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) | Type of interfering signal |
|---|---|---|---|---|---|
| 5 | 15 | G-FR1-A1-7 | -95.6 | -76.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 10 RBs |
| 10, 15, 20, 25, 30, 35 | 15 | G-FR1-A1-1 | -93.7 | -72.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 25 RBs |
| 40, 45, 50 | 15 | G-FR1-A1-4 | -87.3 | -66.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 100 RBs |
| 5 | 30 | G-FR1-A1-8 | -96.3 | -76.4 | DFT-s-OFDM NR signal, 30 kHz SCS, 5 RBs |
| 10, 15, 20, 25, 30, 35 | 30 | G-FR1-A1-2 | -93.8 | -73.4 | DFT-s-OFDM NR signal, 30 kHz SCS, 10 RBs |
| 40, 45, 50, 60, 70, 80, 90, 100 | 30 | G-FR1-A1-5 | -87.6 | -66.4 | DFT-s-OFDM NR signal, 30 kHz SCS, 50 RBs |
| 10, 15, 20, 25, 30, 35 | 60 | G-FR1-A1-9 | -93.2 | -73.4 | DFT-s-OFDM NR signal, 60 kHz SCS, 5 RBs |
| 40, 45, 50, 60, 70, 80, 90, 100 | 60 | G-FR1-A1-6 | -87.7 | -66.6 | DFT-s-OFDM NR signal, 60 kHz SCS, 24 RBs |
| NOTE: Wanted and interfering signal are placed adjacently around $F_c$, where the $F_c$ is defined for *BS channel bandwidth* of the wanted signal according to the table 5.4.2.2-1. The aggregated wanted and interferer signal shall be centred in the *BS channel bandwidth* of the wanted signal. | | | | | |

**Table 7.8.2-2a: Medium Range BS in-channel selectivity for NB-IoT operation in NR in-band**

| *BS channel bandwidth* (MHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) / $BW_{Config}$ | Type of interfering signal |
|---|---|---|---|---|
| 5 | FRC A14-1 in Annex A.14 in TS 36.104 [13] | -119.3 | -76.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 10 RBs |
| 10, 15, 20, 25, 30, 35 | | | -72.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 25 RBs |
| 40, 45, 50 | | | -66.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 100 RBs |
| 5 | FRC A14-2 in Annex A.14 in TS 36.104 [13] | -125.2 | -76.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 10 RBs |
| 10, 15, 20, 25, 30, 35 | | | -72.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 25 RBs |
| 40, 45, 50 | | | -66.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 100 RBs |
| NOTE: Interfering signal is placed in one side of the $F_c$, while the NB-IoT PRB is placed on the other side. Both interfering signal and NB-IoT PRB are placed at the middle of the available PRB locations. The wanted NB-IoT tone is placed at the centre of this NB-IoT PRB. | | | | |

**Table 7.8.2-2b: Medium Range BS in-channel selectivity for band n46**

| *BS channel bandwidth* (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) | Type of interfering signal |
|---|---|---|---|---|---|
| 10 | 15 | G-FR1-A1-12 | -100.0 | -79.5 | CP-OFDM NR signal, 15 kHz SCS, 10 RBs |
| | 30 | G-FR1-A1-13 | -97.7 | -77.4 | CP-OFDM NR signal, 30 kHz SCS, 10 RBs |
| | 60 | G-FR1-A1-9 | -93.2 | -73.4 | DFT-s-OFDM NR signal, 60 kHz SCS, 5 RBs |
| 20 | 15 | G-FR1-A1-14 | -97.1 | -76.4 | CP-OFDM NR signal, 15 kHz SCS, 10 RBs |
| | 30 | G-FR1-A1-15 | -94.1 | -73.4 | CP-OFDM NR signal, 30 kHz SCS, 10 RBs |
| | 60 | G-FR1-A1-9 | -93.2 | -73.4 | DFT-s-OFDM NR signal, 60 kHz SCS, 5 RBs |
| 40 | 15 | G-FR1-A1-16 | -94.0 | -73.2 | CP-OFDM NR signal, 15 kHz SCS, 20 RBs |
| | 30 | G-FR1-A1-17 | -91.0 | -70.2 | CP-OFDM NR signal, 30 kHz SCS, 10 RBs |
| | 60 | G-FR1-A1-6 | -87.7 | -66.6 | DFT-s-OFDM NR signal, 60 kHz SCS, 24 RBs |
| 60 | 30 | G-FR1-A1-18 | -89.4 | -68.4 | CP-OFDM NR signal, 30 kHz SCS, 20 RBs |
| | 60 | G-FR1-A1-6 | -87.7 | -66.6 | DFT-s-OFDM NR signal, 60 kHz SCS, 24 RBs |
| 80 | 30 | G-FR1-A1-19 | -88.1 | -67.1 | CP-OFDM NR signal, 30 kHz SCS, 20 RBs |
| | 60 | G-FR1-A1-6 | -87.7 | -66.6 | DFT-s-OFDM NR signal, 60 kHz SCS, 24 RBs |

NOTE: Wanted and interfering signal are placed adjacently around $F_c$, where the $F_c$ is defined for *BS channel bandwidth* of the wanted signal according to the table 5.4.2.2-1. The aggregated wanted and interferer signal shall be centred in the BS channel bandwidth of the wanted signal.

**Table 7.8.2-2c: Medium Range BS in-channel selectivity for band n96 and n102**

| *BS channel bandwidth* (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) | Type of interfering signal |
|---|---|---|---|---|---|
| 20 | 15 | G-FR1-A1-14 | -96.1 | -75.4 | CP-OFDM NR signal, 15 kHz SCS, 10 RBs |
| | 30 | G-FR1-A1-15 | -93.1 | -72.4 | CP-OFDM NR signal, 30 kHz SCS, 10 RBs |
| | 60 | G-FR1-A1-9 | -92.2 | -72.4 | DFT-s-OFDM NR signal, 60 kHz SCS, 5 RBs |
| 40 | 15 | G-FR1-A1-16 | -93.0 | -72.2 | CP-OFDM NR signal, 15 kHz SCS, 20 RBs |
| | 30 | G-FR1-A1-17 | -90.0 | -69.2 | CP-OFDM NR signal, 30 kHz SCS, 10 RBs |
| | 60 | G-FR1-A1-6 | -86.7 | -65.6 | DFT-s-OFDM NR signal, 60 kHz SCS, 24 RBs |
| 60 | 30 | G-FR1-A1-18 | -88.4 | -67.4 | CP-OFDM NR signal, 30 kHz SCS, 20 RBs |
| | 60 | G-FR1-A1-6 | -86.7 | -65.6 | DFT-s-OFDM NR signal, 60 kHz SCS, 24 RBs |
| 80 | 30 | G-FR1-A1-19 | -87.1 | -66.1 | CP-OFDM NR signal, 30 kHz SCS, 20 RBs |
| | 60 | G-FR1-A1-6 | -86.7 | -65.6 | DFT-s-OFDM NR signal, 60 kHz SCS, 24 RBs |
| NOTE: Wanted and interfering signal are placed adjacently around $F_c$, where the $F_c$ is defined for *BS channel bandwidth* of the wanted signal according to the table 5.4.2.2-1. The aggregated wanted and interferer signal shall be centred in the BS channel bandwidth of the wanted signal. | | | | | |

**Table 7.8.2-2d: Medium Range BS in-channel selectivity for band n104**

| BS channel bandwidth (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) | Type of interfering signal |
|---|---|---|---|---|---|
| 20, 30 | 15 | G-FR1-A1-1 | -92.7 | -71.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 25 RBs |
| 40, 50 | 15 | G-FR1-A1-4 | -86.3 | -65.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 100 RBs |
| 20, 30 | 30 | G-FR1-A1-2 | -92.8 | -72.4 | DFT-s-OFDM NR signal, 30 kHz SCS, 10 RBs |
| 40, 50, 60, 70, 80, 90, 100 | 30 | G-FR1-A1-5 | -86.6 | -65.4 | DFT-s-OFDM NR signal, 30 kHz SCS, 50 RBs |
| 20, 30 | 60 | G-FR1-A1-9 | -92.2 | -72.4 | DFT-s-OFDM NR signal, 60 kHz SCS, 5 RBs |
| 40, 50, 60, 70, 80, 90, 100 | 60 | G-FR1-A1-6 | -86.7 | -65.6 | DFT-s-OFDM NR signal, 60 kHz SCS, 24 RBs |

NOTE: Wanted and interfering signal are placed adjacently around $F_c$, where the $F_c$ is defined for *BS channel bandwidth* of the wanted signal according to the table 5.4.2.2-1. The aggregated wanted and interferer signal shall be centred in the *BS channel bandwidth* of the wanted signal.

**Table 7.8.2-3: Local area BS in-channel selectivity**

| *BS channel bandwidth* (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) | Type of interfering signal |
|---|---|---|---|---|---|
| 5 | 15 | G-FR1-A1-7 | -92.6 | -73.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 10 RBs |
| 10, 15, 20, 25, 30, 35 | 15 | G-FR1-A1-1 | -90.7 | -69.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 25 RBs |
| 40, 45, 50 | 15 | G-FR1-A1-4 | -84.3 | -63.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 100 RBs |
| 5 | 30 | G-FR1-A1-8 | -93.3 | -73.4 | DFT-s-OFDM NR signal, 30 kHz SCS, 5 RBs |
| 10, 15, 20, 25, 30, 35 | 30 | G-FR1-A1-2 | -90.8 | -70.4 | DFT-s-OFDM NR signal, 30 kHz SCS, 10 RBs |
| 40, 45, 50, 60, 70, 80, 90, 100 | 30 | G-FR1-A1-5 | -84.6 | -63.4 | DFT-s-OFDM NR signal, 30 kHz SCS, 50 RBs |
| 10, 15, 20, 25, 30, 35 | 60 | G-FR1-A1-9 | -90.2 | -70.4 | DFT-s-OFDM NR signal, 60 kHz SCS, 5 RBs |
| 40, 45, 50, 60, 70, 80, 90, 100 | 60 | G-FR1-A1-6 | -84.7 | -63.6 | DFT-s-OFDM NR signal, 60 kHz SCS, 24 RBs |
| NOTE: Wanted and interfering signal are placed adjacently around $F_c$, where the $F_c$ is defined for *BS channel bandwidth* of the wanted signal according to the table 5.4.2.2-1. The aggregated wanted and interferer signal shall be centred in the *BS channel bandwidth* of the wanted signal. | | | | | |

**Table 7.8.2-3a: Local Area BS in-channel selectivity for NB-IoT operation in NR in-band**

| *BS channel bandwidth* (MHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) / $BW_{Config}$ | Type of interfering signal |
|---|---|---|---|---|
| 5 | FRC A14-1 in Annex A.14 in TS 36.104 [13] | -116.3 | -73.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 10 RBs |
| 10, 15, 20, 25, 30, 35 | | | -69.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 25 RBs |
| 40, 45, 50 | | | -63.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 100 RBs |
| 5 | FRC A14-2 in Annex A.14 in TS 36.104 [13] | -122.2 | -73.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 10 RBs |
| 10, 15, 20, 25, 30, 35 | | | -69.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 25 RBs |
| 40, 45, 50 | | | -63.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 100 RBs |
| NOTE: Interfering signal is placed in one side of the $F_c$, while the NB-IoT PRB is placed on the other side. Both interfering signal and NB-IoT PRB are placed at the middle of the available PRB locations. The wanted NB-IoT tone is placed at the centre of this NB-IoT PRB. | | | | |

**Table 7.8.2-3b: Local Area BS in-channel selectivity for band n46**

| *BS channel bandwidth* (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) | Type of interfering signal |
|---|---|---|---|---|---|
| 10 | 15 | G-FR1-A1-12 | -97.0 | -76.5 | CP-OFDM NR signal, 15 kHz SCS, 10 RBs |
| | 30 | G-FR1-A1-13 | -94.7 | -74.4 | CP-OFDM NR signal, 30 kHz SCS, 10 RBs |
| | 60 | G-FR1-A1-9 | -90.2 | -70.4 | DFT-s-OFDM NR signal, 60 kHz SCS, 5 RBs |
| 20 | 15 | G-FR1-A1-14 | -94.1 | -73.4 | CP-OFDM NR signal, 15 kHz SCS, 10 RBs |
| | 30 | G-FR1-A1-15 | -91.1 | -70.4 | CP-OFDM NR signal, 30 kHz SCS, 10 RBs |
| | 60 | G-FR1-A1-9 | -90.2 | -70.4 | DFT-s-OFDM NR signal, 60 kHz SCS, 5 RBs |
| 40 | 15 | G-FR1-A1-16 | -91.0 | -70.2 | CP-OFDM NR signal, 15 kHz SCS, 20 RBs |
| | 30 | G-FR1-A1-17 | -88.0 | -67.2 | CP-OFDM NR signal, 30 kHz SCS, 10 RBs |
| | 60 | G-FR1-A1-6 | -84.7 | -63.6 | DFT-s-OFDM NR signal, 60 kHz SCS, 24 RBs |
| 60 | 30 | G-FR1-A1-18 | -86.4 | -65.4 | CP-OFDM NR signal, 30 kHz SCS, 20 RBs |
| | 60 | G-FR1-A1-6 | -84.7 | -63.6 | DFT-s-OFDM NR signal, 60 kHz SCS, 24 RBs |
| 80 | 30 | G-FR1-A1-19 | -85.1 | -64.1 | CP-OFDM NR signal, 30 kHz SCS, 20 RBs |
| | 60 | G-FR1-A1-6 | -84.7 | -63.6 | DFT-s-OFDM NR signal, 60 kHz SCS, 24 RBs |

NOTE: Wanted and interfering signal are placed adjacently around $F_c$, where the $F_c$ is defined for *BS channel bandwidth* of the wanted signal according to the table 5.4.2.2-1. The aggregated wanted and interferer signal shall be centred in the BS channel bandwidth of the wanted signal.

**Table 7.8.2-3c: Local Area BS in-channel selectivity for band n96 and n102**

| *BS channel bandwidth* (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) | Type of interfering signal |
|---|---|---|---|---|---|
| 20 | 15 | G-FR1-A1-14 | -93.1 | -72.4 | CP-OFDM NR signal, 15 kHz SCS, 10 RBs |
| | 30 | G-FR1-A1-15 | -90.1 | -69.4 | CP-OFDM NR signal, 30 kHz SCS, 10 RBs |
| | 60 | G-FR1-A1-9 | -89.2 | -69.4 | DFT-s-OFDM NR signal, 60 kHz SCS, 5 RBs |
| 40 | 15 | G-FR1-A1-16 | -90.0 | -69.2 | CP-OFDM NR signal, 15 kHz SCS, 20 RBs |
| | 30 | G-FR1-A1-17 | -87.0 | -66.2 | CP-OFDM NR signal, 30 kHz SCS, 10 RBs |
| | 60 | G-FR1-A1-6 | -83.7 | -62.6 | DFT-s-OFDM NR signal, 60 kHz SCS, 24 RBs |
| 60 | 30 | G-FR1-A1-18 | -85.4 | -64.4 | CP-OFDM NR signal, 30 kHz SCS, 20 RBs |
| | 60 | G-FR1-A1-6 | -83.7 | -62.6 | DFT-s-OFDM NR signal, 60 kHz SCS, 24 RBs |
| 80 | 30 | G-FR1-A1-19 | -84.1 | -63.1 | CP-OFDM NR signal, 30 kHz SCS, 20 RBs |
| | 60 | G-FR1-A1-6 | -83.7 | -62.6 | DFT-s-OFDM NR signal, 60 kHz SCS, 24 RBs |
| NOTE: Wanted and interfering signal are placed adjacently around $F_c$, where the $F_c$ is defined for *BS channel bandwidth* of the wanted signal according to the table 5.4.2.2-1. The aggregated wanted and interferer signal shall be centred in the BS channel bandwidth of the wanted signal. | | | | | |

**Table 7.8.2-3d: Local area BS in-channel selectivity for band n104**

| BS channel bandwidth (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) | Type of interfering signal |
|---|---|---|---|---|---|
| 20, 30 | 15 | G-FR1-A1-1 | -89.7 | -68.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 25 RBs |
| 40, 50 | 15 | G-FR1-A1-4 | -83.3 | -62.4 | DFT-s-OFDM NR signal, 15 kHz SCS, 100 RBs |
| 20, 30 | 30 | G-FR1-A1-2 | -89.8 | -69.4 | DFT-s-OFDM NR signal, 30 kHz SCS, 10 RBs |
| 40, 50, 60, 70, 80, 90, 100 | 30 | G-FR1-A1-5 | -83.6 | -62.4 | DFT-s-OFDM NR signal, 30 kHz SCS, 50 RBs |
| 20, 30 | 60 | G-FR1-A1-9 | -89.2 | -69.4 | DFT-s-OFDM NR signal, 60 kHz SCS, 5 RBs |
| 40, 50, 60, 70, 80, 90, 100 | 60 | G-FR1-A1-6 | -83.7 | -62.6 | DFT-s-OFDM NR signal, 60 kHz SCS, 24 RBs |
| NOTE: Wanted and interfering signal are placed adjacently around $F_c$, where the $F_c$ is defined for *BS channel bandwidth* of the wanted signal according to the table 5.4.2.2-1. The aggregated wanted and interferer signal shall be centred in the *BS channel bandwidth* of the wanted signal. | | | | | |

# 8 Conducted performance requirements

## 8.1 General

### 8.1.1 Scope and definitions

Conducted performance requirements specify the ability of the *BS type 1-C* or *BS type 1-H* to correctly demodulate signals in various conditions and configurations. Conducted performance requirements are specified at the *antenna connector(s)* (for *BS type 1-C*) and at the *TAB connector(s)* (for *BS type 1-H*).

Conducted performance requirements for the BS are specified for the fixed reference channels defined in annex A and the propagation conditions in annex G. The requirements only apply to those FRCs that are supported by the base station.

Unless stated otherwise, performance requirements apply for a single carrier only. Performance requirements for a BS supporting *carrier aggregation* are defined in terms of single carrier requirements.

For FDD operation the requirements in clause 8 shall be met with the transmitter units associated with *antenna connectors* (for *BS type 1-C*) or *TAB connectors* (for *BS type 1-H*) in the *operating band* turned ON.

NOTE: In normal operating conditions, *antenna connector*s (for *BS type 1-C*) or *TAB connectors* (for *BS type 1-H*) in FDD operation are configured to transmit and receive at the same time. The associated transmitter unit(s) may be OFF for some of the tests as specified in TS 38.141-1 [5].

The SNR used in this clause is specified based on a single carrier and defined as:

SNR = S / N

Where:

S is the total signal energy in the slot on a single *antenna connector* (for *BS type 1-C*) or on a single *TAB connector* (for *BS type 1-H*).

N is the noise energy in a bandwidth corresponding to the *transmission bandwidth* over the same duration where signal energy exists on a single *antenna connector* (for *BS type 1-C*) or on a single *TAB connector* (for *BS type 1-H*).

### 8.1.2 Void

## 8.2 Performance requirements for PUSCH

### 8.2.1 Requirements for PUSCH with transform precoding disabled

#### 8.2.1.1 General

The performance requirement of PUSCH is determined by a minimum required throughput for a given SNR. The required throughput is expressed as a fraction of maximum throughput for the FRCs listed in annex A. The performance requirements assume HARQ retransmissions.

**Table: 8.2.1.1-1 Test parameters for testing PUSCH**

| Parameter | | Value |
|---|---|---|
| Transform precoding | | Disabled |
| Default TDD UL-DL pattern (Note 1) | | 15 kHz SCS: 3D1S1U, S=10D:2G:2U<br>30 kHz SCS: 7D1S2U, S=6D:4G:4U |
| HARQ | Maximum number of HARQ transmissions | 4 |
| | RV sequence | 0, 2, 3, 1 |
| DM-RS | DM-RS configuration type | 1 |
| | DM-RS duration | single-symbol DM-RS |
| | Additional DM-RS position | pos1 |
| | Number of DM-RS CDM group(s) without data | 2 |
| | Ratio of PUSCH EPRE to DM-RS EPRE | -3 dB |
| | DM-RS port | {0}, {0, 1} |
| | DM-RS sequence generation | $N_{ID}^{0}$=0, $n_{SCID}$ =0 |
| Time domain resource assignment | PUSCH mapping type | A, B |
| | Start symbol | 0 |
| | Allocation length | 14 |
| Frequency domain resource assignment | RB assignment | Full applicable test bandwidth |
| | Frequency hopping | Disabled |
| TPMI index for 2Tx two-layer spatial multiplexing transmission | | 0 |
| Code block group based PUSCH transmission | | Disabled |
| NOTE 1: The same requirements are applicable to FDD and TDD with different UL-DL pattern. | | |

## 8.2.1.2 Minimum requirements

The throughput shall be equal to or larger than the fraction of maximum throughput for the FRCs stated in tables 8.2.1.2-1 to 8.2.1.2-18 at the given SNR for 1Tx or for 2Tx two-layer spatial multiplexing transmission. FRCs are defined in annex A.

**Table 8.2.1.2-1: Minimum requirements for PUSCH with 70% of maximum throughput, Type A, 5 MHz channel bandwidth, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-8 | pos1 | -2.3 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-8 | pos1 | 10.1 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-8 | pos1 | 12.3 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-1 | pos1 | 19.1 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-8 | pos1 | -5.8 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-8 | pos1 | 6.2 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-8 | pos1 | 8.8 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-1 | pos1 | 15.5 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-8 | pos1 | -8.7 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-8 | pos1 | 3.0 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-8 | pos1 | 5.6 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-1 | pos1 | 12.4 |
| 2 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-22 | pos1 | 1.0 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-22 | pos1 | 18.2 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-22 | pos1 | -2.3 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-22 | pos1 | 11.0 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-22 | pos1 | -5.3 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-22 | pos1 | 6.8 |

**Table 8.2.1.2-2: Minimum requirements for PUSCH with 70% of maximum throughput, Type A, 10 MHz channel bandwidth, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-9 | pos1 | -2.5 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-9 | pos1 | 10.2 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-9 | pos1 | 12.2 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-2 | pos1 | 19.5 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-9 | pos1 | -6.0 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-9 | pos1 | 6.3 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-9 | pos1 | 8.6 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-2 | pos1 | 15.9 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-9 | pos1 | -8.7 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-9 | pos1 | 3.1 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-9 | pos1 | 5.5 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-2 | pos1 | 12.6 |
| 2 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-23 | pos1 | 1.7 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-23 | pos1 | 18.3 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-23 | pos1 | -2.0 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-23 | pos1 | 11.2 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-23 | pos1 | -5.5 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-23 | pos1 | 6.8 |

**Table 8.2.1.2-3: Minimum requirements for PUSCH with 70% of maximum throughput, Type A, 20 MHz channel bandwidth, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-10 | pos1 | -2.1 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-10 | pos1 | 10.0 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-10 | pos1 | 12.4 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-10 | pos1 | -5.5 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-10 | pos1 | 6.2 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-10 | pos1 | 8.6 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-10 | pos1 | -8.5 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-10 | pos1 | 3.0 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-10 | pos1 | 5.5 |
| 2 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-24 | pos1 | 2.1 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-24 | pos1 | 18.3 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-24 | pos1 | -1.8 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-24 | pos1 | 11.1 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-24 | pos1 | -5.3 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-24 | pos1 | 6.9 |

**Table 8.2.1.2-4: Minimum requirements for PUSCH with 70% of maximum throughput, Type A, 10 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-11 | pos1 | -2.3 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-11 | pos1 | 10.2 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-11 | pos1 | 12.8 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-3 | pos1 | 19.3 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-11 | pos1 | -5.6 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-11 | pos1 | 6.4 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-11 | pos1 | 8.6 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-3 | pos1 | 15.6 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-11 | pos1 | -8.6 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-11 | pos1 | 3.3 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-11 | pos1 | 5.5 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-3 | pos1 | 12.6 |
| 2 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-25 | pos1 | 1.3 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-25 | pos1 | 18.4 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-25 | pos1 | -2.2 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-25 | pos1 | 11.2 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-25 | pos1 | -5.2 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-25 | pos1 | 7.0 |

**Table 8.2.1.2-5: Minimum requirements for PUSCH with 70% of maximum throughput, Type A, 20 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-12 | pos1 | -2.9 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-12 | pos1 | 10.2 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-12 | pos1 | 12.5 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-12 | pos1 | -6.0 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-12 | pos1 | 6.4 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-12 | pos1 | 8.6 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-12 | pos1 | -8.8 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-12 | pos1 | 3.2 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-12 | pos1 | 5.5 |
| 2 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-26 | pos1 | 1.3 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-26 | pos1 | 18.1 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-26 | pos1 | -2.2 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-26 | pos1 | 11.3 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-26 | pos1 | -5.3 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-26 | pos1 | 6.9 |

**Table 8.2.1.2-6: Minimum requirements for PUSCH with 70% of maximum throughput, Type A, 40 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-13 | pos1 | -2.5 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-13 | pos1 | 10.0 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-13 | pos1 | 12.4 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-4 | pos1 | 19.9 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-13 | pos1 | -5.8 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-13 | pos1 | 6.3 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-13 | pos1 | 8.5 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-4 | pos1 | 16.1 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-13 | pos1 | -8.7 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-13 | pos1 | 3.1 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-13 | pos1 | 5.4 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-4 | pos1 | 12.6 |
| 2 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-27 | pos1 | 1.3 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-27 | pos1 | 19.5 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-27 | pos1 | -2.3 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-27 | pos1 | 11.3 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-27 | pos1 | -5.2 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-27 | pos1 | 6.9 |

**Table 8.2.1.2-7: Minimum requirements for PUSCH with 70% of maximum throughput, Type A, 100 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-14 | pos1 | -2.8 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-14 | pos1 | 10.2 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-14 | pos1 | 13.0 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-5 | pos1 | 21.1 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-14 | pos1 | -5.8 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-14 | pos1 | 6.5 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-14 | pos1 | 9.0 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-5 | pos1 | 16.7 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-14 | pos1 | -8.7 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-14 | pos1 | 3.2 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-14 | pos1 | 5.8 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-5 | pos1 | 13.1 |
| 2 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-28 | pos1 | 1.4 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-28 | pos1 | 19.2 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-28 | pos1 | -2.2 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-28 | pos1 | 11.6 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-28 | pos1 | -5.2 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-28 | pos1 | 7.1 |

**Table 8.2.1.2-8: Minimum requirements for PUSCH with 70% of maximum throughput, Type B, 5 MHz channel bandwidth, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-8 | pos1 | -2.3 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-8 | pos1 | 10.2 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-8 | pos1 | 12.5 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-1 | pos1 | 19.1 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-8 | pos1 | -5.7 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-8 | pos1 | 6.3 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-8 | pos1 | 8.9 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-1 | pos1 | 15.5 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-8 | pos1 | -8.7 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-8 | pos1 | 3.0 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-8 | pos1 | 5.7 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-1 | pos1 | 12.3 |
| 2 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-22 | pos1 | 1.5 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-22 | pos1 | 18.3 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-22 | pos1 | -2.3 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-22 | pos1 | 11.1 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-22 | pos1 | -5.4 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-22 | pos1 | 6.8 |

**Table 8.2.1.2-9: Minimum requirements for PUSCH with 70% of maximum throughput, Type B, 10 MHz channel bandwidth, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-9 | pos1 | -2.3 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-9 | pos1 | 10.5 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-9 | pos1 | 12.6 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-2 | pos1 | 19.5 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-9 | pos1 | -5.7 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-9 | pos1 | 6.5 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-9 | pos1 | 8.9 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-2 | pos1 | 15.9 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-9 | pos1 | -9.0 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-9 | pos1 | 3.2 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-9 | pos1 | 5.8 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-2 | pos1 | 12.5 |
| 2 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-23 | pos1 | 2.0 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-23 | pos1 | 18.7 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-23 | pos1 | -2.3 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-23 | pos1 | 11.3 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-23 | pos1 | -5.2 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-23 | pos1 | 7.0 |

**Table 8.2.1.2-10: Minimum requirements for PUSCH with 70% of maximum throughput, Type B, 20 MHz channel bandwidth, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-10 | pos1 | -2.1 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-10 | pos1 | 10.4 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-10 | pos1 | 12.3 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-10 | pos1 | -5.7 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-10 | pos1 | 6.3 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-10 | pos1 | 8.8 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-10 | pos1 | -8.5 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-10 | pos1 | 3.1 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-10 | pos1 | 5.7 |
| 2 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-24 | pos1 | 1.6 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-24 | pos1 | 18.1 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-24 | pos1 | -2.0 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-24 | pos1 | 11.2 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-24 | pos1 | -5.3 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-24 | pos1 | 6.9 |

**Table 8.2.1.2-11: Minimum requirements for PUSCH with 70% of maximum throughput, Type B, 10 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-11 | pos1 | -2.4 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-11 | pos1 | 10.1 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-11 | pos1 | 12.5 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-3 | pos1 | 19.2 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-11 | pos1 | -5.7 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-11 | pos1 | 6.4 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-11 | pos1 | 8.6 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-3 | pos1 | 15.7 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-11 | pos1 | -8.8 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-11 | pos1 | 3.2 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-11 | pos1 | 5.6 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-3 | pos1 | 12.4 |
| 2 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-25 | pos1 | 1.1 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-25 | pos1 | 18.5 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-25 | pos1 | -2.5 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-25 | pos1 | 11.3 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-25 | pos1 | -5.6 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-25 | pos1 | 7.0 |

**Table 8.2.1.2-12: Minimum requirements for PUSCH with 70% of maximum throughput, Type B, 20 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-12 | pos1 | -2.9 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-12 | pos1 | 10.1 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-12 | pos1 | 12.5 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-12 | pos1 | -6.0 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-12 | pos1 | 6.3 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-12 | pos1 | 8.6 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-12 | pos1 | -9.0 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-12 | pos1 | 3.1 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-12 | pos1 | 5.6 |
| 2 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-26 | pos1 | 1.3 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-26 | pos1 | 18.2 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-26 | pos1 | -2.3 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-26 | pos1 | 11.2 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-26 | pos1 | -5.4 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-26 | pos1 | 7.0 |

**Table 8.2.1.2-13: Minimum requirements for PUSCH with 70% of maximum throughput, Type B, 40 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-13 | pos1 | -2.5 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-13 | pos1 | 10.0 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-13 | pos1 | 12.5 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-4 | pos1 | 19.9 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-13 | pos1 | -5.8 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-13 | pos1 | 6.2 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-13 | pos1 | 8.7 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-4 | pos1 | 16.0 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-13 | pos1 | -8.8 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-13 | pos1 | 3.0 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-13 | pos1 | 5.5 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-4 | pos1 | 12.7 |
| 2 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-27 | pos1 | 1.7 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-27 | pos1 | 18.7 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-27 | pos1 | -2.1 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-27 | pos1 | 11.2 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-27 | pos1 | -5.2 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-27 | pos1 | 6.9 |

**Table 8.2.1.2-14: Minimum requirements for PUSCH with 70% of maximum throughput, Type B, 100 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-14 | pos1 | -2.5 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-14 | pos1 | 10.1 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-14 | pos1 | 13.1 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-5 | pos1 | 21.1 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-14 | pos1 | -5.8 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-14 | pos1 | 6.3 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-14 | pos1 | 9.2 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-5 | pos1 | 16.9 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-14 | pos1 | -8.7 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-14 | pos1 | 3.1 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A5-14 | pos1 | 5.9 |
| | | Normal | TDLA30-10 Low | 70 % | G-FR1-A9-5 | pos1 | 13.2 |
| 2 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-28 | pos1 | 1.6 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-28 | pos1 | 19.3 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-28 | pos1 | -2.2 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-28 | pos1 | 11.6 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-28 | pos1 | -5.3 |
| | | Normal | TDLC300-100 Low | 70 % | G-FR1-A4-28 | pos1 | 7.1 |

**Table 8.2.1.2-15: Minimum requirements for PUSCH with 30% of maximum throughput, Type A, 5 MHz channel bandwidth, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLC300-100 Low | 30 % | G-FR1-A4-8 | pos1 | 2.9 |

**Table 8.2.1.2-16: Minimum requirements for PUSCH with 30% of maximum throughput, Type A, 10 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLC300-100 Low | 30 % | G-FR1-A4-11 | pos1 | 2.8 |

**Table 8.2.1.2-17: Minimum requirements for PUSCH with 30% of maximum throughput, Type B, 5 MHz channel bandwidth, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLC300-100 Low | 30 % | G-FR1-A4-8 | pos1 | 2.8 |

**Table 8.2.1.2-18: Minimum requirements for PUSCH with 30% of maximum throughput, Type B, 10 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLC300-100 Low | 30 % | G-FR1-A4-11 | pos1 | 2.9 |

## 8.2.2 Requirements for PUSCH with transform precoding enabled

### 8.2.2.1 General

The performance requirement of PUSCH is determined by a minimum required throughput for a given SNR. The required throughput is expressed as a fraction of maximum throughput for the FRCs listed in annex A. The performance requirements assume HARQ retransmissions.

**Table 8.2.2.1-1: Test parameters for testing PUSCH**

| Parameter | | Value |
|---|---|---|
| Transform precoding | | Enabled |
| Default TDD UL-DL pattern (Note 1) | | 15 kHz SCS: 3D1S1U, S=10D:2G:2U<br>30 kHz SCS: 7D1S2U, S=6D:4G:4U |
| HARQ | Maximum number of HARQ transmissions | 4 |
| | RV sequence | 0, 2, 3, 1 |
| DM-RS | DM-RS configuration type | 1 |
| | DM-RS duration | single-symbol DM-RS |
| | Additional DM-RS position | pos1 |
| | Number of DM-RS CDM group(s) without data | 2 |
| | Ratio of PUSCH EPRE to DM-RS EPRE | -3 dB |
| | DM-RS port(s) | 0 |
| | DM-RS sequence generation | $N_{ID}^0$=0, group hopping and sequence hopping are disabled |
| Time domain resource assignment | PUSCH mapping type | A, B |
| | Start symbol | 0 |
| | Allocation length | 14 |
| Frequency domain resource assignment | RB assignment | 15 kHz SCS: 25 PRBs in the middle of the test bandwidth<br>30 kHz SCS: 24 PRBs in the middle of the test bandwidth |
| | Frequency hopping | Disabled |
| Code block group based PUSCH transmission | | Disabled |
| NOTE 1: The same requirements are applicable to FDD and TDD with different UL-DL patterns. | | |

### 8.2.2.2 Minimum requirements

The throughput shall be equal to or larger than the fraction of maximum throughput for the FRCs stated in tables 8.2.2.2-1 to 8.2.2.2-4 at the given SNR. FRCs are defined in annex A.

**Table 8.2.2.2-1: Minimum requirements for PUSCH with 70% of maximum throughput, Type A, 5 MHz channel bandwidth, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-31 | pos1 | -2.4 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-31 | pos1 | -5.7 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-31 | pos1 | -8.5 |

**Table 8.2.2.2-2: Minimum requirements for PUSCH with 70% of maximum throughput, Type A, 10 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-32 | pos1 | -2.5 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-32 | pos1 | -5.7 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-32 | pos1 | -8.4 |

**Table 8.2.2.2-3: Minimum requirements for PUSCH with 70% of maximum throughput, Type B, 5 MHz channel bandwidth, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-31 | pos1 | -2.3 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-31 | pos1 | -5.8 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-31 | pos1 | -8.6 |

**Table 8.2.2.2-4: Minimum requirements for PUSCH with 70% of maximum throughput, Type B, 10 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-32 | pos1 | -2.7 |
| | 4 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-32 | pos1 | -6.0 |
| | 8 | Normal | TDLB100-400 Low | 70 % | G-FR1-A3-32 | pos1 | -8.8 |

## 8.2.3 Requirements for UCI multiplexed on PUSCH

### 8.2.3.1 General

In the tests for UCI multiplexed on PUSCH, the UCI information only contains CSI part 1 and CSI part 2 information, and there is no HACK/ACK information transmitted.

The CSI part 1 block error probability (BLER) is defined as the probability of incorrectly decoding the CSI part 1 information when the CSI part 1 information is sent as follow:

$$BLER_{CSI\ part\ 1} = \frac{\#(\text{false CSI part 1})}{\#(\text{CSI part 1})}$$

where:

- #(false CSI part 1) denotes the number of incorrectly decoded CSI part 1 information transmitted occasions

- #(CSI part 1) denotes the number of CSI part 1 information transmitted occasions.

The CSI part 2 block error probability is defined as the probability of incorrectly decoding the CSI part 2 information when the CSI part 2 information is sent as follows:

$$BLER_{CSI\ part\ 2} = \frac{\#(\text{false CSI part 2})}{\#(\text{CSI part 2})}$$

where:

- #(false CSI part 2) denotes the number of incorrectly decoded CSI part 2 information transmitted occasions
- #(CSI part 2) denotes the number of CSI part 2 information transmitted occasions.

The number of UCI information bit payload per slot is defined for two cases as follows:

- 5 bits in CSI part 1, 2 bits in CSI part 2
- 20 bits in CSI part 1, 20 bits in CSI part 2

The 7bits UCI case is further defined with the bitmap [c0 c1 c2 c3 c4] = [0 1 0 1 0] for CSI part 1 information, where c0 is mapping to the RI information, and with the bitmap [c0 c1] = [1 0] for CSI part2 information.

The 40bits UCI information case is assumed random information bit selection.

In both tests, PUSCH data, CSI part 1 and CSI part 2 information are transmitted simultaneously.

**Table 8.2.3.1-1: Test parameters for testing UCI on PUSCH**

| Parameter | | Value |
|---|---|---|
| Transform precoding | | Disabled |
| Default TDD UL-DL pattern (Note 1) | | 30 kHz SCS: 7D1S2U, S=6D:4G:4U |
| HARQ | Maximum number of HARQ transmissions | 1 |
| | RV sequence | 0 |
| DM-RS | DM-RS configuration type | 1 |
| | DM-RS duration | Single-symbol DM-RS |
| | Additional DM-RS position | pos1 |
| | Number of DM-RS CDM group(s) without data | 2 |
| | Ratio of PUSCH EPRE to DM-RS EPRE | -3 dB |
| | DM-RS port(s) | {0} |
| | DM-RS sequence generation | $N_{ID}^{0}$=0, $n_{SCID}$=0 |
| Time domain resource assignment | PUSCH mapping type | A,B |
| | Start symbol | 0 |
| | Allocation length | 14 |
| Frequency domain resource assignment | RB assignment | Full applicable test bandwidth |
| | Frequency hopping | Disabled |
| Code block group based PUSCH transmission | | Disabled |
| UCI | Number of CSI part 1 and CSI part 2 information bit payload | {5,2},{20,20} |
| | *scaling* | 1 |
| | *betaOffsetACK-Index1* | 11 |
| | *betaOffsetCSI-Part1-Index1 and betaOffsetCSI-Part1-Index2* | 13 |
| | *betaOffsetCSI-Part2-Index1 and betaOffsetCSI-Part2-Index2* | 13 |
| | UCI partition for frequency hopping | Disabled |
| NOTE 1: The same requirements are applicable to FDD and TDD with different UL-DL patterns. | | |

## 8.2.3.2 Minimum requirements

The CSI part 1 block error probability shall not exceed 0.1% at the SNR in table 8.2.3.2-1 and table 8.2.3.2-2.The CSI part 2 block error probability shall not exceed 1% at the SNR given in table 8.2.3.2-3 and table 8.2.3.2-4.

**Table 8.2.3.2-1: Minimum requirements for UCI multiplexed on PUSCH, Type A, CSI part 1, 10 MHz Channel Bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | UCI bits (CSI part 1, CSI part 2) | Additional DM-RS position | FRC (Annex A) | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLC300-100 Low | 7(5,2) | pos1 | G-FR1-A4-11 | 5.4 |
| | 2 | Normal | TDLC300-100 Low | 40(20,20) | pos1 | G-FR1-A4-11 | 4.3 |

**Table 8.2.3.2-2: Minimum requirements for UCI multiplexed on PUSCH, Type B, CSI part 1, 10 MHz Channel Bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | UCI bits (CSI part 1, CSI part 2) | Additional DM-RS position | FRC (Annex A) | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLC300-100 Low | 7(5,2) | pos1 | G-FR1-A4-11 | 5.8 |
| | 2 | Normal | TDLC300-100 Low | 40(20,20) | pos1 | G-FR1-A4-11 | 4.1 |

**Table 8.2.3.2-3: Minimum requirements for UCI multiplexed on PUSCH, Type A, CSI part 2, 10 MHz Channel Bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | UCI bits (CSI part 1, CSI part 2) | Additional DM-RS position | FRC (Annex A) | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLC300-100 Low | 7(5,2) | pos1 | G-FR1-A4-11 | -0.2 |
| | 2 | Normal | TDLC300-100 Low | 40(20,20) | pos1 | G-FR1-A4-11 | 2.4 |

**Table 8.2.3.2-4: Minimum requirements for UCI multiplexed on PUSCH, Type B, CSI part 2, 10 MHz Channel Bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | UCI bits (CSI part 1, CSI part 2) | Additional DM-RS position | FRC (Annex A) | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLC300-100 Low | 7(5,2) | pos1 | G-FR1-A4-11 | 0.3 |
| | 2 | Normal | TDLC300-100 Low | 40(20,20) | pos1 | G-FR1-A4-11 | 2.6 |

## 8.2.4 Requirements for PUSCH for high speed train

### 8.2.4.1 General

The performance requirement of PUSCH is determined by a minimum required throughput for a given SNR. The required throughput is expressed as a fraction of maximum throughput for the FRCs listed in annex A. The performance requirements assume HARQ retransmissions. The performance requirements for high speed train are optional.

The performance requirements for PUSCH for high speed train only apply to Wide Area Base Stations and Medium Range Base Stations (subject to declaration).

**Table: 8.2.4.1-1 Test parameters for testing high speed train PUSCH**

| Parameter | | Value |
|---|---|---|
| Transform precoding | | Disabled |
| Default TDD UL-DL pattern (Note 1) | | 15 kHz SCS: 3D1S1U, S=10D:2G:2U<br>30 kHz SCS: 7D1S2U, S=6D:4G:4U |
| HARQ | Maximum number of HARQ transmissions | 4 |
| | RV sequence | 0, 2, 3, 1 |
| DM-RS | DM-RS configuration type | 1 |
| | DM-RS duration | single-symbol DM-RS |
| | Additional DM-RS position | pos2 |
| | Number of DM-RS CDM group(s) without data | 2 |
| | Ratio of PUSCH EPRE to DM-RS EPRE | -3 dB |
| | DM-RS port | 0 |
| | DM-RS sequence generation | $N_{ID}^0$=0, $n_{SCID}$ =0 |
| Time domain resource assignment | PUSCH mapping type | A |
| | Start symbol | 0 |
| | Allocation length | 14 |
| Frequency domain resource assignment | RB assignment | Full applicable test bandwidth |
| | Frequency hopping | Disabled |
| Code block group based PUSCH transmission | | Disabled |
| NOTE 1: The same requirements are applicable to FDD and TDD with different UL-DL pattern. | | |

## 8.2.4.2 Minimum requirements

The throughput shall be equal to or larger than the fraction of maximum throughput for the FRCs stated in tables 8.2.4.2-1 to 8.2.4.2-10 at the given SNR for 1Tx. FRCs are defined in annex A. Unless stated otherwise, the MIMO correlation matrices for the gNB are defined in annex G for low correlation.

**Table 8.2.4.2-1: Minimum requirements for PUSCH, Type A, 10 MHz channel bandwidth, 15 kHz SCS, 350km/h**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 1 | Normal | HST Scenario 3-NR350 | 70 % | G-FR1-A3-33 | pos2 | -0.8 |
| | 2 | Normal | HST Scenario 1-NR350 | 70 % | G-FR1-A3-33 | pos2 | -3.7 |
| | | Normal | HST Scenario 1-NR350 | 70 % | G-FR1-A4-29 | pos2 | 8.4 |
| | | Normal | HST Scenario 3-NR350 | 70 % | G-FR1-A3-33 | pos2 | -3.6 |
| | | Normal | HST Scenario 3-NR350 | 70 % | G-FR1-A4-29 | pos2 | 8.7 |
| | 8 | Normal | HST Scenario 1-NR350 | 70 % | G-FR1-A3-33 | pos2 | -9.2 |
| | | Normal | HST Scenario 1-NR350 | 70 % | G-FR1-A4-29 | pos2 | 2.6 |

**Table 8.2.4.2-2: Minimum requirements for PUSCH, Type A, 40 MHz channel bandwidth, 30 kHz SCS, 350km/h**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 1 | Normal | HST Scenario 3-NR350 | 70 % | G-FR1-A3-34 | pos2 | -0.6 |
| | 2 | Normal | HST Scenario 1-NR350 | 70 % | G-FR1-A3-34 | pos2 | -3.7 |
| | | Normal | HST Scenario 1-NR350 | 70 % | G-FR1-A4-30 | pos2 | 8.5 |
| | | Normal | HST Scenario 3-NR350 | 70 % | G-FR1-A3-34 | pos2 | -3.5 |
| | | Normal | HST Scenario 3-NR350 | 70 % | G-FR1-A4-30 | pos2 | 8.8 |
| | 8 | Normal | HST Scenario 1-NR350 | 70 % | G-FR1-A3-34 | pos2 | -9.1 |
| | | Normal | HST Scenario 1-NR350 | 70 % | G-FR1-A4-30 | pos2 | 2.7 |

**Table 8.2.4.2-3: Minimum requirements for PUSCH, Type A, 10 MHz channel bandwidth, 15 kHz SCS, 500km/h**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 1 | Normal | HST Scenario 3-NR500 | 70 % | G-FR1-A3-33 | pos2 | -0.7 |
| | 2 | Normal | HST Scenario 1-NR500 | 70 % | G-FR1-A3-33 | pos2 | -3.9 |
| | | Normal | HST Scenario 1-NR500 | 70 % | G-FR1-A4-29 | pos2 | 8.5 |
| | | Normal | HST Scenario 3-NR500 | 70 % | G-FR1-A3-33 | pos2 | -3.6 |
| | | Normal | HST Scenario 3-NR500 | 70 % | G-FR1-A4-29 | pos2 | 9.2 |
| | 8 | Normal | HST Scenario 1-NR500 | 70 % | G-FR1-A3-33 | pos2 | -9.4 |
| | | Normal | HST Scenario 1-NR500 | 70 % | G-FR1-A4-29 | pos2 | 2.7 |

**Table 8.2.4.2-4: Minimum requirements for PUSCH, Type A, 40 MHz channel bandwidth, 30 kHz SCS, 500km/h**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 1 | Normal | HST Scenario 3-NR500 | 70 % | G-FR1-A3-34 | pos2 | -0.5 |
| | 2 | Normal | HST Scenario 1-NR500 | 70 % | G-FR1-A3-34 | pos2 | -3.9 |
| | | Normal | HST Scenario 1-NR500 | 70 % | G-FR1-A4-30 | pos2 | 8.7 |
| | | Normal | HST Scenario 3-NR500 | 70 % | G-FR1-A3-34 | pos2 | -3.4 |
| | | Normal | HST Scenario 3-NR500 | 70 % | G-FR1-A4-30 | pos2 | 10.2 |
| | 8 | Normal | HST Scenario 1-NR500 | 70 % | G-FR1-A3-34 | pos2 | -9.2 |
| | | Normal | HST Scenario 1-NR500 | 70 % | G-FR1-A4-30 | pos2 | 2.8 |

**Table 8.2.4.2-5: Minimum requirements for PUSCH, Type A, 5 MHz channel bandwidth, 15 kHz SCS, 350km/h**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 1 | Normal | HST Scenario 3-NR350 | 70 % | G-FR1-A3-33A | pos2 | -0.7 |
| | 2 | Normal | HST Scenario 1-NR350 | 70 % | G-FR1-A3-33A | pos2 | -3.7 |
| | | Normal | HST Scenario 1-NR350 | 70 % | G-FR1-A4-29A | pos2 | 8.5 |
| | | Normal | HST Scenario 3-NR350 | 70 % | G-FR1-A3-33A | pos2 | -3.6 |
| | | Normal | HST Scenario 3-NR350 | 70 % | G-FR1-A4-29A | pos2 | 8.6 |
| | 8 | Normal | HST Scenario 1-NR350 | 70 % | G-FR1-A3-33A | pos2 | -9.1 |
| | | Normal | HST Scenario 1-NR350 | 70 % | G-FR1-A4-29A | pos2 | 2.8 |

**Table 8.2.4.2-6: Minimum requirements for PUSCH, Type A, 10 MHz channel bandwidth, 30 kHz SCS, 350km/h**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 1 | Normal | HST Scenario 3-NR350 | 70 % | G-FR1-A3-34A | pos2 | -0.7 |
| | 2 | Normal | HST Scenario 1-NR350 | 70 % | G-FR1-A3-34A | pos2 | -3.6 |
| | | Normal | HST Scenario 1-NR350 | 70 % | G-FR1-A4-30A | pos2 | 8.3 |
| | | Normal | HST Scenario 3-NR350 | 70 % | G-FR1-A3-34A | pos2 | -3.6 |
| | | Normal | HST Scenario 3-NR350 | 70 % | G-FR1-A4-30A | pos2 | 8.6 |
| | 8 | Normal | HST Scenario 1-NR350 | 70 % | G-FR1-A3-34A | pos2 | -9.0 |
| | | Normal | HST Scenario 1-NR350 | 70 % | G-FR1-A4-30A | pos2 | 2.6 |

**Table 8.2.4.2-7: Minimum requirements for PUSCH, Type A, 5 MHz channel bandwidth, 15 kHz SCS, 500km/h**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 1 | Normal | HST Scenario 3-NR500 | 70 % | G-FR1-A3-33A | pos2 | -0.6 |
| | 2 | Normal | HST Scenario 1-NR500 | 70 % | G-FR1-A3-33A | pos2 | -3.6 |
| | | Normal | HST Scenario 1-NR500 | 70 % | G-FR1-A4-29A | pos2 | 8.7 |
| | | Normal | HST Scenario 3-NR500 | 70 % | G-FR1-A3-33A | pos2 | -3.5 |
| | | Normal | HST Scenario 3-NR500 | 70 % | G-FR1-A4-29A | pos2 | 8.8 |
| | 8 | Normal | HST Scenario 1-NR500 | 70 % | G-FR1-A3-33A | pos2 | -9.1 |
| | | Normal | HST Scenario 1-NR500 | 70 % | G-FR1-A4-29A | pos2 | 3.0 |

**Table 8.2.4.2-8: Minimum requirements for PUSCH, Type A, 10 MHz channel bandwidth, 30 kHz SCS, 500km/h**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 1 | Normal | HST Scenario 3-NR500 | 70 % | G-FR1-A3-34A | pos2 | -0.5 |
| | 2 | Normal | HST Scenario 1-NR500 | 70 % | G-FR1-A3-34A | pos2 | -3.6 |
| | | Normal | HST Scenario 1-NR500 | 70 % | G-FR1-A4-30A | pos2 | 8.6 |
| | | Normal | HST Scenario 3-NR500 | 70 % | G-FR1-A3-34A | pos2 | -3.4 |
| | | Normal | HST Scenario 3-NR500 | 70 % | G-FR1-A4-30A | pos2 | 8.6 |
| | 8 | Normal | HST Scenario 1-NR500 | 70 % | G-FR1-A3-34A | pos2 | -8.8 |
| | | Normal | HST Scenario 1-NR500 | 70 % | G-FR1-A4-30A | pos2 | 2.9 |

**Table 8.2.4.2-9: Minimum requirements for PUSCH, Type A, 5 MHz channel bandwidth, 15 kHz SCS, 500km/h, multi-path fading channel requirements with high Doppler value**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLC300-600 | 70 % | G-FR1-A3-33A | pos2 | -1.9 |

**Table 8.2.4.2-10: Minimum requirements for PUSCH, Type A, 10 MHz channel bandwidth, 30 kHz SCS, 500km/h, multi-path fading channel requirements with high Doppler value**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLC300-1200 | 70 % | G-FR1-A3-34A | pos2 | -2.0 |

## 8.2.5 Requirements for UL timing adjustment

The performance requirement of UL timing adjustment is determined by a minimum required throughput for the moving The performance requirement of UL timing adjustment is determined by a minimum required throughput for the moving UE at given SNR. The performance requirements assume HARQ retransmissions. The performance requirements for UL timing adjustment scenario Y and scenario Z defined in Annex G.4 are optional.

In the tests for UL timing adjustment, two signals are configured, one being transmitted by a moving UE and the other being transmitted by a stationary UE. The transmission of SRS from UE is optional. FRC parameters in Table A.4-2B are applied for both UEs. The received power for both UEs is the same. The resource blocks allocated for both UEs are consecutive. In scenario Y and scenario Z Doppler shift is not taken into account.

**Table 8.2.5-1 Test parameters for testing UL timing adjustment**

| Parameter | | Value |
|---|---|---|
| Transform precoding | | Disabled |
| Uplink-downlink allocation for TDD | | 15 kHz SCS:<br>3D1S1U, S=10D:2G:2U<br>30 kHz SCS:<br>7D1S2U, S=6D:4G:4U |
| Channel bandwidth | | 15 kHz SCS: 5Mhz, 10 MHz<br>30 kHz SCS: 10MHz, 40 MHz |
| MCS | | 16 |
| HARQ | Maximum number of HARQ transmissions | 4 |
| | RV sequence | 0, 2, 3, 1 |
| DM-RS | DM-RS configuration type | 1 |
| | DM-RS duration | single-symbol DM-RS |
| | DM-RS position ($l_0$) | 2 |
| | Additional DM-RS position | pos2 |
| | Number of DM-RS CDM group(s) without data | 2 |
| | Ratio of PUSCH EPRE to DM-RS EPRE | -3 dB |
| | DM-RS port | {0} |
| Time domain resource assignment | DM-RS sequence generation | $N_{ID}^0$=0, $n_{SCID}$ =0 for moving UE<br>$N_{ID}^0$=1, $n_{SCID}$ =1 for stationary UE |
| | PUSCH mapping type | Both A and B |
| | Allocation length | 14 |
| Frequency domain resource assignment | RB assignment | 5 MHz CBW/15kHz SCS: 12 RB for each UE<br>10MHz CBW/15kHz SCS: 25 RB for each UE<br>10MHz CBW/30kHz SCS: 12 RB for each UE<br>40MHz CBW/30kHz SCS: 50 RB for each UE |
| | Starting PRB index | Moving UE: 0<br>Stationary UE: 12 for 5MHz, 25 for 10 MHz CBW for SCS 15kHz, and 12 for 10MHz, 50 for 40 MHz CBW for SCS 30kHz |
| | Frequency hopping | Disabled |
| SRS resource allocation | Slots in which sounding RS is transmitted (Note 1) | For FDD: slot #1 in radio frames<br><br>For TDD:<br>- last symbol in slot #3 in radio frames for 15KHz<br>- last symbol in slot #7 in radio frames for 30KHz |
| | SRS resource allocation | 15 kHz SCS:<br>$C_{SRS}$ = 5, $B_{SRS}$ =0, for 20 RB<br>$C_{SRS}$ = 11, $B_{SRS}$ =0, for 40 RB<br>30 kHz SCS:<br>$C_{SRS}$ =5, $B_{SRS}$ =0, for 20 RB<br>$C_{SRS}$ = 21, $B_{SRS}$ =0, for 80 RB |
| NOTE 1. The transmission of SRS is optional. And the transmission comb and SRS periodicity are configured as $K_{TC}$ = 2, and $T_{SRS}$ = 10 for 15 kHz SCS, $T_{SRS}$ = 20 for 30 kHz SCS respectively. | | |

### 8.2.5.1 Minimum requirements for high speed train

The throughput shall be ≥ 70% of the maximum throughput of the reference measurement channel as specified in Annex A for the moving UE at the SNR given in table 8.2.5.1-1 for mapping type A and table 8.2.5.1-2 for mapping type B respectively.

**Table 8.2.5.1-1 Minimum requirements for UL timing adjustment with mapping type A for high speed train**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Channel Bandwidth [MHz] | SCS [kHz] | Moving propagation conditions and correlation matrix (Annex G) | FRC (Annex A) | SNR [dB] |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | 5 | 15 | Scenario Y | G-FR1-A4-31A | 8.2 |
| | | | | | Scenario Z | G-FR1-A4-31A | 8.3 |
| | | | 10 | 15 | Scenario Y | G-FR1-A4-31 | 8.5 |
| | | | | | Scenario Z | G-FR1-A4-31 | 8.4 |
| | | | 10 | 30 | Scenario Y | G-FR1-A4-32A | 8.3 |
| | | | | | Scenario Z | G-FR1-A4-32A | 8.3 |
| | | | 40 | 30 | Scenario Y | G-FR1-A4-32 | 8.4 |
| | | | | | Scenario Z | G-FR1-A4-32 | 8.5 |

**Table 8.2.5.1-2 Minimum requirements for UL timing adjustment with mapping type B for high speed train**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Channel Bandwidth [MHz] | SCS [kHz] | Moving propagation conditions and correlation matrix (Annex G) | FRC (Annex A) | SNR [dB] |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | 5 | 15 | Scenario Y | G-FR1-A4-31A | 8.3 |
| | | | | | Scenario Z | G-FR1-A4-31A | 8.3 |
| | | | 10 | 15 | Scenario Y | G-FR1-A4-31 | 8.5 |
| | | | | | Scenario Z | G-FR1-A4-31 | 8.5 |
| | | | 10 | 30 | Scenario Y | G-FR1-A4-32A | 8.3 |
| | | | | | Scenario Z | G-FR1-A4-32A | 8.4 |
| | | | 40 | 30 | Scenario Y | G-FR1-A4-32 | 8.4 |
| | | | | | Scenario Z | G-FR1-A4-32 | 8.5 |

### 8.2.5.2 Minimum requirements for normal mode

The throughput shall be ≥ 70% of the maximum throughput of the reference measurement channel as specified in Annex A for the moving UE at the SNR given in table 8.2.5.2-1 for mapping type A and table 8.2.5.2-2 for mapping type B respectively.

**Table 8.2.5.2-1 Minimum requirements for UL timing adjustment with mapping type A for normal mode**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Channel Bandwidth [MHz] | SCS [kHz] | Moving propagation conditions and correlation matrix (Annex G) | FRC (Annex A) | SNR [dB] |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | 5 | 15 | Scenario X | G-FR1-A4-31A | 10.6 |
| | | | 10 | 15 | Scenario X | G-FR1-A4-31 | 11.2 |
| | | | 10 | 30 | Scenario X | G-FR1-A4-32A | 10.8 |
| | | | 40 | 30 | Scenario X | G-FR1-A4-32 | 12.0 |

**Table 8.2.5.2-2 Minimum requirements for UL timing adjustment with mapping type B for normal mode**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Channel Bandwidth [MHz] | SCS [kHz] | Moving propagation conditions and correlation matrix (Annex G) | FRC (Annex A) | SNR [dB] |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | 5 | 15 | Scenario X | G-FR1-A4-31A | 10.6 |
| | | | 10 | 15 | Scenario X | G-FR1-A4-31 | 11.3 |
| | | | 10 | 30 | Scenario X | G-FR1-A4-32A | 10.7 |
| | | | 40 | 30 | Scenario X | G-FR1-A4-32 | 12.4 |

## 8.2.6 Requirements for PUSCH 0.001% BLER

### 8.2.6.1 General

The performance requirement of PUSCH is determined by a maximum required transport block error rate (BLER) for a given SNR. The required BLER is defined as the probability of incorrectly decoding the transport block after reaching the maximum number of HARQ transmissions for the FRCs listed in annex A.

**Table: 8.2.6.1-1 Test parameters for testing PUSCH 0.001% BLER**

| Parameter | | Value |
|---|---|---|
| Transform precoding | | Disabled |
| Default TDD UL-DL pattern (Note 1) | | 15 kHz SCS:<br>3D1S1U, S=10D:2G:2U<br>30 kHz SCS:<br>7D1S2U, S=6D:4G:4U |
| HARQ | Maximum number of HARQ transmissions | 1 |
| | RV sequence | 0 |
| DM-RS | DM-RS configuration type | 1 |
| | DM-RS duration | single-symbol DM-RS |
| | Additional DM-RS position | pos1 |
| | Number of DM-RS CDM group(s) without data | 2 |
| | Ratio of PUSCH EPRE to DM-RS EPRE | -3 dB |
| | DM-RS port(s) | {0} |
| | DM-RS sequence generation | $N_{ID}^0$=0, $n_{SCID}$ =0 |
| Time domain resource assignment | PUSCH mapping type | A, B |
| | Start symbol | 0 |
| | Allocation length | 14 |
| Frequency domain resource assignment | RB assignment | Full applicable test bandwidth |
| | Frequency hopping | Disabled |
| Code block group based PUSCH transmission | | Disabled |
| Note 1: The same requirements are applicable to FDD and TDD with different UL-DL patterns. | | |

### 8.2.6.2 Minimum requirements

The BLER shall be equal to or smaller than the BLER stated in tables 8.2.6.2-1 to 8.2.6.2-8 at the given SNR for 1Tx. FRCs are defined in annex A. Unless stated otherwise, the MIMO correlation matrices for the gNB are defined in annex G for low correlation.

**Table 8.2.6.2-1: Minimum requirements for PUSCH with 0.001%BLER, Type A, 5 MHz channel bandwidth, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions (Annex G) | BLER | FRC (annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | AWGN | 0.001% | G-FR1-A3A-1 | pos1 | -5.1 |

**Table 8.2.6.2-2: Minimum requirements for PUSCH with 0.001%BLER, Type A, 10 MHz channel bandwidth, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions (Annex G) | BLER | FRC (annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | AWGN | 0.001% | G-FR1-A3A-2 | pos1 | -5.9 |

**Table 8.2.6.2-3: Minimum requirements for PUSCH with 0.001%BLER, Type A, 10 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions (Annex G) | BLER | FRC (annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | AWGN | 0.001% | G-FR1-A3A-3 | pos1 | -5.4 |

**Table 8.2.6.2-4: Minimum requirements for PUSCH with 0.001%BLER, Type A, 40 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions (Annex G) | BLER | FRC (annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | AWGN | 0.001% | G-FR1-A3A-4 | pos1 | -6.2 |

**Table 8.2.6.2-5: Minimum requirements for PUSCH with 0.001%BLER, Type B, 5 MHz channel bandwidth, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions (Annex G) | BLER | FRC (annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | AWGN | 0.001% | G-FR1-A3A-1 | pos1 | -5.2 |

**Table 8.2.6.2-6: Minimum requirements for PUSCH with 0.001%BLER, Type B, 10 MHz channel bandwidth, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions (Annex G) | BLER | FRC (annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | AWGN | 0.001% | G-FR1-A3A-2 | pos1 | -5.9 |

**Table 8.2.6.2-7: Minimum requirements for PUSCH with 0.001%BLER, Type B, 10 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions (Annex G) | BLER | FRC (annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | AWGN | 0.001% | G-FR1-A3A-3 | pos1 | -5.4 |

**Table 8.2.6.2-8: Minimum requirements for PUSCH with 0.001%BLER, Type B, 40 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions (Annex G) | BLER | FRC (annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | AWGN | 0.001% | G-FR1-A3A-4 | pos1 | -6.2 |

## 8.2.7 Requirements for PUSCH repetition Type A

### 8.2.7.1 General

The performance requirement of PUSCH is determined by a maximum block error probability (BLER) for a given SNR. The BLER is defined as the probability of incorrectly decoding the PUSCH information when the PUSCH information is sent. The performance requirements assume HARQ re-transmissions.

**Table: 8.2.7.1-1 Test parameters for testing PUSCH repetition Type A**

| Parameter | | Value |
|---|---|---|
| Transform precoding | | Disabled |
| Default TDD UL-DL pattern (Note 1) | | 15 kHz SCS: 3D1S1U, S=10D:2G:2U<br>30 kHz SCS: 7D1S2U, S=6D:4G:4U |
| HARQ | Maximum number of HARQ transmissions | 4 |
| | RV sequence | 0, 3, 0, 3 [Note 2] |
| DM-RS | DM-RS configuration type | 1 |
| | DM-RS duration | single-symbol DM-RS |
| | Additional DM-RS position | pos1 |
| | Number of DM-RS CDM group(s) without data | 2 |
| | Ratio of PUSCH EPRE to DM-RS EPRE | -3 dB |
| | DM-RS port | 0 |
| | DM-RS sequence generation | $N_{ID}^0$=0, $n_{SCID}$ =0 |
| Time domain resource assignment | PUSCH mapping type | A, B |
| | Start symbol | 0 |
| | Allocation length | 14 |
| | PUSCH aggregation factor | 30 kHz SCS: n2<br>15 kHz SCS: n2 for FDD and n8 for TDD [Note 3] |
| Frequency domain resource assignment | RB assignment | Full applicable test bandwidth |
| | Frequency hopping | Disabled |
| Code block group based PUSCH transmission | | Disabled |

Note 1: The same requirements are applicable to FDD and TDD with different UL-DL pattern.
Note 2: The effective RV sequence is {0, 2, 3, 1} with slot aggregation.
Note 3: The intention of this configuration is to have two effective transmissions of the transport block. To achieve this for the standard TDD pattern captured in this table, a value of n8 is necessary, while for FDD a value of n2 is necessary.

## 8.2.7.2 Minimum requirements

The BLER shall be equal to or smaller than the required target BLER for the FRCs stated in tables 8.2.7.2-1 to 8.2.7.2-8 at the given SNR for 1Tx. FRCs are defined in annex A.

**Table 8.2.7.2-1: Minimum requirements for PUSCH, Type A, 5 MHz channel bandwidth, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Target BLER | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLB100-400 Low | 1% (Note 1) | G-FR1-A3A-1 | pos1 | -8.4 |

Note 1: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block.

**Table 8.2.7.2-2: Minimum requirements for PUSCH, Type A, 10 MHz channel bandwidth, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Target BLER | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLB100-400 Low | 1% (Note 1) | G-FR1- A3A -2 | pos1 | -10.2 |

Note 1: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block.

**Table 8.2.7.2-3: Minimum requirements for PUSCH, Type A, 10 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Target BLER | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLB100-400 Low | 1% (Note 1) | G-FR1- A3A -3 | pos1 | -10. 8 |
| Note 1: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block. | | | | | | | |

**Table 8.2.7.2-4: Minimum requirements for PUSCH, Type A, 40 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Target BLER | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLB100-400 Low | 1% (Note 1) | G-FR1- A3A -4 | pos1 | -11.5 |
| Note 1: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block. | | | | | | | |

**Table 8.2.7.2-5: Minimum requirements for PUSCH, Type B, 5 MHz channel bandwidth, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Target BLER | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLB100-400 Low | 1% (Note 1) | G-FR1- A3A -1 | pos1 | -8. 2 |
| Note 1: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block. | | | | | | | |

**Table 8.2.7.2-6: Minimum requirements for PUSCH, Type B, 10 MHz channel bandwidth, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Target BLER | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLB100-400 Low | 1% (Note 1) | G-FR1- A3A -2 | pos1 | -10.1 |
| Note 1: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block. | | | | | | | |

**Table 8.2.7.2-7: Minimum requirements for PUSCH, Type B, 10 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Target BLER | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLB100-400 Low | 1% (Note 1) | G-FR1- A3A -3 | pos1 | -10.8 |
| Note 1: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block. | | | | | | | |

**Table 8.2.7.2-8: Minimum requirements for PUSCH, Type B, 40 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Target BLER | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLB100-400 Low | 1% (Note 1) | G-FR1- A3A -4 | pos1 | -11.4 |
| Note 1: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block. | | | | | | | |

## 8.2.8 Requirements for PUSCH mapping Type B with non-slot transmission

### 8.2.8.1 General

The performance requirement of PUSCH is determined by a minimum required throughput for a given SNR. The required throughput is expressed as a fraction of maximum throughput for the FRCs listed in annex A. The performance requirements do not assume HARQ retransmissions.

**Table: 8.2.8.1-1 Test parameters for testing PUSCH mapping Type B with non-slot transmission**

| Parameter | | Value |
|---|---|---|
| Transform precoding | | Disabled |
| Default TDD UL-DL pattern (Note 1) | | 15 kHz SCS:<br>3D1S1U, S=10D:2G:2U<br>30 kHz SCS:<br>7D1S2U, S=6D:4G:4U |
| HARQ | Maximum number of HARQ transmissions | 1 |
| | RV sequence | 0 |
| DM-RS | DM-RS configuration type | 1 |
| | DM-RS duration | single-symbol DM-RS |
| | Additional DM-RS position | Pos0 |
| | Number of DM-RS CDM group(s) without data | 2 |
| | Ratio of PUSCH EPRE to DM-RS EPRE | -3 dB |
| | DM-RS port(s) | {0} |
| | DM-RS sequence generation | $N_{ID}^0$=0, $n_{SCID}$ =0 |
| Time domain resource assignment | PUSCH mapping type | B |
| | Start symbol | 0 |
| | Allocation length | 2 |
| Frequency domain resource assignment | RB assignment | Full applicable test bandwidth |
| | Frequency hopping | Disabled |
| Code block group based PUSCH transmission | | Disabled |
| Note 1: The same requirements are applicable to FDD and TDD with different UL-DL patterns. | | |

### 8.2.8.2 Minimum requirements

The throughput shall be equal to or larger than the fraction of maximum throughput for the FRCs stated in tables 8.2.4.8-1 to 8.2.4.8-4 at the given SNR for 1Tx. FRCs are defined in annex A. Unless stated otherwise, the MIMO correlation matrices for the gNB are defined in annex G for low correlation.

**Table 8.2.8.2-1: Minimum requirements for PUSCH with 2 symbols, Type B, 5 MHz channel bandwidth, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLC300-100 | 70 % | G-FR1-A3B-1 | pos0 | 0.5 |

**Table 8.2.8.2-2: Minimum requirements for PUSCH with 2 symbols, Type B, 10 MHz channel bandwidth, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLC300-100 | 70 % | G-FR1-A3B-2 | pos0 | 0.3 |

**Table 8.2.8.2-3: Minimum requirements for PUSCH with 2 symbols, Type B, 10 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLC300-100 | 70 % | G-FR1-A3B-3 | pos0 | 0.3 |

**Table 8.2.8.2-4: Minimum requirements for PUSCH with 2 symbols, Type B, 40 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLC300-100 | 70 % | G-FR1-A3B-4 | pos0 | 0.0 |

## 8.2.9 Requirements of PUSCH for 2-step RA type

### 8.2.9.1 General

The performance requirement of PUSCH for 2-step RA type is determined by a minimum required block error rate of MsgA received by BS at given SNR for the FRCs listed in Annex A. The performance requirements assume that the precedent preamble of MsgA is correctly detected in a 2-step RA type procedure, and no HARQ retransmissions.

The performance requirements are applicable for wide area and medium range BS that support 2-step RA type.

The performance requirements are not applied for a local area BS that supports 2-step RA type.

**Table 8.2.9.1-1: Test parameters for testing PUSCH for 2-step RA type**

| Parameter | | Value |
|---|---|---|
| Transform precoding | | Disabled |
| Channel bandwidth | | 15 kHz SCS: 10 MHz<br>30 kHz SCS: 40 MHz |
| MCS | | 1 |
| DM-RS | DM-RS configuration type | 1 |
| | DM-RS duration | single-symbol DM-RS |
| | DM-RS position ($l_0$) | 2 |
| | Additional DM-RS position | pos2 or pos1 |
| | Number of DM-RS CDM group(s) without data | 2 |
| | Ratio of PUSCH EPRE to DM-RS EPRE | -3 dB |
| | DM-RS port | {0} |
| | DM-RS sequence generation | $N_{ID}^0$=0, $n_{SCID}$ = 0 |
| Time domain resource assignment | PUSCH mapping type | Both A and B |
| | Allocation length | 14 |
| Frequency domain resource assignment | RB assignment | 2 PRBs |
| | Starting PRB index | 0 |
| | Frequency hopping | Disabled |
| Time offset (TO) Cycling (μs) | start:end | 15k SCS: 0:3.8 |
| | | 30k SCS: 0:2 |
| Test Metric | BLER | 0.01 |
| NOTE 1: A single requirement is defined that is applicable regardless of whether pos1 or pos2 is configured for the additional DM-RS position<br>NOTE 2: The power ratio between preamble and msgA (msgA-DeltaPreamble) is set to be sufficient to achieve 100% preamble detection. The SNR for the requirement is defined on the msgA PUSCH | | |

### 8.2.9.2 Minimum requirements

The block error rate of MsgA for the reference measurement channel as specified in Annex A at the SNR given shall not exceed 1% in table 8.2.9.2-1 for mapping type A and table 8.2.9.2-2 for mapping type B respectively.

**Table 8.2.9.2-1: Minimum requirements of PUSCH for 2-step RA type with mapping type A**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Mapping Type | TO cycling (start:end) [μs] | Channel Bandwidth [MHz] | SCS [kHz] | Propagation conditions and correlation matrix (Annex G) | FRC (Annex A) | SNR [dB] |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | Type A | 0:3.8 | 10 | 15 | TDLC300-100 low | G-FR1-A8-1, or G-FR1-A8-3 | 7.3 |
| | | | | 0:2 | 40 | 30 | TDLC300-100 low | G-FR1-A8-2, or G-FR1-A8-4 | 7.1 |

**Table 8.2.9.2-2: Minimum requirements of PUSCH for 2-step RA type with mappying type B**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Mapping Type | TO cycling (start:end) [μs] | Channel Bandwidth [MHz] | SCS [kHz] | Propagation conditions and correlation matrix (Annex G) | FRC (Annex A) | SNR [dB] |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | Type B | 0:3.8 | 10 | 15 | TDLC300-100 low | G-FR1-A8-1, or G-FR1-A8-3 | 7.0] |
| | | | | 0:2 | 40 | 30 | TDLC300-100 low | G-FR1-A8-2, or G-FR1-A8-4 | 7.6 |

## 8.2.10 Requirements for interlaced PUSCH

### 8.2.10.1 General

The performance requirement of PUSCH with interlace allocation is determined by a minimum required throughput for a given SNR. The required throughput is expressed as a fraction of maximum throughput for the FRCs listed in annex A. The performance requirements assume HARQ retransmissions.

**Table 8.2.10.1-1: Test parameters for testing PUSCH**

| Parameter | | Value |
|---|---|---|
| Transform precoding | | Disabled |
| Default TDD UL-DL pattern (Note 1) | | 15 kHz SCS:<br>3D1S1U, S=10D:2G:2U<br>30 kHz SCS:<br>7D1S2U, S=6D:4G:4U |
| HARQ | Maximum number of HARQ transmissions | 4 |
| | RV sequence | 0, 2, 3, 1 |
| DM-RS | DM-RS configuration type | 1 |
| | DM-RS duration | single-symbol DM-RS |
| | Additional DM-RS position | pos1 |
| | Number of DM-RS CDM group(s) without data | 2 |
| | Ratio of PUSCH EPRE to DM-RS EPRE | -3 dB |
| | DM-RS port(s) | 0 |
| | DM-RS sequence generation | $N_{ID}^0$=0, $n_{SCID}$ =0 |
| Time domain resource assignment | PUSCH mapping type | A, B |
| | Start symbol | 0 |
| | Allocation length | 14 |
| Frequency domain resource assignment | RB assignment | Full applicable test bandwidth. Frist interlace with RBs 0,10,20,…,100 are allocated for tests with 15kHz and first interlace with RBs 0,5,10,…50 are allocated for tests with 30kHz. |
| | Frequency hopping | Disabled |
| Code block group based PUSCH transmission | | Disabled |
| NOTE 1: The same requirements are applicable to FDD and TDD with different UL-DL patterns. | | |

## 8.2.10.2 Minimum requirements

The throughput shall be equal to or larger than the fraction of maximum throughput for the FRCs stated in tables 8.2.10.2-1 to 8.2.10.2-4 at the given SNR. FRCs are defined in annex A.

**Table 8.2.10.2-1: Minimum requirements for PUSCH with 70% of maximum throughput, Type A, 20 MHz channel bandwidth, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-10 Low | 70% | G-FR1-A5-15 | pos1 | 12.3 |

**Table 8.2.10.2-2: Minimum requirements for PUSCH with 70% of maximum throughput, Type A, 20 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-10 Low | 70% | G-FR1-A5-16 | pos1 | 12.2 |

**Table 8.2.10.2-3: Minimum requirements for PUSCH with 70% of maximum throughput, Type B, 20 MHz channel bandwidth, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-10 Low | 70% | G-FR1-A5-15 | pos1 | 12.3 |

**Table 8.2.10.2-4: Minimum requirements for PUSCH with 70% of maximum throughput, Type B, 20 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-10 Low | 70% | G-FR1-A5-16 | pos1 | 12.2 |

## 8.2.11 Performance requirements for CG-UCI multiplexed on interlaced PUSCH

### 8.2.11.1 General

The performance requirement of CG-UCI multiplexed on interlaced PUSCH is determined by the parameter: block error probability (BLER) of CG-UCI. The performance is measured by the required SNR at block error probability of CG-UCI not exceeding 1%.

The CG-UCI BLER is defined as the probability of incorrectly decoding the CG-UCI information when the CG-UCI information is sent.

In the test of UCI multiplexed on interlaced PUSCH, the UCI information only contains CG-UCI information, there is no HACK/ACK, CSI part 1 or CSI part 2 information transmitted.

The number of UCI information bit payload per slot is defined as 18 bits.

The 18 bits UCI information case is further defined with the bitmap [c0 c1 c2 c3 c4 … c17] for CG-UCI information, where

- [c0 c1 c2 c3] = [0 0 0 1] is mapping to the HARQ process number information,
- [c4 c5] = [0 0] is mapping to the RV sequence information.
- [c6] = [1] is mapping to the NDI information.
- [c7 c8 … c17] = [0 0 … 0] is the COT sharing information field.

In the test, PUSCH data and CG-UCI are transmitted simultaneously.

**Table 8.2.11.1-1: Test parameters for testing CG-UCI on interlaced PUSCH**

| Parameter | | Value |
|---|---|---|
| Transform precoding | | Disabled |
| Default TDD UL-DL pattern | | 30 kHz SCS:<br>7D1S2U, S=6D:4G:4U<br><br>15 kHz SCS:<br>3D1S1U S=10D:2G:2U |
| HARQ | Maximum number of HARQ transmissions | 1 |
| | RV sequence | 0 |
| DM-RS | DM-RS configuration type | 1 |
| | DM-RS duration | Single-symbol DM-RS |
| | Additional DM-RS position | pos1 |
| | Number of DM-RS CDM group(s) without data | 2 |
| | Ratio of PUSCH EPRE to DM-RS EPRE | -3 dB |
| | DM-RS port(s) | {0} |
| | DM-RS sequence generation | $N_{ID}^0$=0, $n_{SCID}$ =0 |
| Time domain resource assignment | PUSCH mapping type | A,B |
| | Start symbol | 0 |
| | Allocation length | 14 |
| Frequency domain resource assignment | RB assignment | Full applicable test bandwidth.<br>First interlace with RBs 0,10,20,…,100 are allocated for tests with 15kHz and first interlace with RBs 0,5,10,…50 are allocated for tests with 30kHz. |
| | Frequency hopping | Disabled |
| Code block group based PUSCH transmission | | Disabled |
| PT-RS | | Disabled |
| UCI | Number of information bits | 18 |
| | scaling | 1 |
| | betaOffsetCG-UCI-Index1 | 8 |
| | UCI partition for frequency hopping | Disabled |

## 8.2.11.2 Minimum requirements

The fraction of incorrectly decoded CG-UCI according to clause 8.2.11.1 shall be less than 1 % for the SNR listed in table 8.2.11.2-1 to .8.2.11.2-4.

**Table 8.2.11.2-1: Minimum requirements for CG-UCI multiplexing on interlaced PUSCH, Type A, 20 MHz channel bandwidth, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | CG-UCI bits | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-10 Low | 18 | G-FR1-A5-15 | pos1 | 4.6 |

**Table 8.2.11.2-2: Minimum requirements for CG-UCI multiplexing on interlaced PUSCH, Type A, 20 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | CG-UCI bits | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-10 Low | 18 | G-FR1-A5-16 | pos1 | 4.7 |

**Table 8.2.11.2-3: Minimum requirements for CG-UCI multiplexing on interlaced PUSCH, Type B, 20 MHz channel bandwidth, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | CG-UCI bits | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-10 Low | 18 | G-FR1-A5-15 | pos1 | 4.6 |

**Table 8.2.11.2-4: Minimum requirements for CG-UCI multiplexing on interlaced PUSCH, Type B, 20 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | CG-UCI bits | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-10 Low | 18 | G-FR1-A5-16 | pos1 | 5.0 |

## 8.2.12 Requirements for TB processing over multi-slot PUSCH (TBoMS)

### 8.2.12.1 General

The performance requirement of PUSCH TBoMS is determined by a minimum required throughput for a given SNR. The required throughput is expressed as a fraction of maximum throughput for the FRCs listed in annex A. The performance requirements assume HARQ retransmissions.

**Table: 8.2.12.1-1: Test parameters for testing PUSCH TBoMS**

| Parameter | | Value |
|---|---|---|
| Transform precoding | | Disabled |
| Default TDD UL-DL pattern (Note 1) | | 15 kHz SCS: 3D1S1U, S=10D:2G:2U<br>30 kHz SCS: 7D1S2U, S=6D:4G:4U |
| HARQ | Maximum number of HARQ transmissions | 4 |
| | RV sequence | 0, 2, 3, 1 |
| DM-RS | DM-RS configuration type | 1 |
| | DM-RS duration | single-symbol DM-RS |
| | Additional DM-RS position | pos1 |
| | Number of DM-RS CDM group(s) without data | 2 |
| | Ratio of PUSCH EPRE to DM-RS EPRE | -3 dB |
| | DM-RS port | 0 |
| | DM-RS sequence generation | $N_{ID}^0$=0, $n_{SCID}$ =0 |
| Time domain resource assignment | PUSCH mapping type | A, B |
| | Start symbol | 0 |
| | Allocation length | 14 |
| | Number of slots allocated for TBoMS PUSCH | 4 for FDD<br>2 for TDD |
| | Number of repetitions of a single TBoMS | 1 |
| Frequency domain resource assignment | RB assignment | 5 RBs in the middle of the test bandwidth |
| | Frequency hopping | Disabled |
| Code block group based PUSCH transmission | | Disabled |
| Note 1: The same requirements are applicable to TDD with different UL-DL pattern. | | |

### 8.2.12.2 Minimum requirements

The throughput shall be equal to or larger than the fraction of maximum throughput for the FRCs stated in tables 8.2.12.2-1 to 8.2.12.2-4 at the given SNR. FRCs are defined in annex A.

**Table 8.2.12.2-1: Minimum requirements for PUSCH TBoMS, Type A, 5 MHz channel bandwidth, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Duplex | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | FDD | TDLB100-400 Low | 70% | G-FR1-A3-36 | pos1 | -2.3 |
| 1 | 2 | Normal | TDD | TDLB100-400 Low | 70% | G-FR1-A3-35 | pos1 | -2.5 |

**Table 8.2.12.2-2: Minimum requirements for PUSCH TBoMS, Type A, 10 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Duplex | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | FDD | TDLB100-400 Low | 70% | G-FR1-A3-38 | pos1 | -2.4 |
| 1 | 2 | Normal | TDD | TDLB100-400 Low | 70% | G-FR1-A3-37 | pos1 | -2.5 |

**Table 8.2.12.2-3: Minimum requirements for PUSCH TBoMS, Type B, 5 MHz channel bandwidth, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Duplex | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | FDD | TDLB100-400 Low | 70% | G-FR1-A3-36 | pos1 | -2.1 |
| 1 | 2 | Normal | TDD | TDLB100-400 Low | 70% | G-FR1-A3-35 | pos1 | -2.5 |

**Table 8.2.12.2-4: Minimum requirements for PUSCH TBoMS, Type B, 10 MHz channel bandwidth, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Duplex | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | FDD | TDLB100-400 Low | 70% | G-FR1-A3-38 | pos1 | -2.4 |
| 1 | 2 | Normal | TDD | TDLB100-400 Low | 70% | G-FR1-A3-37 | pos1 | -2.5 |

## 8.2.13 Requirements for PUSCH with DM-RS bundling

### 8.2.13.1 General

The performance requirement of PUSCH is determined by a minimum required throughput for a given SNR. The required throughput is expressed as a fraction of maximum throughput for the FRCs listed in annex A. The performance requirements assume HARQ re-transmissions.

**Table: 8.2.13.1-1: Test parameters for testing PUSCH with DM-RS bundling**

| Parameter | | Value |
|---|---|---|
| Transform precoding | | Disabled |
| Example UL-DL pattern [Note 1] | | 15 kHz SCS: FDD and TDD 7D1S2U, S=6D:4G:4U<br>30 kHz SCS: FDD and TDD 7D1S2U, S=6D:4G:4U |
| HARQ | Maximum number of HARQ transmissions | 4 |
| | RV sequence [Note 2] | 0, 0, 0, 0 for FDD<br>0, 3, 0, 3 for TDD |
| DM-RS | DM-RS configuration type | 1 |
| | DM-RS duration | single-symbol DM-RS |
| | Additional DM-RS position | pos0, pos1 |
| | Number of DM-RS CDM group(s) without data | 2 |
| | Ratio of PUSCH EPRE to DM-RS EPRE | -3 dB |
| | DM-RS port | 0 |
| | DM-RS sequence generation | $N_{ID}^0$=0, $n_{SCID}$ =0 |
| Time domain resource assignment | PUSCH mapping type | A, B |
| | Start symbol | 0 |
| | Allocation length | 14 |
| | PUSCH aggregation factor | n8 for FDD<br>n2 for TDD |
| pusch-TimeDomainWindowLength | | 8 slots for FDD<br>2 slots for TDD |
| Frequency domain resource assignment | RB assignment | Full applicable test bandwidth |
| | Frequency hopping | Disabled |
| Code block group based PUSCH transmission | | Disabled |
| Note 1: The same TDD requirements are applicable to different UL-DL patterns with more than one consecutive UL slots when both pusch-TimeDomainWindowLength and PUSCH aggregation factor are configured as 2 slots. The UL (re)transmission of PUSCH is only scheduled for the actual TDW including 2 consecutive UL slots.<br>Note 2: The effective RV sequence is {0, 2, 3, 1} with slot aggregation. | | |

## 8.2.13.2 Minimum requirements

The throughput shall be equal to or larger than the fraction of maximum throughput for the FRCs stated in tables 8.2.13.2-1 to 8.2.13.2-8 at the given SNR for 1Tx. FRCs are defined in annex A.

**Table 8.2.13.2-1: Minimum requirements for PUSCH, Type A, 5 MHz channel bandwidth, 15 kHz SCS FDD**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-10 Low | 70% | G-FR1-A3B-7 | pos0 | -7.9 |
| | | | | | G-FR1-A3B-5 | pos1 | -9.9 |
| | 4 | Normal | TDLA30-10 Low | 70% | G-FR1-A3B-7 | pos0 | -11.8 |
| | | | | | G-FR1-A3B-5 | pos1 | -12.3 |
| | 8 | Normal | TDLA30-10 Low | 70% | G-FR1-A3B-7 | pos0 | -15.0 |
| | | | | | G-FR1-A3B-5 | pos1 | -15.6 |

**Table 8.2.13.2-2: Minimum requirements for PUSCH, Type A, 5 MHz channel bandwidth, 15 kHz SCS TDD**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-10 Low | 70% | G-FR1-A3B-7 | pos0 | -2.6 |
| | | | | | G-FR1-A3B-5 | pos1 | -3.6 |
| | 4 | Normal | TDLA30-10 Low | 70% | G-FR1-A3B-7 | pos0 | -6.2 |
| | | | | | G-FR1-A3B-5 | pos1 | -6.8 |
| | 8 | Normal | TDLA30-10 Low | 70% | G-FR1-A3B-7 | pos0 | -9.4 |
| | | | | | G-FR1-A3B-5 | pos1 | -10.6 |

**Table 8.2.13.2-3: Minimum requirements for PUSCH, Type A, 10 MHz channel bandwidth, 30 kHz SCS FDD**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-10 Low | 70% | G-FR1-A3B-8 | pos0 | -9.8 |
| | | | | | G-FR1-A3B-6 | pos1 | -9.5 |
| | 4 | Normal | TDLA30-10 Low | 70% | G-FR1-A3B-8 | pos0 | -13.0 |
| | | | | | G-FR1-A3B-6 | pos1 | -13.1 |
| | 8 | Normal | TDLA30-10 Low | 70% | G-FR1-A3B-8 | pos0 | -15.3 |
| | | | | | G-FR1-A3B-6 | pos1 | -15.8 |

**Table 8.2.13.2-4: Minimum requirements for PUSCH, Type A, 10 MHz channel bandwidth, 30 kHz SCS TDD**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-10 Low | 70% | G-FR1-A3B-8 | pos0 | -2.9 |
| | | | | | G-FR1-A3B-6 | pos1 | -3.8 |
| | 4 | Normal | TDLA30-10 Low | 70% | G-FR1-A3B-8 | pos0 | -6.5 |
| | | | | | G-FR1-A3B-6 | pos1 | -7.0 |
| | 8 | Normal | TDLA30-10 Low | 70% | G-FR1-A3B-8 | pos0 | -9.7 |
| | | | | | G-FR1-A3B-6 | pos1 | -10.3 |

**Table 8.2.13.2-5: Minimum requirements for PUSCH, Type B, 5 MHz channel bandwidth, 15 kHz SCS FDD**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-10 Low | 70% | G-FR1-A3B-7 | pos0 | -8.3 |
| | | | | | G-FR1-A3B-5 | pos1 | -9.9 |
| | 4 | Normal | TDLA30-10 Low | 70% | G-FR1-A3B-7 | pos0 | -11.8 |
| | | | | | G-FR1-A3B-5 | pos1 | -12.3 |
| | 8 | Normal | TDLA30-10 Low | 70% | G-FR1-A3B-7 | pos0 | -15.0 |
| | | | | | G-FR1-A3B-5 | pos1 | -15.6 |

**Table 8.2.13.2-6: Minimum requirements for PUSCH, Type B, 5 MHz channel bandwidth, 15 kHz SCS TDD**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-10 Low | 70% | G-FR1-A3B-7 | pos0 | -2.6 |
| | | | | | G-FR1-A3B-5 | pos1 | -3.4 |
| | 4 | Normal | TDLA30-10 Low | 70% | G-FR1-A3B-7 | pos0 | -6.3 |
| | | | | | G-FR1-A3B-5 | pos1 | -6.8 |
| | 8 | Normal | TDLA30-10 Low | 70% | G-FR1-A3B-7 | pos0 | -9.4 |
| | | | | | G-FR1-A3B-5 | pos1 | -10.6 |

**Table 8.2.13.2-7: Minimum requirements for PUSCH, Type B, 10 MHz channel bandwidth, 30 kHz SCS FDD**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-10 Low | 70% | G-FR1-A3B-8 | pos0 | -9.8 |
| | | | | | G-FR1-A3B-6 | pos1 | -9.5 |
| | 4 | Normal | TDLA30-10 Low | 70% | G-FR1-A3B-8 | pos0 | -13.3 |
| | | | | | G-FR1-A3B-6 | pos1 | -13.1 |
| | 8 | Normal | TDLA30-10 Low | 70% | G-FR1-A3B-8 | pos0 | -15.5 |
| | | | | | G-FR1-A3B-6 | pos1 | -15.8 |

**Table 8.2.13.2-8: Minimum requirements for PUSCH, Type B, 10 MHz channel bandwidth, 30 kHz SCS TDD**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-10 Low | 70% | G-FR1-A3B-8 | pos0 | -2.8 |
| | | | | | G-FR1-A3B-6 | pos1 | -3.3 |
| | 4 | Normal | TDLA30-10 Low | 70% | G-FR1-A3B-8 | pos0 | -6.5 |
| | | | | | G-FR1-A3B-6 | pos1 | -7.0 |
| | 8 | Normal | TDLA30-10 Low | 70% | G-FR1-A3B-8 | pos0 | -9.7 |
| | | | | | G-FR1-A3B-6 | pos1 | -10.3 |

# 8.3 Performance requirements for PUCCH

## 8.3.1 DTX to ACK probability

### 8.3.1.1 General

The DTX to ACK probability, i.e. the probability that ACK is detected when nothing was sent:

$$\text{Prob}(\text{PUCCH DTX} \rightarrow \text{Ack bits}) = \frac{\#(false\ ACK\ bits)}{\#(\text{PUCCH DTX}) * \#(ACK/NACK\ bits)}$$

where:

- #(false ACK bits) denotes the number of detected ACK bits.
- #(ACK/NACK bits) denotes the number of encoded bits per slot
- #(PUCCH DTX) denotes the number of DTX occasions

### 8.3.1.2 Minimum requirement

The DTX to ACK probability shall not exceed 1% for all PUCCH formats carrying ACK/NACK bits:

$$\text{Prob}(\text{PUCCH DTX} \rightarrow \text{Ack bits}) \leq 10^{-2}$$

## 8.3.2 Performance requirements for PUCCH format 0

### 8.3.2.1 General

The ACK missed detection probability is the probability of not detecting an ACK when an ACK was sent.

**Table 8.3.2.1-1: Test Parameters**

| Parameter | Test |
|---|---|
| Number of UCI information bits | 1 |
| Number of PRBs | 1 |
| First PRB prior to frequency hopping | 0 |
| Intra-slot frequency hopping | N/A for 1 symbol Enabled for 2 symbols |
| First PRB after frequency hopping | The largest PRB index – (Number of PRBs – 1) |
| Group and sequence hopping | neither |
| Hopping ID | 0 |
| Initial cyclic shift | 0 |
| First symbol | 13 for 1 symbol<br>12 for 2 symbols |

The transient period as specified in TS 38.101-1 [17] clause 6.3.3.1 is not taken into account for performance requirement testing, where the RB hopping is symmetric to the CC centre, i.e. intra-slot frequency hopping is enabled.

### 8.3.2.2 Minimum requirements

The ACK missed detection probability shall not exceed 1% at the SNR given in table 8.3.2.2-1 and in table 8.3.2.2-2.

**Table 8.3.2.2-1: Minimum requirements for PUCCH format 0 and 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Propagation conditions and correlation matrix (Annex G) | Number of OFDM symbols | Channel bandwidth / SNR (dB) | | |
|---|---|---|---|---|---|---|
| | | | | 5 MHz | 10 MHz | 20 MHz |
| 1 | 2 | TDLC300-100 Low | 1 | 9.4 | 8.8 | 9.3 |
| | | | 2 | 2.8 | 3.7 | 3.3 |
| 1 | 4 | TDLC300-100 Low | 1 | 3.0 | 2.9 | 3.2 |
| | | | 2 | -1.0 | -0.5 | -0.8 |
| 1 | 8 | TDLC300-100 Low | 1 | -1.1 | -1.1 | -1.1 |
| | | | 2 | -4.1 | -3.9 | -4.0 |

**Table 8.3.2.2-2: Minimum requirements for PUCCH format 0 and 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Propagation conditions and correlation matrix (Annex G) | Number of OFDM symbols | Channel bandwidth / SNR (dB) | | | |
|---|---|---|---|---|---|---|---|
| | | | | 10 MHz | 20 MHz | 40 MHz | 100 MHz |
| 1 | 2 | TDLC300-100 Low | 1 | 9.8 | 9.8 | 9.5 | 9.2 |
| | | | 2 | 4.2 | 3.6 | 3.8 | 3.5 |
| 1 | 4 | TDLC300-100 Low | 1 | 3.4 | 3.4 | 3.0 | 3.3 |
| | | | 2 | -0.3 | -0.4 | -0.5 | -0.8 |
| 1 | 8 | TDLC300-100 Low | 1 | -1.0 | -1.0 | -1.1 | -1.0 |
| | | | 2 | -3.7 | -3.8 | -4.0 | -3.9 |

## 8.3.3 Performance requirements for PUCCH format 1

### 8.3.3.1 NACK to ACK requirements

#### 8.3.3.1.1 General

The NACK to ACK detection probability is the probability that an ACK bit is falsely detected when an NACK bit was sent on the particular bit position, where the NACK to ACK detection probability is defined as follows:

$$\text{Prob(PUCCH NACK} \rightarrow \text{ACK bits)} = \frac{\#(\text{NACK bits decoded as ACK bits})}{\#(\text{Total NACK bits})},$$

where:

- $\#(\text{Total NACK bits})$ denotes the total number of NACK bits transmitted
- $\#(\text{NACK bits decoded as ACK bits})$ denotes the number of NACK bits decoded as ACK bits at the receiver, i.e. the number of received ACK bits
- NACK bits in the definition do not contain the NACK bits which are mapped from DTX, i.e. NACK bits received when DTX is sent should not be considered.

Random codeword selection is assumed.

**Table 8.3.3.1.1-1: Test Parameters**

| Parameter | Test |
|---|---|
| Number of information bits | 2 |
| Number of PRBs | 1 |
| Number of symbols | 14 |
| First PRB prior to frequency hopping | 0 |
| Intra-slot frequency hopping | enabled |
| First PRB after frequency hopping | The largest PRB index – (nrofPRBs – 1) |
| Group and sequence hopping | neither |
| Hopping ID | 0 |
| Initial cyclic shift | 0 |
| First symbol | 0 |
| Index of orthogonal cover code (*timeDomainOCC*) | 0 |

The transient period as specified in TS 38.101-1 [17] clause 6.3.3.1 is not taken into account for performance requirement testing, where the RB hopping is symmetric to the CC centre, i.e. intra-slot frequency hopping is enabled.

#### 8.3.3.1.2 Minimum requirements

The NACK to ACK probability shall not exceed 0.1% at the SNR given in table 8.3.3.1.2-1 and table 8.3.3.1.2-2.

**Table 8.3.3.1.2-1: Minimum requirements for PUCCH format 1 with 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Channel bandwidth / SNR (dB) | | |
|---|---|---|---|---|---|---|
| | | | | 5 MHz | 10 MHz | 20 MHz |
| 1 | 2 | Normal | TDLC-300-100 Low | -3.8 | -3.6 | -3.6 |
| | 4 | Normal | TDLC-300-100 Low | -8.4 | -7.6 | -8.4 |
| | 8 | Normal | TDLC-300-100 Low | -11.8 | -11.4 | -11.4 |

**Table 8.3.3.1.2-2: Minimum requirements for PUCCH format 1 with 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Channel bandwidth / SNR (dB) | | | |
|---|---|---|---|---|---|---|---|
| | | | | 10 MHz | 20 MHz | 40 MHz | 100 MHz |
| 1 | 2 | Normal | TDLC-300-100 Low | -2.8 | -3.3 | -3.9 | -3.5 |
| | 4 | Normal | TDLC-300-100 Low | -8.1 | -8.3 | -7.5 | -8.0 |
| | 8 | Normal | TDLC-300-100 Low | -11.5 | -11.2 | -11.6 | -11.3 |

## 8.3.3.2 ACK missed detection requirements

### 8.3.3.2.1 General

The ACK missed detection probability is the probability of not detecting an ACK when an ACK was sent. The test parameters in table 8.3.3.1.1-1 are configured.

The transient period as specified in TS 38.101-1 [17] clause 6.3.3.1 is not taken into account for performance requirement testing, where the RB hopping is symmetric to the centre, i.e. intra-slot frequency hopping is enabled.

### 8.3.3.2.2 Minimum requirements

The ACK missed detection probability shall not exceed 1% at the SNR given in table 8.3.3.2.2-1 and in table 8.3.3.2.2-2.

**Table 8.3.3.2.2-1: Minimum requirements for PUCCH format 1 with 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Channel bandwidth / SNR (dB) | | |
|---|---|---|---|---|---|---|
| | | | | 5 MHz | 10 MHz | 20 MHz |
| 1 | 2 | Normal | TDLC-300-100 Low | -5.0 | -4.4 | -5.0 |
| | 4 | Normal | TDLC-300-100 Low | -8.6 | -8.2 | -8.5 |
| | 8 | Normal | TDLC-300-100 Low | -11.6 | -11.5 | -11.5 |

**Table 8.3.3.2.2-2: Minimum requirements for PUCCH format 1 with 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Channel bandwidth / SNR (dB) | | | |
|---|---|---|---|---|---|---|---|
| | | | | 10 MHz | 20 MHz | 40 MHz | 100 MHz |
| 1 | 2 | Normal | TDLC-300-100 Low | -3.9 | -4.4 | -4.4 | -4.2 |
| | 4 | Normal | TDLC-300-100 Low | -8.0 | -8.1 | -8.4 | -8.3 |
| | 8 | Normal | TDLC-300-100 Low | -11.4 | -11.4 | -11.4 | -11.4 |

## 8.3.4 Performance requirements for PUCCH format 2

### 8.3.4.1 ACK missed detection requirements

#### 8.3.4.1.1 General

The ACK missed detection probability is the probability of not detecting an ACK when an ACK was sent.

The ACK missed detection requirement only applies to the PUCCH format 2 with 4 UCI bits.

**Table 8.3.4.1.1-1: Test Parameters**

| Parameter | Value |
|---|---|
| Modulation order | QSPK |
| Starting RB location | 0 |
| Intra-slot frequency hopping | N/A |
| Number of PRBs | 4 |
| Number of symbols | 1 |
| The number of UCI information bits | 4 |
| First symbol | 13 |
| DM-RS sequence generation | $N_{ID}^{0}$=0 |

#### 8.3.4.1.2 Minimum requirements

The ACK missed detection probability shall not exceed 1% at the SNR given in table 8.3.4.1.2-1 and table 8.3.4.1.2-2 for 4UCI bits.

**Table 8.3.4.1.2-1: Minimum requirements for PUCCH format 2 with 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Channel bandwidth / SNR (dB) | | |
|---|---|---|---|---|---|---|
| | | | | 5 MHz | 10 MHz | 20 MHz |
| 1 | 2 | Normal | TDLC300-100 Low | 5.8 | 5.6 | 5.9 |
| | 4 | Normal | TDLC300-100 Low | 0.4 | 0.5 | 0.3 |
| | 8 | Normal | TDLC300-100 Low | -3.5 | -3.5 | -3.5 |

**Table 8.3.4.1.2-2: Minimum requirements for PUCCH format 2 with 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Channel bandwidth / SNR (dB) | | | |
|---|---|---|---|---|---|---|---|
| | | | | 10 MHz | 20 MHz | 40 MHz | 100 MHz |
| 1 | 2 | Normal | TDLC300-100 Low | 5.5 | 5.6 | 5.5 | 5.7 |
| | 4 | Normal | TDLC300-100 Low | 0.3 | 0.2 | 0.3 | 0.4 |
| | 8 | Normal | TDLC300-100 Low | -3.6 | -3.6 | -3.5 | -3.3 |

## 8.3.4.2 UCI BLER performance requirements

### 8.3.4.2.1 General

The UCI block error probability (BLER) is defined as the probability of incorrectly decoding the UCI information when the UCI information is sent. The UCI information does not contain CSI part 2.

The transient period as specified in TS 38.101-1 [17] clause 6.3.3.1 is not taken into account for performance requirement testing, where the RB hopping is symmetric to the CC centre, i.e. intra-slot frequency hopping is enabled.

The UCI block error probability performance requirement only applies to the PUCCH format 2 with 22 UCI bits.

**Table 8.3.4.2.1-1: Test Parameters**

| Parameter | Value |
|---|---|
| Modulation order | QSPK |
| First PRB prior to frequency hopping | 0 |
| Intra-slot frequency hopping | enabled |
| Frist PRB after frequency hopping | The largest PRB index – (Number of PRBs – 1) |
| Number of PRBs | 9 |
| Number of symbols | 2 |
| The number of UCI information bits | 22 |
| First symbol | 12 |
| DM-RS sequence generation | $N_{ID}^{0}$=0 |

### 8.3.4.2.2 Minimum requirements

The UCI block error probability shall not exceed 1% at the SNR given in table 8.3.4.2.2-1 and table 8.3.4.2.2-2 for 22 UCI bits.

**Table 8.3.4.2.2-1: Minimum requirements for PUCCH format 2 with 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Channel bandwidth / SNR (dB) | | |
|---|---|---|---|---|---|---|
| | | | | 5 MHz | 10 MHz | 20 MHz |
| 1 | 2 | Normal | TDLC300-100 Low | 0.2 | 0.8 | 1.2 |
| | 4 | Normal | TDLC300-100 Low | -3.6 | -3.2 | -3.2 |
| | 8 | Normal | TDLC300-100 Low | -6.8 | -6.7 | -6.8 |

**Table 8.3.4.2.2-2: Minimum requirements for PUCCH format 2 with 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Channel bandwidth / SNR (dB) | | | |
|---|---|---|---|---|---|---|---|
| | | | | 10 MHz | 20 MHz | 40 MHz | 100 MHz |
| 1 | 2 | Normal | TDLC300-100 Low | 0.5 | 1.1 | 0.4 | 0.3 |
| | 4 | Normal | TDLC300-100 Low | -3.3 | -2.9 | -3.3 | -3.4 |
| | 8 | Normal | TDLC300-100 Low | -5.8 | -5.8 | -6.7 | -5.9 |

## 8.3.5 Performance requirements for PUCCH format 3

### 8.3.5.1 General

The performance is measured by the required SNR at UCI block error probability not exceeding 1%.

The UCI block error probability is defined as the conditional probability of incorrectly decoding the UCI information when the UCI information is sent. The UCI information does not contain CSI part 2.

The transient period as specified in TS 38.101-1 [17] clause 6.3.3.1 is not taken into account for performance requirement testing, where the RB hopping is symmetric to the centre, i.e. intra-slot frequency hopping is enabled.

**Table 8.3.5.1-1: Test Parameters**

| Parameter | Test 1 | Test 2 |
|---|---|---|
| Modulation order | QPSK | |
| First PRB prior to frequency hopping | 0 | |
| Intra-slot frequency hopping | enabled | |
| First PRB after frequency hopping | The largest PRB index – (Number of PRBs – 1) | |
| Group and sequence hopping | neither | |
| Hopping ID | 0 | |
| Number of PRBs | 1 | 3 |
| Number of symbols | 14 | 4 |
| The number of UCI information bits | 16 | 16 |
| First symbol | 0 | 0 |

### 8.3.5.2 Minimum requirements

The UCI block error probability shall not exceed 1% at the SNR given in Table 8.3.5.2-1 and Table 8.3.5.2-2.

**Table 8.3.5.2-1: Minimum requirements for PUCCH format 3 with 15 kHz SCS**

| Test Number | Number of TX antennas | Number of RX antennas | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Additional DM-RS configuration | Channel bandwidth / SNR (dB) | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | 5 MHz | 10 MHz | 20 MHz |
| 1 | 1 | 2 | Normal | TDLC300-100 Low | No additional DM-RS | 0.2 | 1.1 | 0.3 |
| | | | | | Additional DM-RS | -0.1 | 0.5 | -0.1 |
| | | 4 | Normal | TDLC300-100 Low | No additional DM-RS | -3.8 | -3.3 | -3.8 |
| | | | | | Additional DM-RS | -4.3 | -4.0 | -4.0 |
| | | 8 | Normal | TDLC300-100 Low | No additional DM-RS | -7.0 | -6.7 | -6.9 |
| | | | | | Additional DM-RS | -7.7 | -7.5 | -7.7 |
| 2 | 1 | 2 | Normal | TDLC300-100 Low | No additional DM-RS | 1.4 | 2.2 | 2.0 |
| | | 4 | Normal | TDLC300-100 Low | No additional DM-RS | -3.1 | -2.5 | -2.5 |
| | | 8 | Normal | TDLC300-100 Low | No additional DM-RS | -6.5 | -6.0 | -6.2 |

**Table 8.3.5.2-2: Minimum requirements for PUCCH format 3 with 30 kHz SCS**

| Test Number | Number of TX antennas | Number of RX antennas | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Additional DM-RS configuration | Channel bandwidth / SNR (dB) | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | 10 MHz | 20 MHz | 40 MHz | 100 MHz |
| 1 | 1 | 2 | Normal | TDLC300-100 Low | No additional DM-RS | 0.9 | 0.6 | 0.6 | 0.9 |
| | | | | | Additional DM-RS | 0.5 | 0.3 | 0.0 | 0.1 |
| | | 4 | Normal | TDLC300-100 Low | No additional DM-RS | -3.1 | -3.4 | -3.2 | -3.5 |
| | | | | | Additional DM-RS | -3.7 | -4.1 | -4.0 | -4.2 |
| | | 8 | Normal | TDLC300-100 Low | No additional DM-RS | -6.6 | -6.7 | -6.8 | -6.8 |
| | | | | | Additional DM-RS | -7.5 | -7.6 | -7.6 | -7.7 |
| 2 | 1 | 2 | Normal | TDLC300-100 Low | No additional DM-RS | 1.8 | 2.0 | 2.0 | 1.5 |
| | | 4 | Normal | TDLC300-100 Low | No additional DM-RS | -2.9 | -3.0 | -2.4 | -3.0 |
| | | 8 | Normal | TDLC300-100 Low | No additional DM-RS | -6.4 | -6.0 | -6.4 | -6.2 |

## 8.3.6 Performance requirements for PUCCH format 4

### 8.3.6.1 General

The performance is measured by the required SNR at UCI block error probability not exceeding 1%.

The UCI block error probability is defined as the conditional probability of incorrectly decoding the UCI information when the UCI information is sent. The UCI information does not contain CSI part 2.

The transient period as specified in TS 38.101-1 [17] clause 6.3.3.1 is not taken into account for performance requirement testing, where the RB hopping is symmetric to the centre, i.e. intra-slot frequency hopping is enabled.

**Table 8.3.6.1-1: Test parameters**

| Parameter | Value |
|---|---|
| Modulation order | QPSK |
| First PRB prior to frequency hopping | 0 |
| Number of PRBs | 1 |
| Intra-slot frequency hopping | enabled |
| First PRB after frequency hopping | The largest PRB index – (Number of PRBs – 1) |
| Group and sequence hopping | neither |
| Hopping ID | 0 |
| Number of symbols | 14 |
| The number of UCI information bits | 22 |
| First symbol | 0 |
| Length of the orthogonal cover code | n2 |
| Index of the orthogonal cover code | n0 |

### 8.3.6.2 Minimum requirement

The UCI block error probability shall not exceed 1% at the SNR given in Table 8.3.6.2-1 and Table 8.3.6.2-2.

**Table 8.3.6.2-1: Required SNR for PUCCH format 4 with 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Additional DM-RS configuration | Channel bandwidth / SNR (dB) | | |
|---|---|---|---|---|---|---|---|
| | | | | | 5 MHz | 10 MHz | 20 MHz |
| 1 | 2 | Normal | TDLC300-100 Low | No additional DM-RS | 1.8 | 2.6 | 2.2 |
| | | | | Additional DM-RS | 1.6 | 2.4 | 1.8 |
| | 4 | Normal | TDLC300-100 Low | No additional DM-RS | -2.3 | -1.9 | -2.2 |
| | | | | Additional DM-RS | -2.9 | -2.6 | -2.7 |
| | 8 | Normal | TDLC300-100 Low | No additional DM-RS | -5.9 | -5.7 | -5.8 |
| | | | | Additional DM-RS | -6.6 | -6.4 | -6.3 |

**Table 8.3.6.2-2: Required SNR for PUCCH format 4 with 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Additional DM-RS configuration | Channel bandwidth / SNR (dB) | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | 10 MHz | 20 MHz | 40 MHz | 100 MHz |
| 1 | 2 | Normal | TDLC300-100 Low | No additional DM-RS | 3.1 | 2.8 | 3.1 | 2.8 |
| | | | | Additional DM-RS | 2.8 | 2.3 | 3.1 | 2.2 |
| | 4 | Normal | TDLC300-100 Low | No additional DM-RS | -1.7 | -1.9 | -1.7 | -2.1 |
| | | | | Additional DM-RS | -2.0 | -2.5 | -2.5 | -2.4 |
| | 8 | Normal | TDLC300-100 Low | No additional DM-RS | -5.6 | -5.5 | -5.5 | -5.5 |
| | | | | Additional DM-RS | -6.2 | -6.1 | -6.4 | -6.2 |

## 8.3.7 Performance requirements for multi-slot PUCCH

### 8.3.7.1 General

### 8.3.7.2 Performance requirements for multi-slot PUCCH format 1

#### 8.3.7.2.1 NACK to ACK requirements

##### 8.3.7.2.1.1 General

The NACK to ACK detection probability is the probability that an ACK bit is falsely detected when an NACK bit was sent on the particular bit position, where the NACK to ACK detection probability is defined as follows:

$$\text{Prob}(\text{PUCCH NACK} \rightarrow \text{ACK bits}) = \frac{\#(\text{NACK bits decoded as ACK bits})}{\#(\text{Total NACK bits})},$$

where:

- $\#(\text{Total NACK bits})$ denotes the total number of NACK bits transmitted
- $\#(\text{NACK bits decoded as ACK bits})$ denotes the number of NACK bits decoded as ACK bits at the receiver, i.e. the number of received ACK bits
- NACK bits in the definition do not contain the NACK bits which are mapped from DTX, i.e. NACK bits received when DTX is sent should not be considered.

Random codeword selection is assumed.

**Table 8.3.7.2.1.1-1: Test Parameters for multi-slot PUCCH format 1**

| Parameter | Test |
|---|---|
| Number of information bits | 2 |
| Number of PRBs | 1 |
| Number of symbols | 14 |
| First PRB prior to frequency hopping | 0 |
| Intra-slot frequency hopping | disabled |
| Inter-slot frequency hopping | enabled |
| First PRB after frequency hopping | The largest PRB index – (nrofPRBs – 1) |
| Group and sequence hopping | neither |
| Hopping ID | 0 |
| Initial cyclic shift | 0 |
| First symbol | 0 |
| Index of orthogonal cover code (*timeDomainOCC*) | 0 |
| Number of slots for PUCCH repetition | 2 |

#### 8.3.7.2.1.2 Minimum requirements

The multi-slot NACK to ACK probability shall not exceed 0.1% at the SNR given in table 8.3.7.2.1.2-1.

**Table 8.3.7.2.1.2-1: Minimum requirements for multi-slot PUCCH format 1 with 30kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Channel bandwidth / SNR (dB) |
|---|---|---|---|---|
| | | | | 40 MHz |
| 1 | 2 | Normal | TDLC-300-100 Low | -6.3 |

### 8.3.7.2.2 ACK missed detection requirements

#### 8.3.7.2.2.1 General

The ACK missed detection probability is the probability of not detecting an ACK when an ACK was sent. The test parameters in table 8.3.7.2.1.1-1 are configured.

#### 8.3.7.2.2.2 Minimum requirements

The multi-slot ACK missed detection probability shall not exceed 1% at the SNR given in table 8.3.7.2.2.2-1.

**Table 8.3.7.2.2.2-1: Minimum requirements for multi-slot PUCCH format 1 with 30kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Channel bandwidth / SNR (dB) |
|---|---|---|---|---|
| | | | | 40 MHz |
| 1 | 2 | Normal | TDLC-300-100 Low | -7.6 |

## 8.3.8 Performance requirements for interlaced PUCCH format 0

### 8.3.8.1 General

The ACK missed detection probability is the probability of not detecting an ACK when an ACK was sent.

The ACK missed detection probability performance requirement only apply to PUCCH format 0 with 1 UCI bit. The UCI information only contain ACK information.

The 1bit UCI information is further defined with the bitmap as [1].

**Table 8.3.8.1-1: Test Parameters**

| Parameter | Test |
|---|---|
| Number of UCI information bits | 1 |
| Number of symbols | 1 |
| Intra-slot frequency hopping | N/A |
| Group and sequence hopping | neither |
| Hopping ID | 0 |
| Initial cyclic shift | 0 |
| First symbol | 13 |
| Number of interlaces | 1 |
| Interlace index | 0Note1 |
| NOTE 1: RBs 0,10,20,…, 100 are allocated for 15kHz SCS and RBs 0,5,10,…, 50 are allocated for 30kHz SCS | |

### 8.3.8.2 Minimum requirements

The ACK missed detection probability shall not exceed 1% at the SNR given in table 8.3.8.2-1

**Table 8.3.8.2-1: Minimum requirements for interlaced PUCCH format 0 with 15 kHz SCS, 20MHz channel bandwidth**

| Number of Tx antennas | Number of RX antennas | Propagation conditions and correlation matrix (Annex G) | Number of OFDM symbols | SNR (dB) |
|---|---|---|---|---|
| 1 | 2 | TDLA30-10 Low | 1 | -2.8 |

**Table 8.3.8.2-2: Minimum requirements for interlaced PUCCH format 0 with 30 kHz SCS, 20MHz channel bandwidth**

| Number of Tx antennas | Number of RX antennas | Propagation conditions and correlation matrix (Annex G) | Number of OFDM symbols | SNR (dB) |
|---|---|---|---|---|
| 1 | 2 | TDLA30-10 Low | 1 | -2.0 |

## 8.3.9 Performance requirements for interlaced PUCCH format 1

### 8.3.9.1 NACK to ACK requirements

#### 8.3.9.1.1 General

The NACK to ACK detection probability is the probability that an ACK bit is falsely detected when an NACK bit was sent on the particular bit position, where the NACK to ACK detection probability is defined as follows:

$$\text{Prob}(\text{PUCCH NACK} \rightarrow \text{ACK bits}) = \frac{\#(\text{NACK bits decoded as ACK bits})}{\#(\text{Total NACK bits})},$$

where:

- $\#(\text{Total NACK bits})$ denotes the total number of NACK bits transmitted
- $\#(\text{NACK bits decoded as ACK bits})$ denotes the number of NACK bits decoded as ACK bits at the receiver, i.e. the number of received ACK bits
- NACK bits in the definition do not contain the NACK bits which are mapped from DTX, i.e. NACK bits received when DTX is sent should not be considered.

The NACK to ACK detection probability performance requirement only apply to PUCCH format 1 with 2 UCI bits. The UCI information only contain ACK/NACK information.

The 2bits UCI information is further defined with bitmap as [0 1].

**Table 8.3.9.1.1-1: Test Parameters**

| Parameter | Test |
|---|---|
| Number of information bits | 2 |
| Number of symbols | 14 |
| Intra-slot frequency hopping | N/A |
| Group and sequence hopping | neither |
| Hopping ID | 0 |
| Initial cyclic shift | 0 |
| First symbol | 0 |
| Index of orthogonal cover code (*timeDomainOCC*) | 0 |
| Number of interlace | 1 |
| Interlace index | 0 Note1 |
| NOTE 1: RBs 0,10,20,…, 100 are allocated for 15kHz SCS and RBs 0,5,10,…, 50 are allocated for 30kHz SCS | |

### 8.3.9.1.2 Minimum requirements

The NACK to ACK probability shall not exceed 0.1% at the SNR given in table 8.3.9.1.2-1.

**Table 8.3.9.1.2-1: Minimum requirements for interlaced PUCCH format 1 with 15 kHz SCS, 20MHz channel bandwidth**

| Number of Tx antennas | Number of RX antennas | Cyclic-Prefix | Propagation conditions and correlation matrix (Annex G) | SNR (dB) |
|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-10 Low | -13.8 |

**Table 8.3.9.1.2-2: Minimum requirements for interlaced PUCCH format 1 with 30 kHz SCS, 20MHz channel bandwidth**

| Number of Tx antennas | Number of RX antennas | Cyclic-Prefix | Propagation conditions and correlation matrix (Annex G) | SNR (dB) |
|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-10 Low | -13.3 |

## 8.3.9.2 ACK missed detection requirements

### 8.3.9.2.1 General

The ACK missed detection probability is the probability of not detecting an ACK when an ACK was sent. The test parameters in table 8.3.9.1.1-1 are configured.

The ACK missed detection probability performance requirement only apply to PUCCH format 1 with 2 UCI bits. The UCI information only contain ACK/NACK information.

The 2bits UCI information is further defined with bitmap as [0 1].

### 8.3.9.2.2 Minimum requirements

The ACK missed detection probability shall not exceed 1% at the SNR given in table 8.3.9.2.2-1.

**Table 8.3.9.2.2-1: Minimum requirements for interlaced PUCCH format 1 with 15 kHz SCS, 20MHz channel bandwidth**

| Number of Tx antennas | Number of RX antennas | Cyclic-Prefix | Propagation conditions and correlation matrix (Annex G) | SNR (dB) |
|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-10 Low | -14.4 |

**Table 8.3.9.2.2-2: Minimum requirements for interlaced PUCCH format 1 with 30 kHz SCS, 20MHz channel bandwidth**

| Number of Tx antennas | Number of RX antennas | Cyclic-Prefix | Propagation conditions and correlation matrix (Annex G) | SNR (dB) |
|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-10 Low | -14.1 |

## 8.3.10 Performance requirements for interlaced PUCCH format 2

### 8.3.10.1 General

The performance is measured by the required SNR at UCI block error probability not exceeding 1%.

The UCI block error probability (BLER) is defined as the probability of incorrectly decoding the UCI information when the UCI information is sent. The UCI information does not contain CSI part 2.

The UCI block error probability performance requirement only applies to the PUCCH format 2 with 22 UCI bits.

The 22bits UCI information case is assumed random information bit selection.

**Table 8.3.10.1-1: Test Parameters**

| Parameter | Value |
|---|---|
| Modulation order | QSPK |
| Intra-slot frequency hopping | N/A |
| Number of symbols | 1 |
| The number of UCI information bits | 22 |
| First symbol | 13 |
| DM-RS sequence generation | $N_{ID}^{0}$=0 |
| Number of interlaces | 1 |
| Interlace index | 0(note 1) |
| OCC-length-r16 | Not configured |
| NOTE 1: RBs 0,10,20,…,100 are allocated for 15kHz SCS and RBs 0,5,10,…,50 are allocated for 30kHz SCS | |

### 8.3.10.2 Minimum requirements

The UCI block error probability shall not exceed 1% at the SNR given in table 8.3.10.2-1 and table 8.3.10.2-2 for 22 UCI bits.

**Table 8.3.10.2-1: Minimum requirements for interlaced PUCCH format 2 with 15 kHz SCS, 20 MHz channel bandwidth**

| Number of Tx antennas | Number of RX antennas | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | SNR(dB) |
|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-10 Low | 3.5 |

**Table 8.3.10.2-2: Minimum requirements for interlaced PUCCH format 2 with 30 kHz SCS, 20 MHz channel bandwidth**

| Number of Tx antennas | Number of RX antennas | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | SNR(dB) |
|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-10 Low | 3.9 |

## 8.3.11 Performance requirements for interlaced PUCCH format 3

### 8.3.11.1 General

The ACK missed detection probability is the probability of not detecting an ACK when an ACK was sent.

The ACK missed detection requirement only applies to the PUCCH format 3 with 4 UCI bits. The UCI information only contains ACK information.

The 4 bits UCI information case is further defined with the bitmap as [1 1 1 1].

**Table 8.3.11.1-1: Test Parameters**

| Parameter | Value |
|---|---|
| Modulation order | QPSK |
| Intra-slot frequency hopping | N/A |
| Group and sequence hopping | Neither |
| Hopping ID | 0 |
| Number of symbols | 4 |
| The number of UCI information bits | 4 |
| Index of OCC | Not configured |
| Length of OCC | Not configured |
| Cyclic shift index for DMRS | 0 |
| Number of Interlace | 1 |
| Interlace index | 0(note 1) |
| NOTE 1: RBs 0,10,20,…,90 are allocated for 15kHz SCS and RBs 0,5,10,…,45 are allocated for 30kHz SCS | |

### 8.3.11.2 Minimum requirements

The ACK missed detection probability shall not exceed 1% at the SNR given in table 8.3.11.2-1 and table 8.3.11.2-2 for 4UCI bits.

**Table 8.3.11.2-1: Minimum requirements for interlaced PUCCH format 3 with 15 kHz SCS, 20 MHz channel bandwidth**

| Number of Tx antennas | Number of RX antennas | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Additional DM-RS configuration | SNR(dB) |
|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-10 Low | No additional DM-RS | -6 |

**Table 8.3.11.2-2: Minimum requirements for interlaced PUCCH format 3 with 30 kHz SCS, 20 MHz channel bandwidth**

| Number of Tx antennas | Number of RX antennas | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Additional DM-RS configuration | SNR(dB) |
|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-10 Low | No additional DM-RS | -5.4 |

## 8.3.12 Performance requirements for PUCCH format 1 with DM-RS bundling

### 8.3.12.1 NACK to ACK requirements

#### 8.3.12.1.1 General

The NACK to ACK detection probability is the probability that an ACK bit is falsely detected when an NACK bit was sent on the particular bit position, where the NACK to ACK detection probability is defined as follows:

$$\text{Prob}(\text{PUCCH NACK} \rightarrow \text{ACK bits}) = \frac{\#(\text{NACK bits decoded as ACK bits})}{\#(\text{Total NACK bits})},$$

where:

- $\#(\text{Total NACK bits})$ denotes the total number of NACK bits transmitted
- $\#(\text{NACK bits decoded as ACK bits})$ denotes the number of NACK bits decoded as ACK bits at the receiver, i.e. the number of received ACK bits
- NACK bits in the definition do not contain the NACK bits which are mapped from DTX, i.e. NACK bits received when DTX is sent should not be considered.

The NACK to ACK detection probability performance requirement only apply to PUCCH format 1 with 2 UCI bits. The UCI information only contain ACK/NACK information.

The 2bits UCI information is further defined with bitmap as [0 1].

**Table 8.3.12.1.1-1: Test Parameters**

| Parameter | Test 1 | Test 2 |
|---|---|---|
| Example TDD UL-DL pattern (Note1) | 15 kHz SCS: 7D1S2U, S=6D:4G:4U<br>30 kHz SCS: TDD 7D1S2U, S=6D:4G:4U | 15 kHz SCS: FDD<br>30 kHz SCS: FDD |
| Number of information bits | 2 | 2 |
| Number of PRBs | 1 | 1 |
| Number of symbols | 14 | 14 |
| First PRB prior to frequency hopping | 0 | 0 |
| Intra-slot frequency hopping | disabled | disabled |
| Group and sequence hopping | neither | neither |
| Hopping ID | 0 | 0 |
| Initial cyclic shift | 0 | 0 |
| First symbol | 0 | 0 |
| Index of orthogonal cover code (*timeDomainOCC*) | 0 | 0 |
| Number of slots for PUCCH repetition | 2 | 8 |
| PUCCH-TimeDomainWindowLength | 2 | 8 |
| Note 1: The same TDD requirements are applicable to different UL-DL patterns with more than one consecutive UL slots when both pucch-TimeDomainWindowLength and PUCCH aggregation factor are configured as 2 slots. The UL (re)transmission of PUCCH is only scheduled for the actual TDW including 2 consecutive UL slots. | | |

#### 8.3.12.1.2 Minimum requirements

The NACK to ACK probability shall not exceed 0.1% at the SNR given in table 8.3.12.1.2-1 and table 8.3.12.1.2-2

**Table 8.3.12.1.2-1: Minimum requirements for PUCCH format 1 with DMRS bundling, 15 kHz SCS, 5MHz channel bandwidth,**

| Test number | Number of Tx antennas | Number of RX antennas | Cyclic-Prefix | Propagation conditions and correlation matrix (Annex G) | SNR (dB) |
|---|---|---|---|---|---|
| 1 | 1 | 2 | Normal | TDLA30-10 Low | -3.0 |
| 2 | 1 | 2 | Normal | TDLA30-10 Low | -9.1 |

**Table 8.3.12.1.2-2: Minimum requirements for PUCCH format 1 with DMRS bundling, 30 kHz SCS, 10MHz channel bandwidth,**

| Test number | Number of Tx antennas | Number of RX antennas | Cyclic-Prefix | Propagation conditions and correlation matrix (Annex G) | SNR (dB) |
|---|---|---|---|---|---|
| 1 | 1 | 2 | Normal | TDLA30-10 Low | -3.0 |
| 2 | 1 | 2 | Normal | TDLA30-10 Low | -8.7 |

### 8.3.12.2 ACK missed detection requirements

#### 8.3.12.2.1 General

The ACK missed detection probability is the probability of not detecting an ACK when an ACK was sent. The test parameters in table 8.3.12.1.1-1 are configured.

The ACK missed detection probability performance requirement only apply to PUCCH format 1 with 2 UCI bits. The UCI information only contain ACK/NACK information.

The 2bits UCI information is further defined with bitmap as [0 1].

#### 8.3.12.2.2 Minimum requirements

The ACK missed detection probability shall not exceed 1% at the SNR given in table 8.3.12.2.2-1 and table 8.3.12.2.2-2

**Table 8.3.12.2.2-1: Minimum requirements for PUCCH format 1 with DMRS bundling, 15 kHz SCS, 5MHz channel bandwidth,**

| Test number | Number of Tx antennas | Number of RX antennas | Cyclic-Prefix | Propagation conditions and correlation matrix (Annex G) | SNR (dB) |
|---|---|---|---|---|---|
| 1 | 1 | 2 | Normal | TDLA30-10 Low | -3.9 |
| 2 | 1 | 2 | Normal | TDLA30-10 Low | -8.9 |

**Table 8.3.12.2.2-2: Minimum requirements for PUCCH format 1 with DMRS bundling, 30 kHz SCS, 10MHz channel bandwidth,**

| Test number | Number of Tx antennas | Number of RX antennas | Cyclic-Prefix | Propagation conditions and correlation matrix (Annex G) | SNR (dB) |
|---|---|---|---|---|---|
| 1 | 1 | 2 | Normal | TDLA30-10 Low | -3.8 |
| 2 | 1 | 2 | Normal | TDLA30-10 Low | -8.5 |

## 8.3.13 Performance requirements for PUCCH format 3 with DMRS bundling

### 8.3.13.1 General

The performance is measured by the required SNR at UCI block error probability not exceeding 1%.

The UCI block error probability is defined as the conditional probability of incorrectly decoding the UCI information when the UCI information is sent. The UCI information does not contain CSI part 2.

**Table 8.3.13.1-1: Test Parameters**

| Parameter | Test 1 | Test 2 |
|---|---|---|
| Eample TDD UL-DL pattern (Note1) | 15 kHz SCS:<br>7D1S2U, S=6D:4G:4U<br>30 kHz SCS: TDD<br>7D1S2U, S=6D:4G:4U | 15 kHz SCS: FDD<br>30 kHz SCS: FDD |
| Modulation order | QPSK | QPSK |
| First PRB prior to frequency hopping | 0 | 0 |
| Intra-slot frequency hopping | disabled | disabled |
| Group and sequence hopping | neither | neither |
| Hopping ID | 0 | 0 |
| Number of PRBs | 1 | 1 |
| Number of symbols | 14 | 14 |
| The number of UCI information bits | 16 | 16 |
| First symbol | 0 | 0 |
| Number of slots for PUCCH repetition | 2 | 8 |
| PUCCH-TimeDomainWindowLength | 2 | 8 |
| Note 1: The same TDD requirements are applicable to different UL-DL patterns with more than one consecutive UL slots when both pucch-TimeDomainWindowLength and PUCCH aggregation factor are configured as 2 slots.<br>The UL (re)transmission of PUCCH is only scheduled for the actual TDW including 2 consecutive UL slots. | | |

## 8.3.13.2 Minimum requirements

The UCI block error probability shall not exceed 1% at the SNR given in Table 8.3.13.2-1 and Table 8.3.13.2-2.

**Table 8.3.13.2-1: Minimum requirements for PUCCH format 3 with DMRS bundling, 15 kHz SCS, 5MHz channel bandwidth**

| Test Number | Number of TX antennas | Number of RX antennas | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Additional DM-RS configuration | Channel bandwidth / SNR (dB) |
|---|---|---|---|---|---|---|
| | | | | | | 5 MHz |
| 1 | 1 | 2 | Normal | TDLA30-10 Low | No additional DM-RS | 0.3 |
| | | | | | Additional DM-RS | -0.1 |
| 2 | 1 | 2 | Normal | TDLA30-10 Low | No additional DM-RS | -5.3 |
| | | | | | Additional DM-RS | -5.8 |

**Table 8.3.13.2-2: Minimum requirements for PUCCH format 3 with DMRS bundling, 30 kHz SCS, 10MHz channel bandwidth**

| Test Number | Number of TX antennas | Number of RX antennas | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Additional DM-RS configuration | Channel bandwidth / SNR (dB) |
|---|---|---|---|---|---|---|
| | | | | | | 10 MHz |
| 1 | 1 | 2 | Normal | TDLA30-10 Low | No additional DM-RS | 0.2 |
| | | | | | Additional DM-RS | 0.0 |
| 2 | `1 | 2 | Normal | TDLA30-10 Low | No additional DM-RS | -5.2 |
| | | | | | Additional DM-RS | -5.6 |

## 8.3.14 Performance requirements for sub-slot repetition PUCCH format 0

### 8.3.14.1 General

The ACK missed detection probability is the probability of not detecting an ACK when an ACK was sent.

The ACK missed detection probability performance requirement only apply to PUCCH format 0 with 1 UCI bit. The UCI information only contain ACK information.

The 1bit UCI information is further defined with the bitmap as [1].

**Table 8.3.14.1-1: Test Parameters**

| Parameter | Test |
|---|---|
| Number of UCI information bits | 1 |
| Number of PRBs | 1 |
| Number of PUCCH symbols *(nrofSymobls)* | 2 |
| Number of sub-slot PUCCH repetitions *(nrofSlots)* | 2 |
| Number of Sub-slot symbols *(subslotLengthForPUCCH-r16)* | 7 |
| First symbol of sub-slot *(startingSymbolIndex)* | 5 |
| First PRB prior to frequency hopping | 0 |
| First PRB after frequency hopping | The largest PRB index – (Number of PRBs – 1) |
| Intra-slot frequency hopping | disabled |
| Inter-slot frequency hopping | enabled |
| Group and sequence hopping | neither |
| Hopping ID | 0 |
| Initial cyclic shift | 0 |

The transient period as specified in TS 38.101-1 [17] clause 6.3.3.1 is not taken into account for performance requirement testing, where the RB hopping is symmetric to the CC centre, i.e., inter-slot frequency hopping is enabled.

### 8.3.14.2 Minimum requirements

The ACK missed detection probability shall not exceed 1% at the SNR given in table 8.3.14.2-1

**Table 8.3. 14.2-1: Minimum requirements for sub-slot repetition PUCCH format 0 with 30 kHz SCS, 10MHz channel bandwidth**

| Number of Tx antennas | Number of RX antennas | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | SNR (dB) |
|---|---|---|---|---|
| 1 | 2 | Normal | TDLC300-100 Low | 1.5 |

# 8.4 Performance requirements for PRACH

## 8.4.1 PRACH False alarm probability

### 8.4.1.1 General

The false alarm requirement is valid for any number of receive antennas, for any channel bandwidth.

The false alarm probability is the conditional total probability of erroneous detection of the preamble (i.e. erroneous detection from any detector) when input is only noise.

### 8.4.1.2 Minimum requirement

The false alarm probability shall be less than or equal to 0.1%.

## 8.4.2 PRACH detection requirements

### 8.4.2.1 General

The probability of detection is the conditional probability of correct detection of the preamble when the signal is present. There are several error cases – detecting different preamble than the one that was sent, not detecting a preamble at all or correct preamble detection but with the wrong timing estimation. For AWGN, TDLC300-100 and TDLA30-10, a timing estimation error occurs if the estimation error of the timing of the strongest path is larger than the time error tolerance given in Table 8.4.2.1-1.

The performance requirements for high speed train (table 8.4.23-1 to 8.4.2.3-4) are optional.

**Table 8.4.2.1-1: Time error tolerance for AWGN, TDLC300-100 and TDLA30-10**

| PRACH preamble | PRACH SCS (kHz) | Time error tolerance | | |
|---|---|---|---|---|
| | | AWGN | TDLC300-100 | TDLA30-10 |
| 0 | 1.25 | 1.04 us | 2.55 us | N/A |
| A1, A2, A3, B4, C0, C2 | 15 | 0.52 us | 2.03 us | 0.67 us |
| | 30 | 0.26 us | 1.77 us | 0.41 us |

The test preambles for normal mode are listed in table A.6-1 and the test parameter *msg1-FrequencyStart* is set to 0. The test preambles for high speed train restricted set type A are listed in A.6-3, the test preambles for high speed train restricted set type B are listed in A.6-4, and the test preambles for high speed train short formats are listed in A.6-5. The test parameter *msg1-FrequencyStart* for high speed train is set to 0.

The test preambles for PRACH with LRA=1151 and LRA=571 are listed in table A.6-6.

### 8.4.2.2 Minimum requirements for Normal Mode

The probability of detection shall be equal to or exceed 99% for the SNR levels listed in Tables 8.4.2.2-1 to 8.4.2.2-3.

**Table 8.4.2.2-1: PRACH missed detection requirements for Normal Mode, 1.25 kHz SCS**

| Number of TX antennas | Number of RX antennas | Propagation conditions and correlation matrix (Annex G) | Frequency offset | SNR (dB) |
|---|---|---|---|---|
| | | | | Burst format 0 |
| 1 | 2 | AWGN | 0 | -14.5 |
| | | TDLC300-100 Low | 400 Hz | -6.6 |
| | 4 | AWGN | 0 | -16.7 |
| | | TDLC300-100 Low | 400 Hz | -11.9 |
| | 8 | AWGN | 0 | -18.9 |
| | | TDLC300-100 Low | 400 Hz | -15.8 |

**Table 8.4.2.2-2: PRACH missed detection requirements for Normal Mode, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Propagation conditions and correlation matrix (Annex G) | Frequency offset | SNR (dB) | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | Burst format A1 | Burst format A2 | Burst format A3 | Burst format B4 | Burst format C0 | Burst format C2 |
| 1 | 2 | AWGN | 0 | -9.3 | -12.6 | -14.2 | -16.8 | -6.3 | -12.5 |
| | | TDLC300-100 Low | 400 Hz | -2.1 | -4.8 | -6.6 | -8.8 | 0.8 | -4.9 |
| | 4 | AWGN | 0 | -11.6 | -14.3 | -16.0 | -19.0 | -8.7 | -14.1 |
| | | TDLC300-100 Low | 400 Hz | -7.3 | -10.3 | -11.7 | -13.8 | -4.3 | -10.2 |
| | 8 | AWGN | 0 | -13.8 | -16.7 | -18.2 | -21.2 | -11.1 | -16.6 |
| | | TDLC300-100 Low | 400 Hz | -11.0 | -13.9 | -15.2 | -17.3 | -8.1 | -13.9 |

**Table 8.4.2.2-3: PRACH missed detection requirements for Normal Mode, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Propagation conditions and correlation matrix (Annex G) | Frequency offset | SNR (dB) | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | Burst format A1 | Burst format A2 | Burst format A3 | Burst format B4 | Burst format C0 | Burst format C2 |
| 1 | 2 | AWGN | 0 | -9.1 | -12.0 | -13.8 | -16.5 | -6.1 | -11.9 |
| | | TDLC300-100 Low | 400 Hz | -2.8 | -5.7 | -7.4 | -9.9 | 0.1 | -5.6 |
| | 4 | AWGN | 0 | -11.4 | -14.2 | -15.9 | -19.0 | -8.6 | -14.1 |
| | | TDLC300-100 Low | 400 Hz | -7.2 | -10.4 | -12.0 | -14.5 | -4.5 | -10.4 |
| | 8 | AWGN | 0 | -13.7 | -16.6 | -18.1 | -21.1 | -11.0 | -16.5 |
| | | TDLC300-100 Low | 400 Hz | -10.7 | -13.7 | -15.1 | -17.6 | -7.8 | -13.7 |

**Table 8.4.2.2-4: Void**

**Table 8.4.2.2-5: Void**

### 8.4.2.3 Minimum requirements for high speed train

The probability of detection shall be equal to or exceed 99% for the SNR levels listed in Tables 8.4.2.3-1 to 8.4.2.3-4

**Table 8.4.2.3-1: PRACH missed detection requirements for high speed train, burst format 0, restricted set type A, 1.25 kHz SCS**

| Number of TX antennas | Number of RX antennas | Propagation conditions and correlation matrix (Annex G) | Frequency offset | SNR (dB) |
|---|---|---|---|---|
| | | | | Burst format 0 |
| 1 | 2 | AWGN | 625 Hz | -12.0 |
| | | AWGN | 1340 Hz | -13.8 |
| | | TDLC300-100 Low | 0 Hz | -6.3 |
| | 4 | AWGN | 625 Hz | -14.5 |
| | | AWGN | 1340 Hz | -16.2 |
| | | TDLC300-100 Low | 0 Hz | -11.8 |
| | 8 | AWGN | 625 Hz | -16.5 |
| | | AWGN | 1340 Hz | -18.4 |
| | | TDLC300-100 Low | 0 Hz | -16.2 |

**Table 8.4.2.3-2: PRACH missed detection requirements for high speed train, burst format 0, restricted set type B, 1.25 kHz SCS**

| Number of TX antennas | Number of RX antennas | Propagation conditions and correlation matrix (Annex G) | Frequency offset | SNR (dB) |
|---|---|---|---|---|
| | | | | Burst format 0 |
| 1 | 2 | AWGN | 625 Hz | -11.6 |
| | | AWGN | 2334 Hz | -13.1 |
| | | TDLC300-100 Low | 0 Hz | -6.0 |
| | 4 | AWGN | 625 Hz | -14.0 |
| | | AWGN | 2334 Hz | -15.4 |
| | | TDLC300-100 Low | 0 Hz | -11.7 |
| | 8 | AWGN | 625 Hz | -16.3 |
| | | AWGN | 2334 Hz | -17.4 |
| | | TDLC300-100 Low | 0 Hz | -16.0 |

**Table 8.4.2.3-3: PRACH missed detection requirements for high speed train, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Propagation conditions and correlation matrix (Annex G) | Frequency offset | SNR (dB) | | |
|---|---|---|---|---|---|---|
| | | | | Burst format A2 | Burst format B4 | Burst format C2 |
| **1** | 2 | AWGN | 1740 Hz | -11.3 | -14.3 | -11.1 |
| | 4 | AWGN | 1740 Hz | -13.5 | -16.7 | -13.4 |
| | 8 | AWGN | 1740 Hz | -15.6 | -18.2 | -15.5 |

**Table 8.4.2.3-4: PRACH missed detection requirements for high speed train, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Propagation conditions and correlation matrix (Annex G) | Frequency offset | SNR (dB) | | |
|---|---|---|---|---|---|---|
| | | | | Burst format A2 | Burst format B4 | Burst format C2 |
| **1** | 2 | AWGN | 3334 Hz | -11.2 | -14.6 | -11.0 |
| | 4 | AWGN | 3334 Hz | -13.4 | -16.7 | -13.4 |
| | 8 | AWGN | 3334 Hz | -15.4 | -18.4 | -15.4 |

### 8.4.2.4 Minimum requirements for PRACH with $L_{RA}$=1151 and $L_{RA}$=571

The probability of detection shall be equal to or exceed 99% for the SNR levels listed in Tables 8.4.2.4-1 to 8.4.2.4-2.

**Table 8.4.2.4-1: Missed detection requirements for PRACH with $L_{RA}$=1151, 15 kHz SCS**

| Number of TX antennas | Number of RX antennas | Propagation conditions and correlation matrix (Annex G) | Frequency offset | SNR (dB) | | |
|---|---|---|---|---|---|---|
| | | | | Burst format A2 | Burst format B4 | Burst format C2 |
| 1 | 2 | AWGN | 0 | -21.4 | -25.4 | -21.4 |
| | | TDLA30-10 Low | 400 Hz | -15.1 | -18.3 | -15.2 |

**Table 8.4.2.4-2: Missed detection requirements for PRACH with $L_{RA}$=571, 30 kHz SCS**

| Number of TX antennas | Number of RX antennas | Propagation conditions and correlation matrix (Annex G) | Frequency offset | SNR (dB) | | |
|---|---|---|---|---|---|---|
| | | | | Burst format A2 | Burst format B4 | Burst format C2 |
| 1 | 2 | AWGN | 0 | -18.4 | -22.3 | -18.4 |
| | | TDLA30-10 Low | 400 Hz | -12.1 | -15.8 | -12.1 |

# 9 Radiated transmitter characteristics

## 9.1 General

Radiated transmitter characteristics requirements apply on the *BS type 1-H*, *BS type 1-O*, or *BS type 2-O* including all its functional components active and for all foreseen modes of operation of the BS unless otherwise stated.

## 9.2 Radiated transmit power

### 9.2.1 General

*BS type 1-H*, *BS type 1-O* and *BS type 2-O* are declared to support one or more beams, as per manufacturer's declarations specified in TS 38.141-2 [6]. Radiated transmit power is defined as the EIRP level for a declared beam at a specific *beam peak direction*.

For each beam, the requirement is based on declaration of a beam identity, *reference beam direction pair*, beamwidth, *rated beam EIRP*, *OTA peak directions set*, the *beam direction pairs* at the maximum steering directions and their associated *rated beam EIRP* and beamwidth(s).

For a declared beam and *beam direction pair*, the *rated beam EIRP* level is the maximum power that the base station is declared to radiate at the associated *beam peak direction* during the *transmitter ON period*.

For each *beam peak direction* associated with a *beam direction pair* within the *OTA peak directions set*, a specific *rated beam EIRP* level may be claimed. Any claimed value shall be met within the accuracy requirement as described below. *Rated beam EIRP* is only required to be declared for the *beam direction pairs* subject to conformance testing as detailed in TS 38.141-2 [6].

NOTE 1: *OTA peak directions set* is set of *beam peak directions* for which the EIRP accuracy requirement is intended to be met. The *beam peak directions* are related to a corresponding contiguous range or discrete list of *beam centre directions* by the *beam direction pairs* included in the set.

NOTE 2: A *beam direction pair* is data set consisting of the *beam centre direction* and the related *beam peak direction.*

NOTE 3: A declared EIRP value is a value provided by the manufacturer for verification according to the conformance specification declaration requirements, whereas a claimed EIRP value is provided by the manufacturer to the equipment user for normal operation of the equipment and is not subject to formal conformance testing.

For *operating bands* where the supported *fractional bandwidth* (FBW) is larger than 6%, two rated carrier EIRP may be declared by manufacturer:

- $P_{rated,c,FBWlow}$ for lower supported frequency range, and
- $P_{rated,c,FBWhigh}$ for higher supported frequency range.

For frequencies in between $F_{FBWlow}$ and $F_{FBWhigh}$ the rated carrier EIRP is:

- $P_{rated,c,FBWlow,}$ for the carrier whose carrier frequency is within frequency range $F_{FBWlow} \leq f < (F_{FBWlow} + F_{FBWhigh}) / 2$,
- $P_{rated,c,FBWhigh,}$ for the carrier whose carrier frequency is within frequency range $(F_{FBWlow} + F_{FBWhigh}) / 2 \leq f \leq F_{FBWhigh}$.

For *BS type 1-O* there is no requirement specified for bands n46, n96, n100 and n101 and n102.

### 9.2.2 Minimum requirement for *BS type 1-H* and *BS type 1-O*

For each declared beam, in normal conditions, for any specific *beam peak direction* associated with a *beam direction pair* within the *OTA peak directions set*, a manufacturer claimed EIRP level in the corresponding *beam peak direction* shall be achievable to within ±2.2 dB of the claimed value.

For *BS type 1-O* only, for each declared beam, in extreme conditions, for any specific *beam peak direction* associated with a *beam direction pair* within the *OTA peak directions set*, a manufacturer claimed EIRP level in the corresponding *beam peak direction* shall be achievable to within ±2.7 dB of the claimed value.

Normal and extreme conditions are defined in TS 38.141-2, annex B [6].

In certain regions, the minimum requirement for normal conditions may apply also for some conditions outside the range of conditions defined as normal.

### 9.2.3 Minimum requirement for *BS type 2-O*

For each declared beam, in normal conditions, for any specific *beam peak direction* associated with a *beam direction pair* within the *OTA peak directions set*, a manufacturer claimed EIRP level in the corresponding *beam peak direction* shall be achievable to within ± 3.4 dB of the claimed value.

For each declared beam, in extreme conditions, for any specific *beam peak direction* associated with a *beam direction pair* within the *OTA peak directions set*, a manufacturer claimed EIRP level in the corresponding *beam peak direction* shall be achievable to within ±4.5 dB of the claimed value.

Normal and extreme conditions are defined in TS 38.141-2, annex B [6].

In certain regions, the minimum requirement for normal conditions may apply also for some conditions outside the range of conditions defined as normal.

## 9.3 OTA base station output power

### 9.3.1 General

OTA BS output power is declared as the TRP radiated requirement, with the output power accuracy requirement defined at the RIB during the *transmitter ON period*. TRP does not change with beamforming settings as long as the *beam peak direction* is within the *OTA peak directions set*. Thus the TRP accuracy requirement must be met for any beamforming setting for which the *beam peak direction* is within the *OTA peak directions set*.

The BS *rated carrier TRP output power* for *BS type 1-O* shall be within limits as specified in table 9.3.1-1.

**Table 9.3.1-1: BS *rated carrier TRP output power* limits for *BS type 1-O***

| BS class | $P_{rated,c,TRP}$ |
|---|---|
| Wide Area BS | (note) |
| Medium Range BS | ≤ + 47 dBm |
| Local Area BS | ≤ + 33 dBm |
| NOTE: There is no upper limit for the $P_{rated,c,TRP}$ of the Wide Area Base Station. | |

There is no upper limit for the *rated carrier TRP output power* of *BS type 2-O*.

For Band n41 and n90 operation in Japan, the rated output power, $P_{rated,c,TRP}$, declared by the manufacturer shall be equal to or less than 20 W per 10 MHz bandwidth.

Despite the general requirements for the BS output power described in clauses 9.3.2 – 9.3.3, additional regional requirements might be applicable.

NOTE: In certain regions, power limits corresponding to BS classes may apply for *BS type 2-O*.

### 9.3.2 Minimum requirement for *BS type 1-O*

In normal conditions, the *BS type 1-O maximum carrier TRP output power*, $P_{max,c,TRP}$ measured at the RIB shall remain within ±2 dB of the *rated carrier TRP output power* $P_{rated,c,TRP}$, as declared by the manufacturer.

Normal conditions are defined in TS 38.141-1, annex B [6].

### 9.3.3 Minimum requirement for *BS type 2-O*

In normal conditions, the *BS type 2-O maximum carrier TRP output power*, $P_{max,c,TRP}$ measured at the RIB shall remain within ±3 dB of the *rated carrier TRP output power* $P_{rated,c,TRP}$, as declared by the manufacturer.

Normal conditions are defined in TS 38.141-2, annex B [6].

### 9.3.4 Additional requirements (regional)

*In certain regions, additional regional requirements may apply.*

## 9.4 OTA output power dynamics

### 9.4.1 General

The requirements in clause 9.4 apply during the *transmitter ON period*. Transmit signal quality (as specified in clause 9.6) shall be maintained for the output power dynamics requirements.

The OTA output power requirements are *directional requirements* and apply to the *beam peak directions* over the *OTA peak directions set*.

### 9.4.2 OTA RE power control dynamic range

#### 9.4.2.1 General

The OTA RE power control dynamic range is the difference between the power of an RE and the average RE power for a BS at maximum output power ($P_{max,c,EIRP}$) for a specified reference condition.

This requirement shall apply at each RIB supporting transmission in the *operating band*.

#### 9.4.2.2 Minimum requirement for *BS type 1-O*

The OTA RE power control dynamic range is specified the same as the conducted RE power control dynamic range requirement for *BS type 1-C* and *BS type 1-H* in table 6.3.2.2-1.

### 9.4.3 OTA total power dynamic range

#### 9.4.3.1 General

The OTA total power dynamic range is the difference between the maximum and the minimum transmit power of an OFDM symbol for a specified reference condition.

This requirement shall apply at each RIB supporting transmission in the *operating band*.

NOTE 1: The upper limit of the OTA total power dynamic range is the BS maximum carrier EIRP ($P_{max,c,EIRP}$) when transmitting on all RBs. The lower limit of the OTA total power dynamic range is the average EIRP for single RB transmission in the same direction using the same beam. The OFDM symbol carries PDSCH and not contain RS or SSB.

### 9.4.3.2 Minimum requirement for *BS type 1-O*

OTA total power dynamic range minimum requirement for *BS type 1-O* is specified such as for each NR carrier it shall be larger than or equal to the levels specified for the conducted requirement for *BS type 1-C* and *BS type 1-H* in table 6.3.3.2-1.

### 9.4.3.3 Minimum requirement for *BS type 2-O*

OTA total power dynamic range minimum requirement for *BS type 2-O* is specified such as for each NR carrier it shall be larger than or equal to the levels specified in table 9.4.3.3-1 in FR2-1 and table 9.4.3.3-2 in FR2-2.

**Table 9.4.3.3-1: Minimum requirement for BS type 2-O total power dynamic range in FR2-1**

| SCS (kHz) | OTA total power dynamic range (dB) | | | |
|---|---|---|---|---|
| | 50 MHz | 100 MHz | 200 MHz | 400 MHz |
| 60 | 18.1 | 21.2 | 24.2 | N/A |
| 120 | 15.0 | 18.1 | 21.2 | 24.2 |

**Table 9.4.3.3-2: Minimum requirement for *BS type 2-O* total power dynamic range in FR2-2**

| SCS (kHz) | OTA total power dynamic range (dB) | | | | |
|---|---|---|---|---|---|
| | 100 MHz | 400 MHz | 800 MHz | 1600 MHz | 2000 MHz |
| 120 | 18.1 | 24.2 | N/A | N/A | N/A |
| 480 | N/A | 18.1 | 20.9 | 23.9 | N/A |
| 960 | N/A | 15.1 | 17.9 | 20.9 | 21.7 |

## 9.5 OTA transmit ON/OFF power

### 9.5.1 General

OTA transmit ON/OFF power requirements apply only to TDD operation of NR BS.

### 9.5.2 OTA transmitter OFF power

#### 9.5.2.1 General

OTA transmitter OFF power is defined as the mean power measured over 70/N µs filtered with a square filter of bandwidth equal to the *transmission bandwidth configuration* of the BS ($BW_{Config}$) centred on the assigned channel frequency during the *transmitter OFF period*. N = SCS/15, where SCS is Sub Carrier Spacing in kHz.

For BS supporting intra-band contiguous CA, the OTA transmitter OFF power is defined as the mean power measured over 70/N us filtered with a square filter of bandwidth equal to the *Aggregated BS Channel Bandwidth* $BW_{Channel_CA}$ centred on $(F_{edge,high}+F_{edge,low})/2$ during the *transmitter OFF period*. N = SCS/15, where SCS is the smallest supported Sub Carrier Spacing in kHz in the *Aggregated BS Channel Bandwidth*.

For *BS type 1-O*, the transmitter OFF power is defined as the output power at the *co-location reference antenna* conducted output(s). For *BS type 2-O* the transmitter OFF power is defined as TRP.

For *multi-band RIBs* and *single band RIBs* supporting transmission in multiple bands, the requirement is only applicable during the *transmitter OFF period* in all supported *operating bands*.

#### 9.5.2.2 Minimum requirement for *BS type 1-O*

The total power from all *co-location reference antenna* conducted output(s) shall be less than -106 dBm/MHz.

#### 9.5.2.3 Minimum requirement for *BS type 2-O*

The OTA transmitter OFF TRP spectral density for *BS type 2-O* shall be less than -36 dBm/MHz.

### 9.5.3 OTA transient period

#### 9.5.3.1 General

The OTA *transmitter transient period* is the time period during which the transmitter is changing from the tra*nsmitter OFF period* to the *transmitter ON period* or vice versa. The *transmitter transient period* is illustrated in figure 6.4.2.1-1.

This requirement shall be applied at each RIB supporting transmission in the *operating band*.

#### 9.5.3.2 Minimum requirement for *BS type 1-O*

For *BS type 1-O*, the OTA *transmitter transient period* shall be shorter than the values listed in the minimum requirement table 9.5.3.2-1.

**Table 9.5.3.2-1: Minimum requirement for the OTA *transmitter transient period* for *BS type 1-O***

| Transition | Transient period length (μs) |
|---|---|
| OFF to ON | 10 |
| ON to OFF | 10 |

#### 9.5.3.3 Minimum requirement for *BS type 2-O*

For *BS type 2-O*, the OTA *transmitter transient period* shall be shorter than the values listed in the minimum requirement table 9.5.3.3-1.

**Table 9.5.3.3-1: Minimum requirement for the OTA *transmitter transient period* for *BS type 2-O***

| Transition | Transient period length (μs) |
|---|---|
| OFF to ON | 3 |
| ON to OFF | 3 |

## 9.6 OTA transmitted signal quality

### 9.6.1 OTA frequency error

#### 9.6.1.1 General

The requirements in clause 9.6.1 apply to the *transmitter ON period*.

OTA frequency error is the measure of the difference between the actual BS transmit frequency and the assigned frequency. The same source shall be used for RF frequency and data clock generation.

OTA frequency error requirement is defined as a *directional requirement* at the RIB and shall be met within the *OTA coverage range*.

#### 9.6.1.2 Minimum requirement for *BS type 1-O*

For *BS type 1-O*, the modulated carrier frequency of each NR carrier configured by the BS shall be accurate to within the accuracy range given in table 6.5.1.2-1 observed over 1 ms.

#### 9.6.1.3 Minimum requirement for *BS type 2-O*

For *BS type 2-O*, the modulated carrier frequency of each NR carrier configured by the BS shall be accurate to within the accuracy range given in table 9.6.1.3-1 observed over 1 ms.

**Table 9.6.1.3-1: OTA frequency error minimum requirement**

| BS class | Accuracy |
|---|---|
| Wide Area BS | ±0.05 ppm |
| Medium Range BS | ±0.1 ppm |
| Local Area BS | ±0.1 ppm |

## 9.6.2 OTA modulation quality

### 9.6.2.1 General

Modulation quality is defined by the difference between the measured carrier signal and an ideal signal. Modulation quality can e.g. be expressed as Error Vector Magnitude (EVM). Details about how the EVM is determined are specified in Annex B for FR1 and Annex C for FR2.

OTA modulation quality requirement is defined as a *directional requirement* at the RIB and shall be met within the *OTA coverage range*.

### 9.6.2.2 Minimum Requirement for *BS type 1-O*

For *BS type 1-O*, the EVM levels of each NR carrier for different modulation schemes on PDSCH outlined in table 6.5.2.2-1 shall be met. Requirements shall be the same as clause 6.5.2.2 and follow EVM frame structure from clause 6.5.2.3.

### 9.6.2.3 Minimum Requirement for *BS type 2-O*

For *BS type* 2-O, the EVM levels of each NR carrier for different modulation schemes on PDSCH outlined in table 9.6.2.3-1 shall be met, following the EVM frame structure described in clause 9.6.2.3.1.

**Table 9.6.2.3-1: EVM requirements for *BS type 2-O* carrier**

| Applicability | Modulation scheme for PDSCH | Required EVM (%) |
|---|---|---|
| FR2 | QPSK | 17.5 |
| FR2 | 16QAM | 12.5 |
| FR2 | 64QAM | 8 |
| FR2-1 | 256QAM | 3.5 |

#### 9.6.2.3.1 EVM frame structure for measurement

EVM requirements shall apply for each NR carrier over all allocated resource blocks. Different modulation schemes listed in table 9.6.2.3-1 shall be considered for rank 1.

For NR, for all bandwidths, the EVM measurement shall be performed for each NR carrier over all allocated resource blocks and downlink subframes within 10 ms measurement periods. The boundaries of the EVM measurement periods need not be aligned with radio frame boundaries.

## 9.6.3 OTA time alignment error

### 9.6.3.1 General

This requirement shall apply to frame timing in MIMO transmission, *carrier aggregation* and their combinations.

Frames of the NR signals present in the radiated domain are not perfectly aligned in time. In relation to each other, the RF signals present in the radiated domain may experience certain timing differences.

The TAE is specified for a specific set of signals/transmitter configuration/transmission mode.

For a specific set of signals/transmitter configuration/transmission mode, the OTA Time Alignment Error (OTA TAE) is defined as the largest timing difference between any two different NR signals. The OTA time alignment error requirement is defined as a *directional requirement* at the RIB and shall be met within the *OTA coverage range.*

### 9.6.3.2 Minimum requirement for *BS type 1-O*

For MIMO transmission, at each carrier frequency, OTA TAE shall not exceed 65 ns.

For *intra-band contiguous carrier aggregation*, with or without MIMO, OTA TAE shall not exceed 260 ns.

For *intra-band non-contiguous carrier aggregation*, with or without MIMO, OTA TAE shall not exceed 3 μs.

For inter-band *carrier aggregation*, with or without MIMO, OTA TAE shall not exceed 3 μs.

**Table 9.6.3.2-1: Void**

**Table 9.6.3.2-2: Void**

**Table 9.6.3.2-3: Void**

### 9.6.3.3 Minimum requirement for *BS type 2-O*

The minimum requirement for TAE is given in Table 9.6.3.3-4.

**Table 9.6.3.3-1: Void**

**Table 9.6.3.3-2: Void**

**Table 9.6.3.3-3: Void**

**Table 9.6.3.3-4: TAE requirements for *BS type 2-O***

| Requirement | TAE | | |
|---|---|---|---|
| | 60, 120 kHz SCS (ns) | 480 kHz SCS (ns) | 960 kHz SCS (ns) |
| MIMO transmission | 65 | 32.5 | 32.5 |
| *intra-band contiguous carrier aggregation*, with or without MIMO | 130 | 32.5 | 32.5 |
| *intra-band non-contiguous carrier aggregation*, with or without MIMO (Note) | 260 | N/A | N/A |
| inter-band *carrier aggregation*, with or without MIMO | 3000 | 3000 | 3000 |
| NOTE: *intra-band non-contiguous carrier aggregation* is not supported for FR2-2 in this release. | | | |

## 9.7 OTA unwanted emissions

### 9.7.1 General

Unwanted emissions consist of so-called out-of-band emissions and spurious emissions according to ITU definitions ITU-R SM.329 [2]. In ITU terminology, out of band emissions are unwanted emissions immediately outside the *BS channel bandwidth* resulting from the modulation process and non-linearity in the transmitter but excluding spurious emissions. Spurious emissions are emissions which are caused by unwanted transmitter effects such as harmonics

emission, parasitic emission, intermodulation products and frequency conversion products, but exclude out of band emissions.

The OTA out-of-band emissions requirement for the *BS type 1-O* and *BS type 2-O* transmitter is specified both in terms of Adjacent Channel Leakage power Ratio (ACLR) and operating band unwanted emissions (OBUE). The OTA Operating band unwanted emissions define all unwanted emissions in each supported downlink *operating band* plus the frequency ranges $\Delta f_{OBUE}$ above and $\Delta f_{OBUE}$ below each band. OTA Unwanted emissions outside of this frequency range are limited by an OTA spurious emissions requirement.

The maximum offset of the operating band unwanted emissions mask from the *operating band* edge is $\Delta f_{OBUE}$. The value of $\Delta f_{OBUE}$ is defined in table 9.7.1-1 for *BS type 1-O* and *BS type 2-O* for the NR *operating bands*.

**Table 9.7.1-1: Maximum offset $\Delta f_{OBUE}$ outside the downlink *operating band***

| BS type | *Operating band* characteristics (MHz) | $\Delta f_{OBUE}$ (MHz) |
|---|---|---|
| BS type 1-O | $F_{DL,high} - F_{DL,low} < 100$ | 10 |
| | 100 MHz ≤ $F_{DL,high} - F_{DL,low}$ ≤ 900 | 40 |
| BS type 2-O | $F_{DL,high} - F_{DL,low} \leq 4000$ | 1500 |
| | $4000 < F_{DL,high} - F_{DL,low} \leq 14000$ | 3500 |

The unwanted emission requirements are applied per cell for all the configurations. Requirements for OTA unwanted emissions are captured using TRP, *directional requirements* or co-location requirements as described per requirement.

There is in addition a requirement for occupied bandwidth.

For band n104, the values of $\Delta f_{OBUE}$ are defined in table 9.7.1-1a.

**Table 9.7.1-1a: Maximum offset of OBUE outside the downlink *operating band* for band n104**

| BS type | *Operating band* | $\Delta f_{OBUE}$ (MHz) |
|---|---|---|
| *BS type 1-O* | n104 | 100 |

## 9.7.2 OTA occupied bandwidth

### 9.7.2.1 General

The OTA occupied bandwidth is the width of a frequency band such that, below the lower and above the upper frequency limits, the mean powers emitted are each equal to a specified percentage $\beta/2$ of the total mean transmitted power. See also recommendation ITU-R SM.328 [3].

The value of $\beta/2$ shall be taken as 0.5%.

The OTA occupied bandwidth requirement shall apply during the *transmitter ON period* for a single transmitted carrier. The minimum requirement below may be applied regionally. There may also be regional requirements to declare the OTA occupied bandwidth according to the definition in the present clause.

The OTA occupied bandwidth is defined as a *directional requirement* and shall be met in the manufacturer's declared *OTA coverage range* at the RIB.

### 9.7.2.2 Minimum requirement for *BS type 1-O* and *BS type* 2-O

The OTA occupied bandwidth for each NR carrier shall be less than the *BS channel bandwidth*. For intra-band contiguous CA, the OTA occupied bandwidth shall be less than or equal to the *Aggregated BS Channel Bandwidth*.

## 9.7.3 OTA Adjacent Channel Leakage Power Ratio (ACLR)

### 9.7.3.1 General

OTA Adjacent Channel Leakage power Ratio (ACLR) is the ratio of the filtered mean power centred on the assigned channel frequency to the filtered mean power centred on an adjacent channel frequency. The measured power is TRP.

The requirement shall be applied per RIB during the *transmitter ON period*.

### 9.7.3.2 Minimum requirement for *BS type 1-O*

The ACLR (CACLR) absolute *basic limits* in table 6.6.3.2-2 + X, 6.6.3.2-3a + X (where X = 9 dB) or the ACLR (CACLR) *basic limit* in table 6.6.3.2-1, 6.6.3.2-2a or 6.6.3.2-3, whichever is less stringent, shall apply.

For a *RIB* operating in multi-carrier or contiguous CA, the ACLR requirements in clause 6.6.3.2 shall apply to *BS channel bandwidths* of the outermost carrier for the frequency ranges defined in table 6.6.3.2-1.For a RIB operating in *non-contiguous spectrum*, the ACLR requirement in clause 6.6.3.2 shall apply in *sub-block gaps* for the frequency ranges defined in table 6.6.3.2-2a, while the CACLR requirement in clause 6.6.3.2 shall apply in *sub-block gaps* for the frequency ranges defined in table 6.6.3.2-3.

For a *multi-band RIB*, the ACLR requirement in clause 6.6.3.2 shall apply in *Inter RF Bandwidth gaps* for the frequency ranges defined in table 6.6.3.2-2a, while the CACLR requirement in clause 6.6.3.2 shall apply in *Inter RF Bandwidth gaps* for the frequency ranges defined in table 6.6.3.2-3.

### 9.7.3.3 Minimum requirement for *BS type 2-O*

The OTA ACLR limit is specified in table 9.7.3.3-1.

The OTA ACLR absolute limit is specified in table 9.7.3.3-2.

The OTA ACLR (CACLR) absolute limit in table 9.7.3.3-2 or 9.7.3.3-4a or the ACLR (CACLR) limit in table 9.7.3.3-1, 9.7.3.3-3 or 9.7.3.3-4, whichever is less stringent, shall apply.

For a *RIB* operating in multi-carrier or contiguous CA, the OTA ACLR requirements in table 9.7.3.3-1 shall apply to *BS channel bandwidths* of the outermost carrier for the frequency ranges defined in the table.For a RIB operating in *non-contiguous spectrum*, the OTA ACLR requirement in table 9.7.3.3-3 shall apply in *sub-block gaps* for the frequency ranges defined in the table, while the OTA CACLR requirement in table 9.7.3.3-4 shall apply in *sub-block gaps* for the frequency ranges defined in the table.

The CACLR in a *sub-block gap* is the ratio of:

a) the sum of the filtered mean power centred on the assigned channel frequencies for the two carriers adjacent to each side of the *sub-block gap*, and

b) the filtered mean power centred on a frequency channel adjacent to one of the respective *sub-block* edges.

The assumed filter for the adjacent channel frequency is defined in table 9.7.3.3-4 and the filters on the assigned channels are defined in table 9.7.3.3-5.

For operation in *non-contiguous spectrum*, the CACLR for NR carriers located on either side of the *sub-block gap* shall be higher than the value specified in table 9.7.3.3-4.

**Table 9.7.3.3-1: *BS type 2-O* ACLR limit**

| ***BS channel bandwidth* of *lowest/highest carrier* transmitted $BW_{Channel}$ (MHz)** | **BS adjacent channel centre frequency offset below the *lowest* or above the *highest carrier* centre frequency transmitted** | **Assumed adjacent channel carrier** | **Filter on the adjacent channel frequency and corresponding filter bandwidth** | **ACLR limit (dB)** |
|---|---|---|---|---|
| 50, 100, 200, 400, 800, 1600, 2000 | $BW_{Channel}$ | NR of same BW (Note 2) | Square ($BW_{Config}$) | 28 (Note 3)<br>26 (Note 4)<br>24 (Note 5) |
| NOTE 1: $BW_{Channel}$ and $BW_{Config}$ are the *BS channel bandwidth* and *transmission bandwidth configuration* of the *lowest/highest carrier* transmitted on the assigned channel frequency.<br>NOTE 2: With SCS that provides largest *transmission bandwidth configuration* ($BW_{Config}$).<br>NOTE 3: Applicable to bands defined within the frequency spectrum range of 24.25 – 33.4 GHz<br>NOTE 4: Applicable to bands defined within the frequency spectrum range of 37 – 52.6 GHz<br>NOTE 5: Applicable to bands defined within the frequency spectrum range of 52.6 – 71 GHz. | | | | |

**Table 9.7.3.3-2: *BS type 2-O* ACLR absolute limit**

| **BS class** | **ACLR absolute limit** |
|---|---|
| Wide area BS | -13 dBm/MHz |
| Medium range BS | -20 dBm/MHz |
| Local area BS | -20 dBm/MHz |

**Table 9.7.3.3-3: *BS type 2-O* ACLR limit in non-contiguous spectrum**

| ***BS channel bandwidth* of *carrier* transmitted adjacent to *sub-block gap* (MHz)** | ***Sub-block gap* size ($W_{gap}$) where the limit applies (MHz)** | **BS adjacent channel centre frequency offset below or above the *sub-block* edge (inside the gap)** | **Assumed adjacent channel carrier** | **Filter on the adjacent channel frequency and corresponding filter bandwidth** | **ACLR limit** |
|---|---|---|---|---|---|
| 50, 100 | $W_{gap}$≥ 100 (Note 5)<br>$W_{gap}$≥ 250 (Note 6) | 25 MHz | 50 MHz NR (Note 2) | Square ($BW_{Config}$) | 28 (Note 3)<br>26 (Note 4) |
| 200, 400 | $W_{gap}$≥ 400 (Note 6)<br>$W_{gap}$≥ 250 (Note 5) | 100 MHz | 200 MHz NR (Note 2) | Square ($BW_{Config}$) | 28 (Note 3)<br>26 (Note 4) |
| NOTE 1: $BW_{Config}$ is the *transmission bandwidth configuration* of the assumed adjacent channel carrier.<br>NOTE 2: With SCS that provides largest *transmission bandwidth configuration* ($BW_{Config}$).<br>NOTE 3: Applicable to bands defined within the frequency spectrum range of 24.25 – 33.4 GHz.<br>NOTE 4: Applicable to bands defined within the frequency spectrum range of 37 – 52.6 GHz.<br>NOTE 5: Applicable in case the *BS channel bandwidth* of the NR carrier transmitted at the other edge of the gap is 50 or 100 MHz.<br>NOTE 6: Applicable in case the *BS channel bandwidth* of the NR carrier transmitted at the other edge of the gap is 200 or 400 MHz. | | | | | |

**Table 9.7.3.3-4: *BS type 2-O* CACLR limit in non-contiguous spectrum**

| ***BS channel bandwidth* of *carrier* transmitted adjacent to s*ub-block gap* (MHz)** | ***Sub-block gap* size ($W_{gap}$) where the limit applies (MHz)** | **BS adjacent channel centre frequency offset below or above the *sub-block* edge (inside the gap)** | **Assumed adjacent channel carrier** | **Filter on the adjacent channel frequency and corresponding filter bandwidth** | **CACLR limit** |
|---|---|---|---|---|---|
| 50, 100 | 50 ≤$W_{gap}$< 100 (Note 5)<br>50 ≤$W_{gap}$< 250 (Note 6) | 25 MHz | 50 MHz NR (Note 2) | Square ($BW_{Config}$) | 28 (Note 3)<br>26 (Note 4) |
| 200, 400 | 200 ≤$W_{gap}$< 400 (Note 6)<br>200 ≤$W_{gap}$< 250 (Note 5) | 100 MHz | 200 MHz NR (Note 2) | Square ($BW_{Config}$) | 28 (Note 3)<br>26 (Note 4) |

NOTE 1: $BW_{Config}$ is the transmission bandwidth configuration of the assumed adjacent channel carrier.
NOTE 2: With SCS that provides largest transmission bandwidth configuration ($BW_{Config}$).
NOTE 3: Applicable to bands defined within the frequency spectrum range of 24.25 – 33.4 GHz.
NOTE 4: Applicable to bands defined within the frequency spectrum range of 37 – 52.6 GHz.
NOTE 5: Applicable in case the *BS channel bandwidth* of the NR carrier transmitted at the other edge of the gap is 50 or 100 MHz.
NOTE 6: Applicable in case the *BS channel bandwidth* of the NR carrier transmitted at the other edge of the gap is 200 or 400 MHz.

**Table 9.7.3.3-4a: *BS type 2-O* CACLR absolute limit**

| BS class | CACLR absolute limit |
|---|---|
| Wide area BS | -13 dBm/MHz |
| Medium range BS | -20 dBm/MHz |
| Local area BS | -20 dBm/MHz |

**Table 9.7.3.3-5: Filter parameters for the assigned channel**

| RAT of the carrier adjacent to the *sub-block gap* | Filter on the assigned channel frequency and corresponding filter bandwidth |
|---|---|
| NR | NR of same BW with SCS that provides largest *transmission bandwidth configuration* |

## 9.7.4 OTA operating band unwanted emissions

### 9.7.4.1 General

The OTA limits for operating band unwanted emissions are specified as TRP per RIB unless otherwise stated.

### 9.7.4.2 Minimum requirement for *BS type 1-O*

Out-of-band emissions in FR1 are limited by OTA operating band unwanted emission limits. Unless otherwise stated, the operating band unwanted emission limits in FR1 are defined from $\Delta f_{OBUE}$ below the lowest frequency of each supported downlink *operating band* up to $\Delta f_{OBUE}$ above the highest frequency of each supported downlink *operating band*. The values of $\Delta f_{OBUE}$ are defined in table 9.7.1-1 for the NR *operating bands*.

The requirements shall apply whatever the type of transmitter considered and for all transmission modes foreseen by the manufacturer's specification. For a *RIB* operating in multi-carrier or contiguous CA, the requirements apply to *BS channel bandwidths* of the outermost carrier for the frequency ranges defined in clause 6.6.4.1.

For a *RIB* operating in *non-contiguous spectrum*, the requirements shall apply inside any *sub-block gap* for the frequency ranges defined in clause 6.6.4.1.

For a *multi-band RIB*, the requirements shall apply inside any *Inter RF Bandwidth gap* for the frequency ranges defined in clause 6.6.4.1.

The OTA operating band unwanted emission requirement for *BS type 1-O* is that for each applicable *basic limit* for *BS type 1-H* in clause 6.6.4.2, the power of any unwanted emission shall not exceed an OTA limit specified as the *basic limit* + X, where X = 9 dB.

#### 9.7.4.2.1 Additional requirements

##### 9.7.4.2.1.1 Protection of DTT

In certain regions the following requirement may apply for protection of DTT. For *BS type 1-O* operating in Band n20, the level of emissions in the band 470-790 MHz, measured in an 8 MHz filter bandwidth on centre frequencies $F_{filter}$ according to table 9.7.4.2.1.1-1, shall not exceed the maximum emission TRP level shown in the table. This requirement applies in the frequency range 470-790 MHz even though part of the range falls in the spurious domain.

**Table 9.7.4.2.1.1-1: Declared emissions levels for protection of DTT**

| Case | Measurement filter centre frequency | Condition on BS maximum aggregate TRP / 10 MHz, $P_{TRP_10MHz}$ (NOTE) | Maximum level $P_{TRP,N,MAX}$ | *Measurement bandwidth* |
|---|---|---|---|---|
| A: for DTT frequencies where broadcasting is protected | N*8 + 306 MHz, 21 ≤ N ≤ 60 | $P_{TRP_10MHz} \geq 59$ dBm | 0 dBm | 8 MHz |
| | N*8 + 306 MHz, 21 ≤ N ≤ 60 | $36 \leq P_{TRP_10MHz} < 59$ dBm | $P_{TRP_10MHz} - 59$ dBm | 8 MHz |
| | N*8 + 306 MHz, 21 ≤ N ≤ 60 | $P_{TRP_10MHz} < 36$ dBm | -23 dBm | 8 MHz |
| B: for DTT frequencies where broadcasting is subject to an intermediate level of protection | N*8 + 306 MHz, 21 ≤ N ≤ 60 | $P_{TRP_10MHz} \geq 59$ dBm | 10 dBm | 8 MHz |
| | N*8 + 306 MHz, 21 ≤ N ≤ 60 | $36 \leq P_{TRP_10MHz} < 59$ dBm | $P_{TRP_10MHz} - 49$ dBm | 8 MHz |
| | N*8 + 306 MHz, 21 ≤ N ≤ 60 | $P_{TRP_10MHz} < 36$ dBm | -13 dBm | 8 MHz |
| C: for DTT frequencies where broadcasting is not protected | N*8 + 306 MHz, 21 ≤ N ≤ 60 | N/A | 22 dBm | 8 MHz |
| NOTE: $P_{TRP_10MHz}$ (dBm) is defined by $P_{TRP_10MHz} = P_{10MHz} + G_{ant} + 9$dB, where $G_{ant}$ is 17 dBi. | | | | |

##### 9.7.4.2.1.2 Limits in FCC Title 47

The BS may have to comply with the applicable emission limits established by FCC Title 47 [8], when deployed in regions where those limits are applied, and under the conditions declared by the manufacturer.

### 9.7.4.3 Minimum requirement for *BS type 2-O*

#### 9.7.4.3.1 General

The requirements of either clause 9.7.4.3.2 (Category A limits) or clause 9.7.4.3.3 (Category B limits) shall apply. The application of either Category A or Category B limits shall be the same as for General OTA transmitter spurious emissions requirements (*BS type 2-O*) in clause 9.7.5.3.2. In addition, the limits in clause 9.7.4.3.4 may also apply.

Out-of-band emissions in FR2 are limited by OTA operating band unwanted emission limits. Unless otherwise stated, the OTA operating band unwanted emission limits in FR2 are defined from $\Delta f_{OBUE}$ below the lowest frequency of each supported downlink *operating band* up to $\Delta f_{OBUE}$ above the highest frequency of each supported downlink *operating band*. The values of $\Delta f_{OBUE}$ are defined in table 9.7.1-1 for the NR *operating bands*.

The requirements shall apply whatever the type of transmitter considered and for all transmission modes foreseen by the manufacturer's specification. For a *RIB* operating in multi-carrier or contiguous CA, the requirements apply to the

frequencies ($\Delta f_{OBUE}$) starting from the edge of the *contiguous transmission bandwidth.* In addition, for a *RIB* operating in *non-contiguous spectrum*, the requirements apply inside any *sub-block gap*.

Emissions shall not exceed the maximum levels specified in the tables below, where:

- $\Delta f$ is the separation between the *contiguous transmission bandwidth* edge frequency and the nominal -3dB point of the measuring filter closest to the *contiguous transmission bandwidth* edge.
- f_offset is the separation between the *contiguous transmission bandwidth* edge frequency and the centre of the measuring filter.
- $f_offset_{max}$ is the offset to the frequency $\Delta f_{OBUE}$ outside the downlink *operating band*, where $\Delta f_{OBUE}$ is defined in table 9.7.1-1.
- $\Delta f_{max}$ is equal to $f_offset_{max}$ minus half of the bandwidth of the measuring filter.

In addition, inside any *sub-block gap* for a *RIB* operating in *non-contiguous spectrum*, emissions shall not exceed the cumulative sum of the limits specified for the adjacent *sub-blocks* on each side of the *sub-block gap*. The limit for each *sub-block* is specified in clauses 9.7.4.3.2 and 9.7.4.3.3 below, where in this case:

- $\Delta f$ is the separation between the *sub-block* edge frequency and the nominal -3 dB point of the measuring filter closest to the *sub-block* edge.
- f_offset is the separation between the *sub-block* edge frequency and the centre of the measuring filter.
- $f_offset_{max}$ is equal to the *sub-block gap* bandwidth minus half of the bandwidth of the measuring filter.
- $\Delta f_{max}$ is equal to $f_offset_{max}$ minus half of the bandwidth of the measuring filter.

### 9.7.4.3.2 OTA operating band unwanted emission limits (Category A)

BS unwanted emissions shall not exceed the maximum levels specified in table 9.7.4.3.2-1, 9.7.4.3.2-2, and 9.7.4.3.2-4.

**Table 9.7.4.3.2-1: OBUE limits applicable in the frequency range 24.25 – 33.4 GHz**

| Frequency offset of measurement filter -3B point, $\Delta f$ | Frequency offset of measurement filter centre frequency, f_offset | Limit | *Measurement bandwidth* |
|---|---|---|---|
| 0 MHz ≤ $\Delta f$ < $0.1*BW_{contiguous}$ | 0.5 MHz ≤ f_offset < $0.1*BW_{contiguous}$ +0.5 MHz | Min(-5 dBm, Max($P_{rated,t,TRP}$ – 35 dB, -12 dBm)) | 1 MHz |
| $0.1*BW_{contiguous}$ ≤ $\Delta f$ < $\Delta f_{max}$ | $0.1*BW_{contiguous}$ +0.5 MHz ≤ f_offset < $f_offset_{max}$ | Min(-13 dBm, Max($P_{rated,t,TRP}$ – 43 dB, -20 dBm)) | 1 MHz |
| NOTE 1: For *non-contiguous spectrum* operation within any *operating band* the limit within *sub-block gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* on each side of the *sub-block gap*. | | | |

**Table 9.7.4.3.2-2: OBUE limits applicable in the frequency range 37 – 52.6 GHz**

| Frequency offset of measurement filter -3B point, $\Delta f$ | Frequency offset of measurement filter centre frequency, f_offset | Limit | *Measurement bandwidth* |
|---|---|---|---|
| 0 MHz ≤ $\Delta f$ < $0.1*BW_{contiguous}$ | 0.5 MHz ≤ f_offset < $0.1*BW_{contiguous}$ +0.5 MHz | Min(-5 dBm, Max($P_{rated,t,TRP}$ – 33 dB, -12 dBm)) | 1 MHz |
| $0.1*BW_{contiguous}$ ≤ $\Delta f$ < $\Delta f_{max}$ | $0.1*BW_{contiguous}$ +0.5 MHz ≤ f_offset < $f_offset_{max}$ | Min(-13 dBm, Max($P_{rated,t,TRP}$ – 41 dB, -20 dBm)) | 1 MHz |
| NOTE 1: For *non-contiguous spectrum* operation within any *operating band* the limit within *sub-block gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* on each side of the *sub-block gap*. | | | |

**Table 9.7.4.3.2-3: Void**

**Table 9.7.4.3.2-4: OBUE limits applicable in the frequency range 52.6 – 71 GHz**

| Frequency offset of measurement filter -3B point, Δf | Frequency offset of measurement filter centre frequency, f_offset | Limit | *Measurement bandwidth* |
|---|---|---|---|
| 0 MHz ≤ Δf < 0.1*$BW_{contiguous}$ | 0.5 MHz ≤ f_offset < 0.1* $BW_{contiguous}$ +0.5 MHz | Min(-5 dBm, Max($P_{rated,t,TRP}$ – 31 dB, -12 dBm)) | 1 MHz |
| 0.1*$BW_{contiguous}$ ≤ Δf < $\Delta f_{max}$ | 0.1* $BW_{contiguous}$ +0.5 MHz ≤ f_offset < f_ $offset_{max}$ | Min(-13 dBm, Max($P_{rated,t,TRP}$ – 39 dB, -20 dBm)) | 1 MHz |
| NOTE 1: For *non-contiguous spectrum* operation within any *operating band* the limit within *sub-block gaps* is calculated as a cumulative sum of contributions from adjacent *sub-blocks* on each side of the *sub-block gap*. | | | |

## 9.7.4.3.3 OTA operating band unwanted emission limits (Category B)

BS unwanted emissions shall not exceed the maximum levels specified in table 9.7.4.3.3-1, 9.7.4.3.3-2, and 9.7.4.3.3-3.

**Table 9.7.4.3.3-1: OBUE limits applicable in the frequency range 24.25 – 33.4 GHz**

| Frequency offset of measurement filter -3 dB point, Δf | Frequency offset of measurement filter centre frequency, f_offset | Limit | *Measurement bandwidth* |
|---|---|---|---|
| 0 MHz ≤ Δf < 0.1*$BW_{contiguous}$ | 0.5 MHz ≤ f_offset < 0.1* $BW_{contiguous}$ +0.5 MHz | Min(-5 dBm, Max($P_{rated,t,TRP}$ – 35 dB, -12 dBm)) | 1 MHz |
| 0.1*$BW_{contiguous}$ ≤ Δf < $\Delta f_B$ | 0.1* $BW_{contiguous}$ +0.5 MHz ≤ f_offset < $\Delta f_B$ +0.5 MHz | Min(-13 dBm, Max($P_{rated,t,TRP}$ – 43 dB, -20 dBm)) | 1 MHz |
| $\Delta f_B$ ≤ Δf < $\Delta f_{max}$ | $\Delta f_B$ +5 MHz ≤ f_offset < f_ $offset_{max}$ | Min(-5 dBm, Max($P_{rated,t,TRP}$ – 33 dB, -10 dBm)) | 10 MHz |
| NOTE 1: For non-contiguous spectrum operation within any *operating band* the limit within sub-block gaps is calculated as a cumulative sum of contributions from adjacent sub-blocks on each side of the sub-block gap, where the contribution from the far-end sub-block shall be scaled according to the measurement bandwidth of the near-end sub-block.<br>NOTE 2: $\Delta f_B$ = 2*$BW_{contiguous}$ when $BW_{contiguous}$ ≤ 500 MHz, otherwise $\Delta f_B$ = $BW_{contiguous}$ + 500 MHz. | | | |

**Table 9.7.4.3.3-2: OBUE limits applicable in the frequency range 37 – 52.6 GHz**

| Frequency offset of measurement filter -3 dB point, Δf | Frequency offset of measurement filter centre frequency, f_offset | Limit | *Measurement bandwidth* |
|---|---|---|---|
| 0 MHz ≤ Δf < 0.1*$BW_{contiguous}$ | 0.5 MHz ≤ f_offset < 0.1* $BW_{contiguous}$ +0.5 MHz | Min(-5 dBm, Max($P_{rated,t,TRP}$ – 33 dB, -12 dBm)) | 1 MHz |
| 0.1*$BW_{contiguous}$ ≤ Δf < $\Delta f_B$ | 0.1* $BW_{contiguous}$ +0.5 MHz ≤ f_offset < $\Delta f_B$ +0.5 MHz | Min(-13 dBm, Max($P_{rated,t,TRP}$ – 41 dB, -20 dBm)) | 1 MHz |
| $\Delta f_B$ ≤ Δf < $\Delta f_{max}$ | $\Delta f_B$ +5 MHz ≤ f_offset < f_ $offset_{max}$ | Min(-5 dBm, Max($P_{rated,t,TRP}$ – 31 dB, -10 dBm)) | 10 MHz |
| NOTE 1: For non-contiguous spectrum operation within any *operating band* the limit within sub-block gaps is calculated as a cumulative sum of contributions from adjacent sub-blocks on each side of the sub-block gap, where the contribution from the far-end sub-block shall be scaled according to the measurement bandwidth of the near-end sub-block.<br>NOTE 2: $\Delta f_B$ = 2*$BW_{contiguous}$ when $BW_{contiguous}$ ≤ 500 MHz, otherwise $\Delta f_B$ = $BW_{contiguous}$ + 500 MHz. | | | |

**Table 9.7.4.3.3-3: OBUE limits applicable in the frequency range 52.6 – 71 GHz**

| Frequency offset of measurement filter -3 dB point, Δf | Frequency offset of measurement filter centre frequency, f_offset | Limit | *Measurement bandwidth* |
|---|---|---|---|
| 0 MHz ≤ Δf < 0.1*$BW_{contiguous}$ | 0.5 MHz ≤ f_offset < 0.1* $BW_{contiguous}$ +0.5 MHz | Min(-5 dBm, Max($P_{rated,t,TRP}$ – 31 dB, -12 dBm)) | 1 MHz |
| 0.1*$BW_{contiguous}$ ≤ Δf < $\Delta f_B$ | 0.1* $BW_{contiguous}$ +0.5 MHz ≤ f_offset < $\Delta f_B$ +0.5 MHz | Min(-13 dBm, Max($P_{rated,t,TRP}$ – 39 dB, -20 dBm)) | 1 MHz |
| $\Delta f_B$ ≤ Δf < $\Delta f_{max}$ | $\Delta f_B$ +5 MHz ≤ f_offset < f_ $offset_{max}$ | Min(-5 dBm, Max($P_{rated,t,TRP}$ – 29 dB, -10 dBm)) | 10 MHz |
| NOTE 1: For non-contiguous spectrum operation within any *operating band* the limit within sub-block gaps is calculated as a cumulative sum of contributions from adjacent sub-blocks on each side of the sub-block gap, where the contribution from the far-end sub-block shall be scaled according to the measurement bandwidth of the near-end sub-block.<br>NOTE 2: $\Delta f_B$ = 2*$BW_{contiguous}$ when $BW_{contiguous}$ ≤ 500 MHz, otherwise $\Delta f_B$ = $BW_{contiguous}$ + 500 MHz. | | | |

### 9.7.4.3.4 Additional OTA operating band unwanted emission requirements

#### 9.7.4.3.4.1 Protection of Earth Exploration Satellite Service

For BS operating in the frequency range 24.25 – 27.5 GHz, the power of unwanted emission shall not exceed the limits in table 9.7.4.3.4.1-1.

**Table 9.7.4.3.4.1-1: OBUE limits for protection of Earth Exploration Satellite Service**

| Frequency range | Limit | *Measurement Bandwidth* |
|---|---|---|
| 23.6 – 24 GHz | -3 dBm (Note 1) | 200 MHz |
| 23.6 – 24 GHz | -9 dBm (Note 2) | 200 MHz |
| NOTE 1: This limit applies to BS brought into use on or before 1 September 2027 in countries not adopting EU Decision 2020/590 [26].<br>NOTE 2: This limit applies to BS brought into use after 1 September 2027 or to BS in countries adopting EU Decision 2020/590 [26]. | | |

## 9.7.5 OTA transmitter spurious emissions

### 9.7.5.1 General

Unless otherwise stated, all requirements are measured as mean power.

The OTA spurious emissions limits are specified as TRP per RIB unless otherwise stated.

### 9.7.5.2 Minimum requirement for *BS type 1-O*

#### 9.7.5.2.1 General

The OTA transmitter spurious emission limits for FR1 shall apply from 30 MHz to 12.75 GHz, excluding the frequency range from $\Delta f_{OBUE}$ below the lowest frequency of each supported downlink *operating band*, up to $\Delta f_{OBUE}$ above the highest frequency of each supported downlink *operating band*, where the $\Delta f_{OBUE}$ is defined in table 9.7.1-1. For some FR1 *operating bands*, the upper limit is higher than 12.75 GHz in order to comply with the 5$^{th}$ harmonic limit of the downlink *operating band*, as specified in ITU-R recommendation SM.329 [2].

For *multi-band RIB* each supported *operating band* and $\Delta f_{OBUE}$ MHz around each band are excluded from the OTA transmitter spurious emissions requirements.

The requirements shall apply whatever the type of transmitter considered (single carrier or multi-carrier). It applies for all transmission modes foreseen by the manufacturer's specification.

*BS type 1-O* requirements consists of OTA transmitter spurious emission requirements based on TRP and co-location requirements not based on TRP.

#### 9.7.5.2.2 General OTA transmitter spurious emissions requirements

The Tx spurious emissions requirements for *BS type 1-O* are that for each applicable *basic limit* above 30 MHz in clause 6.6.5.2.1, the TRP of any spurious emission shall not exceed an OTA limit specified as the *basic limit* + X, where X = 9 dB, unless stated differently in regional regulation.

#### 9.7.5.2.3 Protection of the BS receiver of own or different BS

This requirement shall be applied for NR FDD operation in order to prevent the receivers of own or a different BS of the same band being desensitised by emissions from a type 1-O BS.

This requirement is a co-location requirement as defined in clause 4.9, the power levels are specified at the *co-location reference antenna* output.

The total power of any spurious emission from both polarizations of the *co-location reference antenna* connector output shall not exceed the *basic limits* in clause 6.6.5.2.2 + X dB, where X = -21 dB.

#### 9.7.5.2.4 Additional spurious emissions requirements

These requirements may be applied for the protection of systems operating in frequency ranges other than the BS downlink *operating band*. The limits may apply as an optional protection of such systems that are deployed in the same geographical area as the BS, or they may be set by local or regional regulation as a mandatory requirement for an NR *operating band*. It is in some cases not stated in the present document whether a requirement is mandatory or under what exact circumstances that a limit applies, since this is set by local or regional regulation. An overview of regional requirements in the present document is given in clause 4.5.

Some requirements may apply for the protection of specific equipment (UE, MS and/or BS) or equipment operating in specific systems (GSM, CDMA, UTRA, E-UTRA, NR, etc.). The Tx additional spurious emissions requirements for *BS type 1-O* are that for each applicable *basic limit* in clause 6.6.5.2.3, the TRP of any spurious emission shall not exceed an OTA limit specified as the *basic limit* + X, where X = 9 dB.

#### 9.7.5.2.5 Co-location with other base stations

These requirements may be applied for the protection of other BS receivers when GSM900, DCS1800, PCS1900, GSM850, CDMA850, UTRA FDD, UTRA TDD, E-UTRA and/or NR BS are co-located with a BS.

The requirements assume co-location with base stations of the same class.

NOTE: For co-location with UTRA, the requirements are based on co-location with UTRA FDD or TDD base stations.

This requirement is a co-location requirement as defined in clause 4.9, the power levels are specified at the *co-location reference antenna* output(s).

The power sum of any spurious emission is specified over all supported polarizations at the output(s) of the *co-location reference antenna* and shall not exceed the *basic limits* in clause 6.6.5.2.4 + X dB, where X = -21 dB.

For a *multi-band RIB*, the exclusions and conditions in the notes column of table 6.6.5.2.4-1 apply for each supported *operating band*.

### 9.7.5.3 Minimum requirement for *BS type 2-O*

#### 9.7.5.3.1 General

In FR2, the OTA transmitter spurious emission limits apply from 30 MHz to 2nd harmonic of the upper frequency edge of the downlink *operating band*, excluding the frequency range from $\Delta f_{OBUE}$ below the lowest frequency of the downlink *operating band*, up to $\Delta f_{OBUE}$ above the highest frequency of the downlink *operating band*, where the $\Delta f_{OBUE}$ is defined in table 9.7.1-1.

### 9.7.5.3.2 General OTA transmitter spurious emissions requirements

#### 9.7.5.3.2.1 General

The requirements of either clause 9.7.5.3.2.2 (Category A limits) or clause 9.7.5.3.2.3 (Category B limits) shall apply. The application of either Category A or Category B limits shall be the same as for Operating band unwanted emissions in clause 9.7.4.3.

**Table 9.7.5.3.2-1: Void**

NOTE: Table 9.7.5.3.2-1 is moved to clause 9.7.5.3.2.2 as Table 9.7.5.3.2.2-1.

#### 9.7.5.3.2.2 OTA transmitter spurious emissions (Category A)

The power of any spurious emission shall not exceed the limits in table 9.7.5.3.2-1

**Table 9.7.5.3.2.2-1: BS radiated Tx spurious emission limits in FR2**

| Frequency range | Limit | *Measurement Bandwidth* | Note |
|---|---|---|---|
| 30 MHz – 1 GHz | -13 dBm | 100 kHz | Note 1 |
| 1 GHz – 2nd harmonic of the upper frequency edge of the DL *operating band* | | 1 MHz | Note 1, Note 2 |
| NOTE 1: Bandwidth as in ITU-R SM.329 [2], s4.1<br>NOTE 2: Upper frequency as in ITU-R SM.329 [2], s2.5 table 1. | | | |

#### 9.7.5.3.2.3 OTA transmitter spurious emissions (Category B)

The power of any spurious emission shall not exceed the limits in table 9.7.5.3.2.3-1.

**Table 9.7.5.3.2.3-1: BS radiated Tx spurious emission limits in FR2 (Category B)**

| Frequency range (Note 4) | Limit | *Measurement Bandwidth* | Note |
|---|---|---|---|
| 30 MHz ↔ 1 GHz | -36 dBm | 100 kHz | Note 1 |
| 1 GHz ↔ 18 GHz | -30 dBm | 1 MHz | Note 1 |
| 18 GHz ↔ $F_{step,1}$ | -20 dBm | 10 MHz | Note 2 |
| $F_{step,1}$ ↔ $F_{step,2}$ | -15 dBm | 10 MHz | Note 2 |
| $F_{step,2}$ ↔ $F_{step,3}$ | -10 dBm | 10 MHz | Note 2 |
| $F_{step,4}$ ↔ $F_{step,5}$ | -10 dBm | 10 MHz | Note 2 |
| $F_{step,5}$ ↔ $F_{step,6}$ | -15 dBm | 10 MHz | Note 2 |
| $F_{step,6}$ ↔ 2nd harmonic of the upper frequency edge of the DL *operating band* | -20 dBm | 10 MHz | Note 2, Note 3 |
| NOTE 1: Bandwidth as in ITU-R SM.329 [2], s4.1<br>NOTE 2: Limit and bandwidth as in ERC Recommendation 74-01 [19], Annex 2.<br>NOTE 3: Upper frequency as in ITU-R SM.329 [2], s2.5 table 1.<br>NOTE 4: The step frequencies $F_{step,X}$ are defined in Table 9.7.5.3.2.3-2. | | | |

**Table 9.7.5.3.2.3-2: Step frequencies for defining the BS radiated Tx spurious emission limits in FR2 (Category B)**

| Operating band | $F_{step,1}$ (GHz) | $F_{step,2}$ (GHz) | $F_{step,3}$ (GHz) (Note 2) | $F_{step,4}$ (GHz) (Note 2) | $F_{step,5}$ (GHz) | $F_{step,6}$ (GHz) |
|---|---|---|---|---|---|---|
| n257 | 18 | 23.5 | 25 | 31 | 32.5 | 41.5 |
| n258 | 18 | 21 | 22.75 | 29 | 30.75 | 40.5 |
| n259 | 23.5 | 35.5 | 38 | 45 | 47.5 | 59.5 |
| n263 | 18 | 43 | 53.5 | 74.5 | 85 | 127 |
| NOTE 1: $F_{step,X}$ are based on ERC Recommendation 74-01 [19], Annex 2.<br>NOTE 2: $F_{step,3}$ and $F_{step,4}$ are aligned with the values for $\Delta f_{OBUE}$ in Table 9.7.1-1. | | | | | | |

### 9.7.5.3.3 Additional OTA transmitter spurious emissions requirements

These requirements may be applied for the protection of systems operating in frequency ranges other than the BS downlink operating band. The limits may apply as an optional protection of such systems that are deployed in the same geographical area as the BS, or they may be set by local or regional regulation as a mandatory requirement for an NR operating band. It is in some cases not stated in the present document whether a requirement is mandatory or under what exact circumstances that a limit applies, since this is set by local or regional regulation. An overview of regional requirements in the present document is given in clause 4.5.

#### 9.7.5.3.3.1 Limits for protection of Earth Exploration Satellite Service

For BS operating in the frequency range 24.25 – 27.5 GHz, the power of any spurious emissions shall not exceed the limits in Table 9.7.5.3.3.1-1.

**Table 9.7.5.3.3.1-1: Limits for protection of Earth Exploration Satellite Service**

| Frequency range | Limit | *Measurement Bandwidth* | Note |
|---|---|---|---|
| 23.6 – 24 GHz | -3 dBm | 200 MHz | Note 1 |
| 23.6 – 24 GHz | -9 dBm | 200 MHz | Note 2 |
| NOTE 1: This limit applies to BS brought into use on or before 1 September2027 in countries not adopting EU Decision 2020/590 [26].<br>NOTE 2: This limit applies to BS brought into use after 1 September 2027 or to BS in countries adopting EU Decision 2020/590[26]. | | | |

# 9.8 OTA transmitter intermodulation

## 9.8.1 General

The OTA transmitter intermodulation requirement is a measure of the capability of the transmitter unit to inhibit the generation of signals in its non-linear elements caused by presence of the wanted signal and an interfering signal reaching the transmitter unit via the RDN and antenna array from a co-located base station. The requirement shall apply during the *transmitter ON period* and the *transmitter transient period*.

The requirement shall apply at each RIB supporting transmission in the *operating band*.

The transmitter intermodulation level is the *total radiated power* of the intermodulation products when an interfering signal is injected into the *co-location reference antenna*.

The OTA transmitter intermodulation requirement is not applicable for *BS type 2-O*.

## 9.8.2 Minimum requirement for *BS type 1-O*

For *BS type 1-O* the transmitter intermodulation level shall not exceed the TRP unwanted emission limits specified for OTA transmitter spurious emission in clause 9.7.5.2 (except clause 9.7.5.2.3 and clause 9.7.5.2.5), OTA operating band

unwanted emissions in clause 9.7.4.2 and OTA ACLR in clause 9.7.3.2 in the presence of a wanted signal and an interfering signal, defined in table 9.8.2-1.

The requirement is applicable outside the *Base Station RF Bandwidth edges*. The interfering signal offset is defined relative to the *Base Station RF Bandwidth edges* or *Radio Bandwidth* edges.

For RIBs supporting operation in *non-contiguous spectrum*, the requirement is also applicable inside a *sub-block gap* for interfering signal offsets where the interfering signal falls completely within the *sub-block gap*. The interfering signal offset is defined relative to the *sub-block* edges.

For RIBs supporting operation in multiple *operating bands*, the requirement shall apply relative to the *Base Station RF Bandwidth edges* of each *operating band*. In case the *inter RF Bandwidth gap* is less than 3*$BW_{Channel}$ (where $BW_{Channel}$ is the minimal *BS channel bandwidth* of the band), the requirement in the gap shall apply only for interfering signal offsets where the interfering signal falls completely within the *inter RF Bandwidth gap*.

**Table 9.8.2-1: Interfering and wanted signals for the OTA transmitter intermodulation requirement**

| Parameter | Value |
|---|---|
| Wanted signal | NR signal or multi-carrier, or multiple intra-band contiguously or non-contiguously aggregated carriers |
| Interfering signal type | NR signal the minimum *BS channel bandwidth* ($BW_{Channel}$) with 15 kHz SCS of the band defined in clause 5.3.5 |
| Interfering signal power level | min(46 dBm, $P_{rated,t,TRP}$) |
| Interfering signal centre frequency offset from the lower (upper) edge of the wanted signal or edge of *sub-block* inside a gap | $f_{offset} = \pm BW_{Channel}\left(n - \frac{1}{2}\right)$, for n=1, 2 and 3 |
| NOTE 1: Interfering signal positions that are partially or completely outside of any downlink *operating band* of the RIB are excluded from the requirement, unless the interfering signal positions fall within the frequency range of adjacent downlink *operating bands* in the same geographical area. In case that none of the interfering signal positions fall completely within the frequency range of the downlink *operating band*, TS 38.141-2 [6] provides further guidance regarding appropriate test requirements.<br>NOTE 2: In Japan, NOTE 1 is not applied in Band n77, n78, n79.<br>NOTE 3: For *BS type 1-O* with dual polarization, the interfering signal power shall be equally divided between the supported polarizations at the *co-location reference antenna*. | |

## 9.8.3 Additional requirements (regional)

**Table 9.8.3-1: Void**

For Band n79 operation in Japan, the transmitter intermodulation level shall not exceed the TRP unwanted emission limits specified for OTA transmitter spurious emission in clause 9.7.5.2 (except clause 9.7.5.2.3 and clause 9.7.5.2.5), OTA operating band unwanted emissions in clause 9.7.4.2 and OTA ACLR in clause 9.7.3.2 in the presence of a wanted signal and an interfering signal, defined in table 9.8.3-2.

**Table 9.8.3-2: Interfering and wanted signals for the OTA transmitter intermodulation requirement for n79**

| Parameter | Value |
|---|---|
| Wanted signal | NR single carrier |
| Interfering signal type | NR signal of 40 MHz *channel bandwidth* |
| Interfering signal power level | min(46 dBm, $P_{rated,t,TRP}$) |
| Interfering signal centre frequency offset from the lower (upper) edge of the wanted signal | ± 20 MHz<br>± 60 MHz<br>± 100 MHz |

# 10 Radiated receiver characteristics

## 10.1 General

Radiated receiver characteristics are specified at RIB for *BS type 1-H*, *BS type 1-O*, or *BS type 2-O*, with full complement of transceivers for the configuration in normal operating condition.

Unless otherwise stated, the following arrangements apply for the radiated receiver characteristics requirements in clause 10:

- Requirements apply during the BS receive period.
- Requirements shall be met for any transmitter setting.
- For FDD operation the requirements shall be met with the transmitter unit(s) ON.
- Throughput requirements defined for the radiated receiver characteristics do not assume HARQ retransmissions.
- When BS is configured to receive multiple carriers, all the throughput requirements are applicable for each received carrier.
- For ACS, blocking and intermodulation characteristics, the negative offsets of the interfering signal apply relative to the lower *Base Station RF Bandwidth* edge or *sub-block* edge inside a *sub-block gap*, and the positive offsets of the interfering signal apply relative to the upper *Base Station RF Bandwidth* edge or *sub-block* edge inside a *sub-block gap*.
- Each requirement shall be met over the RoAoA specified.

NOTE 1: In normal operating condition the BS in FDD operation is configured to transmit and receive at the same time.

NOTE 2: In normal operating condition the BS in TDD operation is configured to TX OFF power during *receive period*.

For FR1 requirements which are to be met over the *OTA REFSENS RoAoA* absolute requirement values are offset by the following term:

$$\Delta_{OTAREFSENS} = 44.1 - 10*\log_{10}(BeW_{\theta,REFSENS}*BeW_{\varphi,REFSENS}) \text{ dB for the reference direction}$$

and

$$\Delta_{OTAREFSENS} = 41.1 - 10*\log_{10}(BeW_{\theta,REFSENS}*BeW_{\varphi,REFSENS}) \text{ dB for all other directions}$$

For requirements which are to be met over the *minSENS RoAoA* absolute requirement values are offset by the following term:

$$\Delta_{minSENS} = P_{REFSENS} - EIS_{minSENS} \text{ (dB)}$$

For FR2 requirements which are to be met over the *OTA REFSENS RoAoA* absolute requirement values are offset by the following term:

$$\Delta_{FR2_REFSENS} = -3 \text{ dB for the reference direction}$$

and

$$\Delta_{FR2_REFSENS} = 0 \text{ dB for all other directions}$$

# 10.2 OTA sensitivity

## 10.2.1 *BS type 1-H* and *BS type 1-O*

### 10.2.1.1 General

The OTA sensitivity requirement is a *directional requirement* based upon the declaration of one or more *OTA sensitivity direction declarations* (OSDD), related to a *BS type 1-H* and *BS type 1-O* receiver.

The *BS type 1-H* and *BS type 1-O* may optionally be capable of redirecting/changing the *receiver target* by means of adjusting BS settings resulting in multiple *sensitivity RoAoA*. The *sensitivity RoAoA* resulting from the current BS settings is the active *sensitivity RoAoA*.

If the BS is capable of redirecting the *receiver target* related to the OSDD then the OSDD shall include:

- *BS channel bandwidth* and declared minimum EIS level applicable to any active *sensitivity RoAoA* inside the *receiver target redirection range* in the OSDD.
- A declared *receiver target redirection range*, describing all the angles of arrival that can be addressed for the OSDD through alternative settings in the BS.
- Five declared *sensitivity RoAoA* comprising the conformance testing directions as detailed in TS 38.141-2 [6].
- The *receiver target reference direction*.

NOTE 1: Some of the declared *sensitivity RoAoA* may coincide depending on the redirection capability.

NOTE 2: In addition to the declared *sensitivity RoAoA*, several *sensitivity RoAoA* may be implicitly defined by the *receiver target redirection range* without being explicitly declared in the OSDD.

NOTE 3: (Void)

If the BS is not capable of redirecting the *receiver target* related to the OSDD, then the OSDD includes only:

- The set(s) of RAT, *BS channel bandwidth* and declared minimum EIS level applicable to the *sensitivity RoAoA* in the OSDD.
- One declared active *sensitivity RoAoA*.
- The *receiver target reference direction*.

NOTE 4: For BS without target redirection capability, the declared (fixed) *sensitivity RoAoA* is always the active *sensitivity RoAoA*.

The OTA sensitivity EIS level declaration shall apply to each supported polarization, under the assumption of *polarization match*.

### 10.2.1.2 Minimum requirement

For a received signal whose AoA of the incident wave is within the active *sensitivity RoAoA* of an OSDD, the error rate criterion as described in clause 7.2 shall be met when the level of the arriving signal is equal to the minimum EIS level in the respective declared set of EIS level and *BS channel bandwidth*.

## 10.2.2 *BS type 2-O*

There is no OTA sensitivity requirement for FR2, the OTA sensitivity is the same as the OTA reference sensitivity in clause 10.3.

# 10.3 OTA reference sensitivity level

## 10.3.1 General

The OTA REFSENS requirement is a *directional requirement* and is intended to ensure the minimum OTA reference sensitivity level for a declared *OTA REFSENS RoAoA*. The OTA reference sensitivity power level $EIS_{REFSENS}$ is the minimum mean power received at the RIB at which a reference performance requirement shall be met for a specified reference measurement channel.

The OTA REFSENS requirement shall apply to each supported polarization, under the assumption of *polarization match*.

## 10.3.2 Minimum requirement for *BS type 1-O*

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel as specified in the corresponding table and annex A.1 when the OTA test signal is at the corresponding $EIS_{REFSENS}$ level and arrives from any direction within the *OTA REFSENS RoAoA.*

**Table 10.3.2-1: Wide Area BS reference sensitivity levels**

| ***BS channel bandwidth*** **(MHz)** | **Sub-carrier spacing (kHz)** | **Reference measurement channel** | **OTA reference sensitivity level, $EIS_{REFSENS}$ (dBm)** |
|---|---|---|---|
| 5, 10, 15 | 15 | G-FR1-A1-1 | -101.7 - $\Delta_{OTAREFSENS}$ |
| 10, 15 | 30 | G-FR1-A1-2 | -101.8 - $\Delta_{OTAREFSENS}$ |
| 10, 15 | 60 | G-FR1-A1-3 | -98.9 - $\Delta_{OTAREFSENS}$ |
| 20, 25, 30, 35, 40, 45, 50 | 15 | G-FR1-A1-4 | -95.3 - $\Delta_{OTAREFSENS}$ |
| 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100 | 30 | G-FR1-A1-5 | -95.6 - $\Delta_{OTAREFSENS}$ |
| 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100 | 60 | G-FR1-A1-6 | -95.7 - $\Delta_{OTAREFSENS}$ |
| NOTE: $EIS_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*. | | | |

**Table 10.3.2-1a: Wide Area BS reference sensitivity levels for n104**

| ***BS channel bandwidth*** **(MHz)** | **Sub-carrier spacing (kHz)** | **Reference measurement channel** | **OTA reference sensitivity level, $EIS_{REFSENS}$ (dBm)** |
|---|---|---|---|
| 20, 30, 40, 50 | 15 | G-FR1-A1-4 | -94.3 - $\Delta_{OTAREFSENS}$ |
| 20, 30, 40, 50, 60, 70, 80, 90, 100 | 30 | G-FR1-A1-5 | -94.6 - $\Delta_{OTAREFSENS}$ |
| 20, 30, 40, 50, 60, 70, 80, 90, 100 | 60 | G-FR1-A1-6 | -94.7 - $\Delta_{OTAREFSENS}$ |
| NOTE: $EIS_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*. | | | |

**Table 10.3.2-2: Medium Range BS reference sensitivity levels**

| *BS channel bandwidth* (MHz) | Sub-carrier spacing (kHz) | Reference measurement channel | OTA reference sensitivity level, $EIS_{REFSENS}$ (dBm) |
|---|---|---|---|
| 5, 10, 15 | 15 | G-FR1-A1-1 | -96.7 - $\Delta_{OTAREFSENS}$ |
| 10, 15 | 30 | G-FR1-A1-2 | -96.8 - $\Delta_{OTAREFSENS}$ |
| 10, 15 | 60 | G-FR1-A1-3 | -93.9 - $\Delta_{OTAREFSENS}$ |
| 20, 25, 30, 35, 40, 45, 50 | 15 | G-FR1-A1-4 | -90.3 - $\Delta_{OTAREFSENS}$ |
| 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100 | 30 | G-FR1-A1-5 | -90.6 - $\Delta_{OTAREFSENS}$ |
| 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100 | 60 | G-FR1-A1-6 | -90.7 - $\Delta_{OTAREFSENS}$ |
| NOTE: $EIS_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*. | | | |

**Table 10.3.2-2a: Medium Range BS reference sensitivity levels for n104**

| *BS channel bandwidth* (MHz) | Sub-carrier spacing (kHz) | Reference measurement channel | OTA reference sensitivity level, $EIS_{REFSENS}$ (dBm) |
|---|---|---|---|
| 20, 30, 40, 50 | 15 | G-FR1-A1-4 | -89.3 - $\Delta_{OTAREFSENS}$ |
| 20, 30, 40, 50, 60, 70, 80, 90, 100 | 30 | G-FR1-A1-5 | -89.6 - $\Delta_{OTAREFSENS}$ |
| 20, 30, 40, 50, 60, 70, 80, 90, 100 | 60 | G-FR1-A1-6 | -89.7 - $\Delta_{OTAREFSENS}$ |
| NOTE: $EIS_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*. | | | |

**Table 10.3.2-3: Local Area BS reference sensitivity levels**

| *BS channel bandwidth* (MHz) | Sub-carrier spacing (kHz) | Reference measurement channel | OTA reference sensitivity level, $EIS_{REFSENS}$ (dBm) |
|---|---|---|---|
| 5, 10, 15 | 15 | G-FR1-A1-1 | -93.7 - $\Delta_{OTAREFSENS}$ |
| 10, 15 | 30 | G-FR1-A1-2 | -93.8 - $\Delta_{OTAREFSENS}$ |
| 10, 15 | 60 | G-FR1-A1-3 | -90.9 - $\Delta_{OTAREFSENS}$ |
| 20, 25, 30, 35, 40, 45, 50 | 15 | G-FR1-A1-4 | -87.3 - $\Delta_{OTAREFSENS}$ |
| 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100 | 30 | G-FR1-A1-5 | -87.6 - $\Delta_{OTAREFSENS}$ |
| 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100 | 60 | G-FR1-A1-6 | -87.7 - $\Delta_{OTAREFSENS}$ |
| NOTE: $EIS_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*. | | | |

**Table 10.3.2-3a: Local Area BS reference sensitivity levels for n104**

| BS channel bandwidth (MHz) | Sub-carrier spacing (kHz) | Reference measurement channel | OTA reference sensitivity level, $EIS_{REFSENS}$ (dBm) |
|---|---|---|---|
| 20, 30, 40, 50 | 15 | G-FR1-A1-4 | -86.3 - $\Delta_{OTAREFSENS}$ |
| 20, 30, 40, 50, 60, 70, 80, 90, 100 | 30 | G-FR1-A1-5 | -86.6 - $\Delta_{OTAREFSENS}$ |
| 20, 30, 40, 50, 60, 70, 80, 90, 100 | 60 | G-FR1-A1-6 | -86.7 - $\Delta_{OTAREFSENS}$ |
| NOTE: $EIS_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*. | | | |

## 10.3.3 Minimum requirement for *BS type 2-O*

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel as specified in the corresponding table and annex A.1 when the OTA test signal is at the corresponding $EIS_{REFSENS}$ level and arrives from any direction within the *OTA REFSENS RoAoA*.

$EIS_{REFSENS}$ levels are derived from a single declared basis level $EIS_{REFSENS_50M,}$ which is based on a reference measurement channel with 50 MHz *BS channel bandwidth*. $EIS_{REFSENS_50M}$ itself is not a requirement and although it is based on a reference measurement channel with 50 MHz *BS channel bandwidth* it does not imply that BS has to support 50 MHz *BS channel bandwidth*.

For Wide Area BS, $EIS_{REFSENS_50M}$ is an integer value in the range -96 to -119 dBm. The specific value is declared by the vendor.

For Medium Range BS, $EIS_{REFSENS_50M}$ is an integer value in the range -91 to -114 dBm. The specific value is declared by the vendor.

For Local Area BS, $EIS_{REFSENS_50M}$ is an integer value in the range -86 to -109 dBm. The specific value is declared by the vendor.

**Table 10.3.3-1: FR2 OTA reference sensitivity requirement**

| Frequency Range | BS channel Bandwidth (MHz) | Sub-carrier spacing (kHz) | Reference measurement channel | OTA reference sensitivity level, $EIS_{REFSENS}$ (dBm) |
|---|---|---|---|---|
| FR2-1 | 50, 100, 200 | 60 | G-FR2-A1-1 | $EIS_{REFSENS_50M}$ + $\Delta_{FR2_REFSENS}$ |
| | 50 | 120 | G-FR2-A1-2 | $EIS_{REFSENS_50M}$ + $\Delta_{FR2_REFSENS}$ |
| | 100, 200, 400 | 120 | G-FR2-A1-3 | $EIS_{REFSENS_50M}$ + 3 + $\Delta_{FR2_REFSENS}$ |
| FR2-2 | 100,400 | 120 | G-FR2-A1-3 | $EIS_{REFSENS_50M}$ + 3 + $\Delta_{FR2_REFSENS}$ |
| | 400, 800, 1600 | 480 | G-FR2-A1-6 | $EIS_{REFSENS_50M}$ + 9 + $\Delta_{FR2_REFSENS}$ |
| | 400, 800, 1600, 2000 | 960 | G-FR2-A1-7 | $EIS_{REFSENS_50M}$ + 9 + $\Delta_{FR2_REFSENS}$ |
| NOTE 1: $EIS_{REFSENS}$ is the power level of a single instance of the reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.<br>NOTE 2: The declared $EIS_{REFSENS_50M}$ shall be within the range specified above. | | | | |

# 10.4 OTA dynamic range

## 10.4.1 General

The OTA dynamic range is a measure of the capability of the receiver unit to receive a wanted signal in the presence of an interfering signal inside the received *BS channel bandwidth*.

The requirement shall apply at the RIB when the AoA of the incident wave of a received signal and the interfering signal are from the same direction and are within the *OTA REFSENS RoAoA.*

The wanted and interfering signals apply to each supported polarization, under the assumption of *polarization match*.

## 10.4.2 Minimum requirement for *BS type 1-O*

For NR, the throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel.

**Table 10.4.2-1: Wide Area BS OTA dynamic range for NR carrier**

| *BS channel bandwidth* (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) / $BW_{Config}$ | Type of interfering signal |
|---|---|---|---|---|---|
| 5 | 15 | G-FR1-A2-1 | -70.7- $\Delta_{OTAREFSENS}$ | -82.5- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-2 | -71.4- $\Delta_{OTAREFSENS}$ | | |
| 10 | 15 | G-FR1-A2-1 | -70.7- $\Delta_{OTAREFSENS}$ | -79.3- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-2 | -71.4- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-3 | -68.4- $\Delta_{OTAREFSENS}$ | | |
| 15 | 15 | G-FR1-A2-1 | -70.7- $\Delta_{OTAREFSENS}$ | -77.5- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-2 | -71.4- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-3 | -68.4- $\Delta_{OTAREFSENS}$ | | |
| 20 | 15 | G-FR1-A2-4 | -64.5- $\Delta_{OTAREFSENS}$ | -76.2- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -64.5- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -64.8- $\Delta_{OTAREFSENS}$ | | |
| 25 | 15 | G-FR1-A2-4 | -64.5- $\Delta_{OTAREFSENS}$ | -75.2- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -64.5- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -64.8- $\Delta_{OTAREFSENS}$ | | |
| 30 | 15 | G-FR1-A2-4 | -64.5- $\Delta_{OTAREFSENS}$ | -74.4- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -64.5- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -64.8- $\Delta_{OTAREFSENS}$ | | |
| 35 | 15 | G-FR1-A2-4 | -64.5- $\Delta_{OTAREFSENS}$ | -73.7- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -64.5- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -64.8- $\Delta_{OTAREFSENS}$ | | |
| 40 | 15 | G-FR1-A2-4 | -64.5- $\Delta_{OTAREFSENS}$ | -73.1- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -64.5- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -64.8- $\Delta_{OTAREFSENS}$ | | |
| 45 | 15 | G-FR1-A2-4 | -64.5- $\Delta_{OTAREFSENS}$ | -72.6- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -64.5- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -64.8- $\Delta_{OTAREFSENS}$ | | |
| 50 | 15 | G-FR1-A2-4 | -64.5- $\Delta_{OTAREFSENS}$ | -72.1- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -64.5- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -64.8- $\Delta_{OTAREFSENS}$ | | |
| 60 | 30 | G-FR1-A2-5 | -64.5- $\Delta_{OTAREFSENS}$ | -71.3- $\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -64.8- $\Delta_{OTAREFSENS}$ | | |
| 70 | 30 | G-FR1-A2-5 | -64.5- $\Delta_{OTAREFSENS}$ | -70.7- $\Delta_{OTAREFSENS}$ | AWGN |

| | | | | | |
|---|---|---|---|---|---|
| | 60 | G-FR1-A2-6 | -64.8- $\Delta_{OTAREFSENS}$ | | |
| 80 | 30 | G-FR1-A2-5 | -64.5- $\Delta_{OTAREFSENS}$ | -70.1- $\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -64.8- $\Delta_{OTAREFSENS}$ | | |
| 90 | 30 | G-FR1-A2-5 | -64.5- $\Delta_{OTAREFSENS}$ | -69.5- $\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -64.8- $\Delta_{OTAREFSENS}$ | | |
| 100 | 30 | G-FR1-A2-5 | -64.5- $\Delta_{OTAREFSENS}$ | -69.1- $\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -64.8- $\Delta_{OTAREFSENS}$ | | |
| NOTE: The wanted signal mean power is the power level of a single instance of the corresponding reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*. | | | | | |

**Table 10.4.2-1a: Wide Area BS OTA dynamic range for NR carrier of band n104**

| BS channel bandwidth (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) / $BW_{Config}$ | Type of interfering signal |
|---|---|---|---|---|---|
| 20 | 15 | G-FR1-A2-4 | -63.5- $\Delta_{OTAREFSENS}$ | -75.2- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -63.5- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -63.8- $\Delta_{OTAREFSENS}$ | | |
| 30 | 15 | G-FR1-A2-4 | -63.5- $\Delta_{OTAREFSENS}$ | -73.4- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -63.5- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -63.8- $\Delta_{OTAREFSENS}$ | | |
| 40 | 15 | G-FR1-A2-4 | -63.5- $\Delta_{OTAREFSENS}$ | -72.1- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -63.5- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -63.8- $\Delta_{OTAREFSENS}$ | | |
| 50 | 15 | G-FR1-A2-4 | -63.5- $\Delta_{OTAREFSENS}$ | -71.1- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -63.5- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -63.8- $\Delta_{OTAREFSENS}$ | | |
| 60 | 30 | G-FR1-A2-5 | -63.5- $\Delta_{OTAREFSENS}$ | -70.3- $\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -63.8- $\Delta_{OTAREFSENS}$ | | |
| 70 | 30 | G-FR1-A2-5 | -63.5- $\Delta_{OTAREFSENS}$ | -69.7- $\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -63.8- $\Delta_{OTAREFSENS}$ | | |
| 80 | 30 | G-FR1-A2-5 | -63.5- $\Delta_{OTAREFSENS}$ | -69.1- $\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -63.8- $\Delta_{OTAREFSENS}$ | | |
| 90 | 30 | G-FR1-A2-5 | -63.5- $\Delta_{OTAREFSENS}$ | -68.5- $\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -63.8- $\Delta_{OTAREFSENS}$ | | |
| 100 | 30 | G-FR1-A2-5 | -63.5- $\Delta_{OTAREFSENS}$ | -68.1- $\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -63.8- $\Delta_{OTAREFSENS}$ | | |

NOTE 1: The wanted signal mean power is the power level of a single instance of the corresponding reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

**Table 10.4.2-2: Medium Range BS OTA dynamic range for NR carrier**

| *BS channel bandwidth* (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) / $BW_{Config}$ | Type of interfering signal |
|---|---|---|---|---|---|
| 5 | 15 | G-FR1-A2-1 | -65.7- $\Delta_{OTAREFSENS}$ | -77.5- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-2 | -66.4- $\Delta_{OTAREFSENS}$ | | |
| 10 | 15 | G-FR1-A2-1 | -65.7- $\Delta_{OTAREFSENS}$ | -74.3- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-2 | -66.4- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-3 | -63.4- $\Delta_{OTAREFSENS}$ | | |
| 15 | 15 | G-FR1-A2-1 | -65.7- $\Delta_{OTAREFSENS}$ | -72.5- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-2 | -66.4- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-3 | -63.4- $\Delta_{OTAREFSENS}$ | | |
| 20 | 15 | G-FR1-A2-4 | -59.5- $\Delta_{OTAREFSENS}$ | -71.2- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -59.5- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -59.8- $\Delta_{OTAREFSENS}$ | | |
| 25 | 15 | G-FR1-A2-4 | -59.5- $\Delta_{OTAREFSENS}$ | -70.2- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -59.5- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -59.8- $\Delta_{OTAREFSENS}$ | | |
| 30 | 15 | G-FR1-A2-4 | -59.5- $\Delta_{OTAREFSENS}$ | -69.4- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -59.5- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -59.8- $\Delta_{OTAREFSENS}$ | | |
| 35 | 15 | G-FR1-A2-4 | -59.5- $\Delta_{OTAREFSENS}$ | -68.7- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -59.5- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -59.8- $\Delta_{OTAREFSENS}$ | | |
| 40 | 15 | G-FR1-A2-4 | -59.5- $\Delta_{OTAREFSENS}$ | -68.1- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -59.5- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -59.8- $\Delta_{OTAREFSENS}$ | | |
| 45 | 15 | G-FR1-A2-4 | -59.5- $\Delta_{OTAREFSENS}$ | -67.6- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -59.5- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -59.8- $\Delta_{OTAREFSENS}$ | | |
| 50 | 15 | G-FR1-A2-4 | -59.5- $\Delta_{OTAREFSENS}$ | -67.1- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -59.5- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -59.8- $\Delta_{OTAREFSENS}$ | | |
| 60 | 30 | G-FR1-A2-5 | -59.5- $\Delta_{OTAREFSENS}$ | -66.3- $\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -59.8- $\Delta_{OTAREFSENS}$ | | |
| 70 | 30 | G-FR1-A2-5 | -59.5- $\Delta_{OTAREFSENS}$ | -65.7- $\Delta_{OTAREFSENS}$ | AWGN |

| | | | | | |
|---|---|---|---|---|---|
| | 60 | G-FR1-A2-6 | -59.8- $\Delta_{OTAREFSENS}$ | | |
| 80 | 30 | G-FR1-A2-5 | -59.5- $\Delta_{OTAREFSENS}$ | -65.1- $\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -59.8- $\Delta_{OTAREFSENS}$ | | |
| 90 | 30 | G-FR1-A2-5 | -59.5- $\Delta_{OTAREFSENS}$ | -64.5- $\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -59.8- $\Delta_{OTAREFSENS}$ | | |
| 100 | 30 | G-FR1-A2-5 | -59.5- $\Delta_{OTAREFSENS}$ | -64.1- $\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -59.8- $\Delta_{OTAREFSENS}$ | | |

NOTE: The wanted signal mean power is the power level of a single instance of the corresponding reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

**Table 10.4.2-2a: Medium Range BS OTA dynamic range for NR carrier of band n104**

| BS channel bandwidth (MHz) | Subcarrier spacing (kHz) | Reference measurement channel (Note 2) | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) / $BW_{Config}$ | Type of interfering signal |
|---|---|---|---|---|---|
| 20 | 15 | G-FR1-A2-4 | -58.5-$\Delta_{OTAREFSENS}$ | -70.2-$\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -58.5-$\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -58.8-$\Delta_{OTAREFSENS}$ | | |
| 30 | 15 | G-FR1-A2-4 | -58.5-$\Delta_{OTAREFSENS}$ | -68.4-$\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -58.5-$\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -58.8-$\Delta_{OTAREFSENS}$ | | |
| 40 | 15 | G-FR1-A2-4 | -58.5-$\Delta_{OTAREFSENS}$ | -67.1-$\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -58.5-$\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -58.8-$\Delta_{OTAREFSENS}$ | | |
| 50 | 15 | G-FR1-A2-4 | -58.5-$\Delta_{OTAREFSENS}$ | -66.1-$\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -58.5-$\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -58.8-$\Delta_{OTAREFSENS}$ | | |
| 60 | 30 | G-FR1-A2-5 | -58.5-$\Delta_{OTAREFSENS}$ | -65.3-$\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -58.8-$\Delta_{OTAREFSENS}$ | | |
| 70 | 30 | G-FR1-A2-5 | -58.5-$\Delta_{OTAREFSENS}$ | -64.7-$\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -58.8-$\Delta_{OTAREFSENS}$ | | |
| 80 | 30 | G-FR1-A2-5 | -58.5-$\Delta_{OTAREFSENS}$ | -64.1-$\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -58.8-$\Delta_{OTAREFSENS}$ | | |
| 90 | 30 | G-FR1-A2-5 | -58.5-$\Delta_{OTAREFSENS}$ | -63.5-$\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -58.8-$\Delta_{OTAREFSENS}$ | | |
| 100 | 30 | G-FR1-A2-5 | -58.5-$\Delta_{OTAREFSENS}$ | -63.1-$\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -58.8-$\Delta_{OTAREFSENS}$ | | |

NOTE 1: The wanted signal mean power is the power level of a single instance of the corresponding reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

**Table 10.4.2-3: Local Area BS OTA dynamic range for NR carrier**

| *BS channel bandwidth* (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) / $BW_{Config}$ | Type of interfering signal |
|---|---|---|---|---|---|
| 5 | 15 | G-FR1-A2-1 | -62.7- $\Delta_{OTAREFSENS}$ | -74.5- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-2 | -64.4- $\Delta_{OTAREFSENS}$ | | |
| 10 | 15 | G-FR1-A2-1 | -62.7- $\Delta_{OTAREFSENS}$ | -71.3- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-2 | -64.4- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-3 | -60.4- $\Delta_{OTAREFSENS}$ | | |
| 15 | 15 | G-FR1-A2-1 | -62.7- $\Delta_{OTAREFSENS}$ | -69.5- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-2 | -64.4- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-3 | -60.4- $\Delta_{OTAREFSENS}$ | | |
| 20 | 15 | G-FR1-A2-4 | -56.5- $\Delta_{OTAREFSENS}$ | -68.2- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -56.5- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -56.8- $\Delta_{OTAREFSENS}$ | | |
| 25 | 15 | G-FR1-A2-4 | -56.5- $\Delta_{OTAREFSENS}$ | -67.2- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -56.5- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -56.8- $\Delta_{OTAREFSENS}$ | | |
| 30 | 15 | G-FR1-A2-4 | -56.5- $\Delta_{OTAREFSENS}$ | -66.4- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -56.5- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -56.8- $\Delta_{OTAREFSENS}$ | | |
| 35 | 15 | G-FR1-A2-4 | -56.5- $\Delta_{OTAREFSENS}$ | -65.7- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -56.5- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -56.8- $\Delta_{OTAREFSENS}$ | | |
| 40 | 15 | G-FR1-A2-4 | -56.5- $\Delta_{OTAREFSENS}$ | -65.1- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -56.5- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -56.8- $\Delta_{OTAREFSENS}$ | | |
| 45 | 15 | G-FR1-A2-4 | -56.5- $\Delta_{OTAREFSENS}$ | -64.6- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -56.5- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -56.8- $\Delta_{OTAREFSENS}$ | | |
| 50 | 15 | G-FR1-A2-4 | -56.5- $\Delta_{OTAREFSENS}$ | -64.1- $\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -56.5- $\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -56.8- $\Delta_{OTAREFSENS}$ | | |
| 60 | 30 | G-FR1-A2-5 | -56.5- $\Delta_{OTAREFSENS}$ | -63.3- $\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -56.8- $\Delta_{OTAREFSENS}$ | | |
| 70 | 30 | G-FR1-A2-5 | -56.5- $\Delta_{OTAREFSENS}$ | -62.7- $\Delta_{OTAREFSENS}$ | AWGN |

| | | | | | |
|---|---|---|---|---|---|
| | 60 | G-FR1-A2-6 | -56.8- $\Delta_{OTAREFSENS}$ | | |
| 80 | 30 | G-FR1-A2-5 | -56.5- $\Delta_{OTAREFSENS}$ | -62.1- $\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -56.8- $\Delta_{OTAREFSENS}$ | | |
| 90 | 30 | G-FR1-A2-5 | -56.5- $\Delta_{OTAREFSENS}$ | -61.5- $\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -56.8- $\Delta_{OTAREFSENS}$ | | |
| 100 | 30 | G-FR1-A2-5 | -56.5- $\Delta_{OTAREFSENS}$ | -61.1- $\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -56.8- $\Delta_{OTAREFSENS}$ | | |

NOTE: The wanted signal mean power is the power level of a single instance of the corresponding reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

**Table 10.4.2-3a: Local Area BS OTA dynamic range for NR carrier of band n104**

| BS channel bandwidth (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) / $BW_{Config}$ | Type of interfering signal |
|---|---|---|---|---|---|
| 20 | 15 | G-FR1-A2-4 | -55.5-$\Delta_{OTAREFSENS}$ | -67.2-$\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -55.5-$\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -55.8-$\Delta_{OTAREFSENS}$ | | |
| 30 | 15 | G-FR1-A2-4 | -55.5-$\Delta_{OTAREFSENS}$ | -65.4-$\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -55.5-$\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -55.8-$\Delta_{OTAREFSENS}$ | | |
| 40 | 15 | G-FR1-A2-4 | -55.5-$\Delta_{OTAREFSENS}$ | -64.1-$\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -55.5-$\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -55.8-$\Delta_{OTAREFSENS}$ | | |
| 50 | 15 | G-FR1-A2-4 | -55.5-$\Delta_{OTAREFSENS}$ | -63.1-$\Delta_{OTAREFSENS}$ | AWGN |
| | 30 | G-FR1-A2-5 | -55.5-$\Delta_{OTAREFSENS}$ | | |
| | 60 | G-FR1-A2-6 | -55.8-$\Delta_{OTAREFSENS}$ | | |
| 60 | 30 | G-FR1-A2-5 | -55.5-$\Delta_{OTAREFSENS}$ | -62.3-$\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -55.8-$\Delta_{OTAREFSENS}$ | | |
| 70 | 30 | G-FR1-A2-5 | -55.5-$\Delta_{OTAREFSENS}$ | -61.7-$\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -55.8-$\Delta_{OTAREFSENS}$ | | |
| 80 | 30 | G-FR1-A2-5 | -55.5-$\Delta_{OTAREFSENS}$ | -61.1-$\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -55.8-$\Delta_{OTAREFSENS}$ | | |
| 90 | 30 | G-FR1-A2-5 | -55.5-$\Delta_{OTAREFSENS}$ | -60.5-$\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -55.8-$\Delta_{OTAREFSENS}$ | | |
| 100 | 30 | G-FR1-A2-5 | -55.5-$\Delta_{OTAREFSENS}$ | -60.1-$\Delta_{OTAREFSENS}$ | AWGN |
| | 60 | G-FR1-A2-6 | -55.8-$\Delta_{OTAREFSENS}$ | | |

NOTE 1: The wanted signal mean power is the power level of a single instance of the corresponding reference measurement channel. This requirement shall be met for each consecutive application of a single instance of the reference measurement channel mapped to disjoint frequency ranges with a width corresponding to the number of resource blocks of the reference measurement channel each, except for one instance that might overlap one other instance to cover the full *BS channel bandwidth*.

# 10.5 OTA in-band selectivity and blocking

## 10.5.1 OTA adjacent channel selectivity

### 10.5.1.1 General

OTA Adjacent channel selectivity (ACS) is a measure of the receiver's ability to receive an OTA wanted signal at its assigned channel frequency in the presence of an OTA adjacent channel signal with a specified centre frequency offset of the interfering signal to the band edge of a victim system.

### 10.5.1.2 Minimum requirement for *BS type 1-O*

The requirement shall apply at the RIB when the AoA of the incident wave of a received signal and the interfering signal are from the same direction and are within the *minSENS RoAoA*.

The wanted and interfering signals apply to each supported polarization, under the assumption o*f polarization match*.

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel.

For FR1 except for band n104 , the OTA wanted and the interfering signal are specified in table 10.5.1.2-1 and table 10.5.1.2-2 for OTA ACS. For band n104 , the OTA wanted and the interfering signal are specified in table 10.5.1.2-1a and table 10.5.1.2-2 for OTA ACS. The reference measurement channel for the OTA wanted signal is further specified in annex A.1. The characteristics of the interfering signal is further specified in annex D.

The OTA ACS requirement is applicable outside the *Base Station RF Bandwidth* or *Radio Bandwidth*. The OTA interfering signal offset is defined relative to the *Base station RF Bandwidth edges* or *Radio Bandwidth edges*.

For RIBs supporting operation in *non-contiguous spectrum* within any *operating band*, the OTA ACS requirement shall apply in addition inside any *sub-block gap*, in case the *sub-block gap* size is at least as wide as the NR interfering signal in table 10.5.1.2-2. The OTA interfering signal offset is defined relative to the *sub-block* edges inside the *sub-block gap*.

For *multi-band RIBs*, the OTA ACS requirement shall apply in addition inside any *Inter RF Bandwidth gap*, in case the *Inter RF Bandwidth gap* size is at least as wide as the NR interfering signal in table 10.5.1.2-2. The interfering signal offset is defined relative to the *Base Station RF Bandwidth* edges inside the *Inter RF Bandwidth gap*.

**Table 10.5.1.2-1: OTA ACS requirement for *BS type 1-O***

| ***BS channel bandwidth* of the *lowest/highest carrier* received (MHz)** | **Wanted signal mean power (dBm) (Note 2)** | **Interfering signal mean power (dBm)** |
|---|---|---|
| 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80,90, 100 (Note 1) | $EIS_{minSENS}$ + 6 dB | Wide Area BS: -52 – $\Delta_{minSENS}$<br>Medium Range BS: -47– $\Delta_{minSENS}$<br>Local Area BS: -44– $\Delta_{minSENS}$ |
| NOTE 1: The SCS for the *lowest/highest carrier* received is the lowest SCS supported by the BS for that bandwidth<br>NOTE 2: $EIS_{minSENS}$ depends on the *BS channel bandwidth* | | |

**Table 10.5.1.2-1a: OTA ACS requirement for *BS type 1-O* for band n104**

| BS channel bandwidth of the lowest/highest carrier received (MHz) | Wanted signal mean power (dBm) (Note 2) | Interfering signal mean power (dBm) |
|---|---|---|
| 20, 30, 40, 50, 60, 70, 80,90, 100 (Note 1) | $EIS_{minSENS}$ + 6 dB | Wide Area BS: -55 – $\Delta_{minSENS}$<br>Medium Range BS: -50– $\Delta_{minSENS}$<br>Local Area BS: -47– $\Delta_{minSENS}$ |
| NOTE 1: The SCS for the *lowest/highest carrier* received is the lowest SCS supported by the BS for that bandwidth<br>NOTE 2: $EIS_{minSENS}$ depends on the *BS channel bandwidth* | | |

**Table 10.5.1.2-2: OTA ACS interferer frequency offset for *BS type 1-O***

| ***BS channel bandwidth* of the *lowest/highest carrier* received (MHz)** | **Interfering signal centre frequency offset from the lower/upper *Base Station RF Bandwidth* edge or *sub-block* edge inside a *sub-block gap* (MHz)** | **Type of interfering signal** |
|---|---|---|
| 5 | ±2.5025 | 5 MHz DFT-s-OFDM NR signal, 15 kHz SCS, 25 RBs |
| 10 | ±2.5075 | |
| 15 | ±2.5125 | |
| 20 | ±2.5025 | |
| 25 | ±9.4675 | 20 MHz DFT-s-OFDM NR signal, 15 kHz SCS, 100 RBs |
| 30 | ±9.4725 | |
| 35 | ±9.4625 | |
| 40 | ±9.4675 | |
| 45 | ±9.4725 | |
| 50 | ±9.4625 | |
| 60 | ±9.4725 | |
| 70 | ±9.4675 | |
| 80 | ±9.4625 | |
| 90 | ±9.4725 | |
| 100 | ±9.4675 | |

### 10.5.1.3 Minimum requirement for *BS type 2-O*

The requirement shall apply at the RIB when the AoA of the incident wave of a received signal and the interfering signal are from the same direction and are within the *OTA REFSENS RoAoA.*

The wanted and interfering signals apply to each supported polarization, under the assumption o*f polarization match*.

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel.

For FR2, the OTA wanted and the interfering signal are specified in table 10.5.1.3-1 and table 10.5.1.3-2 for OTA ACS. The reference measurement channel for the OTA wanted signal is further specified in annex A.1. The characteristics of the interfering signal is further specified in annex D.

The OTA ACS requirement is applicable outside the *Base Station RF Bandwidth*. The OTA interfering signal offset is defined relative to the Base station *RF Bandwidth edges*.

For RIBs supporting operation in *non-contiguous spectrum* within any *operating band*, the OTA ACS requirement shall apply in addition inside any *sub-block gap*, in case the *sub-block gap* size is at least as wide as the NR interfering signal in table 10.5.1.3-2. The OTA interfering signal offset is defined relative to the *sub-block* edges inside the *sub-block gap*.

**Table 10.5.1.3-1: OTA ACS requirement for *BS type 2-O***

| Frequency Range | BS channel bandwidth of the lowest/highest carrier received (MHz) | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) |
|---|---|---|---|
| FR2-1 | 50, 100, 200, 400 | $EIS_{REFSENS}$ + 6 dB (Note 3) | $EIS_{REFSENS_50M}$ + 27.7 + $\Delta_{FR2_REFSENS}$ (Note 1)<br>$EIS_{REFSENS_50M}$ + 26.7 + $\Delta_{FR2_REFSENS}$ (Note 2) |
| FR2-2 | 100, 400, 800, 1600, 2000 | $EIS_{REFSENS}$ + 6 dB (Note 3) | $EIS_{REFSENS_50M}$ + 28.7 + $\Delta_{FR2_REFSENS}$ (Note 4) |
| NOTE 1: Applicable to bands defined within the frequency spectrum range of 24.25 – 33.4 GHz<br>NOTE 2: Applicable to bands defined within the frequency spectrum range of 37 – 52.6 GHz<br>NOTE 3: $EIS_{REFSENS}$ is given in clause 10.3.3<br>NOTE 4: Applicable to bands defined within the frequency spectrum range of 57 – 71 GHz | | | |

**Table 10.5.1.3-2: OTA ACS interferer frequency offset for *BS type 2-O***

| Frequency Range | BS channel bandwidth of the lowest/highest carrier received (MHz) | Interfering signal centre frequency offset from the lower/upper Base Station RF Bandwidth edge or sub-block edge inside a sub-block gap (MHz) | Type of interfering signal |
|---|---|---|---|
| FR2-1 | 50 | ±24.29 | 50 MHz DFT-s-OFDM NR signal,60 kHz SCS, 64 RBs |
| | 100 | ±24.31 | |
| | 200 | ±24.29 | |
| | 400 | ±24.31 | |
| FR2-2 | 100 | ±48.58 | 100 MHz DFT-s-OFDM NR signal,120 kHz SCS, 64 RBs |
| | 400 | ±48.58 | |
| | 800 | ±48.62 | |
| | 1600 | ±48.58 | |
| | 2000 | ±48.62 | |

## 10.5.2 OTA in-band blocking

### 10.5.2.1 General

The OTA in-band blocking characteristics is a measure of the receiver's ability to receive a OTA wanted signal at its assigned channel in the presence of an unwanted OTA interferer, which is an NR signal for general blocking or an NR signal with one RB for narrowband blocking.

### 10.5.2.2 Minimum requirement for *BS type 1-O*

The requirement shall apply at the RIB when the AoA of the incident wave of a received signal and the interfering signal are from the same direction, and:

- when the wanted signal is based on $EIS_{REFSENS}$: the AoA of the incident wave of a received signal and the interfering signal are within the *OTA REFSENS RoAoA.*
- when the wanted signal is based on $EIS_{minSENS}$: the AoA of the incident wave of a received signal and the interfering signal are within the *minSENS RoAoA.*

The wanted and interfering signals apply to each supported polarization, under the assumption of *polarization match*.

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel, with OTA wanted and OTA interfering signal specified in tables 10.5.2.2-1, table 10.5.2.2-2 and table 10.5.2.2-3 for general OTA and narrowband OTA blocking requirements. Narrowband blocking requirements are not applied for band n104. The reference measurement channel for the OTA wanted signal is identified in clause 10.3.2 and are further specified in annex A.1. The characteristics of the interfering signal is further specified in annex D.

The OTA in-band blocking requirements apply outside the *Base Station RF Bandwidth* or *Radio Bandwidth*. The interfering signal offset is defined relative to the *Base Station RF Bandwidth edges* or *Radio Bandwidth* edges.

For *BS type 1-O* the OTA in-band blocking requirement shall apply in the in-band blocking frequency range, which is from $F_{UL,low}$ - $\Delta f_{OOB}$ to $F_{UL,high}$ + $\Delta f_{OOB}$, excluding the downlink frequency range of the FDD *operating band.* The $\Delta f_{OOB}$ for *BS type 1-O* is defined in table 10.5.2.2-0 except for band n104 and 10.5.2.2-0a for band n104.

**Table 10.5.2.2-0: $\Delta f_{OOB}$ offset for NR *operating bands* in FR1**

| BS type | *Operating band* characteristics | $\Delta f_{OOB}$ (MHz) |
|---|---|---|
| *BS type 1-O* | $F_{UL,high} - F_{UL,low} < 100$ MHz | 20 |
| | 100 MHz ≤ $F_{UL,high} - F_{UL,low} \leq 900$ MHz | 60 |

For band n104, $\Delta f_{OOB}$ for *BS type 1-O* is defined in table 10.5.2.2-0a.

**Table 10.5.2.2-0a: $\Delta f_{OOB}$ offset for NR *operating bands* for band n104**

| BS type | *Operating band* | $\Delta f_{OOB}$ (MHz) |
|---|---|---|
| *BS type 1-O* | n104 | 100 |

For RIBs supporting operation in *non-contiguous spectrum* within any *operating band*, the OTA in-band blocking requirements apply in addition inside any *sub-block gap*, in case the *sub-block gap* size is at least as wide as twice the interfering signal minimum offset in table 10.5.2.2-1. The interfering signal offset is defined relative to the *sub-block* edges inside the *sub-block gap*.

For *multi-band RIBs*, the OTA in-band blocking requirements apply in the in-band blocking frequency ranges for each supported *operating band*. The requirement shall apply in addition inside any *Inter RF Bandwidth gap*, in case the *Inter RF Bandwidth gap* size is at least as wide as twice the interfering signal minimum offset in tables 10.5.2.2-1 and 10.5.2.2-3.

For a RIBs supporting operation in *non-contiguous spectrum* within any *operating band*, the OTA narrowband blocking requirements apply in addition inside any *sub-block gap*, in case the *sub-block gap* size is at least as wide as the interfering signal minimum offset in table 10.5.2.2-3. The interfering signal offset is defined relative to the *sub-block* edges inside the *sub-block gap*.

For a *multi-band RIBs*, the OTA narrowband blocking requirements apply in the narrowband blocking frequency ranges for each supported *operating band*. The requirement shall apply in addition inside any *Inter RF Bandwidth gap*, in case the *Inter RF Bandwidth gap* size is at least as wide as the interfering signal minimum offset in table 10.5.2.2-3.

**Table 10.5.2.2-1: General OTA blocking requirement for *BS type 1-O***

| ***BS channel bandwidth* of the *lowest/highest carrier* received (MHz)** | **Wanted signal mean power (dBm) (Note 1)** | **Interfering signal mean power (dBm)** | **Interfering signal centre frequency minimum offset from the lower/upper *Base Station RF Bandwidth edge* or *sub-block* edge inside a *sub-block gap* (MHz)** | **Type of interfering signal** |
|---|---|---|---|---|
| 5, 10, 15, 20 | $EIS_{REFSENS}$ + x dB | Wide Area BS: -43 - $\Delta_{OTAREFSENS}$<br>Medium Range BS: -38 - $\Delta_{OTAREFSENS}$<br>Local Area BS: -35 - $\Delta_{OTAREFSENS}$ | ±7.5 | 5 MHz DFT-s-OFDM NR signal, 15 kHz SCS, 25 RBs |
| | $EIS_{minSENS}$ + x dB | Wide Area BS: -43 – $\Delta_{minSENS}$<br>Medium Range BS: -38 – $\Delta_{minSENS}$<br>Local Area BS: -35 – $\Delta_{minSENS}$ | ±7.5 | |
| 25 ,30, 35, 40, 45, 50, 60, 70, 80, 90, 100 | $EIS_{REFSENS}$ + x dB | Wide Area BS: -43 - $\Delta_{OTAREFSENS}$<br>Medium Range BS: -38 - $\Delta_{OTAREFSENS}$<br>Local Area BS: -35 - $\Delta_{OTAREFSENS}$ | ±30 | 20 MHz DFT-s-OFDM NR signal, 15 kHz SCS, 100 RBs |
| | $EIS_{minSENS}$ + x dB | Wide Area BS: -43 – $\Delta_{minSENS}$<br>Medium Range BS: -38 – $\Delta_{minSENS}$<br>Local Area BS: -35 – $\Delta_{minSENS}$ | ±30 | |
| NOTE 1: For a BS capable of single band operation only, "x" is equal to 6 dB. For a BS capable of multi-band operation, "x" is equal to 6 dB in case of interfering signals that are in the in-band blocking frequency range of the operating band where the wanted signal is present or in the in-band blocking frequency range of an adjacent or overlapping operating band. For other in-band blocking frequency ranges of the interfering signal for the supported operating bands, "x" is equal to 1.4 dB. | | | | |

**Table 10.5.2.2-2: OTA narrowband blocking requirement for *BS type 1-O***

| ***BS channel bandwidth* of the *lowest/highest carrier* received (MHz)** | **OTA Wanted signal mean power (dBm)** | **OTA Interfering signal mean power (dBm)** |
|---|---|---|
| 5, 10, 15, 20 | $EIS_{REFSENS}$ + 6 dB | Wide Area BS: -49 - $\Delta_{OTAREFSENS}$<br>Medium Range BS: -44 - $\Delta_{OTAREFSENS}$<br>Local Area BS: -41 - $\Delta_{OTAREFSENS}$ |
| | $EIS_{minSENS}$ + 6 dB | Wide Area BS: -49 – $\Delta_{minSENS}$<br>Medium Range BS: -44 – $\Delta_{minSENS}$<br>Local Area BS: -41 – $\Delta_{minSENS}$ |
| 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100 | $EIS_{REFSENS}$ + 6 dB | Wide Area BS: -49 - $\Delta_{OTAREFSENS}$<br>Medium Range BS: -44 - $\Delta_{OTAREFSENS}$<br>Local Area BS: -41 - $\Delta_{OTAREFSENS}$ |
| | $EIS_{minSENS}$ + 6 dB | Wide Area BS: -49 – $\Delta_{minSENS}$<br>Medium Range BS: -44 – $\Delta_{minSENS}$<br>Local Area BS: -41 – $\Delta_{minSENS}$ |
| NOTE 1: The SCS for the *lowest/highest carrier* received is the lowest SCS supported by the BS for that bandwidth.<br>NOTE 2: 7.5 kHz shift is not applied to the wanted signal. | | |

**Table 10.5.2.2-3: OTA narrowband blocking interferer frequency offsets for *BS type 1-O***

| ***BS channel bandwidth* of the *lowest/highest carrier* received (MHz)** | **Interfering RB centre frequency offset to the lower/upper *Base Station RF Bandwidth edge* or *sub-block edge* inside a *sub-block gap* (kHz) (Note 2)** | **Type of interfering signal** |
|---|---|---|
| 5 | ±(350 + m*180),<br>m=0, 1, 2, 3, 4, 9, 14, 19, 24 | 5 MHz DFT-s-OFDM NR signal, 15 kHz SCS, 1 RB |
| 10 | ±(355 + m*180),<br>m=0, 1, 2, 3, 4, 9, 14, 19, 24 | |
| 15 | ±(360 + m*180),<br>m=0, 1, 2, 3, 4, 9, 14, 19, 24 | |
| 20 | ±(350 + m*180),<br>m=0, 1, 2, 3, 4, 9, 14, 19, 24 | |
| 25 | ±(565 + m*180),<br>m=0, 1, 2, 3, 4, 29, 54, 79, 99 | 20 MHz DFT-s-OFDM NR signal, 15 kHz SCS, 1 RB |
| 30 | ±(570 + m*180),<br>m=0, 1, 2, 3, 4, 29, 54, 79, 99 | |
| 35 | ±(560+m*180),<br>m=0, 1, 2, 3, 4, 29, 54, 79, 99 | |
| 40 | ±(565 + m*180),<br>m=0, 1, 2, 3, 4, 29, 54, 79, 99 | |
| 45 | ±(570+m*180),<br>m=0, 1, 2, 3, 4, 29, 54, 79, 99 | |
| 50 | ±(560 + m*180),<br>m=0, 1, 2, 3, 4, 29, 54, 79, 99 | |
| 60 | ±(570 + m*180),<br>m=0, 1, 2, 3, 4, 29, 54, 79, 99 | |
| 70 | ±(565 + m*180),<br>m=0, 1, 2, 3, 4, 29, 54, 79, 99 | |
| 80 | ±(560 + m*180),<br>m=0, 1, 2, 3, 4, 29, 54, 79, 99 | |
| 90 | ±(570 + m*180),<br>m=0, 1, 2, 3, 4, 29, 54, 79, 99 | |
| 100 | ±(565 + m*180),<br>m=0, 1, 2, 3, 4, 29, 54, 79, 99 | |
| NOTE 1: Interfering signal consisting of one resource block is positioned at the stated offset, the channel bandwidth of the interfering signal is located adjacently to the lower/upper *Base Station RF Bandwidth* edge or *sub-block* edge inside a *sub-block gap*.<br>NOTE 2: The centre of the interfering RB refers to the frequency location between the two central subcarriers. | | |

## 10.5.2.3 Minimum requirement for *BS type 2-O*

The requirement shall apply at the RIB when the AoA of the incident wave of a received signal and the interfering signal are from the same direction and are within the *OTA REFSENS RoAoA.*

The wanted and interfering signals apply to each supported polarization, under the assumption o*f polarization match*.

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel.

For *BS type 2-O*, the OTA wanted and OTA interfering signals are provided at RIB using the parameters in table 10.5.2.3-1 for general OTA blocking requirements. The reference measurement channel for the wanted signal is further specified in annex A.1. The characteristics of the interfering signal is further specified in annex D.

The OTA blocking requirements are applicable outside the *Base Station RF Bandwidth*. The interfering signal offset is defined relative to the *Base Station RF Bandwidth* edges.

For *BS type 2-O* the OTA in-band blocking requirement shall apply from $F_{UL_low}$ - $\Delta f_{OOB}$ to $F_{UL_high}$ + $\Delta f_{OOB}$. The $\Delta f_{OOB}$ for *BS type 2-O* is defined in table 10.5.2.3-0.

**Table 10.5.2.3-0: $\Delta f_{OOB}$ offset for NR *operating bands* in FR2**

| BS type | *Frequency Range* | *Operating band* characteristics (MHz) | $\Delta f_{OOB}$ (MHz) |
|---|---|---|---|
| *BS type 2-O* | FR2-1 | $F_{UL_high} - F_{UL_low} \leq 4000$ | 1500 |
| | FR2-2 | $4000 < F_{UL_high} - F_{UL_low} \leq 14000$ | 3500 |

For a RIBs supporting operation in *non-contiguous spectrum* within any *operating band*, the OTA blocking requirements apply in addition inside any *sub-block gap*, in case the *sub-block gap* size is at least as wide as twice the interfering signal minimum offset in table 10.5.2.3-1. The interfering signal offset is defined relative to the *sub-block* edges inside the *sub-block gap*.

**Table 10.5.2.3-1: General OTA blocking requirement for *BS type 2-O***

| Frequency Range | BS channel bandwidth of the lowest/highest carrier received (MHz) | OTA wanted signal mean power (dBm) | OTA interfering signal mean power (dBm) | OTA interfering signal centre frequency offset from the lower/upper Base Station RF Bandwidth edge or sub-block edge inside a sub-block gap (MHz) | Type of OTA interfering signal |
|---|---|---|---|---|---|
| FR2-1 | 50, 100, 200, 400 | $EIS_{REFSENS}$ + 6 dB | $EIS_{REFSENS_50M}$ + 33 + $\Delta_{FR2_REFSENS}$ | ±75 | 50 MHz DFT-s-OFDM NR signal, 60 kHz SCS, 64 RBs |
| FR2-2 | 100, 400, 800, 1600, 2000 | $EIS_{REFSENS}$ + 6 dB | $EIS_{REFSENS_50M}$ + 36 + $\Delta_{FR2_REFSENS}$ | ±150 | 100 MHz DFT-s-OFDM NR signal, 120 kHz SCS, 64 RBs |
| NOTE: $EIS_{REFSENS}$ and $EIS_{REFSENS_50M}$ are given in clause 10.3.3. | | | | | |

# 10.6 OTA out-of-band blocking

## 10.6.1 General

The OTA out-of-band blocking characteristics are a measure of the receiver unit ability to receive a wanted signal at the *RIB* at its assigned channel in the presence of an unwanted interferer.

## 10.6.2 Minimum requirement for *BS type 1-O*

### 10.6.2.1 General minimum requirement

The requirement shall apply at the RIB when the AoA of the incident wave of the received signal and the interfering signal are from the same direction and are within the *minSENS RoAoA*.

The wanted signal applies to each supported polarization, under the assumption of *polarization match.* The interferer shall be *polarization matched* in-band and the polarization maintained for out-of-band frequencies.

For OTA wanted and OTA interfering signals provided at the RIB using the parameters in table 10.6.2.1-1, the following requirements shall be met:

- The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel. The reference measurement channel for the OTA wanted signal is identified in clause 10.3.2 for each *BS channel bandwidth* and further specified in annex A.1.

For a *multi-band RIB*, the OTA out-of-band requirement shall apply for each supported *operating band*, with the exception that the in-band blocking frequency ranges of all supported *operating bands* according to clause 7.4.2.2 shall be excluded from the OTA out-of-band blocking requirement.

For *BS type 1-O* the OTA out-of-band blocking requirement apply from 30 MHz to $F_{UL,low}$ - $\Delta f_{OOB}$ and from $F_{UL,high}$ + $\Delta f_{OOB}$ up to 12750 MHz, including the downlink frequency range of the FDD *operating band* for BS supporting FDD. The $\Delta f_{OOB}$ for *BS type 1-O* is defined in table 10.5.2.2-0.

**Table 10.6.2.1-1: OTA out-of-band blocking performance requirement**

| Wanted signal mean power (dBm) | Interfering signal RMS field-strength (V/m) | Type of interfering Signal |
|---|---|---|
| $EIS_{minSENS}$ + 6 dB (Note 1) | 0.36 | CW carrier |
| NOTE 1: $EIS_{minSENS}$ depends on the *channel bandwidth* as specified in clause 10.2.<br>NOTE 2: The RMS field-strength level in V/m is related to the interferer EIRP level at a distance described as $E=\frac{\sqrt{30 \cdot EIRP}}{r}$, where EIRP is in W and r is in m; for example, 0.36 V/m is equivalent to 36 dBm at fixed distance of 30 m. | | |

### 10.6.2.2 Co-location minimum requirement

This additional OTA out-of-band blocking requirement may be applied for the protection of BS receivers when NR, E-UTRA BS, UTRA BS, CDMA BS or GSM/EDGE BS operating in a different frequency band are co-located with a BS.

The requirement is a co-location requirement. The interferer power levels are specified at the *co-location reference antenna* conducted input. The interfering signal power is specified per supported polarization.

The requirement is valid over the *minSENS RoAoA*.

For OTA wanted and OTA interfering signal provided at the RIB using the parameters in table 10.6.2.1-1, the following requirements shall be met:

- The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel. The reference measurement channel for the OTA wanted signal is identified in clause 10.3.2 for each *BS channel bandwidth* and further specified in annex A.1.

For *BS type 1-O* the OTA blocking requirement for co-location with BS in other frequency bands is applied for all *operating bands* for which co-location protection is provided.

**Table 10.6.2.2-1: OTA blocking requirement for co-location with BS in other frequency bands**

| Frequency range of interfering signal | Wanted signal mean power (dBm) | Interfering signal mean power for WA BS (dBm) | Interfering signal mean power for MR BS (dBm) | Interfering signal mean power for LA BS (dBm) | Type of interfering signal |
|---|---|---|---|---|---|
| Frequency range of co-located downlink *operating band* | $EIS_{minSENS}$ + 6 dB (Note 1) | +46 | +38 | +24 | CW carrier |
| NOTE 1: $EIS_{minSENS}$ depends on the BS class and on the *BS channel bandwidth*, see clause 10.2.<br>NOTE 2: The requirement does not apply when the interfering signal falls within any of the supported uplink *operating band(s)* or in $\Delta f_{OOB}$ immediately outside any of the supported uplink *operating band(s)*. | | | | | |

## 10.6.3 Minimum requirement for *BS type 2-O*

### 10.6.3.1 General minimum requirement

The requirement shall apply at the RIB when the AoA of the incident wave of the received signal and the interfering signal are from the same direction and are within the *OTA REFSENS RoAoA*.

The wanted signal applies to each supported polarization, under the assumption of *polarization match*. The interferer shall be *polarization matched* in-band and the polarization maintained for out-of-band frequencies.

For *BS type 2-O* operating in FR2-1, the OTA out-of-band blocking requirement apply from 30 MHz to $F_{UL,low}$ – 1500 MHz and from $F_{UL,high}$ + 1500 MHz up to 2nd harmonic of the upper frequency edge of the *operating band*.

For *BS type 2-O* operating in FR2-2, the OTA out-of-band blocking requirement apply from 30 MHz to $F_{UL,low}$ – 3500 MHz and from $F_{UL,high}$ + 3500 MHz up to 2nd harmonic of the upper frequency edge of the *operating band*.

For OTA wanted and OTA interfering signals provided at the RIB using the parameters in table 10.6.3.1-1, the following requirements shall be met:

- The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel. The reference measurement channel for the OTA wanted signal is identified in clause 10.3.3 for each *BS channel bandwidth* and further specified in annex A.1.

**Table 10.6.3.1-1: OTA out-of-band blocking performance requirement**

| Frequency Range | Frequency range of interfering signal (MHz) | Wanted signal mean power (dBm) | Interferer RMS field-strength (V/m) | Type of interfering signal |
|---|---|---|---|---|
| FR2-1 | 30 to 12750 | $EIS_{REFSENS}$ + 6 dB | 0.36 | CW |
| | 12750 to $F_{UL,low}$ – 1500 | $EIS_{REFSENS}$ + 6 dB | 0.1 | CW |
| | $F_{UL,high}$ + 1500 to 2nd harmonic of the upper frequency edge of the *operating band* | $EIS_{REFSENS}$ + 6 dB | 0.1 | CW |
| FR2-2 | 30 to 12750 | $EIS_{REFSENS}$ + 6 dB | 0.36 | CW |
| | 12750 to $F_{UL,low}$ – 3500 | $EIS_{REFSENS}$ + 6 dB | 0.1 | CW |
| | $F_{UL,high}$ + 3500 to 2nd harmonic of the upper frequency edge of the *operating band* | $EIS_{REFSENS}$ + 6 dB | 0.1 | CW |

# 10.7 OTA receiver spurious emissions

## 10.7.1 General

The OTA RX spurious emission is the power of the emissions radiated from the antenna array from a receiver unit.

The metric used to capture OTA receiver spurious emissions for *BS type 1-O* and *BS type 2-O* is *total radiated power* (TRP), with the requirement defined at the RIB.

## 10.7.2 Minimum requirement for *BS type 1-O*

For a BS operating in FDD, OTA RX spurious emissions requirement do not apply as they are superseded by the OTA TX spurious emissions requirement. This is due to the fact that TX and RX spurious emissions cannot be distinguished in OTA domain.

For a BS operating in TDD, the OTA RX spurious emissions requirement shall apply during the *transmitter OFF period* only.

For RX only *multi-band RIB*, the OTA RX spurious emissions requirements are subject to exclusion zones in each supported *operating band*.

The OTA RX spurious emissions requirement for *BS type 1-O* is that for each *basic limit* specified in table 10.7.2-1, the power sum of emissions at the RIB shall not exceed limits specified as the *basic limit* + X, where X = 9 dB, unless stated differently in regional regulation.

**Table 10.7.2-1: General BS receiver spurious emission basic limits for *BS type 1-O***

| Spurious frequency range | *Basic limit* (Note 4) | Measurement bandwidth | Notes |
|---|---|---|---|
| 30 MHz – 1 GHz | -36 dBm | 100 kHz | Note 1 |
| 1 GHz – 12.75 GHz | -30 dBm | 1 MHz | Note 1, Note 2 |
| 12.75 GHz – 5th harmonic of the upper frequency edge of the UL *operating band* in GHz | | 1 MHz | Note 1, Note 2, Note 3 |
| 12.75 GHz - 26 GHz | -30 dBm | 1 MHz | Note 1, Note 2, Note 6 |
| NOTE 1: Measurement bandwidths as in ITU-R SM.329 [2], s4.1.<br>NOTE 2: Upper frequency as in ITU-R SM.329 [2], s2.5 table 1.<br>NOTE 3: Applies for Band for which the upper frequency edge of the UL *operating band* is greater than 2.55 GHz and less than or equal to 5.2 GHz.<br>NOTE 4: Additional limits may apply regionally.<br>NOTE 5: The frequency range from $\Delta f_{OBUE}$ below the lowest frequency of the BS transmitter *operating band* to $\Delta f_{OBUE}$ above the highest frequency of the BS transmitter *operating band* may be excluded from the requirement. $\Delta f_{OBUE}$ is defined in clause 9.7.1. For *multi-band RIB*, the exclusion applies for all supported *operating bands*.<br>NOTE 6: Applies for Band for which the upper frequency edge of the UL *operating band* is greater than 5.2 GHz. | | | |

## 10.7.3 Minimum requirement for *BS type 2-O*

The OTA RX spurious emissions requirement shall apply during the *transmitter OFF period* only.

For the *BS type 2-O*, the power of any RX spurious emission shall not exceed the limits in table 10.7.3-1.

**10.7.3-1: Radiated Rx spurious emission limits for *BS type 2-O***

| Spurious frequency range (Note 4) | Limit (Note 5) | Measurement Bandwidth | Note |
|---|---|---|---|
| 30 MHz ↔ 1 GHz | -36 dBm | 100 kHz | Note 1 |
| 1 GHz ↔ 18 GHz | -30 dBm | 1 MHz | Note 1 |
| 18 GHz ↔ $F_{step,1}$ | -20 dBm | 10 MHz | Note 2 |
| $F_{step,1}$ ↔ $F_{step,2}$ | -15 dBm | 10 MHz | Note 2 |
| $F_{step,2}$ ↔ $F_{step,3}$ | -10 dBm | 10 MHz | Note 2 |
| $F_{step,4}$ ↔ $F_{step,5}$ | -10 dBm | 10 MHz | Note 2 |
| $F_{step,5}$ ↔ $F_{step,6}$ | -15 dBm | 10 MHz | Note 2 |
| $F_{step,6}$ ↔ 2nd harmonic of the upper frequency edge of the UL *operating band* | -20 dBm | 10 MHz | Note 2, Note 3 |
| NOTE 1: Bandwidth as in ITU-R SM.329 [2], s4.1.<br>NOTE 2: Limit and bandwidth as in ERC Recommendation 74-01 [19], Annex 2.<br>NOTE 3: Upper frequency as in ITU-R SM.329 [2], s2.5 table 1.<br>NOTE 4: The step frequencies $F_{step,X}$ are defined in table 10.7.3-2.<br>NOTE 5: Additional limits may apply regionally. | | | |

**Table 10.7.3-2: Step frequencies for defining the radiated Rx spurious emission limits for *BS type 2-O***

| Operating band | $F_{step,1}$ (GHz) | $F_{step,2}$ (GHz) | $F_{step,3}$ (GHz) | $F_{step,4}$ (GHz) | $F_{step,5}$ (GHz) | $F_{step,6}$ (GHz) |
|---|---|---|---|---|---|---|
| n257 | 18 | 23.5 | 25 | 31 | 32.5 | 41.5 |
| n258 | 18 | 21 | 22.75 | 29 | 30.75 | 40.5 |
| n259 | 23.5 | 35.5 | 38 | 45 | 47.5 | 59.5 |
| n260 | 25 | 34 | 35.5 | 41.5 | 43 | 52 |
| n261 | 18 | 25.5 | 26.0 | 29.85 | 30.35 | 38.35 |
| n262 | 37.2 | 45.2 | 45.7 | 49.7 | 50.2 | 58.2 |
| n263 | 18 | 43 | 53.5 | 74.5 | 85 | 127 |

In addition to the requirements in Table 10.7.3-1, the requirement for protection of EESS for BS operating in frequency range 24.25 - 27.5 GHz in clause 9.7.5.3.3 may be applied.

# 10.8 OTA receiver intermodulation

## 10.8.1 General

Third and higher order mixing of the two interfering RF signals can produce an interfering signal in the band of the desired channel. Intermodulation response rejection is a measure of the capability of the receiver unit to receive a wanted signal on its assigned channel frequency in the presence of two interfering signals which have a specific frequency relationship to the wanted signal. The requirement is defined as a *directional requirement* at the *RIB*.

## 10.8.2 Minimum requirement for *BS type 1-O*

The requirement shall apply at the RIB when the AoA of the incident wave of a received signal and the interfering signal are from the same direction, and:

- when the wanted signal is based on $EIS_{REFSENS}$: the AoA of the incident wave of a received signal and the interfering signal are within the *OTA REFSENS RoAoA.*
- when the wanted signal is based on $EIS_{minSENS}$: the AoA of the incident wave of a received signal and the interfering signal are within the *minSENS RoAoA.*

The wanted and interfering signals apply to each supported polarization, under the assumption of *polarization match*.

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel, with a wanted signal at the assigned channel frequency and two interfering signals at the RIB with the conditions specified in tables 10.8.2-1 and 10.8.2-2 for intermodulation performance and in tables 10.8.2-3 and 10.8.2-4 for narrowband intermodulation performance. Narrowband intermodulation requirements are not applied for band n104.

The reference measurement channel for the wanted signal is identified in table 10.3.2-1, table 10.3.2-2 and table 10.3.2-3 for each *BS channel bandwidth* and further specified in annex A.1. The characteristics of the interfering signal is further specified in annex D.

The subcarrier spacing for the modulated interfering signal shall be the same as the subcarrier spacing for the wanted signal, except for the case of wanted signal subcarrier spacing 60kHz and *BS channel bandwidth* <=20MHz, for which the subcarrier spacing of the interfering signal shall be 30kHz.

The receiver intermodulation requirement is applicable outside the *Base Station RF Bandwidth* or *Radio Bandwidth edges*. The interfering signal offset is defined relative to the *Base Station RF Bandwidth edges* or *Radio Bandwidth edges*.

For a RIBs supporting operation in non-contiguous spectrum within any *operating band*, the narrowband intermodulation requirement shall apply in addition inside any *sub-block gap* in case the *sub-block gap* is at least as wide as the *BS channel bandwidth* of the NR interfering signal in tables 10.8.2-2 and 10.8.2-4. The interfering signal offset is defined relative to the *sub-block* edges inside the *sub-block gap*.

For *multi-band RIBs*, the intermodulation requirement shall apply in addition inside any *Inter RF Bandwidth gap*, in case the gap size is at least twice as wide as the NR interfering signal centre frequency offset from the *Base Station RF Bandwidth edge*.

For *multi-band RIBs*, the narrowband intermodulation requirement shall apply in addition inside any *Inter RF Bandwidth gap* in case the gap size is at least as wide as the NR interfering signal in tables 10.8.2-2 and 10.8.2-4. The interfering signal offset is defined relative to the *Base Station RF Bandwidth edges* inside the *Inter RF Bandwidth gap*.

**Table 10.8.2-1: General intermodulation requirement**

| BS class | Wanted Signal mean power (dBm) | Mean power of the interfering signals (dBm) | Type of interfering signals |
|---|---|---|---|
| Wide Area BS | $EIS_{REFSENS}$ + 6 dB | -52 - $\Delta_{OTAREFSENS}$ | See Table 10.8.2-2 |
| | $EIS_{minSENS}$ + 6 dB | -52 - $\Delta_{minSENS}$ | |
| Medium Range BS | $EIS_{REFSENS}$ + 6 dB | -47 - $\Delta_{OTAREFSENS}$ | |
| | $EIS_{minSENS}$ + 6 dB | -47 - $\Delta_{minSENS}$ | |
| Local Area BS | $EIS_{REFSENS}$ + 6 dB | -44 - $\Delta_{OTAREFSENS}$ | |
| | $EIS_{minSENS}$ + 6 dB | -44 - $\Delta_{minSENS}$ | |
| NOTE 1: $EIS_{REFSENS}$ and $EIS_{minSENS}$ depend on the BS class and on the *BS channel bandwidth*, see clause 10.3 and 10.2. | | | |

**Table 10.8.2-2: Interfering signals for intermodulation requirement**

| ***BS channel bandwidth*** **of the** ***lowest/highest carrier*** **received (MHz)** | **Interfering signal centre frequency offset from the lower/upper** ***base station RF Bandwidth edge*** **(MHz)** | **Type of interfering signal (Note 3)** |
|---|---|---|
| 5 | ±7.5 | CW |
| | ±17.5 | 5 MHz DFT-s-OFDM NR signal (Note 1) |
| 10 | ±7.465 | CW |
| | ±17.5 | 5 MHz DFT-s-OFDM NR signal (Note 1) |
| 15 | ±7.43 | CW |
| | ±17.5 | 5 MHz DFT-s-OFDM NR signal (Note 1) |
| 20 | ±7.395 | CW |
| | ±17.5 | 5 MHz DFT-s-OFDM NR signal (Note 1) |
| 25 | ±7.465 | CW |
| | ±25 | 20 MHz DFT-s-OFDM NR signal (Note 2) |
| 30 | ±7.43 | CW |
| | ±25 | 20 MHz DFT-s-OFDM NR signal (Note 2) |
| 35 | ±7.44 | CW |
| | ±25 | 20 MHz DFT-s-OFDM NR signal (Note 2) |
| 40 | ±7.45 | CW |
| | ±25 | 20 MHz DFT-s-OFDM NR signal (Note 2) |
| 45 | ±7.37 | CW |
| | ±25 | 20 MHz DFT-s-OFDM NR signal (Note 2) |
| 50 | ±7.38 | CW |
| | ±25 | 20 MHz DFT-s-OFDM NR signal (Note 2) |
| 60 | ±7.49 | CW |
| | ±25 | 20 MHz DFT-s-OFDM NR signal (Note 2) |
| 70 | ±7.42 | CW |
| | ±25 | 20 MHz DFT-s-OFDM NR signal (Note 2) |
| 80 | ±7.44 | CW |
| | ±25 | 20 MHz DFT-s-OFDM NR signal (Note 2) |
| 90 | ±7.46 | CW |
| | ±25 | 20 MHz DFT-s-OFDM NR signal (Note 2) |
| 100 | ±7.48 | CW |
| | ±25 | 20 MHz DFT-s-OFDM NR signal (Note 2) |

NOTE 1: Number of RBs is 25 for 15 kHz subcarrier spacing and 10 for 30 kHz subcarrier spacing.
NOTE 2: Number of RBs is 100 for 15 kHz subcarrier spacing, 50 for 30 kHz subcarrier spacing and 24 for 60 kHz subcarrier spacing.
NOTE 3: The RBs shall be placed adjacent to the transmission bandwidth configuration edge which is closer to the *Base Station RF Bandwidth* edge.

**Table 10.8.2-3: Narrowband intermodulation performance requirement in FR1**

| BS class | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) | Type of interfering signals |
|---|---|---|---|
| Wide Area BS | $EIS_{REFSENS}$ + 6 dB (Note 1) | -52 - $\Delta_{OTAREFSENS}$ | See Table 10.8.2-4 |
| | $EIS_{minSENS}$ + 6 dB (Note 1) | -52 - $\Delta_{minSENS}$ | |
| Medium Range BS | $EIS_{REFSENS}$ + 6 dB (Note 1) | -47 - $\Delta_{OTAREFSENS}$ | |
| | $EIS_{minSENS}$ + 6 dB (Note 1) | -47 - $\Delta_{minSENS}$ | |
| Local Area BS | $EIS_{REFSENS}$ + 6 dB (Note 1) | -44 - $\Delta_{OTAREFSENS}$ | |
| | $EIS_{minSENS}$ + 6 dB (Note 1) | -44 - $\Delta_{minSENS}$ | |
| NOTE 1: $EIS_{REFSENS}$ / $EIS_{minSENS}$ depends on the *BS channel bandwidth*, see clause 10.3 and 10.2. | | | |

**Table 10.8.2-4: Interfering signals for narrowband intermodulation requirement in FR1**

| ***BS channel bandwidth*** **of the** ***lowest/highest carrier*** **received (MHz)** | **Interfering RB centre frequency offset from the lower/upper** ***Base Station RF Bandwidth edge*** **or** ***sub-block*** **edge inside a** ***sub-block gap*** **(kHz) (Note 3)** | **Type of interfering signal** |
|---|---|---|
| 5 | ±360 | CW |
| | ±1420 | 5 MHz DFT-s-OFDM NR signal, 1 RB (NOTE 1) |
| 10 | ±370 | CW |
| | ±1960 | 5 MHz DFT-s-OFDM NR signal, 1 RB (NOTE 1) |
| 15 (NOTE 2) | ±380 | CW |
| | ±1960 | 5 MHz DFT-s-OFDM NR signal, 1 RB (NOTE 1) |
| 20 (NOTE 2) | ±390 | CW |
| | ±2320 | 5 MHz DFT-s-OFDM NR signal, 1 RB (NOTE 1) |
| 25 (NOTE 2) | ±325 | CW |
| | ±2350 | 20 MHz DFT-s-OFDM NR signal, 1 RB (NOTE 1) |
| 30 (NOTE 2) | ±335 | CW |
| | ±2350 | 20 MHz DFT-s-OFDM NR signal, 1 RB (NOTE 1) |
| 35 (NOTE 2) | ±345 | CW |
| | ±2710 | 20 MHz DFT-s-OFDM NR signal, 1 RB (NOTE 1) |
| 40 (NOTE 2) | ±355 | CW |
| | ±2710 | 20 MHz DFT-s-OFDM NR signal, 1 RB (NOTE 1) |
| 45 (NOTE 2) | ±365 | CW |
| | ±2710 | 20 MHz DFT-s-OFDM NR signal, 1 RB (NOTE 1) |
| 50 (NOTE 2) | ±375 | CW |
| | ±2710 | 20 MHz DFT-s-OFDM NR signal, 1 RB (NOTE 1) |
| 60 (NOTE 2) | ±395 | CW |
| | ±2710 | 20 MHz DFT-s-OFDM NR signal, 1 RB (NOTE 1) |
| 70 (NOTE 2) | ±415 | CW |
| | ±2710 | 20 MHz DFT-s-OFDM NR signal, 1 RB (NOTE 1) |
| 80 (NOTE 2) | ±435 | CW |
| | ±2710 | 20 MHz DFT-s-OFDM NR signal, 1 RB (NOTE 1) |
| 90 (NOTE 2) | ±365 | CW |
| | ±2530 | 20 MHz DFT-s-OFDM NR signal, 1 RB (NOTE 1) |
| 100 (NOTE 2) | ±385 | CW |
| | ±2530 | 20 MHz DFT-s-OFDM NR signal, 1 RB (NOTE 1) |

NOTE 1: Interfering signal consisting of one resource block positioned at the stated offset, the *BS channel bandwidth* of the interfering signal is located adjacently to the lower/upper *Base Station RF Bandwidth* edge or *sub-block* edge inside a *sub-block gap*.

NOTE 2: This requirement shall apply only for a G-FRC mapped to the frequency range at the *channel edge* adjacent to the interfering signals.

NOTE 3: The centre of the interfering RB refers to the frequency location between the two central subcarriers.

## 10.8.3 Minimum requirement for *BS type 2-O*

The requirement shall apply at the RIB when the AoA of the incident wave of the received signal and the interfering signal are from the same direction and are within the *OTA REFSENS RoAoA.*

The wanted and interfering signals applies to each supported polarization, under the assumption of *polarization match.*

Throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel, with OTA wanted signal at the assigned channel frequency and two OTA interfering signals provided at the RIB using the parameters in tables 10.8.3-1 and 10.8.3-2. All of the OTA test signals arrive from the same direction, and the requirement is valid if the signals arrive from any direction within the *OTA REFSENS RoAoA.* The reference measurement channel for the wanted signal is identified in table 10.3.3-1 for each *BS channel bandwidth* and further specified in annex A.1. The characteristics of the interfering signal is further specified in annex D.

The subcarrier spacing for the modulated interfering signal shall be the same as the subcarrier spacing for the wanted signal except for FR2-2 with 400MHz, 800MHz, 1600MHz and 2000MHz case.

The receiver intermodulation requirement is applicable outside the *Base Station RF Bandwidth*. The interfering signal offset is defined relative to the *Base Station RF Bandwidth edges*.

**Table 10.8.3-1: General intermodulation requirement**

| BS channel bandwidth of the lowest/highest carrier received (MHz) | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) | Type of interfering signals |
|---|---|---|---|
| 50, 100, 200, 400, 800, 1600, 2000 | $EIS_{REFSENS}$ + 6 | $EIS_{REFSENS_50M}$ + 25 + $\Delta_{FR2_REFSENS}$ (Note 2), $EIS_{REFSENS_50M}$ + 28 + $\Delta_{FR2_REFSENS}$ (Note 3) | See Table 10.8.3-2 |
| NOTE 1: $EIS_{REFSENS}$ and $EIS_{REFSENS_50M}$ are given in clause 10.3.3.<br>NOTE 2: Applicable to bands defined within the frequency spectrum range of FR2-1.<br>NOTE 3: Applicable to bands defined within the frequency spectrum range of FR2-2. | | | |

**Table 10.8.3-2: Interfering signals for intermodulation requirement**

| Frequency Range | BS channel bandwidth of the lowest/highest carrier received (MHz) | Interfering signal centre frequency offset from the lower/upper Base Station RF Bandwidth edge (MHz) | Type of interfering signal |
|---|---|---|---|
| FR2-1 | 50 | ±7.5 | CW |
| | | ±40 | 50MHz DFT-s-OFDM NR signal (Note 1) |
| | 100 | ±6.88 | CW |
| | | ±40 | 50MHz DFT-s-OFDM NR signal (Note 1) |
| | 200 | ±5.64 | CW |
| | | ±40 | 50MHz DFT-s-OFDM NR signal (Note 1) |
| | 400 | ±6.02 | CW |
| | | ±45 | 50MHz DFT-s-OFDM NR signal (Note 1) |
| FR2-2 | 100 | ±7.5 | CW |
| | | ±65 | 100MHz DFT-s-OFDM NR signal (Note 2) |
| | 400 | ±6.28 | CW |
| | | ±70 | 100MHz DFT-s-OFDM NR signal (Note 2) |
| | 800 | [±7.3] | CW |
| | | [±105] | 100MHz DFT-s-OFDM NR signal (Note 2) |
| | 1600 | [±5.86] | CW |
| | | [±145] | 100MHz DFT-s-OFDM NR signal (Note 2) |
| | 2000 | ±7.48 | CW |
| | | ±210 | 100MHz DFT-s-OFDM NR signal (Note 2) |
| NOTE 1: Number of RBs is 64 for the 60 kHz subcarrier spacing, 32 for the 120 kHz subcarrier spacing<br>NOTE 2: Number of RBs is 64 with 120 kHz subcarrier spacing. | | | |

# 10.9 OTA in-channel selectivity

## 10.9.1 General

In-channel selectivity (ICS) is a measure of the receiver ability to receive a wanted signal at its assigned resource block locations in the presence of an interfering signal received at a larger power spectral density. In this condition a throughput requirement shall be met for a specified reference measurement channel. The interfering signal shall be an NR signal as specified in annex A.1 and shall be time aligned with the wanted signal.

## 10.9.2 Minimum requirement for *BS type 1-O*

The requirement shall apply at the RIB when the AoA of the incident wave of the received signal and the interfering signal are the same direction and are within the *minSENS RoAoA*

The wanted and interfering signals applies to each supported polarization, under the assumption of *polarization match*.

For *BS type 1-O*, the throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel as specified in annex A.1 with parameters specified:

- In table 10.9.2-1 for Wide Area BS, in table 10.9.2-2 for Medium Range BS and in table 10.9.2-3 for Local Area BS except in operating band n104.

- In table 10.9.2-1a for Wide Area BS, in table 10.9.2-2a for Medium Range BS and in table 10.9.2-3a for Local Area BS in operating band n104.

The characteristics of the interfering signal is further specified in annex D.

**Table 10.9.2-1: Wide Area BS in-channel selectivity**

| *BS channel bandwidth* (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) | Type of interfering signal |
|---|---|---|---|---|---|
| 5 | 15 | G-FR1-A1-7 | -100.6-$\Delta_{minSENS}$ | -81.4 - $\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 15 kHz SCS, 10 RBs |
| 10, 15, 20, 25, 30, 35 | 15 | G-FR1-A1-1 | -98.7-$\Delta_{minSENS}$ | -77.4 - $\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 15 kHz SCS, 25 RBs |
| 40, 45, 50 | 15 | G-FR1-A1-4 | -92.3-$\Delta_{minSENS}$ | -71.4 - $\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 15 kHz SCS, 100 RBs |
| 5 | 30 | G-FR1-A1-8 | -101.3-$\Delta_{minSENS}$ | -81.4 - $\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 30 kHz SCS, 5 RBs |
| 10, 15, 20, 25, 30, 35 | 30 | G-FR1-A1-2 | -98.8-$\Delta_{minSENS}$ | -78.4 - $\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 30 kHz SCS, 10 RBs |
| 40, 45, 50, 60, 70, 80, 90, 100 | 30 | G-FR1-A1-5 | -92.6-$\Delta_{minSENS}$ | -71.4 - $\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 30 kHz SCS, 50 RBs |
| 10, 15, 20, 25, 30, 35 | 60 | G-FR1-A1-9 | -98.2-$\Delta_{minSENS}$ | -78.4 - $\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 60 kHz SCS, 5 RBs |
| 40, 45, 50, 60, 70, 80, 90, 100 | 60 | G-FR1-A1-6 | -92.7-$\Delta_{minSENS}$ | -71.6 - $\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 60 kHz SCS, 24 RBs |

NOTE: Wanted and interfering signal are placed adjacently around $F_c$, where the $F_c$ is defined for *BS channel bandwidth* of the wanted signal according to the table 5.4.2.2-1 . The aggregated wanted and interferer signal shall be centred in the *BS channel bandwidth* of the wanted signal.

**Table 10.9.2-1a: Wide Area BS in-channel selectivity for band n104**

| BS channel bandwidth (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) | Type of interfering signal |
|---|---|---|---|---|---|
| 20, 30 | 15 | G-FR1-A1-1 | -97.7-$\Delta_{minSENS}$ | -76.4-$\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 15 kHz SCS, 25 RBs |
| 40, 50 | 15 | G-FR1-A1-4 | -91.3-$\Delta_{minSENS}$ | -70.4-$\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 15 kHz SCS, 100 RBs |
| 20, 30 | 30 | G-FR1-A1-2 | -97.8-$\Delta_{minSENS}$ | -77.4-$\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 30 kHz SCS, 10 RBs |
| 40, 50, 60, 70, 80, 90, 100 | 30 | G-FR1-A1-5 | -91.6-$\Delta_{minSENS}$ | -70.4-$\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 30 kHz SCS, 50 RBs |
| 20, 30 | 60 | G-FR1-A1-9 | -97.2-$\Delta_{minSENS}$ | -77.4-$\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 60 kHz SCS, 5 RBs |
| 40, 50, 60, 70, 80, 90, 100 | 60 | G-FR1-A1-6 | -91.7-$\Delta_{minSENS}$ | -70.6-$\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 60 kHz SCS, 24 RBs |
| NOTE 1: Wanted and interfering signal are placed adjacently around $F_c$, where the $F_c$ is defined for *BS channel bandwidth* of the wanted signal according to the table 5.4.2.2-1. The aggregated wanted and interferer signal shall be centred in the *BS channel bandwidth* of the wanted signal. | | | | | |

**Table 10.9.2-2: Medium Range BS in-channel selectivity**

| *BS channel bandwidth* (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) | Type of interfering signal |
|---|---|---|---|---|---|
| 5 | 15 | G-FR1-A1-7 | -95.6-$\Delta_{minSENS}$ | -76.4 - $\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 15 kHz SCS, 10 RBs |
| 10, 15, 20, 25, 30, 35 | 15 | G-FR1-A1-1 | -93.7-$\Delta_{minSENS}$ | -72.4 - $\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 15 kHz SCS, 25 RBs |
| 40, 45, 50 | 15 | G-FR1-A1-4 | -87.3-$\Delta_{minSENS}$ | -66.4 - $\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 15 kHz SCS, 100 RBs |
| 5 | 30 | G-FR1-A1-8 | -96.3-$\Delta_{minSENS}$ | -76.4 - $\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 30 kHz SCS, 5 RBs |
| 10, 15, 20, 25, 30, 35 | 30 | G-FR1-A1-2 | -93.8-$\Delta_{minSENS}$ | -73.4 - $\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 30 kHz SCS, 10 RBs |
| 40, 45, 50, 60, 70, 80, 90, 100 | 30 | G-FR1-A1-5 | -87.6-$\Delta_{minSENS}$ | -66.4 - $\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 30 kHz SCS, 50 RBs |
| 10, 15, 20, 25, 30, 35 | 60 | G-FR1-A1-9 | -93.2-$\Delta_{minSENS}$ | -73.4 - $\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 60 kHz SCS, 5 RBs |
| 40, 45, 50, 60, 70, 80, 90, 100 | 60 | G-FR1-A1-6 | -87.7-$\Delta_{minSENS}$ | -66.6 - $\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 60 kHz SCS, 24 RBs |
| NOTE: Wanted and interfering signal are placed adjacently around $F_c$, where the $F_c$ is defined for *BS channel bandwidth* of the wanted signal according to the table 5.4.2.2-1. The aggregated wanted and interferer signal shall be centred in the *BS channel bandwidth* of the wanted signal. | | | | | |

**Table 10.9.2-2a: Medium Range BS in-channel selectivity for band n104**

| BS channel bandwidth (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) | Type of interfering signal |
|---|---|---|---|---|---|
| 20, 30 | 15 | G-FR1-A1-1 | -92.7-$\Delta_{minSENS}$ | -71.4-$\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 15 kHz SCS, 25 RBs |
| 40, 50 | 15 | G-FR1-A1-4 | -86.3-$\Delta_{minSENS}$ | -65.4-$\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 15 kHz SCS, 100 RBs |
| 20, 30 | 30 | G-FR1-A1-2 | -92.8-$\Delta_{minSENS}$ | -72.4-$\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 30 kHz SCS, 10 RBs |
| 40, 50, 60, 70, 80, 90, 100 | 30 | G-FR1-A1-5 | -86.6-$\Delta_{minSENS}$ | -65.4-$\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 30 kHz SCS, 50 RBs |
| 20, 30 | 60 | G-FR1-A1-9 | -92.2-$\Delta_{minSENS}$ | -72.4-$\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 60 kHz SCS, 5 RBs |
| 40, 50, 60, 70, 80, 90, 100 | 60 | G-FR1-A1-6 | -86.7-$\Delta_{minSENS}$ | -65.6-$\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 60 kHz SCS, 24 RBs |
| NOTE: Wanted and interfering signal are placed adjacently around $F_c$, where the $F_c$ is defined for *BS channel bandwidth* of the wanted signal according to the table 5.4.2.2-1. The aggregated wanted and interferer signal shall be centred in the *BS channel bandwidth* of the wanted signal. | | | | | |

**Table 10.9.2-3: Local area BS in-channel selectivity**

| *BS channel bandwidth* (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) | Type of interfering signal |
|---|---|---|---|---|---|
| 5 | 15 | G-FR1-A1-7 | -92.6-$\Delta_{minSENS}$ | -73.4 - $\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 15 kHz SCS, 10 RBs |
| 10, 15, 20, 25, 30, 35 | 15 | G-FR1-A1-1 | -90.7-$\Delta_{minSENS}$ | -69.4 - $\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 15 kHz SCS, 25 RBs |
| 40, 45, 50 | 15 | G-FR1-A1-4 | -84.3-$\Delta_{minSENS}$ | -63.4 - $\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 15 kHz SCS, 100 RBs |
| 5 | 30 | G-FR1-A1-8 | -93.3-$\Delta_{minSENS}$ | -73.4 - $\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 30 kHz SCS, 5 RBs |
| 10, 15, 20, 25, 30, 35 | 30 | G-FR1-A1-2 | -90.8-$\Delta_{minSENS}$ | -70.4 - $\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 30 kHz SCS, 10 RBs |
| 40, 45, 50, 60, 70, 80, 90, 100 | 30 | G-FR1-A1-5 | -84.6-$\Delta_{minSENS}$ | -63.4 - $\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 30 kHz SCS, 50 RBs |
| 10, 15, 20, 25, 30, 35 | 60 | G-FR1-A1-9 | -90.2-$\Delta_{minSENS}$ | -70.4 - $\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 60 kHz SCS, 5 RBs |
| 40, 45, 50, 60, 70, 80, 90, 100 | 60 | G-FR1-A1-6 | -84.7-$\Delta_{minSENS}$ | -63.6 - $\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 60 kHz SCS, 24 RBs |
| NOTE: Wanted and interfering signal are placed adjacently around $F_c$, where the $F_c$ is defined for *BS channel bandwidth* of the wanted signal according to the table 5.4.2.2-1. The aggregated wanted and interferer signal shall be centred in the *BS channel bandwidth* of the wanted signal. | | | | | |

**Table 10.9.2-3a: Local area BS in-channel selectivity for band n104**

| BS channel bandwidth (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) | Interfering signal mean power (dBm) | Type of interfering signal |
|---|---|---|---|---|---|
| 20, 30 | 15 | G-FR1-A1-1 | -89.7-$\Delta_{minSENS}$ | -68.4-$\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 15 kHz SCS, 25 RBs |
| 40, 50 | 15 | G-FR1-A1-4 | -83.3-$\Delta_{minSENS}$ | -62.4-$\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 15 kHz SCS, 100 RBs |
| 20, 30 | 30 | G-FR1-A1-2 | -89.8-$\Delta_{minSENS}$ | -69.4-$\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 30 kHz SCS, 10 RBs |
| 40, 50, 60, 70, 80, 90, 100 | 30 | G-FR1-A1-5 | -83.6-$\Delta_{minSENS}$ | -62.4-$\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 30 kHz SCS, 50 RBs |
| 20, 30 | 60 | G-FR1-A1-9 | -89.2-$\Delta_{minSENS}$ | -69.4-$\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 60 kHz SCS, 5 RBs |
| 40, 50, 60, 70, 80, 90, 100 | 60 | G-FR1-A1-6 | -83.7-$\Delta_{minSENS}$ | -62.6-$\Delta_{minSENS}$ | DFT-s-OFDM NR signal, 60 kHz SCS, 24 RBs |
| NOTE: Wanted and interfering signal are placed adjacently around $F_c$, where the $F_c$ is defined for *BS channel bandwidth* of the wanted signal according to the table 5.4.2.2-1. The aggregated wanted and interferer signal shall be centred in the *BS channel bandwidth* of the wanted signal. | | | | | |

## 10.9.3 Minimum requirement for *BS type 2-O*

The requirement shall apply at the RIB when the AoA of the incident wave of the received signal and the interfering signal are from the same direction and are within the *OTA REFSENS RoAoA.*

The wanted and interfering signals applies to each supported polarization, under the assumption of *polarization match.*

For *BS type 2-O*, the throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel as specified in annex A.1 with parameters specified in table 10.9.3-1. The characteristics of the interfering signal is further specified in annex D.

**Table 10.9.3-1: OTA in-channel selectivity requirement for *BS type 2-O***

| Frequency Range | BS channel bandwidth (MHz) | Subcarrier spacing (kHz) | Reference measurement channel | Wanted signal mean power (dBm) (Note 2) | Interfering signal mean power (dBm) (Note 2) | Type of interfering signal |
|---|---|---|---|---|---|---|
| FR2-1 | 50 | 60 | G-FR2-A1-4 | $EIS_{REFSENS_50M} + \Delta_{FR2_REFSENS}$ | $EIS_{REFSENS_50M} + 10 + \Delta_{FR2_REFSENS}$ | DFT-s-OFDM NR signal, 60 kHz SCS, 32 RB |
| | 100,200 | 60 | G-FR2-A1-1 | $EIS_{REFSENS_50M} + 3 + \Delta_{FR2_REFSENS}$ | $EIS_{REFSENS_50M} + 13 + \Delta_{FR2_REFSENS}$ | DFT-s-OFDM NR signal, 60 kHz SCS, 64 RB |
| | 50 | 120 | G-FR2-A1-5 | $EIS_{REFSENS_50M} + \Delta_{FR2_REFSENS}$ | $EIS_{REFSENS_50M} + 10 + \Delta_{FR2_REFSENS}$ | DFT-s-OFDM NR signal, 120 kHz SCS, 16 RB |
| | 100,200,400 | 120 | G-FR2-A1-2 | $EIS_{REFSENS_50M} + 3 + \Delta_{FR2_REFSENS}$ | $EIS_{REFSENS_50M} + 13 + \Delta_{FR2_REFSENS}$ | DFT-s-OFDM NR signal, 120 kHz SCS, 32 RB |
| FR2-2 | 100,400 | 120 | G-FR2-A1-2 | $EIS_{REFSENS_50M} + 3 + \Delta_{FR2_REFSENS}$ | $EIS_{REFSENS_50M} + 13 + \Delta_{FR2_REFSENS}$ | DFT-s-OFDM NR signal, 120 kHz SCS, 32 RB |
| | 400 | 480 | G-FR2-A1-8 | $EIS_{REFSENS_50M} + 9 + \Delta_{FR2_REFSENS}$ | $EIS_{REFSENS_50M} + 19+ \Delta_{FR2_REFSENS}$ | DFT-s-OFDM NR signal, 480 kHz SCS, 32 RB |
| | 800, 1600 | 480 | G-FR2-A1-6 | $EIS_{REFSENS_50M} + 12 + \Delta_{FR2_REFSENS}$ | $EIS_{REFSENS_50M} + 22 + \Delta_{FR2_REFSENS}$ | DFT-s-OFDM NR signal, 480 kHz SCS, 54 RB |
| | 400 | 960 | G-FR2-A1-9 | $EIS_{REFSENS_50M} + 9 + \Delta_{FR2_REFSENS}$ | $EIS_{REFSENS_50M} + 19+ \Delta_{FR2_REFSENS}$ | DFT-s-OFDM NR signal, 960 kHz SCS, 16 RB |
| | 800, 1600, 2000 | 960 | G-FR2-A1-7 | $EIS_{REFSENS_50M} + 12 + \Delta_{FR2_REFSENS}$ | $EIS_{REFSENS_50M} + 22+ \Delta_{FR2_REFSENS}$ | DFT-s-OFDM NR signal, 960 kHz SCS, 27 RB |

NOTE 1: Wanted and interfering signal are placed adjacently around $F_c$, where the $F_c$ is defined for *BS channel bandwidth* of the wanted signal according to the table 5.4.2.2-1. The aggregated wanted and interferer signal shall be centred in the *BS channel bandwidth* of the wanted signal.
NOTE 2: $EIS_{REFSENS_50M}$ is defined in clause 10.3.3.

**Table 10.9.3-2: (Void)**

**Table 10.9.3-3: (Void)**

# 11 Radiated performance requirements

## 11.1 General

### 11.1.1 Scope and definitions

Radiated performance requirements specify the ability of the *BS type 1-O* or *BS type 2-O* to correctly demodulate radiated signals in various conditions and configurations. Radiated performance requirements are specified at the RIB.

Radiated performance requirements for the BS are specified for the fixed reference channels defined in annex A and the propagation conditions in annex G. The requirements only apply to those FRCs that are supported by the BS.

The radiated performance requirements for *BS type 1-O* and for the *BS type 2-O* are limited to two OTA *demodulation branches* as described in clause 11.1.2. Conformance requirements can only be tested for 1 or 2 *demodulation branches* depending on the number of polarizations supported by the BS, with the required SNR applied separately per polarization.

NOTE 1: The BS can support more than 2 *demodulation branches*, however OTA conformance testing can only be performed for 1 or 2 *demodulation branches*.

Unless stated otherwise, radiated performance requirements apply for a single carrier only. Radiated performance requirements for a BS supporting CA are defined in terms of single carrier requirements.

For *BS type 1-O* in FDD operation the requirements in clause 8 shall be met with the transmitter units associated with the RIB in the *operating band* turned ON.

NOTE 2: *BS type 1-O* in normal operating conditions in FDD operation is configured to transmit and receive at the same time. The transmitter unit(s) associated with the RIB may be OFF for some of the tests.

In tests performed with signal generators a synchronization signal may be provided from the BS to the signal generator, to enable correct timing of the wanted signal.

Whenever the "RX antennas" term is used for the radiated performance requirements description, it shall refer to the *demodulation branches* (i.e. not physical antennas of the antenna array).

The SNR used in this clause is specified based on a single carrier and defined as:

SNR = S / N

Where:

S is the total signal energy in a slot on a RIB.

N is the noise energy in a bandwidth corresponding to the *transmission bandwidth* over the duration where signal energy exists on a RIB.

### 11.1.2 OTA demodulation branches

Radiated performance requirements are only specified for up to 2 *demodulation branches*.

If the *BS type 1-O*, or the *BS type 2-O* uses polarization diversity and has the ability to maintain isolation between the signals for each of the *demodulation branches*, then radiated performance requirements can be tested for up to two *demodulation branches* (i.e. 1RX or 2RX test setups). When tested for two *demodulation branches*, each demodulation branch maps to one polarization.

If the *BS type 1-O*, or the *BS type 2-O* does not use polarization diversity then radiated performance requirements can only be tested for a single *demodulation branch* (i.e. 1RX test setup).

### 11.1.3 Void

## 11.2 Performance requirements for PUSCH

### 11.2.1 Requirements for *BS type 1-O*

#### 11.2.1.1 Requirements for PUSCH with transform precoding disabled

Apply the requirements defined in clause 8.2.1 for 2Rx.

#### 11.2.1.2 Requirements for PUSCH with transform precoding enabled

Apply the requirements defined in clause 8.2.2 for 2Rx.

#### 11.2.1.3 Requirements for UCI multiplexed on PUSCH

Apply the requirements defined in clause 8.2.3 for 2Rx.

#### 11.2.1.4 Requirements for PUSCH for high speed train

Apply the requirements defined in clause 8.2.4 for 2Rx.

#### 11.2.1.5 Requirements for UL timing adjustment

Apply the requirements defined in clause 8.2.5 for 2Rx.

#### 11.2.1.6 Requirements for PUSCH 0.001% BLER

Apply the requirements defined in clause 8.2.6 for 2Rx.

#### 11.2.1.7 Requirements for PUSCH repetition Type A

Apply the requirements defined in clause 8.2.7 for 2Rx.

#### 11.2.1.8 Requirements for PUSCH mapping Type B with non-slot transmission

Apply the requirements defined in clause 8.2.8 for 2Rx.

#### 11.2.1.9 Requirements for PUSCH for 2-step RA type

Apply the requirements defined in clause 8.2.9 for 2Rx.

#### 11.2.1.10 Requirements for interlaced PUSCH

Apply the requirements defined in clause 8.2.10 for 2Rx.

#### 11.2.1.11 Requirements for CG-UCI multiplexed on interlaced PUSCH

Apply the requirements defined in clause 8.2.11 for 2Rx.

#### 11.2.1.12 Requirements for TB processing over multi-slot PUSCH (TBoMS)

Apply the requirements defined in clause 8.2.12 for 2Rx.

#### 11.2.1.13 Requirements for PUSCH with DM-RS bundling

Apply the requirements for 2Rx defined in clause 8.2.13 for 2Rx.

### 11.2.2 Requirements for *BS type 2-O*

#### 11.2.2.1 Requirements for PUSCH with transform precoding disabled

##### 11.2.2.1.1 General

The performance requirement of PUSCH is determined by a minimum required throughput for a given SNR. The required throughput is expressed as a fraction of maximum throughput for the FRCs listed in annex A. The performance requirements assume HARQ retransmissions.

**Table 11.2.2.1.1-1: Test parameters for testing PUSCH**

| Parameter | | Value |
|---|---|---|
| Transform precoding | | Disabled |
| Default TDD UL-DL pattern (Note 1) | | 60 kHz and 120kHz SCS: 3D1S1U, S=10D:2G:2U<br>480kHz SCS:14D2S4U, S1=12D:2G0U, S2=0D:6G:8U |
| HARQ | Maximum number of HARQ transmissions | 4 |
| | RV sequence | 0, 2, 3, 1 |
| DM-RS | DM-RS configuration type | 1 |
| | DM-RS duration | single-symbol DM-RS |
| | Additional DM-RS symbols | pos0, pos1 |
| | Number of DM-RS CDM group(s) without data | 2 |
| | Ratio of PUSCH EPRE to DM-RS EPRE | -3 dB |
| | DM-RS port(s) | {0}, {0, 1} |
| | DM-RS sequence generation | $N_{ID}$=0, $n_{SCID}$ =0 |
| Time domain resource | PUSCH mapping type | B |
| | Start symbol index | 0 |
| | Allocation length | 10 |
| Frequency domain resource | RB assignment | Full applicable test bandwidth |
| | Frequency hopping | Disabled |
| TPMI index for 2Tx two-layer spatial multiplexing transmission | | 0 |
| Code block group based PUSCH transmission | | Disabled |
| PT-RS configuration | Frequency density ($K_{PT-RS}$) | 2, Disabled |
| | Time density ($L_{PT-RS}$) | 1, Disabled |
| NOTE 1: The same requirements are applicable to TDD with different UL-DL patterns | | |

##### 11.2.2.1.2 Minimum requirements

The throughput shall be equal to or larger than the fraction of maximum throughput stated in the tables 11.2.2.1.2-1 to 11.2.2.1.2-10 at the given SNR for 1Tx and for 2Tx two-layer spatial multiplexing transmission.

**Table 11.2.2.1.2-1: Minimum requirements for PUSCH with 70% of maximum throughput, 50 MHz channel bandwidth, 60 kHz SCS in FR2-1**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | PT-RS | SNR (dB) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-300 Low | 70 % | G-FR2-A3-1 | pos0 | No | -2.0 |
| | | | | | G-FR2-A3-13 | pos1 | No | -2.2 |
| | | Normal | TDLA30-300 Low | 70 % | G-FR2-A4-1 | pos0 | Yes | 12.0 |
| | | | | | | | No | 11.5 |
| | | | | | G-FR2-A4-11 | pos1 | Yes | 10.7 |
| | | | | | | | No | 10.7 |
| | | Normal | TDLA30-75 Low | 70 % | G-FR2-A5-1 | pos0 | Yes | 13.7 |
| | | | | | | | No | 13.1 |
| | | | | | G-FR2-A5-6 | pos1 | Yes | 13.4 |
| | | | | | | | No | 12.9 |
| 2 | | Normal | TDLA30-300 Low | 70 % | G-FR2-A3-6 | pos0 | No | 1.5 |
| | | | | | G-FR2-A3-18 | pos1 | No | 1.2 |
| | | Normal | TDLA30-300 Low | 70 % | G-FR2-A7-1 | pos0 | Yes | 15.2 |
| | | | | | | | No | 14.3 |
| | | | | | G-FR2-A7-6 | pos1 | Yes | 13.8 |
| | | | | | | | No | 13.0 |

**Table 11.2.2.1.2-2: Minimum requirements for PUSCH with 70% of maximum throughput, 100 MHz channel bandwidth, 60 kHz SCS in FR2-1**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | PT-RS | SNR (dB) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-300 Low | 70 % | G-FR2-A3-2 | pos0 | No | -2.1 |
| | | | | | G-FR2-A3-14 | pos1 | No | -2.4 |
| | | Normal | TDLA30-300 Low | 70 % | G-FR2-A4-2 | pos0 | Yes | 12.2 |
| | | | | | | | No | 11.2 |
| | | | | | G-FR2-A4-12 | pos1 | Yes | 11.2 |
| | | | | | | | No | 10.6 |
| | | Normal | TDLA30-75 Low | 70 % | G-FR2-A5-2 | pos0 | Yes | 14.2 |
| | | | | | | | No | 13.3 |
| | | | | | G-FR2-A5-7 | pos1 | Yes | 13.7 |
| | | | | | | | No | 13.1 |
| 2 | | Normal | TDLA30-300 Low | 70 % | G-FR2-A3-7 | pos0 | No | 1.5 |
| | | | | | G-FR2-A3-19 | pos1 | No | 1.2 |
| | | Normal | TDLA30-300 Low | 70 % | G-FR2-A7-2 | pos0 | Yes | 16.0 |
| | | | | | | | No | 14.9 |
| | | | | | G-FR2-A7-7 | pos1 | Yes | 13.8 |
| | | | | | | | No | 13.1 |

**Table 11.2.2.1.2-3: Minimum requirements for PUSCH with 70% of maximum throughput, 50 MHz channel bandwidth, 120 kHz SCS in FR2-1**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | PT-RS | SNR (dB) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-300 Low | 70 % | G-FR2-A3-3 | pos0 | No | -1.8 |
| | | | | | G-FR2-A3-15 | pos1 | No | -2.1 |
| | | Normal | TDLA30-300 Low | 70 % | G-FR2-A4-3 | pos0 | Yes | 11.6 |
| | | | | | | | No | 10.9 |
| | | | | | G-FR2-A4-13 | pos1 | Yes | 10.9 |
| | | | | | | | No | 10.5 |
| | | Normal | TDLA30-75 Low | 70 % | G-FR2-A5-3 | pos0 | Yes | 13.7 |
| | | | | | | | No | 13.1 |
| | | | | | G-FR2-A5-8 | pos1 | Yes | 13.2 |
| | | | | | | | No | 13.0 |
| 2 | | Normal | TDLA30-300 Low | 70 % | G-FR2-A3-8 | pos0 | No | 1.4 |
| | | | | | G-FR2-A3-20 | pos1 | No | 1.3 |
| | | Normal | TDLA30-300 Low | 70 % | G-FR2-A7-3 | pos0 | Yes | 14.2 |
| | | | | | | | No | 13.6 |
| | | | | | G-FR2-A7-8 | pos1 | Yes | 13.9 |
| | | | | | | | No | 13.1 |

**Table 11.2.2.1.2-4: Minimum requirements for PUSCH with 70% of maximum throughput, 100 MHz channel bandwidth, 120 kHz SCS in FR2-1**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | PT-RS | SNR (dB) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-300 Low | 70 % | G-FR2-A3-4 | pos0 | No | -2.4 |
| | | | | | G-FR2-A3-16 | pos1 | No | -2.5 |
| | | Normal | TDLA30-300 Low | 70 % | G-FR2-A4-4 | pos0 | Yes | 11.9 |
| | | | | | | | No | 10.5 |
| | | | | | G-FR2-A4-14 | pos1 | Yes | 11.1 |
| | | | | | | | No | 10.5 |
| | | Normal | TDLA30-75 Low | 70 % | G-FR2-A5-4 | pos0 | Yes | 13.5 |
| | | | | | | | No | 12.9 |
| | | | | | G-FR2-A5-9 | pos1 | Yes | 13.4 |
| | | | | | | | No | 12.8 |
| 2 | | Normal | TDLA30-300 Low | 70 % | G-FR2-A3-9 | pos0 | No | 1.4 |
| | | | | | G-FR21-A3-21 | pos1 | No | 1.2 |
| | | Normal | TDLA30-300 Low | 70 % | G-FR2-A7-4 | pos0 | Yes | 13.9 |
| | | | | | | | No | 13.2 |
| | | | | | G-FR2-A7-9 | pos1 | Yes | 13.5 |
| | | | | | | | No | 12.9 |

**Table 11.2.2.1.2-5: Minimum requirements for PUSCH with 70% of maximum throughput, 200 MHz channel bandwidth, 120 kHz SCS in FR2-1**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | PT-RS | SNR (dB) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-300 Low | 70 % | G-FR2-A3-5 | pos0 | No | -2.1 |
| | | | | | G-FR2-A3-17 | pos1 | No | -2.4 |
| | | Normal | TDLA30-300 Low | 70 % | G-FR2-A4-5 | pos0 | Yes | 11.3 |
| | | | | | | | No | 10.9 |
| | | | | | G-FR2-A4-15 | pos1 | Yes | 11.2 |
| | | | | | | | No | 10.7 |
| | | Normal | TDLA30-75 Low | 70 % | G-FR2-A5-5 | pos0 | Yes | 14.1 |
| | | | | | | | No | 13.4 |
| | | | | | G-FR2-A5-10 | pos1 | Yes | 13.7 |
| | | | | | | | No | 13.3 |
| 2 | | Normal | TDLA30-300 Low | 70 % | G-FR2-A3-10 | pos0 | No | 1.4 |
| | | | | | G-FR2-A3-22 | pos1 | No | 1.1 |
| | | Normal | TDLA30-300 Low | 70 % | G-FR2-A7-5 | pos0 | Yes | 14.0 |
| | | | | | | | No | 13.3 |
| | | | | | G-FR2-A7-10 | pos1 | Yes | 13.6 |
| | | | | | | | No | 13.0 |

**Table 11.2.2.1.2-6: Minimum requirements for PUSCH with 30% of maximum throughput, 50 MHz channel bandwidth, 60 kHz SCS in FR2-1**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | PT-RS | SNR (dB) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-300 Low | 30 % | G-FR2-A4-1 | pos0 | Yes | 4.0 |
| | | | | | | | No | 3.5 |
| | | | | | G-FR2-A4-11 | pos1 | Yes | 3.7 |
| | | | | | | | No | 3.1 |

**Table 11.2.2.1.2-7: Minimum requirements for PUSCH with 30% of maximum throughput, 50 MHz channel bandwidth, 120 kHz SCS in FR2-1**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | PT-RS | SNR (dB) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-300 Low | 30 % | G-FR2-A4-3 | pos0 | Yes | 4.0 |
| | | | | | | | No | 3.6 |
| | | | | | G-FR2-A4-13 | pos1 | Yes | 3.7 |
| | | | | | | | No | 3.2 |

**Table 11.2.2.1.2-8: Test requirements for PUSCH with 70% of maximum throughput, 100 MHz Channel Bandwidth, 120 kHz SCS in FR2-2**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (annex G) | Fraction of maximum throughput | FRC (annex A) | Additional DM-RS position | PT-RS | SNR (dB) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-650 | 70 % | G-FR2-A3B-3 | pos1 | No | -0.2 |
| | | Normal | TDLA30-650 | 70 % | G-FR2-A4-21 | pos1 | Yes | 11.2 |
| | | Normal | TDLD30-200 | 70 % | G-FR2-A5-11 | pos1 | Yes | 12.5 |
| 2 | | Normal | TDLA30-650 | 70 % | G-FR2-A3B-8 | pos1 | No | 3.9 |
| | | Normal | TDLA30-650 | 70 % | G-FR2-A7-11 | pos1 | Yes | 13.6 |

**Table 11.2.2.1.2-9: Test requirements for PUSCH with 70% of maximum throughput, 400 MHz Channel Bandwidth, 120 kHz SCS in FR2-2**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (annex G) | Fraction of maximum throughput | FRC (annex A) | Additional DM-RS position | PT-RS | SNR (dB) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA10-650 | 70 % | G-FR2-A3B-4 | pos1 | No | -0.2 |
| | | Normal | TDLA10-650 | 70 % | G-FR2-A4-22 | pos1 | Yes | 11.2 |
| | | Normal | TDLD10-200 | 70 % | G-FR2-A5-12 | pos1 | Yes | 12.6 |
| 2 | | Normal | TDLA10-650 | 70 % | G-FR2-A3B-9 | pos1 | No | 4 |
| | | Normal | TDLA10-650 | 70 % | G-FR2-A7-12 | pos1 | Yes | 14 |

**Table 11.2.2.1.2-10: Test requirements for PUSCH with 70% of maximum throughput, 400 MHz Channel Bandwidth, 480 kHz SCS in FR2-2**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (annex G) | Fraction of maximum throughput | FRC (annex A) | Additional DM-RS position | PT-RS | SNR (dB) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA10-650 | 70 % | G-FR2-A3B-5 | pos1 | No | -0.5 |
| | | Normal | TDLA10-650 | 70 % | G-FR2-A4-23 | pos1 | Yes | 10.8 |
| | | Normal | TDLD10-200 | 70 % | G-FR2-A5-13 | pos1 | Yes | 12.8 |
| 2 | | Normal | TDLA10-650 | 70 % | G-FR2-A3B-10 | pos1 | No | 3.5 |
| | | Normal | TDLA10-650 | 70 % | G-FR2-A7-13 | pos1 | Yes | 13.8 |

### 11.2.2.2 Requirements for PUSCH with transform precoding enabled

#### 11.2.2.2.1 General

The performance requirement of PUSCH is determined by a minimum required throughput for a given SNR. The required throughput is expressed as a fraction of maximum throughput for the FRCs listed in Annex A. The performance requirements assume HARQ retransmissions.

**Table 11.2.2.2.1-1: Test parameters for testing PUSCH**

| Parameter | | Value |
|---|---|---|
| Transform precoding | | Enabled |
| Default TDD UL-DL pattern (Note 1) | | 60 kHz and 120kHz SCS: 3D1S1U, S=10D:2G:2U<br>480kHz SCS:14D2S4U, S1=12D:2G0U, S2=0D:6G:8U |
| HARQ | Maximum number of HARQ transmissions | 4 |
| | RV sequence | 0, 2, 3, 1 |
| DM-RS | DM-RS configuration type | 1 |
| | DM-RS duration | single-symbol DM-RS |
| | Additional DM-RS position | pos0, pos1 |
| | Number of DM-RS CDM group(s) without data | 2 |
| | Ratio of PUSCH EPRE to DM-RS EPRE | -3 dB |
| | DM-RS port(s) | 0 |
| | DM-RS sequence generation | $N_{ID}^0$=0, group hopping and sequence hopping are disabled |
| Time domain resource assignment | PUSCH mapping type | B |
| | Start symbol | 0 |
| | Allocation length | 10 |
| Frequency domain resource assignment | RB assignment | FR2-1: 30 PRBs in the middle of the test bandwidth<br>FR2-2: 64 PRBs in the middle of the test bandwidth |
| | Frequency hopping | Disabled |
| Code block group based PUSCH transmission | | Disabled |
| PT-RS | | Not configured |
| NOTE 1: The same requirements are applicable to TDD with different UL-DL patterns. | | |

#### 11.2.2.2.2 Minimum requirements

The throughput shall be equal to or larger than the fraction of maximum throughput stated in the tables 11.2.2.2.2-1 to 11.2.2.2.2-4 at the given SNR.

**Table 11.2.2.2.2-1: Minimum requirements for PUSCH with 70% of maximum throughput, Type B, 50 MHz Channel Bandwidth, 60 kHz SCS in FR2-1**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-300 Low | 70 % | G-FR2-A3-11 | pos0 | -1.8 |
| | | | | | G-FR2-A3-23 | pos1 | -1.9 |

**Table 11.2.2.2.2-2: Minimum requirements for PUSCH with 70% of maximum throughput, Type B, 50 MHz Channel Bandwidth, 120 kHz SCS in FR2-1**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-300 Low | 70 % | G-FR2-A3-12 | pos0 | -1.8 |
| | | | | | G-FR2-A3-24 | pos1 | -1.9 |

**Table 11.2.2.2.2-3: Minimum requirements for PUSCH with 70% of maximum throughput, Type B, 100 MHz Channel Bandwidth, 120 kHz SCS in FR2-2**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-650 | 70 % | G-FR2-A3B-6 | pos1 | 0.1 |

**Table 11.2.2.2.2-4: Minimum requirements for PUSCH with 70% of maximum throughput, Type B, 400 MHz Channel Bandwidth, 480 kHz SCS in FR2-2**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA10-650 | 70 % | G-FR2-A3B-7 | pos1 | 0.2 |

## 11.2.2.3 Requirements for UCI multiplexed on PUSCH

### 11.2.2.3.1 General

In the tests for UCI multiplexed on PUSCH, the UCI information only contains CSI part 1 and CSI part 2 information, and there is no HACK/ACK information transmitted.

The CSI part 1 block error probability (BLER) is defined as the probability of incorrectly decoding the CSI part 1 information when the CSI part 1 information is sent as follow:

$$BLER_{CSI\ part\ 1} = \frac{\#(\text{false CSI part 1})}{\#(\text{CSI part 1})}$$

where:

- #(false CSI part 1) denotes the number of incorrectly decoded CSI part 1 information transmitted occasions
- #(CSI part 1) denotes the number of CSI part 1information transmitted occasions.

The CSI part 2 block error probability is defined as the probability of incorrectly decoding the CSI part 2 information when the CSI part 2 information is sent as follows:

$$BLER_{CSI\ part\ 2} = \frac{\#(\text{false CSI part 2})}{\#(\text{CSI part 2})}$$

where:

- #(false CSI part 2) denotes the number of incorrectly decoded CSI part 2 information transmitted occasions
- #(CSI part 2) denotes the number of CSI part 2 information transmitted occasions.

The number of UCI information bit payload per slot is defined for two cases as follows:

- 5 bits in CSI part 1, 2 bits in CSI part 2
- 20 bits in CSI part 1, 20 bits in CSI part 2

The 7bits UCI case is further defined with the bitmap [c0 c1 c2 c3 c4] = [0 1 0 1 0] for CSI part 1 information, where c0 is mapping to the RI information, and with the bitmap [c0 c1] = [1 0] for CSI part2 information.

The 40bits UCI information case is assumed random information bits selection.

In both tests, PUSCH data, CSI part 1 and CSI part 2 information are transmitted simultaneously.

**Table 11.2.2.3.1-1: Test parameters for testing UCI multiplexed on PUSCH**

| Parameter | | Value | |
|---|---|---|---|
| Transform precoding | | Disabled | |
| Default TDD UL-DL pattern (Note 1) | | 120 kHz SCS: 3D1S1U, S=10D:2G:2U | |
| HARQ | Maximum number of HARQ transmissions | 1 | |
| | RV sequence | 0 | |
| DM-RS | DM-RS configuration type | 1 | |
| | DM-RS duration | single-symbol DM-RS | |
| | Additional DM-RS position | pos0,pos1 | |
| | Number of DM-RS CDM group(s) without data | 2 | |
| | Ratio of PUSCH EPRE to DM-RS EPRE | -3 dB | |
| | DM-RS port(s) | {0} | |
| | DM-RS sequence generation | $N_{ID}^{0}$=0,$n_{SCID}$=0 | |
| Time domain resource assignment | PUSCH mapping type | B | |
| | Start symbol | 0 | |
| | Allocation length | 10 | |
| Frequency domain resource assignment | RB assignment | Full applicable test bandwidth | |
| | Frequency hopping | Disabled | |
| Code block group based PUSCH transmission | | Disabled | |
| PT-RS configuration | PT-RS | Disabled, | Enabled |
| | Frequency density ($K_{PT-RS}$) | N/A. | 2 |
| | Time density ($L_{PT-RS}$) | N/A. | 1 |
| UCI | Number of CSI part 1 and CSI part 2 information bit payload | {5,2},{20,20} | |
| | *scaling* | 1 | |
| | *betaOffsetACK-Index1* | 11 | |
| | *betaOffsetCSI-Part1-Index1 and betaOffsetCSI-Part1-Index2* | 13 | |
| | *betaOffsetCSI-Part2-Index1 and betaOffsetCSI-Part2-Index2* | 13 | |
| | UCI partition for frequency hopping | Disabled | |
| NOTE 1: The same requirements are applicable to TDD with different UL-DL patterns. | | | |

### 11.2.2.3.2 Minimum requirements

The CSI part 1 block error probability shall not exceed 0.1% at the SNR given in table 11.2.2.3.2-1 and table 11.2.2.3.2-2. The CSI part 2 block error probability shall not exceed 1% at the SNR given in table 11.2.2.3.2-3 and table 11.2.2.3.2-4.

**Table 11.2.2.3.2-1: Minimum requirements for UCI multiplexed on PUSCH, Type B, with PT-RS, CSI part 1, 50 MHz Channel Bandwidth, 120 kHz SCS**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | UCI bits (CSI part 1, CSI part 2) | Additional DM-RS position | FRC (Annex A) | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-300 Low | 7(5,2) | pos0 | G-FR2-A4-3 | 7.2 |
| | 2 | Normal | TDLA30-300 Low | 40(20,20) | pos0 | G-FR2-A4-3 | 5.8 |
| | 2 | Normal | TDLA30-300 Low | 7(5,2) | pos1 | G-FR2-A4-13 | 7.8 |
| | 2 | Normal | TDLA30-300 Low | 40(20,20) | pos1 | G-FR2-A4-13 | 5.9 |

**Table 11.2.2.3.2-2: Minimum requirements for UCI multiplexed on PUSCH, Type B, Without PTRS, CSI part 1, 50 MHz Channel Bandwidth, 120 kHz SCS**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | UCI bits (CSI part 1, CSI part 2) | Additional DM-RS position | FRC (Annex A) | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-300 LOW | 7(5,2) | pos0 | G-FR2-A4-3 | 7.1 |
| | 2 | Normal | TDLA30-300 LOW | 40(20,20) | pos0 | G-FR2-A4-3 | 5.8 |
| | 2 | Normal | TDLA30-300 LOW | 7(5,2) | pos1 | G-FR2-A4-13 | 7.3 |
| | 2 | Normal | TDLA30-300 LOW | 40(20,20) | pos1 | G-FR2-A4-13 | 5.5 |

**Table 11.2.2.3.2-3: Minimum requirements for UCI multiplexed on PUSCH, Type B, with PTRS, CSI part 2, 50 MHz Channel Bandwidth, 120 kHz SCS**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | UCI bits (CSI part 1, CSI part 2) | Additional DM-RS position | FRC (Annex A) | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-300 Low | 7(5,2) | pos0 | G-FR2-A4-3 | 1.1 |
| | 2 | Normal | TDLA30-300 Low | 40(20,20) | pos0 | G-FR2-A4-3 | 4.0 |
| | 2 | Normal | TDLA30-300 Low | 7(5,2) | pos1 | G-FR2-A4-13 | 1.3 |
| | 2 | Normal | TDLA30-300 Low | 40(20,20) | pos1 | G-FR2-A4-13 | 4.0 |

**Table 11.2.2.3.2-3: Minimum requirements for UCI multiplexed on PUSCH, Type B, with PTRS, CSI part 2, 50 MHz Channel Bandwidth, 120 kHz SCS**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | UCI bits (CSI part 1, CSI part 2) | Additional DM-RS position | FRC (Annex A) | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-300 Low | 7(5,2) | pos0 | G-FR2-A4-3 | 1.1 |
| | 2 | Normal | TDLA30-300 Low | 40(20,20) | pos0 | G-FR2-A4-3 | 4.0 |
| | 2 | Normal | TDLA30-300 Low | 7(5,2) | pos1 | G-FR2-A4-13 | 1.3 |
| | 2 | Normal | TDLA30-300 Low | 40(20,20) | pos1 | G-FR2-A4-13 | 4.0 |

**Table 11.2.2.3.2-4: Minimum requirements for UCI multiplexed on PUSCH, Type B, Without PTRS, CSI part 2, 50 MHz Channel Bandwidth, 120 kHz SCS**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | UCI bits (CSI part 1, CSI part 2) | Additional DM-RS position | FRC (Annex A) | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-300 Low | 7(5,2) | pos0 | G-FR2-A4-3 | 1.1 |
| | 2 | Normal | TDLA30-300 Low | 40(20,20) | pos0 | G-FR2-A4-3 | 3.9 |
| | 2 | Normal | TDLA30-300 Low | 7(5,2) | pos1 | G-FR2-A4-13 | 1.2 |
| | 2 | Normal | TDLA30-300 Low | 40(20,20) | pos1 | G-FR2-A4-13 | 3.7 |

## 11.2.2.4 Requirements for PUSCH for 2-step RA type

### 11.2.2.4.1 General

The performance requirement of PUSCH for 2-step RA type is determined by a minimum required block error rate of MsgA for a given SNR for the FRCs listed in annex A. The performance requirements assume that the precedent preamble of MsgA is correctly detected in a 2-step RA type procedure and no HARQ retransmissions.

The performance requirements are applicable for wide area and medium range BS that support 2-step RA type. The performance requirements are not applied for a local area BS that supports 2-step RA type.

**Table 11.2.2.4.1-1: Test parameters for testing PUSCH for 2-step RA type**

| Parameter | | Value |
|---|---|---|
| Transform precoding | | Disabled |
| Channel bandwidth | | 60 kHz SCS: 50 MHz<br>120 kHz SCS: 100 MHz |
| MCS | | 2 |
| DM-RS | DM-RS configuration type | 1 |
| | DM-RS duration | single-symbol DM-RS |
| | Additional DM-RS symbols | pos1 |
| | Number of DM-RS CDM group(s) without data | 2 |
| | Ratio of PUSCH EPRE to DM-RS EPRE | -3 dB |
| | DM-RS port(s) | {0} |
| | DM-RS sequence generation | $N_{ID}$=0, $n_{SCID}$ =0 |
| Time domain resource | PUSCH mapping type | B |
| | Start symbol | 0 |
| | Allocation length | 10 |

| | | |
|---|---|---|
| Frequency domain resource | RB assignment | 2 PRB |
| | Starting PRB index | 0 |
| | Frequency hopping | Disabled |
| Time Offset (TO) | | 60 kHz SCS: 0 to 0.6<br>120kHz SCS: 0 to 0.5 |
| Note1: The power ratio between preamble and msgA is assumed to be sufficient to achieve 100% preamble detection. The SNR for the requirement is defined on the msgA PUSCH. | | |

#### 11.2.2.4.2 Minimum requirements

The block error rate of MsgA shall be equal to or smaller than 1% at the given SNR.

**Table 11.2.2.4.2-1 Minimum requirements of PUSCH for 2-step RA type with mappying type B**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Channel Bandwidth [MHz] | SCS [kHz] | Propagation conditions and correlation matrix (Annex G) | FRC (Annex A) | SNR [dB] |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | 50 | 60 | TDLA30-300 low | G-FR2-A3-25 | 8. 7 |
| | | | 100 | 120 | TDLA30-300 low | G-FR2-A3-26 | 8. 3 |

### 11.2.2.5 Requirements for PUSCH repetition Type A

#### 11.2.2.5.1 General

The performance requirement of PUSCH is determined by a maximum block error rate (BLER) for a given SNR. The BLER is defined as the probability of incorrectly decoding the PUSCH information when the PUSCH information is sent. The performance requirements assume HARQ retransmissions.

**Table 11.2.2.5.1-1: Test parameters for testing PUSCH repetition Type A**

| Parameter | | Value |
|---|---|---|
| Transform precoding | | Disabled |
| Default TDD UL-DL pattern (Note 1) | | 60 kHz and 120kHz SCS: 3D1S1U, S=10D:2G:2U |
| HARQ | Maximum number of HARQ transmissions | 4 |
| | RV sequence | 0, 3, 0,3 |
| DM-RS | DM-RS configuration type | 1 |
| | DM-RS duration | single-symbol DM-RS |
| | Additional DM-RS symbols | Pos1 |
| | Number of DM-RS CDM group(s) without data | 2 |
| | Ratio of PUSCH EPRE to DM-RS EPRE | -3 dB |
| | DM-RS port(s) | 0 |
| | DM-RS sequence generation | $N_{ID}$=0, $n_{SCID}$ =0 |
| Time domain resource | PUSCH mapping type | B |
| | Start symbol index | 0 |
| | Allocation length | 10 |
| | PUSCH aggregation factor (Note 2) | n8 |
| Frequency domain resource | RB assignment | Full applicable test bandwidth |
| | Frequency hopping | Disabled |
| Code block group based PUSCH transmission | | Disabled |
| PT-RS configuration | Frequency density ($K_{PT-RS}$) | Disabled |
| | Time density ($L_{PT-RS}$) | Disabled |
| NOTE 1: The effective RV sequence is {0,2,3,1} with slot aggregation<br>NOTE 2: The intention of this configuration is to have two effective transmissions of the transport block. To achieve this for the standard TDD pattern captured in this table, a value of n8 is necessary | | |

#### 11.2.2.5.2 Minimum requirements

The BLER shall be equal to or smaller than the required target BLER for the FRCs stated in tables 11.2.2.5.2-1 to 11.2.2.5.2-4 at the given SNR for 1Tx. FRCs are defined in annex A. Unless stated otherwise, the MIMO correlation matrices for the gNB are defined in annex G for low correlation.

**Table 11.2.2.5.2-1: Minimum requirements for PUSCH, TypeB, 50 MHz channel bandwidth, 60 kHz SCS**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Target BLER | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-300 | 1% (Note1) | G-FR2-A3A-5 | Pos1 | -11.9 |
| NOTE 1: BLER is defined as residual BLER, i.e. ratio of incorrectly received transport blocks/sent transport blocks, independently of the number of HARQ transmission(s) for each transport block | | | | | | | |

**Table 11.2.2.5.2-2: Minimum requirements for PUSCH, TypeB, 100 MHz channel bandwidth, 60 kHz SCS**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Target BLER | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-300 | 1% (Note1) | G-FR2-A3A-6 | Pos1 | -12.2 |
| NOTE 1: BLER is defined as residual BLER, i.e. ratio of incorrectly received transport blocks/sent transport blocks, independently of the number of HARQ transmission(s) for each transport block | | | | | | | |

**Table 11.2.2.5.2-3: Minimum requirements for PUSCH, TypeB, 50 MHz channel bandwidth, 120 kHz SCS**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Target BLER | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-300 | 1% (Note1) | G-FR2-A3A-7 | Pos1 | -11.2 |
| NOTE 1: BLER is defined as residual BLER, i.e. ratio of incorrectly received transport blocks/sent transport blocks, independently of the number of HARQ transmission(s) for each transport block | | | | | | | |

**Table 11.2.2.5.2-4: Minimum requirements for PUSCH, TypeB, 100 MHz channel bandwidth, 120 kHz SCS**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Target BLER | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-300 | 1% (Note1) | G-FR2-A3A-8 | Pos1 | -11.7 |
| NOTE 1: BLER is defined as residual BLER, i.e. ratio of incorrectly received transport blocks/sent transport blocks, independently of the number of HARQ transmission(s) for each transport block | | | | | | | |

## 11.2.2.6 Requirements for PUSCH mapping Type B with non-slot transmission

### 11.2.2.6.1 General

The performance requirement of PUSCH is determined by a minimum required throughtput for a given SNR. The required throughput is expressed as a fraction of maximum throughput for the FRCs listed in annex A. The performance requirements do not assume HARQ retransmissions.

**Table 11.2.2.6.1-1: Test parameters for testing PUSCH mapping Type B with non-slot transmission**

| Parameter | | Value |
|---|---|---|
| Transform precoding | | Disabled |
| Default TDD UL-DL pattern (Note 1) | | 60 kHz and 120kHz SCS: 3D1S1U, S=10D:2G:2U |
| HARQ | Maximum number of HARQ transmissions | 1 |
| | RV sequence | 0 |
| DM-RS | DM-RS configuration type | 1 |
| | DM-RS duration | single-symbol DM-RS |
| | Additional DM-RS symbols | Pos0 |
| | Number of DM-RS CDM group(s) without data | 2 |
| | Ratio of PUSCH EPRE to DM-RS EPRE | -3 dB |
| | DM-RS port(s) | 0 |
| | DM-RS sequence generation | $N_{ID}$=0, $n_{SCID}$ =0 |
| Time domain resource | PUSCH mapping type | B |
| | Start symbol index | 0 |
| | Allocation length | 4 |
| Frequency domain resource | RB assignment | Full applicable test bandwidth |
| | Frequency hopping | Disabled |
| Code block group based PUSCH transmission | | Disabled |
| PT-RS configuration | Frequency density ($K_{PT-RS}$) | Disabled |
| | Time density ($L_{PT-RS}$) | Disabled |
| NOTE 1: The same requirements are applicable to TDD with different UL-DL patterns | | |

### 11.2.2.6.2 Minimum requirements

The throughput shall be equal to or larger than the fraction of maximum throughput for the FRCs stated in tables 11.2.2.6.2-1 to 11.2.2.6.2-4 at the given SNR for 1Tx. FRCs are defined in an annex A. Unless stated otherwise, the MIMO correlation matrices for the gNB are defined in annex G for low correlation.

**Table 11.2.2.6.2-1: Minimum requirements for PUSCH, Type B, 50 MHz channel bandwidth, 60 kHz SCS**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-300 | 70% | G-FR2-A3A-1 | Pos0 | -4.4 |

**Table 11.2.2.6.2-2: Minimum requirements for PUSCH, Type B, 100 MHz channel bandwidth, 60 kHz SCS**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-300 | 70% | G-FR2-A3A-2 | Pos0 | -5.0 |

**Table 11.2.2.6.2-3: Minimum requirements for PUSCH, Type B, 50 MHz channel bandwidth, 120 kHz SCS**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-300 | 70% | G-FR2-A3A-3 | Pos0 | -4.1 |

**Table 11.2.2.6.2-4: Minimum requirements for PUSCH, Type B, 100 MHz channel bandwidth, 120 kHz SCS**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-300 | 70% | G-FR2-A3A-4 | Pos0 | -5.1 |

## 11.2.2.7 Requirements for PUSCH for high speed train

### 11.2.2.7.1 General

The performance requirement of PUSCH is determined by a minimum required throughput for a given SNR. The required throughput is expressed as a fraction of maximum throughput for the FRCs listed in annex A. The performance requirements assume HARQ retransmissions. The performance requirements for high speed train are optional and only applicable for FR2-1 below 30GHz.

The performance requirements for PUSCH high speed train apply to Wide Area Base Stations and Medium Range Base Stations (subject to declaration).

**Table 11.2.2.7.1-1: Test parameters for testing high speed train**

| Parameter | | Value |
|---|---|---|
| Transform precoding | | Disabled |
| Default TDD UL-DL pattern (Note 1) | | 3D1S1U, S=10D:2G:2U |
| HARQ | Maximum number of HARQ transmissions | 4 |
| | RV sequence | 0, 2, 3, 1 |
| DM-RS | DM-RS configuration type | 1 |
| | DM-RS duration | single-symbol DM-RS |
| | Additional DM-RS symbols | Pos0 or Pos1 or Pos2 |
| | Number of DM-RS CDM group(s) without data | 2 |
| | Ratio of PUSCH EPRE to DM-RS EPRE | -3 dB |
| | DM-RS port(s) | 0 |
| | DM-RS sequence generation | $N_{ID}$=0, $n_{SCID}$ =0 |
| Time domain resource | PUSCH mapping type | B |
| | Start symbol index | 0 |
| | Allocation length | 10 |
| Frequency domain resource | RB assignment | Full applicable test bandwidth |
| | Frequency hopping | Disabled |
| Code block group based PUSCH transmission | | Disabled |
| PT-RS configuration | Frequency density ($K_{PT-RS}$) | 2 |
| | Time density ($L_{PT-RS}$) | 1 |
| NOTE 1: The same requirements are applicable to TDD with different UL-DL patterns | | |

### 11.2.2.7.2 Minimum requirements

The throughput shall be equal to or larger than the fraction of maximum throughput for the FRCs stated in tables 11.2.2.7.2-1 to 11.2.2.7.2-4 at the given SNR for 1Tx. FRCs are defined in an annex A. Unless stated otherwise, the MIMO correlation matrices for the gNB are defined in annex G for low correlation.

**Table 11.2.2.7.2-1: Minimum requirements for PUSCH, Type B, 50 MHz channel bandwidth, 120 kHz SCS**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | Scenario 4-BI-NR350, FR2 | 70% | G-FR2-A10-1 | pos0 | 12.9 |
| 1 | 2 | Normal | Scenario 4-BI-NR350, FR2 | 70% | G-FR2-A10-3, G-FR2-A10-5 | pos1, pos2 | 12.5 |

**Table 11.2.2.7.2-2: Minimum requirements for PUSCH, Type B, 200 MHz channel bandwidth, 120 kHz SCS**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | Scenario 4-BI-NR350, FR2 | 70% | G-FR2-A10-2 | pos0 | 12.8 |
| 1 | 2 | Normal | Scenario 4-BI-NR350, FR2 | 70% | G-FR2-A10-4, G-FR2-A10-6 | pos1, pos2 | 12.3 |

## 11.2.2.8 Requirements for UL timing adjustment

### 11.2.2.8.1 General

The performance requirement of UL timing adjustment is determined by a minimum required throughput for the moving UE at given SNR. The performance requirements assume HARQ retransmissions. The performance requirements for UL timing adjustment scenario Y defined in Annex G.4 are optional and only applicable for FR2-1 below 30GHz.

In the tests for UL timing adjustment, two signals are configured, one being transmitted by a moving UE and the other being transmitted by a stationary UE. The transmission of SRS from UE is optional. FRC parameters in Tables A.10-4, A.10-5, and A.10-6 are applied for both UEs. The received power for both UEs is the same. The resource blocks allocated for both UEs are consecutive. In scenario Y, Doppler shift is not taken into account.

**Table 11.2.2.8.1-1 Test parameters for testing UL timing adjustment**

| Parameter | | Value |
|---|---|---|
| Transform precoding | | Disabled |
| Uplink-downlink allocation for TDD | | 120 kHz SCS:<br>3D1S1U, S=10D:2G:2U |
| Channel bandwidth | | 120 kHz SCS: 50MHz, 200 MHz |
| HARQ | Maximum number of HARQ transmissions | 4 |
| | RV sequence | 0, 2, 3, 1 |
| DM-RS | DM-RS configuration type | 1 |
| | DM-RS duration | single-symbol DM-RS |
| | DM-RS position ($l_0$) | 2 |
| | Additional DM-RS position | pos0, pos1, pos2 |
| | Number of DM-RS CDM group(s) without data | 2 |
| | Ratio of PUSCH EPRE to DM-RS EPRE | -3 dB |
| | DM-RS port | {0} |
| Time domain resource assignment | DM-RS sequence generation | $N_{ID}^0$=0, $n_{SCID}$ =0 for moving UE<br>$N_{ID}^0$=1, $n_{SCID}$ =1 for stationary UE |
| | PUSCH mapping type | B |
| | Allocation length | 10 |
| Frequency domain resource assignment | RB assignment | 50 MHz CBW/120kHz SCS: 16 RB for each UE<br>200MHz CBW/120kHz SCS: 66 RB for each UE |
| | Starting PRB index | Moving UE: 0<br>Stationary UE: 16 for 50MHz CBW, 66 for 200MHz CBW for SCS 120kHz |
| | Frequency hopping | Disabled |
| SRS resource allocation | Slots in which sounding RS is transmitted (Note 1) | For TDD:<br>- last symbol in slot #3 in radio frames for 120KHz |
| | SRS resource allocation | 120 kHz SCS:<br>- $C_{SRS}$ = 9, $B_{SRS}$ =0, for 32 RB<br>- $C_{SRS}$ = 33, $B_{SRS}$ =0, for 132 RB |
| NOTE 1. The transmission of SRS is optional. And the transmission comb and SRS periodic are configured as $K_{TC}$ = 2, and $T_{SRS}$ = 80 respectively. | | |

### 11.2.2.8.2 Minimum requirements for high speed train

The throughput shall be ≥ 70% of the maximum throughput of the reference measurement channel as specified in Annex A for the moving UE at the SNR given in table 11.2.2.8.2-1 for mapping type B.

nex A for the moving UE at the SNR given in table 11.2.2.8.2-1 for mapping type B.

**Table 11.2.2.8.2-1 Minimum requirements for UL timing adjustment with mapping type B for high speed train**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Channel Bandwidth [MHz] | SCS [kHz] | Moving propagation conditions and correlation matrix (Annex G) | FRC (Annex A) | SNR [dB] |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | 50 | 120 | Scenario Y | G-FR2-A10-7 | 9.1 |
| | | | | | Scenario Y | G-FR2-A10-9, G-FR2-A10-11 | 8.8 |
| | | | 200 | 120 | Scenario Y | G-FR2-A10-8 | 9.0 |
| | | | | | Scenario Y | G-FR2-A10-10, G-FR2-A10-12 | 8.9 |

## 11.2.2.9 Requirements for TB processing over multi-slot PUSCH (TBoMS)

### 11.2.2.9.1 General

The performance requirement of PUSCH TBoMS is determined by a minimum required throughput for a given SNR. The required throughput is expressed as a fraction of maximum throughput for the FRCs listed in annex A. The performance requirements assume HARQ retransmissions.

**Table 11.2.2.9.1-1: Test parameters for testing PUSCH TBoMS**

| Parameter | | Value |
|---|---|---|
| Transform precoding | | Disabled |
| Default TDD UL-DL pattern (Note 1) | | 60 kHz and 120kHz SCS: 3D1S1U, S=10D:2G:2U |
| HARQ | Maximum number of HARQ transmissions | 4 |
| | RV sequence | 0, 2, 3, 1 |
| DM-RS | DM-RS configuration type | 1 |
| | DM-RS duration | single-symbol DM-RS |
| | Additional DM-RS position | pos1 |
| | Number of DM-RS CDM group(s) without data | 2 |
| | Ratio of PUSCH EPRE to DM-RS EPRE | -3 dB |
| | DM-RS port | 0 |
| | DM-RS sequence generation | $N_{ID}^0$=0, $n_{SCID}$ =0 |
| Time domain resource assignment | PUSCH mapping type | B |
| | Start symbol | 0 |
| | Allocation length | 10 |
| | Number of slots allocated for TBoMS PUSCH | 2 |
| | Number of repetitions of a single TBoMS | 1 |
| Frequency domain resource assignment | RB assignment | 5 RBs in the middle of the test bandwidth |
| | Frequency hopping | Disabled |
| Code block group based PUSCH transmission | | Disabled |
| PT-RS configuration | Frequency density ($K_{PT-RS}$) | Disabled |
| | Time density ($L_{PT-RS}$) | Disabled |
| Note 1: The same requirements are applicable to TDD with different UL-DL pattern. | | |

### 11.2.2.9.2 Minimum requirements

The throughput shall be equal to or larger than the fraction of maximum throughput for the FRCs stated in tables 11.2.2.9.2-1 to 11.2.2.9.2-2 at the given SNR. FRCs are defined in annex A.

**Table 11.2.2.9.2-1: Minimum requirements for PUSCH TBoMS, Type B, 50 MHz channel bandwidth, 60 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Duplex | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDD | TDLA30-300 Low | 70% | G-FR2-A3-27 | pos1 | -2.6 |

**Table 11.2.2.9.2-2: Minimum requirements for PUSCH TBoMS, Type B, 50 MHz channel bandwidth, 120 kHz SCS**

| Number of TX antennas | Number of RX antennas | Cyclic prefix | Duplex | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDD | TDLA30-300 Low | 70% | G-FR2-A3-28 | pos1 | -2.4 |

## 11.2.2.10 Requirements for PUSCH with DM-RS bundling

### 11.2.2.10.1 General

The performance requirement of PUSCH is determined by a minimum required throughput for a given SNR. The required throughput is expressed as a fraction of maximum throughput for the FRCs listed in annex A. The performance requirements assume HARQ retransmissions.

**Table 11.2.2.10.1-1: Test parameters for testing PUSCH with DM-RS bundling**

| Parameter | | Value |
|---|---|---|
| Transform precoding | | Disabled |
| Example TDD UL-DL pattern (Note 1) | | 60 kHz and 120kHz SCS: DDSUU or DSUUU, S=10D:2G:2U |
| HARQ | Maximum number of HARQ transmissions | 4 |
| | RV sequence | 0, 3, 0,3 (Note 2) |
| DM-RS | DM-RS configuration type | 1 |
| | DM-RS duration | single-symbol DM-RS |
| | Additional DM-RS symbols | pos1 |
| | Number of DM-RS CDM group(s) without data | 2 |
| | Ratio of PUSCH EPRE to DM-RS EPRE | -3 dB |
| | DM-RS port(s) | 0 |
| | DM-RS sequence generation | $N_{ID}$=0, $n_{SCID}$ =0 |
| Time domain resource | PUSCH mapping type | B |
| | Start symbol index | 0 |
| | Allocation length | 14 |
| | PUSCH aggregation factor | n2 |
| pusch-TimeDomainWindowLength | | 2 slots |
| Frequency domain resource | RB assignment | Full applicable test bandwidth |
| | Frequency hopping | Disabled |
| Code block group based PUSCH transmission | | Disabled |
| PT-RS configuration | Frequency density ($K_{PT-RS}$) | Disabled |
| | Time density ($L_{PT-RS}$) | Disabled |
| NOTE 1: The same TDD requirements are applicable to different UL-DL patterns with more than one consecutive UL slots when both pusch-TimeDomainWindowLength and PUSCH aggregation factor are configured as 2 slots. The UL (re)transmission of PUSCH is only scheduled for the actual TDW including 2 consecutive UL slots.<br>NOTE 2: The effective RV sequence is {0,2,3,1} with slot aggregation. | | |

### 11.2.2.10.2 Minimum requirements

The throughput shall be equal to or larger than the fraction of maximum throughput stated in the tables 11.2.2.10.2-1 and 11.2.2.10.2-2 at the given SNR for 1Tx. FRCs are defined in annex A. Unless stated otherwise, the MIMO correlation matrices for the gNB are defined in annex G for low correlation.

**Table 11.2.2.10.2-1: Minimum requirements for PUSCH, TypeB, 50 MHz channel bandwidth, 60 kHz SCS**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-75 | 70% | G-FR2-A3B-1 | pos1 | -3.9 |

**Table 11.2.2.10.2-2: Minimum requirements for PUSCH, TypeB, 50 MHz channel bandwidth, 120 kHz SCS**

| Number of TX antennas | Number of demodulation branches | Cyclic prefix | Propagation conditions and correlation matrix (Annex G) | Fraction of maximum throughput | FRC (Annex A) | Additional DM-RS position | SNR (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-75 | 70% | G-FR2-A3B-2 | pos1 | -3.5 |

# 11.3 Performance requirements for PUCCH

## 11.3.1 Requirements for *BS type 1-O*

### 11.3.1.1 DTX to ACK probability

Apply the requirements defined in clause 8.3.1

### 11.3.1.2 Performance requirements for PUCCH format 0

Apply the requirements defined in clause 8.3.2 for 2 Rx.

### 11.3.1.3 Performance requirements for PUCCH format 1

Apply the requirements defined in sub-clause 8.3.3 for 2Rx.

### 11.3.1.4 Performance requirements for PUCCH format 2

Apply the requirements defined in clause 8.3.4 for 2Rx.

### 11.3.1.5 Performance requirements for PUCCH format 3

Apply the requirements defined in clause 8.3.5 for 2Rx.

### 11.3.1.6 Performance requirements for PUCCH format 4

Apply the requirements defined in clause 8.3.6 for 2Rx.

### 11.3.1.7 Performance requirements for multi-slot PUCCH

Apply the requirements defined in clause 8.3.7 for 2Rx.

### 11.3.1.8 Performance requirements for PUCCH format 1 with DMRS bundling

Apply the requirements defined in clause 8.3.12 for 2Rx.

#### 11.3.1.9 Performance requirements for PUCCH format 3 with DMRS bundling

Apply the requirements defined in clause 8.3.13 for 2Rx.

#### 11.3.1.10 Performance requirements for sub-slot repetition PUCCH format 0

Apply the requirements defined in clause 8.3.14 for 2Rx.

### 11.3.2 Requirements for *BS type 2-O*

#### 11.3.2.1 DTX to ACK probability

Apply the requirements defined in clause 8.3.1.

#### 11.3.2.2 Performance requirements for PUCCH format 0

##### 11.3.2.2.1 General

The ACK missed detection probability is the probability of not detecting an ACK when an ACK was sent.

**Table 11.3.2.2.1-1: Test Parameters**

| Parameter | Test |
|---|---|
| Number of UCI information bits | 1 |
| Number of PRBs | FR2-1: 1.<br>FR2-2:1,16 |
| First PRB prior to frequency hopping | 0 |
| Intra-slot frequency hopping | N/A for 1 symbol<br>Enabled for 2 symbols |
| First PRB after frequency hopping | The largest PRB index – (Number of PRBs - 1) |
| Group and sequence hopping | neither |
| Hopping ID | 0 |
| Initial cyclic shift | 0 |
| First symbol | 13 for 1 symbol<br>12 for 2 symbols |

The transient period as specified in TS 38.101-1 [17] clause 6.3.3.1 and TS 38.101-2 [18] clause 6.3.3.1 is not taken into account for performance requirement testing, where the RB hopping is symmetric to the CC centre, i.e. intra-slot frequency hopping is enabled.

##### 11.3.2.2.2 Minimum requirements

The ACK missed detection probability shall not exceed 1% at the SNR given in table 11.3.2.2.2-1 to 11.3.2.2.2-4.

**Table 11.3.2.2.2-1: Minimum requirements for PUCCH format 0 and 60 kHz SCS in FR2-1**

| Number of TX antennas | Number of demodulation branches | Propagation conditions and correlation matrix (Annex G) | Number of OFDM symbols | Channel bandwidth / SNR (dB) | |
|---|---|---|---|---|---|
| | | | | 50 MHz | 100 MHz |
| 1 | 2 | TDLA30-300 Low | 1 | 9.3 | 9.0 |
| | | | 2 | 4.2 | 4.0 |

**Table 11.3.2.2.2-2: Minimum requirements for PUCCH format 0 and 120 kHz SCS in FR2-1**

| Number of TX antennas | Number of demodulation branches | Propagation conditions and correlation matrix (Annex G) | Number of OFDM symbols | Channel bandwidth / SNR (dB) | | |
|---|---|---|---|---|---|---|
| | | | | 50 MHz | 100 MHz | 200 MHz |
| 1 | 2 | TDLA30-300 Low | 1 | 9.5 | 9.2 | 9.7 |
| | | | 2 | 4.1 | 3.8 | 4.0 |

**Table 11.3.2.2.2-3: Minimum requirements for PUCCH format 0 and 120 kHz SCS in FR2-2**

| Number of TX antennas | Number of demodulation branches | Propagation conditions and correlation matrix (Annex G) | Number of OFDM symbols | Number of PRBs | Channel bandwidth / SNR (dB) |
|---|---|---|---|---|---|
| | | | | | 100 MHz |
| 1 | 2 | TDLA30-650 Low | 1 | 1 | 9.6 |
| | | | 2 | 16 | -5.8 |

**Table 11.3.2.2.2-4: Minimum requirements for PUCCH format 0 and 480 kHz SCS in FR2-2**

| Number of TX antennas | Number of demodulation branches | Propagation conditions and correlation matrix (Annex G) | Number of OFDM symbols | Number of PRBs | Channel bandwidth / SNR (dB) |
|---|---|---|---|---|---|
| | | | | | 400 MHz |
| 1 | 2 | TDLA10-650 Low | 1 | 1 | 9.6 |
| | | | 2 | 16 | -4.8 |

## 11.3.2.3 Performance requirements for PUCCH format 1

### 11.3.2.3.1 NACK to ACK requirements

#### 11.3.2.3.1.1 General

The NACK to ACK detection probability is the probability that an ACK bit is falsely detected when an NACK bit was sent on the particular bit position, where the NACK to ACK detection probability is defined as follows:

$$\text{Prob}(\text{PUCCH NACK} \rightarrow \text{ACK bits}) = \frac{\#(\text{NACK bits decoded as ACK bits})}{\#(\text{Total NACK bits})}$$

where:

- $\#(\text{Total NACK bits})$ denotes the total number of NACK bits transmitted
- $\#(\text{NACK bits decoded as ACK bits})$ denotes the number of NACK bits decoded as ACK bits at the receiver, i.e. the number of received ACK bits
- NACK bits in the definition do not contain the NACK bits which are mapped from DTX, i.e. NACK bits received when DTX is sent should not be considered.

Random codeword selection is assumed.

**Table 11.3.2.3.1.1-1: Test Parameters**

| Parameter | Test |
|---|---|
| Number of information bits | 2 |
| Number of PRBs | FR2-1:1<br>FR2-2:1,16 |
| Number of symbols | 14 |
| First PRB prior to frequency hopping | 0 |
| Intra-slot frequency hopping | enabled |
| First PRB after frequency hopping | The largest PRB index – (nrofPRBs – 1) |
| Group and sequence hopping | neither |
| Hopping ID | 0 |
| Initial cyclic shift | 0 |
| First symbol | 0 |
| Index of orthogonal cover code (*timeDomainOCC*) | 0 |

The transient period as specified in TS 38.101-1 [17] and TS 38.101-2 [18] clause 6.3.3.1 is not taken into account for performance requirement testing, where the RB hopping is symmetric to the CC centre, i.e. intra-slot frequency hopping is enabled.

### 11.3.2.3.1.2 Minimum requirements

The NACK to ACK probability shall not exceed 0.1% at the SNR given in Table 11.3.2.3.1.2-1 and Table 11.3.2.3.1.2-2.

**Table 11.3.2.3.1.2-1: Minimum requirements for PUCCH format 1 with 60 kHz SCS in FR2-1**

| Number of TX antennas | Number of Demodulation Branches | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Channel bandwidth / SNR (dB) | |
|---|---|---|---|---|---|
| | | | | 50 MHz | 100 MHz |
| 1 | 2 | Normal | TDLA30-300 Low | -1.2 | -4.2 |

**Table 11.3.2.3.1.2-2: Minimum requirements for PUCCH format 1 with 120 kHz SCS in FR2-1**

| Number of TX antennas | Number of Demodulation Branches | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Channel bandwidth / SNR (dB) | | |
|---|---|---|---|---|---|---|
| | | | | 50 MHz | 100 MHz | 200 MHz |
| 1 | 2 | Normal | TDLA30-300 Low | -3.9 | -3.9 | -3.0 |

**Table 11.3.2.3.1.2-3: Minimum requirements for PUCCH format 1 and 120 kHz SCS in FR2-2**

| Number of TX antennas | Number of demodulation branches | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Number of RBs | Channel bandwidth / SNR (dB) |
|---|---|---|---|---|---|
| | | | | | 100 MHz |
| 1 | 2 | Normal | TDLA30-650 Low | 1 | -2.7 |
| | | | | 16 | -13.1 |

**Table 11.3.2.3.1.2-4: Minimum requirements for PUCCH format 1 and 480 kHz SCS in FR2-2**

| Number of TX antennas | Number of demodulation branches | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Number of RBs | Channel bandwidth / SNR (dB) |
|---|---|---|---|---|---|
| | | | | | 400 MHz |
| 1 | 2 | Normal | TDLA10-650 Low | 1 | -2.9 |
| | | | | 16 | -13.8 |

## 11.3.2.3.2 ACK missed detection requirements

### 11.3.2.3.2.1 General

The ACK missed detection probability is the probability of not detecting an ACK when an ACK was sent. The test parameters in Table 11.3.2.3.1.1-1 are configured.

The transient period as specified in TS 38.101-1 [17] and TS 38.101-2 [18] clause 6.3.3.1 is not taken into account for performance requirement testing, where the RB hopping is symmetric to the CC centre, i.e. intra-slot frequency hopping is enabled.

### 11.3.2.3.2.2 Minimum requirements

The ACK missed detection probability shall not exceed 1% at the SNR given in Table 11.3.2.3.2.2-1 to Table 11.3.2.3.2.2-4.

**Table 11.3.2.3.2.2-1: Minimum requirements for PUCCH format 1 with 60 kHz SCS in FR2-1**

| Number of TX antennas | Number of Demodulation Branches | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Channel bandwidth / SNR (dB) | |
|---|---|---|---|---|---|
| | | | | 50 MHz | 100 MHz |
| 1 | 2 | Normal | TDLA30-300 Low | -3.9 | -4.2 |

**Table 11.3.2.3.2.2-2: Minimum requirements for PUCCH format 1 with 120 kHz SCS in FR2-1**

| Number of TX antennas | Number of Demodulation Branches | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Channel bandwidth / SNR (dB) | | |
|---|---|---|---|---|---|---|
| | | | | 50 MHz | 100 MHz | 200 MHz |
| 1 | 2 | Normal | TDLA30-300 Low | -4.7 | -4.6 | -4.6 |

**Table 11.3.2.3.2.2-3: Minimum requirements for PUCCH format 1 and 120 kHz SCS in FR2-2**

| Number of TX antennas | Number of demodulation branches | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Number of RBs | Channel bandwidth / SNR (dB) |
|---|---|---|---|---|---|
| | | | | | 100 MHz |
| 1 | 2 | Normal | TDLA30-650 Low | 1 | -4.1 |
| | | | | 16 | -14.8 |

**Table 11.3.2.3.2.2-4: Minimum requirements for PUCCH format 1 and 480 kHz SCS in FR2-2**

| Number of TX antennas | Number of demodulation branches | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Number of RBs | Channel bandwidth / SNR (dB) |
|---|---|---|---|---|---|
| | | | | | 400 MHz |
| 1 | 2 | Normal | TDLA10-650 Low | 1 | -4.2 |
| | | | | 16 | -14.4 |

## 11.3.2.4 Performance requirements for PUCCH format 2

### 11.3.2.4.1 ACK missed detection requirements

#### 11.3.2.4.1.1 General

The ACK missed detection probability is the probability of not detecting an ACK when an ACK was sent.

The ACK missed detection requirement only applies to the PUCCH format 2 with 4 UCI bits.

**Table 11.3.2.4.1.1-1: Test Parameters**

| Parameter | Value |
|---|---|
| Modulation order | QSPK |
| Starting RB location | 0 |
| Intra-slot frequency hopping | N/A |
| Number of PRBs | 4 |
| Number of symbols | 1 |
| The number of UCI information bits | 4 |
| First symbol | 13 |
| DM-RS sequence generation | $N_{ID}^{0}$=0 |

#### 11.3.2.4.1.2 Minimum requirements

The ACK missed detection probability shall not exceed 1% at the SNR given in table 11.3.2.4.1.2-1 to 11.3.2.4.1.2-4 for 4UCI bits.

**Table 11.3.2.4.1.2-1: Minimum requirements for PUCCH format 2 with 60 kHz SCS in FR2-1**

| Number of TX antennas | Number of Demodulation Branches | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Channel bandwidth / SNR (dB) | |
|---|---|---|---|---|---|
| | | | | 50 MHz | 100 MHz |
| 1 | 2 | Normal | TDLA30-300 Low | 6.7 | 7.2 |

**Table 11.3.2.4.1.2-2: Minimum requirements for PUCCH format 2 with 120 kHz SCS in FR2-1**

| Number of TX antennas | Number of Demodulation Branches | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Channel bandwidth / SNR (dB) | | |
|---|---|---|---|---|---|---|
| | | | | 50 MHz | 100 MHz | 200 MHz |
| 1 | 2 | Normal | TDLA30-300 Low | 6.6 | 6.3 | 6.6 |

**Table 11.3.2.4.1.2-3: Minimum requirements for PUCCH format 2 and 120 kHz SCS in FR2-2**

| Number of TX antennas | Number of demodulation branches | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Channel bandwidth / SNR (dB) |
|---|---|---|---|---|
| | | | | 100 MHz |
| 1 | 2 | Normal | TDLA30-650 Low | 5.4 |

**Table 11.3.2.4.1.2-4: Minimum requirements for PUCCH format 2 and 480 kHz SCS in FR2-2**

| Number of TX antennas | Number of demodulation branches | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Channel bandwidth / SNR (dB) |
|---|---|---|---|---|
| | | | | 400 MHz |
| 1 | 2 | Normal | TDLA10-650 Low | 5.4 |

### 11.3.2.4.2 UCI BLER performance requirements

#### 11.3.2.4.2.1 General

The UCI block error probability (BLER) is defined as the probability of incorrectly decoding the UCI information when the UCI information is sent. The UCI information does not contain CSI part 2.

The transient period as specified in TS 38.101-1 [17] and TS 38.101-2 [18] clause 6.3.3.1 is not taken into account for performance requirement testing, where the RB hopping is symmetric to the CC centre, i.e. intra-slot frequency hopping is enabled.

The UCI performance only applies to the PUCCH format 2 with 22 UCI bits.

**Table 11.3.2.4.2.1-1: Test Parameters**

| Parameter | Value |
|---|---|
| Modulation order | QSPK |
| First PRB prior to frequency hopping | 0 |
| Intra-slot frequency hopping | enabled |
| First PRB after frequency hopping | The largest PRB index – (Number of PRBs – 1) |
| Number of PRBs | 9 |
| Number of symbols | 2 |
| The number of UCI information bits | 22 |
| First symbol | 12 |
| DM-RS sequence generation | $N_{ID}^{0}$=0 |

#### 11.3.2.4.2.2 Minimum requirements

The UCI block error probability shall not exceed 1% at the SNR given in table 11.3.2.4.2.2-1 to 11.3.2.4.2.2-4 for 22 UCI bits.

**Table 11.3.2.4.2.2-1: Minimum requirements for PUCCH format 2 with 60 kHz SCS in FR2-1**

| Number of TX antennas | Number of Demodulation Branches | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Channel bandwidth / SNR (dB) | |
|---|---|---|---|---|---|
| | | | | 50 MHz | 100 MHz |
| 1 | 2 | Normal | TDLA30-300 Low | 2.6 | 1.1 |

**Table 11.3.2.4.2.2-2: Minimum requirements for PUCCH format 2 with 120 kHz SCS in FR2-1**

| Number of TX antennas | Number of Demodulation Branches | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Channel bandwidth / SNR (dB) | | |
|---|---|---|---|---|---|---|
| | | | | 50 MHz | 100 MHz | 200 MHz |
| 1 | 2 | Normal | TDLA30-300 Low | 1.2 | 1.2 | 1.1 |

**Table 11.3.2.4.2.2-3: Minimum requirements for PUCCH format 2 and 120 kHz SCS in FR2-2**

| Number of TX antennas | Number of demodulation branches | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Channel bandwidth / SNR (dB) |
|---|---|---|---|---|
| | | | | 100 MHz |
| 1 | 2 | Normal | TDLA30-650 Low | 2.1 |

**Table 11.3.2.4.2.2-4: Minimum requirements for PUCCH format 2 and 480 kHz SCS in FR2-2**

| Number of TX antennas | Number of demodulation branches | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Channel bandwidth / SNR (dB) |
|---|---|---|---|---|
| | | | | 400 MHz |
| 1 | 2 | Normal | TDLA10-650 Low | 2.6 |

## 11.3.2.5 Performance requirements for PUCCH format 3

### 11.3.2.5.1 General

The performance is measured by the required SNR at UCI block error probability not exceeding 1%.

The UCI block error probability is defined as the conditional probability of incorrectly decoding the UCI information when the UCI information is sent. The UCI information does not contain CSI part 2.

The transient period as specified in TS 38.101-2 [18] clause 6.3.3.1 is not taken into account for performance requirement testing, where the RB hopping is symmetric to the CC centre, i.e. intra-slot frequency hopping is enabled.

**Table 11.3.2.5.1-1: Test parameters**

| Parameter | Test 1 | Test 2 |
|---|---|---|
| Modulation order | QPSK | |
| First PRB prior to frequency hopping | 0 | |
| Intra-slot frequency hopping | enabled | |
| First PRB after frequency hopping | The largest PRB index – (Number of PRBs - 1) | |
| Group and sequence hopping | neither | |
| Hopping ID | 0 | |
| Number of PRBs | 1 | 3 |
| Number of symbols | 14 | 4 |
| The number of UCI information bits | 16 | 16 |
| First symbol | 0 | 0 |

### 11.3.2.5.2 Minimum requirements

The UCI block error probability shall not exceed 1% at the SNR given in Table 11.3.2.5.2-1 to 11.3.2.5.2-4.

**Table 11.3.2.5.2-1: Required SNR for PUCCH format 3 with 60kHz SCS in FR2-1**

| Test Number | Number of TX antennas | Number of demodulation branches | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Additional DM-RS configuration | Channel Bandwidth / SNR (dB) | |
|---|---|---|---|---|---|---|---|
| | | | | | | 50 MHz | 100 MHz |
| 1 | 1 | 2 | Normal | TDLA30-300 Low | No additional DM-RS | 1.6 | 0.7 |
| | | | | | Additional DM-RS | 1.3 | 0.9 |
| 2 | 1 | 2 | Normal | TDLA30-300 Low | No additional DM-RS | 3.0 | 2.4 |

**Table 11.3.2.5.2-2: Required SNR for PUCCH format 3 with 120kHz SCS in FR2-1**

| Test Number | Number of TX antennas | Number of demodulation branches | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Additional DM-RS configuration | Channel Bandwidth / SNR (dB) | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | 50 MHz | 100 MHz | 200 MHz |
| 1 | 1 | 2 | Normal | TDLA30-300 Low | No additional DM-RS | 1.4 | 0.7 | 0.7 |
| | | | | | Additional DM-RS | 1.3 | 1.4 | 0.9 |
| 2 | 1 | 2 | Normal | TDLA30-300 Low | No additional DM-RS | 1.1 | 2.9 | 1.4 |

**Table 11.3.2.5.2-3: Minimum requirements for PUCCH format 3 and 120 kHz SCS in FR2-2**

| Test Number | Number of TX antennas | Number of Demodulation branches | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Additional DM-RS configuration | Channel bandwidth / SNR (dB) |
|---|---|---|---|---|---|---|
| | | | | | | 100 |
| 1 | 1 | 2 | Normal | TDLA30-650 Low | No additional DM-RS | 0.6 |
| | | | | | Additional DM-RS | 0.7 |
| 2 | 1 | 2 | Normal | TDLA30-650 Low | No additional DM-RS | 1.8 |

**Table 11.3.2.5.2-4: Minimum requirements for PUCCH format 3 and 480 kHz SCS in FR2-2**

| Test Number | Number of TX antennas | Number of Demodulation branches | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Additional DM-RS configuration | Channel bandwidth / SNR (dB) |
|---|---|---|---|---|---|---|
| | | | | | | 400 |
| 1 | 1 | 2 | Normal | TDLA10-650 Low | No additional DM-RS | 0.5 |
| | | | | | Additional DM-RS | 0.5 |
| 2 | 1 | 2 | Normal | TDLA10-650 Low | No additional DM-RS | 2 |

## 11.3.2.6 Performance requirements for PUCCH format 4

### 11.3.2.6.1 General

The performance is measured by the required SNR at UCI block error probability not exceeding 1%.

The UCI block error probability is defined as the conditional probability of incorrectly decoding the UCI information when the UCI information is sent. The UCI information does not contain CSI part 2.

The transient period as specified in TS 38.101-2 [18] clause 6.3.3.1 is not taken into account for performance requirement testing, where the RB hopping is symmetric to the CC centre, i.e. intra-slot frequency hopping is enabled.

**Table 11.3.2.6.1-1: Test parameters**

| Parameter | Value |
|---|---|
| Modulation order | QPSK |
| First PRB prior to frequency hoppingstartingPRB | 0 |
| Number of PRBs | For tests with FR2-1 : 1.<br>For tests 2 with FR2-2:16 |
| Intra-slot frequency hopping | enabled |
| First PRB after frequency hopping | The largest PRB index – (Number of PRBs – 1) |
| Group and sequence hopping | neither |
| Hopping ID | 0 |
| Number of symbols | 14 |
| The number of UCI information bits | 22 |
| First symbol | 0 |
| Length of the orthogonal cover code | n2 |
| Index of the orthogonal cover code | n0 |

### 11.3.2.6.2 Minimum requirements

The UCI block error probability shall not exceed 1% at the SNR given in Table 11.3.2.6.2-1 to 11.3.2.6.2-4.

**Table 11.3.2.6.2-1: Required SNR for PUCCH format 4 with 60 kHz SCS in FR2-1**

| Number of TX antennas | Number of demodulation branches | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Additional DM-RS configuration | Channel Bandwidth / SNR (dB) | |
|---|---|---|---|---|---|---|
| | | | | | 50 MHz | 100 MHz |
| 1 | 2 | Normal | TDLA30-300 Low | No additional DM-RS | 3.0 | 2.7 |
| | | | | Additional DM-RS | 3.1 | 3.5 |

**Table 11.3.2.6.2-2: Required SNR for PUCCH format 4 with 120 kHz SCS in FR2-1**

| Number of TX antennas | Number of demodulation branches | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Additional DM-RS configuration | Channel Bandwidth / SNR (dB) | | |
|---|---|---|---|---|---|---|---|
| | | | | | 50 MHz | 100 MHz | 200 MHz |
| 1 | 2 | Normal | TDLA30-300 Low | No additional DM-RS | 2.8 | 2.8 | 3.5 |
| | | | | Additional DM-RS | 3.6 | 3.8 | 3.2 |

**Table 11.3.2.6.2-3: Minimum requirements for PUCCH format 4 and 120 kHz SCS in FR2-2**

| Number of TX antennas | Number of demodulation branches | Cyclic Prefix | Propagation conditions and correlation matrix (annex G) | Number of PRBs | Additional DM-RS configuration | Channel bandwidth / SNR (dB) |
|---|---|---|---|---|---|---|
| | | | | | | 100 MHz |
| 1 | 2 | Normal | TDLA30-650 Low | 1 | No additional DM-RS | 3.0 |
| | | | | | Additional DM-RS | 3.0 |
| 1 | 2 | Normal | TDLA30-650 Low | 16 | No additional DM-RS | -9.0 |
| | | | | | Additional DM-RS | -8.9 |

**Table 11.3.2.6.2-4: Minimum requirements for PUCCH format 4 and 480 kHz SCS in FR2-2**

| Number of TX antennas | Number of demodulation branches | Cyclic Prefix | Propagation conditions and correlation matrix (annex G) | Number of PRBs | Additional DM-RS configuration | Channel bandwidth / SNR (dB) |
|---|---|---|---|---|---|---|
| | | | | | | 400 MHz |
| 1 | 2 | Normal | TDLA10-650 Low | 1 | No additional DM-RS | 3.6 |
| | | | | | Additional DM-RS | [-3.4] |
| 1 | 2 | Normal | TDLA10-650 Low | 16 | No additional DM-RS | -9.1 |
| | | | | | Additional DM-RS | [-9.0] |

## 11.3.2.7 Performance requirements for PUCCH format 1 with DMRS bundling

### 11.3.2.7.1 NACK to ACK requirements

#### 11.3.2.7.1.1 General

The NACK to ACK detection probability is the probability that an ACK bit is falsely detected when an NACK bit was sent on the particular bit position, where the NACK to ACK detection probability is defined as follows:

$$\text{Prob}(\text{PUCCH NACK} \rightarrow \text{ACK bits}) = \frac{\#(\text{NACK bits decoded as ACK bits})}{\#(\text{Total NACK bits})},$$

where:

- $\#(\text{Total NACK bits})$ denotes the total number of NACK bits transmitted
- $\#(\text{NACK bits decoded as ACK bits})$ denotes the number of NACK bits decoded as ACK bits at the receiver, i.e. the number of received ACK bits
- NACK bits in the definition do not contain the NACK bits which are mapped from DTX, i.e. NACK bits received when DTX is sent should not be considered.

The NACK to ACK detection probability performance requirement only apply to PUCCH format 1 with 2 UCI bits. The UCI information only contain ACK/NACK information.

The 2bits UCI information is further defined with bitmap as [0 1].

**Table 11.3.2.7.1.1-1: Test Parameters**

| Parameter | Test |
|---|---|
| Example TDD UL-DL pattern (Note1) | 60 kHz SCS: DDSUU, S=10D:2G:2U or DSUUU<br>120 kHz SCS: DDSUU, S=10D:2G:2U, DSUUU or DSUUU |
| Number of information bits | 2 |
| Number of PRBs | 1 |
| Number of symbols | 14 |
| First PRB prior to frequency hopping | 0 |
| Intra-slot frequency hopping | disabled |
| Group and sequence hopping | neither |
| Hopping ID | 0 |
| Initial cyclic shift | 0 |
| First symbol | 0 |
| Index of orthogonal cover code (*timeDomainOCC*) | 0 |
| Number of slots for PUCCH repetition | 2 |
| PUCCH-TimeDomainWindowLength | 2 |
| Note 1: The same TDD requirements are applicable to different UL-DL patterns with more than one consecutive UL slots when both pucch-TimeDomainWindowLength and PUCCH aggregation factor are configured as 2 slots. The UL (re)transmission of PUCCH is only scheduled for the actual TDW including 2 consecutive UL slots. | |

#### 11.3.2.7.1.2 Minimum requirements

The NACK to ACK probability shall not exceed 0.1% at the SNR given in table 11.3.2.7.1.2-1 and table 11.3.2.7.1.2-2

**Table 11.3.2.7.1.2-1: Minimum requirements for PUCCH format 1 with DMRS bundling, 60 kHz SCS, 50MHz channel bandwidth**

| Number of Tx antennas | Number of demodulation branches | Cyclic-Prefix | Propagation conditions and correlation matrix (Annex G) | SNR (dB) |
|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-75 Low | -3.3 |

**Table 11.3.2.7.1.2-2: Minimum requirements for PUCCH format 1 with DMRS bundling, 120 kHz SCS, 50MHz channel bandwidth**

| Number of Tx antennas | Number of demodulation branches | Cyclic-Prefix | Propagation conditions and correlation matrix (Annex G) | SNR (dB) |
|---|---|---|---|---|
| 1 | 2 | Normal | TDLA30-75 Low | -3.2 |

### 11.3.2.7.2 ACK missed detection requirements

#### 11.3.2.7.2.1 General

The ACK missed detection probability is the probability of not detecting an ACK when an ACK was sent. The test parameters in table 11.3.2.7.2.1-1 are configured.

The ACK missed detection probability performance requirement only apply to PUCCH format 1 with 2 UCI bits. The UCI information only contain ACK/NACK information.

The 2bits UCI information is further defined with bitmap as [0 1].

#### 11.3.2.7.2.2 Minimum requirements

The ACK missed detection probability shall not exceed 1% at the SNR given in table 11.3.2.7.2.2-1 to table 11.3.2.7.2.2-2

**Table 11.3.2.7.2.2-1: Minimum requirements for PUCCH format 1 with JCE, 60 kHz SCS, 50MHz channel bandwidth**

| Test Number | Number of Tx antennas | Number of demodulation branches | Cyclic-Prefix | Propagation conditions and correlation matrix (Annex G) | SNR (dB) |
|---|---|---|---|---|---|
| 1 | 1 | 2 | Normal | TDLA30-75 Low | -4.8 |

**Table 11.3.2.7.2.2-1: Minimum requirements for PUCCH format 1 with JCE, 120 kHz SCS, 100MHz channel bandwidth**

| Test Number | Number of Tx antennas | Number of demodulation branches | Cyclic-Prefix | Propagation conditions and correlation matrix (Annex G) | SNR (dB) |
|---|---|---|---|---|---|
| 1 | 1 | 2 | Normal | TDLA30-75 Low | -4.9 |

### 11.3.2.8 Performance requirements for PUCCH format 3 with DMRS bunding

#### 11.3.2.8.1 General

The performance is measured by the required SNR at UCI block error probability not exceeding 1%.

The UCI block error probability is defined as the conditional probability of incorrectly decoding the UCI information when the UCI information is sent. The UCI information does not contain CSI part 2.

**Table 11.3.2.8.1-1: Test Parameters**

| Parameter | Test |
|---|---|
| Example TDD UL-DL pattern (Note 1) | 60 kHz SCS:<br>DDSUU, S=10D:2G:2U or DSUUU<br>120 kHz SCS:<br>DDSUU, S=10D:2G:2U or DSUUU |
| Modulation order | QPSK |
| First PRB prior to frequency hopping | 0 |
| Intra-slot frequency hopping | disabled |
| Group and sequence hopping | neither |
| Hopping ID | 0 |
| Number of PRBs | 1 |
| Number of symbols | 14 |
| The number of UCI information bits | 16 |
| First symbol | 0 |
| Number of slots for PUCCH repetition | 2 |
| PUCCH-TimeDomainWindowLength | 2 |
| Note 1: The same TDD requirements are applicable to different UL-DL patterns with more than one consecutive UL slots when both pucch-TimeDomainWindowLength and PUCCH aggregation factor are configured as 2 slots.<br>The UL (re)transmission of PUCCH is only scheduled for the actual TDW including 2 consecutive UL slots. | |

#### 11.3.2.8.2 Minimum requirements

The UCI block error probability shall not exceed 1% at the SNR given in Table 11.3.2.8.2-1 and Table 11.3.2.8.2-2.

**Table 11.3.2.8.2-1: Minimum requirements for PUCCH format 3 with JCE, 60 kHz SCS, 50MHz channel bandwidth**

| Test Number | Number of TX antennas | Number of demodulation branches | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Additional DM-RS configuration | Channel bandwidth / SNR (dB) |
|---|---|---|---|---|---|---|
| | | | | | | 50 MHz |
| 1 | 1 | 2 | Normal | TDLA30-75 Low | No additional DM-RS | 0.1 |
| | | | | | Additional DM-RS | -0.2 |

**Table 11.3.2.8.2-2: Minimum requirements for PUCCH format 3 with JCE, 120 kHz SCS, 50MHz channel bandwidth**

| Test Number | Number of TX antennas | Number of demodulation branches | Cyclic Prefix | Propagation conditions and correlation matrix (Annex G) | Additional DM-RS configuration | Channel bandwidth / SNR (dB) |
|---|---|---|---|---|---|---|
| | | | | | | 50 MHz |
| 1 | 1 | 2 | Normal | TDLA30-75 Low | No additional DM-RS | 0.7 |
| | | | | | Additional DM-RS | -0.1 |

## 11.4 Performance requirements for PRACH

### 11.4.1 Requirements for *BS type 1-O*

#### 11.4.1.1 PRACH False alarm probability

Apply the requirements defined in clause 8.4.1 for 2Rx.

#### 11.4.1.2 PRACH detection requirements

Apply the requirements defined in clause 8.4.2 for 2Rx.

### 11.4.2 Requirements for *BS type 2-O*

#### 11.4.2.1 PRACH False alarm probability

##### 11.4.2.1.1 General

The false alarm requirement is valid for any number of receive antennas, for any channel bandwidth.

The false alarm probability is the conditional total probability of erroneous detection of the preamble (i.e. erroneous detection from any detector) when input is only noise.

##### 11.4.2.1.2 Minimum requirement

The false alarm probability shall be less than or equal to 0.1%.

## 11.4.2.2 PRACH detection requirements

### 11.4.2.2.1 General

The probability of detection is the conditional probability of correct detection of the preamble when the signal is present. There are several error cases – detecting different preamble than the one that was sent, not detecting a preamble at all or correct preamble detection but with the wrong timing estimation. For AWGN and TDLA30-300, a timing estimation error occurs if the estimation error of the timing of the strongest path is larger than the time error tolerance given in Table 11.4.2.2-1.

The performance requirements for high speed train (table 11.4.2.2.3-1) are optional and only applicable for FR2-1 below 30GHz.

**Table 11.4.2.2-1: Time error tolerance for AWGN, TDLA30-300, and TDLA10-650**

| PRACH preamble | PRACH SCS (kHz) | Time error tolerance | | |
|---|---|---|---|---|
| | | AWGN | TDLA30-300 | TDLA10-650 |
| A1, A2, A3, B4, C0, C2 | 60 | 0.13 us | 0.28 us | N/A |
| | 120 | 0.07 us | 0.22 us | N/A |
| | 480 | 18 ns | N/A | 68 ns |

The test preambles for normal mode are listed in table A.6-2 and the test parameter *msg1-FrequencyStart* is set to 0. The test preambles for high speed train short formats are listed in table A.6-7 and the test parameter *msg1-FrequencyStart* for high speed train is set to 0.

### 11.4.2.2.2 Minimum requirements for Normal mode

The probability of detection shall be equal to or exceed 99% for the SNR levels listed in Tables 11.4.2.2.2-1 to 11.4.2.2.2-4.

**Table 11.4.2.2.2-1: PRACH missed detection requirements for Normal Mode, 60 kHz SCS in FR2-1**

| Number of TX antennas | Number of demodulation branches | Propagation conditions and correlation matrix (Annex G) | Frequency offset | SNR (dB) | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | Burst format A1 | Burst format A2 | Burst format A3 | Burst format B4 | Burst format C0 | Burst format C2 |
| 1 | 2 | AWGN | 0 | -8.9 | -11.9 | -13.5 | -15.8 | -6.0 | -11.8 |
| | | TDLA30-300 Low | 4000 Hz | -1.6 | -3.8 | -4.8 | -6.9 | 1.1 | -3.9 |

**Table 11.4.2.2.2-2: PRACH missed detection requirements for Normal Mode, 120 kHz SCS in FR2-1**

| Number of TX antennas | Number of demodulation branches | Propagation conditions and correlation matrix (Annex G) | Frequency offset | SNR (dB) | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | Burst format A1 | Burst format A2 | Burst format A3 | Burst format B4 | Burst format C0 | Burst format C2 |
| 1 | 2 | AWGN | 0 | -8.7 | -11.5 | -13.3 | -15.8 | -5.8 | -11.4 |
| | | TDLA30-300 Low | 4000 Hz | -1.7 | -4.4 | -5.8 | -7.5 | 1.2 | -4.2 |

**Table 11.4.2.2.2-3: PRACH missed detection test requirements for Normal Mode, 120 kHz SCS in FR2-2**

| Number of TX antennas | Number of demodulation branches | Propagation conditions and correlation matrix (Annex G) | Frequency offset | SNR (dB) | | |
|---|---|---|---|---|---|---|
| | | | | Burst format A2 | Burst format B4 | Burst format C2 |
| 1 | 2 | AWGN | 0 | -11.8 | -16 | -11.8 |
| | | TDLA30-650 Low | 7100 Hz | -4.3 | -7.3 | -4.2 |

**Table 11.4.2.2.22-4: PRACH missed detection test requirements for Normal Mode, 480 kHz SCS in FR2-2**

| Number of TX antennas | Number of demodulation branches | Propagation conditions and correlation matrix (Annex G) | Frequency offset | SNR (dB) | | |
|---|---|---|---|---|---|---|
| | | | | Burst format A2 | Burst format B4 | Burst format C2 |
| 1 | 2 | AWGN | 0 | -11.7 | -15.9 | -11.8 |
| | | TDLA10-650 Low | 7100 Hz | -4.5 | -9.3 | -4.5 |

### 11.4.2.2.3 Minimum requirements for high speed train

The probability of detection shall be equal to or exceed 99% for the SNR levels listed in Table 11.4.2.2.3-1.

**Table 11.4.2.2.3-1: PRACH missed detection requirements for high speed train, 120 kHz SCS**

| Number of TX antennas | Number of RX antennas | Propagation conditions (Annex G) | Frequency offset | SNR (dB) |
|---|---|---|---|---|
| | | | | Burst format C2 |
| 1 | 2 | AWGN | 19444 Hz | -10.4 |

### 11.4.2.2.4 Minimum requirements for PRACH with $L_{RA}$=1151 and $L_{RA}$=571

The probability of detection shall be equal to or exceed 99% for the SNR levels listed in Tables 11.4.2.2.4-1 to 11.4.2.2.4-3.

**Table 11.4.2.2.4-1: Missed detection requirements for PRACH with $L_{RA}$=571, 120 kHz SCS in FR2-2**

| Number of TX antennas | Number of demodulation branches | Propagation conditions and correlation matrix (Annex G) | Frequency offset | SNR (dB) | | |
|---|---|---|---|---|---|---|
| | | | | Burst format A2 | Burst format B4 | Burst format C2 |
| 1 | 2 | AWGN | 0 | -17.9 | -22.1 | -17.9 |
| | | TDLA30-650 Low | 7100 Hz | -10.9 | -13.8 | -10.7 |

**Table 11.4.2.2.4-2: Missed detection requirements for PRACH with $L_{RA}$=1151, 120 kHz SCS in FR2-2**

| Number of TX antennas | Number of demodulation branches | Propagation conditions and correlation matrix (Annex G) | Frequency offset | SNR (dB) | | |
|---|---|---|---|---|---|---|
| | | | | Burst format A2 | Burst format B4 | Burst format C2 |
| 1 | 2 | AWGN | 0 | -20.9 | -25.0 | -20.8 |
| | | TDLA10-650 | 7100 Hz | -14.0 | -16.9 | -14.0 |

**Table 11.4.2.2.4-3: Missed detection requirements for PRACH with $L_{RA}$=571, 480 kHz SCS in FR2-2**

| Number of TX antennas | Number of demodulation branches | Propagation conditions and correlation matrix (Annex G) | Frequency offset | SNR (dB) | | |
|---|---|---|---|---|---|---|
| | | | | Burst format A2 | Burst format B4 | Burst format C2 |
| 1 | 2 | AWGN | 0 | -17.9 | -22.0 | -17.9 |
| | | TDLA10-650 | 7100 Hz | -11.1 | -15.0 | -11.1 |

# Annex A (normative): Reference measurement channels

## A.1 Fixed Reference Channels for reference sensitivity level, ACS, in-band blocking, out-of-band blocking, receiver intermodulation and in-channel selectivity (QPSK, R=1/3)

The parameters for the reference measurement channels are specified in table A.1-1 for FR1 reference sensitivity level, ACS, in-band blocking, out-of-band blocking, receiver intermodulation, in-channel selectivity, OTA sensitivity, OTA reference sensitivity level, OTA ACS, OTA in-band blocking, OTA out-of-band blocking, OTA receiver intermodulation and OTA in-channel selectivity. The parameters for the band n46, n96 and n102 reference measurement channels are specified in table A.1-1a and A.1-1b for reference sensitivity level, ACS, in-band blocking, out-of-band blocking, receiver intermodulation, in-channel selectivity.

The parameters for the reference measurement channels are specified in table A.1-2 for FR2 OTA reference sensitivity level, OTA ACS, OTA in-band blocking, OTA out-of-band blocking, OTA receiver intermodulation and OTA in-channel selectivity.

Reference measurement channels G-FR2-A1-1 to G-FR2-A1-5 are used for FR2-1 requirements. Reference measurement channels G-FR2-A1-2, G-FR2-A1-3, G-FR2-A1-6 to G-FR2-A1-9 are used for FR2-2 requirements.

**Table A.1-1: FRC parameters for FR1 reference sensitivity level, ACS, in-band blocking, out-of-band blocking, receiver intermodulation, in-channel selectivity, OTA sensitivity, OTA reference sensitivity level, OTA ACS, OTA in-band blocking, OTA out-of-band blocking, OTA receiver intermodulation and OTA in-channel selectivity**

| Reference channel | G-FR1-A1-1 | G-FR1-A1-2 | G-FR1-A1-3 | G-FR1-A1-4 | G-FR1-A1-5 | G-FR1-A1-6 | G-FR1-A1-7 | G-FR1-A1-8 | G-FR1-A1-9 | G-FR1-A1-10 | G-FR1-A1-11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Subcarrier spacing (kHz) | 15 | 30 | 60 | 15 | 30 | 60 | 15 | 30 | 60 | 15 | 15 |
| Allocated resource blocks | 25 | 11 | 11 | 106 | 51 | 24 | 15 | 6 | 6 | 24 | 105 |
| CP-OFDM Symbols per slot (Note 1) | 12 | 12 | 12 | 12 | 12 | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Code rate (Note 2) | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size (bits) | 2152 | 984 | 984 | 9224 | 4352 | 2088 | 1320 | 528 | 528 | 2088 | 8968 |
| Transport block CRC (bits) | 16 | 16 | 16 | 24 | 24 | 16 | 16 | 16 | 16 | 16 | 24 |
| Code block CRC size (bits) | - | - | - | 24 | - | - | - | - | - | - | 24 |
| Number of code blocks - C | 1 | 1 | 1 | 2 | 1 | 1 | 1 | 1 | 1 | 1 | 2 |
| Code block size including CRC (bits) (Note 3) | 2168 | 1000 | 1000 | 4648 | 4376 | 2104 | 1336 | 544 | 544 | 2104 | 4520 |
| Total number of bits per slot | 7200 | 3168 | 3168 | 30528 | 14688 | 6912 | 4320 | 1728 | 1728 | 6912 | 30240 |
| Total symbols per slot | 3600 | 1584 | 1584 | 15264 | 7344 | 3456 | 2160 | 864 | 864 | 3456 | 15120 |

NOTE 1: *UL-DMRS-config-type* = 1 with *UL-DMRS-max-len* = 1, *UL-DMRS-add-pos* = 1 with $l_0$= 2, $l$ = 11 as per table 6.4.1.1.3-3 of TS 38.211 [9].

NOTE 2: MCS index 4 and target coding rate = 308/1024 are adopted to calculate payload size for receiver sensitivity and in-channel selectivity

NOTE 3: Code block size including CRC (bits) equals to $K'$ in sub-clause 5.2.2 of TS 38.212 [15].NOTE 2: MCS index 4 and target coding rate = 308/1024 are adopted to calculate payload size for receiver sensitivity and in-channel selectivity

NOTE 3: Code block size including CRC (bits) equals to $K'$ in sub-clause 5.2.2 of TS 38.212 [15].

**Table A.1-1a: FRC parameters for band n46, n96 and n102 reference sensitivity level, ACS, in-band blocking, out-of-band blocking, receiver intermodulation, in-channel selectivity**

| Reference channel | G-FR1-A1-12 | G-FR1-A1-13 | G-FR1-A1-14 | G-FR1-A1-15 | G-FR1-A1-16 | G-FR1-A1-17 | G-FR1-A1-18 | G-FR1-A1-19 |
|---|---|---|---|---|---|---|---|---|
| Channel bandwidth (MHz) | 10 | 10 | 20 | 20 | 40 | 40 | 60 | 80 |
| Subcarrier spacing (kHz) | 15 | 30 | 15 | 30 | 15 | 30 | 30 | 30 |
| Allocated resource blocks | 5 | 4 | 10 | 10 | 21 | 21 | 32 | 43 |
| CP-OFDM Symbols per slot (Note 1) | 12 | 12 | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Code rate (Note 2) | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size (bits) | 432 | 352 | 888 | 888 | 1864 | 1864 | 2792 | 3752 |
| Transport block CRC (bits) | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 |
| Code block CRC size (bits) | - | - | - | - | - | - | - | - |
| Number of code blocks - C | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| Code block size including CRC (bits) (Note 3) | 448 | 368 | 904 | 904 | 1880 | 1880 | 2808 | 3768 |
| Total number of bits per slot | 1440 | 1152 | 2880 | 2880 | 6048 | 6048 | 9216 | 12384 |
| Total symbols per slot | 720 | 576 | 1440 | 1440 | 3024 | 3024 | 4608 | 6192 |

NOTE 1: *UL-DMRS-config-type* = 1 with *UL-DMRS-max-len* = 1, *UL-DMRS-add-pos* = 1 with $l_0$= 2, $l$ = 11 as per table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: MCS index 4 and target coding rate = 308/1024 are adopted to calculate payload size for receiver sensitivity and in-channel selectivity
NOTE 3: Code block size including CRC (bits) equals to $K'$ in sub-clause 5.2.2 of TS 38.212 [15].
NOTE 4: For reference channel A1-12, the allocated RB's are uniformly spaced over the channel bandwidth at RB index N, N+10, N+20, N+30, N+40 where N={0,1,2,3,4,…,9}.
NOTE 5: For reference channel A1-13, the allocated RB's are uniformly spaced over the channel bandwidth at RB index N, N+5, N+10, N+15 where N={0,1,2,3,4}.
NOTE 7: For reference channel A1-14, the allocated RB's are uniformly spaced over the channel bandwidth at RB index N, N+10,N+20,..N+90 where N={0,1,2,3,...,9}.
NOTE 8: For reference channel A1-15, the allocated RB's are uniformly spaced over the channel bandwidth at RB index N, N+5,N+10,..,N+45 where N={0,1,2,3,4}.
NOTE 10: For reference channel A1-16, the allocated RB's are uniformly spaced over the channel bandwidth at RB index N, N+10,N+20,...,N+200 where N={0,1,2,3,4,...,9}.
NOTE 11: For reference channel A1-17, the allocated RB's are uniformly spaced over the channel bandwidth at RB index N, N+5, N+10, ..., N+100 where N={0,1,2,3,4}.
NOTE 12: For reference channel A1-18, the allocated RB's are uniformly spaced over the channel bandwidth at RB index N, N+5,N+10,...,N+155 where N={0,1,2,3,4}.
NOTE 13: For reference channel A1-19, the allocated RB's are uniformly spaced over the channel bandwidth at RB index N, N+5,N+10,...,N+210 where N={0,1,2,3,4}.

**Table A.1-2: FRC parameters for FR2 OTA reference sensitivity level, OTA ACS, OTA in-band blocking, OTA out-of-band blocking, OTA receiver intermodulation and OTA in-channel selectivity**

| Reference channel | G-FR2-A1-1 | G-FR2-A1-2 | G-FR2-A1-3 | G-FR2-A1-4 | G-FR2-A1-5 | G-FR2-A1-6 | G-FR2-A1-7 | G-FR2-A1-8 | G-FR2-A1-9 |
|---|---|---|---|---|---|---|---|---|---|
| Subcarrier spacing (kHz) | 60 | 120 | 120 | 60 | 120 | 480 | 960 | 480 | 960 |
| Allocated resource blocks | 66 | 32 | 66 | 33 | 16 | 66 | 33 | 33 | 17 |
| CP-OFDM Symbols per slot (Note 1) | 12 | 12 | 12 | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Code rate (Note 2) | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size (bits) | 5632 | 2792 | 5632 | 2856 | 1416 | 5632 | 2856 | 2856 | 1480 |
| Transport block CRC (bits) | 24 | 16 | 24 | 16 | 16 | 24 | 16 | 16 | 16 |
| Code block CRC size (bits) | - | - | - | - | - | - | - | - | - |
| Number of code blocks - C | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| Code block size including CRC (bits) (Note 3) | 5656 | 2808 | 5656 | 2872 | 1432 | 5656 | 2872 | 2872 | 1496 |
| Total number of bits per slot | 19008 | 9216 | 19008 | 9504 | 4608 | 19008 | 9504 | 9504 | 4896 |
| Total symbols per slot | 9504 | 4608 | 9504 | 4752 | 2304 | 9504 | 4752 | 4752 | 2448 |

NOTE 1: DM-RS configuration type = 1 with DM-RS duration = single-symbol DM-RS, additional DM-RS position = pos1 with $l_0$ = 2, $l$ = 11 as per table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: MCS index 4 and target coding rate = 308/1024 are adopted to calculate payload size.
NOTE 3: Code block size including CRC (bits) equals to $K'$ in sub-clause 5.2.2 of TS 38.212 [15].

# A.2 Fixed Reference Channels for dynamic range (16QAM, R=2/3)

The parameters for the reference measurement channels are specified in table A.2-1 for FR1 dynamic range and OTA dynamic range. The parameters for the band n46, n96 and n102 reference measurement channels are specified in table A.2-1a and A.2-1b for band n46, n96 and n102 dynamic range.

**Table A.2-1: FRC parameters for FR1 dynamic range and OTA dynamic range**

| Reference channel | G-FR1-A2-1 | G-FR1-A2-2 | G-FR1-A2-3 | G-FR1-A2-4 | G-FR1-A2-5 | G-FR1-A2-6 |
|---|---|---|---|---|---|---|
| Subcarrier spacing (kHz) | 15 | 30 | 60 | 15 | 30 | 60 |
| Allocated resource blocks | 25 | 11 | 11 | 106 | 51 | 24 |
| CP-OFDM Symbols per slot (Note 1) | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM |
| Code rate (Note 2) | 2/3 | 2/3 | 2/3 | 2/3 | 2/3 | 2/3 |
| Payload size (bits) | 9224 | 4032 | 4032 | 38936 | 18960 | 8968 |
| Transport block CRC (bits) | 24 | 24 | 24 | 24 | 24 | 24 |
| Code block CRC size (bits) | 24 | - | - | 24 | 24 | 24 |
| Number of code blocks - C | 2 | 1 | 1 | 5 | 3 | 2 |
| Code block size including CRC (bits) (Note 3) | 4648 | 4056 | 4056 | 7816 | 6352 | 4520 |
| Total number of bits per slot | 14400 | 6336 | 6336 | 61056 | 29376 | 13824 |
| Total symbols per slot | 3600 | 1584 | 1584 | 15264 | 7344 | 3456 |
| NOTE 1: DM-RS configuration type = 1 with DM-RS duration = single-symbol DM-RS, additional DM-RS position = pos1 with $l_0$= 2, $l$ = 11 as per table 6.4.1.1.3-3 of TS 38.211 [9].<br>NOTE 2: MCS index 16 and target coding rate = 658/1024 are adopted to calculate payload size.<br>NOTE 3: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15]. | | | | | | |

**Table A.2-1a: FRC parameters for dynamic range for band n46, n96 and n102**

| Reference channel | G-FR1-A2-7 | G-FR1-A2-8 | G-FR1-A2-9 | G-FR1-A2-10 | G-FR1-A2-11 | G-FR1-A2-12 | G-FR1-A2-13 | G-FR1-A2-14 |
|---|---|---|---|---|---|---|---|---|
| Channel bandwidth (MHz) | 10 | 10 | 20 | 20 | 40 | 40 | 60 | 80 |
| Subcarrier spacing (kHz) | 15 | 30 | 15 | 30 | 15 | 30 | 30 | 30 |
| Allocated resource blocks | 5 | 4 | 10 | 10 | 21 | 21 | 32 | 43 |
| CP-OFDM Symbols per slot (Note 1) | 12 | 12 | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM |
| Code rate (Note 2) | 2/3 | 2/3 | 2/3 | 2/3 | 2/3 | 2/3 | 2/3 | 2/3 |
| Payload size (bits) | 1864 | 1480 | 3752 | 3752 | 7808 | 7808 | 11784 | 15880 |
| Transport block CRC (bits) | 16 | 16 | 16 | 16 | 24 | 24 | 24 | 24 |
| Code block CRC size (bits) | - | - | - | - | - | - | 24 | 24 |
| Number of code blocks - C | 1 | 1 | 1 | 1 | 1 | 1 | 2 | 2 |
| Code block size including CRC (bits) (Note 3) | 1880 | 1496 | 3768 | 3768 | 7832 | 7832 | 5928 | 7976 |
| Total number of bits per slot | 2880 | 2304 | 5760 | 5760 | 12096 | 12096 | 18432 | 24768 |
| Total symbols per slot | 720 | 576 | 1440 | 1440 | 3024 | 3024 | 4608 | 6192 |

NOTE 1: DM-RS configuration type = 1 with DM-RS duration = single-symbol DM-RS, additional DM-RS position = pos1 with l0= 2, l = 11 as per table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: MCS index 16 and target coding rate = 658/1024 are adopted to calculate payload size.
NOTE 3: Code block size including CRC (bits) equals to K' in sub-clause 5.2.2 of TS 38.212 [15].
NOTE 4: For reference channel A2-7, the allocated RB's are uniformly spaced over the channel bandwidth at RB index N, N+10, N+20, N+30, N+40 where N={0,1,2,3,4,5,6,7,8,9}.
NOTE 5: For reference channel A2-8, the allocated RB's are uniformly spaced over the channel bandwidth at RB index N, N+5, N+10, N+15 where N={0,1,2,3,4}.
NOTE 6: For reference channel A2-9, the allocated RB's are uniformly spaced over the channel bandwidth at RB index N, N+10,N+20,..N+90 where N={0,1,2,3,...,9}.
NOTE 7: For reference channel A2-10, the allocated RB's are uniformly spaced over the channel bandwidth at RB index N, N+5,N+10,.., N+45 where N={0,1,2,3,4}.
NOTE 8: For reference channel A2-11, the allocated RB's are uniformly spaced over the channel bandwidth at RB index N, N+10,N+20,...,N+200 where N={0,1,2,3,4,...,9}.
NOTE 9: For reference channel A2-12, the allocated RB's are uniformly spaced over the channel bandwidth at RB index N, N+5, N+10, ..., N+100 where N={0,1,2,3,4}.
NOTE 10: For reference channel A2-13, the allocated RB's are uniformly spaced over the channel bandwidth at RB index N, N+5,N+10,..., N+155 where N={0,1,2,3,4}.
NOTE 11: For reference channel A2-14, the allocated RB's are uniformly spaced over the channel bandwidth at RB index N, N+5,N+10,..., N+210 where N={0,1,2,3,4}.

# A.3 Fixed Reference Channels for performance requirements (QPSK, R=193/1024)

The parameters for the reference measurement channels are specified in table A.3-2, table A.3-2A, table A.3-4, and table A.3-6 for FR1 PUSCH performance requirements:

- FRC parameters are specified in table A.3-2 for FR1 PUSCH with transform precoding disabled, *Additional DM-RS position = pos1* and 1 transmission layer.

- FRC parameters are specified in table A.3-2A for FR1 PUSCH with transform precoding disabled, *Additional DM-RS position = pos2* and 1 transmission layer.

- FRC parameters are specified in table A.3-4 for FR1 PUSCH with transform precoding disabled, *Additional DM-RS position = pos1* and 2 transmission layers.

- FRC parameters are specified in table A.3-6 for FR1 PUSCH with transform precoding enabled, *Additional DM-RS position = pos1* and 1 transmission layer.

The parameters for the reference measurement channels are specified in table A.3-14 for FR1 PUSCH performance requirements for TB processing over multi-slot PUSCH:

- FRC parameters are specified in table A.3-14 for FR1 PUSCH with transform precoding disabled, *Additional DM-RS position = pos1* and 1 transmission layer.

The parameters for the reference measurement channels are specified in table A.3-7 to table A.3-12 for FR2 PUSCH performance requirements:

- FRC parameters are specified in table A.3-7 for FR2 PUSCH with transform precoding disabled, *Additional DM-RS position = pos0* and 1 transmission layer.
- FRC parameters are specified in table A.3-8 for FR2 PUSCH with transform precoding disabled, *Additional DM-RS position = pos0* and 2 transmission layers.
- FRC parameters are specified in table A.3-9 for FR2 PUSCH with transform precoding enabled, *Additional DM-RS position = pos0* and 1 transmission layer.
- FRC parameters are specified in table A.3-10 for FR2 PUSCH with transform precoding disabled, *Additional DM-RS position = pos1* and 1 transmission layer.
- FRC parameters are specified in table A.3-11 for FR2 PUSCH with transform precoding disabled, *Additional DM-RS position = pos1* and 2 transmission layers.
- FRC parameters are specified in table A.3-12 for FR2 PUSCH with transform precoding enabled, *Additional DM-RS position = pos1* and 1 transmission layer.

The parameters for the reference measurement channels are specified in table A.3-13 for FR2 PUSCH performance requirements for 2-step RA type:

- FRC parameters are specified in table A.3-13 for FR2 PUSCH with transform precoding disabled, *Additional DM-RS position = pos1* and 1 transmission layer.

The parameters for the reference measurement channels are specified in table A.3-15 for FR2 PUSCH performance requirements for TBoMS:

- FRC parameters are specified in table A.3-15 for FR2 PUSCH with transform precoding disabled, *Additional DM-RS position = pos1* and 1 transmission layer.

**Table A.3-1: Void**

**Table A.3-2: FRC parameters for FR1 PUSCH performance requirements, transform precoding disabled, *Additional DM-RS position* = *pos1* and 1 transmission layer (QPSK, R=193/1024)**

| Reference channel | G-FR1-A3-8 | G-FR1-A3-9 | G-FR1-A3-10 | G-FR1-A3-11 | G-FR1-A3-12 | G-FR1-A3-13 | G-FR1-A3-14 |
|---|---|---|---|---|---|---|---|
| Subcarrier spacing [kHz] | 15 | 15 | 15 | 30 | 30 | 30 | 30 |
| Allocated resource blocks | 25 | 52 | 106 | 24 | 51 | 106 | 273 |
| CP-OFDM Symbols per slot (Note 1) | 12 | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Code rate (Note 2) | 193/1024 | 193/1024 | 193/1024 | 193/1024 | 193/1024 | 193/1024 | 193/1024 |
| Payload size (bits) | 1352 | 2856 | 5768 | 1320 | 2792 | 5768 | 14856 |
| Transport block CRC (bits) | 16 | 16 | 24 | 16 | 16 | 24 | 24 |
| Code block CRC size (bits) | - | - | 24 | - | - | 24 | 24 |
| Number of code blocks - C | 1 | 1 | 2 | 1 | 1 | 2 | 4 |
| Code block size including CRC (bits) (Note 2) | 1368 | 2872 | 2920 | 1336 | 2808 | 2920 | 3744 |
| Total number of bits per slot | 7200 | 14976 | 30528 | 6912 | 14688 | 30528 | 78624 |
| Total symbols per slot | 3600 | 7488 | 15264 | 3456 | 7344 | 15264 | 39312 |
| NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos1*, $l_0$= 2 and $l$ =11 for PUSCH mapping type A, $l_0$= 0 and $l$ =10 for PUSCH mapping type B as per table 6.4.1.1.3-3 of TS 38.211 [9].<br>NOTE 2: Code block size including CRC (bits) equals to *K'* in clause 5.2.2 of TS 38.212 [15]. | | | | | | | |

**Table A.3-2A: FRC parameters for FR1 PUSCH performance requirements, transform precoding disabled, *Additional DM-RS position* = *pos2* and 1 transmission layer (QPSK, R=193/1024)**

| Reference channel | G-FR1-A3-33 | G-FR1-A3-33A | G-FR1-A3-34 | G-FR1-A3-34A |
|---|---|---|---|---|
| Subcarrier spacing [kHz] | 15 | 15 | 30 | 30 |
| Allocated resource blocks | 52 | 25 | 106 | 24 |
| Data bearing CP-OFDM Symbols per slot (Note 1) | 11 | 11 | 11 | 11 |
| Modulation | QPSK | QPSK | QPSK | QPSK |
| Code rate (Note 2) | 193/1024 | 193/1024 | 193/1024 | 193/1024 |
| Payload size (bits) | 2600 | 1256 | 5256 | 1192 |
| Transport block CRC (bits) | 16 | 16 | 24 | 16 |
| Code block CRC size (bits) | - | - | 24 | - |
| Number of code blocks - C | 1 | 1 | 2 | 1 |
| Code block size including CRC (bits) (Note 2) | 2616 | 1272 | 2664 | 1208 |
| Total number of bits per slot | 13728 | 6600 | 27984 | 6336 |
| Total resource elements per slot | 6864 | 3300 | 13992 | 3168 |
| NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos2*, and $l_0$= 2 or 3 for PUSCH mapping type A, as per table 6.4.1.1.3-3 of TS 38.211 [5].<br>NOTE 2: Code block size including CRC (bits) equals to *K'* in clause 5.2.2 of TS 38.212 [15]. | | | | |

**Table A.3-3: Void**

**Table A.3-4: FRC parameters for FR1 PUSCH performance requirements, transform precoding disabled, *Additional DM-RS position = pos1* and 2 transmission layers (QPSK, R=193/1024)**

| Reference channel | G-FR1-A3-22 | G-FR1-A3-23 | G-FR1-A3-24 | G-FR1-A3-25 | G-FR1-A3-26 | G-FR1-A3-27 | G-FR1-A3-28 |
|---|---|---|---|---|---|---|---|
| Subcarrier spacing [kHz] | 15 | 15 | 15 | 30 | 30 | 30 | 30 |
| Allocated resource blocks | 25 | 52 | 106 | 24 | 51 | 106 | 273 |
| CP-OFDM Symbols per slot (Note 1) | 12 | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Code rate (Note 2) | 193/1024 | 193/1024 | 193/1024 | 193/1024 | 193/1024 | 193/1024 | 193/1024 |
| Payload size (bits) | 2728 | 5640 | 11528 | 2600 | 5512 | 11528 | 29736 |
| Transport block CRC (bits) | 16 | 24 | 24 | 16 | 24 | 24 | 24 |
| Code block CRC size (bits) | - | 24 | 24 | - | 24 | 24 | 24 |
| Number of code blocks - C | 1 | 2 | 4 | 1 | 2 | 4 | 8 |
| Code block size including CRC (bits) (Note 2) | 2744 | 2856 | 2912 | 2616 | 2792 | 2912 | 3744 |
| Total number of bits per slot | 14400 | 29952 | 61056 | 13824 | 29376 | 61056 | 157248 |
| Total symbols per slot | 7200 | 14976 | 30528 | 6912 | 14688 | 30528 | 78624 |

NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos1*, $l_0$= 2 and $l$=11 for PUSCH mapping type A, $l_0$ = 0 and $l$ =10 for PUSCH mapping type B as per table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: Code block size including CRC (bits) equals to *K'* in clause 5.2.2 of TS 38.212 [15].

**Table A.3-5: Void**

**Table A.3-6: FRC parameters for FR1 PUSCH performance requirements, transform precoding enabled, *Additional DM-RS position = pos1* and 1 transmission layer (QPSK, R=193/1024)**

| Reference channel | G-FR1-A3-31 | G-FR1-A3-32 |
|---|---|---|
| Subcarrier spacing [kHz] | 15 | 30 |
| Allocated resource blocks | 25 | 24 |
| DFT-s-OFDM Symbols per slot (Note 1) | 12 | 12 |
| Modulation | QPSK | QPSK |
| Code rate (Note 2) | 193/1024 | 193/1024 |
| Payload size (bits) | 1352 | 1320 |
| Transport block CRC (bits) | 16 | 16 |
| Code block CRC size (bits) | - | - |
| Number of code blocks - C | 1 | 1 |
| Code block size including CRC (bits) (Note 2) | 1368 | 1336 |
| Total number of bits per slot | 7200 | 6912 |
| Total symbols per slot | 3600 | 3456 |

NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos1*, $l_0$= 2 and $l$=11 for PUSCH mapping type A, $l_0$= 0 and $l$ =10 for PUSCH mapping type B as per Table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15].

**Table A.3-7: FRC parameters for FR2 PUSCH performance requirements, transform precoding disabled, *Additional DM-RS position = pos0* and 1 transmission layer (QPSK, R=193/1024)**

| Reference channel | G-FR2-A3-1 | G-FR2-A3-2 | G-FR2-A3-3 | G-FR2-A3-4 | G-FR2-A3-5 |
|---|---|---|---|---|---|
| Subcarrier spacing [kHz] | 60 | 60 | 120 | 120 | 120 |
| Allocated resource blocks | 66 | 132 | 32 | 66 | 132 |
| CP-OFDM Symbols per slot (Note 1) | 9 | 9 | 9 | 9 | 9 |
| Modulation | QPSK | QPSK | QPSK | QPSK | QPSK |
| Code rate (Note 2) | 193/1024 | 193/1024 | 193/1024 | 193/1024 | 193/1024 |
| Payload size (bits) | 2664 | 5384 | 1320 | 2664 | 5384 |
| Transport block CRC (bits) | 16 | 24 | 16 | 16 | 24 |
| Code block CRC size (bits) | - | 24 | - | - | 24 |
| Number of code blocks - C | 1 | 2 | 1 | 1 | 2 |
| Code block size including CRC (bits) (Note 2) | 2680 | 2728 | 1336 | 2680 | 2728 |
| Total number of bits per slot | 14256 | 28512 | 6912 | 14256 | 28512 |
| Total symbols per slot | 7128 | 14256 | 3456 | 7128 | 14256 |

NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos0* with $l_0$= 0 as per Table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15].

**Table A.3-8: FRC parameters for FR2 PUSCH performance requirements, transform precoding disabled, *Additional DM-RS position = pos0* and 2 transmission layers (QPSK, R=193/1024)**

| Reference channel | G-FR2-A3-6 | G-FR2-A3-7 | G-FR2-A3-8 | G-FR2-A3-9 | G-FR2-A3-10 |
|---|---|---|---|---|---|
| Subcarrier spacing [kHz] | 60 | 60 | 120 | 120 | 120 |
| Allocated resource blocks | 66 | 132 | 32 | 66 | 132 |
| CP-OFDM Symbols per slot (Note 1) | 9 | 9 | 9 | 9 | 9 |
| Modulation | QPSK | QPSK | QPSK | QPSK | QPSK |
| Code rate (Note 2) | 193/1024 | 193/1024 | 193/1024 | 193/1024 | 193/1024 |
| Payload size (bits) | 5384 | 10752 | 2600 | 5384 | 10752 |
| Transport block CRC (bits) | 24 | 24 | 16 | 24 | 24 |
| Code block CRC size (bits) | 24 | 24 | - | 24 | 24 |
| Number of code blocks - C | 2 | 3 | 1 | 2 | 3 |
| Code block size including CRC (bits) (Note 2) | 2728 | 3616 | 2616 | 2728 | 3616 |
| Total number of bits per slot | 28512 | 57024 | 13824 | 28512 | 57024 |
| Total symbols per slot | 14256 | 28512 | 6912 | 14256 | 28512 |

NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos0* with $l_0$= 0 as per Table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15].

**Table A.3-9: FRC parameters for FR2 PUSCH performance requirements, transform precoding enabled, *Additional DM-RS position = pos0* and 1 transmission layer (QPSK, R=193/1024)**

| Reference channel | G-FR2-A3-11 | G-FR2-A3-12 |
|---|---|---|
| Subcarrier spacing [kHz] | 60 | 120 |
| Allocated resource blocks | 30 | 30 |
| DFT-s-OFDM Symbols per slot (Note 1) | 9 | 9 |
| Modulation | QPSK | QPSK |
| Code rate (Note 2) | 193/1024 | 193/1024 |
| Payload size (bits) | 1224 | 1224 |
| Transport block CRC (bits) | 16 | 16 |
| Code block CRC size (bits) | - | - |
| Number of code blocks - C | 1 | 1 |
| Code block size including CRC (bits) (Note 2) | 1240 | 1240 |
| Total number of bits per slot | 6480 | 6480 |
| Total symbols per slot | 3240 | 3240 |
| NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos0* with $l_0$= 0 as per Table 6.4.1.1.3-3 of TS 38.211 [9]. | | |
| NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15]. | | |

**Table A.3-10: FRC parameters for FR2 PUSCH performance requirements, transform precoding disabled, *Additional DM-RS position = pos1* and 1 transmission layer (QPSK, R=193/1024)**

| Reference channel | G-FR2-A3-13 | G-FR2-A3-14 | G-FR2-A3-15 | G-FR2-A3-16 | G-FR2-A3-17 |
|---|---|---|---|---|---|
| Subcarrier spacing [kHz] | 60 | 60 | 120 | 120 | 120 |
| Allocated resource blocks | 66 | 132 | 32 | 66 | 132 |
| CP-OFDM Symbols per slot (Note 1) | 8 | 8 | 8 | 8 | 8 |
| Modulation | QPSK | QPSK | QPSK | QPSK | QPSK |
| Code rate (Note 2) | 193/1024 | 193/1024 | 193/1024 | 193/1024 | 193/1024 |
| Payload size (bits) | 2408 | 4744 | 1160 | 2408 | 4744 |
| Transport block CRC (bits) | 16 | 24 | 16 | 16 | 24 |
| Code block CRC size (bits) | - | 24 | - | - | 24 |
| Number of code blocks - C | 1 | 2 | 1 | 1 | 2 |
| Code block size including CRC (bits) (Note 2) | 2424 | 2408 | 1176 | 2424 | 2408 |
| Total number of bits per slot | 12672 | 25344 | 6144 | 12672 | 25344 |
| Total symbols per slot | 6336 | 12672 | 3072 | 6336 | 12672 |
| NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos1* with $l_0$= 0 and $l$ =8 as per Table 6.4.1.1.3-3 of TS 38.211 [9]. | | | | | |
| NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15]. | | | | | |

**Table A.3-11: FRC parameters for FR2 PUSCH performance requirements, transform precoding disabled, *Additional DM-RS position = pos1* and 2 transmission layers (QPSK, R=193/1024)**

| Reference channel | G-FR2-A3-18 | G-FR2-A3-19 | G-FR2-A3-20 | G-FR2-A3-21 | G-FR2-A3-22 |
|---|---|---|---|---|---|
| Subcarrier spacing [kHz] | 60 | 60 | 120 | 120 | 120 |
| Allocated resource blocks | 66 | 132 | 32 | 66 | 132 |
| CP-OFDM Symbols per slot (Note 1) | 8 | 8 | 8 | 8 | 8 |
| Modulation | QPSK | QPSK | QPSK | QPSK | QPSK |
| Code rate (Note 2) | 193/1024 | 193/1024 | 193/1024 | 193/1024 | 193/1024 |
| Payload size (bits) | 4744 | 9480 | 2408 | 4744 | 9480 |
| Transport block CRC (bits) | 24 | 24 | 16 | 24 | 24 |
| Code block CRC size (bits) | 24 | 24 | - | 24 | 24 |
| Number of code blocks - C | 2 | 3 | 1 | 2 | 3 |
| Code block size including CRC (bits) (Note 2) | 2408 | 3192 | 2424 | 2408 | 3192 |
| Total number of bits per slot | 25344 | 50688 | 12288 | 25344 | 50688 |
| Total symbols per slot | 12672 | 25344 | 6144 | 12672 | 25344 |
| NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos1* with $l_0$= 0 and *l* =8 as per Table 6.4.1.1.3-3 of TS 38.211 [9]. | | | | | |
| NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15]. | | | | | |

**Table A.3-12: FRC parameters for FR2 PUSCH performance requirements, transform precoding enabled, *Additional DM-RS position = pos1* and 1 transmission layer (QPSK, R=193/1024)**

| Reference channel | G-FR2-A3-23 | G-FR2-A3-24 |
|---|---|---|
| Subcarrier spacing [kHz] | 60 | 120 |
| Allocated resource blocks | 30 | 30 |
| DFT-s-OFDM Symbols per slot (Note 1) | 8 | 8 |
| Modulation | QPSK | QPSK |
| Code rate (Note 2) | 193/1024 | 193/1024 |
| Payload size (bits) | 1128 | 1128 |
| Transport block CRC (bits) | 16 | 16 |
| Code block CRC size (bits) | - | - |
| Number of code blocks - C | 1 | 1 |
| Code block size including CRC (bits) (Note 2) | 1144 | 1144 |
| Total number of bits per slot | 5760 | 5760 |
| Total symbols per slot | 2880 | 2880 |
| NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos1* with $l_0$= 0 and *l* =8 as per Table 6.4.1.1.3-3 of TS 38.211 [9]. | | |
| NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15]. | | |

**Table A.3-13: FRC parameters for FR2 PUSCH performance requirements, transform precoding disabled, Additional DM-RS position = pos1 and 1 transmission layer (QPSK, R=193/1024)**

| Reference channel | G-FR2-A3-25 | G-FR2-A3-26 |
|---|---|---|
| Subcarrier spacing [kHz] | 60 | 120 |
| Allocated resource blocks | 2 | 2 |
| CP-OFDM Symbols per slot (Note 1) | 8 | 8 |
| Modulation | QPSK | QPSK |
| Code rate (Note 2) | 193/1024 | 193/1024 |
| Payload size (bits) | 72 | 72 |
| Transport block CRC (bits) | 16 | 16 |
| Code block CRC size (bits) | 0 | 0 |
| Number of code blocks - C | 1 | 1 |
| Code block size including CRC (bits) (Note 2) | 88 | 88 |
| Total number of bits per slot | 384 | 384 |
| Total number of symbols per slot | 192 | 192 |
| NOTE 1: DM-RS configuration type = 1 with DM-RS duration = single-symbol DM-RS and the number of DM-RS CDM groups without data is 2, Additional DM-RS position = pos1 with l0= 2 as per Table 6.4.1.1.3-3 of TS 38.211 [9].<br>NOTE 2: Code block size including CRC (bits) equals to K' in sub-clause 5.2.2 of TS 38.212 [15]. | | |

**Table A.3-14: FRC parameters for FR1 PUSCH performance requirements, transform precoding disabled, Additional DM-RS position = pos1 and 1 transmission layer (QPSK, R=193/1024)**

| Reference channel | G-FR1-A3-35 | G-FR1-A3-36 | G-FR1-A3-37 | G-FR1-A3-38 |
|---|---|---|---|---|
| Number of TBoMS slots | 2 | 4 | 2 | 4 |
| Subcarrier spacing [kHz] | 15 | 15 | 30 | 30 |
| Allocated resource blocks per slot | 5 | 5 | 5 | 5 |
| Data bearing CP-OFDM Symbols per slot (Note 1) | 12 | 12 | 12 | 12 |
| Modulation | QPSK | QPSK | QPSK | QPSK |
| Code rate (Note 2) | 193/1024 | 193/1024 | 193/1024 | 193/1024 |
| Payload size (bits) | 552 | 1128 | 552 | 1128 |
| Transport block CRC (bits) | 16 | 16 | 16 | 16 |
| Code block CRC size (bits) | - | - | - | - |
| Number of code blocks - C | 1 | 1 | 1 | 1 |
| Code block size including CRC (bits) (Note 2) | 568 | 1144 | 568 | 1144 |
| Total number of bits over all TBoMS slots | 2880 | 5760 | 2880 | 5760 |
| Total resource elements over all TBoMS slots | 1440 | 2880 | 1440 | 2880 |
| NOTE 1: DM-RS configuration type = 1 with DM-RS duration = single-symbol DM-RS and the number of DM-RS CDM groups without data is 2, Additional DM-RS position = pos1 with l0= 2 as per Table 6.4.1.1.3-3 of TS 38.211 [9].<br>NOTE 2: Code block size including CRC (bits) equals to K' in sub-clause 5.2.2 of TS 38.212 [15]. | | | | |

**Table A.3-15: FRC parameters for FR2 PUSCH performance requirements, transform precoding disabled, Additional DM-RS position = pos1 and 1 transmission layer (QPSK, R=193/1024)**

| Reference channel | G-FR2-A3-27 | G-FR2-A3-28 |
|---|---|---|
| Number of TBoMS slots | 2 | 2 |
| Subcarrier spacing [kHz] | 60 | 120 |
| Allocated resource blocks per slot | 5 | 5 |
| Data bearing CP-OFDM Symbols per slot (Note 1) | 8 | 8 |
| Modulation | QPSK | QPSK |
| Code rate (Note 2) | 193/1024 | 193/1024 |
| Payload size (bits) | 368 | 368 |
| Transport block CRC (bits) | 16 | 16 |
| Code block CRC size (bits) | - | - |
| Number of code blocks - C | 1 | 1 |
| Code block size including CRC (bits) (Note 2) | 384 | 384 |
| Total number of bits over all TBoMS slots | 1920 | 1920 |
| Total resource elements over all TBoMS slots | 960 | 960 |

NOTE 1: DM-RS configuration type = 1 with DM-RS duration = single-symbol DM-RS and the number of DM-RS CDM groups without data is 2, Additional DM-RS position = pos1 with , l0 = 0 and l = 8 for PUSCH mapping type B as per table 6.4.1.1.3-3 of TS 38.211 [9].

NOTE 2: Code block size including CRC (bits) equals to K' in sub-clause 5.2.2 of TS 38.212 [15].

# A.3A Fixed Reference Channels for performance requirements (QPSK, R=99/1024)

The parameters for the reference measurement channel are specified in table A.3A-1 for FR1 PUSCH performance requirements:

- FRC parameters are specified in table A.3A-1 for FR1 PUSCH with transform precoding disabled, additional DM-RS position = pos1 and 1 transmission layer.

The parameters for the reference measurement channels are specified in table A.3A-2 to A.3A-3 for FR2 PUSCH performance requirements:

- FRC parameters are specified in table A.3A-2 for FR2 PUSCH with transform precoding disabled, additional DM-RS position = pos0 and 1 transmission layer.

- FRC parameters are specified in table A.3A-3 for FR2 PUSCH with transform precoding disabled, additional DM-RS position = pos1 and 1 transmission layer.

**Table A.3A-1: FRC parameters for FR1 PUSCH performance requirements, transform precoding disabled, additional DM-RS position = pos1 and 1 transmission layer (QPSK, R=99/1024)**

| Reference channel | G-FR1-A3A-1 | G-FR1-A3A-2 | G-FR1-A3A-3 | G-FR1-A3A-4 |
|---|---|---|---|---|
| Subcarrier spacing (kHz) | 15 | 15 | 30 | 30 |
| Allocated resource blocks | 25 | 52 | 24 | 106 |
| Data beraing CP-OFDM Symbols per slot (Note 1) | 12 | 12 | 12 | 12 |
| Modulation | QPSK | QPSK | QPSK | QPSK |
| Code rate (Note 2) | 99/1024 | 99/1024 | 99/1024 | 99/1024 |
| Payload size (bits) | 704 | 1480 | 672 | 2976 |
| Transport block CRC (bits) | 16 | 16 | 16 | 16 |
| Code block CRC size (bits) | - | - | - | - |
| Number of code blocks - C | 1 | 1 | 1 | 1 |
| Code block size including CRC (bits) (Note 2) | 720 | 1496 | 688 | 2992 |
| Total number of bits per slot | 7200 | 14976 | 6912 | 30528 |
| Total resource elements per slot | 3600 | 7488 | 3456 | 15264 |
| NOTE 1: DM-RS configuration type = 1 with DM-RS duration = single-symbol DM-RS and the number of DM-RS CDM groups without data is 2, additional DM-RS position = pos1, $l_0$ = 2 and $l$ = 11 for PUSCH mapping type A, $l_0$ = 0 and $l$ = 10 for PUSCH mapping type B as per table 6.4.1.1.3-3 of TS 38.211 [9].<br>NOTE 2: Code block size including CRC (bits) equals to $K'$ in clause 5.2.2 of TS 38.212 [15]. | | | | |

**Table A.3A-2: FRC parameters for FR2 PUSCH performance requirements, transform precoding disabled, additional DM-RS position = pos0 and 1 transmission layer (QPSK, R=99/1024)**

| Reference channel | G-FR2-A3A-1 | G-FR2-A3A-2 | G-FR2-A3A-3 | G-FR2-A3A-4 |
|---|---|---|---|---|
| Subcarrier spacing [kHz] | 60 | 60 | 120 | 120 |
| Allocated resource blocks | 66 | 132 | 32 | 66 |
| Data bearing CP-OFDM Symbols per slot (Note 1) | 3 | 3 | 3 | 3 |
| Modulation | QPSK | QPSK | QPSK | QPSK |
| Code rate (Note 2) | 99/1024 | 99/1024 | 99/1024 | 99/1024 |
| Payload size (bits) | 456 | 928 | 224 | 456 |
| Transport block CRC (bits) | 16 | 16 | 16 | 16 |
| Code block CRC size (bits) | - | - | - | - |
| Number of code blocks - C | 1 | 1 | 1 | 1 |
| Code block size including CRC (bits) (Note 2) | 472 | 944 | 240 | 472 |
| Total number of bits per slot | 4752 | 9504 | 2304 | 4752 |
| Total symbols per slot | 2376 | 4752 | 1152 | 2376 |
| NOTE 1: DM-RS configuration type = 1 with DM-RS duration = single-symbol DM-RS and the number of DM-RS CDM groups without data is 2, additional DM-RS position = pos0, $l_0$= 0 as per table 6.4.1.1.3-3 of TS 38.211 [9].<br>NOTE 2: Code block size including CRC (bits) equals to $K'$ in clause 5.2.2 of TS 38.212 [15]. | | | | |

**Table A.3A-3: FRC parameters for FR2 PUSCH performance requirements, transform precoding disabled, additional DM-RS position = pos1 and 1 transmission layer (QPSK, R=99/1024)**

| Reference channel | G-FR2-A3A-5 | G-FR2-A3A-6 | G-FR2-A3A-7 | G-FR2-A3A-8 |
|---|---|---|---|---|
| Subcarrier spacing [kHz] | 60 | 60 | 120 | 120 |
| Allocated resource blocks | 66 | 132 | 32 | 66 |
| Data bearing CP-OFDM Symbols per slot (Note 1) | 8 | 8 | 8 | 8 |
| Modulation | QPSK | QPSK | QPSK | QPSK |
| Code rate (Note 2) | 99/1024 | 99/1024 | 99/1024 | 99/1024 |
| Payload size (bits) | 1224 | 2472 | 608 | 1224 |
| Transport block CRC (bits) | 16 | 16 | 16 | 16 |
| Code block CRC size (bits) | - | - | - | - |
| Number of code blocks - C | 1 | 1 | 1 | 1 |
| Code block size including CRC (bits) (Note 2) | 1240 | 2488 | 624 | 1240 |
| Total number of bits per slot | 12672 | 25344 | 6144 | 12672 |
| Total symbols per slot | 6336 | 12672 | 3072 | 6336 |

NOTE 1: DM-RS configuration type = 1 with DM-RS duration = single-symbol DM-RS and the number of DM-RS CDM groups without data is 2, additional DM-RS position = pos1, l0= 0 and l = 8 as per table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: Code block size including CRC (bits) equals to K' in clause 5.2.2 of TS 38.212 [15].

# A.3B Fixed Reference Channels for performance requirements (QPSK, R=308/1024)

The parameters for the reference measurement channel are specified in table A.3B-1 for FR1 PUSCH performance requirements:

- FRC parameters are specified in table A.3B-1 for FR1 PUSCH with transform precoding disabled, additional DM-RS position = pos0 and 1 transmission layer.

The parameters for the reference measurement channel are specified in table A.3B-2 and table A.3B-3 for FR1 PUSCH performance requirements with DM-RS bundling:

- FRC parameters are specified in table A.3B-2 for FR1 PUSCH with transform precoding disabled, additional DM-RS position = pos1 and 1 transmission layer.

- FRC parameters are specified in table A.3B-3 for FR1 PUSCH with transform precoding disabled, additional DM-RS position = pos0 and 1 transmission layer.

The parameters for the reference measurement channel are specified in table A.3B-4 for FR2-1 PUSCH performance requirements with DM-RS bundling:

- FRC parameters are specified in table A.3B-4 for FR2-1 PUSCH with transform precoding disabled, additional DM-RS position = pos1 and 1 transmission layer.

The parameters for the reference measurement channel are specified in table A.3B-5 to table A.3B-7 for FR2-2 PUSCH performance requirements:

- FRC parameters are specified in table A.3B-5 for FR2-2 PUSCH with transform precoding disabled, *Additional DM-RS position = pos1* and 1 transmission layer.

- FRC parameters are specified in table A.3B-6 for FR2-2 PUSCH with transform precoding enabled, *Additional DM-RS position = pos1* and 1 transmission layer.

- FRC parameters are specified in table A.3B-7 for FR2-2 PUSCH with transform precoding disabled, *Additional DM-RS position = pos1* and 2 transmission layers.

**Table A.3B-1: FRC parameters for FR1 PUSCH performance requirements, transform precoding disabled, additional DM-RS position = pos0 and 1 transmission layer (QPSK, R=308/1024)**

| Reference channel | G-FR1-A3B-1 | G-FR1-A3B-2 | G-FR1-A3B-3 | G-FR1-A3B-4 |
|---|---|---|---|---|
| Subcarrier spacing (kHz) | 15 | 15 | 30 | 30 |
| Allocated resource blocks | 25 | 52 | 24 | 106 |
| Data bearing CP-OFDM Symbols per slot (Note 1) | 1 | 1 | 1 | 1 |
| Modulation | QPSK | QPSK | QPSK | QPSK |
| Code rate (Note 2) | 308/1024 | 308/1024 | 308/1024 | 308/1024 |
| Payload size (bits) | 176 | 368 | 168 | 768 |
| Transport block CRC (bits) | 16 | 16 | 16 | 16 |
| Code block CRC size (bits) | - | - | - | - |
| Number of code blocks - C | 1 | 1 | 1 | 1 |
| Code block size including CRC (bits) (Note 2) | 192 | 384 | 184 | 784 |
| Total number of bits per slot | 600 | 1248 | 576 | 2544 |
| Total resource elements per slot | 300 | 624 | 288 | 1272 |

NOTE 1: DM-RS configuration type = 1 with DM-RS duration = single-symbol DM-RS and the number of DM-RS CDM groups without data is 2, additional DM-RS position = pos0, $l_0$ = 0 for PUSCH mapping type B as per table 6.4.1.1.3-3 of TS 38.211 [9].

NOTE 2: Code block size including CRC (bits) equals to *K'* in clause 5.2.2 of TS 38.212 [15].

**Table A.3B-2: FRC parameters for FR1 PUSCH performance requirements, transform precoding disabled, *Additional DM-RS position = pos1* and 1 transmission layer (QPSK, R=308/1024)**

| Reference channel | G-FR1-A3B-5 | G-FR1-A3B-6 |
|---|---|---|
| Subcarrier spacing [kHz] | 15 | 30 |
| Allocated resource blocks | 25 | 24 |
| CP-OFDM Symbols per slot (Note 1) | 12 | 12 |
| Modulation | QPSK | QPSK |
| Code rate (Note 2) | 308/1024 | 308/1024 |
| Payload size (bits) | 2152 | 2088 |
| Transport block CRC (bits) | 16 | 16 |
| Code block CRC size (bits) | - | - |
| Number of code blocks - C | 1 | 1 |
| Code block size including CRC (bits) (Note 2) | 2168 | 2104 |
| Total number of bits per slot | 7200 | 6912 |
| Total resource elements per slot | 3600 | 3456 |

NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos1*, $l_0$ = 2 and *l* = 11 for PUSCH mapping type A, $l_0$= 0 and *l* =10 for PUSCH mapping type B as per Table 6.4.1.1.3-3 of TS 38.211 [9].

NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15].

**Table A.3B-3: FRC parameters for FR1 PUSCH performance requirements, transform precoding disabled, *Additional DM-RS position = pos0* and 1 transmission layer (QPSK, R=308/1024)**

| Reference channel | G-FR1-A3B-7 | G-FR1-A3B-8 |
|---|---|---|
| Subcarrier spacing [kHz] | 15 | 30 |
| Allocated resource blocks | 25 | 24 |
| CP-OFDM Symbols per slot (Note 1) | 13 | 13 |
| Modulation | QPSK | QPSK |
| Code rate (Note 2) | 308/1024 | 308/1024 |
| Payload size (bits) | 2408 | 2280 |
| Transport block CRC (bits) | 16 | 16 |
| Code block CRC size (bits) | - | - |
| Number of code blocks - C | 1 | 1 |
| Code block size including CRC (bits) (Note 2) | 2424 | 2296 |
| Total number of bits per slot | 7800 | 7488 |
| Total resource elements per slot | 3900 | 3744 |
| NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos0* l0 = 2 and l = 11 for PUSCH mapping type A, l0 = 0 and l = 10 for PUSCH mapping type B as per table 6.4.1.1.3-3 of TS 38.211 [9]. | | |
| NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15]. | | |

**Table A.3B-4: FRC parameters for FR2-1 PUSCH performance requirements, transform precoding disabled, *Additional DM-RS position = pos1* and 1 transmission layer (QPSK, R=308/1024)**

| Reference channel | G-FR2-A3B-1 | G-FR2-A3B-2 |
|---|---|---|
| Subcarrier spacing [kHz] | 60 | 120 |
| Allocated resource blocks | 66 | 32 |
| CP-OFDM Symbols per slot (Note 1) | 12 | 12 |
| Modulation | QPSK | QPSK |
| Code rate (Note 2) | 308/1024 | 308/1024 |
| Payload size (bits) | 5632 | 2792 |
| Transport block CRC (bits) | 24 | 16 |
| Code block CRC size (bits) | - | - |
| Number of code blocks - C | 1 | 1 |
| Code block size including CRC (bits) (Note 2) | 5656 | 2808 |
| Total number of bits per slot | 19008 | 9216 |
| Total resource elements per slot | 9504 | 4608 |
| NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos1* with $l_0$= 0 and *l* =10 as per Table 6.4.1.1.3-3 of TS 38.211 [9]. | | |
| NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15]. | | |

**Table A.3B-5: FRC parameters for FR2-2 PUSCH performance requirements, transform precoding disabled, *Additional DM-RS position* = *pos1* and 1 transmission layer (QPSK, R=308/1024)**

| Reference channel | G-FR2-A3B-3 | G-FR2-A3B-4 | G-FR2-A3B-5 |
|---|---|---|---|
| Subcarrier spacing [kHz] | 120 | 120 | 480 |
| Allocated resource blocks | 66 | 264 | 66 |
| CP-OFDM Symbols per slot (Note 1) | 8 | 8 | 8 |
| Modulation | QPSK | QPSK | QPSK |
| Code rate (Note 2) | 308/1024 | 308/1024 | 308/1024 |
| Payload size (bits) | 3824 | 15112 | 3824 |
| Transport block CRC (bits) | 16 | 24 | 16 |
| Code block CRC size (bits) | - | 24 | - |
| Number of code blocks - C | 1 | 2 | 1 |
| Code block size including CRC (bits) (Note 2) | 3840 | 7592 | 3840 |
| Total number of bits per slot without PT-RS | 12672 | 50688 | 12672 |
| Total symbols per slot without PT-RS | 6336 | 25344 | 6336 |

NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos1* with $l_0$= 0 and $l$ =8 as per Table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15].
NOTE 3: The calculation of the "Total number of bits per slot" and "Total symbols per slot" fields include the REs taken up by CSI part 1 and CSI part 2, if present.

**Table A.3B-6: FRC parameters for FR2-2 PUSCH performance requirements, transform precoding enabled, *Additional DM-RS position* = *pos1* and 1 transmission layer (QPSK, R=308/1024)**

| Reference channel | G-FR2-A3B-6 | G-FR2-A3B-7 |
|---|---|---|
| Subcarrier spacing [kHz] | 120 | 480 |
| Allocated resource blocks | 64 | 64 |
| DFT-s-OFDM Symbols per slot (Note 1) | 8 | 8 |
| Modulation | QPSK | QPSK |
| Code rate (Note 2) | 308/1024 | 308/1024 |
| Payload size (bits) | 3752 | 3752 |
| Transport block CRC (bits) | 16 | 16 |
| Code block CRC size (bits) | - | - |
| Number of code blocks - C | 1 | 1 |
| Code block size including CRC (bits) (Note 2) | 3768 | 3768 |
| Total number of bits per slot without PT-RS | 12288 | 12288 |
| Total symbols per slot without PT-RS | 6144 | 6144 |

NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos1* with $l_0$= 0 and $l$ =8 as per Table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15].
NOTE 3: The calculation of the "Total number of bits per slot" and "Total symbols per slot" fields include the REs taken up by CSI part 1 and CSI part 2, if present.

**Table A.3B-7: FRC parameters for FR2-2 PUSCH performance requirements, transform precoding disabled, *Additional DM-RS position = pos1* and 2 transmission layers (QPSK, R=308/1024)**

| Reference channel | G-FR2-A3B-8 | G-FR2-A3B-9 | G-FR2-A3B-10 |
|---|---|---|---|
| Subcarrier spacing [kHz] | 120 | 120 | 480 |
| Allocated resource blocks | 66 | 264 | 66 |
| CP-OFDM Symbols per slot (Note 1) | 8 | 8 | 8 |
| Modulation | QPSK | QPSK | QPSK |
| Code rate (Note 2) | 308/1024 | 308/1024 | 308/1024 |
| Payload size (bits) | 7552 | 30728 | 7552 |
| Transport block CRC (bits) | 24 | 24 | 24 |
| Code block CRC size (bits) | - | 24 | - |
| Number of code blocks - C | 1 | 4 | 1 |
| Code block size including CRC (bits) (Note 2) | 7576 | 7712 | 7576 |
| Total number of bits per slot without PT-RS | 25344 | 101376 | 25344 |
| Total symbols per slot without PT-RS | 12672 | 50688 | 12672 |

NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration = single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position = pos1* with $l_0$= 0 and $l$ =8 as per Table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15].
NOTE 3: The calculation of the "Total number of bits per slot" and "Total symbols per slot" fields include the REs taken up by CSI part 1 and CSI part 2, if present.

# A.4 Fixed Reference Channels for performance requirements (16QAM, R=658/1024)

The parameters for the reference measurement channels are specified in table A.4-2, table A.4-2A, table A.4-2B and table A.4-4 for FR1 PUSCH performance requirements:

- FRC parameters are specified in table A.4-2 for FR1 PUSCH with transform precoding disabled, *Additional DM-RS position = pos1* and 1 transmission layer.
- FRC parameters are specified in table A.4-2A for FR1 PUSCH with transform precoding disabled, additional DM-RS position = pos 2 and 1 transmission layer.
- FRC parameters are specified in table A.4-2B for FR1 PUSCH with transform-precoding disabled, *Additional DM-RS position = pos2* and 1 transmission layer
- FRC parameters are specified in table A.4-4 for FR1 PUSCH with transform precoding disabled, *Additional DM-RS position = pos1* and 2 transmission layers.

The parameters for the reference measurement channels are specified in table A.4-5 to table A.4-8 for FR2-1 PUSCH performance requirements:

- FRC parameters are specified in table A.4-5 for FR2-1 PUSCH with transform precoding disabled, *Additional DM-RS position = pos0* and 1 transmission layer.
- FRC parameters are specified in table A.4-6 for FR2-1 PUSCH with transform precoding disabled, *Additional DM-RS position = pos0* and 2 transmission layers.
- FRC parameters are specified in table A.4-7 for FR2-1 PUSCH with transform precoding disabled, *Additional DM-RS position = pos1* and 1 transmission layer.
- FRC parameters are specified in table A.4-8 for FR2-1 PUSCH with transform precoding disabled, *Additional DM-RS position = pos1* and 2 transmission layers.

The parameters for the reference measurement channels are specified in table A.4-7A for FR2-2 PUSCH performance requirements:

- FRC parameters are specified in table A.4-7A for FR2-2 PUSCH with transform precoding disabled, *Additional DM-RS position = pos1* and 1 transmission layer.

**Table A.4-1: Void**

**Table A.4-2: FRC parameters for FR1 PUSCH performance requirements, transform precoding disabled, *Additional DM-RS position = pos1* and 1 transmission layer (16QAM, R=658/1024)**

| Reference channel | G-FR1-A4-8 | G-FR1-A4-9 | G-FR1-A4-10 | G-FR1-A4-11 (Note 3) | G-FR1-A4-12 | G-FR1-A4-13 | G-FR1-A4-14 |
|---|---|---|---|---|---|---|---|
| Subcarrier spacing [kHz] | 15 | 15 | 15 | 30 | 30 | 30 | 30 |
| Allocated resource blocks | 25 | 52 | 106 | 24 | 51 | 106 | 273 |
| CP-OFDM Symbols per slot (Note 1) | 12 | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM |
| Code rate (Note 2) | 658/1024 | 658/1024 | 658/1024 | 658/1024 | 658/1024 | 658/1024 | 658/1024 |
| Payload size (bits) | 9224 | 19464 | 38936 | 8968 | 18960 | 38936 | 100392 |
| Transport block CRC (bits) | 24 | 24 | 24 | 24 | 24 | 24 | 24 |
| Code block CRC size (bits) | 24 | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks - C | 2 | 3 | 5 | 2 | 3 | 5 | 12 |
| Code block size including CRC (bits) (Note 2) | 4648 | 6520 | 7816 | 4520 | 6352 | 7816 | 8392 |
| Total number of bits per slot | 14400 | 29952 | 61056 | 13824 | 29376 | 61056 | 157248 |
| Total symbols per slot | 3600 | 7488 | 15264 | 3456 | 7344 | 15264 | 39312 |

NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos1*, $l_0$= 2 and $l$=11 for PUSCH mapping type A, $l_0$= 0 and $l$ =10 for PUSCH mapping type B as per table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: Code block size including CRC (bits) equals to *K'* in clause 5.2.2 of TS 38.212 [15].
NOTE 3: The calculation of the "Total number of bits per slot" and "Total symbols per slot" fields include the REs taken up by CSI part 1 and CSI part 2, if present.

**Table A.4-2A: FRC parameters for FR1 PUSCH performance requirements, transform precoding disabled, *Additional DM-RS position = pos2* and 1 transmission layer (16QAM, R=658/1024)**

| Reference channel | G-FR1-A4-29 | G-FR1-A4-29A | G-FR1-A4-30 | G-FR1-A4-30A |
|---|---|---|---|---|
| Subcarrier spacing [kHz] | 15 | 15 | 30 | 30 |
| Allocated resource blocks | 52 | 25 | 106 | 24 |
| Data bearing CP-OFDM Symbols per slot (Note 1) | 11 | 11 | 11 | 11 |
| Modulation | 16QAM | 16QAM | 16QAM | 16QAM |
| Code rate (Note 2) | 658/1024 | 658/1024 | 658/1024 | 658/1024 |
| Payload size (bits) | 17424 | 8456 | 35856 | 8064 |
| Transport block CRC (bits) | 24 | 24 | 24 | 24 |
| Code block CRC size (bits) | 24 | 24 | 24 | - |
| Number of code blocks - C | 3 | 2 | 5 | 1 |
| Code block size including CRC (bits) (Note 2) | 5840 | 4264 | 7200 | 8088 |
| Total number of bits per slot | 27456 | 13200 | 55968 | 12672 |
| Total resource elements per slot | 6864 | 3300 | 13992 | 3168 |

NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos2*, and $l_0$= 2 or 3 for PUSCH mapping type A, as per table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: Code block size including CRC (bits) equals to *K'* in clause 5.2.2 of TS 38.212 [15].

**Table A.4-2B: FRC parameters for FR1 PUSCH performance requirements, transform precoding disabled, *Additional DM-RS position = pos2* and 1 transmission layer (16QAM, R=658/1024)**

| Reference channel | G-FR1-A4-31A | G-FR1-A4-31 | G-FR1-A4-32A | G-FR1-A4-32 |
|---|---|---|---|---|
| Subcarrier spacing [kHz] | 15 | 15 | 30 | 30 |
| Allocated resource blocks | 12 | 25 | 12 | 50 |
| CP-OFDM Symbols per slot (Note 1) | 11 | 11 | 11 | 11 |
| Modulation | 16QAM | 16QAM | 16QAM | 16QAM |
| Code rate (Note 2) | 658/1024 | 658/1024 | 658/1024 | 658/1024 |
| Payload size (bits) | 4032 | 8456 | 4032 | 16896 |
| Transport block CRC (bits) | 24 | 24 | 24 | 24 |
| Code block CRC size (bits) | - | 24 | - | 24 |
| Number of code blocks - C | 1 | 2 | 1 | 3 |
| Code block size including CRC (bits) (Note 2) | 4056 | 4264 | 4056 | 5664 |
| Total number of bits per slot | 6336 | 13200 | 6336 | 26400 |
| Total symbols per slot | 1584 | 3300 | 1584 | 6600 |

NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos2*, $l_0$ = 2 for PUSCH mapping type A, $l_0$ = 2 for PUSCH mapping type B, as per table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: Code block size including CRC (bits) equals to *K'* in clause 5.2.2 of TS 38.212 [15].

**Table A.4-3: Void**

**Table A.4-4: FRC parameters for FR1 PUSCH performance requirements, transform precoding disabled, *Additional DM-RS position = pos1* and 2 transmission layers (16QAM, R=658/1024)**

| Reference channel | G-FR1-A4-22 | G-FR1-A4-23 | G-FR1-A4-24 | G-FR1-A4-25 | G-FR1-A4-26 | G-FR1-A4-27 | G-FR1-A4-28 |
|---|---|---|---|---|---|---|---|
| Subcarrier spacing [kHz] | 15 | 15 | 15 | 30 | 30 | 30 | 30 |
| Allocated resource blocks | 25 | 52 | 106 | 24 | 51 | 106 | 273 |
| CP-OFDM Symbols per slot (Note 1) | 12 | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM |
| Code rate (Note 2) | 658/1024 | 658/1024 | 658/1024 | 658/1024 | 658/1024 | 658/1024 | 658/1024 |
| Payload size (bits) | 18432 | 38936 | 77896 | 17928 | 37896 | 77896 | 200808 |
| Transport block CRC (bits) | 24 | 24 | 24 | 24 | 24 | 24 | 24 |
| Code block CRC size (bits) | 24 | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks - C | 3 | 5 | 10 | 3 | 5 | 10 | 24 |
| Code block size including CRC (bits) (Note 2) | 6176 | 7816 | 7816 | 6008 | 7608 | 7816 | 8392 |
| Total number of bits per slot | 28800 | 59904 | 122112 | 27648 | 58752 | 122112 | 314496 |
| Total symbols per slot | 7200 | 14976 | 30528 | 6912 | 14688 | 30528 | 78624 |

NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos1*, $l_0$= 2 and *l*=11 for PUSCH mapping type A, $l_0$= 0 and *l* =10 for PUSCH mapping type B as per table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: Code block size including CRC (bits) equals to *K'* in clause 5.2.2 of TS 38.212 [15].

**Table A.4-5: FRC parameters for FR2-1 PUSCH performance requirements, transform precoding disabled, *Additional DM-RS position = pos0* and 1 transmission layer (16QAM, R=658/1024)**

| Reference channel | G-FR2-A4-1 | G-FR2-A4-2 | G-FR2-A4-3 (Note 3) | G-FR2-A4-4 | G-FR2-A4-5 |
|---|---|---|---|---|---|
| Subcarrier spacing [kHz] | 60 | 60 | 120 | 120 | 120 |
| Allocated resource blocks | 66 | 132 | 32 | 66 | 132 |
| CP-OFDM Symbols per slot (Note 1) | 9 | 9 | 9 | 9 | 9 |
| Modulation | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM |
| Code rate (Note 2) | 658/1024 | 658/1024 | 658/1024 | 658/1024 | 658/1024 |
| Payload size (bits) | 18432 | 36896 | 8968 | 18432 | 36896 |
| Transport block CRC (bits) | 24 | 24 | 24 | 24 | 24 |
| Code block CRC size (bits) | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks - C | 3 | 5 | 2 | 3 | 5 |
| Code block size including CRC (bits) (Note 2) | 6176 | 7408 | 4520 | 6176 | 7408 |
| Total number of bits per slot without PT-RS | 28512 | 57024 | 13824 | 28512 | 57024 |
| Total number of bits per slot with PT-RS (Note 4) | 27324 | 54648 | 13248 | 27324 | 54648 |
| Total symbols per slot without PT-RS | 7128 | 14256 | 3456 | 7128 | 14256 |
| Total symbols per slot with PT-RS (Note 4) | 6831 | 13662 | 3312 | 6831 | 13662 |

NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos0* with $l_0$= 0 as per Table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15].
NOTE 3: The calculation of the "Total number of bits per slot" and "Total symbols per slot" fields include the REs taken up by CSI part 1 and CSI part 2, if present.
NOTE 4: PT-RS configuration $K_{PT\text{-}RS}$ =2, $L_{PT\text{-}RS}$ =1.

**Table A.4-6: FRC parameters for FR2-1 PUSCH performance requirements, transform precoding disabled, *Additional DM-RS position = pos0* and 2 transmission layers (16QAM, R=658/1024)**

| Reference channel | G-FR2-A4-6 | G-FR2-A4-7 | G-FR2-A4-8 | G-FR2-A4-9 | G-FR2-A4-10 |
|---|---|---|---|---|---|
| Subcarrier spacing [kHz] | 60 | 60 | 120 | 120 | 120 |
| Allocated resource blocks | 66 | 132 | 32 | 66 | 132 |
| CP-OFDM Symbols per slot (Note 1) | 9 | 9 | 9 | 9 | 9 |
| Modulation | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM |
| Code rate (Note 2) | 658/1024 | 658/1024 | 658/1024 | 658/1024 | 658/1024 |
| Payload size (bits) | 36896 | 73776 | 17928 | 36896 | 73776 |
| Transport block CRC (bits) | 24 | 24 | 24 | 24 | 24 |
| Code block CRC size (bits) | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks - C | 5 | 9 | 3 | 5 | 9 |
| Code block size including CRC (bits) (Note 2) | 7408 | 8224 | 6008 | 7408 | 8224 |
| Total number of bits per slot without PT-RS | 57024 | 114048 | 27648 | 57024 | 114048 |
| Total number of bits per slot with PT-RS (Note 3) | 54648 | 109296 | 26496 | 54648 | 109296 |
| Total symbols per slot without PT-RS | 14256 | 28512 | 6912 | 14256 | 28512 |
| Total symbols per slot with PT-RS (Note 3) | 13662 | 27324 | 6624 | 13662 | 27324 |

NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos0* with $l_0$= 0 as per Table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15].
NOTE 3: PT-RS configuration $K_{PT\text{-}RS}$ =2, $L_{PT\text{-}RS}$ =1.

**Table A.4-7: FRC parameters for FR2-1 PUSCH performance requirements, transform precoding disabled, *Additional DM-RS position = pos1* and 1 transmission layer (16QAM, R=658/1024)**

| Reference channel | G-FR2-A4-11 | G-FR2-A4-12 | G-FR2-A4-13 (Note 3) | G-FR2-A4-14 | G-FR2-A4-15 |
|---|---|---|---|---|---|
| Subcarrier spacing [kHz] | 60 | 60 | 120 | 120 | 120 |
| Allocated resource blocks | 66 | 132 | 32 | 66 | 132 |
| CP-OFDM Symbols per slot (Note 1) | 8 | 8 | 8 | 8 | 8 |
| Modulation | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM |
| Code rate (Note 2) | 658/1024 | 658/1024 | 658/1024 | 658/1024 | 658/1024 |
| Payload size (bits) | 16392 | 32776 | 7936 | 16392 | 32776 |
| Transport block CRC (bits) | 24 | 24 | 24 | 24 | 24 |
| Code block CRC size (bits) | 24 | 24 | - | 24 | 24 |
| Number of code blocks - C | 2 | 4 | 1 | 2 | 4 |
| Code block size including CRC (bits) (Note 2) | 8232 | 8224 | 7960 | 8232 | 8224 |
| Total number of bits per slot without PT-RS | 25344 | 50688 | 12288 | 25344 | 50688 |
| Total number of bits per slot with PT-RS (Note 4) | 24288 | 48576 | 11776 | 24288 | 48576 |
| Total symbols per slot without PT-RS | 6336 | 12672 | 3072 | 6336 | 12672 |
| Total symbols per slot with PT-RS (Note 4) | 6072 | 12144 | 2944 | 6072 | 12144 |

NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos1* with $l_0$= 0 and *l* =8 as per Table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15].
NOTE 3: The calculation of the "Total number of bits per slot" and "Total symbols per slot" fields include the REs taken up by CSI part 1 and CSI part 2, if present.
NOTE 4: PT-RS configuration $K_{PT\text{-}RS}$ =2, $L_{PT\text{-}RS}$ =1.

**Table A.4-7A: FRC parameters for FR2-2 PUSCH performance requirements, transform precoding disabled, *Additional DM-RS position = pos1* and 1 transmission layer (16QAM, R=658/1024)**

| Reference channel | G-FR2-A4-21 | G-FR2-A4-22 | G-FR2-A4-23 |
|---|---|---|---|
| Subcarrier spacing [kHz] | 120 | 120 | 480 |
| Allocated resource blocks | 66 | 264 | 66 |
| CP-OFDM Symbols per slot (Note 1) | 8 | 8 | 8 |
| Modulation | 16QAM | 16QAM | 16QAM |
| Code rate (Note 2) | 658/1024 | 658/1024 | 658/1024 |
| Payload size (bits) | 16392 | 65576 | 16392 |
| Transport block CRC (bits) | 24 | 24 | 24 |
| Code block CRC size (bits) | 24 | 24 | 24 |
| Number of code blocks - C | 2 | 8 | 2 |
| Code block size including CRC (bits) (Note 2) | 8232 | 8224 | 8232 |
| Total number of bits per slot without PT-RS | 25344 | 101376 | 25344 |
| Total number of bits per slot with PT-RS (Note 4) | 24288 | 97152 | 24288 |
| Total symbols per slot without PT-RS | 6336 | 25344 | 6336 |
| Total symbols per slot with PT-RS (Note 4) | 6072 | 24288 | 6072 |

NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos1* with $l_0$= 0 and *l* =8 as per Table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15].
NOTE 3: The calculation of the "Total number of bits per slot" and "Total symbols per slot" fields include the REs taken up by CSI part 1 and CSI part 2, if present.
NOTE 4: PT-RS configuration $K_{PT\text{-}RS}$ =2, $L_{PT\text{-}RS}$ =1.

**Table A.4-8: FRC parameters for FR2-1 PUSCH performance requirements, transform precoding disabled, *Additional DM-RS position = pos1* and 2 transmission layers (16QAM, R=658/1024)**

| Reference channel | G-FR2-A4-16 | G-FR2-A4-17 | G-FR2-A4-18 | G-FR2-A4-19 | G-FR2-A4-20 |
|---|---|---|---|---|---|
| Subcarrier spacing [kHz] | 60 | 60 | 120 | 120 | 120 |
| Allocated resource blocks | 66 | 132 | 32 | 66 | 132 |
| CP-OFDM Symbols per slot (Note 1) | 8 | 8 | 8 | 8 | 8 |
| Modulation | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM |
| Code rate (Note 2) | 658/1024 | 658/1024 | 658/1024 | 658/1024 | 658/1024 |
| Payload size (bits) | 32776 | 65576 | 15880 | 32776 | 65576 |
| Transport block CRC (bits) | 24 | 24 | 24 | 24 | 24 |
| Code block CRC size (bits) | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks - C | 4 | 8 | 2 | 4 | 8 |
| Code block size including CRC (bits) (Note 2) | 8224 | 8224 | 7976 | 8224 | 8224 |
| Total number of bits per slot without PT-RS | 50688 | 101376 | 24576 | 50688 | 101376 |
| Total number of bits per slot with PT-RS (Note 3) | 48576 | 97152 | 23552 | 48576 | 97152 |
| Total symbols per slot without PT-RS | 12672 | 25344 | 6144 | 12672 | 25344 |
| Total symbols per slot with PT-RS (Note 3) | 12144 | 24288 | 5888 | 12144 | 24288 |
| NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos1* with $l_0$= 0 and $l$ =8 as per Table 6.4.1.1.3-3 of TS 38.211 [9]. | | | | | |
| NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15]. | | | | | |
| NOTE 3: PT-RS configuration $K_{PT-RS}$ =2, $L_{PT-RS}$ =1. | | | | | |

**Table A.4-8A: Void**

# A.5 Fixed Reference Channels for performance requirements (64QAM, R=567/1024)

The parameters for the reference measurement channels are specified in table A.5-2 and table A.5-5 for FR1 PUSCH performance requirements:

- FRC parameters are specified in table A.5-2 for FR1 PUSCH with transform precoding disabled, *Additional DM-RS position = pos1* and 1 transmission layer.
- FRC parameters are specified in table A.5-5 for FR1 interlaced PUSCH with transform precoding disabled, *Additional DM-RS position = pos1* and 1 transmission layer.

The parameters for the reference measurement channels are specified in table A.5-3 to table A.5-4 for FR2-1 PUSCH performance requirements:

- FRC parameters are specified in table A.5-3 for FR2-1 PUSCH with transform precoding disabled, *Additional DM-RS position = pos0* and 1 transmission layer.
- FRC parameters are specified in table A.5-4 for FR2-1 PUSCH with transform precoding disabled, *Additional DM-RS position = pos1* and 1 transmission layer.

The parameters for the reference measurement channels are specified in table A.5-4A for FR2-2 PUSCH performance requirements:

- FRC parameters are specified in table A.5-4A for FR2-2 PUSCH with transform precoding disabled, *Additional DM-RS position = pos1* and 1 transmission layer.

**Table A.5-1: Void**

**Table A.5-2: FRC parameters for FR1 PUSCH performance requirements, transform precoding disabled, *Additional DM-RS position = pos1* and 1 transmission layer (64QAM, R=567/1024)**

| Reference channel | G-FR1-A5-8 | G-FR1-A5-9 | G-FR1-A5-10 | G-FR1-A5-11 | G-FR1-A5-12 | G-FR1-A5-13 | G-FR1-A5-14 |
|---|---|---|---|---|---|---|---|
| Subcarrier spacing [kHz] | 15 | 15 | 15 | 30 | 30 | 30 | 30 |
| Allocated resource blocks | 25 | 52 | 106 | 24 | 51 | 106 | 273 |
| CP-OFDM Symbols per slot (Note 1) | 12 | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | 64QAM | 64QAM | 64QAM | 64QAM | 64QAM | 64QAM | 64QAM |
| Code rate (Note 2) | 567/1024 | 567/1024 | 567/1024 | 567/1024 | 567/1024 | 567/1024 | 567/1024 |
| Payload size (bits) | 12040 | 25104 | 50184 | 11528 | 24576 | 50184 | 131176 |
| Transport block CRC (bits) | 24 | 24 | 24 | 24 | 24 | 24 | 24 |
| Code block CRC size (bits) | 24 | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks - C | 2 | 3 | 6 | 2 | 3 | 6 | 16 |
| Code block size including CRC (bits) (Note 2) | 6056 | 8400 | 8392 | 5800 | 8224 | 8392 | 8224 |
| Total number of bits per slot | 21600 | 44928 | 91584 | 20736 | 44064 | 91584 | 235872 |
| Total symbols per slot | 3600 | 7488 | 15264 | 3456 | 7344 | 15264 | 39312 |

NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos1*, $l_0$= 2 and *l* =11 for PUSCH mapping type A, $l_0$= 0 and *l* =10 for PUSCH mapping type B as per table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: Code block size including CRC (bits) equals to *K'* in clause 5.2.2 of TS 38.212 [15].

**Table A.5-3: FRC parameters for FR2-1 PUSCH performance requirements, transform precoding disabled, *Additional DM-RS position = pos0* and 1 transmission layer (64QAM, R=567/1024)**

| Reference channel | G-FR2-A5-1 | G-FR2-A5-2 | G-FR2-A5-3 | G-FR2-A5-4 | G-FR2-A5-5 |
|---|---|---|---|---|---|
| Subcarrier spacing [kHz] | 60 | 60 | 120 | 120 | 120 |
| Allocated resource blocks | 66 | 132 | 32 | 66 | 132 |
| CP-OFDM Symbols per slot (Note 1) | 9 | 9 | 9 | 9 | 9 |
| Modulation | 64QAM | 64QAM | 64QAM | 64QAM | 64QAM |
| Code rate (Note 2) | 567/1024 | 567/1024 | 567/1024 | 567/1024 | 567/1024 |
| Payload size (bits) | 23568 | 47112 | 11528 | 23568 | 47112 |
| Transport block CRC (bits) | 24 | 24 | 24 | 24 | 24 |
| Code block CRC size (bits) | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks - C | 3 | 6 | 2 | 3 | 6 |
| Code block size including CRC (bits) (Note 2) | 7888 | 7880 | 5800 | 7888 | 7880 |
| Total number of bits per slot without PT-RS | 42768 | 85536 | 20736 | 42768 | 85536 |
| Total number of bits per slot with PT-RS (Note 3) | 40986 | 81972 | 19872 | 40986 | 81972 |
| Total symbols per slot without PT-RS | 7128 | 14256 | 3456 | 7128 | 14256 |
| Total symbols per slot with PT-RS (Note 3) | 6831 | 13662 | 3312 | 6831 | 13662 |

NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos0* with $l_0$= 0 as per Table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15].
NOTE 3: PT-RS configuration $K_{PT\text{-}RS}$ =2, $L_{PT\text{-}RS}$ =1.

**Table A.5-4: FRC parameters for FR2-1 PUSCH performance requirements, transform precoding disabled, *Additional DM-RS position = pos1* and 1 transmission layer (64QAM, R=567/1024)**

| Reference channel | G-FR2-A5-6 | G-FR2-A5-7 | G-FR2-A5-8 | G-FR2-A5-9 | G-FR2-A5-10 |
|---|---|---|---|---|---|
| Subcarrier spacing [kHz] | 60 | 60 | 120 | 120 | 120 |
| Allocated resource blocks | 66 | 132 | 32 | 66 | 132 |
| CP-OFDM Symbols per slot (Note 1) | 8 | 8 | 8 | 8 | 8 |
| Modulation | 64QAM | 64QAM | 64QAM | 64QAM | 64QAM |
| Code rate (Note 2) | 567/1024 | 567/1024 | 567/1024 | 567/1024 | 567/1024 |
| Payload size (bits) | 21000 | 42016 | 10248 | 21000 | 42016 |
| Transport block CRC (bits) | 24 | 24 | 24 | 24 | 24 |
| Code block CRC size (bits) | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks - C | 3 | 5 | 2 | 3 | 5 |
| Code block size including CRC (bits) (Note 2) | 7032 | 8432 | 5160 | 7032 | 8432 |
| Total number of bits per slot without PT-RS | 38016 | 76032 | 18432 | 38016 | 76032 |
| Total number of bits per slot with PT-RS (Note 3) | 36432 | 72864 | 17664 | 36432 | 72864 |
| Total symbols per slot without PT-RS | 6336 | 12672 | 3072 | 6336 | 12672 |
| Total symbols per slot with PT-RS (Note 3) | 6072 | 12144 | 2944 | 6072 | 12144 |

NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos1* with $l_0$= 0 and $l$ =8 as per Table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15].
NOTE 3: PT-RS configuration $K_{PT\text{-}RS}$ =2, $L_{PT\text{-}RS}$ =1.

**Table A.5-4A: FRC parameters for FR2-2 PUSCH performance requirements, transform precoding disabled, *Additional DM-RS position = pos1* and 1 transmission layer (64QAM, R=567/1024)**

| Reference channel | G-FR2-A5-11 | G-FR2-A5-12 | G-FR2-A5-13 |
|---|---|---|---|
| Subcarrier spacing [kHz] | 120 | 120 | 480 |
| Allocated resource blocks | 66 | 264 | 66 |
| CP-OFDM Symbols per slot (Note 1) | 8 | 8 | 8 |
| Modulation | 64QAM | 64QAM | 64QAM |
| Code rate (Note 2) | 567/1024 | 567/1024 | 567/1024 |
| Payload size (bits) | 21000 | 83976 | 21000 |
| Transport block CRC (bits) | 24 | 24 | 24 |
| Code block CRC size (bits) | 24 | 24 | 24 |
| Number of code blocks - C | 3 | 10 | 3 |
| Code block size including CRC (bits) (Note 2) | 7032 | 8424 | 7032 |
| Total number of bits per slot without PT-RS | 38016 | 152064 | 38016 |
| Total number of bits per slot with PT-RS (Note 3) | 36432 | 145728 | 36432 |
| Total symbols per slot without PT-RS | 6336 | 25344 | 6336 |
| Total symbols per slot with PT-RS (Note 3) | 6072 | 24288 | 6072 |

NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos1* with $l_0$= 0 and $l$ =8 as per Table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15].
NOTE 3: PT-RS configuration $K_{PT\text{-}RS}$ =2, $L_{PT\text{-}RS}$ =1.

**Table A.5-5: FRC parameters for FR1 interlaced PUSCH performance requirements, transform precoding disabled, *additional DM-RS position* = *pos1* and 1 transmission layer (64QAM, R=567/1024)**

| Reference channel | G-FR1-A5-15 | G-FR1-A5-16 |
|---|---|---|
| Subcarrier spacing [kHz] | 15 | 30 |
| Allocated resource blocks | 11 | 11 |
| CP-OFDM Symbols per slot (Note 1) | 12 | 12 |
| Modulation | 64QAM | 64QAM |
| Code rate | 567/1024 | 567/1024 |
| Payload size (bits) | 5248 | 5248 |
| Transport block CRC (bits) | 24 | 24 |
| Code block CRC size (bits) | - | - |
| Number of code blocks - C | 1 | 1 |
| Code block size including CRC (bits) (Note 2) | 5272 | 5272 |
| Total number of bits per slot (Note 3) | 9504 | 9504 |
| Total symbols per slot (Note 3) | 1584 | 1584 |
| NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos1*, $l_0$= 2 and $l$ =11 for PUSCH mapping type A, $l_0$= 0 and $l$ =10 for PUSCH mapping type B as per table 6.4.1.1.3-3 of TS 38.211 [9]. | | |
| NOTE 2: Code block size including CRC (bits) equals to *K'* in clause 5.2.2 of TS 38.212 [15]. | | |
| NOTE 3: The calculation of the "Total number of bits per slot" and "Total symbols per slot" fields include the REs taken up by CG-UCI, if present. | | |

# A.6 PRACH Test preambles

**Table A.6-1: Test preambles for Normal Mode in FR1**

| Burst format | SCS (kHz) | Ncs | Logical sequence index | v |
|---|---|---|---|---|
| 0 | 1.25 | 13 | 22 | 32 |
| A1, A2, A3, B4, C0, C2 | 15 | 23 | 0 | 0 |
| | 30 | 46 | 0 | 0 |

**Table A.6-2: Test preambles for Normal Mode in FR2**

| Burst format | SCS (kHz) | Ncs | Logical sequence index | v |
|---|---|---|---|---|
| A1, A2, A3, B4, C0, C2 | 60 | 69 | 0 | 0 |
| | 120 | 69 | 0 | 0 |

**Table A.6-3: Test preambles for high speed train restricted set type A**

| Burst format | SCS (kHz) | Ncs | Logical sequence index | v |
|---|---|---|---|---|
| 0 | 1.25 | 15 | 384 | 0 |

**Table A.6-4: Test preambles for high speed train restricted set type B**

| Burst format | SCS (kHz) | Ncs | Logical sequence index | v |
|---|---|---|---|---|
| 0 | 1.25 | 15 | 30 | 30 |

**Table A.6-5: Test preambles for high speed train short formats in FR1**

| Burst format | SCS (kHz) | Ncs | Logical sequence index | v |
|---|---|---|---|---|
| A2, B4, C2 | 15 | 23 | 0 | 0 |
| | 30 | 46 | 0 | 0 |

**Table A.6-6: Test preambles for PRACH with $L_{RA}$=1151 and $L_{RA}$=571**

| Burst format | SCS (kHz) | Ncs | Logical sequence index | v |
|---|---|---|---|---|
| A2, B4, C2 | 15 | 164 | 0 | 0 |
| | 30 | 190 | 0 | 0 |

**Table A.6-7: Test preambles for high speed train short formats in FR2**

| Burst format | SCS (kHz) | Ncs | Logical sequence index | v |
|---|---|---|---|---|
| C2 | 120 | 0 | 0 | 0 |

**Table A.6-8: Test preambles for PRACH with $L_{RA}$=139, $L_{RA}$=571 and $L_{RA}$=1151 for 120 kHZ and 480 kHz SCS**

| Burst format | SCS (kHz) | $L_{RA}$ | Ncs | Logical sequence index | v |
|---|---|---|---|---|---|
| A2<br>B4, C2 | 120 | 571 | 285 | 0 | 0 |
| | 120 | 1151 | 575 | 0 | 0 |
| | 480 | 139 | 69 | 0 | 0 |
| | 480 | 571 | 285 | 0 | 0 |

# A.7 Fixed Reference Channels for performance requirements (16QAM, R=434/1024)

The parameters for the reference measurement channels are specified in table A.7-1 for FR2-1 PUSCH performance requirements with transform precoding disabled, additional DM-RS position = pos0 and 2 transmission layers.

The parameters for the reference measurement channels are specified in table A.7-2 for FR2-1 PUSCH performance requirements with transform precoding disabled, additional DM-RS position = pos1 and 2 transmission layers.

The parameters for the reference measurement channels are specified in table A.7-3 for FR2-2 PUSCH performance requirements with transform precoding disabled, additional DM-RS position = pos1 and 2 transmission layers.

**Table A.7-1: FRC parameters for FR2-1 PUSCH performance requirements, transform precoding disabled, Additional DM-RS position = pos0 and 2 transmission layers (16QAM, R=434/1024)**

| Reference channel | G-FR2-A7-1 | G-FR2-A7-2 | G-FR2-A7-3 | G-FR2-A7-4 | G-FR2-A7-5 |
|---|---|---|---|---|---|
| Subcarrier spacing [kHz] | 60 | 60 | 120 | 120 | 120 |
| Allocated resource blocks | 66 | 132 | 32 | 66 | 132 |
| CP-OFDM Symbols per slot (Note 1) | 9 | 9 | 9 | 9 | 9 |
| Modulation | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM |
| Code rate (Note 2) | 434/1024 | 434/1024 | 434/1024 | 434/1024 | 434/1024 |
| Payload size (bits) | 24072 | 48168 | 11784 | 24072 | 48168 |
| Transport block CRC (bits) | 24 | 24 | 24 | 24 | 24 |
| Code block CRC size (bits) | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks - C | 3 | 6 | 2 | 3 | 6 |
| Code block size including CRC (bits) (Note 2) | 8056 | 8056 | 5928 | 8056 | 8056 |
| Total number of bits per slot without PT-RS | 57024 | 114048 | 27648 | 57024 | 114048 |
| Total number of bits per slot with PT-RS (Note 3) | 54648 | 109296 | 26496 | 54648 | 109296 |
| Total symbols per slot without PT-RS | 14256 | 28512 | 6912 | 14256 | 28512 |
| Total symbols per slot with PT-RS (Note 3) | 13662 | 27324 | 6624 | 13662 | 27324 |

NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos0* with $l_0$= 0 as per Table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15].
NOTE 3: PT-RS configuration $K_{PT\text{-}RS}$ =2, $L_{PT\text{-}RS}$ =1.

**Table A.7-2: FRC parameters for FR2-1 PUSCH performance requirements, transform precoding disabled, Additional DM-RS position = pos1 and 2 transmission layers (16QAM, R=434/1024)**

| Reference channel | G-FR2-A7-6 | G-FR2-A7-7 | G-FR2-A7-8 | G-FR2-A7-9 | G-FR2-A7-10 |
|---|---|---|---|---|---|
| Subcarrier spacing [kHz] | 60 | 60 | 120 | 120 | 120 |
| Allocated resource blocks | 66 | 132 | 32 | 66 | 132 |
| CP-OFDM Symbols per slot (Note 1) | 8 | 8 | 8 | 8 | 8 |
| Modulation | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM |
| Code rate (Note 2) | 434/1024 | 434/1024 | 434/1024 | 434/1024 | 434/1024 |
| Payload size (bits) | 21504 | 43032 | 10504 | 21504 | 43032 |
| Transport block CRC (bits) | 24 | 24 | 24 | 24 | 24 |
| Code block CRC size (bits) | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks - C | 3 | 6 | 2 | 3 | 6 |
| Code block size including CRC (bits) (Note 2) | 7200 | 7200 | 5288 | 7200 | 7200 |
| Total number of bits per slot without PT-RS | 50688 | 101376 | 24576 | 50688 | 101376 |
| Total number of bits per slot with PT-RS (Note 3) | 48576 | 97152 | 23552 | 48576 | 97152 |
| Total symbols per slot without PT-RS | 12672 | 25344 | 6144 | 12672 | 25344 |
| Total symbols per slot with PT-RS (Note 3) | 12144 | 24288 | 5888 | 12144 | 24288 |

NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos1* with $l_0$ = 0 and $l$ = 8 as per Table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: Code block size including CRC (bits) equals to *K'* in clause 5.2.2 of TS 38.212 [15].
NOTE 3: PT-RS configuration $K_{PT\text{-}RS}$ =2, $L_{PT\text{-}RS}$ =1.

**Table A.7-3: FRC parameters for FR2-2 PUSCH performance requirements, transform precoding disabled, *Additional DM-RS position* = *pos1* and 2 transmission layers (16QAM, R=434/1024)**

| Reference channel | G-FR2-A7-11 | G-FR2-A7-12 | G-FR2-A7-13 |
|---|---|---|---|
| Subcarrier spacing [kHz] | 120 | 120 | 480 |
| Allocated resource blocks | 66 | 264 | 66 |
| CP-OFDM Symbols per slot (Note 1) | 8 | 8 | 8 |
| Modulation | 16QAM | 16QAM | 16QAM |
| Code rate (Note 2) | 434/1024 | 434/1024 | 434/1024 |
| Payload size (bits) | 21504 | 86040 | 21504 |
| Transport block CRC (bits) | 24 | 24 | 24 |
| Code block CRC size (bits) | 24 | 24 | 24 |
| Number of code blocks - C | 3 | 11 | 3 |
| Code block size including CRC (bits) (Note 2) | 7200 | 7848 | 7200 |
| Total number of bits per slot without PT-RS | 50688 | 202752 | 50688 |
| Total number of bits per slot with PT-RS (Note 4) | 48576 | 194304 | 48576 |
| Total symbols per slot without PT-RS | 12672 | 50688 | 12672 |
| Total symbols per slot with PT-RS (Note 4) | 12144 | 48576 | 12144 |
| NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos1* with $l_0$= 0 and $l$ =8 as per Table 6.4.1.1.3-3 of TS 38.211 [9].<br>NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15].<br>NOTE 3: The calculation of the "Total number of bits per slot" and "Total symbols per slot" fields include the REs taken up by CSI part 1 and CSI part 2, if present.<br>NOTE 4: PT-RS configuration $K_{PT\text{-}RS}$ =2, $L_{PT\text{-}RS}$ =1. | | | |

# A.8 Fixed Reference Channels for performance requirements (QPSK, R=157/1024)

Note: Different FRC numbers are assigned in TS 38.141-1 [5] for the FRCs in this annex.

The parameters for the reference measurement channels are specified in table A.8-1 and A.8-2 for FR1 PUSCH performance requirements for 2-step RA type with *Additional DM-RS position* equals to *pos2* and *pos1* respectively.

**Table A.8-1: FRC parameters for FR1 PUSCH performance requirements, transform precoding disabled, Additional DM-RS position = pos2 and 1 transmission layer (QPSK, R=157/1024)**

| Reference channel | G-FR1-A8-1 | G-FR1-A8-2 |
|---|---|---|
| Subcarrier spacing [kHz] | 15 | 30 |
| Allocated resource blocks | 2 | 2 |
| CP-OFDM Symbols per slot (Note 1) | 11 | 11 |
| Modulation | QPSK | QPSK |
| Code rate (Note 2) | 157/1024 | 157/1024 |
| Payload size (bits) | 80 | 80 |
| Transport block CRC (bits) | 16 | 16 |
| Code block CRC size (bits) | 0 | 0 |
| Number of code blocks - C | 1 | 1 |
| Code block size including CRC (bits) (Note 2) | 96 | 96 |
| Total number of bits per slot | 528 | 528 |
| Total symbols per slot | 264 | 264 |
| NOTE 1: DM-RS configuration type = 1 with DM-RS duration = single-symbol DM-RS and the number of DM-RS CDM groups without data is 2, Additional DM-RS position = pos2 with l0= 2 as per Table 6.4.1.1.3-3 of TS 38.211 [9].<br>NOTE 2: Code block size including CRC (bits) equals to K' in clause 5.2.2 of TS 38.212 [15]. | | |

**Table A.8-2: FRC parameters for FR1 PUSCH performance requirements, transform precoding disabled, Additional DM-RS position = pos1 and 1 transmission layer (QPSK, R=157/1024)**

| Reference channel | G-FR1-A8-3 | G-FR1-A8-4 |
|---|---|---|
| Subcarrier spacing [kHz] | 15 | 30 |
| Allocated resource blocks | 2 | 2 |
| CP-OFDM Symbols per slot (Note 1) | 12 | 12 |
| Modulation | QPSK | QPSK |
| Code rate (Note 2) | 157/1024 | 157/1024 |
| Payload size (bits) | 88 | 88 |
| Transport block CRC (bits) | 16 | 16 |
| Code block CRC size (bits) | 0 | 0 |
| Number of code blocks - C | 1 | 1 |
| Code block size including CRC (bits) (Note 2) | 104 | 104 |
| Total number of bits per slot | 576 | 576 |
| Total symbols per slot | 288 | 288 |
| NOTE 1: DM-RS configuration type = 1 with DM-RS duration = single-symbol DM-RS and the number of DM-RS CDM groups without data is 2, Additional DM-RS position = pos1 with l0= 2 as per Table 6.4.1.1.3-3 of TS 38.211 [9].<br>NOTE 2: Code block size including CRC (bits) equals to K' in clause 5.2.2 of TS 38.212 [15]. | | |

# A.9 Fixed Reference Channels for performance requirements (256QAM, R=682.5/1024)

Note: Different FRC numbers are assigned in TS 38.141-1 [5] for the FRCs in this annex.

The parameters for the reference measurement channels are specified in table A.9-1 for FR1 PUSCH performance requirements:

- FRC parameters are specified in table A.9-1 for FR1 PUSCH with transform precoding disabled, *Additional DM-RS position = pos1* and 1 transmission layer.

**Table A.9-1: FRC parameters for FR1 PUSCH performance requirements, transform precoding disabled, Additional DM-RS position = pos1 and 1 transmission layer (256QAM, R=682.5/1024)**

| Reference channel | G-FR1-A9-1 | G-FR1-A9-2 | G-FR1-A9-3 | G-FR1-A9-4 | G-FR1-A9-5 |
|---|---|---|---|---|---|
| Subcarrier spacing [kHz] | 15 | 15 | 30 | 30 | 30 |
| Allocated resource blocks | 25 | 52 | 24 | 106 | 273 |
| CP-OFDM Symbols per slot (Note 1) | 12 | 12 | 12 | 12 | 12 |
| Modulation | 256QAM | 256QAM | 256QAM | 256QAM | 256QAM |
| Code rate (Note 2) | 682/1024 | 682/1024 | 682/1024 | 682/1024 | 682/1024 |
| Payload size (bits) | 18960 | 39936 | 18432 | 81976 | 208976 |
| Transport block CRC (bits) | 24 | 24 | 24 | 24 | 24 |
| Code block CRC size (bits) | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks - C | 3 | 5 | 3 | 10 | 25 |
| Code block size including CRC (bits) (Note 2) | 6352 | 8016 | 6176 | 8224 | 8384 |
| Total number of bits per slot | 28800 | 59904 | 27648 | 122112 | 314496 |
| Total symbols per slot | 3600 | 7488 | 3456 | 15264 | 39312 |
| NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos1*, $l_0$= 2 and *l* =11 for PUSCH mapping type A, $l_0$= 0 and *l* =10 for PUSCH mapping type B as per table 6.4.1.1.3-3 of TS 38.211 [5].<br>NOTE 2: Code block size including CRC (bits) equals to *K'* in clause 5.2.2 of TS 38.212 [15]. | | | | | |

# A.10 Fixed Reference Channels for performance requirements (64QAM, R=517/1024)

The parameters for the reference measurement channels are specified in table A.10-1, A.10-2 and A.10-3 for FR2 PUSCH performance requirements:

- FRC parameters are specified in table A.10-1 for FR2 PUSCH with transform precoding disabled, *Additional DM-RS position = pos0* and 1 transmission layer.
- FRC parameters are specified in table A.10-2 for FR2 PUSCH with transform precoding disabled, *Additional DM-RS position = pos1* and 1 transmission layer.
- FRC parameters are specified in table A.10-3 for FR2 PUSCH with transform precoding disabled, *Additional DM-RS position = pos2* and 1 transmission layer.

The parameters for the reference measurement channels are specified in table A.10-4, table A.10-5 and table A.10-6 for FR2 PUSCH UL timing adjustment perfromance requirements:

- FRC parameters are specified in table A.10-4 for FR2 UL timing adjustment requirements, PUSCH with transform precoding disabled, *Additional DM-RS position = pos0* and 1 transmission layer.
- FRC parameters are specified in table A.10-5 for FR2 UL timing adjustment requirements, PUSCH with transform precoding disabled, *DM-RS position = pos1* and 1 transmission layer.
- FRC parameters are specified in table A.10-6 for FR2 UL timing adjustment requirements, PUSCH with transform precoding disabled, *Additional Additional DM-RS position = pos2* and 1 transmission layer.

**Table A.10-1: FRC parameters for FR2 PUSCH performance requirements, transform precoding disabled, Additional DM-RS position = pos0 and 1 transmission layer (64QAM, R=517/1024)**

| Reference channel | G-FR2-A10-1 | G-FR2-A10-2 |
|---|---|---|
| Subcarrier spacing [kHz] | 120 | 120 |
| Allocated resource blocks | 32 | 132 |
| Data bearing CP-OFDM Symbols per slot (Note 1) | 9 | 9 |
| Modulation | 64QAM | 64QAM |
| Code rate (Note 2) | 517/1024 | 517/1024 |
| Payload size (bits) | 10504 | 43032 |
| Transport block CRC (bits) | 24 | 24 |
| Code block CRC size (bits) | 24 | 24 |
| Number of code blocks - C | 2 | 6 |
| Code block size including CRC (bits) (Note 2) | 5288 | 7200 |
| Total number of bits per slot without PT-RS | 20736 | 85536 |
| Total number of bits per slot with PT-RS (Note 3) | 19872 | 81972 |
| Total resource elements per slot without PT-RS | 3456 | 14256 |
| Total resource elements per slot with PT-RS (Note 3) | 3312 | 13662 |
| NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos0* with $l_0$= 0 as per Table 6.4.1.1.3-3 of TS 38.211 [9].<br>NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15].<br>NOTE 3: PT-RS configuration $K_{PT\text{-}RS}$ =2, $L_{PT\text{-}RS}$ =1. | | |

**Table A.10-2: FRC parameters for FR2 PUSCH performance requirements, transform precoding disabled, Additional DM-RS position = pos1 and 1 transmission layer (64QAM, R=517/1024)**

| Reference channel | G-FR2-A10-3 | G-FR2-A10-4 |
|---|---|---|
| Subcarrier spacing [kHz] | 120 | 120 |
| Allocated resource blocks | 32 | 132 |
| Data bearing CP-OFDM Symbols per slot (Note 1) | 8 | 8 |
| Modulation | 64QAM | 64QAM |
| Code rate (Note 2) | 517/1024 | 517/1024 |
| Payload size (bits) | 9224 | 37896 |
| Transport block CRC (bits) | 24 | 24 |
| Code block CRC size (bits) | 24 | 24 |
| Number of code blocks - C | 2 | 5 |
| Code block size including CRC (bits) (Note 2) | 4648 | 7608 |
| Total number of bits per slot without PT-RS | 18432 | 76032 |
| Total number of bits per slot with PT-RS (Note 3) | 17664 | 72864 |
| Total resource elements per slot without PT-RS | 3072 | 12672 |
| Total resource elements per slot with PT-RS (Note 3) | 2944 | 12144 |
| NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos1* with $l_0$= 0 and *l* =8 as per Table 6.4.1.1.3-3 of TS 38.211 [9].<br>NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15].<br>NOTE 3: PT-RS configuration $K_{PT\text{-}RS}$ =2, $L_{PT\text{-}RS}$ =1. | | |

**Table A.10-3: FRC parameters for FR2 PUSCH performance requirements, transform precoding disabled, Additional DM-RS position = pos2 and 1 transmission layer (64QAM, R=517/1024)**

| Reference channel | G-FR2-A10-5 | G-FR2-A10-6 |
|---|---|---|
| Subcarrier spacing [kHz] | 120 | 120 |
| Allocated resource blocks | 32 | 132 |
| Data bearing CP-OFDM Symbols per slot (Note 1) | 7 | 7 |
| Modulation | 64QAM | 64QAM |
| Code rate (Note 2) | 517/1024 | 517/1024 |
| Payload size (bits) | 8064 | 33816 |
| Transport block CRC (bits) | 24 | 24 |
| Code block CRC size (bits) | - | 24 |
| Number of code blocks - C | 1 | 5 |
| Code block size including CRC (bits) (Note 2) | 8088 | 6792 |
| Total number of bits per slot without PT-RS | 16128 | 66528 |
| Total number of bits per slot with PT-RS (Note 3) | 15456 | 63756 |
| Total resource elements per slot without PT-RS | 2688 | 11088 |
| Total resource elements per slot with PT-RS (Note 3) | 2576 | 10626 |

NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos2* with $l_0$= 0 and $l$ =4,8 as per Table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15].
NOTE 3: PT-RS configuration $K_{PT\text{-}RS}$ =2, $L_{PT\text{-}RS}$ =1.

**Table A.10-4: FRC parameters for FR2 UL timing adjustment requirements, PUSCH with transform precoding disabled, Additional DM-RS position = pos0 and 1 transmission layer (64QAM, R=517/1024)**

| Reference channel | G-FR2-A10-7 | G-FR2-A10-8 |
|---|---|---|
| Subcarrier spacing [kHz] | 120 | 120 |
| Allocated resource blocks | 16 | 66 |
| Data bearing CP-OFDM Symbols per slot (Note 1) | 9 | 9 |
| Modulation | 64QAM | 64QAM |
| Code rate (Note 2) | 517/1024 | 517/1024 |
| Payload size (bits) | 5248 | 21504 |
| Transport block CRC (bits) | 24 | 24 |
| Code block CRC size (bits) | - | 24 |
| Number of code blocks - C | 1 | 3 |
| Code block size including CRC (bits) (Note 2) | 5272 | 7200 |
| Total number of bits per slot without PT-RS | 10368 | 42768 |
| Total number of bits per slot with PT-RS (Note 3) | 9936 | 40986 |
| Total resource elements per slot without PT-RS | 1728 | 7128 |
| Total resource elements per slot with PT-RS (Note 3) | 1656 | 6831 |

NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos0* with $l_0$= 0 as per Table 6.4.1.1.3-3 of TS 38.211 [9].
NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15].
NOTE 3: PT-RS configuration $K_{PT\text{-}RS}$ =2, $L_{PT\text{-}RS}$ =1.

**Table A.10-5: FRC parameters for FR2 UL timing adjustment requirements, PUSCH with transform precoding disabled, Additional DM-RS position = pos1 and 1 transmission layer (64QAM, R=517/1024)**

| Reference channel | G-FR2-A10-9 | G-FR2-A10-10 |
|---|---|---|
| Subcarrier spacing [kHz] | 120 | 120 |
| Allocated resource blocks | 16 | 66 |
| Data bearing CP-OFDM Symbols per slot (Note 1) | 8 | 8 |
| Modulation | 64QAM | 64QAM |
| Code rate (Note 2) | 517/1024 | 517/1024 |
| Payload size (bits) | 4608 | 18960 |
| Transport block CRC (bits) | 24 | 24 |
| Code block CRC size (bits) | - | 24 |
| Number of code blocks - C | 1 | 3 |
| Code block size including CRC (bits) (Note 2) | 4632 | 6352 |
| Total number of bits per slot without PT-RS | 9216 | 38016 |
| Total number of bits per slot with PT-RS (Note 3) | 8832 | 36432 |
| Total resource elements per slot without PT-RS | 1536 | 6336 |
| Total resource elements per slot with PT-RS (Note 3) | 1472 | 6072 |
| NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos1* with $l_0$= 0 and $l$ =8 as per Table 6.4.1.1.3-3 of TS 38.211 [9]. | | |
| NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15]. | | |
| NOTE 3: PT-RS configuration $K_{PT-RS}$ =2, $L_{PT-RS}$ =1. | | |

**Table A.10-6: FRC parameters for FR2 UL timing adjustment requirements, PUSCH with transform precoding disabled, Additional DM-RS position = pos2 and 1 transmission layer (64QAM, R=517/1024)**

| Reference channel | G-FR2-A10-11 | G-FR2-A10-12 |
|---|---|---|
| Subcarrier spacing [kHz] | 120 | 120 |
| Allocated resource blocks | 16 | 66 |
| Data bearing CP-OFDM Symbols per slot (Note 1) | 7 | 7 |
| Modulation | 64QAM | 64QAM |
| Code rate (Note 2) | 517/1024 | 517/1024 |
| Payload size (bits) | 4032 | 16896 |
| Transport block CRC (bits) | 24 | 24 |
| Code block CRC size (bits) | - | 24 |
| Number of code blocks - C | 1 | 3 |
| Code block size including CRC (bits) (Note 2) | 4056 | 5664 |
| Total number of bits per slot without PT-RS | 8064 | 33264 |
| Total number of bits per slot with PT-RS (Note 3) | 7728 | 31878 |
| Total resource elements per slot without PT-RS | 1344 | 5544 |
| Total resource elements per slot with PT-RS (Note 3) | 1288 | 5313 |
| NOTE 1: *DM-RS configuration type* = 1 with *DM-RS duration* = *single-symbol DM-RS* and the number of DM-RS CDM groups without data is 2, *Additional DM-RS position* = *pos2* with $l_0$= 0 and $l$ =4,8 as per Table 6.4.1.1.3-3 of TS 38.211 [9]. | | |
| NOTE 2: Code block size including CRC (bits) equals to *K'* in sub-clause 5.2.2 of TS 38.212 [15]. | | |
| NOTE 3: PT-RS configuration $K_{PT-RS}$ =2, $L_{PT-RS}$ =1. | | |

# Annex B (normative): Error Vector Magnitude (FR1)

## B.1 Reference point for measurement

The EVM shall be measured at the point after the FFT and a zero-forcing (ZF) equalizer in the receiver, as depicted in figure B.1-1 below.

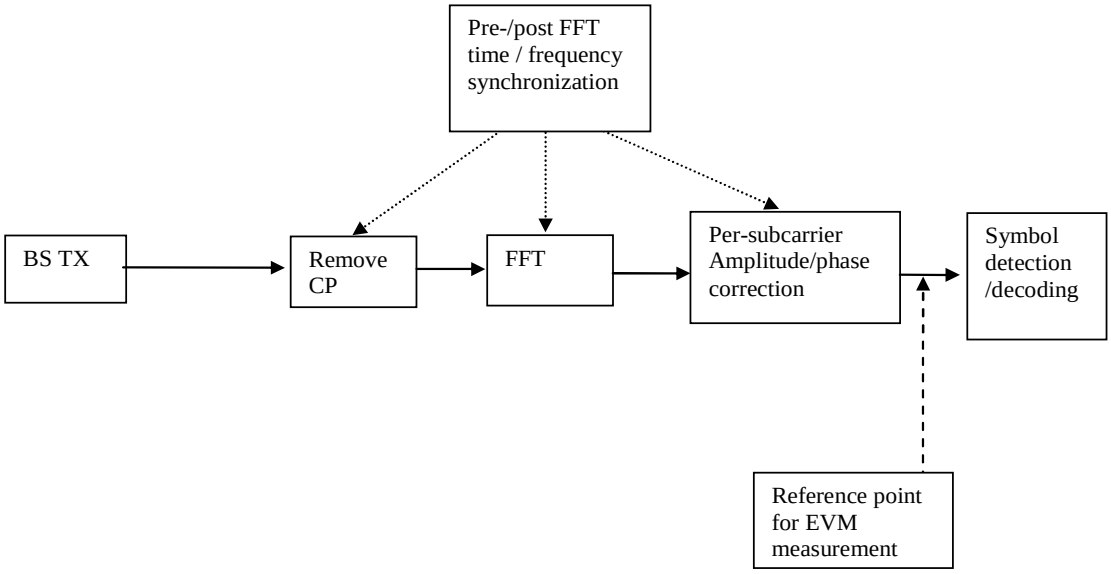



**Figure B.1-1: Reference point for EVM measurement**

## B.2 Basic unit of measurement

The basic unit of EVM measurement is defined over one slot in the time domain and $N_{BW}^{RB}$ subcarriers in the frequency domain:

$$EVM = \sqrt{\frac{\sum_{t \in T} \sum_{f \in F(t)} \left| Z'(t, f) - I(t, f) \right|^2}{\sum_{t \in T} \sum_{f \in F(t)} \left| I(t, f) \right|^2}}$$

where

$T$ is the set of symbols with the considered modulation scheme being active within the slot,

$F(t)$ is the set of subcarriers within the $N_{\mathrm{BW}}^{\mathrm{RB}}$ subcarriers with the considered modulation scheme being active in symbol $t$,

$I(t, f)$ is the ideal signal reconstructed by the measurement equipment in accordance with relevant Tx models,

$Z'(t, f)$ is the modified signal under test defined in annex B.3.

NOTE: Although the basic unit of measurement is one slot, the equalizer is calculated over 10 ms measurement interval to reduce the impact of noise in the reference signals. The boundaries of the 10 ms measurement intervals need not be aligned with radio frame boundaries.

# B.3 Modified signal under test

Implicit in the definition of EVM is an assumption that the receiver is able to compensate a number of transmitter impairments. The signal under test is equalized and decoded according to:

$$Z'(t,f) = \frac{FFT\left\{z(v-\Delta\tilde{t})\cdot e^{-j2\pi\Delta\tilde{f}v}\right\}e^{j2\pi f\Delta\tilde{t}}}{\tilde{a}(f)\cdot e^{j\tilde{\varphi}(f)}}$$

where

$z(v)$ is the time domain samples of the signal under test.

$\Delta\tilde{t}$ is the sample timing difference between the FFT processing window in relation to nominal timing of the ideal signal. Note that two timing offsets are determined, the corresponding EVM is measured and the maximum used as described in annex B.7.

$\Delta\tilde{f}$ is the RF frequency offset.

$\tilde{\varphi}(f)$ is the phase response of the TX chain.

$\tilde{a}(f)$ is the amplitude response of the TX chain.

# B.4 Estimation of frequency offset

The observation period for determining the frequency offset $\Delta\tilde{f}$ shall be 1 slot.

# B.5 Estimation of time offset

## B.5.1 General

The observation period for determining the sample timing difference $\Delta\tilde{t}$ shall be 1 slot.

In the following $\Delta\tilde{c}$ represents the middle sample of the EVM window of length *W* (defined in annex B.5.2) or the last sample of the first window half if *W* is even.

$\Delta\tilde{c}$ is estimated so that the EVM window of length *W* is centred on the measured cyclic prefix of the considered OFDM symbol. To minimize the estimation error the timing shall be based on demodulation reference signals. To limit time distortion of any transmit filter the reference signals in the 1 outer RBs are not taken into account in the timing estimation

Two values for $\Delta\tilde{t}$ are determined:

$\Delta\tilde{t}_l = \Delta\tilde{c} + \alpha - \left\lfloor \frac{W}{2} \right\rfloor$ and

$\Delta\tilde{t}_h = \Delta\tilde{c} + \left\lfloor \frac{W}{2} \right\rfloor$ where $\alpha = 0$ if *W* is odd and $\alpha = 1$ if *W* is even.

When the cyclic prefix length varies from symbol to symbol then *T* shall be further restricted to the subset of symbols with the considered modulation scheme being active and with the considered cyclic prefix length type.

## B.5.2 Window length

Table B.5.2-1, B.5.2-2, B.5.2-3 specify the EVM window length (*W*) for normal CP.

**Table B.5.2-1: EVM window length for normal CP, FR1, 15 kHz SCS**

| Channel bandwidth (MHz) | FFT size | CP length for symbols 1-6 and 8-13 in FFT samples | EVM window length *W* | Ratio of *W* to total CP length for symbols 1-6 and 8-13 (Note) (%) |
|---|---|---|---|---|
| 5 | 512 | 36 | 14 | 40 |
| 10 | 1024 | 72 | 28 | 40 |
| 15 | 1536 | 108 | 44 | 40 |
| 20 | 2048 | 144 | 58 | 40 |
| 25 | 2048 | 144 | 72 | 50 |
| 30 | 3072 | 216 | 108 | 50 |
| 35 | 3072 | 216 | 108 | 50 |
| 40 | 4096 | 288 | 144 | 50 |
| 45 | 4096 | 288 | 144 | 50 |
| 50 | 4096 | 288 | 144 | 50 |
| NOTE: These percentages are informative and apply to a slot's symbols 1 to 6 and 8 to 13. Symbols 0 and 7 have a longer CP and therefore a lower percentage. | | | | |

**Table B.5.2-2: EVM window length for normal CP, FR1, 30 kHz SCS**

| Channel bandwidth (MHz) | FFT size | CP length for symbols 1-13 in FFT samples | EVM window length *W* | Ratio of *W* to total CP length for symbols 1-13 (Note) (%) |
|---|---|---|---|---|
| 5 | 256 | 18 | 8 | 40 |
| 10 | 512 | 36 | 14 | 40 |
| 15 | 768 | 54 | 22 | 40 |
| 20 | 1024 | 72 | 28 | 40 |
| 25 | 1024 | 72 | 36 | 50 |
| 30 | 1536 | 108 | 54 | 50 |
| 35 | 1536 | 108 | 54 | 50 |
| 40 | 2048 | 144 | 72 | 50 |
| 45 | 2048 | 144 | 72 | 50 |
| 50 | 2048 | 144 | 72 | 50 |
| 60 | 3072 | 216 | 130 | 60 |
| 70 | 3072 | 216 | 130 | 60 |
| 80 | 4096 | 288 | 172 | 60 |
| 90 | 4096 | 288 | 172 | 60 |
| 100 | 4096 | 288 | 172 | 60 |
| NOTE: These percentages are informative and apply to a slot's symbols 1 through 13. Symbol 0 has a longer CP and therefore a lower percentage. | | | | |

**Table B.5.2-3: EVM window length for normal CP, FR1, 60 kHz SCS**

| Channel bandwidth (MHz) | FFT size | CP length in FFT samples | EVM window length *W* | Ratio of *W* to total CP length (Note) (%) |
|---|---|---|---|---|
| 10 | 256 | 18 | 8 | 40 |
| 15 | 384 | 27 | 11 | 40 |
| 20 | 512 | 36 | 14 | 40 |
| 25 | 512 | 36 | 18 | 50 |
| 30 | 768 | 54 | 26 | 50 |
| 35 | 768 | 54 | 26 | 50 |
| 40 | 1024 | 72 | 36 | 50 |
| 45 | 1024 | 72 | 36 | 50 |
| 50 | 1024 | 72 | 36 | 50 |
| 60 | 1536 | 108 | 64 | 60 |
| 70 | 1536 | 108 | 64 | 60 |
| 80 | 2048 | 144 | 86 | 60 |
| 90 | 2048 | 144 | 86 | 60 |
| 100 | 2048 | 144 | 86 | 60 |
| NOTE: These percentages are informative and apply to all OFDM symbols within subframe except for symbol 0 of slot 0 and slot 2. Symbol 0 of slot 0 and slot 2 may have a longer CP and therefore a lower percentage. | | | | |

Table B.5.2-4 below specifies the EVM window length (*W*) for extended CP. The number of CP samples excluded from the EVM window is the same as for normal CP length.

**Table B.5.2-4: EVM window length for extended CP, FR1, 60 kHz SCS**

| Channel bandwidth (MHz) | FFT size | CP length in FFT samples | EVM window length *W* | Ratio of *W* to total CP length (Note) (%) |
|---|---|---|---|---|
| 10 | 256 | 64 | 54 | 84 |
| 15 | 384 | 96 | 80 | 83 |
| 20 | 512 | 128 | 106 | 83 |
| 25 | 512 | 128 | 110 | 85.9 |
| 30 | 768 | 192 | 164 | 85.9 |
| 35 | 768 | 192 | 164 | 85.9 |
| 40 | 1024 | 256 | 220 | 85.9 |
| 45 | 1024 | 256 | 220 | 85.9 |
| 50 | 1024 | 256 | 220 | 85.9 |
| 60 | 1536 | 384 | 340 | 88.6 |
| 70 | 1536 | 384 | 340 | 88.7 |
| 80 | 2048 | 512 | 454 | 88.7 |
| 90 | 2048 | 512 | 454 | 88.7 |
| 100 | 2048 | 512 | 454 | 88.7 |
| NOTE: These percentages are informative. | | | | |

# B.6 Estimation of TX chain amplitude and frequency response parameters

The equalizer coefficients $\tilde{a}(f)$ and $\tilde{\varphi}(f)$ are determined as follows:

1. Calculate the complex ratios (amplitude and phase) of the post-FFT acquired signal $Z'(t, f)$ and the post-FFT ideal signal $I_2(t, f)$, for each reference signal, over 10ms measurement interval. This process creates a set of complex ratios:

$$a(t, f).e^{j\varphi(t,f)} = \frac{Z'(t, f)}{I_2(t, f)}$$

Where the post-FFT ideal signal $I_2(t, f)$ is constructed by the measuring equipment according to the relevant TX specifications, using the following parameters: i.e. nominal demodulation reference signals, (all other modulation symbols are set to 0 V), nominal carrier frequency, nominal amplitude and phase for each applicable subcarrier, nominal timing.

2. Perform time averaging at each reference signal subcarrier of the complex ratios, the time-averaging length is 10ms measurement interval. Prior to the averaging of the phases $\varphi(t_i, f)$ an unwrap operation must be performed according to the following definition: The unwrap operation corrects the radian phase angles of $\varphi(t_i, f)$ by adding multiples of 2*PI when absolute phase jumps between consecutive time instances $t_i$ are greater than or equal to the jump tolerance of PI radians. This process creates an average amplitude and phase for each reference signal subcarrier (i.e. every second subcarrier).

$$a(f) = \frac{\sum_{i=1}^{N} a(t_i, f)}{N}$$

$$\varphi(f) = \frac{\sum_{i=1}^{N} \varphi(t_i, f)}{N}$$

Where $N$ is the number of reference signal; time-domain locations $t_i$ from $Z'(t, f)$ for each reference signal subcarrier $f$.

3. The equalizer coefficients for amplitude and phase $\hat{a}(f)$ and $\hat{\varphi}(f)$ at the reference signal subcarriers are obtained by computing the moving average in the frequency domain of the time-averaged reference signal subcarriers, i.e. every second subcarrier. The moving average window size is 19 and averaging is over the DM-RS subcarriers in the allocated RBs. For DM-RS subcarriers at or near the edge of the channel , or when the number of available DM-RS subcarriers within a set of contiguously allocated RBs is smaller than the moving average window size, the window size is reduced accordingly as per figure B.6-1.

4. Perform linear interpolation from the equalizer coefficients $\hat{a}(f)$ and $\hat{\varphi}(f)$ to compute coefficients $\tilde{a}(f)$, $\tilde{\varphi}(f)$ for each subcarrier.

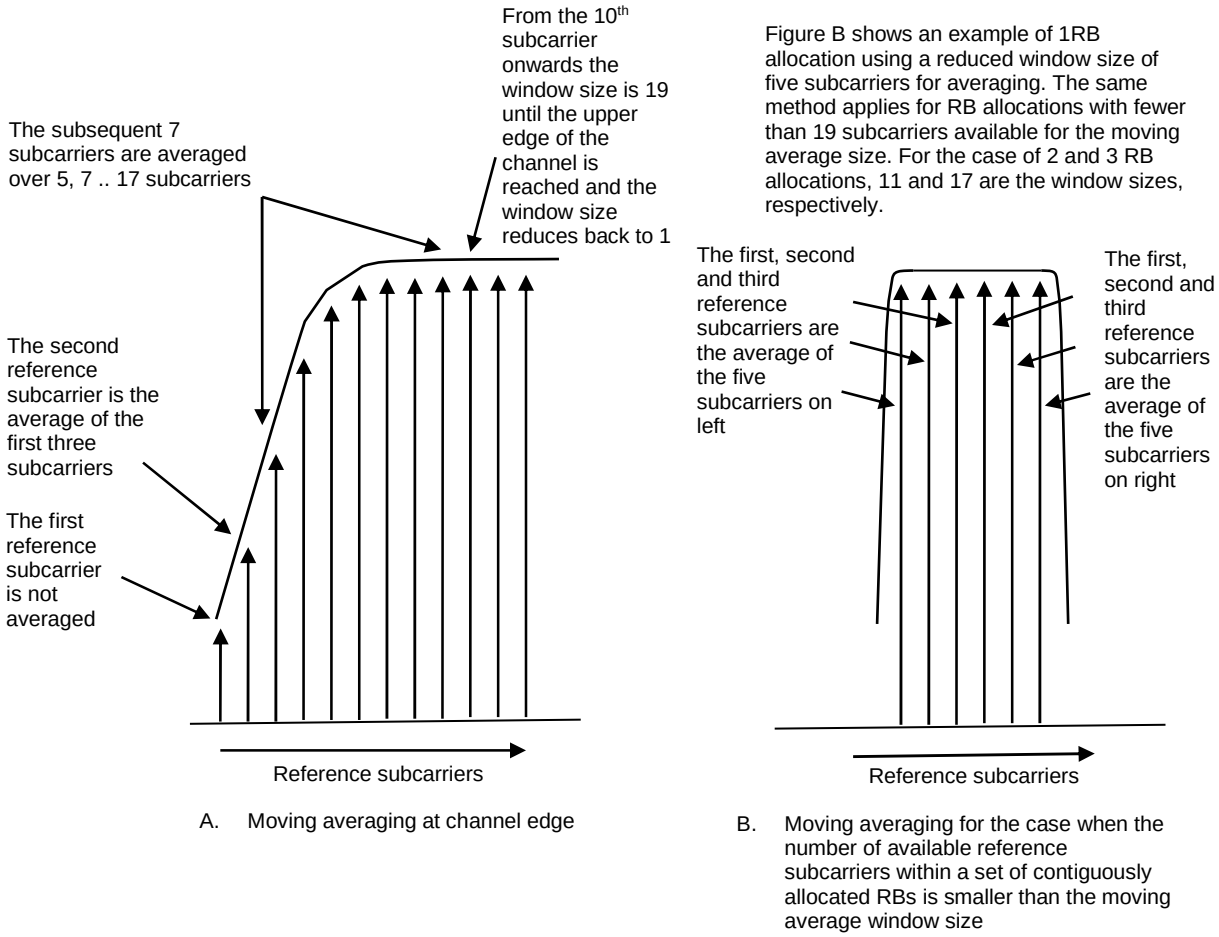



**Figure B.6-1: Reference subcarrier smoothing in the frequency domain**

# B.7 Averaged EVM

EVM is averaged over all allocated downlink resource blocks with the considered modulation scheme in the frequency domain, and a minimum of $N_{dl}$ slots where $N_{dl}$ is the number of slots in a 10 ms measurement interval.

For FDD the averaging in the time domain equals the $N_{dl}$ slot duration of the 10 ms measurement interval from the equalizer estimation step.

$$\overline{EVM_{frame}} = \sqrt{\frac{1}{\sum_{i=1}^{N_{dl}} Ni} \sum_{i=1}^{N_{dl}} \sum_{j=1}^{Ni} EVM_{i,j}^{2}}$$

- Where *Ni* is the number of resource blocks with the considered modulation scheme in slot *i*.
- The EVM requirements shall be tested against the maximum of the RMS average at the window *W* extremities of the EVM measurements:
- Thus $\overline{\mathrm{EVM}}_{\mathrm{frame,l}}$ is calculated using $\Delta\tilde{t} = \Delta\tilde{t}_l$ in the expressions above and $\overline{\mathrm{EVM}}_{frame,h}$ is calculated using $\Delta\tilde{t} = \Delta\tilde{t}_h$ in the $\overline{EVM_{frame}}$ calculation.
- Thus we get:

$$\overline{EVM} = \max\left(\overline{EVM}_{\mathrm{frame,l}}, \overline{EVM}_{\mathrm{frame,h}}\right)$$

For TDD, let $N_{dl}^{TDD}$ be the number of slots with downlink symbols within a 10 ms measurement interval, the averaging in the time domain can be calculated from $N_{dl}^{TDD}$ slots of different 10 ms measurement intervals and should have a minimum of $N_{dl}$ slots averaging length where $N_{dl}$ is the number of slots in a 10 ms measurement interval.

- $\overline{EVM}_{\text{frame}}$ is derived by: Square the EVM results in each 10 ms measurement interval. Sum the squares, divide the sum by the number of EVM relevant locations, square-root the quotient (RMS).

$$\overline{EVM}_{\text{frame}} = \sqrt{\frac{1}{\sum_{i=1}^{N_{dl}^{TDD}} N_i} \sum_{i=1}^{N_{dl}^{TDD}} \sum_{j=1}^{N_i} EVM_{i,j}^2}$$

- Where $N_i$ is the number of resource blocks with the considered modulation scheme in slot *i*.

- The $EVM_{\text{frame}}$ is calculated, using the maximum of $\overline{EVM}_{\text{frame}}$ at the window *W* extremities. Thus $\overline{EVM}_{\text{frame,l}}$ is calculated using $\tilde{t} = \Delta\tilde{t}_l$ and $\overline{EVM}_{\text{frame,h}}$ is calculated using $\tilde{t} = \Delta\tilde{t}_h$ (*l* and *h*, low and high; where low is the timing $(\Delta c - W/2)$ and and high is the timing $(\Delta c + W/2)$).

$$EVM_{\text{frame}} = \max\left(\overline{EVM}_{\text{frame,l}}, \overline{EVM}_{\text{frame,h}}\right)$$

- In order to unite at least $N_{dl}$ slots, consider the minimum integer number of 10 ms measurement intervals, where $N_{frame}$ is determined by.

$$N_{frame} = \left\lceil \frac{10 \times N_{slot}}{N_{dl}^{TDD}} \right\rceil$$

and $N_{slot} = 1$ for 15 kHz SCS, $N_{slot} = 2$ for 30 kHz SCS and $N_{slot} = 4$ for 60 kHz SCS normal CP.

- Unite by RMS.

$$\overline{EVM} = \sqrt{\frac{1}{N_{frame}} \sum_{k=1}^{N_{frame}} EVM_{frame,k}^2}$$

# Annex C (normative): Error Vector Magnitude (FR2)

## C.1 Reference point for measurement

The EVM shall be measured at the point after the FFT and a zero-forcing (ZF) equalizer in the receiver, as depicted in figure C.1-1 below.

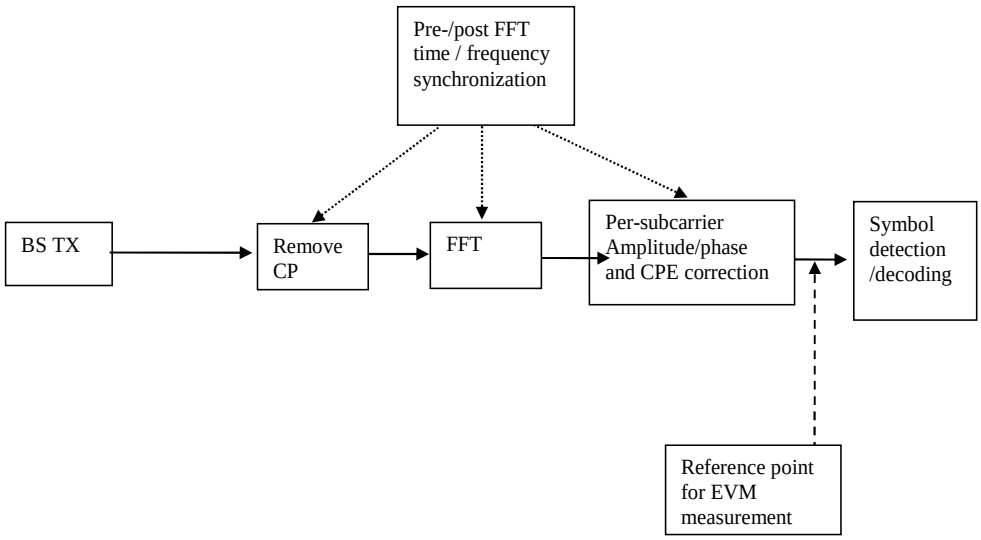



**Figure C.1-1: Reference point for EVM measurement**

## C.2 Basic unit of measurement

The basic unit of EVM measurement is defined over one slot in the time domain and $N_{BW}^{RB}$ subcarriers in the frequency domain:

$$EVM = \sqrt{\frac{\sum_{t \in T} \sum_{f \in F(t)} \left| Z'(t, f) - I(t, f) \right|^2}{\sum_{t \in T} \sum_{f \in F(t)} \left| I(t, f) \right|^2}}$$

where

$T$ is the set of symbols with the considered modulation scheme being active within the slot,

$F(t)$ is the set of subcarriers within the $N_{\mathrm{BW}}^{\mathrm{RB}}$ subcarriers with the considered modulation scheme being active in symbol *t*,

$I(t, f)$ is the ideal signal reconstructed by the measurement equipment in accordance with relevant Tx models,

$Z'(t, f)$ is the modified signal under test defined in C.3.

NOTE: Although the basic unit of measurement is one slot, the equalizer is calculated over 10 ms measurement intervals to reduce the impact of noise in the reference signals. The boundaries of the 10 ms measurement intervals need not be aligned with radio frame boundaries.

# C.3 Modified signal under test

Implicit in the definition of EVM is an assumption that the receiver is able to compensate a number of transmitter impairments. The signal under test is equalized and decoded according to:

$$Z'(t,f) = \frac{FFT\left\{z(v-\Delta\tilde{t})\cdot e^{-j2\pi\Delta\tilde{f}v}\right\}e^{j2\pi f\Delta\tilde{t}}}{\tilde{a}(f)\cdot e^{j\tilde{\varphi}(f)}}$$

where

$z(v)$ is the time domain samples of the signal under test.

$\Delta\tilde{t}$ is the sample timing difference between the FFT processing window in relation to nominal timing of the ideal signal. Note that two timing offsets are determined, the corresponding EVM is measured and the maximum used as described in C.7.

$\Delta\tilde{f}$ is the RF frequency offset.

$\tilde{\varphi}(f)$ is the phase response of the TX chain.

$\tilde{a}(f)$ is the amplitude response of the TX chain.

# C.4 Estimation of frequency offset

The observation period for determining the frequency offset $\Delta\tilde{f}$ shall be 1 slot.

# C.5 Estimation of time offset

## C.5.1 General

The observation period for determining the sample timing difference $\Delta\tilde{t}$ shall be 1 slot.

In the following $\Delta\tilde{c}$ represents the middle sample of the EVM window of length $W$ (defined in C.5.2) or the last sample of the first window half if $W$ is even.

$\Delta\tilde{c}$ is estimated so that the EVM window of length $W$ is centred on the measured cyclic prefix of the considered OFDM symbol. To minimize the estimation error the timing shall be based on the reference signals. To limit time distortion of any transmit filter the reference signals in the 1 outer RBs are not taken into account in the timing estimation

Two values for $\Delta\tilde{t}$ are determined:

$$\Delta\tilde{t}_l = \Delta\tilde{c} + \alpha - \left\lfloor \frac{W}{2} \right\rfloor \text{ and}$$

$$\Delta\tilde{t}_h = \Delta\tilde{c} + \left\lfloor \frac{W}{2} \right\rfloor \text{ where } \alpha = 0 \text{ if } W \text{ is odd and } \alpha = 1 \text{ if } W \text{ is even.}$$

When the cyclic prefix length varies from symbol to symbol then $T$ shall be further restricted to the subset of symbols with the considered modulation scheme being active and with the considered cyclic prefix length type.

## C.5.2 Window length

Table C.5.2-1, Table C.5.2-2, Table C.5.2-2a and Table C.5.2-2b specify the EVM window length (*W*) for normal CP for FR2.

**Table C.5.2-1: EVM window length for normal CP, FR2, 60 kHz SCS**

| Channel bandwidth (MHz) | FFT size | CP length in FFT samples | EVM window length W | Ratio of *W* to total CP length (Note) (%) |
|---|---|---|---|---|
| 50 | 1024 | 72 | 36 | 50 |
| 100 | 2048 | 144 | 72 | 50 |
| 200 | 4096 | 288 | 144 | 50 |
| NOTE: These percentages are informative and apply to all OFDM symbols within subframe except for symbol 0 of slot 0 and slot 2. Symbol 0 of slot 0 and slot 2 may have a longer CP and therefore a lower percentage. | | | | |

**Table C.5.2-2: EVM window length for normal CP, FR2, 120 kHz SCS**

| Channel bandwidth (MHz) | FFT size | CP length in FFT samples | EVM window length W | Ratio of *W* to total CP length (Note) (%) |
|---|---|---|---|---|
| 50 | 512 | 36 | 18 | 50 |
| 100 | 1024 | 72 | 36 | 50 |
| 200 | 2048 | 144 | 72 | 50 |
| 400 | 4096 | 288 | 144 | 50 |
| NOTE 1: These percentages are informative and apply to all OFDM symbols within subframe except for symbol 0 of slot 0 and slot 4. Symbol 0 of slot 0 and slot 4 may have a longer CP and therefore a lower percentage. | | | | |

**Table C.5.2-2a: EVM window length for normal CP, FR2-2, 480 kHz SCS**

| Channel bandwidth (MHz) | FFT size | CP length in FFT samples | EVM window length W | Ratio of *W* to total CP length (Note) (%) |
|---|---|---|---|---|
| 400 | 1024 | 72 | [36] | [50] |
| 800 | 2048 | 144 | [72] | [50] |
| 1600 | 4096 | 288 | [144] | [50] |
| NOTE 1: These percentages are informative and apply to all OFDM symbols within subframe except for symbol 0 of slot 0 and slot 4. Symbol 0 of slot 0 and slot 4 may have a longer CP and therefore a lower percentage. | | | | |

**Table C.5.2-2b: EVM window length for normal CP, FR2-2, 960 kHz SCS**

| Channel bandwidth (MHz) | FFT size | CP length in FFT samples | EVM window length W | Ratio of *W* to total CP length (Note) (%) |
|---|---|---|---|---|
| 400 | 512 | 36 | [18] | [50] |
| 800 | 1024 | 72 | [36] | [50] |
| 1600 | 2048 | 144 | [72] | [50] |
| 2000 | [2048] | [144] | [72] | [50] |
| NOTE 1: These percentages are informative and apply to all OFDM symbols within subframe except for symbol 0 of slot 0 and slot 4. Symbol 0 of slot 0 and slot 4 may have a longer CP and therefore a lower percentage. | | | | |

Table C.5.2-3 below specifies the EVM window length (*W*) for extended CP. The number of CP samples excluded from the EVM window is the same as for normal CP length.

**Table C.5.2-3: EVM window length for extended CP, FR2, 60 kHz SCS**

| Channel bandwidth (MHz) | FFT size | CP length in FFT samples | EVM window length W | Ratio of *W* to total CP length (Note) (%) |
|---|---|---|---|---|
| 50 | 1024 | 256 | 220 | 85.9 |
| 100 | 2048 | 512 | 440 | 85.9 |
| 200 | 4096 | 1024 | 880 | 85.9 |
| NOTE: These percentages are informative. | | | | |

# C.6 Estimation of TX chain amplitude and frequency response parameters

The equalizer coefficients $\tilde{a}(f)$ and $\tilde{\varphi}(f)$ are determined as follows:

1. Calculate the complex ratios (amplitude and phase) of the post-FFT acquired signal $Z'(t, f)$ and the post-FFT ideal signal $I_2(t, f)$, for each reference signal, over 10ms measurement intervals. This process creates a set of complex ratios:

$$a(t, f).e^{j\varphi(t,f)} = \frac{Z'(t, f)}{I_2(t, f)}$$

Where the post-FFT ideal signal $I_2(t, f)$ is constructed by the measuring equipment according to the relevant TX specifications, using the following parameters:

- nominal demodulation reference signals and nominal PT-RS if present (all other modulation symbols are set to 0 V),
- nominal carrier frequency,
- nominal amplitude and phase for each applicable subcarrier,
- nominal timing.

2. Perform time averaging at each reference signal subcarrier of the complex ratios, the time-averaging length is 10ms measurement interval. Prior to the averaging of the phases $\varphi(t_i, f)$ an unwrap operation must be performed according to the following definition: The unwrap operation corrects the radian phase angles of $\varphi(t_i, f)$ by adding multiples of 2*PI when absolute phase jumps between consecutive time instances $t_i$ are greater than or equal to the jump tolerance of PI radians. This process creates an average amplitude and phase for each reference signal subcarrier (i.e. every second subcarrier).

$$a(f) = \frac{\sum_{i=1}^{N} a(t_i, f)}{N}$$
$$\varphi(f) = \frac{\sum_{i=1}^{N} \varphi(t_i, f)}{N}$$

Where $N$ is the number of reference signal time-domain locations $t_i$ from $Z'(t, f)$ for each reference signal subcarrier $f$.

3. The equalizer coefficients for amplitude and phase $\hat{a}(f)$ and $\hat{\varphi}(f)$ at the reference signal subcarriers are obtained by computing the moving average in the frequency domain of the time-averaged reference signal subcarriers, i.e. every second subcarrier. The moving average window size is 19 and averaging is over the DM-RS subcarriers in allocated RBs. For DM-RS subcarriers at or near the edge of the channel, or when the number

of available DM-RS subcarriers within a set of contiguously allocated RBs is smaller than the moving average window size, the window size is reduced accordingly as per figure C.6-1.

4. Perform linear interpolation from the equalizer coefficients $\hat{a}(f)$ and $\hat{\varphi}(f)$ to compute coefficients $\tilde{a}(f)$, $\tilde{\varphi}(f)$ for each subcarrier. To account for the common phase error (CPE) experienced in millimetre wave frequencies, $\bar{\varphi}(f)$, in the estimated coefficients contain phase rotation due to the CPE, $\theta$, in addition to the phase of the equalizer coefficient $\tilde{\varphi}(f)$, that is

$$\bar{\varphi}(f) = \tilde{\varphi}(f) + \theta(t)$$

For OFDM symbols where PT-RS does not exist, $\theta(t)$ can be estimated by performing linear interpolation from neighboring symbols where PT-RS is present.

In order to separate component of the CPE, $\theta$, contained in, $\bar{\varphi}(f)$, estimation and compensation of the CPE needs to follow. $\theta(t)$ is the common phase error (CPE), that rotates all the subcarriers of the OFDM symbol at time $t$.

Estimate of the CPE, $\theta(t)$, at OFDM symbol time, $t$, can then be obtained from using the PT-RS employing the expression

$$\tilde{\theta}(t) = arg\left\{\sum_{f \in f^{ptrs}} \left(\frac{Z'(t,f)}{I_{ptrs}(t,f)}\right)\left(\tilde{a}(f)e^{-j\bar{\varphi}(f)}\right)\right\}$$

In the above equation, $f^{ptrs}$ is the set of subcarriers where PT-RS are mapped, $t \in t^{ptrs}$ where $t^{ptrs}$ is the set of OFDM symbols where PT-RS are mapped while $Z'(t,f)$ and $I_{ptrs}(t,f)$ are is the post-FFT acquired signal and the ideal PT-RS signal respectively. That is, estimate of the CPE at a given OFDM symbol is obtained from frequency correlation of the complex ratios at the PT-RS positions with the conjugate of the estimated equalizer complex coefficients. The estimated CPE can be subtracted from $\bar{\varphi}(f)$ to remove influence of the CPE, and obtain estimate of the complex coefficient's phase

$$\tilde{\varphi}(f) = \bar{\varphi}(f) - \tilde{\theta}(\mathrm{t})$$

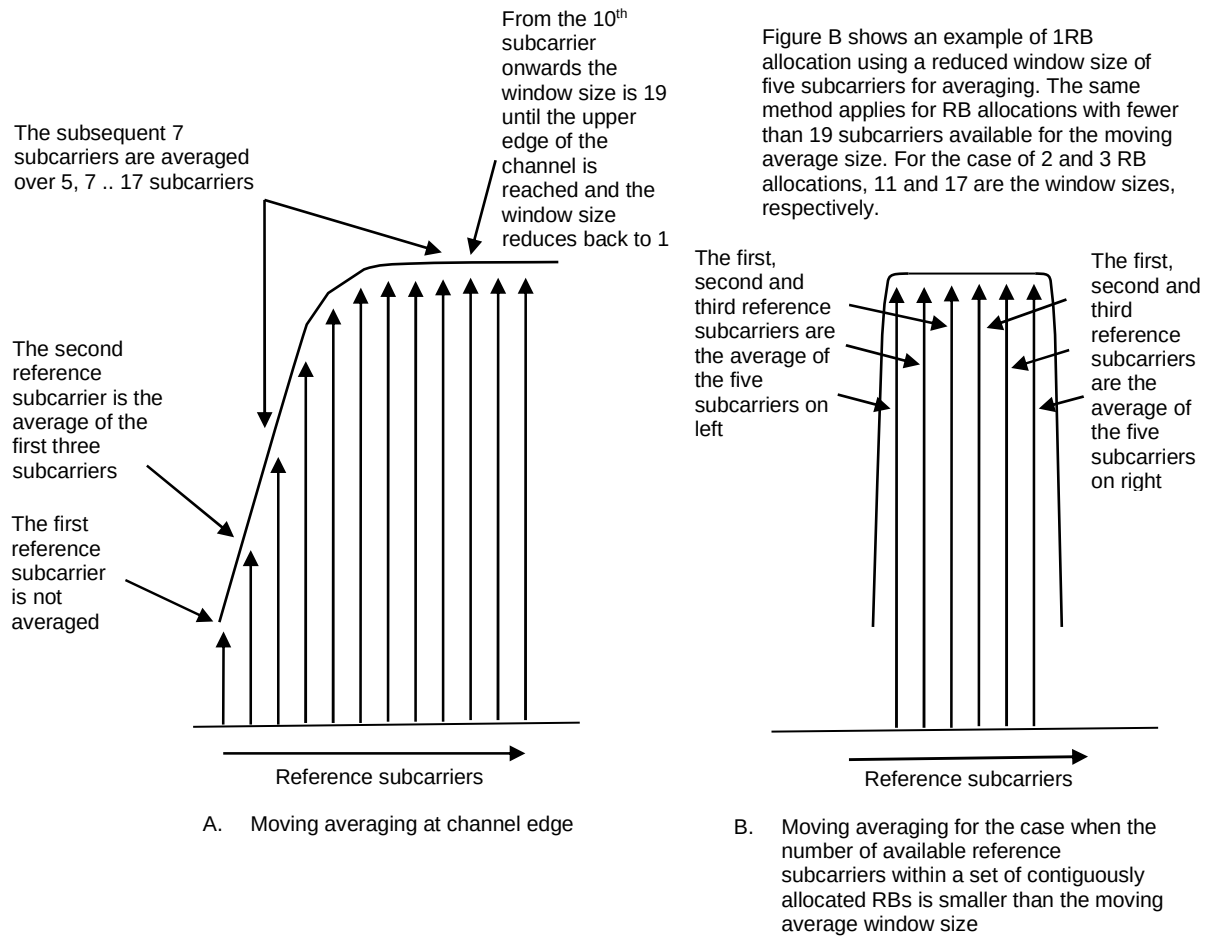



**Figure C.6-1: Reference subcarrier smoothing in the frequency domain**

# C.7 Averaged EVM

EVM is averaged over all allocated downlink resource blocks with the considered modulation scheme in the frequency domain, and a minimum of $N_{dl}$ slots where $N_{dl}$ is the number of slots in a 10 ms measurement interval.

For TDD, let $N_{dl}^{TDD}$ be the number of slots with downlink symbols within a 10 ms measurement interval, the averaging in the time domain can be calculated from $N_{dl}^{TDD}$ slots of different 10 ms measurement intervals and should have a minimum of $N_{dl}$ slots averaging length where $N_{dl}$ is the number of slots in a 10 ms measurement interval.

- $\overline{EVM}_{\text{frame}}$ is derived by: Square the EVM results in each 10 ms measurement intervals. Sum the squares, divide the sum by the number of EVM relevant locations, square-root the quotient (RMS).

$$\overline{EVM}_{\text{frame}} = \sqrt{\frac{1}{\sum_{i=1}^{N_{dl}^{TDD}} N_i} \sum_{i=1}^{N_{dl}^{TDD}} \sum_{j=1}^{N_i} EVM_{i,j}^2}$$

- Where $N_i$ is the number of resource blocks with the considered modulation scheme in slot *i*.

- The $EVM_{\text{frame}}$ is calculated, using the maximum of $\overline{EVM}_{\text{frame}}$ at the window *W* extremities. Thus $\overline{EVM}_{\text{frame,l}}$ is calculated using $\tilde{t} = \Delta\tilde{t}_l$ and $\overline{EVM}_{\text{frame,h}}$ is calculated using $\tilde{t} = \Delta\tilde{t}_h$ (*l* and *h*, low and high; where low is the timing $(\Delta c - W/2)$ and and high is the timing $(\Delta c + W/2)$).

$$EVM_{\text{frame}} = \max\left(\overline{EVM}_{\text{frame,l}}, \overline{EVM}_{\text{frame,h}}\right)$$

- In order to unite at least $N_{dl}$ slots, consider the minimum integer number of 10 ms measurement intervals, where $N_{frame}$ is determined by.

$$N_{frame} = \left\lceil \frac{10 \times N_{slot}}{N_{dl}^{TDD}} \right\rceil$$

and $N_{slot} = 4$ for 60 kHz SCS and $N_{slot} = 8$ for 120 kHz SCS.

- Unite by RMS.

$$\overline{EVM} = \sqrt{\frac{1}{N_{frame}} \sum_{k=1}^{N_{frame}} EVM_{frame,k}^2}$$

# Annex D (normative): Characteristics of the interfering signals

The interfering signal shall be a PUSCH containing data and DM-RS symbols. Normal cyclic prefix is used. The data content shall be uncorrelated to the wanted signal and modulated according to clause 6 of TS38.211 [9]. Mapping of PUSCH modulation to receiver requirement are specified in table D-1.

**Table D-1: Modulation of the interfering signal**

| Receiver requirement | Modulation |
|---|---|
| In-channel selectivity | 16QAM |
| Adjacent channel selectivity and narrow-band blocking | QPSK |
| General blocking | QPSK |
| Receiver intermodulation | QPSK |

# Annex E:
# Void

# Annex F (normative): Relationship between EIRP based regulatory requirements and 3GPP requirements

## F.1 General

This annex applies to FR1 *BS type 1-C*, *BS type 1-H* and *BS type 1-O*.

Some regional requirements are defined per effective isotropic radiated power (EIRP), which is a combination of the transmitted power (or in some cases spectral density) and the effective antenna gain which is a site-specific condition. Such requirements may be applied per antenna, per cell, or per base station. It shall be noted that the definition of BS or cell may differ between regulations.

The regulations are based on the assumption on *BS type 1-C* conducted requirements and a passive antenna and must be interpreted for active antenna systems that have active beamforming. This annex describes how the power per connector and sum power over *TAB connectors* can be related to such requirements.

Where the regulator prescribes a method for EIRP calculation, that method supersedes the proposed assessment in this annex.

## F.2 Relationship between EIRP based regulatory requirements and conducted requirements

When 3GPP specifications mandate manufacturer declarations of the (conducted) output power or power spectral density per connector for the base station under the reference conditions stated as a way to accommodate the referred regional requirements without putting requirements on the local site conditions.

For the case when the base station manufacturer maximum output power or unwanted emission declarations apply per connector, the maximum EIRP can be estimated using the following formulas:

EIRP per antenna (applicable for *BS type 1-C*): $P_{EIRP} = P_{Tx} + G_{Ant}$

EIRP per cell or per BS (applicable for *BS type 1-H*): $P_{EIRPcell} = 10 * \log (\sum 10^{PEIRPn/10})$

In case the EIRP requirement is set per polarization, the summation shall be made per polarization.

- "$P_{EIRP}$" is the resulting effective isotropic radiated power (or radiated power spectral density) resulting from the power (or power spectral density) declared by the manufacturer in dBm (or dBm/measurement BW).
- "$P_{Tx}$" is the conducted power or power spectral density declared by the manufacturer in dBm (or dBm/measurement BW).
- "$G_{Ant}$" is the effective antenna gain, calculated as the antenna gain (dBi) minus the loss of the site infrastructure connecting the BS antenna connector with the antenna (dB) for the applied frequency. The antenna nominal gain is only applicable within a certain frequency range. For *BS type 1-H*, $G_{Ant}$ shall be an assumption on the gain of a passive antenna system in order to provide a total power emissions level comparable to the level obtained when a *BS type 1-C* is connected to a passive antenna. A typical example of a passive antenna gain, as used for *BS type 1-O*, is 17 dBi.
- "n" is the index number of the co-located antennas illuminating the same cell. $P_{EIRPn}$ is the $P_{EIRP}$ of the *n*th antenna.
- "Cell" is in this annex used in the sense that it is the limited geographical area covered by the carrier transmitted from one site.

# F.3 Relationship between EIRP based regulatory requirements and OTA requirements

The regulations set an EIRP limit considering a passive antenna BS. Although the gain of passive antennas may vary somewhat, the variation is in the order of a few dBs. The gain variation of a *BS type 1-O* may be much larger. However, *BS type 1-O* unwanted emissions requirements are defined as TRP, since TRP impacts co-existence properties.

In order to relate the EIRP values in the specifications to TRP, a fixed assumption has been made on the gain of a typical passive BS antenna.

Thus, the maximum TRP can be estimated using the following formulas:

TRP limit per antenna: $P_{TRP,\ antenna} = P_{EIRP} - G_{Ant}$

TRP limit per cell or per BS: $P_{TRP} = P_{TRP,antenna} + 9\ dB$

It is noted that the *BS type 1-O* architecture assumes that a BS subject to OTA requirements will have at least 8 antennas.

In case the TRP requirement is set per polarization, the summation shall be made per polarization.

- "$P_{EIRP}$" is the effective isotropic radiated power (or radiated power spectral density) set in the regulation (assuming a passive BS antenna) in dBm (or dBm/measurement BW).
- "$G_{Ant}$" is the effective antenna gain, the antenna gain (dBi) is a fixed reference value of 17 dBi. Directivity value should be used in above equations, however with all antenna losses are assumed zero then we can use effective antenna gain.

# Annex G (Normative): Propagation conditions

## G.1 Static propagation condition

The propagation for the static performance measurement is an Additive White Gaussian Noise (AWGN) environment. No fading or multi-paths exist for this propagation model.

## G.2 Multi-path fading propagation conditions

The multipath propagation conditions consist of several parts:

- A delay profile in the form of a "tapped delay-line", characterized by a number of taps at fixed positions on a sampling grid. The profile can be further characterized by the r.m.s. delay spread and the maximum delay spanned by the taps.
- A combination of channel model parameters that include the Delay profile and the Doppler spectrum that is characterized by a classical spectrum shape and a maximum Doppler frequency.
- Different models are used for FR1, FR2(24.25 GHz – 71 GHz).

### G.2.1 Delay profiles

The delay profiles are simplified from the TR 38.901 [16] TDL models. The simplification steps are shown below for information. These steps are only used when new delay profiles are created. Otherwise, the delay profiles specified in G.2.1.1, G.2.1.2 and G.2.1.3 can be used as such.

Step 1: Use the original TDL model from TR 38.901 [16].

Step 2: Re-order the taps in ascending delays.

Step 3: Perform delay scaling according to the procedure described in clause 7.7.3 in TR 38.901 [16].

Step 4: Apply the quantization to the delay resolution 5 ns. This is done simply by rounding the tap delays to the nearest multiple of the delay resolution.

Step 5: If multiple taps are rounded to the same delay bin, merge them by calculating their linear power sum.

Step 6: If there are more than 12 taps in the quantized model, merge the taps as follows

- Find the weakest tap from all taps (both merged and unmerged taps are considered)
  - If there are two or more taps having the same value and are the weakest, select the tap with the smallest delay as the weakest tap.
- When the weakest tap is the first delay tap, merge taps as follows
  - Update the power of the first delay tap as the linear power sum of the weakest tap and the second delay tap.
  - Remove the second delay tap.
- When the weakest tap is the last delay tap, merge taps as follows
  - Update the power of the last delay tap as the linear power sum of the second-to-last tap and the last tap.
  - Remove the second-to-last tap.

- Otherwise
  - For each side of the weakest tap, identify the neighbour tap that has the smaller delay difference to the weakest tap.
    - When the delay difference between the weakest tap and the identified neighbour tap on one side equals the delay difference between the weakest tap and the identified neighbour tap on the other side.
      - Select the neighbour tap that is weaker in power for merging.
    - Otherwise, select the neighbour tap that has smaller delay difference for merging.
  - To merge, the power of the merged tap is the linear sum of the power of the weakest tap and the selected tap.
  - When the selected tap is the first tap, the location of the merged tap is the location of the first tap. The weakest tap is removed.
  - When the selected tap is the last tap, the location of the merged tap is the location of the last tap. The weakest tap is removed.
  - Otherwise, the location of the merged tap is based on the average delay of the weakest tap and selected tap. If the average delay is on the sampling grid, the location of the merged tap is the average delay. Otherwise, the location of the merged tap is rounded towards the direction of the selected tap (e.g. 10 ns & 20 ns → 15 ns, 10 ns & 25 ns → 20 ns, if 25 ns had higher or equal power; 15 ns, if 10 ns had higher power) . The weakest tap and the selected tap are removed.
  - Repeat step 6 until the final number of taps is 12.

Step 7: Round the amplitudes of taps to one decimal (e.g. -8.78 dB → -8.8 dB)

Step 8: If the delay spread has slightly changed due to the tap merge, adjust the final delay spread by increasing or decreasing the power of the last tap so that the delay spread is corrected.

Step 9: Re-normalize the highest tap to 0 dB.

Note 1: Some values of the delay profile created by the simplification steps may differ from the values in tables G.2.1.1-2, G.2.1.1-3, G.2.1.1-4, G.2.1.2-2, G.2.1.2-3, G.2.1.2-4 and G.2.1.2-5 for the corresponding model.

Note 2: For Step 5 and Step 6, the power values are expressed in the linear domain using 6 digits of precision. The operations are in the linear domain.

## G.2.1.1 Delay profiles for FR1

The delay profiles for FR1 are selected to be representative of low, medium and high delay spread environment. The resulting model parameters are specified in table G.2.1.1-1 and the tapped delay line models are specified in tables G.2.1.1-2 ~ G.2.1.1-4.

**Table G.2.1.1-1: Delay profiles for NR channel models**

| Model | Number of channel taps | Delay spread (r.m.s.) | Maximum excess tap delay (span) | Delay resolution |
|---|---|---|---|---|
| TDLA30 | 12 | 30 ns | 290 ns | 5 ns |
| TDLB100 | 12 | 100 ns | 480 ns | 5 ns |
| TDLC300 | 12 | 300 ns | 2595 ns | 5 ns |

**Table G.2.1.1-2: TDLA30 (DS = 30 ns)**

| Tap # | Delay (ns) | Power (dB) | Fading distribution |
|---|---|---|---|
| 1 | 0 | -15.5 | Rayleigh |
| 2 | 10 | 0 | Rayleigh |
| 3 | 15 | -5.1 | Rayleigh |
| 4 | 20 | -5.1 | Rayleigh |
| 5 | 25 | -9.6 | Rayleigh |
| 6 | 50 | -8.2 | Rayleigh |
| 7 | 65 | -13.1 | Rayleigh |
| 8 | 75 | -11.5 | Rayleigh |
| 9 | 105 | -11.0 | Rayleigh |
| 10 | 135 | -16.2 | Rayleigh |
| 11 | 150 | -16.6 | Rayleigh |
| 12 | 290 | -26.2 | Rayleigh |

**Table G.2.1.1-3: TDLB100 (DS = 100 ns)**

| Tap # | Delay (ns) | Power (dB) | Fading distribution |
|---|---|---|---|
| 1 | 0 | 0 | Rayleigh |
| 2 | 10 | -2.2 | Rayleigh |
| 3 | 20 | -0.6 | Rayleigh |
| 4 | 30 | -0.6 | Rayleigh |
| 5 | 35 | -0.3 | Rayleigh |
| 6 | 45 | -1.2 | Rayleigh |
| 7 | 55 | -5.9 | Rayleigh |
| 8 | 120 | -2.2 | Rayleigh |
| 9 | 170 | -0.8 | Rayleigh |
| 10 | 245 | -6.3 | Rayleigh |
| 11 | 330 | -7.5 | Rayleigh |
| 12 | 480 | -7.1 | Rayleigh |

**Table G.2.1.1-4: TDLC300 (DS = 300 ns)**

| Tap # | Delay (ns) | Power (dB) | Fading distribution |
|---|---|---|---|
| 1 | 0 | -6.9 | Rayleigh |
| 2 | 65 | 0 | Rayleigh |
| 3 | 70 | -7.7 | Rayleigh |
| 4 | 190 | -2.5 | Rayleigh |
| 5 | 195 | -2.4 | Rayleigh |
| 6 | 200 | -9.9 | Rayleigh |
| 7 | 240 | -8.0 | Rayleigh |
| 8 | 325 | -6.6 | Rayleigh |
| 9 | 520 | -7.1 | Rayleigh |
| 10 | 1045 | -13.0 | Rayleigh |
| 11 | 1510 | -14.2 | Rayleigh |
| 12 | 2595 | -16.0 | Rayleigh |

## G.2.1.2 Delay profiles for FR2

The delay profiles for FR2 are specified in table G.2.1.2-1 and the tapped delay line models are specified in table G.2.1.2-2 - G.2.1.2-5.

**Table G.2.1.2-1: Delay profiles for NR channel models**

| Model | Number of channel taps | Delay spread (r.m.s.) | Maximum excess tap delay (span) | Delay resolution |
|---|---|---|---|---|
| TDLA30 | 12 | 30 ns | 290 ns | 5 ns |
| TDLA10 | 16 | 10 ns | 96 ns | 2 ns |
| TDLD10 | 10 | 10 ns | 126 ns | 2 ns |
| TDLD30 | 10 | 30 ns | 375 ns | 5 ns |

**Table G.2.1.2-2: TDLA30 (DS = 30 ns)**

| Tap # | Delay (ns) | Power (dB) | Fading distribution |
|---|---|---|---|
| 1 | 0 | -15.5 | Rayleigh |
| 2 | 10 | 0 | Rayleigh |
| 3 | 15 | -5.1 | Rayleigh |
| 4 | 20 | -5.1 | Rayleigh |
| 5 | 25 | -9.6 | Rayleigh |
| 6 | 50 | -8.2 | Rayleigh |
| 7 | 65 | -13.1 | Rayleigh |
| 8 | 75 | -11.5 | Rayleigh |
| 9 | 105 | -11.0 | Rayleigh |
| 10 | 135 | -16.2 | Rayleigh |
| 11 | 150 | -16.6 | Rayleigh |
| 12 | 290 | -26.2 | Rayleigh |

**Table G.2.1.2-3: TDLA10 (DS = 10 ns)**

| Tap # | Delay (ns] | Power (dB) | Fading distribution |
|---|---|---|---|
| 1 | 0 | -16.1 | Rayleigh |
| 2 | 4 | 0 | |
| 3 | 6 | -4 | |
| 4 | 8 | -10.2 | |
| 5 | 16 | -18.6 | |
| 6 | 18 | -9.3 | |
| 7 | 22 | -13.7 | |
| 8 | 24 | -17.9 | |
| 9 | 26 | -13.5 | |
| 10 | 30 | -14 | |
| 11 | 40 | -15.4 | |
| 12 | 44 | -18.9 | |
| 13 | 46 | -21.0 | |
| 14 | 48 | -21.6 | |
| 15 | 50 | -19.3 | |
| 16 | 96 | -25.9 | |

**Table G.2.1.2-4: TDLD10 (DS = 10 ns)**

| Tap # | Delay (ns] | Power (dB) | Fading distribution |
|---|---|---|---|
| 1 | 0 | -0.2 | LOS |
| | 0 | -12.4 | Rayleigh |
| 2 | 6 | -21.1 | |
| 3 | 14 | -16.7 | |
| 4 | 18 | -18.3 | |
| 5 | 26 | -22 | |
| 6 | 40 | -27.9 | |
| 7 | 80 | -23.7 | |
| 8 | 94 | -24.9 | |
| 9 | 98 | -30.0 | |
| 10 | 126 | -27.7 | |
| Note 1: Tap #1 follows a Ricean distribution. | | | |

**Table G.2.1.2-5: TDLD30 (DS = 30 ns)**

| Tap # | Delay (ns] | Power (dB) | Fading distribution |
|---|---|---|---|
| 1 | 0 | -0.2 | LOS |
| | 0 | -12.4 | Rayleigh |
| 2 | 20 | -21 | |
| 3 | 40 | -16.7 | |
| 4 | 55 | -18.3 | |
| 5 | 80 | -21.9 | |
| 6 | 120 | -27.8 | |
| 7 | 240 | -23.6 | |
| 8 | 285 | -24.8 | |
| 9 | 290 | -30.0 | |
| 10 | 375 | -27.6 | |
| Note 1: Tap #1 follows a Ricean distribution. | | | |

## G.2.2 Combinations of channel model parameters

The propagation conditions used for the performance measurements in multi-path fading environment are indicated as a combination of a channel model name and a maximum Doppler frequency, i.e., TDLA<DS>-<Doppler>, TDLB<DS>-<Doppler> or TDLC<DS>-<Doppler> where '<DS>' indicates the desired delay spread and '<Doppler>' indicates the maximum Doppler frequency (Hz).

Table G.2.2-1 and G.2.2-2 show the propagation conditions that are used for the performance measurements in multi-path fading environment for low, medium and high Doppler frequencies for FR1 and FR2 (24.25 GHz – 71 GHz), respectively.

**Table G.2.2-1: Channel model parameters for FR1**

| Combination name | Tapped delay line model | Maximum Doppler frequency |
|---|---|---|
| TDLA30-5 | TDLA30 | 5 Hz |
| TDLA30-10 | TDLA30 | 10 Hz |
| TDLB100-400 | TDLB100 | 400 Hz |
| TDLC300-100 | TDLC300 | 100 Hz |
| TDLC300-600 | TDLC300 | 600 Hz |
| TDLC300-1200 | TDLC300 | 1200 Hz |

**Table G.2.2-2: Channel model parameters for FR2**

| Combination name | Tapped delay line model | Maximum Doppler frequency |
|---|---|---|
| TDLA30-75 | TDLA30 | 75 Hz |
| TDLA30-300 | TDLA30 | 300 Hz |
| TDLA10-650 | TDLA10 | 650 Hz |
| TDLA30-650 | TDLA30 | 650 Hz |
| TDLD10-200 | TDLD10 | 200 Hz |
| TDLD30-200 | TDLD30 | 200 Hz |

# G.2.3 MIMO Channel Correlation Matrices

The MIMO channel correlation matrices defined in G.2.3 apply for the antenna configuration using uniform linear arrays at both gNB and UE and for the antenna configuration using cross polarized antennas.

## G.2.3.1 MIMO Correlation Matrices using Uniform Linear Array (ULA)

The MIMO channel correlation matrices defined in G.2.3.1 apply for the antenna configuration using uniform linear array (ULA) at both gNB and UE.

### G.2.3.1.1 Definition of MIMO Correlation Matrices

Table G.2.3.1.1-1 defines the correlation matrix for the gNB:

**Table G.2.3.1.1-1: gNB correlation matrix**

| | One antenna | Two antennas | Four antennas | Eight antennas |
|---|---|---|---|---|
| **gNode B Correlation** | $R_{gNB}=1$ | $R_{gNB}=\begin{pmatrix}1 & \alpha\\ \alpha^{*} & 1\end{pmatrix}$ | $R_{gNB}=\begin{pmatrix}1 & \alpha^{1/9} & \alpha^{4/9} & \alpha\\ \alpha^{1/9*} & 1 & \alpha^{1/9} & \alpha^{4/9}\\ \alpha^{4/9*} & \alpha^{1/9*} & 1 & \alpha^{1/9}\\ \alpha^{*} & \alpha^{4/9*} & \alpha^{1/9*} & 1\end{pmatrix}$ | $R_{gNB}=\begin{pmatrix}1 & \alpha^{1/49} & \alpha^{4/49} & \alpha^{9/49} & \alpha^{16/49} & \alpha^{25/49} & \alpha^{36/49} & \alpha\\ \alpha^{1/49*} & 1 & \alpha^{1/49} & \alpha^{4/49} & \alpha^{9/49} & \alpha^{16/49} & \alpha^{25/49} & \alpha^{36/49}\\ \alpha^{4/49*} & \alpha^{1/49*} & 1 & \alpha^{1/49} & \alpha^{4/49} & \alpha^{9/49} & \alpha^{16/49} & \alpha^{25/49}\\ \alpha^{9/49*} & \alpha^{4/49*} & \alpha^{1/49*} & 1 & \alpha^{1/49} & \alpha^{4/49} & \alpha^{9/49} & \alpha^{16/49}\\ \alpha^{16/49*} & \alpha^{9/49*} & \alpha^{4/49*} & \alpha^{1/49*} & 1 & \alpha^{1/49} & \alpha^{4/49} & \alpha^{9/49}\\ \alpha^{25/49*} & \alpha^{16/49*} & \alpha^{9/49*} & \alpha^{4/49*} & \alpha^{1/49*} & 1 & \alpha^{1/49} & \alpha^{4/49}\\ \alpha^{36/49*} & \alpha^{25/49*} & \alpha^{16/49*} & \alpha^{9/49*} & \alpha^{4/49*} & \alpha^{1/49*} & 1 & \alpha^{1/49}\\ \alpha^{*} & \alpha^{36/49*} & \alpha^{25/49*} & \alpha^{16/49*} & \alpha^{9/49*} & \alpha^{4/49*} & \alpha^{1/49*} & 1\end{pmatrix}$ |

Table G.2.3.1.1-2 defines the correlation matrix for the UE:

**Table G.2.3.1.1-2: UE correlation matrix**

| | One antenna | Two antennas | Four antennas |
|---|---|---|---|
| **UE Correlation** | $R_{UE} = 1$ | $R_{UE} = \begin{pmatrix} 1 & \beta \\ \beta^* & \mathbf{1} \end{pmatrix}$ | $R_{UE} = \begin{pmatrix} 1 & \beta^{1/9} & \beta^{4/9} & \beta \\ \beta^{1/9*} & 1 & \beta^{1/9} & \beta^{4/9} \\ \beta^{4/9*} & \beta^{1/9*} & 1 & \beta^{1/9} \\ \beta^* & \beta^{4/9*} & \beta^{1/9*} & 1 \end{pmatrix}$ |

Table G.2.3.1.1-3 defines the channel spatial correlation matrix $R_{spat}$ . The parameters $\alpha$ and $\beta$ in Table G.2.3.1.1-3 defines the spatial correlation between the antennas at the gNB and UE respectively.

**Table G.2.3.1.1-3: $R_{spat}$ correlation matrices**

| | |
|---|---|
| **1x2 case** | $$R_{spat} = R_{gNB} = \begin{pmatrix} 1 & \alpha \\ \alpha^{*} & 1 \end{pmatrix}$$ |
| **1x4 case** | $$R_{spat} = R_{gNB} = \begin{pmatrix} 1 & \alpha^{1/9} & \alpha^{4/9} & \alpha \\ \alpha^{1/9^{*}} & 1 & \alpha^{1/9} & \alpha^{4/9} \\ \alpha^{4/9^{*}} & \alpha^{1/9^{*}} & 1 & \alpha^{1/9} \\ \alpha^{*} & \alpha^{4/9^{*}} & \alpha^{1/9^{*}} & 1 \end{pmatrix}$$ |
| **1x8 case** | $$R_{spat} = R_{gNB} = \begin{pmatrix} 1 & \alpha^{1/49} & \alpha^{4/49} & \alpha^{9/49} & \alpha^{16/49} & \alpha^{25/49} & \alpha^{36/49} & \alpha \\ \alpha^{1/49^{*}} & 1 & \alpha^{1/49} & \alpha^{4/49} & \alpha^{9/49} & \alpha^{16/49} & \alpha^{25/49} & \alpha^{36/49} \\ \alpha^{4/49^{*}} & \alpha^{1/49^{*}} & 1 & \alpha^{1/49} & \alpha^{4/49} & \alpha^{9/49} & \alpha^{16/49} & \alpha^{25/49} \\ \alpha^{9/49^{*}} & \alpha^{4/49^{*}} & \alpha^{1/49^{*}} & 1 & \alpha^{1/49} & \alpha^{4/49} & \alpha^{9/49} & \alpha^{16/49} \\ \alpha^{16/49^{*}} & \alpha^{9/49^{*}} & \alpha^{4/49^{*}} & \alpha^{1/49^{*}} & 1 & \alpha^{1/49} & \alpha^{4/49} & \alpha^{9/49} \\ \alpha^{25/49^{*}} & \alpha^{16/49^{*}} & \alpha^{9/49^{*}} & \alpha^{4/49^{*}} & \alpha^{1/49^{*}} & 1 & \alpha^{1/49} & \alpha^{4/49} \\ \alpha^{36/49^{*}} & \alpha^{25/49^{*}} & \alpha^{16/49^{*}} & \alpha^{9/49^{*}} & \alpha^{4/49^{*}} & \alpha^{1/49^{*}} & 1 & \alpha^{1/49} \\ \alpha^{*} & \alpha^{36/49^{*}} & \alpha^{25/49^{*}} & \alpha^{16/49^{*}} & \alpha^{9/49^{*}} & \alpha^{4/49^{*}} & \alpha^{1/49^{*}} & 1 \end{pmatrix}$$ |
| **2x2 case** | $$R_{spat} = R_{UE} \otimes R_{gNB} = \begin{bmatrix} 1 & \beta \\ \beta^{*} & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & \alpha \\ \alpha^{*} & 1 \end{bmatrix} = \begin{bmatrix} 1 & \alpha & \beta & \beta\alpha \\ \alpha^{*} & 1 & \beta\alpha^{*} & \beta \\ \beta^{*} & \beta^{*}\alpha & 1 & \alpha \\ \beta^{*}\alpha^{*} & \beta^{*} & \alpha^{*} & 1 \end{bmatrix}$$ |
| **2x4 case** | $$R_{spat} = R_{UE} \otimes R_{gNB} = \begin{bmatrix} 1 & \beta \\ \beta^{*} & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & \alpha^{1/9} & \alpha^{4/9} & \alpha \\ \alpha^{1/9^{*}} & 1 & \alpha^{1/9} & \alpha^{4/9} \\ \alpha^{4/9^{*}} & \alpha^{1/9^{*}} & 1 & \alpha^{1/9} \\ \alpha^{*} & \alpha^{4/9^{*}} & \alpha^{1/9^{*}} & 1 \end{bmatrix}$$ |
| **2x8 case** | $$R_{spat} = R_{UE} \otimes R_{gNB} = \begin{bmatrix} 1 & \beta \\ \beta^{*} & 1 \end{bmatrix} \otimes \begin{pmatrix} 1 & \alpha^{1/49} & \alpha^{4/49} & \alpha^{9/49} & \alpha^{16/49} & \alpha^{25/49} & \alpha^{36/49} & \alpha \\ \alpha^{1/49^{*}} & 1 & \alpha^{1/49} & \alpha^{4/49} & \alpha^{9/49} & \alpha^{16/49} & \alpha^{25/49} & \alpha^{36/49} \\ \alpha^{4/49^{*}} & \alpha^{1/49^{*}} & 1 & \alpha^{1/49} & \alpha^{4/49} & \alpha^{9/49} & \alpha^{16/49} & \alpha^{25/49} \\ \alpha^{9/49^{*}} & \alpha^{4/49^{*}} & \alpha^{1/49^{*}} & 1 & \alpha^{1/49} & \alpha^{4/49} & \alpha^{9/49} & \alpha^{16/49} \\ \alpha^{16/49^{*}} & \alpha^{9/49^{*}} & \alpha^{4/49^{*}} & \alpha^{1/49^{*}} & 1 & \alpha^{1/49} & \alpha^{4/49} & \alpha^{9/49} \\ \alpha^{25/49^{*}} & \alpha^{16/49^{*}} & \alpha^{9/49^{*}} & \alpha^{4/49^{*}} & \alpha^{1/49^{*}} & 1 & \alpha^{1/49} & \alpha^{4/49} \\ \alpha^{36/49^{*}} & \alpha^{25/49^{*}} & \alpha^{16/49^{*}} & \alpha^{9/49^{*}} & \alpha^{4/49^{*}} & \alpha^{1/49^{*}} & 1 & \alpha^{1/49} \\ \alpha^{*} & \alpha^{36/49^{*}} & \alpha^{25/49^{*}} & \alpha^{16/49^{*}} & \alpha^{9/49^{*}} & \alpha^{4/49^{*}} & \alpha^{1/49^{*}} & 1 \end{pmatrix}$$ |

| | |
|---|---|
| **4x4 case** | $R_{spat} = R_{UE} \otimes R_{gNB} = \begin{pmatrix} 1 & \beta^{1/9} & \beta^{4/9} & \beta \\ \beta^{1/9*} & 1 & \beta^{1/9} & \beta^{4/9} \\ \beta^{4/9*} & \beta^{1/9*} & 1 & \beta^{1/9} \\ \beta^{*} & \beta^{4/9*} & \beta^{1/9*} & 1 \end{pmatrix} \otimes \begin{bmatrix} 1 & \alpha^{1/9} & \alpha^{4/9} & \alpha \\ \alpha^{1/9^{*}} & 1 & \alpha^{1/9} & \alpha^{4/9} \\ \alpha^{4/9^{*}} & \alpha^{1/9^{*}} & 1 & \alpha^{1/9} \\ \alpha^{*} & \alpha^{4/9^{*}} & \alpha^{1/9^{*}} & 1 \end{bmatrix}$ |

For cases with more antennas at either gNB or UE or both, the channel spatial correlation matrix can still be expressed as the Kronecker product of $R_{UE}$ and $R_{gNB}$ according to $R_{spat} = R_{UE} \otimes R_{gNB}$.

### G.2.3.1.2 MIMO Correlation Matrices at High, Medium and Low Level

The $\alpha$ and $\beta$ for different correlation types are given in Table G.2.3.1.2-1.

**Table G.2.3.1.2-1: Correlation for High Medium and Low Level**

| Low correlation | | Medium Correlation | | High Correlation | |
|---|---|---|---|---|---|
| α | β | α | β | α | β |
| 0 | 0 | 0.9 | 0.3 | 0.9 | 0.9 |

The correlation matrices for high, medium and low correlation are defined in Table G.2.3.1.2-2, G.2.3.1.2-3 and G.2.3.1.2-4 as below.

The values in Table G.2.3.1.2-2 have been adjusted for the 2x4 and 4x4 high correlation cases to insure the correlation matrix is positive semi-definite after round-off to 4-digit precision. This is done using the equation:

$$\mathbf{R}_{high} = [\mathbf{R}_{spatial} + aI_n]/(1+a)$$

Where the value "a" is a scaling factor such that the smallest value is used to obtain a positive semi-definite result. For the 2x4 high correlation case, a=0.00010. For the 4x4 high correlation case, a=0.00012.

The same method is used to adjust the 4x4 medium correlation matrix in Table G.2.3.1.2-3 to insure the correlation matrix is positive semi-definite after round-off to 4-digit precision with a =0.00012.

**Table G.2.3.1.2-2: MIMO correlation matrices for high correlation**

| | |
|---|---|
| **1x2 case** | $R_{high} = \begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix}$ |
| **2x2 case** | $R_{high} = \begin{pmatrix} 1 & 0.9 & 0.9 & 0.81 \\ 0.9 & 1 & 0.81 & 0.9 \\ 0.9 & 0.81 & 1 & 0.9 \\ 0.81 & 0.9 & 0.9 & 1 \end{pmatrix}$ |
| **2x4 case** | $R_{high} = \begin{bmatrix} 1.0000 & 0.9883 & 0.9542 & 0.8999 & 0.8999 & 0.8894 & 0.8587 & 0.8099 \\ 0.9883 & 1.0000 & 0.9883 & 0.9542 & 0.8894 & 0.8999 & 0.8894 & 0.8587 \\ 0.9542 & 0.9883 & 1.0000 & 0.9883 & 0.8587 & 0.8894 & 0.8999 & 0.8894 \\ 0.8999 & 0.9542 & 0.9883 & 1.0000 & 0.8099 & 0.8587 & 0.8894 & 0.8999 \\ 0.8999 & 0.8894 & 0.8587 & 0.8099 & 1.0000 & 0.9883 & 0.9542 & 0.8999 \\ 0.8894 & 0.8999 & 0.8894 & 0.8587 & 0.9883 & 1.0000 & 0.9883 & 0.9542 \\ 0.8587 & 0.8894 & 0.8999 & 0.8894 & 0.9542 & 0.9883 & 1.0000 & 0.9883 \\ 0.8099 & 0.8587 & 0.8894 & 0.8999 & 0.8999 & 0.9542 & 0.9883 & 1.0000 \end{bmatrix}$ |
| **4x4 case** | $R_{high} = \begin{bmatrix} 1.0000 & 0.9882 & 0.9541 & 0.8999 & 0.9882 & 0.9767 & 0.9430 & 0.8894 & 0.9541 & 0.9430 & 0.9105 & 0.8587 & 0.8999 & 0.8894 & 0.8587 & 0.8099 \\ 0.9882 & 1.0000 & 0.9882 & 0.9541 & 0.9767 & 0.9882 & 0.9767 & 0.9430 & 0.9430 & 0.9541 & 0.9430 & 0.9105 & 0.8894 & 0.8999 & 0.8894 & 0.8587 \\ 0.9541 & 0.9882 & 1.0000 & 0.9882 & 0.9430 & 0.9767 & 0.9882 & 0.9767 & 0.9105 & 0.9430 & 0.9541 & 0.9430 & 0.8587 & 0.8894 & 0.8999 & 0.8894 \\ 0.8999 & 0.9541 & 0.9882 & 1.0000 & 0.8894 & 0.9430 & 0.9767 & 0.9882 & 0.8587 & 0.9105 & 0.9430 & 0.9541 & 0.8099 & 0.8587 & 0.8894 & 0.8999 \\ 0.9882 & 0.9767 & 0.9430 & 0.8894 & 1.0000 & 0.9882 & 0.9541 & 0.8999 & 0.9882 & 0.9767 & 0.9430 & 0.8894 & 0.9541 & 0.9430 & 0.9105 & 0.8587 \\ 0.9767 & 0.9882 & 0.9767 & 0.9430 & 0.9882 & 1.0000 & 0.9882 & 0.9541 & 0.9767 & 0.9882 & 0.9767 & 0.9430 & 0.9430 & 0.9541 & 0.9430 & 0.9105 \\ 0.9430 & 0.9767 & 0.9882 & 0.9767 & 0.9541 & 0.9882 & 1.0000 & 0.9882 & 0.9430 & 0.9767 & 0.9882 & 0.9767 & 0.9105 & 0.9430 & 0.9541 & 0.9430 \\ 0.8894 & 0.9430 & 0.9767 & 0.9882 & 0.8999 & 0.9541 & 0.9882 & 1.0000 & 0.8894 & 0.9430 & 0.9767 & 0.9882 & 0.8587 & 0.9105 & 0.9430 & 0.9541 \\ 0.9541 & 0.9430 & 0.9105 & 0.8587 & 0.9882 & 0.9767 & 0.9430 & 0.8894 & 1.0000 & 0.9882 & 0.9541 & 0.8999 & 0.9882 & 0.9767 & 0.9430 & 0.8894 \\ 0.9430 & 0.9541 & 0.9430 & 0.9105 & 0.9767 & 0.9882 & 0.9767 & 0.9430 & 0.9882 & 1.0000 & 0.9882 & 0.9541 & 0.9767 & 0.9882 & 0.9767 & 0.9430 \\ 0.9105 & 0.9430 & 0.9541 & 0.9430 & 0.9430 & 0.9767 & 0.9882 & 0.9767 & 0.9541 & 0.9882 & 1.0000 & 0.9882 & 0.9430 & 0.9767 & 0.9882 & 0.9767 \\ 0.8587 & 0.9105 & 0.9430 & 0.9541 & 0.8894 & 0.9430 & 0.9767 & 0.9882 & 0.8999 & 0.9541 & 0.9882 & 1.0000 & 0.8894 & 0.9430 & 0.9767 & 0.9882 \\ 0.8999 & 0.8894 & 0.8587 & 0.8099 & 0.9541 & 0.9430 & 0.9105 & 0.8587 & 0.9882 & 0.9767 & 0.9430 & 0.8894 & 1.0000 & 0.9882 & 0.9541 & 0.8999 \\ 0.8894 & 0.8999 & 0.8894 & 0.8587 & 0.9430 & 0.9541 & 0.9430 & 0.9105 & 0.9767 & 0.9882 & 0.9767 & 0.9430 & 0.9882 & 1.0000 & 0.9882 & 0.9541 \\ 0.8587 & 0.8894 & 0.8999 & 0.8894 & 0.9105 & 0.9430 & 0.9541 & 0.9430 & 0.9430 & 0.9767 & 0.9882 & 0.9767 & 0.9541 & 0.9882 & 1.0000 & 0.9882 \\ 0.8099 & 0.8587 & 0.8894 & 0.8999 & 0.8587 & 0.9105 & 0.9430 & 0.9541 & 0.8894 & 0.9430 & 0.9767 & 0.9882 & 0.8999 & 0.9541 & 0.9882 & 1.0000 \end{bmatrix}$ |

**Table G.2.3.1.2-3: MIMO correlation matrices for medium correlation**

| | |
|---|---|
| **1x2 case** | [N/A] |
| **2x2 case** | $R_{medium} = \begin{pmatrix} 1.0000 & 0.9000 & 0.3000 & 0.2700 \\ 0.9000 & 1.0000 & 0.2700 & 0.3000 \\ 0.3000 & 0.2700 & 1.0000 & 0.9000 \\ 0.2700 & 0.3000 & 0.9000 & 1.0000 \end{pmatrix}$ |
| **2x4 case** | $R_{medium} = \begin{pmatrix} 1.0000 & 0.9884 & 0.9543 & 0.9000 & 0.3000 & 0.2965 & 0.2863 & 0.2700 \\ 0.9884 & 1.0000 & 0.9884 & 0.9543 & 0.2965 & 0.3000 & 0.2965 & 0.2863 \\ 0.9543 & 0.9884 & 1.0000 & 0.9884 & 0.2863 & 0.2965 & 0.3000 & 0.2965 \\ 0.9000 & 0.9543 & 0.9884 & 1.0000 & 0.2700 & 0.2863 & 0.2965 & 0.3000 \\ 0.3000 & 0.2965 & 0.2863 & 0.2700 & 1.0000 & 0.9884 & 0.9543 & 0.9000 \\ 0.2965 & 0.3000 & 0.2965 & 0.2863 & 0.9884 & 1.0000 & 0.9884 & 0.9543 \\ 0.2863 & 0.2965 & 0.3000 & 0.2965 & 0.9543 & 0.9884 & 1.0000 & 0.9884 \\ 0.2700 & 0.2863 & 0.2965 & 0.3000 & 0.9000 & 0.9543 & 0.9884 & 1.0000 \end{pmatrix}$ |
| **4x4 case** | $R_{medium} = \begin{pmatrix} 1.0000 & 0.9882 & 0.9541 & 0.8999 & 0.8747 & 0.8645 & 0.8347 & 0.7872 & 0.5855 & 0.5787 & 0.5588 & 0.5270 & 0.3000 & 0.2965 & 0.2862 & 0.2700 \\ 0.9882 & 1.0000 & 0.9882 & 0.9541 & 0.8645 & 0.8747 & 0.8645 & 0.8347 & 0.5787 & 0.5855 & 0.5787 & 0.5588 & 0.2965 & 0.3000 & 0.2965 & 0.2862 \\ 0.9541 & 0.9882 & 1.0000 & 0.9882 & 0.8347 & 0.8645 & 0.8747 & 0.8645 & 0.5588 & 0.5787 & 0.5855 & 0.5787 & 0.2862 & 0.2965 & 0.3000 & 0.2965 \\ 0.8999 & 0.9541 & 0.9882 & 1.0000 & 0.7872 & 0.8347 & 0.8645 & 0.8747 & 0.5270 & 0.5588 & 0.5787 & 0.5855 & 0.2700 & 0.2862 & 0.2965 & 0.3000 \\ 0.8747 & 0.8645 & 0.8347 & 0.7872 & 1.0000 & 0.9882 & 0.9541 & 0.8999 & 0.8747 & 0.8645 & 0.8347 & 0.7872 & 0.5855 & 0.5787 & 0.5588 & 0.5270 \\ 0.8645 & 0.8747 & 0.8645 & 0.8347 & 0.9882 & 1.0000 & 0.9882 & 0.9541 & 0.8645 & 0.8747 & 0.8645 & 0.8347 & 0.5787 & 0.5855 & 0.5787 & 0.5588 \\ 0.8347 & 0.8645 & 0.8747 & 0.8645 & 0.9541 & 0.9882 & 1.0000 & 0.9882 & 0.8347 & 0.8645 & 0.8747 & 0.8645 & 0.5588 & 0.5787 & 0.5855 & 0.5787 \\ 0.7872 & 0.8347 & 0.8645 & 0.8747 & 0.8999 & 0.9541 & 0.9882 & 1.0000 & 0.7872 & 0.8347 & 0.8645 & 0.8747 & 0.5270 & 0.5588 & 0.5787 & 0.5855 \\ 0.5855 & 0.5787 & 0.5588 & 0.5270 & 0.8747 & 0.8645 & 0.8347 & 0.7872 & 1.0000 & 0.9882 & 0.9541 & 0.8999 & 0.8747 & 0.8645 & 0.8347 & 0.7872 \\ 0.5787 & 0.5855 & 0.5787 & 0.5588 & 0.8645 & 0.8747 & 0.8645 & 0.8347 & 0.9882 & 1.0000 & 0.9882 & 0.9541 & 0.8645 & 0.8747 & 0.8645 & 0.8347 \\ 0.5588 & 0.5787 & 0.5855 & 0.5787 & 0.8347 & 0.8645 & 0.8747 & 0.8645 & 0.9541 & 0.9882 & 1.0000 & 0.9882 & 0.8347 & 0.8645 & 0.8747 & 0.8645 \\ 0.5270 & 0.5588 & 0.5787 & 0.5855 & 0.7872 & 0.8347 & 0.8645 & 0.8747 & 0.8999 & 0.9541 & 0.9882 & 1.0000 & 0.7872 & 0.8347 & 0.8645 & 0.8747 \\ 0.3000 & 0.2965 & 0.2862 & 0.2700 & 0.5855 & 0.5787 & 0.5588 & 0.5270 & 0.8747 & 0.8645 & 0.8347 & 0.7872 & 1.0000 & 0.9882 & 0.9541 & 0.8999 \\ 0.2965 & 0.3000 & 0.2965 & 0.2862 & 0.5787 & 0.5855 & 0.5787 & 0.5588 & 0.8645 & 0.8747 & 0.8645 & 0.8347 & 0.9882 & 1.0000 & 0.9882 & 0.9541 \\ 0.2862 & 0.2965 & 0.3000 & 0.2965 & 0.5588 & 0.5787 & 0.5855 & 0.5787 & 0.8347 & 0.8645 & 0.8747 & 0.8645 & 0.9541 & 0.9882 & 1.0000 & 0.9882 \\ 0.2700 & 0.2862 & 0.2965 & 0.3000 & 0.5270 & 0.5588 & 0.5787 & 0.5855 & 0.7872 & 0.8347 & 0.8645 & 0.8747 & 0.8999 & 0.9541 & 0.9882 & 1.0000 \end{pmatrix}$ |

**Table G.2.3.1.2-4: MIMO correlation matrices for low correlation**

| | |
|---|---|
| **1x2 case** | $R_{low} = \mathbf{I}_2$ |
| **1x4 case** | $R_{low} = \mathbf{I}_4$ |
| **1x8 case** | $R_{low} = \mathbf{I}_8$ |
| **2x2 case** | $R_{low} = \mathbf{I}_4$ |
| **2x4 case** | $R_{low} = \mathbf{I}_8$ |
| **2x8 case** | $R_{low} = \mathbf{I}_{16}$ |
| **4x4 case** | $R_{low} = \mathbf{I}_{16}$ |

In Table G.2.3.1.2-4, $\mathbf{I}_d$ is a $d \times d$ identity matrix.

NOTE: For completeness, the correlation matrices were defined for high, medium and low correlation but performance requirements exist only for low correlation.

## G.2.3.2 Multi-Antenna channel models using cross polarized antennas

The MIMO channel correlation matrices defined in G.2.3.2 apply to two cases as presented below:

- One TX antenna and multiple RX antennas case, with cross polarized antennas used at gNB
- Multiple TX antennas and multiple RX antennas case, with cross polarized antennas used at both UE and gNB

The cross-polarized antenna elements with +/-45 degrees polarization slant angles are deployed at gNB. For one TX antenna case, antenna element with +90 degree polarization slant angle is deployed at UE. For multiple TX antennas case, cross-polarized antenna elements with +90/0 degrees polarization slant angles are deployed at UE.

For the cross-polarized antennas, the N antennas are labelled such that antennas for one polarization are listed from 1 to N/2 and antennas for the other polarization are listed from N/2+1 to N, where N is the number of TX or RX antennas.

### G.2.3.2.1 Definition of MIMO Correlation Matrices using cross polarized antennas

For the channel spatial correlation matrix, the following is used:

$$R_{Spat} = P_{UL}\left(R_{UE} \otimes \Gamma_{UL} \otimes R_{gNB}\right)P_{UL}^{T}$$

Where

- $R_{UE}$ is the spatial correlation matrix at the UE with same polarization,
- $R_{gNB}$ is the spatial correlation matrix at the gNB with same polarization,
- $\Gamma_{UL}$ is a polarization correlation matrix,
- $P_{UL}$ is a permutation matrix, and
- $(\bullet)^{T}$ denotes transpose.

Table G.2.3.2.1-1 defines the polarization correlation matrix.

**Table G.2.3.2.1-1: Polarization correlation matrix**

| | One TX antenna | Multiple TX antennas |
|---|---|---|
| **Polarization correlation matrix** | $\Gamma_{UL} = \begin{bmatrix} 1 & -\gamma \\ -\gamma & 1 \end{bmatrix}$ | $\Gamma_{UL} = \begin{bmatrix} 1 & -\gamma & 0 & 0 \\ -\gamma & 1 & 0 & 0 \\ 0 & 0 & 1 & \gamma \\ 0 & 0 & \gamma & 1 \end{bmatrix}$ |

The matrix $P_{UL}$ is defined as

$$\mathbf{P}_{UL}(a,b) = \begin{cases} 1 & for\ a = (j-1)Nr + i\ and\ b = 2(j-1)Nr + i, \quad i = 1,\cdots,Nr,\ j = 1,\cdots,\lceil Nt/2 \rceil \\ 1 & for\ a = (j-1)Nr + i\ and\ b = 2(j - Nt/2)Nr - Nr + i, \quad i = 1,\cdots,Nr,\ j = \lceil Nt/2 \rceil + 1,...,Nt \\ 0 & otherwise \end{cases}$$

where $Nt$ and $Nr$ is the number of TX and RX antennas respectively, and $\lceil \bullet \rceil$ is the ceiling operator.

The matrix $P_{UL}$ is used to map the spatial correlation coefficients in accordance with the antenna element labelling system described in G.2.3.2.

### G.2.3.2.2 Spatial Correlation Matrices at UE and gNB sides

#### G.2.3.2.2.1 Spatial Correlation Matrices at UE side

For 1-antenna transmitter, $R_{UE} = 1$.

For 2-antenna transmitter using one pair of cross-polarized antenna elements, $R_{UE} = 1$.

For 4-antenna transmitter using two pairs of cross-polarized antenna elements, $R_{UE} = \begin{pmatrix} 1 & \beta \\ \beta^* & 1 \end{pmatrix}$.

#### G.2.3.2.2.2 Spatial Correlation Matrices at gNB side

For 2-antenna receiver using one pair of cross-polarized antenna elements, $R_{gNB} = 1$.

For 4-antenna receiver using two pairs of cross-polarized antenna elements, $R_{gNB} = \begin{pmatrix} 1 & \alpha \\ \alpha^* & 1 \end{pmatrix}$.

For 8-antenna receiver using four pairs of cross-polarized antenna elements, $R_{gNB} = \begin{pmatrix} 1 & \alpha^{1/9} & \alpha^{4/9} & \alpha \\ \alpha^{1/9*} & 1 & \alpha^{1/9} & \alpha^{4/9} \\ \alpha^{4/9*} & \alpha^{1/9*} & 1 & \alpha^{1/9} \\ \alpha^* & \alpha^{4/9*} & \alpha^{1/9*} & 1 \end{pmatrix}$.

### G.2.3.2.3 MIMO Correlation Matrices using cross polarized antennas

The values for parameters $\alpha$, $\beta$ and $\gamma$ for low spatial correlation are given in Table G.2.3.2.3-1.

**Table G.2.3.2.3-1: Values for parameters α, β and γ**

| Low spatial correlation | | |
|---|---|---|
| α | β | γ |
| 0 | 0 | 0 |
| Note 1: Value of $\alpha$ applies when more than one pair of cross-polarized antenna elements at gNB side. | | |
| Note 2: Value of $\beta$ applies when more than one pair of cross-polarized antenna elements at UE side. | | |

The correlation matrices for low spatial correlation are defined in Table G.2.3.2.3-2 as below.

**Table G.2.3.2.3-2: MIMO correlation matrices for low spatial correlation**

| | |
|---|---|
| 1x8 case | $R_{low} = \mathbf{I}_8$ |
| 2x8 case | $R_{low} = \mathbf{I}_{16}$ |

In Table G.2.3.2.3-2, $\mathbf{I}_d$ is a $d \times d$ identity matrix.

## G.3 High speed train condition

High speed train conditions are as follows:

- Scenario 1-NR350 / Scenario 1-NR500: Open space
- Scenario 3-NR350 / Scenario 3-NR500: Tunnel
- Scenario 4-BI-NR350, FR2: Open space

The high speed train conditions for the test of the baseband performance are three non-fading propagation channels. For BS with Rx diversity, the Doppler shift time variation is the same for each antenna at each time instant.

Doppler shift for all three scenarios is given by:

$$f_s(t) = f_d \cos\theta(t) \quad \text{(G.3.1)}$$

where $f_s(t)$ is the Doppler shift and $f_d$ is the maximum Doppler frequency.

For Sceanrio 1 and Scenario 3, he cosine of angle $\theta(t)$ is given by:

$$\cos\theta(t)=\frac{D_s/2-vt}{\sqrt{D_{\min}{}^2+(D_s/2-vt)^2}},\ 0\le t\le D_s/v \quad \text{(G.3.2)}$$

$$\cos\theta(t)=\frac{-1.5D_s+vt}{\sqrt{D_{\min}{}^2+(-1.5D_s+vt)^2}},\ D_s/v < t \le 2D_s/v \quad \text{(G.3.3)}$$

$$\cos\theta(t)=\cos\theta(t \mod (2D_s/v)),\ t > 2D_s/v \quad \text{(G.3.4)}$$

where $D_s/2$ is the initial distance of the train from BS, and $D_{\min}$ is BS-Railway track distance, both in meters; $v$ is the velocity of the train in m/s, $t$ is time in seconds.

For Sceanrio 4, the cosine of angle $\theta(t)$ is given by:

$$cos\ \theta(t) = \frac{D_s - vt}{\sqrt{D_{min}^2+(D_s-vt)^2}},\quad 0 < t \le (0.5 * D_s)/v \quad \text{(G.3.5)}$$

$$cos\ \theta(t) = -\frac{vt}{\sqrt{D_{min}^2+(vt)^2}},\quad (0.5 * D_s)/v < t \le D_s/v \quad \text{(G.3.6)}$$

$$cos\ \theta(t) = cos\theta\left(t\ mod\left(\frac{D_s}{v}\right)\right),\quad t > D_s/v \quad \text{(G.3.7)}$$

where $D_s$ is the initial distance of the train from BS, and $D_{min}$ is BS-Railway track distance, both in meters; $v$ is the velocity of the train in m/s; $t$ is time in seconds.

The required input parameters are listed in table G.3-1 and G.3-2. The resulting time varying Doppler shift is shown in Figure G.3-1, G.3-2, G.3-3, G.3-4 and G.3-9 for 350km/h scenarios, and in Figure G.3-5, G.3-6, G.3-7 and G.3-8 for 500km/h scenarios. For 350km/h scenarios, the Doppler shift was derived such that it corresponds to a velocity of around 350km/h for band n1 for the 15kHz SCS, for band n77 for the 30kHz SCS, and for band n257 for the 120kHz SCS. For 500km/h scenarios, the Doppler shift was derived such that it corresponds to a velocity of around 500km/h for band n3 for the 15kHz SCS and for band n77 for the 30kHz SCS. However, the same Doppler shift requirement shall be applied regardless of the frequency of operation of the basestation and thus for lower frequencies, the supported speed is higher.

**Table G.3-1: Parameters for high speed train conditions for UE velocity 350 km/h**

| Parameter | Value | | |
|---|---|---|---|
| | **Scenario 1-NR350** | **Scenario 3-NR350** | **Scenario 4-BI-NR350, FR2** |
| $D_s$ | 700 m | 300 m | 700 m |
| $D_{\min}$ | 150 m | 2 m | 150 m |
| $v$ | 350 km/h | 350 km/h | 350 km/h |
| $f_d$ | 1340 Hz for 15kHz SCS<br>2334 Hz for 30kHz SCS | 1340 Hz for 15kHz SCS<br>2334 Hz for 30kHz SCS | 19444 Hz for 120kHz SCS |

**Table G.3-2: Parameters for high speed train conditions for UE velocity 500 km/h**

| Parameter | Value | |
|---|---|---|
| | Scenario 1-NR500 | Scenario 3-NR500 |
| $D_s$ | 700 m | 300 m |
| $D_{\min}$ | 150 m | 2 m |
| $v$ | 500 km/h | 500 km/h |
| $f_d$ | 1740 Hz for 15kHz SCS<br>3334 Hz for 30kHz SCS | 1740 Hz for 15kHz SCS<br>3334 Hz for 30kHz SCS |

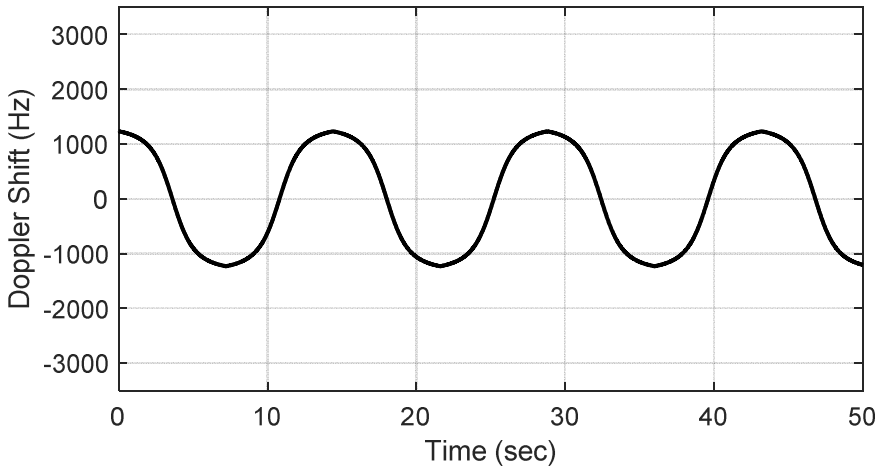



**Figure G.3-1: Doppler shift trajectory for scenario 1-NR350 (15 kHz SCS)**

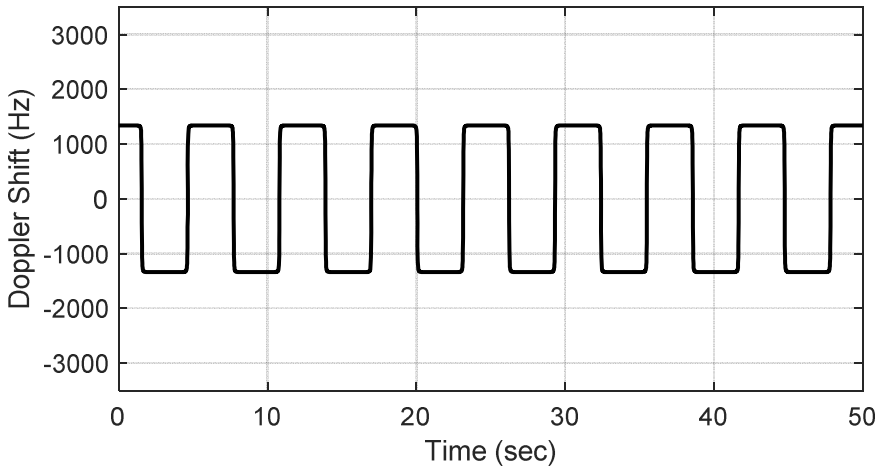



**Figure G.3-2: Doppler shift trajectory for scenario 3-NR350 (15 kHz SCS)**

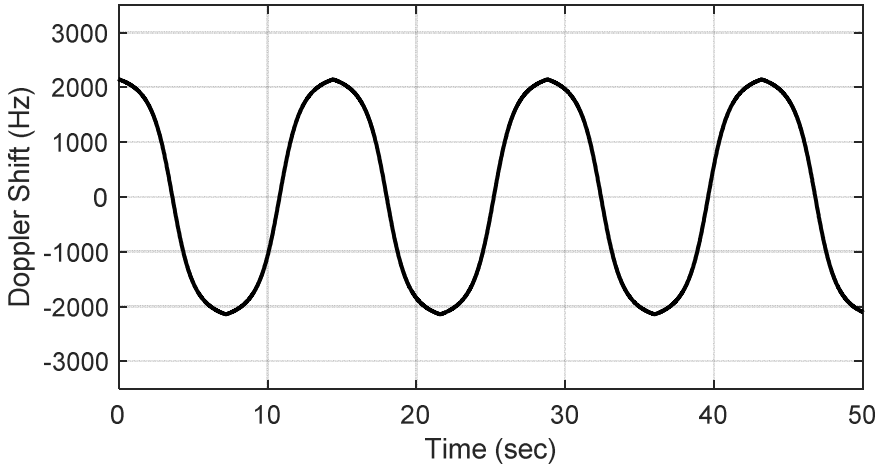



**Figure G.3-3: Doppler shift trajectory for scenario 1-NR350 (30 kHz SCS)**

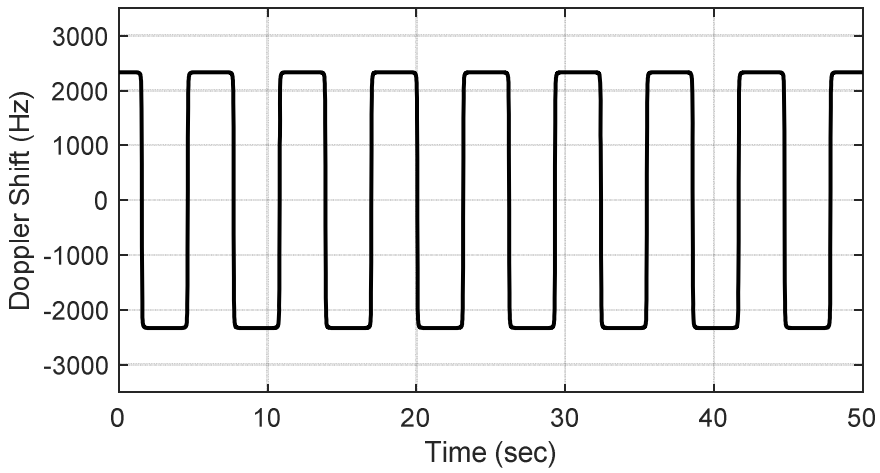



**Figure G.3-4: Doppler shift trajectory for scenario 3-NR350 (30 kHz SCS)**

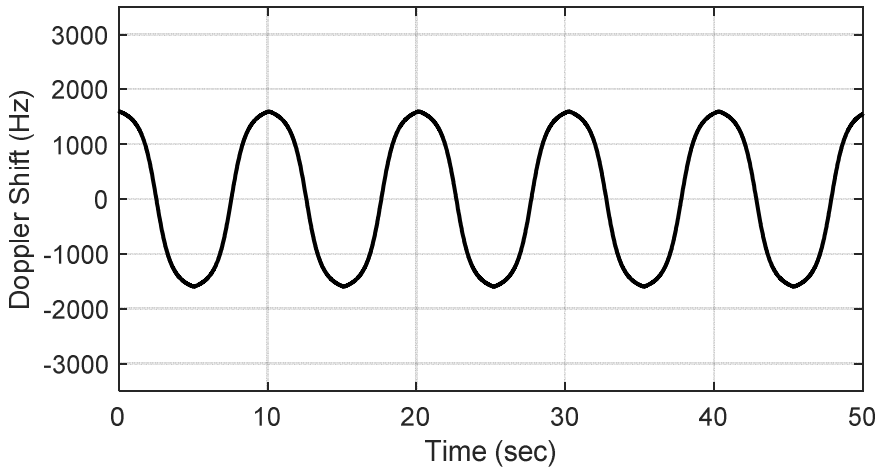



**Figure G.3-5: Doppler shift trajectory for scenario 1-NR500 (15 kHz SCS)**

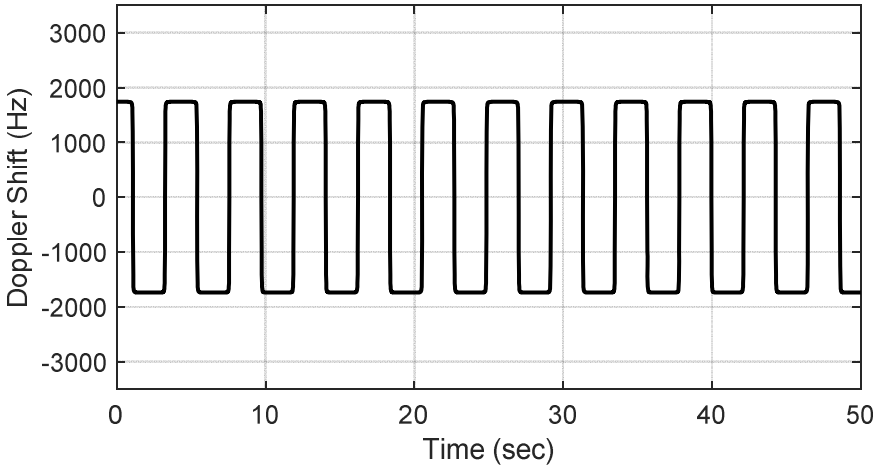



**Figure G.3-6: Doppler shift trajectory for scenario 3-NR500 (15 kHz SCS)**

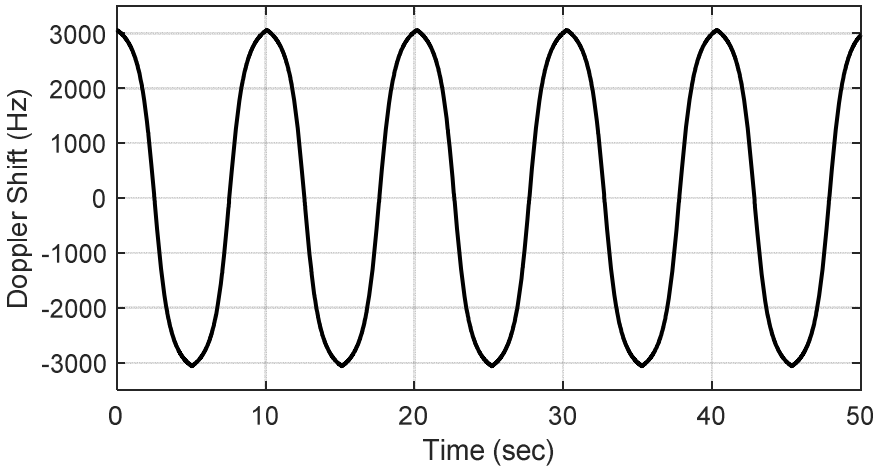



**Figure G.3-7: Doppler shift trajectory for scenario 1-NR500 (30 kHz SCS)**

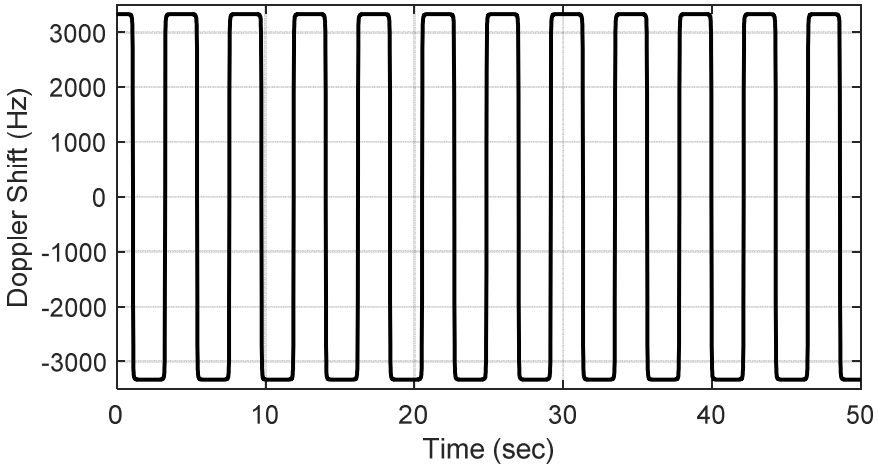



**Figure G.3-8: Doppler shift trajectory for scenario 3-NR500 (30 kHz SCS)**

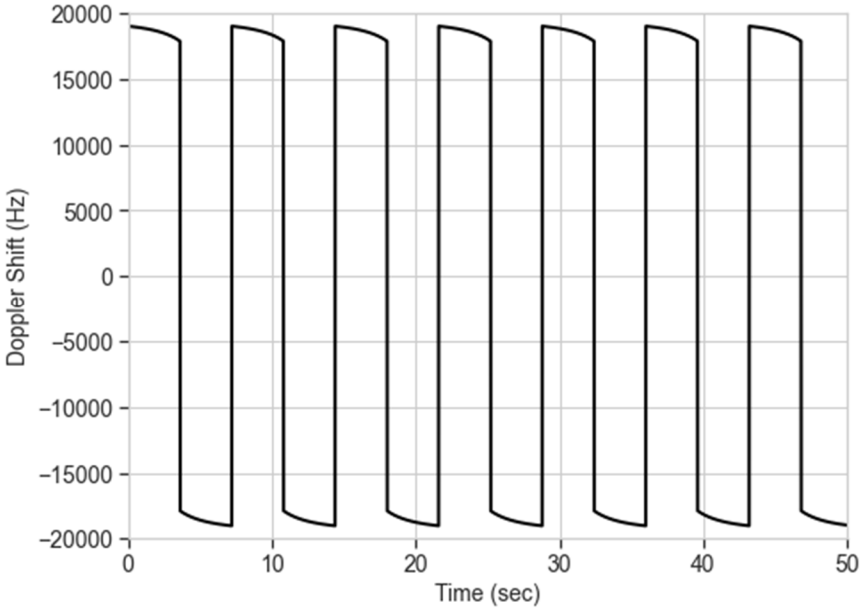



**Figure G.3-9: Doppler shift trajectory for scenario 4-BI-NR350, FR2 (120 kHz SCS)**

# G.4 Moving propagation conditions

Figure G.4-1 illustrates the moving propagation conditions for the test of the UL timing adjustment performance. The time difference between the reference timing and the first tap is according Equation (G.4-1). The timing difference between moving UE and stationary UE is equal to $\Delta\tau$ - $(T_A - 31)\times 16\times 64T_c$ for 15kHz SCS, $\Delta\tau$ - $(T_A - 31)\times 16\times 32T_c$ for 30kHz SCS and $\Delta\tau$ - $(T_A - 31)\times 16\times 8T_c$ for 120kHz SCS. The relative timing among all taps is fixed. The parameters for the moving propagation conditions are shown in Table G.4-1.

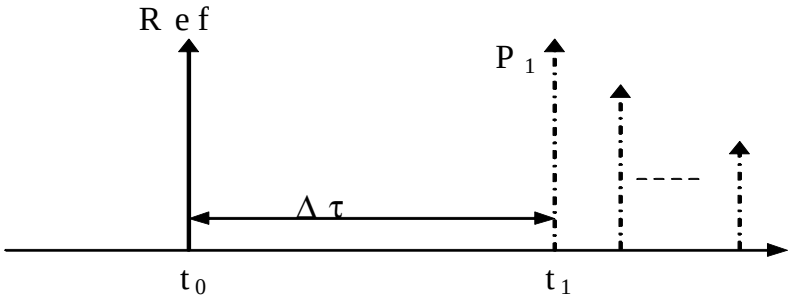



**Figure G.4-1: Moving propagation conditions**

$$\Delta\tau = \frac{A}{2}\cdot\sin(\Delta\omega\cdot t) \qquad \text{(G.4-1)}$$

**Table G.4-1: Parameters for UL timing adjustment**

| Parameter | Scenario X | Scenario Y | Scenario Z |
|---|---|---|---|
| Channel model | Stationary UE: AWGN<br>Moving UE: TDLC300-400 | Stationary UE: AWGN<br>Moving UE: AWGN | Stationary UE: AWGN<br>Moving UE: AWGN |
| UE speed | 120 km/h | 350 km/h | 500 km/h |
| CP length | Normal | Normal | Normal |
| A | 15 kHz: 10 μs<br>30 kHz: 5 μs | 15 kHz: 10 μs<br>30 kHz: 5 μs<br>120 kHz: 1.25 μs | 15 kHz: 10 μs<br>30 kHz: 5 μs |
| $\Delta\omega$ | 15 kHz: 0.04 $s^{-1}$<br>30 kHz: 0.08 $s^{-1}$ | 15 kHz: 0.13 $s^{-1}$<br>30 kHz: 0.26 $s^{-1}$<br>120 kHz: 1.04 $s^{-1}$ | 15 kHz: 0.18 $s^{-1}$<br>30 kHz: 0.36 $s^{-1}$ |

NOTE 1: Doppler shift is not taken into account in UL TA scenario Y and Z.

# Annex H (informative): Change history

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Date** | **Meeting** | **TDoc** | **CR** | **Rev** | **Cat** | **Subject/Comment** | **New version** |
| 2017-05 | RAN4#83 | R4-1704619 | | | | Specification skeleton | 0.0.1 |
| 2017-05 | RAN4#83 | R4-1705332 | | | | Specification skeleton (revised) | 0.0.2 |
| 2017-05 | RAN4#83 | R4-1706228 | | | | Specification skeleton (revised) | 0.0.3 |
| 2017-07 | RAN4-NR AH #2 | R4-1706983 | | | | Agreed Text Proposal in RAN4 NR AH #2:<br>**R4-1706955**, "TP to TS 38.104: BS classification for NR BS" | 0.1.0 |
| 2018-08 | RAN4#84 | R4-1709212 | | | | Agreed Text Proposal in RAN4 #84:<br>**R4-1708872**, "TP to TS 38.104 BS transmitter transient period" | 0.2.0 |
| 2018-10 | RAN4#84 bis | R4-1711970 | | | | Agreed Text Proposal in RAN4 #84bis:<br>**R4-1710199**, "TP for TS 38.104: out of band blocking (10.4)"<br>**R4-1710587**, "TP for TS 38.104: Relationship with other core specifications (4.1)"<br>**R4-1710588**, "TP for TS 38.104: Relationship between minimum requirements and test requirements (4.2)"<br>**R4-1710589**, "TP for TS 38.104: Regional requirements (4.5)"<br>**R4-1710591**, "TP for TS 38.104: Conducted transmitter characteristics (general) (6.1)"<br>**R4-1710593**, "TP for TS 38.104: Operating band unwanted emissions (conducted) (6.6.4)"<br>**R4-1710594**, "TP for TS 38.104: Conducted receiver characteristics (General) (7.1)"<br>**R4-1710595**, "TP for TS 38.104: Radiated transmitter characteristics (General) (9.1)"<br>**R4-1710598**, "TP for TS 38.104: Radiated receiver characteristics (General) (10.1)"<br>**R4-1711325**, "TP to TS38.104: OTA Output power dynamics (9.4)"<br>**R4-1711363**, "TP to TS 38.104 - Occupied bandwidth (6.6.2)"<br>**R4-1711745**, "TP to TS 38.104 - Conducted and radiated requirement reference points (4.3)"<br>**R4-1711746**, "TP for TS 38.104: Adding applicability table to clause 4.6"<br>**R4-1711747**, "TP for TS 38.104: Operating bands and channel arrangements. (5)"<br>**R4-1711748**, "TP to TS38.104: conducted NR BS output power (6.2)"<br>**R4-1711750**, "TP for TS 38.104: Transmit ON/OFF power (6.4)"<br>**R4-1711753**, "TP for TS 38.104: Time alignment error requirements (6.5)"<br>**R4-1711754**, "TP for TS 38.104: Unwanted emissions, General (Conducted) (6.6.1)"<br>**R4-1711755**, "TP to TS 38.104: Occupied bandwidth for FR1 and FR2 NR BS (9.7)"<br>**R4-1711756**, "TP to TS 38.104: Transmitter spurious emissions (conducted) (6.6.5)"<br>**R4-1711757**, "TP for TS 38.104:Conducted BS transmitter intermodulation for FR1 (section 6.7)"<br>**R4-1711758**, "TP to TS 38.104: Reference Sensitivity (conducted) (7.2)"<br>**R4-1711759**, "TP to TS 38.104: NR BS conducted ACLR requirement in FR1 (6.6.3)"<br>**R4-1711760**, "TP to TS38.104: conducted NR BS receiver spurious emissions (7.6)"<br>**R4-1711761**, "TP to TS38.104: Radiated NR BS transmit power; FR1 (9.2)"<br>**R4-1711762**, "TP to TS38.104: OTA base station output power, FR1 (9.3)"<br>**R4-1711763**, "TP for TS 38.104: OTA Transmit ON/OFF power (9.5)"<br>**R4-1711764**, "TP to TS 38.104 - OTA ACLR"<br>**R4-1711765**, "TP for TS 38.104: OTA Operating band unwanted emissions and Spectrum emissions mask (9.7.4)"<br>**R4-1711766**, "TP for TS 38.104: OTA Spurious emission (9.7.5)"<br>**R4-1711767**, "TP for TS 38.104: Adding specification text for OTA TX IMD requirement in clause 9.8"<br>**R4-1711768**, "TP to TS 38.104: OTA Sensitivity (10.2)"<br>**R4-1711771**, "TP to TS38.104: OTA receiver spurious emissions, FR1 (10.7)"<br>**R4-1711772**, "TP to TS 38.104: Receiver Intermodulation (10.8)"<br>**R4-1711811**, "TP to TS 38.104: NR BS conducted in-band selectivity and blocking requirements in FR1 (7.4)"<br>**R4-1711950**, "TP to TS 38.104: Modulation Quality Skeleton (6.5)"<br>**R4-1711951**, "TP to TS38.104: frequency error for FR1 NR BS (6.5&9.6)"<br>**R4-1711952**, "TP to TS 38.104: OTA reference sensitivity (10.3)" | 0.3.0 |

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| 2017-11 | RAN4#84 bis | R4-1711971 | | | | Alignment of structure, terminology, and definitions between clauses. | 0.4.0 |
| 2017-12 | RAN4#85 | R4-1714544 | | | | Agreed Text Proposal in RAN4 #85:<br>**R4-1712614**, "TP to TS 38.104 - OTA sensitivity (10.2)"<br>**R4-1712648**, "TP to TS 38.104: corrections for the applicability of "BS type" and "requirement set" definitions"<br>**R4-1712964**, "TP for TS 38.104: out of band blocking (7.5)"<br>**R4-1713631**, "TP to 38.104 on introduction of n71"<br>**R4-1713632**, "TP to 38.104, clause 4.7 (Requirements for contiguous and non-contiguous spectrum)"<br>**R4-1713633**, "TP to 38.104, clause 4.8 (Requirements for BS capable of multi-band operation)"<br>**R4-1713634**, "TP to 38.104, clause 6.6.4.2.6 (basic limits for additional requirements for operating band unwanted emissions)"<br>**R4-1714116**, "TP to TS 38.104: Revision of the TRP definition"<br>**R4-1714117**, "TP to TS 38.104: Radiated NR BS transmit power; 2-O (9.2.3)"<br>**R4-1714121**, "TP to TS 38.104: OTA Output power dynamics (9.4)"<br>**R4-1714125**, "TP to TS 38.104 v0.4.0: OTA TDD Off power"<br>**R4-1714127**, "TP for TS 38.104: OTA frequency error (9.6.1)"<br>**R4-1714129**, "TP to TS 38.104: NR BS conducted CACLR requirements in FR1 (6.6.3)"<br>**R4-1714134**, "TP to TS 38.104v0.4.0: Absolute levels for FR2 ACLR absolute levels for NR BS"<br>**R4-1714136**, "TP for TS 38.104: Update of OTA TX IM requirement for sub-clause 4.9 and sub-clause 9.8"<br>**R4-1714141**, "TP to TS 38.104: Reference Sensitivity (conducted) (7.2)"<br>**R4-1714142**, "TP to TS 38.104: NR BS FRCs for receiver requirements"<br>**R4-1714150**, "TP to TS 38.104 - OTA out of band blocking FR1 (10.6)"<br>**R4-1714306**, "TP for TS 38.104: Adding of TRP in terminology in clause 3"<br>**R4-1714307**, "TP to TS 38.104 - Conducted and radiated requirement reference points (4.3)"<br>**R4-1714308**, "TP for TS 38.104: Base station classes (4.4)"<br>**R4-1714310**, "TP to TS 38.104: Directional and TRP requirements identification (directional vs. TRP)"<br>**R4-1714312**, "TP for TS 38.104: Update of applicability table in sub-clause 4.6"<br>**R4-1714313**, "TP to TS 38.104: Operating bands (5.1-5.3)"<br>R4-1714315, "TP to TS38.104: frequency error for NR BS (6.5&9.6)"<br>**R4-1714316**, "TP for TS 38.104: Adding text for clause 6.5.2 Modulation quality"<br>**R4-1714317**, "TP to TS 38.104: Dynamic Range for FR1 (conducted)"<br>**R4-1714318**, "TP to TS38.104: ICS requirement (7.8&10.9)"<br>**R4-1714319**, "TP for TS 38.104: Adding text for clause 9.6.4 Modulation quality"<br>**R4-1714320**, "TP for TS 38.104: OTA Spurious emission (9.7.5)"<br>**R4-1714321**, "TP for TS 38.104: OTA Dynamic range (10.4)"<br>**R4-1714390**, "TP to TS 38.104: FR2 RX IM OTA, 10.8.3"<br>**R4-1714428**, "TP to TS 38.104 v0.4.0: Time alignment for CA"<br>**R4-1714430**, "TP to TS 38.104: Transmitter spurious emissions (conducted) (6.6.5)"<br>**R4-1714432**, "TP to TS 38.104: Output Power Dynamics for FR1 (conducted)"<br>**R4-1714433**, "TP to TS 38.104: OTA Rx spurious emissions for BS type O 2 (10.7.3)"<br>**R4-1714435**, "TP to TS 38.104: FR2 REFSENS"<br>**R4-1714437**, "TP for TS 38.104: Conducted Adjacent Channel Leakage Power Ratio (ACLR) (6.6.3)"<br>**R4-1714439**, "TP for TS 38.104: Receiver spurious emission (7.6)"<br>**R4-1714476**, "TP to TR 38.104: Channel arrangement (5.4)"<br>**R4-1714493**, "TP for TS 38.104: Operating band unwanted emissions (6.6.4)"<br>**R4-1714515**, "TP for TS 38.104: OTA Out-of-band emissions (9.7.4)"<br>**R4-1714517**, "TP to TS 38.104: OTA base station output power, 2-O (9.3.3)"<br>**R4-1714518**, "TP to TS 38.104: ACS and blocking update"<br>**R4-1714520**, "Draft TP to TS 38.104: OTA In-band selectivity and blocking (10.5)"<br>**R4-1714525**, "TP to TS 38.104: FR1 RX IM conducted 7.7"<br>**R4-1714526**, "TP to TS 38.104: FR1 RX IM OTA 10.8.2" | 0.5.0 |
| 2017-12 | RAN#78 | RP-172268 | | | | Presented to TSG RAN for approval. | 1.0.0 |
| 2017-12 | RAN#78 | | | | | Approved by plenary – Rel-15 spec under change control | 15.0.0 |

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| 2018-03 | RAN#79 | RP-180264 | 0004 | | F | TS 38.104 Combined updates (NSA) from RAN4 #86 and AH-1801 | 15.1.0 |
| 2018-06 | RAN#80 | RP-181076 | 0005 | | F | TS 38.104 Combined updates (NSA) from RAN4 #86bis and RAN4 #87 | 15.2.0 |
| 2018-09 | RAN#81 | RP-181896 | 0008 | | F | TS 38.104 Combined updates from RAN4 #88 | 15.3.0 |
| 2018-12 | RAN#82 | RP-182837 | 0016 | 1 | F | CR to TS 38.104 on Combined updates from RAN4 #88bis and #89 (including 7.5 kHz carrier shift in UL for remaining bands) | 15.4.0 |
| 2018-12 | RAN#82 | RP-182362 | 0017 | | B | CR to 38.104 on Combined CRs for BS Demodulation performance | 15.4.0 |
| 2019-03 | RAN#83 | RP-190403 | 0019 | | F | CR to TS 38.104 on Combined updates from RAN4 #90<br><br>This document combines the proposed changes in the following Draft CRs from RAN4 #90:<br>- R4-1900284, "Draft CR on NR PUCCH format2 performance requirements for TS 38.104"<br>- R4-1900763, "Draft CR to TS 38.104: Update of performance requirement numbers for DFT-s-OFDM based PUSCH"<br>- R4-1900876, "Draft CR to TS 38.104: On RX spurious emissions requirement"<br>- R4-1900968, "Draft CR for 38.104: Performance requirements for NR PUCCH format 1"<br>- R4-1901329, "Draft CR to 38.104: Annex C.6 correction"<br>- R4-1901330, "Draft CR to 38.104: Abbreviations addition"<br>- R4-1901387, "Draft CR to TS 38.104 BS demodulation PUCCH format 0 requirements"<br>- R4-1901474, "Draft CR to TS 38.104: Corrections on transmitter co-existence and co-location requirements"<br>- R4-1901483, "Draft CR to TS 38.104: Corrections on general intermodulation requirement"<br>- R4-1902239, "Draft CR to TS 38.104: Addition of missing EIRP/EIS definitions in terminology in clause 3.1"<br>- R4-1902241, "Draft CR to 38.104; clarification of BS power limits"<br>- R4-1902245, "Draft CR to 38.104: Correction to FR2 OTA Interfering signal mean power units"<br>- R4-1902246, "Draft CR to 38.104; Correction to definition of OTA reference sensitivity"<br>- R4-1902260, "draft CR to TS 38.104 - update emissions scaling"<br>- R4-1902338, "Draft CR: Update on FR1 range extension for TS38.104"<br>- R4-1902389, "draftCR for 38.104 on PUSCH requirements with CP-OFDM and FR1"<br>- R4-1902394, "Draft CR to TS 38.104 – PUSCH requirements with CP-OFDM for FR2"<br>- R4-1902396, "CR: Updates to PUCCH formats 3 and 4 performance requirements in TS 38.104"<br>- R4-1902444, "Draft CR to TS 38.104: Editorial CR for BS demodulation requirements"<br>- R4-1902561, "Draft CR for updating PRACH performance requirements in TS38.104"<br>- R4-1902571, "Corrections to 38.104 Delay profile calculation"<br>- R4-1902642, "Draft CR to TS 38.104: Correction on multi-band operation related requirements" | 15.5.0 |

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| 2019-06 | RAN#84 | RP-191240 | 0029 | | F | CR to TS 38.104 Combined updates from RAN4 #90bis and RAN4#91<br><br>From RAN4 #90bis:<br>- R4-1903105, "Draft CR to TS 38.104: Corrections on terminologies and editorial errors"<br>- R4-1903319, "Draft CR to TS 38.104: removal of unused definition: "minimum EIRP level under extreme condition""<br>- R4-1903320, "Draft CR to TS 38.104: OSDD information correction"<br>- R4-1903457, "Draft CR to TS 38.104: Removal of FFS for FR2 TDD OFF power level requirement in clause 9.5"<br>- R4-1903499, "Draft CR to 38.104: Correction to unwanted emissions mask for bands n7 and n38"<br>- R4-1903836, "Draft CR to TS 38.104: Correction on description on multi-band operation in section 4.8"<br>- R4-1904024, "Draft CR to TS 38.104 Applicability rules for BS demodulation"<br>- R4-1904234, "draftCR: Correlation matrix for 8Rx in TS 38.104"<br>- R4-1904723, "Draft CR to TS 38.104: Update of performance requirements for DFT-s-OFDM based PUSCH"<br>- R4-1904726, "draftCR for 38.104 on PUSCH requirements with CP-OFDM and FR1"<br>- R4-1904729, "Draft CR on PRACH performance requirements in TS38.104"<br>- R4-1904734, "Draft CR on TS 38.104 Performance requirement for PUCCH format 1"<br>- R4-1904735, "Draft CR on NR PUCCH format2 performance requirements for TS 38.104"<br>- R4-1904739, "Draft CR to TS 38.104 BS demodulation PUCCH format 0 requirements"<br>- R4-1904745, "draftCR: Updates to PUCCH formats 3 and 4 performance requirements in TS 38.104"<br>- R4-1904799, "Draft CR to TS 38.104: FRC update for PUSCH FR1 mapping type B and FR2 DMRS 1+1"<br>- R4-1904816, "Draft CR : Clarification on step 5 and step 6 for delay profiles calculation (38.104)"<br>- R4-1904842, "Draft CR to TS 38.104 BS demodulation CP-OFDM PUSCH FR2 requirements"<br>- R4-1905126, "draft CR to 38.104 for TAE requirements"<br>- R4-1905139, "draft CR to TS 38.104 on EVM measurement (Annex B and C)"<br>- R4-1905140, "Draft CR: editorial correction on FR1 spurious emission requirement in TS38.104"<br>- R4-1905143, "Draft CR for TS 38.104: Addition of NOTE for transmitter intermodulation requirements in certain regions"<br>- R4-1905144, "Draft CR to TS 38.104: FRC reference corrections for the Rx requirements"<br>- R4-1905145, "Draft CR to TS 38.104: Clarification on application of interfering signal offsets for ACS, blocking and intermodulation requirements"<br>- R4-1905148, "Draft CR to TS 38.104: Corrections on out-of-band blocking requirement"<br><br>From RAN4 #91:<br>- R4-1906002, "Draft CR to 38.104: Subclause 6.7 and 9.8 transmitter intermodulation – correction of interfering signal type"<br>- R4-1906096, "Draft CR to 38.104: Correction of frequency range for OTA spurious emissions"<br>- R4-1906311, "Draft CR to 38.104: Correction on FRC (Annex A)"<br>- R4-1906346, "Removal of n65 in Rel-15 38.104"<br>- R4-1906915, "Draft CR to TS 38.104: Clarification on application of interfering signal offsets for OTA ACS, blocking and intermodulation requirements"<br>- R4-1906918, "Draft CR to TS 38.104: Clarification on type of interfering signal for ACS, in-band blocking and ICS requirements"<br>- R4-1907110, "Draft CR to TS 38.104: correction of the fundamental frequency limit of 2.55GHz for the spurious emissions"<br>- R4-1907246, "Draft CR to TS 38.104: Update of performance requirements for DFT-s-OFDM based PUSCH"<br>- R4-1907249, "Draft CR to TS 38.104: Correction on the terminology in PUSCH FRC tables"<br>- R4-1907252, "Draft CR to TS38.104: Updates of PRACH performance requirements"<br>- R4-1907255, "Draft CR on NR PUCCH format2 performance requirements for TS 38.104" | 15.6.0 |

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| | | | | | | - R4-1907258, "Draft CR on NR UCI on PUSCH performance requirements for TS 38.104"<br>- R4-1907261, "draftCR: Updates to PUCCH formats 3 and 4 performance requirements in TS 38.104"<br>- R4-1907266, "Draft CR on TS 38.104 Performance requirement for PUCCH format 1"<br>- R4-1907267, "Draft CR on TS 38.104 Performance requirement for multi-slot PUCCH format 1"<br>- R4-1907272, "Draft CR to TS 38.104 BS demodulation PUCCH format 0 requirements"<br>- R4-1907275, "Draft CR to TS 38.104 BS demodulation CP-OFDM PUSCH FR2 requirements"<br>- R4-1907277, "draftCR for 38.104 on PUSCH requirements with CP-OFDM and FR1"<br>- R4-1907629, "Draft CR to 38.104: Term “reference signal” replacing by term “ideal signal” in EVM context"<br>- R4-1907634, "Draft CR to 38.104: corrections to the EVM annex"<br>- R4-1907659, "Draft CR to TS 38.104 on Spurious emission Category B in FR2"<br>- R4-1907661, "Draft CR to 38.104 Definition of contiguous transmission bandwidth"<br>- R4-1907662, "Draft CR to 38.104: BS TAE requirements"<br>- R4-1907664, "Draft CR to 38.104: Clarification of interferer RB frequency for narrowband blocking"<br>- R4-1907672, "Draft CR for TS 38.104: Correction on EVM"<br>- R4-1907689, "Correction to CA carrier spacing" | |
| 2019-06 | RAN#84 | RP-191252 | 0024 | 1 | B | CR to TS38.104 to introducing spectrum sharing on band n41 | 16.0.0 |
| 2019-06 | RAN#84 | RP-191242 | 0025 | | B | Introduction of band n14 - CR to TS 38.104 | 16.0.0 |
| 2019-06 | RAN#84 | RP-191246 | 0026 | | B | Introduction of band n30 - CR to TS 38.104 | 16.0.0 |
| 2019-06 | RAN#84 | RP-191244 | 0028 | | B | introduce n18 into TS38.104 | 16.0.0 |
| 2019-06 | RAN#84 | RP-191250 | 0030 | 1 | B | n65 introduction to 38.104 | 16.0.0 |
| 2019-06 | RAN#84 | RP-191251 | 0031 | | B | Addition channel bandwidth of 30MHz for n50 in TS 38.104 | 16.0.0 |
| 2019-06 | RAN#84 | RP-191248 | 0032 | | B | CR to 38.104: Introduction of n48 | 16.0.0 |

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| 2019-09 | RAN#85 | RP-192049 | 0034 | | A | CR to T 38.104: Implementation of endorsed draft CRs from RAN4#92 (Rel-16)<br>(Mirrors changes in R4-1908440 for Rel-15 TS 38.104)<br>- R4-1907940, "Draft CR to TS 38.104: Correction on the terminology in FRC tables in A.1 and A.2"<br>- R4-1908307, "Draft CR to TS 38.104: Clarification on application of OTA receiver requirements for BS supporting polarization"<br>- R4-1908387, "Draft CR for TS38.104: editorial correction for reference meausrement channel"<br>- R4-1908619, "Draft CR to TS38.104: Correction on interferer frequency offset values for ACS"<br>- R4-1908629, "Draft CR to TS38.104: Corrections on EVM window length (Annex B.5.2, C.5.2)"<br>- R4-1908774, "DraftCR to 38.104: Editorial Corrections to redudant units in clause 10.8.3"<br>- R4-1908805, "Draft CR to 38.104: Limits in FCC title 47 for OTA operating band unwanted emissions (9.7)"<br>- R4-1909270, "Draft CR to TS 38.104: Receiver spurious emissions frequency correction"<br>- R4-1909309, "DraftCR to TS 38.104: text corrections, Rel-15"<br>- R4-1909310, "DraftCR to 38.104: correction of TAB connectors mapping to TAB connector TX min cell group, Rel-15"<br>- R4-1909416, "draft CR 38.104 - correct reference to annex F"<br>- R4-1910066, "Draft CR to TS 38.104: Update of performance requirements for DFT-s-OFDM based PUSCH"<br>- R4-1910069, "Draft CR on NR PUCCH format2 performance requirements for TS 38.104"<br>- R4-1910072, "Draft CR on NR UCI on PUSCH performance requirements for TS 38.104"<br>- R4-1910075, "draftCR for 38.104 on PUSCH requirements with CP-OFDM and FR1"<br>- R4-1910078, "Draft CR to TS38.104: Updates to NR PRACH performance requirements"<br>- R4-1910081, "Draft CR to TS 38.104 BS demodulation PUCCH format 0 requirements"<br>- R4-1910084, "Draft CR to TS 38.104 BS demodulation CP-OFDM PUSCH FR2 requirements"<br>- R4-1910088, "Draft CR for 38.104: Performance requirements for NR PUCCH format 1"<br>- R4-1910089, "Draft CR for 38.104: Performance requirements for NR multi-slot PUCCH"<br>- R4-1910094, "draftCR: Updates to PUCCH formats 3 and 4 performance requirements in TS 38.104"<br>- R4-1910431, "Corrections to EVM calculations in 38.141-1 annex B"<br>- R4-1910462, "Draft CR to 38.104: Correction on regional requirements (4.5)"<br>- R4-1910493, "Draft CR to TS 38.104 correction to Annex C.7"<br>- R4-1910606, "Draft CR for TS 38.104: Channel spacing for adjacent NR carriers" | 16.1.0 |
| 2019-09 | RAN#85 | RP-192032 | 0035 | 1 | B | CR to introduce 30MHz bandwidth of n41 to TS 38.104 | 16.1.0 |
| 2019-09 | RAN#85 | RP-192029 | 0036 | | B | CR on Introduction and Protection of SUL band n89 into TS 38.104 | 16.1.0 |
| 2019-09 | RAN#85 | RP-192031 | 0037 | | B | CR for TS 38.104: adding wider channel bandwidths in Band n7 | 16.1.0 |
| 2019-09 | RAN#85 | RP-192034 | 0039 | | B | n29 introduction to 38.104 | 16.1.0 |
| 2019-12 | RAN#86 | RP-192999 | 0049 | 1 | A | CR to TS 38.104: Update of performance requirements for DFT-s-OFDM based PUSCH (Rel-16) | 16.2.0 |
| 2019-12 | RAN#86 | RP-193013 | 0050 | 1 | B | Introduction of 2010-2025MHz SUL band into Rel-16 TS 38.104 | 16.2.0 |
| 2019-12 | RAN#86 | RP-193034 | 0055 | | A | Sync raster to SSB resource element mapping | 16.2.0 |
| 2019-12 | RAN#86 | RP-192999 | 0057 | | A | CR on correction of NR PUCCH format2 performance requirements (Rel-16) for TS 38.104 | 16.2.0 |
| 2019-12 | RAN#86 | RP-192999 | 0059 | 1 | A | CR on correction of NR UCI on PUSCH performance requirements (Rel-16) for TS 38.104 | 16.2.0 |
| 2019-12 | RAN#86 | RP-192999 | 0061 | | A | CR on correction on FRC table for FR1 PUSCH performance requirements (Rel-16) for TS 38.104 | 16.2.0 |
| 2019-12 | RAN#86 | RP-192999 | 0063 | 1 | A | CR for 38.104 on PUSCH requirements with CP-OFDM and FR1 | 16.2.0 |
| 2019-12 | RAN#86 | RP-193034 | 0065 | | A | CR for TS38.104: Corrections on channel bandwdith for band n34 | 16.2.0 |
| 2019-12 | RAN#86 | RP-193035 | 0067 | | A | CR to TS38.104: Editorial corrections | 16.2.0 |
| 2019-12 | RAN#86 | RP-193035 | 0070 | 1 | A | CR to 38.104 on Editorial corrections (Rel-16) | 16.2.0 |
| 2019-12 | RAN#86 | RP-193034 | 0074 | | A | CR to 38.104 on Receiver spurious emission requirements | 16.2.0 |
| 2019-12 | RAN#86 | RP-193021 | 0075 | 1 | F | CR to remove square brackets for n90 in TS38.104 | 16.2.0 |
| 2019-12 | RAN#86 | RP-192999 | 0077 | | A | Updates to PRACH requirements in TS 38.104 for Rel-16 | 16.2.0 |
| 2019-12 | RAN#86 | RP-193034 | 0081 | | A | CR to TS 38.104: Correction on interference level of receiver dynamic range requirement | 16.2.0 |
| 2019-12 | RAN#86 | RP-193034 | 0083 | | A | CR to TS 38.104: Finalization of interfering RB centre frequency offsets in receiver narrowband blocking requirement | 16.2.0 |

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| 2019-12 | RAN#86 | RP-193034 | 0085 | | A | CR to TR 38.104: Correction of table reference of interfering signal for ACS requirement | 16.2.0 |
| 2019-12 | RAN#86 | RP-193034 | 0087 | | A | CR to TS 38.104: Correction on interfering signal frequency offsets for receiver intermodulation requirements | 16.2.0 |
| 2019-12 | RAN#86 | RP-193034 | 0089 | | A | CR to TS 38.104 on corrections to channel raster entries for NR band (Rel-16) | 16.2.0 |
| 2019-12 | RAN#86 | RP-193035 | 0097 | | A | CR to TS38.104: further updates on the abbreviations (section 3.3)-R16 | 16.2.0 |
| 2019-12 | RAN#86 | RP-193023 | 0098 | 1 | B | CR to TS 38.104 - NB-IoT introduction | 16.2.0 |
| 2019-12 | RAN#86 | RP-192999 | 0100 | | F | CR to TS 38.104 BS demodulation PUCCH format 0 requirements | 16.2.0 |
| 2019-12 | RAN#86 | RP-192999 | 0102 | 1 | F | CR to TS 38.104 BS demodulation CP-OFDM PUSCH FR2 requirements | 16.2.0 |
| 2019-12 | RAN#86 | RP-193017 | 0103 | | B | CR to 38.104 - Band n75 - wider CBW | 16.2.0 |
| 2019-12 | RAN#86 | RP-193018 | 0105 | | B | CR for TS 38.104: adding wider channel bandwidths in Band n77/n78 | 16.2.0 |
| 2019-12 | RAN#86 | RP-193016 | 0106 | | B | CR for TS 38.104: Addition channel bandwidth of 40MHz for n38 | 16.2.0 |
| 2019-12 | RAN#86 | RP-193035 | 0108 | | A | CR to 38.104: Correction on FR2 Category B OBUE mask | 16.2.0 |
| 2019-12 | RAN#86 | RP-193002 | 0110 | | A | CR Frame averaging for EVM Annex B and C in TS 38.104 | 16.2.0 |
| 2019-12 | RAN#86 | RP-193035 | 0112 | | A | CR to TS 38.104: OTA TAE correction, Rel-16 | 16.2.0 |
| 2019-12 | RAN#86 | RP-193035 | 0114 | | A | CR to TS 38.104: MR BS class terminology correction, Rel-16 | 16.2.0 |
| 2019-12 | RAN#86 | RP-192999 | 0116 | 1 | F | CR: Updates for PUCCH formats 3 and 4 performance requirements in TS 38.104 (Rel-16) | 16.2.0 |
| 2019-12 | RAN#86 | RP-193035 | 0118 | | A | Correction of limit for TX spurios BS type 1-H | 16.2.0 |
| 2019-12 | RAN#86 | RP-192999 | 0119 | | A | CR for 38.104: Performance requirements for NR PUCCH format 1 | 16.2.0 |
| 2019-12 | RAN#86 | RP-192999 | 0120 | | A | CR for 38.104: Performance requirements for NR multi-slot PUCCH | 16.2.0 |
| 2019-12 | RAN#86 | RP-193035 | 0122 | | A | CR to TS 38.104: Editorial corrections | 16.2.0 |
| 2019-12 | RAN#86 | RP-193035 | 0124 | | A | CR to TS 38.104: Clarification for the number of interfering signals | 16.2.0 |
| 2020-03 | RAN#87 | RP-200381 | 0126 | | B | Introduction of n26 | 16.3.0 |
| 2020-03 | RAN#87 | RP-200382 | 0127 | 1 | B | Introduction of n53 | 16.3.0 |
| 2020-03 | RAN#87 | RP-200384 | 0131 | | B | Introducing new channel bandwidth for band n28 | 16.3.0 |
| 2020-03 | RAN#87 | RP-200410 | 0132 | | F | CR to TS 38.104: Corrections on NB-IoT operation in NR channel bandwidth | 16.3.0 |
| 2020-03 | RAN#87 | RP-200398 | 0134 | 1 | A | CR to TS 38.104: Corrections on rated carrier output power symbols | 16.3.0 |
| 2020-03 | RAN#87 | RP-200379 | 0136 | 1 | C | CR for 38.104: new FRC tables for FR2 PUSCH 2T2R MCS12 | 16.3.0 |
| 2020-03 | RAN#87 | RP-200407 | 0137 | 2 | B | CR for 38.104: introduction of UL timing adjustment | 16.3.0 |
| 2020-03 | RAN#87 | RP-200407 | 0138 | 2 | B | CR for 38.104: Appendix for UL timing adjustment | 16.3.0 |
| 2020-03 | RAN#87 | RP-200386 | 0139 | | B | CR for TS 38.104: adding wider channel bandwidths for n66 | 16.3.0 |
| 2020-03 | RAN#87 | RP-200398 | 0142 | | A | CR to TS 38.104: Regional requirements | 16.3.0 |
| 2020-03 | RAN#87 | RP-200392 | 0143 | | F | Maintenance on the BS BW for n92 and n94 | 16.3.0 |
| 2020-03 | RAN#87 | RP-200398 | 0145 | | A | IntraSlot frequency hopping applicability in the one OFDM symbol test case | 16.3.0 |
| 2020-03 | RAN#87 | RP-200383 | 0146 | 1 | B | CR to 38.104 Band n1 - wider CBW - Additional Channel BW | 16.3.0 |
| 2020-03 | RAN#87 | RP-200385 | 0148 | | B | CR to 38.104 Band n38 - wider CBW - Additional Channel BW | 16.3.0 |
| 2020-03 | RAN#87 | RP-200398 | 0149 | | A | CR to TS 38.104 editorial correction R16 catA | 16.3.0 |
| 2020-03 | RAN#87 | RP-200407 | 0156 | 1 | B | CR for TS 38.104: Introduction of PRACH demodulation requirements for NR HST | 16.3.0 |
| 2020-03 | RAN#87 | RP-200407 | 0157 | 2 | B | CR for 38.104: HST PUSCH demodulation requirements introduction | 16.3.0 |
| 2020-03 | RAN#87 | RP-200407 | 0158 | 2 | B | CR for 38.104: HST PUSCH demodulation Annex including both FRC and channel model | 16.3.0 |
| 2020-03 | RAN#87 | RP-200379 | 0159 | 1 | C | CR for 38.104: Performance requirements for FR2 PUSCH 2T2R 16QAM | 16.3.0 |
| 2020-06 | RAN#88 | RP-200986 | 0164 | | A | CR for 38.104: Performance requirements clarification of PUSCH BS Type O-2 PT-RS configuration for MCS 2 | 16.4.0 |
| 2020-06 | RAN#88 | RP-201043 | 0167 | | F | CR for 38.104: Performance requirements for FR2 PUSCH 2T2R 16QAM | 16.4.0 |
| 2020-06 | RAN#88 | RP-200986 | 0176 | | A | CR to TS 38.104: Correction to out-of-band blocking requirements in subclause 7.5 and subclause 10.6 | 16.4.0 |
| 2020-06 | RAN#88 | RP-200986 | 0178 | | A | CR to TS 38.104: Correction on the CA nominal channel spacing | 16.4.0 |
| 2020-06 | RAN#88 | RP-200986 | 0180 | | A | 30k SSB SCS for n50 | 16.4.0 |
| 2020-06 | RAN#88 | RP-200986 | 0182 | | A | Addition of 30k SSB SCS for Band n38 | 16.4.0 |
| 2020-06 | RAN#88 | RP-200972 | 0185 | | B | CR to TS 38.104: Introduction of FR2 DL 256QAM | 16.4.0 |
| 2020-06 | RAN#88 | RP-200977 | 0190 | | B | CR for TS 38.104: adding 50 MHz CBW for n1 | 16.4.0 |
| 2020-06 | RAN#88 | RP-200986 | 0195 | | A | CR to 38.104: Adding missing clause on Radiated Performance requirements for multi-slot PUCCH (11.3.1) | 16.4.0 |
| 2020-06 | RAN#88 | RP-200980 | 0198 | | B | CR to TS 38.104 - Add 40 MHz CBW in band n3 | 16.4.0 |
| 2020-06 | RAN#88 | RP-200982 | 0199 | | B | CR to TS 38.104 - Add 50 MHz CBW in band n65 | 16.4.0 |
| 2020-06 | RAN#88 | RP-200986 | 0209 | | A | CR to 38.104 on Receiver spurious emissions exclusion band (Rel-16) | 16.4.0 |
| 2020-06 | RAN#88 | RP-200986 | 0211 | | A | CR to 38.104 on EESS protection for bands n257 and n258 (Rel-16) | 16.4.0 |
| 2020-06 | RAN#88 | RP-200986 | 0213 | | A | TS38.104 draft CR on 30KHz SSB SCS for n40 (Rel-16) | 16.4.0 |
| 2020-06 | RAN#88 | RP-200986 | 0207 | 1 | A | CR to 38.104 on Removal of brackets and TBD (Rel-16) | 16.4.0 |
| 2020-06 | RAN#88 | RP-200975 | 0165 | 1 | B | CR for 38.104: HST PUSCH demodulation requirements | 16.4.0 |

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| 2020-06 | RAN#88 | RP-200975 | 0166 | 1 | B | CR for 38.104: HST PUSCH demodulation FRC and channel model annexes | 16.4.0 |
| 2020-06 | RAN#88 | RP-200975 | 0186 | 1 | F | CR for 38.104 Introduction of PRACH demodulation requirements for NR HST | 16.4.0 |
| 2020-06 | RAN#88 | RP-200975 | 0187 | 1 | B | CR for 38.104: Performance requirements for UL timing adjustment | 16.4.0 |
| 2020-06 | RAN#88 | RP-201043 | 0168 | 1 | B | CR for TS 38.104: Introduce PUSCH performance requirements at 30% throughput test point | 16.4.0 |
| 2020-06 | RAN#88 | RP-200986 | 0172 | 1 | F | UCI multiplexed on PUSCH requirement | 16.4.0 |
| 2020-06 | RAN#88 | RP-200978 | 0205 | 1 | B | CR to 38.104 for Introduction of band n259 | 16.4.0 |
| 2020-09 | RAN#89 | RP-201497 | 0218 | 1 | B | CR for 38.104: HST PUSCH demodulation requirements | 16.5.0 |
| 2020-09 | RAN#89 | RP-201497 | 0219 | 1 | B | CR for 38.104: HST PUSCH demodulation FRC and channel model annexes | 16.5.0 |
| 2020-09 | RAN#89 | RP-201502 | 0220 | 1 | B | Introduction of LTE/NR spectrum sharing in band 48/n48 frequency range | 16.5.0 |
| 2020-09 | RAN#89 | RP-201512 | 0222 | | A | CR to TS 38.104: Correction of co-location requirement in subclause 7.5.3 | 16.5.0 |
| 2020-09 | RAN#89 | RP-201512 | 0224 | | A | CR to TS38.104: Add 30k SSB SCS for Band n34 and n39 | 16.5.0 |
| 2020-09 | RAN#89 | RP-201512 | 0227 | 1 | A | CR to TS 38.104: OTA receiver spurious requirements for EESS protection (rel-16) | 16.5.0 |
| 2020-09 | RAN#89 | RP-201499 | 0228 | | F | CR for 38.104: Performance requirements for UL timing adjustment | 16.5.0 |
| 2020-09 | RAN#89 | RP-201507 | 0229 | | F | Correction to NB-IoT Bands with n26 | 16.5.0 |
| 2020-09 | RAN#89 | RP-201512 | 0233 | 1 | A | CR to 38.104: Annex B and C clarification on equlisation calculation (B.6, C.6) | 16.5.0 |
| 2020-09 | RAN#89 | RP-202101 | 0238 | 1 | C | 7.5 kHz UL shift for LTE/NR spectrum sharing in Band 38/n38 | 16.5.0 |
| 2020-09 | RAN#89 | RP-202113 | 0239 | 1 | B | CR to TS 38.104: Introduction of NR-U into BS core specification | 16.5.0 |
| 2020-12 | RAN#90 | RP-202422 | 0242 | 1 | F | CR for 38.104: HST PUSCH demodulation requirements | 16.6.0 |
| 2020-12 | RAN#90 | RP-202416 | 0243 | 1 | B | CR for 38.104: Ultra high reliability BS demodulation requirements | 16.6.0 |
| 2020-12 | RAN#90 | RP-202416 | 0244 | 1 | B | CR for 38.104: Low latency BS demodulation requirements | 16.6.0 |
| 2020-12 | RAN#90 | RP-202415 | 0245 | 2 | B | CR to TS 38.104 BS demodulation requirements for 2-step RACH | 16.6.0 |
| 2020-12 | RAN#90 | RP-202414 | 0247 | 1 | F | CR to TS 38.104 with NR-U remaining open issues updates | 16.6.0 |
| 2020-12 | RAN#90 | RP-202415 | 0248 | 1 | B | CR on BS demodulation requirements for 2-step RACH for FR2 | 16.6.0 |
| 2020-12 | RAN#90 | RP-202416 | 0249 | 1 | B | CR to TS 38.104: Addition of BS performance requirements for URLLC PUSCH repetition Type A | 16.6.0 |
| 2020-12 | RAN#90 | RP-202422 | 0250 | 1 | B | CR for 38.104: Introduction of performance requirements for NR HST PRACH under fading channel | 16.6.0 |
| 2020-12 | RAN#90 | RP-202423 | 0257 | 1 | F | Adding FRC table description in Annex in TS38.104 v16.5.0 | 16.6.0 |
| 2020-12 | RAN#90 | RP-202488 | 0261 | | A | CR to 38.104 on Category B OTA spurious emissions for and n257 | 16.6.0 |
| 2020-12 | RAN#90 | RP-202451 | 0240 | | B | Introduction of 1880-1920MHz SUL band into Rel-17 TS 38.104 | 17.0.0 |
| 2020-12 | RAN#90 | RP-202452 | 0241 | | B | Introduction of 2300-2400MHz SUL band into Rel-17 TS 38.104 | 17.0.0 |
| 2020-12 | RAN#90 | RP-202448 | 0253 | | B | CR to TS 38.104: introduction of NR band n13 | 17.0.0 |
| 2020-12 | RAN#90 | RP-202453 | 0258 | 1 | B | Big CR to 38.104 - Additional Channel BW | 17.0.0 |
| 2020-12 | RAN#90 | RP-202456 | 0214 | 3 | B | LTE/NR spectrum sharing in Band 40/n40 | 17.0.0 |
| 2021-03 | RAN#91e | RP-210096 | 0262 | 2 | B | CR for TS 38.104 Introduction of SUL for UL of NR band n24 | 17.1.0 |
| 2021-03 | RAN#91e | RP-210078 | 0264 | | A | CR for 38.104: HST PUSCH demodulation requirements | 17.1.0 |
| 2021-03 | RAN#91e | RP-210068 | 0266 | | A | CR for 38.104: Low latency FR1 BS demodulation requirements | 17.1.0 |
| 2021-03 | RAN#91e | RP-210068 | 0268 | | A | CR for 38.104: Ultra high reliability BS demodulation requirements | 17.1.0 |
| 2021-03 | RAN#91e | RP-210068 | 0270 | | A | CR on PUSCH repetition type A and PUSCH mapping type B radiated performance requirement for TS 38.104 | 17.1.0 |
| 2021-03 | RAN#91e | RP-210117 | 0273 | | A | CR to TS 38.104: Additions of regional requirements for n41 and n90 in Japan, Rel-17 | 17.1.0 |
| 2021-03 | RAN#91e | RP-210073 | 0275 | | A | CR on correction 2-step RACH performance requirements for FR2 in 38.104 (Rel-17) | 17.1.0 |
| 2021-03 | RAN#91e | RP-210078 | 0277 | | A | CR for 38.104: Cleanup for performance requirements of NR HST PRACH under | 17.1.0 |
| 2021-03 | RAN#91e | RP-210068 | 0279 | | A | CR to TS 38.104: Correction of BS performance requirements for URLLC PUSCH repetition Type A | 17.1.0 |
| 2021-03 | RAN#91e | RP-210084 | 0282 | | A | CR to TS 38.104: system parameters maintenance for NR-U | 17.1.0 |
| 2021-03 | RAN#91e | RP-210084 | 0284 | | A | CR to TS 38.104: corrections of NR-U BS Tx requirements | 17.1.0 |
| 2021-03 | RAN#91e | RP-210084 | 0286 | | A | CR to TS 38.104: corrections of NR-U BS Rx requirements | 17.1.0 |
| 2021-03 | RAN#91e | RP-210117 | 0289 | | A | CR to TS 38.104: EESS protection requirement correction | 17.1.0 |
| 2021-03 | RAN#91e | RP-210189 | 0291 | | B | Big CR to 38.104 - Additional Channel BW | 17.1.0 |
| 2021-03 | RAN#91e | RP-210117 | 0294 | | A | CR to TS38.104: Correction on the Aggregated Channel Bandwidth | 17.1.0 |
| 2021-03 | RAN#91e | RP-210097 | 0295 | | B | CR to 38.104: Introduction of n24 | 17.1.0 |
| 2021-03 | RAN#91e | RP-210084 | 0297 | | A | CR to TS 38.104 corrections to NR-U BS RF Tx requirements | 17.1.0 |
| 2021-03 | RAN#91e | RP-210084 | 0299 | | A | CR to TS 38.104 corrections to NR-U BS RF Rx requirements | 17.1.0 |
| 2021-06 | RAN#92 | RP-211117 | 0305 | | A | Correction of the channel raster of n259 for TS 38.104 | 17.2.0 |
| 2021-06 | RAN#92 | RP-211095 | 0307 | | A | CR to TS 38.104: Corrections on frequency offset symbols for spectrum emission mask for non-transmitted channels | 17.2.0 |
| 2021-06 | RAN#92 | RP-211079 | 0310 | | F | CR to TS 38.104 on BS channel bandwidth per operating band | 17.2.0 |
| 2021-06 | RAN#92 | RP-211105 | 0314 | | A | CR on correction of PUSCH repetition type A and PUSCH mapping type B radiated performance requirements for TS 38.104 | 17.2.0 |
| 2021-06 | RAN#92 | RP-211121 | 0318 | 1 | B | CR to TS 38.104: Introduction of band n262 | 17.2.0 |

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| 2021-06 | RAN#92 | RP-211115 | 0319 | | B | Big CR to TS 38.104: Adding channel BW support in existing NR bands | 17.2.0 |
| 2021-06 | RAN#92 | RP-211116 | 0320 | 1 | B | CR to TS 38.104: Introduction of band n67 | 17.2.0 |
| 2021-06 | RAN#92 | RP-211116 | 0321 | 1 | B | CR to TS 38.104: Introduction of band n85 | 17.2.0 |
| 2021-06 | RAN#92 | RP-211095 | 0323 | | A | CR to 38.104 with correction of NR-U 60 MHz and 80 MHz channels | 17.2.0 |
| 2021-06 | RAN#92 | RP-211094 | 0325 | | A | Big CR for NR-U BS demodulation requirements in TS 38.104 (Rel-17) | 17.2.0 |
| 2021-06 | RAN#92 | RP-211108 | 0329 | | A | CR for 38.104: HST PUSCH demodulation requirements and spec maintenance | 17.2.0 |
| 2021-06 | RAN#92 | RP-211108 | 0331 | | A | CR for 38.104: Low latency and ultra-low BLER FR1 BS demodulation requirements | 17.2.0 |
| 2021-06 | RAN#92 | RP-211090 | 0333 | | A | CR to 38.104: In-band blocking for multi-band Base Stations | 17.2.0 |
| 2021-09 | RAN#93 | RP-211907 | 0343 | | B | Big CR to TS 38.104: Adding channel BW support in existing NR bands | 17.3.0 |
| 2021-09 | RAN#93 | RP-211909 | 0341 | | B | CR for TS 38.104: introduction of channel bandwidths 35MHz and 45MHz<br>NOTE The CR is not implemented | 17.3.0 |
| 2021-09 | RAN#93 | RP-211910 | 0348 | | B | Introduction of the UL 7.5kHz shift for NR TDD band n34 and n39 in 38.104 R17 | 17.3.0 |
| 2021-09 | RAN#93 | RP-211926 | 0351 | | F | Big CR for TS 38.104 Maintenance Demod part (Rel-17, CAT F) | 17.3.0 |
| 2021-09 | RAN#93 | RP-211926 | 0354 | | A | Big CR for TS 38.104 Maintenance RF part (Rel-17, CAT A) | 17.3.0 |
| 2021-12 | RAN#94 | RP-212835 | 0356 | | B | Big CR to TS 38.104: Adding channel BW support in existing NR bands | 17.4.0 |
| 2021-12 | RAN#94 | RP-212836 | 0357 | | F | CR to TS 38.104 - Fix to band n85 introduction | 17.4.0 |
| 2021-12 | RAN#94 | RP-212825 | 0355 | 2 | B | CR for TS 38.104: 1024QAM | 17.4.0 |
| 2021-12 | RAN#94 | RP-212824 | 0360 | | F | Big CR for TS 38.104 Maintenance RF part (Rel-17, CAT F) | 17.4.0 |
| 2021-12 | RAN#94 | RP-212850 | 0362 | | F | Big CR for TS 38.104 Maintenance Demod part (Rel-17, CAT F) | 17.4.0 |
| 2021-12 | RAN#94 | RP-213147 | 0363 | | B | CR for TS 38.104: introduction of channel bandwidths 35MHz and 45MHz<br>NOTE The CR is a resubmission of CR R4-2114918 of RP-211909 | 17.4.0 |
| 2022-03 | RAN#95-e | RP-220377 | 0364 | 1 | B | CR for TS 38.104: capturing HAPS requirements | 17.5.0 |
| 2022-03 | RAN#95-e | RP-220347 | 0365 | 1 | B | CR to TS38.104 on introduction of upper 700MHz A block | 17.5.0 |
| 2022-03 | RAN#95-e | RP-220357 | 0369 | 1 | B | CR for 38.104 to introduce n102 | 17.5.0 |
| 2022-03 | RAN#95-e | RP-220361 | 0372 | | B | BigCR for TS38.104: Introduction of conformance testing requirements for FR1 PUSCH 256QAM | 17.5.0 |
| 2022-03 | RAN#95-e | RP-220334 | 0374 | | F | Big CR for TS 38.104 Maintenance RF part (Rel-17, CAT F) | 17.5.0 |
| 2022-03 | RAN#95-e | RP-220334 | 0376 | | A | Big CR for TS 38.104 Maintenance Demod part (Rel-17, CAT A) | 17.5.0 |
| 2022-03 | RAN#95-e | RP-220376 | 0378 | | B | Big CR to 38.104: RMR 1900MHz band n101 introduction | 17.5.0 |
| 2022-06 | RAN#96 | RP-221680 | 0379 | | B | Big CR on FR2 HST BS demodulation requirement for TS 38.104 | 17.6.0 |
| 2022-06 | RAN#96 | RP-221684 | 0380 | 1 | B | CR to TS 38.104 - Tx requirements: RMR 900MHz band introduction | 17.6.0 |
| 2022-06 | RAN#96 | RP-221671 | 0381 | | B | Big CR to TS 38.104: Adding channel BW support in existing NR bands | 17.6.0 |
| 2022-06 | RAN#96 | RP-221684 | 0382 | 1 | B | CR to 38.104 on introduction of n100 (system parameters) | 17.6.0 |
| 2022-06 | RAN#96 | RP-221673 | 0383 | 1 | B | CR to TS38.104 the introduction of 6425-7125MHz | 17.6.0 |
| 2022-06 | RAN#96 | RP-221676 | 0384 | | B | Big CR to 38.104 for Rel-17 NR extension up to 71 GHz introduction | 17.6.0 |
| 2022-06 | RAN#96 | RP-221684 | 0385 | | B | CR to TS 38.104: RMR900 Rx requirements for band n100, Rel-17 | 17.6.0 |
| 2022-06 | RAN#96 | RP-221676 | 0386 | | B | Big CR on extending NR to 71GHz for TS 38.104 | 17.6.0 |
| 2022-06 | RAN#96 | RP-221655 | 0389 | | F | Big CR for TS 38.104 Maintenance RF part (Rel-17, CAT F) | 17.6.0 |
| 2022-06 | RAN#96 | RP-221659 | 0391 | | A | Big CR for TS 38.104 Maintenance Demod part (Rel-17, CAT A) | 17.6.0 |
| 2022-09 | RAN#97 | RP-222038 | 0394 | 1 | F | CR to 38.104 on n100 corrections | 17.7.0 |
| 2022-09 | RAN#97 | RP-222041 | 0395 | | F | Big CR on FR2 HST BS demodulation requirement for TS 38.104 | 17.7.0 |
| 2022-09 | RAN#97 | RP-222050 | 0401 | 1 | B | CR for 38.104: Add 5 MHz channel BW for n41 | 17.7.0 |
| 2022-09 | RAN#97 | RP-222026 | 0404 | | F | Big CR for TS 38.104 Maintenance RF part (Rel-17, CAT F) | 17.7.0 |
| 2022-09 | RAN#97 | RP-222026 | 0407 | | F | Big CR for TS 38.104 Maintenance Demod part (Rel-17, CAT F) | 17.7.0 |
| 2022-09 | RAN#97 | RP-222051 | 0408 | | F | Big CR to 38.104 for Rel-17 NR extension up to 71 GHz maintenance (Rel-17, CAT F) | 17.7.0 |
| 2022-09 | RAN#97 | RP-222051 | 0410 | | B | Big CR of TS38.104 on system parameter updates for FR2-2 | 17.7.0 |
| 2022-12 | RAN#98-e | RP-223307 | 0411 | | F | CR to TS 38.104 on reference to FRCs | 17.8.0 |
| 2022-12 | RAN#98-e | RP-223307 | 0412 | | F | CR to TS 38.104: Correction of guardband for FR2-2 in sub-clause 5.3.3 | 17.8.0 |
| 2022-12 | RAN#98-e | RP-223300 | 0413 | | B | Big CR for coverage enhancement performance requirements for TS38.104 | 17.8.0 |
| 2022-12 | RAN#98-e | RP-223308 | 0414 | 1 | B | Corrections for 100MHz CBW inclusion for n46 n96 and n102 | 17.8.0 |
| 2022-12 | RAN#98-e | RP-223309 | 0415 | | F | CR to TS 38.104 on tables for BS channel bandwidths | 17.8.0 |
| 2022-12 | RAN#98-e | RP-223297 | 0419 | | F | CR for TS38.104 correction on normal PUSCH requirements (Rel-17) | 17.8.0 |
| 2022-12 | RAN#98-e | RP-223309 | 0424 | 2 | F | CR to 38.104: Spurious emission requirements, Rel-17 | 17.8.0 |
| 2022-12 | RAN#98-e | RP-223302 | 0426 | | B | Big CR for TS 38.104 Demodulation requirements for Enhanced IIOT and URLLC support (Rel-17, CAT B) | 17.8.0 |
| 2022-12 | RAN#98-e | RP-223298 | 0429 | | F | CR for 38.104: define home class as one kind of LA BS | 17.8.0 |

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| 2022-12 | RAN#98-e | RP-223524 | 0432 | 1 | A | CR on R17 TS 38.104 to add channel raster exception for band n28 [n28_BS40MHz_raster] | 17.8.0 |
| 2023-03 | RAN#99 | RP-230513 | 0434 | | F | Correction of coverage enhancement BS demodulation requirement in 38.104 (R17 F) | 17.9.0 |
| 2023-03 | RAN#99 | RP-230510 | 0438 | | B | Big CR to 38.104: demodulation requirements introduction for FR2-2 | 17.9.0 |
| 2023-03 | RAN#99 | RP-230515 | 0440 | | F | CR for TS 38.104 Demodulation requirements for Enhanced IIOT and URLLC support (Rel-17, CAT F) | 17.9.0 |
| 2023-03 | RAN#99 | RP-230506 | 0448 | 1 | F | CR to 38.104: FRC number corrections (Rel-17) | 17.9.0 |
| 2023-03 | RAN#99 | RP-230513 | 0451 | | F | CR to 38.104: BS Conformance, OBUE correction for 6G licensed band | 17.9.0 |
| 2023-03 | RAN#99 | RP-230514 | 0458 | | F | CR to TS 38.104 on table references for OTA operating band unwanted emission limits | 17.9.0 |
| 2023-03 | RAN#99 | RP-230514 | 0459 | | D | CR to TS 38.104: NR-ARFCN table reference correction for band n263, Rel-17 | 17.9.0 |
| 2023-03 | RAN#99 | RP-230514 | 0463 | | F | CR to 38.104: Correction on Rx intermodulation requirements for FR2-2 (Rel-17) | 17.9.0 |
| 2023-06 | RAN#100 | RP-231347 | 0465 | 3 | F | CR 38.104: PUSCH requirements for FR2-2 | 17.10.0 |
| 2023-06 | RAN#100 | RP-231347 | 0467 | 1 | F | CR to TS 38.104 on corrections of table references for extending current NR operation to 71GHz | 17.10.0 |
| 2023-06 | RAN#100 | RP-231348 | 0469 | | F | CR to TS 38.104 on corrections and clarifications of operating band unwanted emission limits for band n104 | 17.10.0 |
| 2023-06 | RAN#100 | RP-231347 | 0476 | | F | CR for TS38.104 FRC tables for FR2-2 PUSCH demodulation_Rel-17 | 17.10.0 |
| 2023-06 | RAN#100 | RP-231350 | 0478 | 1 | F | CR for TS38.104 on FRC table for PUSCH TBoMS_Rel-17 | 17.10.0 |
| 2023-06 | RAN#100 | RP-231352 | 0481 | 1 | A | CR for TS 38.104: Operating band unwanted emissions for Single RAT BS supporting multi-band operation | 17.10.0 |
| 2023-06 | RAN#100 | RP-231339 | 0484 | | F | CR to TS 38.104 - Maintenance related to bands n100 and n101 | 17.10.0 |
| 2023-06 | RAN#100 | RP-231352 | 0487 | | A | CR on 38.104: Updates to FRC of PUSCH requirements(Rel-17) | 17.10.0 |
| 2023-06 | RAN#100 | RP-231342 | 0489 | | F | CR on 38.104 Updates to the requirements for the PF0 requirements with sub-slot repetition | 17.10.0 |
| 2023-06 | RAN#100 | RP-231356 | 0493 | 1 | F | CR to 38.104: Correction to ACLR and CACLR requirement | 17.10.0 |
| 2023-06 | RAN#100 | RP-231356 | 0495 | | A | CR to 38.104: Correction to ACLR and CACLR requirement | 17.10.0 |
| 2023-09 | RAN#101 | RP-232488 | 0504 | | F | [ CR for configuration of FR1 PUSCH TBoMS demodulation requirement | 17.11.0 |
| 2023-09 | RAN#101 | RP-232504 | 0507 | | A | CR to 38.104: Correction to ACLR and CACLR requirement | 17.11.0 |
| 2023-09 | RAN#101 | RP-232490 | 0513 | | F | [NR_ext_to_71GHz-Perf] CR on 38.104: Clean up the brackets for FR2-2 PUSCH requirements | 17.11.0 |
| 2023-09 | RAN#101 | RP-232505 | 0502 | 1 | F | Update to table format for enabling automated data scraping | 17.11.0 |
| 2023-09 | RAN#101 | RP-232496 | 0517 | 1 | F | [NR_RAIL_EU_900MHz-Core, NR_RAIL_EU_1900MHz_TDD-Core] CR to TS 38.104: corrections of RMR-specific BS requirements for band n100 and n101, Rel-17 | 17.11.0 |
| 2023-09 | RAN#101 | RP-232486 | 0509 | 1 | F | [NR_6GHz-Core] CR to 38.104: applicability note for n104 | 17.11.0 |
| 2023-12 | RAN#102 | RP-233342 | 0520 | | A | [NR_unlic] CR for 38.104: Removal of applicability rule (Rel-17, Cat A) | 17.12.0 |
| 2023-12 | RAN#102 | RP-233340 | 0523 | | A | [NR_RF_FR1-Core] CR to TS 38.141-1 on correction of transmitter spurious emissions for protection of Band n20 | 17.12.0 |
| 2023-12 | RAN#102 | RP-233345 | 0529 | | F | Removing 20MHz channel raster points for 5925-5945MHz in the lower 6GHz bands | 17.12.0 |
| 2023-12 | RAN#102 | RP-233339 | 0535 | | A | Addition of 30 KHz SCS for Sync Raster for Band n53 | 17.12.0 |
| 2023-12 | RAN#102 | RP-233345 | 0548 | | F | Removing 20MHz channel raster points for 5925-5945MHz in the lower 6GHz bands | 17.12.0 |
| 2024-03 | RAN#103 | RP-240552 | 0589 | | F | (TEI17) CR to TS 38.104: Clean up of operating band definitions tables in subclause 5.2 | 17.13.0 |
| 2024-03 | RAN#103 | RP-240553 | 0587 | 1 | F | (TEI17) CR to TS 38.104 - BS spurious receiver protection note generalization R17 | 17.13.0 |
| 2024-03 | RAN#103 | RP-240556 | 0571 | | A | (MB_MSR_RF) CR to 38.104: clarification on requirements for BS capable of multi-band operation | 17.13.0 |
| 2024-03 | RAN#103 | RP-240557 | 0550 | | F | (NR_6GHz-Core) CR for TS 38.104, Correction on reference of PREFSENS for in-band blocking and out-of-band blocking for band n104 | 17.13.0 |
| 2024-03 | RAN#103 | RP-240563 | 0557 | | F | CR to TS 38.104 for editorial corrections of operating bands (Rel-17) | 17.13.0 |
| 2024-03 | RAN#103 | RP-240565 | 0582 | | A | CR to 38.104: Correction on transmitter intermodulation additional requirements (Rel-17) | 17.13.0 |
| 2024-03 | RAN#103 | RP-240565 | 0554 | 1 | F | (NR_newRAT-Core) Correction for TS 38.104 R17 | 17.13.0 |
| 2024-03 | RAN#103 | RP-240567 | 0560 | | A | CR to TS 38.104 on correction of radiated performance requirement (Rel-17) | 17.13.0 |
| 2024-03 | RAN#103 | RP-240572 | 0562 | | F | (NR_RAIL_EU_900MHz-Core) CR to TS 38.104 on additional unwanted emission limits for band n100 | 17.13.0 |
| 2024-03 | RAN#103 | RP-240572 | 0595 | 1 | F | (NR_RAIL_EU_900MHz-Core, NR_RAIL_EU_1900MHz_TDD-Core) CR to TS 38.104: correction on n100/n101 deployment aspects and additional n100 OBUE requirement, Rel-17 | 17.13.0 |
| 2024-03 | RAN#103 | RP-240574 | 0574 | | A | (NR_RF_FR1) CR to TS 38.104: Channel raster to resource element mapping | 17.13.0 |

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| 2024-06 | RAN#104 | RP-241384 | 0609 | | F | [NR_newRAT-Perf] CR to TS 38.104 (Corrections on Clause 8.3.4.1.1 and 11.3.2.6.2) (rel 17) | 17.14.0 |
| 2024-06 | RAN#104 | RP-241396 | 0619 | | F | CR for 38.104 on FR2-2 PUSCH demodualtion requirements | 17.14.0 |
| 2024-06 | RAN#104 | RP-241492 | 0628 | | A | (NR_unlic_enh-Core) CR to TS 38.104 NR-U Nominal channel spacing | 17.14.0 |
| 2024-06 | RAN#104 | RP-241383 | 0603 | 1 | F | (NR_newRAT-Core) CR for Total power dynamic range (TPDR) | 17.14.0 |
| 2024-06 | RAN#104 | RP-241398 | 0633 | 1 | F | CR to TS 38.104: Clarification on multiple carrier operation for n100/n101, Rel 17 | 17.14.0 |
| 2024-06 | RAN#104 | RP-241398 | 0636 | 1 | F | CR to TS 38.104: Clarifications on RMR terminology and related operating bands | 17.14.0 |
| 2024-06 | RAN#104 | RP-241393 | 0643 | 1 | F | (NR_bands_R17_BWs-Core) CR for Tx intermodulation core requirements in certain region | 17.14.0 |
| 2024-06 | RAN#104 | RP-241383 | 0639 | 2 | F | CR to TS 38.104: Alignment with UE specification for n30 and n77 notes in Table 5.2-1 | 17.14.0 |
| 2024-09 | RAN#105 | RP-242155 | 0649 | 1 | F | (NR_6GHz_unlic_EU-Core) CR to 38.104 on ACLR and CACLR in non-contiguous spectrum | 17.15.0 |
| 2024-09 | RAN#105 | RP-242183 | 0654 | | F | (NR_RAIL_EU_900MHz-Core, NR_RAIL_EU_1900MHz_TDD-Core) CR to TS 38.104 on clarification on multiple carrier operation for bands n100 and n101 | 17.15.0 |
| 2024-09 | RAN#105 | RP-242189 | 0658 | | F | (NR_SUL_UL_n24-Core) Missing reference to FCC Order DA 20-48 | 17.15.0 |
| 2024-09 | RAN#105 | RP-242174 | 0660 | | D | (NR_newRAT-Core) Fixing notes numbering in Table 5.2-1 | 17.15.0 |
| 2024-09 | RAN#105 | RP-242193 | 0662 | 1 | B | (TEI17) CR to TS 38.104 - BS spurious receiver protection note [MSR_BSRF_RX] | 17.15.0 |
| 2024-12 | RAN#106 | RP-243054 | 0667 | | A | (NR_newRAT-Core) CR to TS 38104 - EESS protection CEPT | 17.16.0 |
| 2024-12 | RAN#106 | RP-243030 | 0672 | | A | [LTE410_Europe_PPDR-Core] CR to 38.104 on correction of Band 87 and 88 co-existence and co-location requirements (Rel-17) | 17.16.0 |
| 2025-03 | RAN#107 | RP-250589 | 0682 | | F | (NR_6GHz_unlic_EU-Core) CR for TS38.104: Correction on channel raster for band n102 | 17.17.0 |
| 2025-03 | RAN#107 | RP-250595 | 0691 | 1 | F | (NR_RAIL_EU_900MHz-Core, NR_RAIL_EU_1900MHz_TDD-Core) NB-IoT operation modes clarification for FRMCS BS in multi-carrier operation | 17.17.0 |
| 2025-06 | RAN#108 | RP-250916 | 0703 | | A | (NR_unlic-Perf) CR to 38.104: Correction to FRC parameters for FR1 interlaced PUSCH performance requirements | 17.18.0 |
| 2025-06 | RAN#108 | RP-250919 | 0706 | 1 | F | (NR_HST_FR1_enh-Perf) CR to 38.104: Correction to FRC parameters for HST FR1 PUSCH performance requirements | 17.18.0 |
| 2025-06 | RAN#108 | RP-250918 | 0709 | 1 | F | (NR_ext_to_71GHz-Perf) CR to 38.104: Correction to FRC parameters for FR2-2 PUSCH performance requirements with transform precoding enabled | 17.18.0 |
| 2025-09 | RAN#109 | RP-252380 | 0730 | | F | (NR_6GHz-Core) CR to 38.104 correction on the band definition for n104 | 17.19.0 |
| 2025-09 | RAN#109 | RP-252380 | 0722 | 1 | F | (NR_6GHz) CR to TS 38.104: Correction of requirement parameter for band n104 in Clause 10.3.2 | 17.19.0 |
| 2025-12 | RAN#110 | RP-253672 | 0744 | | F | (NR_ext_to_71GHz-Perf) Correction to Table G.2.1.2-4 | 17.20.0 |
| 2025-12 | RAN#110 | RP-253684 | 0746 | 1 | F | (TEI17) Correction of Operating band unwanted emissions for multi-band connector | 17.20.0 |
| 2026-03 | RAN#111 | RP-260456 | 0759 | | A | (NR_NewRAT-Core) CR to TS 38.104 on Rx IMD | 17.21.0 |
| 2026-06 | RAN#112 | RP-261408 | 0781 | | A | CR to TS 38.104: [] removal for NB-IoT FRCs (G-FR1-A1-10, G-FR1-A1-11) | 17.22.0 |
| 2026-06 | RAN#112 | RP-261412 | 0785 | | A | CR to TS 38.104: removal od doubled entry of the Note on SCell restriction [Scell_note] | 17.22.0 |

# History

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